

# **Stylus Common UI Database Object Information**

**Product Version 26.10  
May 2026**

© 2026 Cadence Design Systems, Inc.  
Printed in the United States of America.

Cadence Design Systems, Inc. (Cadence), 2655 Seely Ave., San Jose, CA 95134, USA.

Open SystemC, Open SystemC Initiative, OSCI, SystemC, and SystemC Initiative are trademarks or registered trademarks of Open SystemC Initiative, Inc. in the United States and other countries and are used with permission.

**Trademarks:** Trademarks and service marks of Cadence Design Systems, Inc. (Cadence) contained in this document are attributed to Cadence with the appropriate symbol. For queries regarding Cadence's trademarks, contact the corporate legal department at the address shown above or call 800.862.4522.

All other trademarks are the property of their respective holders.

**Restricted Permission:** This publication is protected by copyright law and international treaties and contains trade secrets and proprietary information owned by Cadence. Unauthorized reproduction or distribution of this publication, or any portion of it, may result in civil and criminal penalties. Except as expressly permitted below, this publication may not be copied, reproduced, modified, published, uploaded, posted, transmitted, or distributed in any way, including automated processes, training or services involving a large language model, foundation model, deep machine learning, generative artificial intelligence, or any other similar process or technology, without prior written permission from Cadence. Unless otherwise agreed to by Cadence in writing, customers are granted permission to print one (1) hard copy of this publication subject to the following conditions:

1. The publication may be used only in accordance with a written agreement between Cadence and its customer.
2. The publication may not be modified in any way.
3. Any authorized copy of the publication or portion thereof must include all original copyright, trademark, and other proprietary notices and this permission statement.
4. The information contained in this document cannot be used in the development of like products or software, whether for internal or external use, and shall not be used for the benefit of any other party, whether or not for consideration.

**Disclaimer:** Information in this publication is subject to change without notice and does not represent a commitment on the part of Cadence. Except as may be explicitly set forth in such agreement, Cadence does not make, and expressly disclaims, any representations or warranties as to the completeness, accuracy or usefulness of the information contained in this document. Cadence does not warrant that use of such information will not infringe any third party rights, nor does Cadence assume any liability for damages or costs of any kind that may result from use of such information. Cadence is committed to using respectful language in our code and communications. We are also active in the removal and replacement of inappropriate language from existing content. This product documentation may however contain material that is no longer considered appropriate but still reflects long-standing industry terminology. Such content will be addressed at a time when the related software can be updated without end-user impact.

**Restricted Rights:** Use, duplication, or disclosure by the Government is subject to restrictions as set forth in FAR52.227-14 and DFAR252.227-7013 et seq. or its successor.

---

# Contents

---

|                         |     |
|-------------------------|-----|
| 1                       | 7   |
| Types and Definitions   | 7   |
| 2                       | 12  |
| Database Objects        | 12  |
| analysis_view           | 12  |
| antenna_data            | 15  |
| antenna_model           | 16  |
| arc                     | 19  |
| attribute               | 30  |
| base_cell               | 34  |
| base_pin                | 52  |
| boundary                | 66  |
| bump                    | 67  |
| bump_grid               | 71  |
| bump_pin                | 72  |
| bus                     | 74  |
| bus_guide               | 75  |
| bus_sink_group          | 77  |
| clock                   | 78  |
| clock_spine             | 90  |
| clock_tree              | 90  |
| clock_tree_source_group | 107 |
| constraint_mode         | 108 |
| context_data            | 110 |
| context_top_clock       | 111 |
| def_scan_chain          | 112 |
| delay_corner            | 114 |
| delay_noise_obj         | 118 |
| density_shape           | 121 |
| design                  | 122 |
| flexible_htrie          | 167 |
| flow                    | 176 |
| flow_step               | 178 |

|                   |     |
|-------------------|-----|
| foreign_cell      | 181 |
| gcell             | 182 |
| gcell_grid        | 185 |
| group             | 186 |
| gui_line          | 192 |
| gui_polygon       | 193 |
| gui_rect          | 195 |
| gui_text          | 196 |
| hinst             | 197 |
| hnet              | 213 |
| hpin              | 217 |
| hport             | 228 |
| inst              | 232 |
| inst_obs          | 288 |
| inst_obs_shape    | 288 |
| io_constraint     | 290 |
| layer             | 294 |
| layer_rule        | 301 |
| layer_shape       | 302 |
| lef_set           | 304 |
| lib_arc           | 304 |
| lib_cell          | 309 |
| lib_pin           | 332 |
| library           | 348 |
| library_set       | 354 |
| marker            | 355 |
| message           | 358 |
| module            | 360 |
| net               | 373 |
| net_group         | 408 |
| obj_type          | 410 |
| opcond            | 411 |
| package_component | 413 |
| package_object    | 414 |
| partition         | 416 |
| patch_wire        | 421 |
| path_group        | 423 |

|                       |      |
|-----------------------|------|
| pg_base_pin           | 424  |
| pg_pin                | 430  |
| physical_pin          | 433  |
| pin                   | 434  |
| pin_blockage          | 507  |
| pin_group             | 508  |
| pin_guide             | 511  |
| place_blockage        | 512  |
| port                  | 514  |
| port_shape            | 566  |
| power_domain          | 568  |
| preferred_cell_stripe | 576  |
| rc_corner             | 577  |
| resistor              | 581  |
| resize_blockage       | 585  |
| root                  | 586  |
| route_blockage        | 1143 |
| route_rule            | 1147 |
| route_type            | 1148 |
| row                   | 1153 |
| sdp                   | 1155 |
| shape                 | 1158 |
| shape_via             | 1159 |
| site                  | 1161 |
| skew_group            | 1163 |
| special_via           | 1166 |
| special_wire          | 1170 |
| stack_via_rule        | 1174 |
| tech                  | 1174 |
| text                  | 1177 |
| timing_condition      | 1179 |
| timing_path           | 1180 |
| timing_point          | 1208 |
| track_pattern         | 1214 |
| trim_grid             | 1216 |
| via                   | 1218 |
| via_def               | 1221 |

|                                |      |
|--------------------------------|------|
| via_def_rule                   | 1225 |
| virtual_wire                   | 1228 |
| what_if_via                    | 1229 |
| what_if_wire                   | 1230 |
| wire                           | 1232 |
| port_bus                       | 1234 |
| hport_bus                      | 1236 |
| hpin_bus                       | 1237 |
| test_signal                    | 1239 |
| formal_verification_constraint | 1245 |
| test_clock                     | 1246 |
| scan_chain                     | 1249 |
| actual_scan_chain              | 1254 |
| violation                      | 1258 |
| dft_configuration_mode         | 1261 |
| tap_distance_rule              | 1263 |
| tap_layer_distance             | 1264 |
| interposer_route_rule          | 1265 |
| seq_function                   | 1266 |
| test_clock_domain              | 1267 |
| scan_segment                   | 1268 |
| actual_scan_segment            | 1276 |
| isolation_rule                 | 1282 |
| level_shifter_rule             | 1285 |
| retention_rule                 | 1287 |
| power_switch_rule              | 1290 |
| ndr_eol_spacing                | 1292 |
| inst_physical_pin              | 1292 |

# Types and Definitions

The following is the list of allowed data types and their descriptions:

| Type   | Description                                                               | Accepted Input (from TCL)                                                                                                                                                                                                                       | Output Format (to TCL)                                                                                     |
|--------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| bool   | A boolean value.                                                          | true: true/1/on/yes<br>false: false/0/off/no                                                                                                                                                                                                    | true or false                                                                                              |
| int    | Signed 32-bit integer corresponding to Tcl integer.                       | Any integer value.                                                                                                                                                                                                                              | 26                                                                                                         |
| double | Double-precision (64-bit) signed float.                                   | Any numeric value.                                                                                                                                                                                                                              | 26.4<br>A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1). |
| string | A string. Unlike other types, the empty string is a valid value.          | Any Tcl string value, like "abc" or {abc}.                                                                                                                                                                                                      | abc                                                                                                        |
| enum   | One of a fixed set of string values (provided in the type qualification). | If "true" or "false" are enum values, these are interpreted as booleans.<br>For example: enum (true false auto) accepts<br>- "0", "no" and "off" as synonyms for "false"<br>- "1", "yes" and "on" as synonyms for "true"<br>- "auto" for "auto" | auto                                                                                                       |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| obj      | <p>A database object pointer. It is followed by the list of object types allowed for this attribute. It is optionally followed by</p> <ul style="list-style-type: none"> <li>* object list of 0 or more</li> <li>+ object list of 1 or more</li> <li>? object link may be empty</li> </ul> <p>For example, obj(inst hinst)+ is a list of 1 or more inst or hinst objects. A mixture of both inst and hinst objects is possible.</p> | <p>A dual-ported Tcl object or object list. The string form for netlist objects is &lt;obj_type&gt;:<br/>&lt;design_name&gt;/&lt;object_name&gt;, for example:</p> <p>inst:my_design/i1/i2</p> <p>For technology or library objects it is just &lt;obj_type&gt;:&lt;name&gt;, for example:</p> <p>layer:metal1</p> <p>For objects with no useful name (like a wire), it is just the hex memory pointer, for example:</p> <p>wire:0x123300012</p> <p>Type "man get_db" for more details on dual-ported Tcl objects.</p> | <p>layer:metal1</p> <p>inst:my_design/i1/i2</p> <p>wire:0x123300012</p>                                               |
| in_file  | A file name that will be read in.                                                                                                                                                                                                                                                                                                                                                                                                   | A Unix path name (relative or absolute). The file must exist and be readable.                                                                                                                                                                                                                                                                                                                                                                                                                                          | my_dir/myfile.tcl                                                                                                     |
| out_file | A file that will be written to.                                                                                                                                                                                                                                                                                                                                                                                                     | A Unix path name (relative or absolute). The file must be writable.                                                                                                                                                                                                                                                                                                                                                                                                                                                    | my_dir/my_file.tcl                                                                                                    |
| in_dir   | A directory to read from.                                                                                                                                                                                                                                                                                                                                                                                                           | A Unix path name (relative or absolute). The directory must exist and be readable.                                                                                                                                                                                                                                                                                                                                                                                                                                     | my_dir                                                                                                                |
| out_dir  | A directory to write into.                                                                                                                                                                                                                                                                                                                                                                                                          | A Unix path name (relative or absolute). The directory must be writable.                                                                                                                                                                                                                                                                                                                                                                                                                                               | my_dir                                                                                                                |
| coord    | A single, linear distance in microns that type double..                                                                                                                                                                                                                                                                                                                                                                             | Any numerical value in units of microns.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>26.4</p> <p>A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).</p> |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                 |                                             |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| point | <p>A point in 2D space, expressed as a two-element Tcl list of coord types: {&lt;x&gt; &lt;y&gt;}.<br/>Has sub types:<br/>.x - the first coordinate<br/>.y - the second coordinate</p>                                                                                                                                                                                                                                                             | <p>Any list format that can be translated into two coordinates. For example, the following are allowed:<br/>{1.0 2.2}<br/>{{1} {2.2}}</p>                                                                                                                                                                                       | {1.0 2.2}                                   |
| line  | <p>A line in 2D space, expressed as a two-element Tcl list of points: {{&lt;x1&gt; &lt;y1&gt;} {&lt;x2&gt; &lt;y2&gt;}}.<br/>Has sub types:<br/>.begin - first point<br/>.end - second point<br/>.length - length of the line<br/>.dx - difference in x coordinates<br/>.dy - difference in y coordinates</p>                                                                                                                                      | <p>Any list format that can be translated into two points. For example, the following are allowed as input.<br/>{{1.0 2} {3 4.0}}<br/>{1 2 3 4}<br/>{{1} {2} {3} {4.0}}</p>                                                                                                                                                     | {{1.0 2.2} {3.0 4.0}}                       |
| rect  | <p>A rectangle in 2D space, expressed as an ordered list of coordinates {&lt;left_x&gt; &lt;lower_y&gt; &lt;right_x&gt; &lt;upper_y&gt;}.<br/>Has sub types:<br/>.ll - lower left point<br/>.ur - upper right point<br/>.dx - difference in x coordinates<br/>.dy - difference in y coordinates<br/>.area - area of the rectangle<br/>.width - shorter of dx &amp; dy<br/>.length - longer of dx and dy<br/>.perimeter - perimeter of the rect</p> | <p>Any list format that can be translated into two points. For example, the following are equivalent:<br/>{{1 2} {3 4}}<br/>{1 2 3 4}<br/>The two X coordinates and two Y coordinates can be input in either order. They will be ordered as lower-left for the first point, and upper-right as the second point for output.</p> | A list of four coords:<br>{1.0 2.0 3.0 4.0} |

|                    |                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                      |                                                                                                                                                                       |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>polygon</p>     | <p>A polygon in 2D space, expressed as a list of points {{&lt;x1&gt; y1&gt;}&lt;x2&gt; &lt;y2&gt;} {&lt;x3&gt; &lt;y3&gt;}...}.</p> <p>Has sub types:</p> <ul style="list-style-type: none"> <li>.bbox - bounding box (rect type)</li> <li>.area - area of the rectangle</li> <li>.perimeter - perimeter of the rectangle</li> </ul> | <p>Any list format that can be translated into multiple points. For example, the following are equivalent:</p> <p style="margin-left: 20px;">{{1 2} {3 4} {5 6}}</p> <p style="margin-left: 20px;">{1 2 3 4 5 6}</p> | <p>A list of points: {{1.0 2.0} {3.0 4.0} {5.0 6.0}}.</p>                                                                                                             |
| <p>area</p>        | <p>An area in units of micron squared that is type double.</p>                                                                                                                                                                                                                                                                       | <p>Any numerical value in units of microns squared.</p> <p>26.4</p>                                                                                                                                                  | <p>Value in micron squared without units. Example: 26.4</p> <p>A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).</p> |
| <p>delay</p>       | <p>A time duration that is type double.</p>                                                                                                                                                                                                                                                                                          | <p>Any numerical value in user units. The units can be queried with "get_db timing_time_unit".</p>                                                                                                                   | <p>26.4</p> <p>A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).</p>                                                 |
| <p>resistance</p>  | <p>A resistance that is type double.</p>                                                                                                                                                                                                                                                                                             | <p>Any numerical value in user units. The units can be queried with "get_resistance_unit".</p>                                                                                                                       | <p>26.4</p> <p>A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).</p>                                                 |
| <p>capacitance</p> | <p>A capacitance that is type double.</p>                                                                                                                                                                                                                                                                                            | <p>Any numerical value in user units. The units can be queried with "get_db timing_cap_unit".</p>                                                                                                                    | <p>26.4</p> <p>A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).</p>                                                 |

**Stylus Common UI Database Object Information**  
Types and Definitions

---

|         |                                |                                        |                                                                                                           |
|---------|--------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------|
| voltage | A voltage that is type double. | Any numerical value in units of volts. | 1.4<br>A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1). |
|---------|--------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------|

---

# Database Objects

---

## analysis\_view

### Parent Objects

[design](#), [root](#)

### Definition

An analysis view binds together a constraint mode with a delay corner, providing all the information needed to control a single MMMC analysis. Use the `create_analysis_view` and `update_analysis_view` commands to create and modify analysis\_views.

| Attribute       | Description                                                                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| constraint_mode |                                                                                                                                                                      |
|                 | The constraint_mode object associated with the analysis_view.<br><b>Type:</b> <a href="#">obj(constraint_mode)</a><br><b>Default:</b> ""<br><b>Edit:</b> No          |
| delay_corner    |                                                                                                                                                                      |
|                 | The delay_corner object associated with the analysis_view.<br><b>Type:</b> <a href="#">obj(delay_corner)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                |
| is_active       |                                                                                                                                                                      |
|                 | Indicates that the analysis_view is part of any leakage/dynamic/setup/hold designation<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No |

|                 |                                                                                                                                                                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_dormant      |                                                                                                                                                                                                                                        |
|                 | <p>Indicates that the analysis_view is not being timing analyzed, but a minimum set of data is loaded to keep the view in synch with potential design changes.</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| is_drv          |                                                                                                                                                                                                                                        |
|                 | <p>Indicates that the analysis_view is used as drv view.</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                       |
| is_dynamic      |                                                                                                                                                                                                                                        |
|                 | <p>Indicates that the analysis_view is dynamic power view.</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                     |
| is_em           |                                                                                                                                                                                                                                        |
|                 | <p>Indicates that the analysis_view is used for electromigration analysis.</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                     |
| is_hold         |                                                                                                                                                                                                                                        |
|                 | <p>Indicates that the analysis_view is active for hold analysis</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                |
| is_hold_default |                                                                                                                                                                                                                                        |
|                 | <p>Indicates that the analysis_view is the default view for hold analysis</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                      |
| is_ir           |                                                                                                                                                                                                                                        |

|                  |                                                                                                                                                                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <p>Indicates that the analysis_view is IR view.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                                            |
| is_leakage       |                                                                                                                                                                                                                                                                                                                                    |
|                  | <p>Indicates that the analysis_view is leakage power view.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                                 |
| is_setup         |                                                                                                                                                                                                                                                                                                                                    |
|                  | <p>Indicates that the analysis_view is active for setup analysis</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                           |
| is_setup_default |                                                                                                                                                                                                                                                                                                                                    |
|                  | <p>Indicates that the analysis_view is the default view for setup analysis</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                 |
| latency_file     |                                                                                                                                                                                                                                                                                                                                    |
|                  | <p>Specifies an optional file containing view-specific set_clock_latency constraints used for balancing post_CTS IO timing. This file is normally generated automatically as part of the clock tree implementation flow.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p> |
| name             |                                                                                                                                                                                                                                                                                                                                    |
|                  | <p>Provides the name of this analysis_view object as specified by the create_analysis_view command.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                      |
| obj_type         |                                                                                                                                                                                                                                                                                                                                    |

|             |                                                                                                                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">analysis_view</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| power_modes |                                                                                                                                                                                                              |
|             | Specifies an optional list of power_mode objects defined by the power intent specification.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                  |

## antenna\_data

### Parent Objects

[base\\_pin](#), [port](#)

### Definition

Antenna information for terminals

| Attribute | Description                                                                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| area      |                                                                                                                                                                                              |
|           | Area value for non *Car type cases, 0 value used for *Car type cases as area is not applicable in those cases.<br><b>Type:</b> <a href="#">area</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| layer     |                                                                                                                                                                                              |
|           | The layer of antenna data. If layer is null(0x0), data applies to all layers.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">layer</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No         |
| model     |                                                                                                                                                                                              |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <p>Oxide model, none is used for cases where oxide model does not apply.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none oxide1 oxide2 oxide3 oxide4</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                       |
| obj_type |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|          | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">antenna_data</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                             |
| ratio    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|          | <p>Ratio value for *Car type cases. 0 value used for non *Car enums as ratio is not applicable in those cases.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                           |
| type     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|          | <p>Type of antenna data. Equivalent to LEF MACRO PIN ANTENNA* constructs.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> lib_gate_area lib_diff_area lib_partial_metal_area lib_partial_metal_side_area lib_max_area_car lib_max_side_area_car lib_partial_cut_area lib_max_cut_car top_partial_metal_area top_partial_metal_side_area top_gate_area top_diff_area top_max_area_car top_max_side_area_car top_partial_cut_area top_max_cut_car</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

## antenna\_model

### Parent Objects

[layer](#)

### Definition

Antenna model information for one layer & oxide



| Attribute                 | Description                                                                                                                                                                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| area_factor               |                                                                                                                                                                                                                                                                                  |
|                           | <p>Specifies the multiply factor for the antenna metal area calculation (default value 1.0), LEF(ANTENNAAREAFACITOR)</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                   |
| area_factor_diff_use_only |                                                                                                                                                                                                                                                                                  |
|                           | <p>Specifies that the current antenna area factor should only be used when the corresponding layer is connected to the diffusion, LEF(DIFFUSEONLY)</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                       |
| area_ratio                |                                                                                                                                                                                                                                                                                  |
|                           | <p>Specifies the maximum legal antenna ratio, using the area of the metal wire that is not connected to the diffusion diode (0 indicates that the attribute does not apply), LEF(ANTENNAAREARATIO)</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>     |
| cum_area_ratio            |                                                                                                                                                                                                                                                                                  |
|                           | <p>Specifies the cumulative antenna ratio, using the area of the wire that is not connected to the diffusion diode (0 indicates that the attribute does not apply), LEF(ANTENNACUMAREARATIO)</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>           |
| cum_routing_plus_cut      |                                                                                                                                                                                                                                                                                  |
|                           | <p>Indicates that the cumulative ratio rules (ANTENNACUMAREARATIO and ANTENNACUMDIFFAREARATIO) accumulate with the previous cut layer instead of the previous metal layer, LEF(ANTENNACUMROUTINGPLUSCUT)</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| cum_side_area_ratio       |                                                                                                                                                                                                                                                                                  |

|                                |                                                                                                                                                                                                                                                                                                              |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Specifies the cumulative antenna ratio, using the side wall area of the metal wire that is not connected to the diffusion diode (0 indicates that the attribute does not apply), LEF(ANTENNACUMSIDEAREARATIO)</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| gate_minus_diff                |                                                                                                                                                                                                                                                                                                              |
|                                | <p>Indicates that the antenna ratio metal area should subtract the diffusion area connected to it (0 indicates that the attribute does not apply), LEF(ANTENNAAREAMINUSDIFF)</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                     |
| obj_type                       |                                                                                                                                                                                                                                                                                                              |
|                                | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">antenna_model</a>)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                    |
| side_area_factor               |                                                                                                                                                                                                                                                                                                              |
|                                | <p>Specifies the multiply factor for the antenna metal side wall area calculation (default value 1.0), LEF(ANTENNASIDEAREAFACITOR)</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                               |
| side_area_factor_diff_use_only |                                                                                                                                                                                                                                                                                                              |
|                                | <p>Specifies that the current antenna side area factor should only be used when the corresponding layer is connected to the diffusion, LEF(DIFFUSEONLY)</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                            |
| side_area_ratio                |                                                                                                                                                                                                                                                                                                              |
|                                | <p>Specifies the antenna ratio, using the side wall area of the metal wire that is not connected to the diffusion diode (0 indicates that the attribute does not apply), LEF(ANTENNASIDEAREARATIO)</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>               |

# arc

## Parent Objects

inst, pin, net

## Definition

cte timing arc

| Attribute                            | Description                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| aocv_derate_capture_clock_early_fall |                                                                                                                                                                                                                                                                                                                                    |
|                                      | Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early capture clock paths with fall sink pin transitions. You can use -index to return the value for a specific view.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| aocv_derate_capture_clock_early_rise |                                                                                                                                                                                                                                                                                                                                    |
|                                      | Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early capture clock paths with rise sink pin transitions. You can use -index to return the value for a specific view<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| aocv_derate_capture_clock_late_fall  |                                                                                                                                                                                                                                                                                                                                    |
|                                      | Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late capture clock paths with fall sink pin transitions. You can use -index to return the value for a specific view<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No   |
| aocv_derate_capture_clock_late_rise  |                                                                                                                                                                                                                                                                                                                                    |

|                                     |                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late capture clock paths with rise sink pin transitions. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| aocv_derate_data_early_fall         |                                                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>Returns AOCV derate values for a timing arc on an early data path with fall sink pin transitions. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |
| aocv_derate_data_early_rise         |                                                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>Returns AOCV derate values for a timing arc on an early data path with rise sink pin transitions. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |
| aocv_derate_data_late_fall          |                                                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>Returns AOCV derate values for a timing arc on a late data path with fall sink pin transitions. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                        |
| aocv_derate_data_late_rise          |                                                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>Returns AOCV derate values for a timing arc on a late data path with rise sink pin transitions. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                        |
| aocv_derate_launch_clock_early_fall |                                                                                                                                                                                                                                                                                                                                                          |

|                                      |                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early launch clock paths with fall sink pin transitions. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| aocv_derate_launch_clock_early_rise  |                                                                                                                                                                                                                                                                                                                                                                    |
|                                      | <p>Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early launch clock paths with rise sink pin transitions. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| aocv_derate_launch_clock_late_fall   |                                                                                                                                                                                                                                                                                                                                                                    |
|                                      | <p>Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late launch clock paths with fall sink pin transitions. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| aocv_derate_launch_clock_late_rise   |                                                                                                                                                                                                                                                                                                                                                                    |
|                                      | <p>Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late launch clock paths with rise sink pin transitions. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| aocv_stage_count_capture_clock_early |                                                                                                                                                                                                                                                                                                                                                                    |
|                                      | <p>Returns graph based AOCV stage count values in AOCV mode for a given timing arc that is a part of early capture clock paths. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                           |

|                                     |                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| aocv_stage_count_capture_clock_late |                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Returns graph based AOCV stage count values in AOCV mode for a given timing arc that is a part of late capture clock paths. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| aocv_stage_count_data_early         |                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Returns AOCV stage count values for a timing arc on an early data path. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                     |
| aocv_stage_count_data_late          |                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Returns AOCV stage count values for a timing arc on a late data path. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                       |
| aocv_stage_count_launch_clock_early |                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Returns AOCV stage count values in AOCV mode for a given timing arc that is a part of early launch clock paths. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>             |
| aocv_stage_count_launch_clock_late  |                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Returns AOCV stage count values in AOCV mode for a given timing arc that is a part of late launch clock paths. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>              |
| aocv_weight                         |                                                                                                                                                                                                                                                                                                                               |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>Returns the AOCV stage weight for this arc. By default, all cells and arcs have default stage weight of 1.0. The aocv_weight property is specified as a user-defined library attribute in the Liberty timing library explicitly - or, by asserting it via command. This attribute is inherited from the associated library arc. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| delay_max_fall      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     | <p>Returns the largest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                             |
| delay_max_rise      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     | <p>Returns the largest rising delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                              |
| delay_mean_max_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     | <p>In SOCV analysis mode, this returns the mean component of the largest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                           |
| delay_mean_max_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     | <p>In SOCV analysis mode, this returns the mean component of the largest rising delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                            |

|                      |                                                                                                                                                                                                                                                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| delay_mean_min_fall  |                                                                                                                                                                                                                                                                                                                                              |
|                      | <p>In SOCV analysis mode, this returns the mean component of the smallest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>     |
| delay_mean_min_rise  |                                                                                                                                                                                                                                                                                                                                              |
|                      | <p>In SOCV analysis mode, this returns the mean component of the smallest rising delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| delay_min_fall       |                                                                                                                                                                                                                                                                                                                                              |
|                      | <p>Returns the smallest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                       |
| delay_min_rise       |                                                                                                                                                                                                                                                                                                                                              |
|                      | <p>Returns the smallest rising delay through the arc across all concurrent MMMC views can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                             |
| delay_sigma_max_fall |                                                                                                                                                                                                                                                                                                                                              |
|                      | <p>In SOCV analysis mode, this returns the variation component of the largest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |



|                      |                                                                                                                                                                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| delay_sigma_max_rise |                                                                                                                                                                                                                                                                                                                                               |
|                      | <p>In SOCV analysis mode, this returns the variation component of the largest rising delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| delay_sigma_min_fall |                                                                                                                                                                                                                                                                                                                                               |
|                      | <p>In SOCV analysis mode, this returns the variation component of the smallest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| delay_sigma_min_rise |                                                                                                                                                                                                                                                                                                                                               |
|                      | <p>In SOCV analysis mode, this returns the variation component of the smallest rising delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| delta_delay_max_fall |                                                                                                                                                                                                                                                                                                                                               |
|                      | <p>Returns the delta/SI delay component of the largest falling delay through this arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                        |
| delta_delay_max_rise |                                                                                                                                                                                                                                                                                                                                               |
|                      | <p>Returns the delta/SI delay component of the largest rising delay through this arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                         |

|                              |                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| delta_delay_min_fall         | <p>Returns the delta/SI delay component of the smallest falling delay through this arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                          |
| delta_delay_min_rise         | <p>Returns the delta/SI delay component of the smallest rising delay through this arc across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                           |
| dynamic_delta_delay_max_fall | <p>Returns the dynamic delay component of the largest falling delay through this arc across all concurrent MMMC views. This is computed if dynamic voltages have been specified. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| dynamic_delta_delay_max_rise | <p>Returns the dynamic delay component of the largest rising delay through this arc across all concurrent MMMC views. This is computed if dynamic voltages have been specified. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| dynamic_delta_delay_min_fall | <p>Returns the dynamic delay component of the smallest falling delay through this arc across all concurrent MMMC views. This is computed if dynamic voltages have been specified. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

|                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dynamic_delta_delay_min_rise |                                                                                                                                                                                                                                                                                                                                                                                 |
|                              | <p>Returns the dynamic delay component of the smallest rising delay through this arc across all concurrent MMMC views. This is computed if dynamic voltages have been specified. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| from_pin                     |                                                                                                                                                                                                                                                                                                                                                                                 |
|                              | <p>Returns a pin object for the beginning pin of this timing arc</p> <p><b>Type:</b> obj(pin)* obj(hpin)* obj(hport)*</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                   |
| has_mis_fall                 |                                                                                                                                                                                                                                                                                                                                                                                 |
|                              | <p>Return library attribute has_mis for fall output transition.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                |
| has_mis_rise                 |                                                                                                                                                                                                                                                                                                                                                                                 |
|                              | <p>Return library attribute has_mis for rise output transition.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                |
| is_cell_arc                  |                                                                                                                                                                                                                                                                                                                                                                                 |
|                              | <p>Returns true if the arc is a cell arc.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                      |
| is_disabled                  |                                                                                                                                                                                                                                                                                                                                                                                 |

|              |                                                                                                                                                                                                                                                                                      |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <p>Returns a value of true if this library timing arc has been explicitly disabled by the user via a set_disable_timing constraint</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| lib_arcs     |                                                                                                                                                                                                                                                                                      |
|              | <p>Returns a list of the associated lib_arc objects. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">obj(lib_arc)*</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| mode         |                                                                                                                                                                                                                                                                                      |
|              | <p>If the associated lib_arc is defined as part of a Liberty mode group, this attribute will return the name of the library group</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| obj_type     |                                                                                                                                                                                                                                                                                      |
|              | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum (arc)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                  |
| sdf_cond     |                                                                                                                                                                                                                                                                                      |
|              | <p>Returns the value of the Liberty sdf_cond attribute if specified for the associated lib_arc</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| sdf_cond_end |                                                                                                                                                                                                                                                                                      |
|              | <p>Returns the value of the Liberty sdf_cond_end attribute if specified for the associated lib_arc</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                |

|                 |                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sdf_cond_start  |                                                                                                                                                                                                                                                                                                                                                   |
|                 | <p>Returns the value of the Liberty sdf_cond_start attribute if specified for the associated lib_arc</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                           |
| sense           |                                                                                                                                                                                                                                                                                                                                                   |
|                 | <p>Returns the Liberty timing_sense value associated with this arc. This value is one of: positive_unate, negative_unate, or non_unate. This is inherited from the lib_arc.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                    |
| ssi_derate_fall |                                                                                                                                                                                                                                                                                                                                                   |
|                 | <p>Returns the fall ssi_derate for the arc.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                    |
| ssi_derate_rise |                                                                                                                                                                                                                                                                                                                                                   |
|                 | <p>Returns the rise ssi_derate for the arc.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                    |
| timing_type     |                                                                                                                                                                                                                                                                                                                                                   |
|                 | <p>Returns the Liberty timing_type associated with this arc. You can consult the Liberty documentation for the list of possible values for this attribute. This is inherited from the lib_arc.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| to_pin          |                                                                                                                                                                                                                                                                                                                                                   |

|                     |                                                                                                                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>Returns a pin object for the terminating pin of this timing arc<br/> <b>Type:</b> <a href="#">obj(pin)* obj(hpin)* obj(hport)*</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>    |
| when                |                                                                                                                                                                                                                                            |
|                     | <p>Returns the value of the Liberty 'when' attribute if specified for the related lib_arc<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>       |
| when_end            |                                                                                                                                                                                                                                            |
|                     | <p>Returns the value of the Liberty 'when_end' attribute if specified for the related lib_arc<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>   |
| when_library_string |                                                                                                                                                                                                                                            |
|                     | <p>Returns the exact value of the Liberty 'when' attribute if specified for the related lib_arc<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| when_start          |                                                                                                                                                                                                                                            |
|                     | <p>Returns the value of the Liberty 'when_start' attribute if specified for the related lib_arc<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |

## attribute

### Parent Objects

[root](#), [obj\\_type](#)

## Definition

| Attribute        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| additional_help  |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                  | <p>Range info or other additional help.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                         |
| base_name        |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                  | <p>The name without the obj_type (e.g. place_status rather than inst/place_status).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                             |
| category         |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                  | <p>Defines the category of the attribute. Categories group attributes that perform similar functions whereas object types describe where in the design an attribute is valid. You can specify any category name: both new and existing category names are valid. (examples: physical, timing, etc). Note: the value also be an empty string "".</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| check_function   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                  | <p>Specifies a previously defined Tcl procedure's name in order to ensure that the newly defined attribute is valid. Procedure argument list: object value</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                      |
| compute_function |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                  | <p>Specifies a previously defined Tcl procedure's name in order to get the newly defined attribute's value later command. Procedure argument list: object</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                       |
| data_type        |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                 |                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <p>data type attribute</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                              |
| default_value   |                                                                                                                                                                                                                                                                                                                                                     |
|                 | <p>Specifies a default value for the attribute, most attributes do not have default values and will use "".</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                         |
| help            |                                                                                                                                                                                                                                                                                                                                                     |
|                 | <p>Specifies the help text for the attribute</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                        |
| indices         |                                                                                                                                                                                                                                                                                                                                                     |
|                 | <p>List of object type names that represent indices for an attribute value.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> analysis_view clock delay_corner clock_tree skew_group power_domain layer</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                      |
| is_computed     |                                                                                                                                                                                                                                                                                                                                                     |
|                 | <p>Specifies whether the defined attribute will be computed. For .? use, the value will not be displayed if it needs to be recomputed. For direct access of the attribute the value will be returned or computed as needed (0 = static, 1 = computed)</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_settable     |                                                                                                                                                                                                                                                                                                                                                     |
|                 | <p>whether the defined attribute is a read-only or settable attribute (0 = read-only, 1 = settable)</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                   |
| is_user_defined |                                                                                                                                                                                                                                                                                                                                                     |



|                 |                                                                                                                                                                                                                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <p>Specifies whether the defined attribute was defined by the user via Tcl (define_attribute) or is a system defined attribute (0 = system defined, 1 = user defined)</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                             |
| name            |                                                                                                                                                                                                                                                                                                                                             |
|                 | <p>name of attribute</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                            |
| obj_type        |                                                                                                                                                                                                                                                                                                                                             |
|                 | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">attribute</a>)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                       |
| parent          |                                                                                                                                                                                                                                                                                                                                             |
|                 | <p>The attribute's parent object</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">obj_type</a>)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                         |
| possible_values |                                                                                                                                                                                                                                                                                                                                             |
|                 | <p>enum possible values attribute</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                               |
| set_function    |                                                                                                                                                                                                                                                                                                                                             |
|                 | <p>Specifies a previously defined Tcl procedure's name. This option allows you to override user-defined values provided it conforms to the parameters in the Tcl procedure you created. Procedure argument list: object new_value current_value</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| skip_in_db      |                                                                                                                                                                                                                                                                                                                                             |

|  |                                                                                                                                  |
|--|----------------------------------------------------------------------------------------------------------------------------------|
|  | Prevents write_db command to write out the defined<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
|--|----------------------------------------------------------------------------------------------------------------------------------|

## base\_cell

### Parent Objects

[lib\\_cell](#), [inst](#), [pin\\_group](#), [pg\\_base\\_pin](#), [tech](#), [bump](#), [pin\\_guide](#), [power\\_domain](#), [root](#), [base\\_pin](#)

### Definition

base cell

| Attribute             | Description                                                                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| antenna_lef_file_name |                                                                                                                                                                                                              |
|                       | Specifies the LEF file name of the base_cell terminals' antenna data<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                         |
| aocv_weight           |                                                                                                                                                                                                              |
|                       | Returns the AOCV stage weight specified for the cell either explicitly in the library or as a user-defined library attribute<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| area                  |                                                                                                                                                                                                              |
|                       | The area of the cell<br><b>Type:</b> <a href="#">area</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                           |
| base_class            |                                                                                                                                                                                                              |

|                  |                                                                                                                                                                                                                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <p>The base class is the prefix of the .class enum value. It is useful to separate the major cell categories. See the .class description for the definition of each enum value</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none cover block pad core corner</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| base_name        |                                                                                                                                                                                                                                                                                                                                          |
|                  | <p>Base name of the cell.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                |
| base_pins        |                                                                                                                                                                                                                                                                                                                                          |
|                  | <p>The base_pins for this cell that are part of the logical netlist. This does not include pg_base_pins, or must_join_base_pins, or internal_base_pins.</p> <p><b>Type:</b> <a href="#">obj(base_pin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                          |
| bbox             |                                                                                                                                                                                                                                                                                                                                          |
|                  | <p>The bounding box of the overlap rects that define the placement area used by this cell.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                 |
| bottom_edge_type |                                                                                                                                                                                                                                                                                                                                          |
|                  | <p>Name of cell edge type for the bottom edge of the cell (R0/N orientation), used to indicate which cells need extra spacing to other cells.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                            |
| bottom_padding   |                                                                                                                                                                                                                                                                                                                                          |
|                  | <p>The placer will leave this much extra space to the bottom side of the cell (in r0 orientation). It is only valid for standard cells. It is in units of site width.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                       |
| class            |                                                                                                                                                                                                                                                                                                                                          |

All the LEF CLASS and PROPERTY LEF58\_CLASS values (and equivalent OpenAccess values). Refer to the LEF documentation for the complete list and descriptions. They are separated into 5 classes with a unique prefix based on their usage as described here:

CLASS COVER types all start with cover.

CLASS RING or BLOCK start with block.

CLASS PAD start with pad.

CLASS CORE start with core.

CLASS ENDCAP that are corner cells (TOPLEFT, TOPRIGHT, BOTTOMLEFT, BOTTOMRIGHT) start with corner.

CLASS ENDCAP that are not corner cells all start with core because they are all placed in the core rows like CLASS CORE cells.

No CLASS means the value is none.

Modifications are valid only for the current session.

**Type:** `enum`

**Enum Values:** none cover cover\_bump cover\_fill block\_ring block block\_blackbox block\_soft pad pad\_input pad\_output pad\_inout pad\_power pad\_spacer pad\_area\_io core core\_feedthru core\_tie\_high core\_tie\_low core\_spacer core\_antenna core\_welltap core\_endcap\_pre core\_endcap\_post corner\_top\_left corner\_top\_right corner\_bottom\_left corner\_bottom\_right core\_endcap\_top\_edge core\_endcap\_bottom\_edge core\_endcap\_left\_edge core\_endcap\_right\_edge core\_endcap\_left\_top\_edge core\_endcap\_right\_top\_edge core\_endcap\_left\_bottom\_edge core\_endcap\_right\_bottom\_edge core\_endcap\_left\_top\_corner core\_endcap\_right\_top\_corner core\_endcap\_left\_bottom\_corner core\_endcap\_right\_bottom\_corner core\_endcap\_left\_even\_site\_edge core\_endcap\_left\_odd\_site\_edge core\_endcap\_right\_even\_site\_edge core\_endcap\_right\_odd\_site\_edge core\_endcap\_left\_top\_even\_site\_edge core\_endcap\_left\_top\_odd\_site\_edge core\_endcap\_right\_top\_even\_site\_edge core\_endcap\_right\_top\_odd\_site\_edge core\_endcap\_left\_bottom\_even\_site\_edge core\_endcap\_left\_bottom\_odd\_site\_edge core\_endcap\_right\_bottom\_even\_site\_edge core\_endcap\_right\_bottom\_odd\_site\_edge core\_endcap\_left\_top\_even\_site\_corner core\_endcap\_left\_top\_odd\_site\_corner core\_endcap\_right\_top\_even\_site\_corner core\_endcap\_right\_top\_odd\_site\_corner core\_endcap\_left\_bottom\_even\_site\_corner core\_endcap\_left\_bottom\_odd\_site\_corner core\_endcap\_right\_bottom\_even\_site\_corner core\_endcap\_right\_bottom\_odd\_site\_corner core\_endcap\_left\_edge\_bottom\_border core\_endcap\_left\_edge\_top\_border core\_endcap\_right\_edge\_bottom\_border core\_endcap\_right\_edge\_top\_border core\_endcap\_left\_bottom\_edge\_neighbor core\_endcap\_left\_top\_edge\_neighbor core\_endcap\_right\_bottom\_edge\_neighbor core\_endcap\_right\_top\_edge\_neighbor core\_endcap\_left\_bottom\_corner\_neighbor core\_endcap\_left\_top\_corner\_neighbor core\_endcap\_right\_bottom\_corner\_neighbor core\_endcap\_right\_top\_corner\_neighbor

**Default:** ""

**Edit:** Yes

## cts\_cell\_halo\_x

Specifies the clock halo distance in the x direction. The default value of this attribute is auto.

The following attributes can be used to assign x direction clock halos within CCOpt:

cts\_cell\_halo\_x

cts\_cell\_density

cts\_cell\_halo\_sites

Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which:

. If cts\_cell\_halo\_x is set to a non-auto value, then this defines the x direction clock halo. The attributes cts\_cell\_density and cts\_cell\_halo\_sites have no effect.

. If cts\_cell\_halo\_x is set to auto and cts\_cell\_density is set to a non-auto value then cts\_cell\_density defines the clock halo in the x direction. The attribute cts\_cell\_halo\_sites has no effect.

. If both cts\_cell\_halo\_x and cts\_cell\_density are set to auto then cts\_cell\_halo\_sites defines the clock halo in the x direction.

See also:

. cts\_cell\_density

. cts\_cell\_halo\_sites

**Type:** [string](#)

**Allowed -index values:** clock\_tree power\_domain

**Default:** auto

**Edit:** Yes

## cts\_cell\_halo\_y

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>Specifies the clock halo distance in the y direction. The default value of this attribute is auto.</p> <p>The following attributes can be used to assign y direction clock halos within CCOpt:</p> <ul style="list-style-type: none"> <li>cts_cell_halo_y</li> <li>cts_adjacent_rows_legal</li> <li>cts_cell_halo_rows</li> </ul> <p>Only one of these attributes is used to determine the clock halo in the y direction. The following rules determine which:</p> <ul style="list-style-type: none"> <li>. If cts_cell_halo_y is set to a non-auto value, then this defines the y direction clock halo. The attributes cts_adjacent_rows_legal and cts_cell_halo_rows have no effect.</li> <li>. If cts_cell_halo_y is set to auto and cts_adjacent_rows_legal is set to a non-auto value then cts_adjacent_rows_legal defines the clock halo in the y direction. The attribute cts_cell_halo_rows has no effect.</li> <li>. If both cts_cell_halo_y and cts_adjacent_rows_legal are set to auto then cts_cell_halo_rows defines the clock halo in the y direction.</li> </ul> <p>See also:</p> <ul style="list-style-type: none"> <li>. cts_adjacent_rows_legal</li> <li>. cts_cell_halo_rows</li> </ul> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> clock_tree power_domain<br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes</p> |
| dont_touch     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                | <p>This attribute says any inst of this base_cell cannot be modified during optimization. This is the effective dont_touch value for all lib_cells. It is set to the worst case of the lib_cells during init_design and can only be updated by set_db / set_dont_touch after that (subsequent library reads will not affect). This attribute will get restored back to the state during write_db regardless if the library files have been altered.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| dont_use       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                | <p>This attribute says do not use this base_cell during optimization. This is the effective dont_use value for all lib_cells. It is set to the worst case of the lib_cells during init_design and can only be updated by set_db / set_dont_use after that (subsequent library reads will not affect). This attribute is saved by write_db, and restored by read_db regardless if the library files have been altered.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| drive_strength |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                          |                                                                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>This attribute allows the user to assign a drive strength to each cell (X1, X2, etc.) to be used for metric capture.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| eeq_cells                |                                                                                                                                                                                                                      |
|                          | <p>Electrically equivalent base_cells. Same as LEF MACRO EEQ statement.</p> <p><b>Type:</b> <a href="#">obj(base_cell)</a>*<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                         |
| eeq_variant              |                                                                                                                                                                                                                      |
|                          | <p>The LEF EEQ cell variant number from the LEF58_EEQ property 'EEQ macroName VARIANT num'</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> no_value<br/> <b>Edit:</b> Yes</p>                           |
| escaped_name             |                                                                                                                                                                                                                      |
|                          | <p>Escaped name of the cell.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                             |
| foreign_cells            |                                                                                                                                                                                                                      |
|                          | <p>List of foreign references. Equivalent to LEF MACRO FOREIGN.</p> <p><b>Type:</b> <a href="#">obj(foreign_cell)</a>*<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                              |
| has_pin_access_violation |                                                                                                                                                                                                                      |
|                          | <p>Returns True if the cell has pin access violation. Note the attribute is valid after check_library is executed.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> no_value<br/> <b>Edit:</b> No</p>   |
| ib_cell_type             |                                                                                                                                                                                                                      |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Returns Filler type of the base_cell.</p> <p><b>Type:</b> <code>enum</code></p> <p><b>Enum Values:</b> normal_cell ib_cell IBN IBP IBNN IBPP</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                     |
| internal_base_pins     |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                        | <p>Internal pins from .lib that do not exist in the logical netlist. They do not appear in Verilog or DEF files, but may appear in timing reports.</p> <p><b>Type:</b> <code>obj(base_pin)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                  |
| is_always_on           |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                        | <p>Specifies the cell is an always-on cell. An always-on cell normally has two power pins. One is primary which aligns with the normal cell power-rail, and the other is the secondary which actually powers the cell, even when the primary power is off. This attribute can be set by liberty files, or by CPF commands.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p> |
| is_black_box           |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                        | <p>Returns a value of true if the cell is a black box cell.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                   |
| is_buffer              |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                        | <p>This cell is a buffer.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                     |
| is_combinational       |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                        | <p>Returns a value of true if the cell is a combinational cell (not a sequential cell).</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                       |
| is_fall_edge_triggered |                                                                                                                                                                                                                                                                                                                                                                                                                          |



|                            |                                                                                                                                                        |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | Returns a value of true if the cell is triggered by the falling edge of the clock.<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No       |
| is_fixed_mask              |                                                                                                                                                        |
|                            | Indicates the cell has FIXEDMASK keyword in LEF.<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                         |
| is_flop                    |                                                                                                                                                        |
|                            | Returns true if the cell is recognized as flip-flop/register type cell<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                   |
| is_generic                 |                                                                                                                                                        |
|                            | This cell is Synthesis Generic cell.<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                     |
| is_integrated_clock_gating |                                                                                                                                                        |
|                            | Returns true if the cell has the Liberty clock_gating_integrated_cell set to true<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No        |
| is_interface_timing        |                                                                                                                                                        |
|                            | Returns a value of true if a cell has the Liberty interface_timing attribute set to true<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No |
| is_inverter                |                                                                                                                                                        |
|                            | This cell is an inverter.<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                |
| is_iso_nor                 |                                                                                                                                                        |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Specifies the cell is an ISONOR cell. An ISONOR cell is a kind of isolation cell, which has only one primary power pin and one primary ground pin. An ISONOR cell is defined by library files. In cell library, it has permit_power_down true for primary power pin, alive_during_power_up true for input signal pin, and alive_during_partial_power_down true for enable pin and output signal pin. The attribute should be queried after read and commit power intent.</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| is_isolation_cell    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                      | <p>Specifies the cell is an isolation cell. An isolation cell is used to clamp the signal to high or low when its input is shutoff(unknown). This attribute can be set by liberty files, or by CPF commands.</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                             |
| is_latch             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                      | <p>Returns true if the cell is recognized as latch type cell</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                |
| is_level_shifter     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                      | <p>Specifies the cell is a level shifter cell. A level-shifter cell is used to shift the signal voltage from low(high) to high(low). This attribute can be set by liberty files, or by CPF commands.</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                     |
| is_macro             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                      | <p>Returns a value of true if the Liberty attribute is_macro_cell is set true</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                               |
| is_master_slave_flop |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                      | <p>Returns true if the cell is recognized as a master/slave flip-flop type cell</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                             |

|                             |                                                                                                                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_master_slave_lssd_flop   |                                                                                                                                                                                                                                                                                       |
|                             | <p>Returns true if this cell has been recognized as a master/slave LSSD type cell</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                       |
| is_memory                   |                                                                                                                                                                                                                                                                                       |
|                             | <p>Returns true or false depending on whether the associated library cell is recognized as a Liberty memory cell. Cells which include a Liberty memory group definition are recognized as memory cells.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_negative_level_sensitive |                                                                                                                                                                                                                                                                                       |
|                             | <p>Returns a value of true if the cell is identified as a negative level-sensitive latch</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                |
| is_od_cell                  |                                                                                                                                                                                                                                                                                       |
|                             | <p>Returns True if this base_cell is od cell</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                            |
| is_pad                      |                                                                                                                                                                                                                                                                                       |
|                             | <p>Returns a value of true if Liberty attribute is_pad is set to true</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                   |
| is_physical_defined         |                                                                                                                                                                                                                                                                                       |
|                             | <p>This cell has a physical abstract loaded from LEF or OA so the various physical attributes have been set</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                             |
| is_pll                      |                                                                                                                                                                                                                                                                                       |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>Returns a value of true if the Liberty library is_pll_cell attribute is set to true for this cell</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                             |
| is_positive_level_sensitive |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                             | <p>Returns a value of true if the cell is identified as a positive level-sensitive latch</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                         |
| is_power_on_bottom          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                             | <p>Indicates this standard cell has a power pin along the bottom of the cell. This is derived from the power and ground pin information in the cell. It is used by the placer to align multi-height cells properly to the rows. It is not meaningful for non standard cells. Modifications are not saved and are only valid for the current session.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                         |
| is_power_switch             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                             | <p>Specifies the cell is a power switch cell. The power switch cell is used to switch off the power/ground during shutoff. This attribute can be set by liberty files, or by CPF commands.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                    |
| is_quick_abstract           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                             | <p>Indicates that the base_cell was created from reading an OpenAccess layout view and inferred abstract (LEF MACRO equivalent) information from that view. The oa_abstract_lib and oa_abstract_view attributes indicate which layout view was read. For most uses, the inferred abstracts are intended for floorplanning only and are not recommended for final implementation.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_retention_cell           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                        |                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Specifies the cell is a state-retention cell. A state-retention cell is used to retain its state during shutoff. It has a secondary power pin which powers the cell and retains its state, even when the primary power is off. This attribute can be set by liberty files, or by CPF commands.</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> No</p> |
| is_rise_edge_triggered |                                                                                                                                                                                                                                                                                                                                                                              |
|                        | <p>Returns a value of true if the cell is triggered by the rising edge of the clock</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                  |
| is_scan_cell           |                                                                                                                                                                                                                                                                                                                                                                              |
|                        | <p>Indicates the base_cell is a scan cell. A scan cell has test_cell defined in liberty files.</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                       |
| is_sequential          |                                                                                                                                                                                                                                                                                                                                                                              |
|                        | <p>This cell is sequential. Derived from .lib information</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                            |
| is_synchronizer        |                                                                                                                                                                                                                                                                                                                                                                              |
|                        | <p>The lib_cell or base_cell is a synchronizer. Library cell having back to back connected flops is considered as synchronizer.</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> No</p>                                                                                                                                                                   |
| is_timing_defined      |                                                                                                                                                                                                                                                                                                                                                                              |
|                        | <p>This cell has a .lib file definition loaded, so the various timing attributes have been set</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                       |
| is_tristate            |                                                                                                                                                                                                                                                                                                                                                                              |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <p>Returns a value of true if the cell definition includes the Liberty attribute three_state set to true .</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                               |
| is_vdd_on_bottom |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                  | <p>Indicates this standard cell has a power pin along the bottom of the cell. This is derived from the power and ground pin information in the cell. It is used by the placer to align multi-height cells properly to the rows. It is not meaningful for non standard cells. Modifications are not saved and are only valid for the current session.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> |
| lde_data_left    |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                  | <p>The COPDE depth to the left side of the cell.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                       |
| lde_data_right   |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                  | <p>The COPDE depth to the right side of the cell.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                      |
| lef_file_name    |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                  | <p>Specifies the LEF file name of the base_cell</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                        |
| left_edge_type   |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                  | <p>Name of cell edge type for the left edge of the cell (R0/N orientation), used to indicate which cells need extra spacing to other cells.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                            |
| left_padding     |                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>The placer will leave this much extra space to the left side of the cell (in r0 orientation). It is only valid for standard cells. It is in units of site width.</p> <p><b>Type:</b> <code>int</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                              |
| legal_avg_pass_ratio |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Returns the cell's legal average pass ratio. Range: [0,1]. Note the attribute is valid after check_library is executed.</p> <p><b>Type:</b> <code>double</code><br/> <b>Default:</b> no_value<br/> <b>Edit:</b> No</p>                                                                                                                                                                                              |
| lib_cells            |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Returns a list of lib_cell objects which are associated with this base_cell</p> <p><b>Type:</b> <code>obj(lib_cell)*</code><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                        |
| logic_function       |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>This attribute allows the user to assign a function to each cell (AND, OR, etc.) to be used for metric capture.</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                           |
| must_join_base_pins  |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Must join pins are physical-only pins that do not exist in the logical netlist. They do not appear in Verilog or DEF files. The router will connect the must-join base_pin to its corresponding logical pin. See the base_pin .must_join_pins attribute, and the LEF MACRO PIN MUSTJOIN statement for more details</p> <p><b>Type:</b> <code>obj(base_pin)*</code><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| name                 |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Name of the cell.</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                          |
| num_base_pins        |                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                  |                                                                                                                                                                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <p>Number of signal base_pins for this cell. It does not include pg_base_pins or must_join_base_pins or internal_base_pins.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                     |
| num_drivers      |                                                                                                                                                                                                                                                                          |
|                  | <p>The number of driver base_pins for this base_cell. Signal base_pins that have direction = out or inout are considered drivers.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                               |
| num_loads        |                                                                                                                                                                                                                                                                          |
|                  | <p>The number of load base_pins for this base_cell. Signal base_pins that have direction = in or inout are considered loads.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                    |
| oa_abstract_lib  |                                                                                                                                                                                                                                                                          |
|                  | <p>OpenAccess library name of the physical abstract for the cell if read from OA.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                            |
| oa_abstract_view |                                                                                                                                                                                                                                                                          |
|                  | <p>OpenAccess view name for the physical abstract for this cell if read from OA (equivalent of LEF MACRO data). The OA cell name is the same as this cell's 'name' attribute</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| oa_layout_lib    |                                                                                                                                                                                                                                                                          |
|                  | <p>OpenAccess library name of physical layout for the cell if read from OA.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                  |
| oa_layout_view   |                                                                                                                                                                                                                                                                          |



|                  |                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <p>OpenAccess view name for the physical layout for this cell if read from OA (equivalent of GDSII data). The layout view data can optionally be displayed instead of the abstract view data, but otherwise this layout data is not used in Innovus. The OA cell name is the same as this cell's 'name' attribute</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| obj_type         |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                  | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">base_cell</a>)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                         |
| obs_layer_shapes |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                  | <p>A list of cell obstruction layer_shapes (LEF OBS or OA abstract blockages that are not vias). Note, use obs_shape_vias for obstructions that are from vias (LEF OBS VIA).</p> <p><b>Type:</b> <a href="#">obj(layer_shape)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                           |
| obs_shape_vias   |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                  | <p>A list of cell obstruction shape_vias (LEF OBS VIA or OA abstract via blockages). Use obs_layer_shapes for obstructions on a single layer.</p> <p><b>Type:</b> <a href="#">obj(shape_via)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                            |
| pg_base_pins     |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                  | <p>The power/ground base_pins for this cell.</p> <p><b>Type:</b> <a href="#">obj(pg_base_pin)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                           |
| right_edge_type  |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                  | <p>Name of cell edge type for the right edge of the cell (R0/N orientation), used to indicate which cells need extra spacing to other cells.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                      |
| right_padding    |                                                                                                                                                                                                                                                                                                                                                                                                               |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <p>The placer will leave this much extra space to the right side of the cell (in r0 orientation). It is only valid for standard cells. It is in units of site width.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                   |
| site     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|          | <p>The site for this cell. Modifications are valid only for the current session.</p> <p><b>Type:</b> <a href="#">obj(site)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| symmetry |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|          | <p>The allowed orientations for the placer to try. none means r0 only, x means ok to flip about X axis (r0, mx), y means ok to flip about Y axis (r0, my), xy means ok to flip about X or Y axis (r0, mx, my, r180), any means all orientations are allowed. Equivalent to LEF MACRO SYMMETRY or OA oaSymmetry. LEF values of R90, X R90, Y R90, and X Y R90 are all equivalent to the value = any. Modifications are not saved and are only valid for the current session.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none x y xy any<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| tap_type |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|          | <p>Specifies the name of a well tap type for this cell. Various rules for well taps are grouped together for each tap_type. See the LEF documentation on the TAPTYPE keyword for more details.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                     |
| tap_wall |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|          | <p>Specifies a special well tap cell (LEF CLASS CORE WELLTAP) or a special end_cap (LEF CLASS ENDCAP ...) cell that can be used for a tap wall purpose, which is used to break OD diffusion and aligned vertically to form a tap wall. See the LEF docs about the keyword TAPWALL for more details.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                               |
| tech     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                          |                                                                                                                                                                                                                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>The tech for this base_cell.</p> <p><b>Type:</b> obj(tech)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                              |
| timing_model_type        |                                                                                                                                                                                                                                                 |
|                          | <p>Returns the Liberty model type for a given cell or instance. The supported values are abstracted , extracted , and qtm .</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                     |
| top_edge_type            |                                                                                                                                                                                                                                                 |
|                          | <p>Name of cell edge type for the top edge of the cell (R0/N orientation), used to indicate which cells need extra spacing to other cells.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                      |
| top_padding              |                                                                                                                                                                                                                                                 |
|                          | <p>The placer will leave this much extra space to the top side of the cell (in r0 orientation). It is only valid for standard cells. It is in units of site width.</p> <p><b>Type:</b> int</p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> |
| voltage_threshold_group  |                                                                                                                                                                                                                                                 |
|                          | <p>This attribute allows the user to assign a base_cell to a voltage threshold group for metric capture. The threshold group name can be any valid string.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>     |
| worst_padding_area_of_od |                                                                                                                                                                                                                                                 |
|                          | <p>Returns the worst_padding_area of the OD cell. If the cell is not OD cell, the value will be set to -1.</p> <p><b>Type:</b> area</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> No</p>                                                  |

# base\_pin

## Parent Objects

[base\\_cell](#), [bump\\_pin](#), [lib\\_pin](#), [pin\\_blockage](#), [hpin](#), [inst\\_physical\\_pin](#), [partition](#), [pin\\_group](#), [root](#), [tech](#), [pin](#)

## Definition

base pin

| Attribute                         | Description                                                                                                                                                                                                                                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| antenna_data                      |                                                                                                                                                                                                                                                                   |
|                                   | The various process antenna_data values for this pin, including area of gate, diffusion, metal, cut, and cumulative area ratios for metal, cut attached to this pin.<br><b>Type:</b> <a href="#">obj(antenna_data)</a> *<br><b>Default:</b> ""<br><b>Edit:</b> No |
| base_cell                         |                                                                                                                                                                                                                                                                   |
|                                   | The base_cell of base pin.<br><b>Type:</b> <a href="#">obj(base_cell)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                |
| base_name                         |                                                                                                                                                                                                                                                                   |
|                                   | The name without the base_cell name (e.g. out rather than and2/out).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                              |
| cts_max_fanout                    |                                                                                                                                                                                                                                                                   |
|                                   | The maximum fanout at any point in the clock tree.<br>Valid values: integer ranged between 2 and 1000 inclusive<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 100<br><b>Edit:</b> Yes                                                                    |
| cts_max_source_to_sink_net_length |                                                                                                                                                                                                                                                                   |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe.<br/>This constraint can be applied to either a pin, a clock tree, or a net type.<br/>By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length.<br/>Valid values: double<br/><b>Type:</b> <a href="#">string</a><br/><b>Allowed -index values:</b> clock_tree<br/><b>Default:</b> top auto trunk auto leaf auto<br/><b>Edit:</b> Yes</p> |
| cts_max_source_to_sink_net_length_leaf  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                         | <p>The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe.<br/>This constraint can be applied to either a pin, a clock tree, or a net type.<br/>By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length.<br/>Valid values: double<br/><b>Type:</b> <a href="#">string</a><br/><b>Allowed -index values:</b> clock_tree<br/><b>Default:</b> auto<br/><b>Edit:</b> Yes</p>                          |
| cts_max_source_to_sink_net_length_top   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                         | <p>The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe.<br/>This constraint can be applied to either a pin, a clock tree, or a net type.<br/>By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length.<br/>Valid values: double<br/><b>Type:</b> <a href="#">string</a><br/><b>Allowed -index values:</b> clock_tree<br/><b>Default:</b> auto<br/><b>Edit:</b> Yes</p>                          |
| cts_max_source_to_sink_net_length_trunk |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe.<br/>This constraint can be applied to either a pin, a clock tree, or a net type.<br/>By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length.<br/>Valid values: double<br/><b>Type:</b> <a href="#">string</a><br/><b>Allowed -index values:</b> clock_tree<br/><b>Default:</b> auto<br/><b>Edit:</b> Yes</p> |
| cts_spec_config_base_pin_trace_through_to |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

Clock tree definition will, by default, not continue through certain types of cell arc (for instance, the clock to Q arc in a DFF). This attribute allows you to override this default behavior, permitting the clock tree to trace through all instances of such a cell.

This attribute serves the same function as `trace_through_to`, except that here the clock path is specified at the level of the library cell.

The attribute should be configured on the input library pin at which the clock will arrive. The value of the attribute specifies the output library pin to which the clock should propagate. The specified output pin must be another pin on the same library cell. The output pin may be specified either by its fully qualified name (i.e. inclusive of the cell name), or else simply by its local (cell-relative) name.

There must be a pre-existing (library-defined) chain of one or more delay arcs that connect the input and output pins together. It is not possible to use `library_trace_through_to` to synthesize delay arcs.

If multiple input pins are annotated on a given library cell, the value of `library_trace_through_to` at each of those pins must select the same output pin: i.e. the configuration must identify a single clock output for the cell. If multiple clock outputs are necessary then `library_trace_through_to` should not be used: instead for each instance of the library cell, define a generated clock tree at each of the clock-carrying outputs.

If the configuration of `library_trace_through_to` settings for a given library cell does not meet these requirements, a warning will be issued and the settings for that cell will be ignored.

All instances of the library cell will be affected by this setting. If both `trace_through_to` and `library_trace_through_to` are applicable at a given instance pin, the `trace_through_to` value will take precedence.

Valid values: `base_pin`

**Type:** `string`  
**Default:** `""`  
**Edit:** Yes

`cts_stack_via_rule`

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>The preferred stack via rule for terminal connections.<br/>This attribute helps guide the choice of stack via rule (via pillar) used for connecting routes to netlist terminals. If the specified value names a valid candidate for terminal in question (it is a member of list the candidate rules associated with the terminal's cell pin),<br/>then it will used as the preferred stack via rule for connecting to that terminal.<br/>Valid values: 'auto' or stack via rule name<br/><b>Type:</b> <a href="#">string</a><br/><b>Allowed -index values:</b> clock_tree<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p> |
| cts_stack_via_rule_leaf          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                  | <p>The preferred stack via rule for terminal connections.<br/>This attribute helps guide the choice of stack via rule (via pillar) used for connecting routes to netlist terminals. If the specified value names a valid candidate for terminal in question (it is a member of list the candidate rules associated with the terminal's cell pin),<br/>then it will used as the preferred stack via rule for connecting to that terminal.<br/>Valid values: 'auto' or stack via rule name<br/><b>Type:</b> <a href="#">string</a><br/><b>Allowed -index values:</b> clock_tree<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p> |
| cts_stack_via_rule_required      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                  | <p>The pin-specific required field for stack via rule connections.<br/><b>Type:</b> <a href="#">bool</a><br/><b>Allowed -index values:</b> clock_tree<br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| cts_stack_via_rule_required_leaf |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                  | <p>The pin-specific required field for stack via rule connections.<br/><b>Type:</b> <a href="#">bool</a><br/><b>Allowed -index values:</b> clock_tree<br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| cts_stack_via_rule_required_top  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>The pin-specific required field for stack via rule connections.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> clock_tree</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                        |
| cts_stack_via_rule_required_trunk |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                   | <p>The pin-specific required field for stack via rule connections.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> clock_tree</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                        |
| cts_stack_via_rule_top            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                   | <p>The preferred stack via rule for terminal connections.</p> <p>This attribute helps guide the choice of stack via rule (via pillar) used for connecting routes to netlist terminals. If the specified value names a valid candidate for terminal in question (it is a member of list the candidate rules associated with the terminal's cell pin), then it will used as the preferred stack via rule for connecting to that terminal.</p> <p>Valid values: 'auto' or stack via rule name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> clock_tree</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_stack_via_rule_trunk          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                   | <p>The preferred stack via rule for terminal connections.</p> <p>This attribute helps guide the choice of stack via rule (via pillar) used for connecting routes to netlist terminals. If the specified value names a valid candidate for terminal in question (it is a member of list the candidate rules associated with the terminal's cell pin), then it will used as the preferred stack via rule for connecting to that terminal.</p> <p>Valid values: 'auto' or stack via rule name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> clock_tree</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_target_max_capacitance        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto   double</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> delay_corner clock_tree<br/> <b>Default:</b> top auto trunk auto leaf auto<br/> <b>Edit:</b> Yes</p> |
| cts_target_max_capacitance_leaf |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto   double</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> delay_corner clock_tree<br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes</p>                          |
| cts_target_max_capacitance_top  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto   double</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> delay_corner clock_tree<br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes</p> |
| cts_target_max_capacitance_trunk |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                  | <p>The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto   double</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> delay_corner clock_tree<br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes</p> |
| direction                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Base_pins's direction from .lib if available, otherwise from LEF/OA. It can be in, out, inout or internal. Internal means it is an internal pin from a .lib file for the timing model, and is not part of the netlist. If there is no .lib for this cell, then the direction comes from the LEF PIN DIRECTION or equivalent OA oaTermType. The LEF DIRECTION values (and equivalent oaTermType values) are mapped this way: INPUT = in, OUTPUT = out, OUTPUT TRISTATE = out, INOUT = inout, FEEDTHRU = inout, and the oaTermType unknown = inout.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> in out inout internal<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| effective_stack_via_rule |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>The stack_via_rule that is expected, but not required, to be used by the router for connecting to the instance pins instantiated from this base_pin. The actual stack via rule used (if any) may be effected by other pin and base_pin attributes, or by choices made by the software (optimization, clock tree synthesis, the router etc.)</p> <p><b>Type:</b> <a href="#">obj(stack_via_rule)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                       |
| is_always_on             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>The always on pin of an always on cell.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| is_analog                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Specifies the pin is an analog signal. This attribute can be set by liberty files.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| is_iq_function           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Output is an IQ pin.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| is_iqn_function          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                          |                                                                                                                                                                                                                                                                        |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | Output is an IQN pin.<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                    |
| is_isolated              |                                                                                                                                                                                                                                                                        |
|                          | Specifies the pin is isolated internally in the cell. It is used for cells where some pins are internally isolated and some are not. This attribute can be set by liberty files.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> No                      |
| is_isolation_cell_enable |                                                                                                                                                                                                                                                                        |
|                          | Identifies the pin is an isolation enable pin. This pin is used to control when to clamp the output and isolate it from the input. This attribute can be set by liberty files, or by CPF commands.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> No    |
| is_level_shifter_enable  |                                                                                                                                                                                                                                                                        |
|                          | Identifies the pin is a level shifter enable pin. This pin is used to control when to clamp the output and isolate it from the input. This attribute can be set by liberty files, or by CPF commands.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> No |
| is_must_join_all_ports   |                                                                                                                                                                                                                                                                        |
|                          | Indicates that the port has the LEF MUSTJOINALLPORTS attribute set.<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                      |
| is_power_switch_enable   |                                                                                                                                                                                                                                                                        |
|                          | Identifies the pin is a power switch enable pin. This pin is used to control when to turn on/off the power switch. This attribute can be set by liberty files, or by CPF commands.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> No                    |
| is_retention_cell_enable |                                                                                                                                                                                                                                                                        |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>Identifies the pin as a retention cell enable pin. This pin is used to control when to retain the state and ignore other inputs. This attribute can be set by liberty files, or by CPF commands.</p> <p><b>Type:</b> <code>bool</code><br/> <b>Default:</b> <code>false</code><br/> <b>Edit:</b> No</p>                                                                                                                             |
| is_unconnected     |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                    | <p>Specifies the pin is floating internally. This is used for cells where some of the inputs or outputs are unused by the cell. This attribute can be set by liberty files.</p> <p><b>Type:</b> <code>bool</code><br/> <b>Default:</b> <code>false</code><br/> <b>Edit:</b> No</p>                                                                                                                                                     |
| is_via_in_pin_only |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                    | <p>Indicates that the pin has a LEF VIAINPINONLY property. It means that vias must be dropped inside the original pin shapes to connect to the pin. In some advanced nodes, the pin shapes can be extended for metal alignment purposes. However, via insertion is not allowed in that extended portion if this attribute is true.</p> <p><b>Type:</b> <code>bool</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p> |
| layer              |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                    | <p>Layer of the base_pin. For base_pins with more than one shape, it is the layer of the first shape (which is the same shape used for the .location value).</p> <p><b>Type:</b> <code>obj(layer)</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                                 |
| must_join_pins     |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                    | <p>The must join base_pin</p> <p><b>Type:</b> <code>obj(base_pin)</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                 |
| name               |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                    | <p>Terminal name</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                 |
| obj_type           |                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">base_pin</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                    |
| pg_type            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | <p>Specifies the type of the power or ground pin from liberty data. Signal pins and PG pins with no liberty entry will have "invalid".</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> primary_power primary_ground backup_power backup_ground internal_power internal_ground pwell nwell deepnpwell deepnwell invalid</p> <p><b>Default:</b> invalid</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                        |
| physical_direction |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | <p>Direction of this pin.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> input output inout unknown feedthrough tristate</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                          |
| physical_pins      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | <p>Physical pins for the base_pin. One logical base_pin can have multiple physical_pins. Each physical_pin had a list of strongly connected shapes (equivalent to LEF or DEF PORT or OA pin). The internal connection between physical_pins is weak (e.g. through poly or other high resistance paths)</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">physical_pin</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                          |
| pin_edge           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | <p>If this base_pin is for a partition, or inst that is a black-box, and the base_pin is assigned, the edge value indicates along which edge of the boundary polygon the base_pin is assigned. The edge number starts from the lowest Y, then left-most X vertex, starting with 0, and then counting clock-wise. See the set_pin_constraint command document for a figure showing the edge numbering. If the base_pin is not assigned, or not for a partition, or black-box inst, the value of -1 is returned.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| related_ground_pin |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Specifies which ground pin drives this signal pin. It must be one of the ground pins defined for this cell. It can be set by CPF, Liberty, or LEF/OA, with CPF having highest precedence, then Liberty, then LEF/OA. It is often only set when there is more than one ground pin for the cell.</p> <p><b>Type:</b> <a href="#">obj(base_pin)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| related_power_pin       |                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>Specifies which power pin drives this signal pin. It must be one of the power pins defined for this cell. It can be set by CPF, Liberty, or LEF/OA, with CPF having highest precedence, then Liberty, then LEF/OA. It is often only set when there is more than one power pin for the cell.</p> <p><b>Type:</b> <a href="#">obj(base_pin)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>    |
| stack_via_required      |                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>Sets whether stack vias are required for this term.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                 |
| stack_via_rule_list     |                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>List of <a href="#">stack_via_rule</a> that are valid choices for connecting to the instance pins instantiated from this <a href="#">base_pin</a>. If the list is empty, no stack via is allowed.</p> <p><b>Type:</b> <a href="#">obj(stack_via_rule)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                      |
| stack_via_rule_required |                                                                                                                                                                                                                                                                                                                                                                                                      |



|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <p>Specifies whether a stack via is required when connecting to the instance pins instantiated from this base_pin. If set to 0, a stack via is optional. If set to 1, the min one of the stack_via_rule from stack_via_rules must be used to generate a stack via even if a design rule violation occurs. If set to 2, the min one the stack_via_rule from stack_via_rules is used to generate a stack via if a design rule violation does not occur. If set to 3, tool enforces multiple tries from large to min one of the stack_via_rule from stack_via_rules to generate a stack via even if a design rule violation occurs. If set to 4, tool performs multiple tries from large to min one the stack_via_rule from stack_via_rules to generate a stack via if a design rule violation does not occur.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> 0 1 2 3 4<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| taper_rule |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|            | <p>The taper route_rule for the pin. By default, if tapered routing is needed to access the pin, the default route_rule will be used unless this attribute is set (see LEF MACRO PIN TAPERRULE).</p> <p><b>Type:</b> <a href="#">obj(route_rule)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| tech       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|            | <p>The tech for this base_pin.</p> <p><b>Type:</b> <a href="#">obj(tech)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| tied_to    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|            | <p>Specifies the PG pin name or 'empty' which the PG pin tied to.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| use        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Indicates how this pin is used from the LEF USE value or OA equivalent. The legal values are: signal analog power ground clock. Note that timing analysis does not use these values, it uses the .lib data instead (e.g. is\_clock, is\_analog, etc.). So the signal, clock or analog values are not normally used by applications. The power/ground values are used by many applications.

**Type:** [enum](#)

**Enum Values:** signal analog power ground clock

**Default:** ""

**Edit:** No

## boundary

### Parent Objects

[hinst](#), [design](#), [root](#)

### Definition

A boundary object can only be attached to an hinst, normally when the hinst is a partition. It is used to constrain the placement of all the insts of the hinst. See the group object for placement constraint information attached to the group object (e.g. a list of insts, hinsts, or groups).

| Attribute | Description                                                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------|
| area      |                                                                                                                                      |
|           | Sum of the area of the .rects that define the boundary<br><b>Type:</b> <a href="#">area</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| bbox      |                                                                                                                                      |
|           | Bounding box of the boundary rects.<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> No                    |
| hinst     |                                                                                                                                      |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>The parent hinst. This boundary affects all the insts inside the parent hinst.</p> <p><b>Type:</b> <a href="#">obj(hinst)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                       |
| is_floating |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|             | <p>Only affects boundary with .type = fence or region. If true, the global placer can move the fence or region. The .bbox and .rects value must also be set. The global placer will not change the size of the rect, but may move it.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                      |
| obj_type    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|             | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">boundary</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                     |
| rects       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|             | <p>List of non-overlapping rectangles that defines the shape of the boundary</p> <p><b>Type:</b> <a href="#">rect*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                  |
| type        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|             | <p>fence: all the insts are inside the boundary, and only these insts can be inside.<br/> region: all the insts are inside, but other insts can also be inside.<br/> guide: lower cost for insts to be inside boundary, but not required.<br/> cluster: keep insts near each other (the .rects field is empty).<br/> none: no affect on placement.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> guide region fence cluster none</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |

## bump

## Parent Objects

[marker](#), [bump\\_pin](#), [design](#), [io\\_constraint](#), [root](#), [port](#)

## Definition

Bump instance

| Attribute           | Description                                                                                                                                                                                                                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_cell           |                                                                                                                                                                                                                                                                                                                       |
|                     | The base_cell master of the bump.<br><b>Type:</b> <a href="#">obj(base_cell)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                             |
| base_name           |                                                                                                                                                                                                                                                                                                                       |
|                     | The base_name of the bump<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                             |
| bbox                |                                                                                                                                                                                                                                                                                                                       |
|                     | Bounding box of the bump shape for a package connection, which is the widest shape found on the top routing layer in the base_cell. If there is more than one wide shape of the same size, the bbox of all the wide shapes is returned.<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| bump_connect_target |                                                                                                                                                                                                                                                                                                                       |
|                     | The string property added to the bump by create_bump_connect_target_constraint. The value takes the format like inst[ :pin [ :port ]].<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                |
| bump_pins           |                                                                                                                                                                                                                                                                                                                       |

|                         |                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>The pins of the bump. A bump with ALLPINSCONNECTED LEF syntax can have multiple bump_pins which are internally connected.</p> <p><b>Type:</b> <code>obj(bump_pin)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                      |
| center                  |                                                                                                                                                                                                                                                                                                                                        |
|                         | <p>The center of the bump shape for a package connection, which is the center of the .bbox attribute. This is not the same as the .location of the bump which is normally at the lower-left corner of the cell.</p> <p><b>Type:</b> <code>point</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                            |
| escaped_name            |                                                                                                                                                                                                                                                                                                                                        |
|                         | <p>The escaped name of the bump</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                           |
| foreign_excluded_center |                                                                                                                                                                                                                                                                                                                                        |
|                         | <p>The center of the bump. This is not the same as bump shape center which is the center point of its bounding box. This attribute is calculated from .point which is the lower left point of the bump by considering the FOREIGN offsets.</p> <p><b>Type:</b> <code>point</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_fixed_assignment     |                                                                                                                                                                                                                                                                                                                                        |
|                         | <p>Indicates that the bump is connected to one port</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                         |
| location                |                                                                                                                                                                                                                                                                                                                                        |
|                         | <p>Location of the bump.</p> <p><b>Type:</b> <code>point</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                  |
| name                    |                                                                                                                                                                                                                                                                                                                                        |

|              |                                                                                                                                                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <p>Name of the bump<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                              |
| net          |                                                                                                                                                                                                                                                            |
|              | <p>The net connected to the bump.<br/> <b>Type:</b> <a href="#">obj(net)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                              |
| obj_type     |                                                                                                                                                                                                                                                            |
|              | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br/> <b>Type:</b> <a href="#">enum (bump)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                             |
| orient       |                                                                                                                                                                                                                                                            |
|              | <p>Bump orientation.<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> r0 r90 r180 r270 mx mx90 my my90<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                    |
| parent       |                                                                                                                                                                                                                                                            |
|              | <p>The parent of the bump, which is either a hinst for a bump within hierarchy or the design object for a bump at the top level.<br/> <b>Type:</b> <a href="#">obj(design)</a> <a href="#">obj(hinst)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| place_status |                                                                                                                                                                                                                                                            |
|              | <p>Placement status of bump.<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> unplaced placed fixed cover soft_fixed<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                      |
| ports        |                                                                                                                                                                                                                                                            |

|  |                                                                                                                        |
|--|------------------------------------------------------------------------------------------------------------------------|
|  | The ports the bump is assigned to.<br><b>Type:</b> <a href="#">obj(port)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
|--|------------------------------------------------------------------------------------------------------------------------|

## bump\_grid

### Parent Objects

[design](#), [root](#)

### Definition

The grid to guide bump placement.

| Attribute | Description                                                                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| layers    |                                                                                                                                                                                                          |
|           | List of layers. User assign relative bump pin layer to bump grid.<br><b>Type:</b> <a href="#">obj(layer)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                   |
| name      |                                                                                                                                                                                                          |
|           | Bump grid name.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                          |
| obj_type  |                                                                                                                                                                                                          |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">bump_grid</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| origin    |                                                                                                                                                                                                          |

|         |                                                                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | The coordinates of the first bump grid.<br><b>Type:</b> point<br><b>Default:</b> ""<br><b>Edit:</b> No                                                   |
| pitch_x |                                                                                                                                                          |
|         | The spacing from the center of the bump to the center of the neighbor bump on the x-axis.<br><b>Type:</b> coord<br><b>Default:</b> ""<br><b>Edit:</b> No |
| pitch_y |                                                                                                                                                          |
|         | The spacing from the center of the bump to the center of the neighbor bump on the y-axis.<br><b>Type:</b> coord<br><b>Default:</b> ""<br><b>Edit:</b> No |

## bump\_pin

### Parent Objects

bump, port

### Definition

PIN in bump LEF

| Attribute | Description                                                                                          |
|-----------|------------------------------------------------------------------------------------------------------|
| base_name |                                                                                                      |
|           | The bump_pin name without bump path.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No |
| base_pin  |                                                                                                      |



|              |                                                                                                                                                                                                                                                                     |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <p>The equivalent base_pin.</p> <p><b>Type:</b> <a href="#">obj(base_pin)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                  |
| bump         |                                                                                                                                                                                                                                                                     |
|              | <p>The bump that the bump_pin belongs to.</p> <p><b>Type:</b> <a href="#">obj(bump)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                        |
| direction    |                                                                                                                                                                                                                                                                     |
|              | <p>bump_pin's direction from the corresponding base_pin.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> in out inout</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                      |
| escaped_name |                                                                                                                                                                                                                                                                     |
|              | <p>The bump_pin name including bump path.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                           |
| layer        |                                                                                                                                                                                                                                                                     |
|              | <p>The layer of the bump_pin. For bump_pin with more than one shape, it is the layer of the first shape (which is the same shape used for the .location value).</p> <p><b>Type:</b> <a href="#">obj(layer)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| location     |                                                                                                                                                                                                                                                                     |
|              | <p>The location of the bump_pin. For bump_pin with more than one shape, it is the location of the first shape (which is the same shape used for the .layer value).</p> <p><b>Type:</b> <a href="#">point</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| name         |                                                                                                                                                                                                                                                                     |

|          |                                                                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | The bump_pin name including bump path.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                  |
| obj_type |                                                                                                                                                                                                         |
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">bump_pin</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| port     |                                                                                                                                                                                                         |
|          | The port that the bump_pin connects to.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">port</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                           |

## bus

### Parent Objects

[design](#), [net](#), [root](#)

### Definition

The Verilog bus definitions in the top Verilog module.

| Attribute | Description                                                                                                                                                                     |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bits      |                                                                                                                                                                                 |
|           | The nets of the bus. All the bits of one bus will have the same obj_type.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">net</a> )*<br><b>Default:</b> ""<br><b>Edit:</b> No |
| lsb       |                                                                                                                                                                                 |

|          |                                                                                                                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Bottom bit index of bus.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                              |
| msb      |                                                                                                                                                                                                    |
|          | Top bit index of bus.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                 |
| name     |                                                                                                                                                                                                    |
|          | Non-vector portion of the bus name (eg. A for bus A[7:0])<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                          |
| obj_type |                                                                                                                                                                                                    |
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">bus</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

## bus\_guide

### Parent Objects

[design](#)

### Definition

Bus guide

| Attribute | Description |
|-----------|-------------|
| area      |             |

|              |                                                                                                                                                                                        |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | Area of the bus_guide as defined by the LEF MACRO SIZE or OVERLAP information<br><b>Type:</b> <a href="#">area</a><br><b>Default:</b> ""<br><b>Edit:</b> No                            |
| bottom_layer |                                                                                                                                                                                        |
|              | The bottom layer of allowed layer range<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                           |
| net_group    |                                                                                                                                                                                        |
|              | The net group with list of nets to be routed within the bus_guide<br><b>Type:</b> <a href="#">obj(net_group)</a> *<br><b>Default:</b> ""<br><b>Edit:</b> No                            |
| obj_type     |                                                                                                                                                                                        |
|              | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (bus_guide)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| rect         |                                                                                                                                                                                        |
|              | Rectangel that defines the bus_guide shape.<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                              |
| top_layer    |                                                                                                                                                                                        |
|              | The top layer of allowed layer range<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                              |
| type         |                                                                                                                                                                                        |

The type of bus\_guide.  
hard: Specifies the bus\_guide as a hard constraint for routing. eGR, NR and NRHF should obey the path of bus guide.  
soft: Specifies the bus\_guide as a soft constraint for routing. eGR, NR and NRHF should be guided by the route path. But tool can route the net out of the bus guide.  
**Type:** [enum](#)  
**Enum Values:** soft hard  
**Default:** ""  
**Edit:** Yes

## bus\_sink\_group

### Parent Objects

[design](#), [root](#)

### Definition

A group of sinks (loads) that some floorplan and routing commands use to control adding buffers and routing for a bus. See 'help \*bus\_sink\*' for a list of commands related to this object.

| Attribute | Description                                                                                                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name      |                                                                                                                                                                                                               |
|           | The name of the bus_sink_group.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> {}<br><b>Edit:</b> No                                                                                               |
| obj_type  |                                                                                                                                                                                                               |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">bus_sink_group</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| sinks     |                                                                                                                                                                                                               |
|           | A list of sinks (pins of insts, or ports of the design) for this bus.<br><b>Type:</b> <a href="#">obj(pin)*</a> <a href="#">obj(port)*</a><br><b>Default:</b> {}<br><b>Edit:</b> No                           |

# clock

## Parent Objects

[context\\_data](#), [pin](#), [timing\\_path](#), [root](#), [port](#)

## Definition

cte clock

| Attribute              | Description                                                                                                                                                                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name              |                                                                                                                                                                                                                                                                                                         |
|                        | The base_name of the clock.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                             |
| child_clocks           |                                                                                                                                                                                                                                                                                                         |
|                        | Returns a list of clock object which are the immediate generated clocks derived from this master clock.<br><b>Type:</b> <a href="#">obj(clock)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                            |
| clock_hold_uncertainty |                                                                                                                                                                                                                                                                                                         |
|                        | clock_hold_uncertainty<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                  |
| clock_network_pins     |                                                                                                                                                                                                                                                                                                         |
|                        | Returns a list of pin and hpin objects in the extended fanout of the clock source. The extended fanout includes the fanout of generated clocks which the current clock is the master for.<br><b>Type:</b> <a href="#">obj(port)*</a> <a href="#">obj(pin)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No |

|                         |                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| clock_setup_uncertainty |                                                                                                                                         |
|                         | clock_setup_uncertainty<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                 |
| comment                 |                                                                                                                                         |
|                         | Returns the -comment option of a clock.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                 |
| delay_max_fall          |                                                                                                                                         |
|                         | Returns the maximum falling delay value for the clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No                   |
| delay_max_rise          |                                                                                                                                         |
|                         | Returns the maximum rising delay value for the clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No                    |
| delay_min_fall          |                                                                                                                                         |
|                         | Returns the minimum falling delay value for the clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No                   |
| delay_min_rise          |                                                                                                                                         |
|                         | Returns the minimum rising delay value for the clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No                    |
| divide_by               |                                                                                                                                         |
|                         | Returns the create_generated_clock -divide_by option of a generated clock.<br><b>Type:</b> int<br><b>Default:</b> ""<br><b>Edit:</b> No |

|                      |                                                                                                                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| driving_network_pins |                                                                                                                                                                                                                       |
|                      | <p>Returns a list of pin object which are present between target pins of generated clock and master clock.</p> <p><b>Type:</b> <code>obj(port)* obj(pin)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| duty_cycle           |                                                                                                                                                                                                                       |
|                      | <p>Returns the create_generated_clock -duty_cycle option of a generated clock.</p> <p><b>Type:</b> <code>double</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                           |
| edge_shift           |                                                                                                                                                                                                                       |
|                      | <p>Returns the create_generated_clock -edge_shift option of a generated clock.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                           |
| edges                |                                                                                                                                                                                                                       |
|                      | <p>Returns the create_generated_clock -edges option of a generated clock.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                |
| exceptions           |                                                                                                                                                                                                                       |
|                      | <p>Reports the path exceptions</p> <p><b>Type:</b> <code>string</code></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                  |
| generated_clocks     |                                                                                                                                                                                                                       |
|                      | <p>Returns a list of clock object which are the generated clocks derived from this master clock.</p> <p><b>Type:</b> <code>obj(clock)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                    |
| hierarchical_name    |                                                                                                                                                                                                                       |



|                              |                                                                                                                                                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>hierarchical_name</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                     |
| ideal_transition_max_fall    |                                                                                                                                                                                          |
|                              | <p>Returns the set_clock_transition -max -fall asserted on the clock and used during ideal mode analysis.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| ideal_transition_max_rise    |                                                                                                                                                                                          |
|                              | <p>Returns the set_clock_transition -max -rise asserted on the clock and used during ideal mode analysis.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| ideal_transition_min_fall    |                                                                                                                                                                                          |
|                              | <p>Returns the set_clock_transition -min -fall asserted on the clock and used during ideal mode analysis.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| ideal_transition_min_rise    |                                                                                                                                                                                          |
|                              | <p>Returns the set_clock_transition -min -rise asserted on the clock and used during ideal mode analysis.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_active                    |                                                                                                                                                                                          |
|                              | <p>Returns true if the clock has been set active by the set_active_clocks constraint.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                      |
| is_combinational_source_path |                                                                                                                                                                                          |

|                    |                                                                                                                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>Returns true if the create_generated_clock -combinational option was specified for the creation of this generated clock.</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                     |
| is_context_mapped  |                                                                                                                                                                                                                         |
|                    | <p>Returns true if the clock is mapped in the block level SDC in the timing context reading session</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                             |
| is_generated       |                                                                                                                                                                                                                         |
|                    | <p>Returns true if this clock object represents a generated clock</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                               |
| is_inverted        |                                                                                                                                                                                                                         |
|                    | <p>Returns true if the create_generated_clock -invert option was specified for the creation of this generated clock.</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                            |
| is_library_created |                                                                                                                                                                                                                         |
|                    | <p>Returns true if this generated_clock was specified in a Liberty generated_clock group and not created by a create_generated_clock constraint</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| is_propagated      |                                                                                                                                                                                                                         |
|                    | <p>Returns true if set_propagated_clock was asserted explicitly on this clock object.</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                           |
| is_source_inverted |                                                                                                                                                                                                                         |

|                                 |                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>Returns true if the create_generated_clock -source_invert option was specified for the creation of this generated clock.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                 |
| is_virtual                      |                                                                                                                                                                                                                                                                                                                       |
|                                 | <p>Returns true if this clock is virtual</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                    |
| master_clock                    |                                                                                                                                                                                                                                                                                                                       |
|                                 | <p>Returns the master clock of this generated clock as specified by the create_generated_clock -master_clock option when this generated clock was created.</p> <p><b>Type:</b> <a href="#">obj(clock)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                           |
| master_source                   |                                                                                                                                                                                                                                                                                                                       |
|                                 | <p>Returns the master clock source pin of this generated clock as specified by the create_generated_clock -source option when this generated clock was created.</p> <p><b>Type:</b> <a href="#">obj(pin)*</a> <a href="#">obj(hpin)*</a> <a href="#">obj(hport)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| max_capacitance_clock_path_fall |                                                                                                                                                                                                                                                                                                                       |
|                                 | <p>Returns the value asserted by set_max_capacitance -clock_path -fall on this clock.</p> <p><b>Type:</b> <a href="#">delay</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                      |
| max_capacitance_clock_path_rise |                                                                                                                                                                                                                                                                                                                       |
|                                 | <p>Returns the value asserted by set_max_capacitance -clock_path -rise on this clock.</p> <p><b>Type:</b> <a href="#">delay</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                      |
| max_capacitance_data_path_fall  |                                                                                                                                                                                                                                                                                                                       |

|                                |                                                                                                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | Returns the value asserted by set_max_capacitance -data_path -fall on this clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No |
| max_capacitance_data_path_rise |                                                                                                                                                  |
|                                | Returns the value asserted by set_max_capacitance -data_path -rise on this clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No |
| max_transition_clock_path_fall |                                                                                                                                                  |
|                                | Returns the value asserted by set_max_transition -clock_path -fall on this clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No |
| max_transition_clock_path_rise |                                                                                                                                                  |
|                                | Returns the value asserted by set_max_transition -clock_path -rise on this clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No |
| max_transition_data_path_fall  |                                                                                                                                                  |
|                                | Returns the value asserted by set_max_transition -data_path -fall on this clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| max_transition_data_path_rise  |                                                                                                                                                  |
|                                | Returns the value asserted by set_max_transition -data_path -rise on this clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| min_pulse_width_high           |                                                                                                                                                  |
|                                | Returns the minimum pulse width high requirement set by set_min_pulse_width .<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No    |
| min_pulse_width_low            |                                                                                                                                                  |

|                          |                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | Returns the minimum pulse width low requirement set by set_min_pulse_width.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                      |
| multiply_by              |                                                                                                                                                                  |
|                          | Returns the create_generated_clock -multiply_by option of a generated clock.<br><b>Type:</b> int<br><b>Default:</b> ""<br><b>Edit:</b> No                        |
| name                     |                                                                                                                                                                  |
|                          | The name of the clock.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                           |
| network_latency_fall_max |                                                                                                                                                                  |
|                          | Returns the maximum fall insertion delay specified by an explicit set_clock_latency on the clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No |
| network_latency_fall_min |                                                                                                                                                                  |
|                          | Returns the minimum fall insertion delay specified by an explicit set_clock_latency on the clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No |
| network_latency_rise_max |                                                                                                                                                                  |
|                          | Returns the maximum rise insertion delay specified by an explicit set_clock_latency on the clock.<br><b>Type:</b> delay<br><b>Default:</b> ""<br><b>Edit:</b> No |
| network_latency_rise_min |                                                                                                                                                                  |

|                              |                                                                                                                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Returns the minimum rise insertion delay specified by an explicit set_clock_latency on the clock.</p> <p><b>Type:</b> <a href="#">delay</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                              |
| num_exceptions               |                                                                                                                                                                                                                                                                                  |
|                              | <p>Reports the number of path exceptions</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                   |
| obj_type                     |                                                                                                                                                                                                                                                                                  |
|                              | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">clock</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                            |
| period                       |                                                                                                                                                                                                                                                                                  |
|                              | <p>Returns the period of the clock. The period is either specified directly with create_clock or derived from a generated clock.</p> <p><b>Type:</b> <a href="#">delay</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                  |
| source_jitter_early_fall_max |                                                                                                                                                                                                                                                                                  |
|                              | <p>Returns the amount of the maximum early fall source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_jitter_early_fall_min |                                                                                                                                                                                                                                                                                  |
|                              | <p>Returns the amount of the minimum early fall source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_jitter_early_rise_max |                                                                                                                                                                                                                                                                                  |

|                               |                                                                                                                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | Returns the amount of the maximum early rise source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |
| source_jitter_early_rise_min  |                                                                                                                                                                                                                                               |
|                               | Returns the amount of the minimum early rise source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |
| source_jitter_late_fall_max   |                                                                                                                                                                                                                                               |
|                               | Returns the amount of the maximum late fall source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| source_jitter_late_fall_min   |                                                                                                                                                                                                                                               |
|                               | Returns the amount of the minimum late fall source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| source_jitter_late_rise_max   |                                                                                                                                                                                                                                               |
|                               | Returns the amount of the maximum late rise source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| source_jitter_late_rise_min   |                                                                                                                                                                                                                                               |
|                               | Returns the amount of the minimum late rise source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| source_latency_early_fall_max |                                                                                                                                                                                                                                               |

|                               |                                                                                                                                                                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Returns the maximum early fall source insertion delay specified by an explicit set_clock_latency on the clock.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_latency_early_fall_min |                                                                                                                                                                                                  |
|                               | <p>Returns the minimum early fall source insertion delay specified by an explicit set_clock_latency on the clock.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_latency_early_rise_max |                                                                                                                                                                                                  |
|                               | <p>Returns the maximum early rise source insertion delay specified by an explicit set_clock_latency on the clock.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_latency_early_rise_min |                                                                                                                                                                                                  |
|                               | <p>Returns the minimum early rise source insertion delay specified by an explicit set_clock_latency on the clock.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_latency_late_fall_max  |                                                                                                                                                                                                  |
|                               | <p>Returns the maximum late fall source insertion delay specified by an explicit set_clock_latency on the clock.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| source_latency_late_fall_min  |                                                                                                                                                                                                  |
|                               | <p>Returns the minimum late fall source insertion delay specified by an explicit set_clock_latency on the clock.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| source_latency_late_rise_max  |                                                                                                                                                                                                  |



|                              |                                                                                                                                                                                                                                                                                                                 |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Returns the maximum late rise source insertion delay specified by an explicit set_clock_latency on the clock.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                 |
| source_latency_late_rise_min |                                                                                                                                                                                                                                                                                                                 |
|                              | <p>Returns the minimum late rise source insertion delay specified by an explicit set_clock_latency on the clock.</p> <p><b>Type:</b> delay</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                 |
| source_latency_pins          |                                                                                                                                                                                                                                                                                                                 |
|                              | <p>Returns a list of pin/port object(s) which are present between target pins of generated clock and master clock.</p> <p><b>Type:</b> obj(port)* obj(pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                |
| sources                      |                                                                                                                                                                                                                                                                                                                 |
|                              | <p>Returns the pin or port objects where the clock is attached to the design - as specified by create_clock or create_generated_clock</p> <p><b>Type:</b> obj(port)* obj(pin)* obj(hport)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                 |
| view_name                    |                                                                                                                                                                                                                                                                                                                 |
|                              | <p>The name of the analysis view for the clock.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                 |
| waveform                     |                                                                                                                                                                                                                                                                                                                 |
|                              | <p>Returns a list of the sequence of rising and falling edge times of a single period of the clock. The first list entry is always the first rising edge time, and the second entry is always the subsequent falling edge time.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

# clock\_spine

## Parent Objects

## Definition

clock tree spine

| Attribute | Description                                                                                                                                                                                                |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name      |                                                                                                                                                                                                            |
|           | name of clock_spine<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                        |
| obj_type  |                                                                                                                                                                                                            |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">clock_spine</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

# clock\_tree

## Parent Objects

[clock\\_tree\\_source\\_group](#), [pin](#), [root](#), [port](#)

## Definition

clock tree in ccopt

| Attribute               | Description |
|-------------------------|-------------|
| clock_tree_source_group |             |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>the clock_tree_source_group this clock_trees is a member of</p> <p><b>Type:</b> obj(clock_tree_source_group)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| cts_buffer_cells      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                       | <p>Specifies the buffer cells for CTS. If none are specified CCOpt will choose buffers from the libraries.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>Different buffer cells may be specified for any combination of clock tree and power domain.</p> <p>To use different buffers for each net type set the cts_buffer_cells_top and cts_buffer_cells_leaf attributes .</p> <p>Some examples follow:</p> <p>To specify buffer cells for all clock trees and all power domains:</p> <pre>set_db cts_buffer_cells {bufAX* bufBX*}</pre> <p>To specify buffer cells for a particular clock tree and all power domains:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_buffer_cells {bufX20 bufX18}</pre> <p>To specify buffer cells for a particular clock tree and power domain:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_buffer_cells -index {power_domain &lt;pd&gt;} {bufX12 bufX8}</pre> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_buffer_cells_leaf |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Specifies the buffer cells available for CTS to use on leaf nets. If none are specified CCOpt will use the same buffers as on trunk nets (as specified in the <code>cts_buffer_cells</code> attribute).</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any <code>dont_use</code> settings for the cells specified.</p> <p>Different leaf buffer cells may be specified for any combination of clock tree and power domain.</p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> <code>power_domain</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> Yes</p> |
| <code>cts_buffer_cells_top</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                   | <p>Specifies the buffers cells available for CTS to use on top nets. If none are specified CCOpt will use the same buffers as on trunk nets (as specified in the <code>cts_buffer_cells</code> attribute).</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any <code>dont_use</code> settings for the cells specified.</p> <p>Different top buffer cells may be specified for any combination of clock tree and power domain.</p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> <code>power_domain</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> Yes</p>  |
| <code>cts_cell_density</code>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Specifies the clock halo in the x direction.  
Any x direction clock halo defined by this attribute is proportional to the cell width.  
The constant of proportionality is defined by the attribute value. For example:  
. If `cts_cell_density` = 0.25 then the x direction clock halo equals 3 \* cell width.  
. If `cts_cell_density` = 0.5 then the x direction clock halo equals cell width.  
. If `cts_cell_density` = 0.75 then the x direction clock halo equals cell width / 3.  
. If `cts_cell_density` = 1 then the x direction clock halo equals zero.  
This attribute can specify the x direction clock halo for all clock trees via:  
`set_db cts_cell_density 0.8`  
This attribute can specify the x direction clock halo per-clock tree via:  
`set_db clock_tree:<clk> .cts_cell_density 0.9`  
The following attributes can be used to assign x direction clock halos within CCOpt:  
`cts_cell_halo_x`  
`cts_cell_density`  
`cts_cell_halo_sites`  
Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which:  
. If `cts_cell_halo_x` is set to a non-auto value, then this defines the x direction clock halo. The attributes `cts_cell_density` and `cts_cell_halo_sites` have no effect.  
. If `cts_cell_halo_x` is set to auto and `cts_cell_density` is set to a non-auto value then `cts_cell_density` defines the clock halo in the x direction. The attribute `cts_cell_halo_sites` has no effect.  
. If both `cts_cell_halo_x` and `cts_cell_density` are set to auto then `cts_cell_halo_sites` defines the clock halo in the x direction.  
Valid values: 0.01 to 1  
See also:  
. `cts_cell_halo_x`  
. `cts_cell_halo_sites`  
**Type:** [string](#)  
**Default:** 0.75  
**Edit:** Yes

`cts_cell_halo_x`

Specifies the clock halo distance in the x direction. The default value of this attribute is auto.

The following attributes can be used to assign x direction clock halos within CCOpt:

- cts\_cell\_halo\_x
- cts\_cell\_density
- cts\_cell\_halo\_sites

Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which:

- . If cts\_cell\_halo\_x is set to a non-auto value, then this defines the x direction clock halo. The attributes cts\_cell\_density and cts\_cell\_halo\_sites have no effect.
- . If cts\_cell\_halo\_x is set to auto and cts\_cell\_density is set to a non-auto value then cts\_cell\_density defines the clock halo in the x direction. The attribute cts\_cell\_halo\_sites has no effect.
- . If both cts\_cell\_halo\_x and cts\_cell\_density are set to auto then cts\_cell\_halo\_sites defines the clock halo in the x direction.

See also:

- . cts\_cell\_density
- . cts\_cell\_halo\_sites

**Type:** [string](#)  
**Allowed -index values:** power\_domain  
**Default:** auto  
**Edit:** Yes

cts\_cell\_halo\_y

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Specifies the clock halo distance in the y direction. The default value of this attribute is auto.</p> <p>The following attributes can be used to assign y direction clock halos within CCOpt:</p> <ul style="list-style-type: none"><li>cts_cell_halo_y</li><li>cts_adjacent_rows_legal</li><li>cts_cell_halo_rows</li></ul> <p>Only one of these attributes is used to determine the clock halo in the y direction. The following rules determine which:</p> <ul style="list-style-type: none"><li>. If cts_cell_halo_y is set to a non-auto value, then this defines the y direction clock halo. The attributes cts_adjacent_rows_legal and cts_cell_halo_rows have no effect.</li><li>. If cts_cell_halo_y is set to auto and cts_adjacent_rows_legal is set to a non-auto value then cts_adjacent_rows_legal defines the clock halo in the y direction. The attribute cts_cell_halo_rows has no effect.</li><li>. If both cts_cell_halo_y and cts_adjacent_rows_legal are set to auto then cts_cell_halo_rows defines the clock halo in the y direction.</li></ul> <p>See also:</p> <ul style="list-style-type: none"><li>. cts_adjacent_rows_legal</li><li>. cts_cell_halo_rows</li></ul> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> |
| cts_clock_gating_cells |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Specifies the clock gates for CTS. If none are specified CCOpt will choose clock gates from the libraries.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>A list of lists can be used to specify separate clock gate sub-lists. All CCOpt transforms will resize clock gates within the clock gate sub-list they are initially part of. If an incoming clock gate is not part of any clock gate sub-list, it will be resized to the first clock gate sub-list.</p> <p>Different clock gates may be specified for any combination of clock tree and power domain.</p> <p>Some examples follow:</p> <p>To specify clock gates for all clock trees and all power domains:<br/> <code>set_db cts_clock_gating_cells {cgAX* cgBX*}</code></p> <p>To specify clock gates for a particular clock tree and all power domains:<br/> <code>set_db clock_tree:&lt;clk&gt; .cts_clock_gating_cells {cgX20 cgX18}</code></p> <p>To specify clock gates for a particular clock tree and power domain:<br/> <code>set_db clock_tree:&lt;clk&gt; .cts_clock_gating_cells -index {power_domain &lt;pd&gt;} {cgX12 cgX8}</code></p> <p>To specify separate clock gate sub-lists:<br/> <code>set_db .cts_clock_gating_cells {{cgX12A1 cgX8A1} {cgX12A2 cgX8A2}}</code></p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names, a list of lists to specify separate clock gate sub-lists</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_clock_source_cells |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                        | <p>Specifies the cells available for CTS to size clock sources if the cts_size_clock_sources attribute is set to true. If none are specified the tool will choose cells from the libraries.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, the tool will ignore any dont_use settings for the cells specified.</p> <p>Different cells may be specified for clock trees or power domains. Only clock sources that are buffers, inverters, logic and clock gating cells with a single output can be resized.</p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| cts_clock_tree_parents |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>The list of parent clock trees from which this clock tree is generated, if any.<br/>Valid values: list cts_clock_tree<br/><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                  |
| cts_clock_tree_source_driver                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                 | <p>Specifies the base_pin which is assumed to drive this clock tree. It is either a single base_pin (in which case all arcs to that base_pin shall be used when timing the clock tree root) or a pair of base_pins (in which case only arcs from the first specified pin to the second will be considered).<br/>By default this is generated from clock tree extraction<br/>Valid values: base_pin   {base_pin base_pin}<br/><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p> |
| cts_clock_tree_source_group                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                 | <p>Specifies the clock tree source group to which this clock tree belongs.<br/>Valid values: list cts_clock_tree_source_group<br/><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                             |
| cts_clock_tree_source_group_clock_trees         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                 | <p>A list of the clock trees relevant to this source group.<br/>Valid values: list cts_clock_trees<br/><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                       |
| cts_clock_tree_source_input_max_transition_time |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                 | <p>The slew which will be assumed at the input of the root driver.<br/>Valid values: double<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> delay_corner<br/><b>Default:</b> 0<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                |
| cts_clock_tree_source_max_capacitance           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | <p>The maximum capacitive load which this clock tree is permitted to drive.<br/>Valid values: double   auto<br/>Auto: from clock tree extraction<br/><b>Type:</b> string<br/><b>Default:</b> auto<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                       |
| cts_clock_tree_source_output_max_transition_time |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                  | <p>If non-zero, the slew which will be assumed at the output of the root driver.<br/>This overrides the value from SDC.<br/>Valid values: double<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> delay_corner<br/><b>Default:</b> 0<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                           |
| cts_declone_clock_gates                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                  | <p>This attribute controls clock gate decloning (merging) in both CTS-only and full clock_opt_design (or ECF) flows.<br/>Valid values:<br/>false - decloning is disabled in all flows<br/>cts_only - decloning is enabled for CTS-only flows<br/>ccopt_only - decloning is enabled for full flows<br/>true - decloning is enabled for all flows<br/>See also attribute cts_declone_clock_logic.<br/><b>Type:</b> string<br/><b>Default:</b> ccopt_only<br/><b>Edit:</b> Yes</p>  |
| cts_declone_clock_logic                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                  | <p>This attribute controls clock logic decloning (merging) in both CTS-only and full clock_opt_design (or ECF) flows.<br/>Valid values:<br/>false - decloning is disabled in all flows<br/>cts_only - decloning is enabled for CTS-only flows<br/>ccopt_only - decloning is enabled for full flows<br/>true - decloning is enabled for all flows<br/>See also attribute cts_declone_clock_gates.<br/><b>Type:</b> string<br/><b>Default:</b> ccopt_only<br/><b>Edit:</b> Yes</p> |
| cts_delay_cells                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>Specifies the delay cells available for CTS. If none are specified CCOpt will not use delay cells.</p> <p>Setting this attribute to the string 'auto' means that CCOpt will choose delay cells from the libraries to use.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>Different delay cells may be specified for any combination of clock tree and power domain, or by omitting those arguments a global setting can be applied.</p> <p>Some examples follow:</p> <p>To specify delay cells for all clock trees and power domains:<br/> <code>set_db cts_delay_cells {delayAX* delayBX*}</code></p> <p>To specify delay cells for a particular clock tree and all power domains:<br/> <code>set_db clock_tree:&lt;clk&gt; .cts_delay_cells {delayX1 delayX2}</code></p> <p>To specify delay cells for a particular clock tree and power domain:<br/> <code>set_db clock_tree:&lt;clk&gt; .cts_delay_cells -index {power_domain &lt;pd&gt;} {delayX2 delayX3}</code></p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names, or the string 'auto'</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_inverter_cells |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Specifies the inverter cells available for CTS. If none are specified CCOpt will choose inverters from the libraries.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>Different inverter cells may be specified for any combination of clock tree and power domain.</p> <p>To use different inverters for each net type set the <code>cts_inverter_cells_top</code> and <code>cts_inverter_cells_leaf</code> attributes .</p> <p>Some examples follow:</p> <p>To specify inverter cells for all clock trees and all power domains:</p> <pre>set_db cts_inverter_cells {invAX* invBX*}</pre> <p>To specify inverter cells for a particular clock tree and all power domains:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_inverter_cells {invX20 invX18}</pre> <p>To specify inverter cells for a particular clock tree and power domain:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_inverter_cells -index {power_domain &lt;pd&gt;} {invX12 invX8}</pre> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_inverter_cells_leaf |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                         | <p>Specifies the inverter cells available for CTS to use on leaf nets. If none are specified CCOpt will use the same inverters as on trunk nets (as specified in the <code>cts_inverter_cells</code> attribute).</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>Different leaf inverter cells may be specified for any combination of clock tree and power domain.</p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| cts_inverter_cells_top  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Specifies the inverter cells available for CTS to use on top nets. If none are specified CCOpt will use the same inverters as on trunk nets (as specified in the <code>cts_inverter_cells</code> attribute).</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any <code>dont_use</code> settings for the cells specified.</p> <p>Different top inverter cells may be specified for any combination of clock tree and power domain.</p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> <code>power_domain</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> Yes</p> |
| <code>cts_load_capacitance_cells</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                         | <p>Specifies the load capacitance cells available for CTS. CTS will use cells from this collection for load capacitance optimizations.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> <code>power_domain</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>cts_logic_cells</code>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>Specifies the clock logics for CTS. If none are specified CCOpt will choose clock logics from the libraries.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>Different logic cells may be specified for any combination of clock tree and power domain.</p> <p>Some examples follow:</p> <p>To specify logic cells for all clock trees and all power domains:</p> <pre>set_db cts_logic_cells {and* mux*}</pre> <p>To specify logic cells for a particular clock tree and all power domains:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_logic_cells {andX20 andX18}</pre> <p>To specify logic cells for a particular clock tree and power domain:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_logic_cells -index {power_domain &lt;pd&gt;} {andX12 andX8}</pre> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                  |
| cts_route_type_leaf |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                     | <p>Specifies the route type. Setting this attribute binds an existing user-defined route_type to one or more types of clock tree nets. Binding a route_type to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that route_type.</p> <p>In the most common usage, the route_type is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional -net_type argument. Omitting the -net_type argument causes the route_type to be bound to all three types of clock tree nets. The optional -clock_tree &lt;pattern&gt; argument limits the binding to the clock trees whose name matches &lt;pattern&gt;. Omitting the -clock_tree argument causes the binding to apply to all clock trees.</p> <p>For a route_type to be used in CTS, it must be bound to at least one net type. If net type is not bound to any route_type, a default route_type will be created for that net type at the start of CTS.</p> <p>Valid values: names of route_types created with create_route_type</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> |
| cts_route_type_top  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Specifies the route type. Setting this attribute binds an existing user-defined route_type to one or more types of clock tree nets. Binding a route_type to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that route_type.</p> <p>In the most common usage, the route_type is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional -net_type argument. Omitting the -net_type argument causes the route_type to be bound to all three types of clock tree nets. The optional -clock_tree &lt;pattern&gt; argument limits the binding to the clock trees whose name matches &lt;pattern&gt;. Omitting the -clock_tree argument causes the binding to apply to all clock trees.</p> <p>For a route_type to be used in CTS, it must be bound to at least one net type. If net type is not bound to any route_type, a default route_type will be created for that net type at the start of CTS.</p> <p>Valid values: names of route_types created with create_route_type</p> <p><b>Type:</b> string<br/><b>Default:</b> default<br/><b>Edit:</b> Yes</p> |
| cts_route_type_trunk |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                      | <p>Specifies the route type. Setting this attribute binds an existing user-defined route_type to one or more types of clock tree nets. Binding a route_type to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that route_type.</p> <p>In the most common usage, the route_type is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional -net_type argument. Omitting the -net_type argument causes the route_type to be bound to all three types of clock tree nets. The optional -clock_tree &lt;pattern&gt; argument limits the binding to the clock trees whose name matches &lt;pattern&gt;. Omitting the -clock_tree argument causes the binding to apply to all clock trees.</p> <p>For a route_type to be used in CTS, it must be bound to at least one net type. If net type is not bound to any route_type, a default route_type will be created for that net type at the start of CTS.</p> <p>Valid values: names of route_types created with create_route_type</p> <p><b>Type:</b> string<br/><b>Default:</b> default<br/><b>Edit:</b> Yes</p> |
| cts_source_latency   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Specifies a delay value between the global clock source and this clock tree. This additional delay will be included in all timing analysis involving skew groups for which this clock tree is a source. The default is 0.<br/>Valid values: double<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> delay_corner<br/><b>Default:</b> 0<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                 |
| cts_target_max_transition_time      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                     | <p>The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner.<br/>Valid values: default   auto   ignore   double<br/><b>Type:</b> string<br/><b>Allowed -index values:</b> delay_corner power_domain<br/><b>Default:</b> top default trunk default leaf default<br/><b>Edit:</b> Yes</p> |
| cts_target_max_transition_time_leaf |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                     | <p>The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner.<br/>Valid values: default   auto   ignore   double<br/><b>Type:</b> string<br/><b>Allowed -index values:</b> delay_corner power_domain<br/><b>Default:</b> default<br/><b>Edit:</b> Yes</p>                                |
| cts_target_max_transition_time_sdc  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>If non-zero, the target slew used for clock tree synthesis, overriding the SDC. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow, in library units obtained from SDC.<br/>Valid values: double<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> delay_corner<br/><b>Default:</b> 0<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                       |
| cts_target_max_transition_time_top   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      | <p>The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner.<br/>Valid values: default   auto   ignore   double<br/><b>Type:</b> string<br/><b>Allowed -index values:</b> delay_corner power_domain<br/><b>Default:</b> default<br/><b>Edit:</b> Yes</p> |
| cts_target_max_transition_time_trunk |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      | <p>The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner.<br/>Valid values: default   auto   ignore   double<br/><b>Type:</b> string<br/><b>Allowed -index values:</b> delay_corner power_domain<br/><b>Default:</b> default<br/><b>Edit:</b> Yes</p> |
| cts_top_fanout_threshold             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|          |                                                                                                                                                                                                                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <p>Minimum number of transitive fanout in the clock tree for a net to be routed as a top net. Nets with at least this many sinks in their transitive fanout in the clock tree will have the special routing rules applied to them.<br/>Valid values: integer<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> unset<br/><b>Edit:</b> Yes</p> |
| insts    |                                                                                                                                                                                                                                                                                                                                                         |
|          | <p>instances in the clock tree<br/><b>Type:</b> <a href="#">obj(inst)*</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                               |
| name     |                                                                                                                                                                                                                                                                                                                                                         |
|          | <p>name of clock_tree<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                            |
| nets     |                                                                                                                                                                                                                                                                                                                                                         |
|          | <p>nets in the clock tree<br/><b>Type:</b> <a href="#">obj(net)*</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                     |
| obj_type |                                                                                                                                                                                                                                                                                                                                                         |
|          | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br/><b>Type:</b> <a href="#">enum (clock_tree)</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                       |
| sinks    |                                                                                                                                                                                                                                                                                                                                                         |
|          | <p>sinks (pins or ports) of the clock tree<br/><b>Type:</b> <a href="#">obj(pin)*</a> <a href="#">obj(port)*</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                         |
| source   |                                                                                                                                                                                                                                                                                                                                                         |

|  |                                                                                                                                                                                     |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>The source pin for this cts_clock_tree.<br/>Valid values: pin<br/><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(port)</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## clock\_tree\_source\_group

### Parent Objects

[clock\\_tree](#), [root](#)

### Definition

clock tree source group

| Attribute   | Description                                                                                                                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| clock_trees |                                                                                                                                                                                                                                                                                                    |
|             | <p>clock_tree_source_group member clock_trees<br/><b>Type:</b> <a href="#">obj(clock_tree)*</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                    |
| is_active   |                                                                                                                                                                                                                                                                                                    |
|             | <p>Determines whether the clock tree source group is active or not. Only active source groups will take part in sink allocation.<br/>Valid values: true false<br/><b>Type:</b> <a href="#">bool</a><br/><b>Allowed -index values:</b> clock_tree<br/><b>Default:</b> true<br/><b>Edit:</b> Yes</p> |
| name        |                                                                                                                                                                                                                                                                                                    |
|             | <p>name of clock_tree_source_group<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                          |
| obj_type    |                                                                                                                                                                                                                                                                                                    |

The obj\_type name for this object. To see all obj\_type names, go to [Contents](#)  
**Type:** [enum](#) ([clock\\_tree\\_source\\_group](#))  
**Default:** ""  
**Edit:** No

## constraint\_mode

### Parent Objects

[analysis\\_view](#), [design](#), [root](#)

### Definition

Associates a list of SDC constraint files with a specified constraint mode name, for multi-mode multi-corner analysis. This constraint mode name can be referred to later when creating analysis views. Use the `create_constraint_mode` and `update_constraint_mode` commands to create and modify constraint\_modes.

| Attribute     | Description                                                                                                                                                                                            |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ilm_sdc_files |                                                                                                                                                                                                        |
|               | Specifies an optional list of constraint files to use when using ILM mode<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                              |
| is_active     |                                                                                                                                                                                                        |
|               | Indicates that the constraint_mode is associated with an analysis_view that is used in the active analysis_view.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No         |
| is_dynamic    |                                                                                                                                                                                                        |
|               | Indicates that the constraint_mode is associated with an analysis_view that is used in the active dynamic analysis_view.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No |

|            |                                                                                                                                                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_hold    |                                                                                                                                                                                                                                 |
|            | <p>Indicates that the constraint_mode is associated with an analysis_view that is active for hold analysis.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| is_leakage |                                                                                                                                                                                                                                 |
|            | <p>Indicates that the constraint_mode is associated with an analysis_view that is used in the active leakage analysis_view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| is_setup   |                                                                                                                                                                                                                                 |
|            | <p>Indicates that the constraint_mode is associated with an analysis_view that is active for setup analysis.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                      |
| name       |                                                                                                                                                                                                                                 |
|            | <p>Provides the name of this constraint_mode object as specified by create_constraint_mode.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                     |
| obj_type   |                                                                                                                                                                                                                                 |
|            | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">constraint_mode</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| sdc_files  |                                                                                                                                                                                                                                 |
|            | <p>Specifies an optional list of constraint files to associate with the mode</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                    |
| tcl_vars   |                                                                                                                                                                                                                                 |

Specifies an optional list of user-specified Tcl variable name and value pairs - e.g: {{my\_var1 true} {my\_var2 0.1}} - which are set before the constraint files are read, and can be used inside the constraint files.

**Type:** [string](#)

**Default:** ""

**Edit:** No

## context\_data

### Parent Objects

### Definition

context data objects

| Attribute        | Description                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all_block_clocks |                                                                                                                                                                                           |
|                  | Reports all the mapped and unmapped block clocks of property context<br><b>Type:</b> <a href="#">obj(clock)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                 |
| all_top_clocks   |                                                                                                                                                                                           |
|                  | Reports all the mapped and unmapped top clocks of property context<br><b>Type:</b> <a href="#">obj(context_top_clock)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                       |
| obj_type         |                                                                                                                                                                                           |
|                  | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (context_data)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| top_instance     |                                                                                                                                                                                           |

|                       |                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Reports the top view of proeprty context_data<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| top_view              |                                                                                                                                                                      |
|                       | Reports the top view of proeprty context_data<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| unmapped_block_clocks |                                                                                                                                                                      |
|                       | Reports the unmapped block clocks of property context<br><b>Type:</b> <a href="#">obj(clock)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                           |
| unmapped_top_clocks   |                                                                                                                                                                      |
|                       | Reports the mapped top clocks of property context<br><b>Type:</b> <a href="#">obj(context_top_clock)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                   |

## context\_top\_clock

### Parent Objects

[context\\_data](#)

### Definition

context top clocks

| Attribute         | Description |
|-------------------|-------------|
| hierarchical_name |             |

|          |                                                                                                                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Reports the name of top clock<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                    |
| obj_type |                                                                                                                                                                                                                  |
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">context_top_clock</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

## def\_scan\_chain

### Parent Objects

[design](#), [root](#)

### Definition

A single scan-chain equivalent to one DEF SCANCHAIN. It is used by the placer to reorder scan-chain elements to reduce routing and congestion.

| Attribute      | Description                                                                                                                                                                                                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| end            |                                                                                                                                                                                                                                                                                           |
|                | Specifies the last input_pin pin or an output port of the scan chain.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">pin</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                |
| floating_insts |                                                                                                                                                                                                                                                                                           |
|                | Specifies a list of floating insts that can be re-ordered by scan-chain ordering commands. The format is a list of insts, where each inst is a sub-list of the scan-in pin, scan-out pin, and bit_length.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| max_bits       |                                                                                                                                                                                                                                                                                           |



|                |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | Specify a value that is at least as large as the size of the current chain. When specified with chains that include a partition_name, sets the maximum bit length (flip-flop bit count) that the chain can grow to in the partition.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                  |
| name           |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                | The name of the def_scan_chain.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                                                                                                    |
| obj_type       |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">def_scan_chain</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                      |
| ordered_insts  |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                | : Specifies one or more ordered lists of insts that should not be re-ordered by scan-chain ordering commands. The format is a list of <ordered lists> where each ordered list has an optional "name <string>", and then an "ordered <list>" where <list> has each inst as a sub-list of scan-in pin, scan-out pin, and bit length.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| scan_partition |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                | Specifies a partition name (same as DEF SCANCHAIN PARTITION <name>).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                                                               |
| scan_pins      |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                | The list of all the scan-in and scan-out pins or ports in scan-order that make up this def_scan_chain. The first object can be a scan-in pin or external input port, and the last object can be a scan-out pin or external output port. All the intermediate objects must be scan-in/scan-out pins.<br><b>Type:</b> <a href="#">obj(pin)*</a> <a href="#">obj(port)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No  |

|       |                                                                                                                                                                             |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| start |                                                                                                                                                                             |
|       | <p>Specifies the first output_pin pin or an input port of the scan chain.</p> <p><b>Type:</b> <a href="#">obj(pin)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

## delay\_corner

### Parent Objects

[analysis\\_view](#), [design](#), [root](#)

### Definition

The delay\_corner object provides references to rc\_corner objects which define the interconnect corner and timing\_condition-to-domain bindings which define device operating corners - providing the majority of the configuration information needed to drive delay calculation. An analysis\_view object will reference the delay\_corner and combine it with a constraint\_mode to complete the configuration. Use the create\_delay\_corner and update\_delay\_corner commands to create and modify delay\_corner objects.

| Attribute                      | Description                                                                                                                                                                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| default_early_timing_condition |                                                                                                                                                                                                                                                      |
|                                | <p>Specifies the timing condition to be used for early path analysis if one has not be explicitly specified for a given power domain.</p> <p><b>Type:</b> <a href="#">obj(timing_condition)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| default_late_timing_condition  |                                                                                                                                                                                                                                                      |
|                                | <p>Specifies the timing condition to be used for late path analysis if one has not be explicitly specified for a given power domain.</p> <p><b>Type:</b> <a href="#">obj(timing_condition)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| early_irdrop_files             |                                                                                                                                                                                                                                                      |

|                               |                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Specifies a file with instance-specific voltage data for early path analysis. Instance voltage data can be either exported from power rail analysis tools such as Voltus, or specified with set_instance_voltage constraints</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                        |
| early_pg_net_voltages         |                                                                                                                                                                                                                                                                                                                                                    |
|                               | <p>Specifies the power/ground net voltages to use for early analysis</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                   |
| early_rc_corner               |                                                                                                                                                                                                                                                                                                                                                    |
|                               | <p>Specifies the associated rc_corner object for early path analysis</p> <p><b>Type:</b> <a href="#">obj(rc_corner)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                           |
| early_temperature_files       |                                                                                                                                                                                                                                                                                                                                                    |
|                               | <p>Specifies a file with instance-specific temperature data for early path analysis</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                    |
| early_timing_condition        |                                                                                                                                                                                                                                                                                                                                                    |
|                               | <p>Provides the early timing_condition object associated with the specified power_domain index. If no power_domain index is given, the default early timing_condition will be returned.</p> <p><b>Type:</b> <a href="#">obj(timing_condition)</a><br/> <b>Allowed -index values:</b> power_domain<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| early_timing_condition_string |                                                                                                                                                                                                                                                                                                                                                    |
|                               | <p>Specifies the list of power_domain to timing_condition bindings for early path analysis - using &lt;power_domain&gt;@&lt;timing_condition&gt; list syntax, for example 'defaultTC PD1@TC1 PD2@TC2'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                 |
| is_active                     |                                                                                                                                                                                                                                                                                                                                                    |

|                   |                                                                                                                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>Indicates that the delay_corner is associated with an analysis_view that is used in the active analysis_view</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>         |
| is_dynamic        |                                                                                                                                                                                                       |
|                   | <p>Indicates that the delay_corner is associated with an analysis_view that is used in the active dynamic analysis_view</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_hold           |                                                                                                                                                                                                       |
|                   | <p>Indicates that the delay_corner is associated with an analysis_view that is active for hold analysis</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                 |
| is_leakage        |                                                                                                                                                                                                       |
|                   | <p>Indicates that the delay_corner is associated with an analysis_view that is used in the active leakage analysis_view</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_setup          |                                                                                                                                                                                                       |
|                   | <p>Indicates that the delay_corner is associated with an analysis_view that is active for setup analysis</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                |
| is_si_enabled     |                                                                                                                                                                                                       |
|                   | <p>Indicates that analysis_views associated with this delay_corner should have SI analysis performed on them</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>            |
| late_irdrop_files |                                                                                                                                                                                                       |

|                              |                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Specifies a file with instance-specific voltage data for late path analysis. Instance voltage data can be either exported from power rail analysis tools such as Voltus, or specified with set_instance_voltage constraints</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                       |
| late_pg_net_voltages         |                                                                                                                                                                                                                                                                                                                                                  |
|                              | <p>Specifies the power/ground net voltages to use for late analysis</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                  |
| late_rc_corner               |                                                                                                                                                                                                                                                                                                                                                  |
|                              | <p>Specifies the associated rc_corner object for late path analysis</p> <p><b>Type:</b> <a href="#">obj(rc_corner)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                          |
| late_temperature_files       |                                                                                                                                                                                                                                                                                                                                                  |
|                              | <p>Specifies a file with instance-specific temperature data for late path analysis</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                   |
| late_timing_condition        |                                                                                                                                                                                                                                                                                                                                                  |
|                              | <p>Provides the late timing_condition object associated with the specified power_domain index. If no power_domain index is given, the default late timing_condition will be returned.</p> <p><b>Type:</b> <a href="#">obj(timing_condition)</a><br/> <b>Allowed -index values:</b> power_domain<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| late_timing_condition_string |                                                                                                                                                                                                                                                                                                                                                  |
|                              | <p>Specifies the list of power_domain to timing_condition bindings for late path analysis - using &lt;power_domain&gt;@&lt;timing_condition&gt; list syntax, for example 'defaultTC PD1@TC1 PD2@TC2'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                |
| name                         |                                                                                                                                                                                                                                                                                                                                                  |

|                 |                                                                                                                                                                                                                                       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | Provides the name of this delay_corner object as specified by create_delay_corner. This name can referenced by other commands to refer to this object<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| obj_type        |                                                                                                                                                                                                                                       |
|                 | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">delay_corner</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                           |
| pg_net_voltages |                                                                                                                                                                                                                                       |
|                 | Specifies the power/ground net voltages to use for both early/late analysis<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                           |

## delay\_noise\_obj

### Parent Objects

[timing\\_point](#)

### Definition

description

| Attribute        | Description                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------|
| annotated_glitch |                                                                                                  |
|                  | annotated_glitch<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| attackers        |                                                                                                  |

|                         |                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|                         | Returns the attackers property for a delay_noise_obj<br><b>Type:</b> obj(delay_noise_obj)*<br><b>Default:</b> ""<br><b>Edit:</b> No |
| capacitance             |                                                                                                                                     |
|                         | capacitance<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                         |
| coupling_capacitance    |                                                                                                                                     |
|                         | coupling_capacitance<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                |
| glitch_area             |                                                                                                                                     |
|                         | glitch_area<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                         |
| glitch_width            |                                                                                                                                     |
|                         | glitch_width<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                        |
| hierarchical_name       |                                                                                                                                     |
|                         | Returns the hierarchical_name property for delay_noise_obj<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| input_failure_threshold |                                                                                                                                     |
|                         | input_failure_threshold<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                             |
| input_peak              |                                                                                                                                     |

|                      |                                                                                                                                                                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | input_peak<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                   |
| input_peak_slack     |                                                                                                                                                                              |
|                      | input_peak_slack<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                             |
| level                |                                                                                                                                                                              |
|                      | Returns the level property for delay_noise_obj<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                               |
| num_active_attackers |                                                                                                                                                                              |
|                      | num_active_attackers<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                         |
| num_attackers        |                                                                                                                                                                              |
|                      | num_attackers<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                |
| obj_type             |                                                                                                                                                                              |
|                      | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> enum (delay_noise_obj)<br><b>Default:</b> ""<br><b>Edit:</b> No |
| propagated_glitch    |                                                                                                                                                                              |
|                      | Reports the propagated glitch seen at the victim driver output<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                               |
| receiver_peak        |                                                                                                                                                                              |



|                         |                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------|
|                         | receiver_peak<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                |
| receiver_peak_slack     |                                                                                                                              |
|                         | Returns the receiver_peak_slack property for delay_noise_obj<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No |
| receiver_peak_threshold |                                                                                                                              |
|                         | receiver_peak_threshold<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                      |
| resistance              |                                                                                                                              |
|                         | Returns the resistance property for delay_noise_obj<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| transition              |                                                                                                                              |
|                         | Returns the transition property for delay_noise_obj<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| view_type               |                                                                                                                              |
|                         | Returns the view_type property for delay_noise_obj<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No           |

## density\_shape

### Parent Objects

[place\\_blockage](#)

## Definition

Rectangle with metal/cut or placement density information

| Attribute | Description                                                                                                                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| density   |                                                                                                                                                                                                              |
|           | Density value (range: 0.0-1.0<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                |
| obj_type  |                                                                                                                                                                                                              |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">density_shape</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| rect      |                                                                                                                                                                                                              |
|           | Rectangle. For cellDensity, the coordinates are local to cell, not relative to the design.<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                     |

## design

### Parent Objects

[port\\_bus](#), [hinst](#), [inst](#), [module](#), [pin\\_group](#), [bump](#), [pin\\_guide](#), [root](#), [port](#)

## Definition

top cell

| Attribute          | Description |
|--------------------|-------------|
| actual_scan_chains |             |

|                      |                                                                                                                                                                                                                                |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | List of all actual_scan_chains<br><b>Type:</b> <a href="#">obj(actual_scan_chain)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                |
| actual_scan_segments |                                                                                                                                                                                                                                |
|                      | List of all actual_scan_segments<br><b>Type:</b> <a href="#">obj(actual_scan_segment)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                            |
| analysis_type        |                                                                                                                                                                                                                                |
|                      | Returns analysis type for the session as single, bestcase-worsecase or on-chip-variation.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| analysis_views       |                                                                                                                                                                                                                                |
|                      | Returns the information about the analysis views in the design.<br><b>Type:</b> <a href="#">obj(analysis_view)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                   |
| arch_filename        |                                                                                                                                                                                                                                |
|                      | Source file that contains the specified module (Verilog) or architecture (VHDL).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                              |
| arch_name            |                                                                                                                                                                                                                                |
|                      | The name of the specified module (Verilog) or architecture (VHDL).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                            |
| area                 |                                                                                                                                                                                                                                |

|                         |                                                                                                                                                                                                                                                                                                |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>The area of all the insts contained inside this design and below it. This does not include phys_insts. Use 'get_db \$design .boundary.area' if you want the area of the physical design boundary.</p> <p><b>Type:</b> <a href="#">area</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| base_name               |                                                                                                                                                                                                                                                                                                |
|                         | <p>Name of design</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                  |
| bbox                    |                                                                                                                                                                                                                                                                                                |
|                         | <p>Bounding box of the design boundary,</p> <p><b>Type:</b> <a href="#">rect</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                              |
| boundaries              |                                                                                                                                                                                                                                                                                                |
|                         | <p>All the boundary objects (fence, region, etc.) in the design.</p> <p><b>Type:</b> <a href="#">obj(boundary)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                           |
| boundary                |                                                                                                                                                                                                                                                                                                |
|                         | <p>Polygon that define the shape of the floorplan boundary</p> <p><b>Type:</b> <a href="#">polygon*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                       |
| boundary_model_analysis |                                                                                                                                                                                                                                                                                                |
|                         | <p>Returns true if the design is boundary model</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                     |
| boundary_type           |                                                                                                                                                                                                                                                                                                |

|                           |                                                                                                                                                                                                                                                                                                                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Specifies the IEEE boundary-scan architecture {IEEE_11491   IEEE_11496}.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> Yes</p>                                                                                                                                 |
| bump_grids                |                                                                                                                                                                                                                                                                                                                        |
|                           | <p>List of bump_grid object.</p> <p><b>Type:</b> <code>obj(bump_grid)*</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                                        |
| bumps                     |                                                                                                                                                                                                                                                                                                                        |
|                           | <p>List of bumps in the cell</p> <p><b>Type:</b> <code>obj(bump)*</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                                             |
| bus_guides                |                                                                                                                                                                                                                                                                                                                        |
|                           | <p>List of bus guide objects.</p> <p><b>Type:</b> <code>obj(bus_guide)*</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                                       |
| bus_sink_groups           |                                                                                                                                                                                                                                                                                                                        |
|                           | <p>A group of sinks (loads) that some floorplan and routing commands use to control adding buffers and routing for a bus. See 'help *bus_sink*' for a list of commands related to this object.</p> <p><b>Type:</b> <code>obj(bus_sink_group)*</code></p> <p><b>Default:</b> <code>{}</code></p> <p><b>Edit:</b> No</p> |
| busses                    |                                                                                                                                                                                                                                                                                                                        |
|                           | <p>The Verilog bus definitions for nets in the top Verilog module.</p> <p><b>Type:</b> <code>obj(bus)*</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                        |
| capacitance_unit_in_farad |                                                                                                                                                                                                                                                                                                                        |

|                                             |                                                                                                                                                                                                                                  |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>Returns session units for capacitance in farad</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| chip_size                                   |                                                                                                                                                                                                                                  |
|                                             | <p>Returns the chip size that will be used for bounding box computation</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| clock_gating_add_obs_port                   |                                                                                                                                                                                                                                  |
|                                             | <p>Set clock-gating observation point.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                         |
| clock_gating_auto_cost_group_initial_target |                                                                                                                                                                                                                                  |
|                                             | <p>The initial target slack for automatic clock gate cost groups.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                               |
| clock_gating_auto_cost_grouping             |                                                                                                                                                                                                                                  |
|                                             | <p>Automatically creates a cost group (per clock) for clock gate enable paths.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                  |
| clock_gating_auto_path_adjust               |                                                                                                                                                                                                                                  |
|                                             | <p>Controls automatic timing adjustment on clock gate enable paths.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none fixed variable</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p>              |
| clock_gating_auto_path_adjust_fixed_delay   |                                                                                                                                                                                                                                  |

|                                          |                                                                                                                                                                                                  |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Fixes the path adjustment automatically added to clock gate enable pins.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p>                    |
| clock_gating_auto_path_adjust_modes      |                                                                                                                                                                                                  |
|                                          | <p>The timing modes in which clock gate enable path adjusts will be applied.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                    |
| clock_gating_auto_path_adjust_multiplier |                                                                                                                                                                                                  |
|                                          | <p>Scales the automatic load-dependent path adjustment added to clock gate enable pins.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p>        |
| clock_gating_control_point               |                                                                                                                                                                                                  |
|                                          | <p>Set clock-gating control point.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none precontrol postcontrol</p> <p><b>Default:</b> precontrol</p> <p><b>Edit:</b> Yes</p> |
| clock_gating_exclude                     |                                                                                                                                                                                                  |
|                                          | <p>Exclude this design from clock-gating.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                      |
| clock_gating_extract_common_enable       |                                                                                                                                                                                                  |
|                                          | <p>Inserts clock-gating by extracting common logic from synchronous enable logic of flops.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>      |
| clock_gating_hierarchical                |                                                                                                                                                                                                  |
|                                          | <p>Create hierarchical clock gates in this design.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                             |

|                            |                                                                                                                                                                                                                                                                                                                              |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| clock_gating_max_flops     |                                                                                                                                                                                                                                                                                                                              |
|                            | <p>Set fanout limit to clock-gating cell.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                |
| clock_gating_min_flops     |                                                                                                                                                                                                                                                                                                                              |
|                            | <p>Set minimum number of flops needed to add a clock-gating cell.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 3</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                               |
| clock_gating_module        |                                                                                                                                                                                                                                                                                                                              |
|                            | <p>Module used for clock gating circuitry.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                  |
| clock_gating_style         |                                                                                                                                                                                                                                                                                                                              |
|                            | <p>Set clock-gating style.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none latch ff</p> <p><b>Default:</b> latch</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                        |
| clock_gating_test_signal   |                                                                                                                                                                                                                                                                                                                              |
|                            | <p>Set specified test signal to connect up test logic in clock-gating circuitry. Specify use_shift_enable to derive test logic from shift enable of the gated flops, or use_none to disable test connection for the design.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| constraint_modes           |                                                                                                                                                                                                                                                                                                                              |
|                            | <p>Returns the information about the constraint modes in the design.</p> <p><b>Type:</b> <a href="#">obj(constraint_mode)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                          |
| control_logic_optimization |                                                                                                                                                                                                                                                                                                                              |



|                |                                                                                                                                                                                                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>Controls the effort level for optimization of control logic on this module (basic/advanced/none/inherited).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic advanced none inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p> |
| core_bbox      |                                                                                                                                                                                                                                                                               |
|                | <p>Rectangle that defines the core row area</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                     |
| core_boundary  |                                                                                                                                                                                                                                                                               |
|                | <p>List of polygons that define the shape of the core boundary.</p> <p><b>Type:</b> <a href="#">polygon*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                             |
| core_site      |                                                                                                                                                                                                                                                                               |
|                | <p>The site object to use during floorplan creation</p> <p><b>Type:</b> <a href="#">obj(site)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                        |
| core_to_bottom |                                                                                                                                                                                                                                                                               |
|                | <p>Distance between core edge and its die/io box</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                               |
| core_to_left   |                                                                                                                                                                                                                                                                               |
|                | <p>Distance between core edge and its die/io box</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                               |
| core_to_right  |                                                                                                                                                                                                                                                                               |
|                | <p>Distance between core edge and its die/io box</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                               |

|                                       |                                                                                                                                                                                          |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| core_to_top                           |                                                                                                                                                                                          |
|                                       | <p>Distance between core edge and its die/io box</p> <p><b>Type:</b> coord</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                          |
| cpf_macro_inherit_parent_power_domain |                                                                                                                                                                                          |
|                                       | <p>Specify if the macro instances in CPF will inherit their power domain from parent instance.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>         |
| cpi_inverter_name_prefix              |                                                                                                                                                                                          |
|                                       | <p>Prefix for inverters added by 'commit_power_intent'.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> cpi_inv</p> <p><b>Edit:</b> Yes</p>                                            |
| cpi_output_net_name_prefix            |                                                                                                                                                                                          |
|                                       | <p>The prefix for net name on output side of an instance inserted by 'commit_power_intent'.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>             |
| current_unit_in_amp                   |                                                                                                                                                                                          |
|                                       | <p>Returns session units for current in ampere.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| def_scan_chains                       |                                                                                                                                                                                          |
|                                       | <p>List of all def_scan_chains in the design</p> <p><b>Type:</b> obj(def_scan_chain)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                               |
| delay_corners                         |                                                                                                                                                                                          |

|                                        |                                                                                                                                                                                                       |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | Returns the information about the delay corners in the design.<br><b>Type:</b> <a href="#">obj(delay_corner)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                            |
| delete_unloaded_seqs                   |                                                                                                                                                                                                       |
|                                        | Control deletion of unloaded sequential instances.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> 1 0 true false inherited<br><b>Default:</b> inherited<br><b>Edit:</b> Yes              |
| design_verification_effort             |                                                                                                                                                                                                       |
|                                        | Specifies design verification effort level for LEC. Can be used to disable problematic optimizations in bulk.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> inherited<br><b>Edit:</b> Yes |
| dft_boundary_scan_exists               |                                                                                                                                                                                                       |
|                                        | Attribute that indicates if boundary scan has been inserted into the design.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                        |
| dft_clock_edge_for_head_of_scan_chains |                                                                                                                                                                                                       |
|                                        | Capture clock edge of scan chain (leading).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                          |
| dft_clock_edge_for_tail_of_scan_chains |                                                                                                                                                                                                       |
|                                        | Launch clock edge of scan chain (leading   trailing).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                |
| dft_configuration_modes                |                                                                                                                                                                                                       |
|                                        | List of all dft_configuration_modes<br><b>Type:</b> <a href="#">obj(dft_configuration_mode)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                             |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dft_connect_scan_data_pins_during_mapping       |                                                                                                                                                                                                                                                                                                                                                                        |
|                                                 | <p>DFT scan connection mode {loopback   floating   ground   high}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> loopback floating ground high</p> <p><b>Default:</b> loopback</p> <p><b>Edit:</b> Yes</p>                                                                                                                                       |
| dft_connect_shift_enable_during_mapping         |                                                                                                                                                                                                                                                                                                                                                                        |
|                                                 | <p>DFT shift enable connection mode {tie_off   floating}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> tie_off floating</p> <p><b>Default:</b> tie_off</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                              |
| dft_core_max_pipe_depth                         |                                                                                                                                                                                                                                                                                                                                                                        |
|                                                 | <p>Maximum pipeline depth of core.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                        |
| dft_dont_scan                                   |                                                                                                                                                                                                                                                                                                                                                                        |
|                                                 | <p>Exclude this design from scan-synthesis {inherited   true   false}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                                                                                                           |
| dft_lockup_element_type                         |                                                                                                                                                                                                                                                                                                                                                                        |
|                                                 | <p>Type of lockup element in scan chains {level_sensitive   edge_sensitive   preferred_level_sensitive   preferred_edge_sensitive}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> level_sensitive edge_sensitive preferred_level_sensitive preferred_edge_sensitive</p> <p><b>Default:</b> preferred_level_sensitive</p> <p><b>Edit:</b> Yes</p> |
| dft_lockup_element_type_for_tail_of_scan_chains |                                                                                                                                                                                                                                                                                                                                                                        |
|                                                 | <p>Launch element type for scan chains (level_sensitive   edge_sensitive).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> level_sensitive</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                               |

|                                     |                                                                                                                                                                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dft_max_length_of_scan_chains       |                                                                                                                                                                                                                               |
|                                     | <p>Default max length for any scan chain (unless overridden by chain-specific max length).</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                |
| dft_min_number_of_scan_chains       |                                                                                                                                                                                                                               |
|                                     | <p>Minimum number of total scan chains to be created.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                     |
| dft_mix_clock_edges_in_scan_chains  |                                                                                                                                                                                                                               |
|                                     | <p>Allow mixing in a same scan chain flip-flops triggered by either edge of a same clock.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                   |
| dft_scan_map_mode                   |                                                                                                                                                                                                                               |
|                                     | <p>DFT scan conversion mode {tdrc_pass   preserve   force_all}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> tdrc_pass preserve force_all</p> <p><b>Default:</b> tdrc_pass</p> <p><b>Edit:</b> Yes</p> |
| dft_scan_output_preference          |                                                                                                                                                                                                                               |
|                                     | <p>DFT scan output preference {auto   non_inverted   inverted}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto non_inverted inverted</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p>        |
| dft_shared_wrapper_input_threshold  |                                                                                                                                                                                                                               |
|                                     | <p>Specifies the maximum number of scannable flops which can be shared in wrapper cells for an input port.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 9999</p> <p><b>Edit:</b> Yes</p>                    |
| dft_shared_wrapper_output_threshold |                                                                                                                                                                                                                               |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Specifies the maximum number of scannable flops which can be shared in wrapper cells for an output port.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 9999<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| dft_tap_tck_period   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                      | <p>Tap TCK period.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 50000<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| dont_touch           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                      | <p>This attributes defines the user preservation status of the design during optimization. This setting will apply to all insts within the design unless overridden at a lower level hinst or on the inst object itself. The dont_touch_effective attribute on each child inst and hinst will return the resolved value.</p> <p>Use 'help inst dont_touch' to see the enum value definitions.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none false true delete_ok const_prop_size_delete_ok const_prop_delete_ok size_delete_ok size_ok size_same_height_ok size_same_footprint_ok<br/> <b>Default:</b> none<br/> <b>Edit:</b> Yes</p> |
| dont_touch_effective |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                      | <p>This attribute defines the effective preservation status of the design during optimization from the dont_touch_sources values. If the read_only source is 'true' then the effective value for the hinst is 'true'. Otherwise, the user value has precedence.</p> <p>See help on inst dont_touch attribute for details on possible values.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none false true delete_ok const_prop_size_delete_ok const_prop_delete_ok size_delete_ok size_ok size_same_height_ok size_same_footprint_ok<br/> <b>Default:</b> false<br/> <b>Edit:</b> No</p>                                                  |
| dont_touch_file      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                      | <p>Specifies the sourced file to set dont_touch.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                          |                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dont_touch_sources       |                                                                                                                                                                                                                                                                                                                                                                                     |
|                          | <p>Dictionary of {source &lt;value&gt;} pairs contributing to the dont_touch_effective attribute for this object :</p> <p>{user &lt;value&gt;} {read_only &lt;value&gt;}</p> <p>user # This hinst dont_touch value</p> <p>read_only # The design is read_only (set_module_view -top_read_only true)</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| dont_use_cells           |                                                                                                                                                                                                                                                                                                                                                                                     |
|                          | <p>List of cell names (wildcards supported) to disallow for this design during optimization. This can be overridden at a lower hinst level. Overrides any library dont_use values.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                 |
| dont_use_cells_effective |                                                                                                                                                                                                                                                                                                                                                                                     |
|                          | <p>The resolved list of all cell names to disallow during optimization for the design based on the library dont_use and the dont_use_cells and use_cells attributes of this design. The precedence is: use_cells of this hinst), then dont_use_cells of this design.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                |
| dp_csa                   |                                                                                                                                                                                                                                                                                                                                                                                     |
|                          | <p>Controls CSA operation on design/subdesign in 'synthesize -to_generic' (basic/none/inherited).</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> basic none inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                                                                                                             |
| dp_csa_factorize         |                                                                                                                                                                                                                                                                                                                                                                                     |
|                          | <p>Controls datapath csa_factorize on a design in 'syn_generic'.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> basic none inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                              |
| dp_opt_power_opt         |                                                                                                                                                                                                                                                                                                                                                                                     |

|                               |                                                                                                                                                                                                                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Controls datapath power optimization flow on a given subdesign.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                       |
| dp_rename                     |                                                                                                                                                                                                                                                                                    |
|                               | <p>Attribute to control the datapath rewriting optimization level on a subdesign (basic/advanced/none/inherited).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic advanced none inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>   |
| dp_sharing                    |                                                                                                                                                                                                                                                                                    |
|                               | <p>To control the datapath sharing on a design/subdesign in 'synthesize -to_generic' (advanced/none/inherited).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> advanced none inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>           |
| dp_speculation                |                                                                                                                                                                                                                                                                                    |
|                               | <p>Controls datapath speculation on a design in 'syn_generic'.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic none inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                               |
| early_clk_cell_derate_factor  |                                                                                                                                                                                                                                                                                    |
|                               | <p>Returns the derating factor for early clock paths specified through the set_timing_derate command with the -early parameter.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_data_cell_derate_factor |                                                                                                                                                                                                                                                                                    |



|                                     |                                                                                                                                                                                                                                                                              |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Returns the derating factor for early paths specified through the set_timing_derate command with the -early parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                 |
| early_fall_cell_check_derate_factor |                                                                                                                                                                                                                                                                              |
|                                     | <p>Returns the early cell check derating factor specified through the set_timing_derate command with the -fall parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                 |
| early_fall_clk_cell_derate_factor   |                                                                                                                                                                                                                                                                              |
|                                     | <p>Returns the early clock path derating factor specified through the set_timing_derate command with the -fall parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                 |
| early_fall_data_cell_derate_factor  |                                                                                                                                                                                                                                                                              |
|                                     | <p>Returns the early data cell check derating factor specified through the set_timing_derate command with the -fall parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>            |
| early_rise_cell_check_derate_factor |                                                                                                                                                                                                                                                                              |
|                                     | <p>Returns the early cell check derating factor specified through the set_timing_derate command with the -rise and -cell_check parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_rise_clk_cell_derate_factor   |                                                                                                                                                                                                                                                                              |

|                                    |                                                                                                                                                                                                                                                                    |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Returns the early clock cell check derating factor specified through the set_timing_derate command with the -rise parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_rise_data_cell_derate_factor |                                                                                                                                                                                                                                                                    |
|                                    | <p>Returns the early data cell check derating factor specified through the set_timing_derate command with the -rise parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| embedded_script                    |                                                                                                                                                                                                                                                                    |
|                                    | <p>Stores the SDC command script embedded in the source file that contains the module (Verilog) or entity (VHDL) specification for this design.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                   |
| entity_filename                    |                                                                                                                                                                                                                                                                    |
|                                    | <p>Source file that contains the specified module (Verilog) or entity (VHDL).</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                     |
| entity_name                        |                                                                                                                                                                                                                                                                    |
|                                    | <p>Name of the specified module (Verilog) or entity (VHDL).</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                        |
| exceptions                         |                                                                                                                                                                                                                                                                    |
|                                    | <p>Reports the path exceptions</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                            |
| formal_verification_constraints    |                                                                                                                                                                                                                                                                    |

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <p>List of all formal_verification_constraints</p> <p><b>Type:</b> <a href="#">obj(formal_verification_constraint)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                            |
| gcell_grids  |                                                                                                                                                                                                         |
|              | <p>All the gcell_grid objects in the design.</p> <p><b>Type:</b> <a href="#">obj(gcell_grid)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                  |
| gcells       |                                                                                                                                                                                                         |
|              | <p>The gcells for the current_design. The gcells are only created after global route has occurred.</p> <p><b>Type:</b> <a href="#">obj(gcell)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| groups       |                                                                                                                                                                                                         |
|              | <p>List of groups</p> <p><b>Type:</b> <a href="#">obj(group)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                   |
| gui_lines    |                                                                                                                                                                                                         |
|              | <p>All the gui_line objects in the design.</p> <p><b>Type:</b> <a href="#">obj(gui_line)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |
| gui_polygons |                                                                                                                                                                                                         |
|              | <p>All the gui_polygon objects in the design.</p> <p><b>Type:</b> <a href="#">obj(gui_polygon)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                |
| gui_rects    |                                                                                                                                                                                                         |
|              | <p>All the gui_rect objects in the design.</p> <p><b>Type:</b> <a href="#">obj(gui_rect)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |

|                     |                                                                                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gui_shapes          |                                                                                                                                                                                                               |
|                     | <p>All the gui_shape objects in the design.</p> <p><b>Type:</b> <code>obj(gui_rect)* obj(gui_line)* obj(gui_polygon)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                             |
| gui_texts           |                                                                                                                                                                                                               |
|                     | <p>All the gui_text objects in the design.</p> <p><b>Type:</b> <code>obj(gui_text)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                               |
| has_dp_rewrite_gint |                                                                                                                                                                                                               |
|                     | <p>This design/subdesign has rewritten instance.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                |
| has_dp_shared_gint  |                                                                                                                                                                                                               |
|                     | <p>This design/subdesign has shared instance.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                   |
| hdl_all_filelist    |                                                                                                                                                                                                               |
|                     | <p>Attribute to get all the files along with the included hdl files and package files for a design or subdesign.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| hdl_config_name     |                                                                                                                                                                                                               |
|                     | <p>Name of the configuration used to build this module.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                         |
| hdl_cw_list         |                                                                                                                                                                                                               |

|                         |                                                                                                                                                                                                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>List of CW components used by a design.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                          |
| hdl_elab_command_params |                                                                                                                                                                                                                                                                      |
|                         | <p>List of {parameter value} pair that are passed to 'elaborate -parameters' command.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                |
| hdl_filelist            |                                                                                                                                                                                                                                                                      |
|                         | <p>List of HDL files read prior to the elaboration of a design.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                     |
| hdl_parameters          |                                                                                                                                                                                                                                                                      |
|                         | <p>List of {parameter value cmdset is_default} used to 'elaborate' a design. 'cmdset' is '1' only if the parameter is set using the 'elaborate -parameters' command.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| hdl_pipeline_comp       |                                                                                                                                                                                                                                                                      |
|                         | <p>Specifies whether the design contains a \$pipeline component.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                    |
| hdl_user_name           |                                                                                                                                                                                                                                                                      |
|                         | <p>Name of the module (Verilog) or entity (VHDL) for this design.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                   |
| hdl_vdp_list            |                                                                                                                                                                                                                                                                      |
|                         | <p>Information about datapath function primitives used in a module.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                  |

|                    |                                                                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hierarchical_name  |                                                                                                                                                                                  |
|                    | Returns the hierarchical name of the design<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| hinsts             |                                                                                                                                                                                  |
|                    | all hinsts inside this design<br><b>Type:</b> <a href="#">obj(hinst)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                               |
| hnets              |                                                                                                                                                                                  |
|                    | all hnets inside this design<br><b>Type:</b> <a href="#">obj(hnet)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                 |
| hpin_busses        |                                                                                                                                                                                  |
|                    | Returns all the hpin busses of the design.<br><b>Type:</b> <a href="#">obj(hpin_bus)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                               |
| hpins              |                                                                                                                                                                                  |
|                    | all hpins inside this design<br><b>Type:</b> <a href="#">obj(hpin)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                 |
| hport_busses       |                                                                                                                                                                                  |
|                    | Returns all the hport busses of the design.<br><b>Type:</b> <a href="#">obj(hport_bus)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                             |
| ignore_library_drc |                                                                                                                                                                                  |

|                                     |                                                                                                                                                                           |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | Do not use design rule constraints from the library.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                    |
| ignore_library_max_fanout           |                                                                                                                                                                           |
|                                     | Do not use max_fanout rule constraints from the library.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                |
| incrsyn_force_not_replace           |                                                                                                                                                                           |
|                                     | Disable cache replacement, and force resynthesis of this module during incremental run.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes |
| insert_pmbist_without_liberty_files |                                                                                                                                                                           |
|                                     | Ignore liberty file information.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                        |
| insts                               |                                                                                                                                                                           |
|                                     | all the logical insts inside this design. Does not include physical-insts.<br><b>Type:</b> <a href="#">obj(inst)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No            |
| io_bbox                             |                                                                                                                                                                           |
|                                     | Rectangle that defines the IO area<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                          |
| io_constraints                      |                                                                                                                                                                           |
|                                     | List of io objects<br><b>Type:</b> <a href="#">obj(io_constraint)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                           |
| is_clock_synthesized                |                                                                                                                                                                           |

|                  |                                                                                                                                                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | Design status: Clock synthesized<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                        |
| is_core_to_io    |                                                                                                                                                                                                                           |
|                  | Indicates whether core2* attributes are measured between core edge (core_bbox) and design boundary (box) or between core edge and io box edge (io_bbox)<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes |
| is_detail_routed |                                                                                                                                                                                                                           |
|                  | Design status: detail routed, true if route_design was run.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                             |
| is_dont_touch    |                                                                                                                                                                                                                           |
|                  | Returns true if design is marked dont touch<br><b>Type:</b> bool<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                                                            |
| is_generic       |                                                                                                                                                                                                                           |
|                  | Determine whether the design is generic. The attribute is set true after elaborate_design.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> No                                                               |
| is_io_placed     |                                                                                                                                                                                                                           |
|                  | Design status: Is IO placed<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                 |
| is_mapped        |                                                                                                                                                                                                                           |



|                            |                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Determine whether the design is mapped. The attribute is set true after synthesize_design.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                 |
| is_max_transition_asserted |                                                                                                                                                                                                                |
|                            | <p>Returns true if max transition constraint has been applied on design</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_min_transition_asserted |                                                                                                                                                                                                                |
|                            | <p>Returns true if min transition constraint has been applied on design</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_od_info_saved           |                                                                                                                                                                                                                |
|                            | <p>Indicates if OD info was saved in the DB.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                  |
| is_placed                  |                                                                                                                                                                                                                |
|                            | <p>Design status: Placed</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                     |
| is_proto_model_committed   |                                                                                                                                                                                                                |
|                            | <p>Indicates that the design has committed FlexModels</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                            |
| is_proto_model_specified   |                                                                                                                                                                                                                |

|                               |                                                                                                                                                                                                    |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | Indicates that the design has specified FlexModels<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                   |
| is_rc_extracted               |                                                                                                                                                                                                    |
|                               | Design status: Parasitic extracted<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                               |
| is_retimed                    |                                                                                                                                                                                                    |
|                               | Indicates that the design has been retimed.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                      |
| is_routed                     |                                                                                                                                                                                                    |
|                               | Design status: routed. true if design has global routes from place_design/route_early_global or detail routes from route_design.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes |
| is_scan_optimized             |                                                                                                                                                                                                    |
|                               | Design status: Scan optimized<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                    |
| isolate_zero_pin_retention    |                                                                                                                                                                                                    |
|                               | Insert isolation cell on zero pin SRPG cells.<br><b>Type:</b> bool<br><b>Default:</b> true<br><b>Edit:</b> Yes                                                                                     |
| late_cell_check_derate_factor |                                                                                                                                                                                                    |

|                                    |                                                                                                                                                                                                                                                                   |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Returns the derating factor for late paths specified through the set_timing_derate command with the -late parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| late_clk_cell_derate_factor        |                                                                                                                                                                                                                                                                   |
|                                    | <p>Returns the derating factor for late clock paths specified through the set_timing_derate command with the -late parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| late_data_cell_derate_factor       |                                                                                                                                                                                                                                                                   |
|                                    | <p>Returns the derating factor for late data paths specified through the set_timing_derate command with the -late parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| late_fall_cell_check_derate_factor |                                                                                                                                                                                                                                                                   |
|                                    | <p>Returns the late cell check derating factor specified through the set_timing_derate command with the -fall parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| late_fall_clk_cell_derate_factor   |                                                                                                                                                                                                                                                                   |
|                                    | <p>Returns the late clock cell check derating factor specified through the set_timing_derate command with the -fall parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| late_fall_data_cell_derate_factor  |                                                                                                                                                                                                                                                                   |

|                                    |                                                                                                                                                                                                                                                                   |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Returns the late data cell check derating factor specified through the set_timing_derate command with the -fall parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| late_rise_cell_check_derate_factor |                                                                                                                                                                                                                                                                   |
|                                    | <p>Returns the late cell check derating factor specified through the set_timing_derate command with the -rise parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| late_rise_clk_cell_derate_factor   |                                                                                                                                                                                                                                                                   |
|                                    | <p>Returns the late clock cell check derating factor specified through the set_timing_derate command with the -rise parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| late_rise_data_cell_derate_factor  |                                                                                                                                                                                                                                                                   |
|                                    | <p>Returns the late data cell check derating factor specified through the set_timing_derate command with the -rise parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| library_name                       |                                                                                                                                                                                                                                                                   |
|                                    | <p>Name of the (logical) library in which module (Verilog) or entity (VHDL) reside.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                               |
| local_hinsts                       |                                                                                                                                                                                                                                                                   |
|                                    | <p>List of hinsts in the current level of this design (e.g. from inside the design looking down at the hinsts at this level)</p> <p><b>Type:</b> obj(hinst)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                 |

|                               |                                                                                                                                                                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| local_hnets                   |                                                                                                                                                                                                                                       |
|                               | <p>List of hnets in the current level of this design (e.g. from inside the design looking down at the hnets at this level)</p> <p><b>Type:</b> <code>obj(hnet)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| local_hpins                   |                                                                                                                                                                                                                                       |
|                               | <p>List of hpins in the current level of this design (e.g. from inside the design looking down at the hpins of hinsts at this level)</p> <p><b>Type:</b> <code>obj(hpin)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| local_insts                   |                                                                                                                                                                                                                                       |
|                               | <p>List of insts in the current level of this design (e.g. from inside the design looking down at the insts at this level)</p> <p><b>Type:</b> <code>obj(inst)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| local_pins                    |                                                                                                                                                                                                                                       |
|                               | <p>List of pins in the current level of this design (e.g. from inside the design looking down at the pins of the insts at this level)</p> <p><b>Type:</b> <code>obj(pin)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| lp_clock_tree_buffers         |                                                                                                                                                                                                                                       |
|                               | <p>Specify the buffers or inverters to be used in clock tree synthesis.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                 |
| lp_clock_tree_leaf_max_fanout |                                                                                                                                                                                                                                       |
|                               | <p>Specify the number of flops driven by leaf clock buffer.</p> <p><b>Type:</b> <code>int</code></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                 |
| lp_pso_aware_tcf              |                                                                                                                                                                                                                                       |

|                                        |                                                                                                                                                                                                          |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | Specify that asserted TCF is generated by simulating the design while taking into account when power domains are shutoff.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> Yes |
| markers                                |                                                                                                                                                                                                          |
|                                        | All the marker objects in the design.<br><b>Type:</b> <code>obj(marker)*</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                 |
| max_capacitance                        |                                                                                                                                                                                                          |
|                                        | Returns max capacitance value applied on design<br><b>Type:</b> <code>double</code><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                        |
| max_fanout                             |                                                                                                                                                                                                          |
|                                        | Returns max fanout value applied on design<br><b>Type:</b> <code>double</code><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                             |
| max_transition                         |                                                                                                                                                                                                          |
|                                        | Returns max transition value applied on design<br><b>Type:</b> <code>double</code><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                         |
| mbist_enable_shared_library_domain_set |                                                                                                                                                                                                          |
|                                        | Library domains across which memories can be shared.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                       |
| min_capacitance                        |                                                                                                                                                                                                          |

|                |                                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | Returns min capacitance value applied on design<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| min_fanout     |                                                                                                                                                                                      |
|                | Returns min fanout value applied on design<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No      |
| min_transition |                                                                                                                                                                                      |
|                | Returns min transition value applied on design<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| modules        |                                                                                                                                                                                      |
|                | all modules inside this design<br><b>Type:</b> <a href="#">obj(module)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                 |
| name           |                                                                                                                                                                                      |
|                | Name of design<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                       |
| net_groups     |                                                                                                                                                                                      |
|                | List of net groups<br><b>Type:</b> <a href="#">obj(net_group)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                          |
| nets           |                                                                                                                                                                                      |

|                |                                                                                                                                                                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>All the nets inside this design, including logical Verilog nets, physical-only nets, Verilog supply0/supply1, and Verilog 1'b0/1'b1 nets.</p> <p><b>Type:</b> <a href="#">obj(net)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| num_core_rows  |                                                                                                                                                                                                                                                 |
|                | <p>Number of rows within the core area</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                             |
| num_exceptions |                                                                                                                                                                                                                                                 |
|                | <p>Reports the number of path exceptions</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                  |
| num_insts      |                                                                                                                                                                                                                                                 |
|                | <p>Number of instances in the cell</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                 |
| num_nets       |                                                                                                                                                                                                                                                 |
|                | <p>Number of canonical (flat) nets in the cell</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                     |
| num_pg_nets    |                                                                                                                                                                                                                                                 |
|                | <p>Number of physical nets in the cell</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                             |
| num_phys_insts |                                                                                                                                                                                                                                                 |
|                | <p>Number of physical instances in the cell</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                        |



|                                        |                                                                                                                                                                                                                        |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oa_design_lib                          |                                                                                                                                                                                                                        |
|                                        | <p>Library name for the design in OpenAccess cellview</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                  |
| oa_design_view                         |                                                                                                                                                                                                                        |
|                                        | <p>View name for the design in OpenAccess cellview</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                     |
| obj_type                               |                                                                                                                                                                                                                        |
|                                        | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">design</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| operating_condition_max                |                                                                                                                                                                                                                        |
|                                        | <p>Returns max operating condition of design</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                  |
| operating_condition_min                |                                                                                                                                                                                                                        |
|                                        | <p>Returns min operating condition of design</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                  |
| opt_retime_clocks                      |                                                                                                                                                                                                                        |
|                                        | <p>Perform retiming on these clocks.</p> <p><b>Type:</b> <a href="#">object</a>*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                 |
| opt_retime_move_mux_loop_with_register |                                                                                                                                                                                                                        |

|                    |                                                                                                                                                                                                                                                                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>Move the feedback mux loop along with the register during retiming. Module level control.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> <code>true</code></p> <p><b>Edit:</b> Yes</p>                                                                                                                                    |
| package_components |                                                                                                                                                                                                                                                                                                                                            |
|                    | <p>List of other chips in the package</p> <p><b>Type:</b> <code>obj(package_component)*</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                                           |
| package_objects    |                                                                                                                                                                                                                                                                                                                                            |
|                    | <p>List of objects in the package</p> <p><b>Type:</b> <code>obj(package_object)*</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                                                  |
| partitions         |                                                                                                                                                                                                                                                                                                                                            |
|                    | <p>all partitions inside this design</p> <p><b>Type:</b> <code>obj(partition)*</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                                                    |
| pg_nets            |                                                                                                                                                                                                                                                                                                                                            |
|                    | <p>List of power ground nets in the design. This include physical-only PG nets, and Verilog supply0/supply1 nets. Note that Verilog nets assigned to 1'b0/1'b1 are returned by the nets attribute, not the pg_nets attribute.</p> <p><b>Type:</b> <code>obj(net)*</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p> |
| pg_pins            |                                                                                                                                                                                                                                                                                                                                            |
|                    | <p>All PG pins inside this design.</p> <p><b>Type:</b> <code>obj(pg_pin)*</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                                                         |
| pg_ports           |                                                                                                                                                                                                                                                                                                                                            |

|                              |                                                                                                                                         |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
|                              | List of power/ground terminals in the cell<br><b>Type:</b> <code>obj(port)*</code><br><b>Default:</b> ""<br><b>Edit:</b> No             |
| phys_insts                   |                                                                                                                                         |
|                              | List of physical instances in the cell<br><b>Type:</b> <code>obj(inst)*</code><br><b>Default:</b> ""<br><b>Edit:</b> No                 |
| phys_ports                   |                                                                                                                                         |
|                              | List of physical (unused) signal terminals in the cell<br><b>Type:</b> <code>obj(port)*</code><br><b>Default:</b> ""<br><b>Edit:</b> No |
| pi_relax_map_iso_cell_checks |                                                                                                                                         |
|                              | Relax sanity checks on map_isolation_cell list.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> Yes          |
| pi_relax_map_ls_cell_checks  |                                                                                                                                         |
|                              | Relax sanity checks on map_level_shifter_cell list.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> Yes      |
| pin_blockages                |                                                                                                                                         |
|                              | List of partition pin blockage objects<br><b>Type:</b> <code>obj(pin_blockage)*</code><br><b>Default:</b> ""<br><b>Edit:</b> No         |
| pin_groups                   |                                                                                                                                         |
|                              | List of all the pin groups<br><b>Type:</b> <code>obj(pin_group)*</code><br><b>Default:</b> ""<br><b>Edit:</b> No                        |
| pin_guides                   |                                                                                                                                         |

|                        |                                                                                                                                                                                                                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>List of all the pin guide object</p> <p><b>Type:</b> <code>obj(pin_guide)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                            |
| pin_to_corner_tracks   |                                                                                                                                                                                                                                                                                      |
|                        | <p>List of distance constraints (in tracks) of pins from design/partition corners where the lower left corner is listed first and the remaining corners are listed in clockwise order.</p> <p><b>Type:</b> <code>string*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pins                   |                                                                                                                                                                                                                                                                                      |
|                        | <p>Short-cut for [get_db current_design .pins]</p> <p><b>Type:</b> <code>obj(pin)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                       |
| place_blockages        |                                                                                                                                                                                                                                                                                      |
|                        | <p>List of placement blockages.</p> <p><b>Type:</b> <code>obj(place_blockage)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                           |
| pmbist_hri_async_reset |                                                                                                                                                                                                                                                                                      |
|                        | <p>Asynchronous reset port/pin along with the active state for repair used by programmable MBIST specified as {&lt;pin/port name&gt; active_[high low]}.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                               |
| port_busses            |                                                                                                                                                                                                                                                                                      |
|                        | <p>The Verilog bus definitions for ports in the top Verilog module.</p> <p><b>Type:</b> <code>obj(port_bus)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                             |
| port_shapes            |                                                                                                                                                                                                                                                                                      |

|                                |                                                                                                                                                                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>These are all the individual physical_pin shapes of the top-level ports including signal and PG ports.</p> <p><b>Type:</b> obj(port_shape)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| ports                          |                                                                                                                                                                                                     |
|                                | <p>ports of design. Does not include pg_ports, unless PG port is explicitly in the Verilog netlist.</p> <p><b>Type:</b> obj(port)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>             |
| power_domains                  |                                                                                                                                                                                                     |
|                                | <p>List of power domains (pd) in the design</p> <p><b>Type:</b> obj(power_domain)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                             |
| power_modes                    |                                                                                                                                                                                                     |
|                                | <p>List of the power mode names in the design.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                      |
| preserve_power_domain_boundary |                                                                                                                                                                                                     |
|                                | <p>Preserve value for power domain boundaries.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                  |
| process_max                    |                                                                                                                                                                                                     |
|                                | <p>Returns max process value set on design</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                 |
| process_min                    |                                                                                                                                                                                                     |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Returns min process value set on design</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| read_db_design_name    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                        | <p>The design name of the restored design.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| read_db_tool_name      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                        | <p>The tool name save the restored design in current session.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| read_db_tool_version   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                        | <p>The tool version number save the restored design in current session.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| read_only              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                        | <p>This attribute is set by set_module_view to identify if the top_level is read only or not. It means the partition cannot be optimized, and cells inside will not be moved. Setting this attribute will set the dont_touch_effective attribute on all insts and hinsts within the top level partition unless overridden at a lower level partition. It cannot be overridden by other hinst or inst values.</p> <p><b>Supported values:</b></p> <ul style="list-style-type: none"> <li>false: The top level is allowed to be optimized.</li> <li>true: The top level is read_only.</li> <li>none: No constraint.</li> </ul> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none false true</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p> |
| resistance_unit_in_ohm |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                          |                                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Returns session units for resistance in ohm</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                  |
| resize_blockages         |                                                                                                                                                                                                                          |
|                          | <p>List of resize_blockage objects</p> <p><b>Type:</b> <a href="#">obj(resize_blockage)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                        |
| retime                   |                                                                                                                                                                                                                          |
|                          | <p>Retime this design during synthesis.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                |
| retime_period_percentage |                                                                                                                                                                                                                          |
|                          | <p>Adjust target clock period in retiming.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p>                                                                             |
| retime_ssw_sync_enable   |                                                                                                                                                                                                                          |
|                          | <p>Directs retiming to use SAT-sweep to find equivalence classes for synchronous enable signals. Module level control.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> |
| route_blockages          |                                                                                                                                                                                                                          |
|                          | <p>List of routing blockages</p> <p><b>Type:</b> <a href="#">obj(route_blockage)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                               |
| route_halo_bottom_layer  |                                                                                                                                                                                                                          |
|                          | <p>The bottom layer for which routing halo will be created</p> <p><b>Type:</b> <a href="#">obj(layer)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                           |

|                        |                                                                                                                                                                                                                                                                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_halo_to_boundary |                                                                                                                                                                                                                                                                                                        |
|                        | <p>Specifies routing halo inside the design boundary (honored by the signal router). Only positive values are used and indicate the halo is inside of the design boundary.</p> <p><b>Type:</b> coord</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                       |
| route_halo_top_layer   |                                                                                                                                                                                                                                                                                                        |
|                        | <p>The top layer for which routing halo will be created</p> <p><b>Type:</b> obj(layer)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                            |
| row_flip               |                                                                                                                                                                                                                                                                                                        |
|                        | <p>Specification of floorplan row creation, none = no flipping; first = first row is flipped, other rows alternate; second = first row is not flipped, other rows alternate</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> none second first</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| row_spacing            |                                                                                                                                                                                                                                                                                                        |
|                        | <p>Specification of the floorplan row spacing</p> <p><b>Type:</b> coord</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                    |
| row_spacing_type       |                                                                                                                                                                                                                                                                                                        |
|                        | <p>Indicates whether the rowSpacing is applied between each row (1) or between each pair of rows (2)</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> 0 1 2</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                   |
| rows                   |                                                                                                                                                                                                                                                                                                        |
|                        | <p>List of rows</p> <p><b>Type:</b> obj(row)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                     |
| scan_chains            |                                                                                                                                                                                                                                                                                                        |



|                              |                                                                                                                                                                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>List of all scan_chains</p> <p><b>Type:</b> <a href="#">obj(scan_chain)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                             |
| scan_in_pipeline_clock_edge  |                                                                                                                                                                                                                                                  |
|                              | <p>Edge of scan in pipeline flop {rise   fall}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> rise fall</p> <p><b>Default:</b> rise</p> <p><b>Edit:</b> Yes</p>                                                            |
| scan_out_pipeline_clock_edge |                                                                                                                                                                                                                                                  |
|                              | <p>Edge of scan out pipeline flop {rise   fall}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> rise fall</p> <p><b>Default:</b> rise</p> <p><b>Edit:</b> Yes</p>                                                           |
| scan_segments                |                                                                                                                                                                                                                                                  |
|                              | <p>List of all scan_segments</p> <p><b>Type:</b> <a href="#">obj(scan_segment)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                         |
| sdp_files                    |                                                                                                                                                                                                                                                  |
|                              | <p>List of SDP files that have been read in.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> No</p>                                                                                               |
| sdp_type                     |                                                                                                                                                                                                                                                  |
|                              | <p>Specify the SDP file type read in for the design to create SDP Constraints. Valid values are {datapath   ff_column   both   no_value}.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p> |
| seq_reason_deleted           |                                                                                                                                                                                                                                                  |

|                                  |                                                                                                                                                                                           |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | Deleted sequential instances for a design with reason.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> {}<br><b>Edit:</b> Yes                                                   |
| setup_views                      |                                                                                                                                                                                           |
|                                  | Returns the information about the setup analysis views in the design.<br><b>Type:</b> <a href="#">obj(analysis_view)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                        |
| slew_threshold_percent_fall_high |                                                                                                                                                                                           |
|                                  | Returns fall percentage value of high slew threshold<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| slew_threshold_percent_fall_low  |                                                                                                                                                                                           |
|                                  | Returns fall percentage value of low slew threshold<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| slew_threshold_percent_rise_high |                                                                                                                                                                                           |
|                                  | Returns rise percentage value of high slew threshold<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| slew_threshold_percent_rise_low  |                                                                                                                                                                                           |
|                                  | Returns rise percentage value of low slew threshold<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| symmetry                         |                                                                                                                                                                                           |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>The allowed orientations for this design. This value only affects the SYMMETRY value written out by write_lef_abstract. none means r0 only (no SYMMETRY statement), x means ok to flip about X axis (SYMMETRY X), y means ok to flip about Y axis (SYMMETRY Y), xy means ok to flip about X or Y axis (SYMMETRY X Y), any means all orientations are allowed (SYMMETRY X Y R90).</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none x y xy any<br/> <b>Default:</b> any<br/> <b>Edit:</b> Yes</p> |
| tech               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                    | <p>The tech for this design.</p> <p><b>Type:</b> <a href="#">obj(tech)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                              |
| temperature_max    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                    | <p>Returns max temperature value set on design</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                        |
| temperature_min    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                    | <p>Returns min temperature value set on design</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                        |
| test_clock_domains |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                    | <p>List of all test_clock_domains</p> <p><b>Type:</b> <a href="#">obj(test_clock_domain)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                           |
| test_clocks        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                    | <p>List of all test_clocks</p> <p><b>Type:</b> <a href="#">obj(test_clock)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                         |
| test_signals       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                           |                                                                                                                                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>List of all test_signals</p> <p><b>Type:</b> <a href="#">obj(test_signal)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                |
| texts                     |                                                                                                                                                                                                       |
|                           | <p>All the text objects in the design.</p> <p><b>Type:</b> <a href="#">obj(text)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                            |
| time_unit_in_second       |                                                                                                                                                                                                       |
|                           | <p>Returns session units for time in second.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| timing_context_analysis   |                                                                                                                                                                                                       |
|                           | <p>Returns true if the design is context analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| top_sdps                  |                                                                                                                                                                                                       |
|                           | <p>List of all top level structured data path (sdp) objects.</p> <p><b>Type:</b> <a href="#">obj(sdp)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                       |
| total_coverage_mode_depth |                                                                                                                                                                                                       |
|                           | <p>This property is obsoleted</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                |
| track_patterns            |                                                                                                                                                                                                       |

|                                 |                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>All the track_pattern objects in the design.</p> <p><b>Type:</b> obj(track_pattern)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                     |
| tree_type_max                   |                                                                                                                                                                                                                                                                                                                                  |
|                                 | <p>Returns max tree type set on design through operating condition</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                      |
| tree_type_min                   |                                                                                                                                                                                                                                                                                                                                  |
|                                 | <p>Returns min tree type set on design through operating condition</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                      |
| trim_grids                      |                                                                                                                                                                                                                                                                                                                                  |
|                                 | <p>All the trim_grid objects in the design from the LEF TRIMMETALTRACK statement.</p> <p><b>Type:</b> obj(trim_grid)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                       |
| use_cells                       |                                                                                                                                                                                                                                                                                                                                  |
|                                 | <p>List of cell names to allow for this design during optimization. This can be overridden at a lower hinst level. All lib_cells of each base_cell will be allowed. Overrides cells in the dont_use_cells list and any library dont_use values.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| use_only_on_power_critical_nets |                                                                                                                                                                                                                                                                                                                                  |
|                                 | <p>Use library cells from this base cell set only on power critical nets.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                       |
| verification_directory          |                                                                                                                                                                                                                                                                                                                                  |

|                           |                                                                                                                                                                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Directory in which verification files are written. The directory is created using the 'verification_directory_naming_style' attribute if it did not already exist.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| violations                |                                                                                                                                                                                                                                                        |
|                           | <p>List of all violations</p> <p><b>Type:</b> obj(violation)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                     |
| voltage_max               |                                                                                                                                                                                                                                                        |
|                           | <p>Return max voltage set on design</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                           |
| voltage_min               |                                                                                                                                                                                                                                                        |
|                           | <p>Returns min voltage set on design</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                          |
| voltage_unit_in_volt      |                                                                                                                                                                                                                                                        |
|                           | <p>Returns session units for voltage in volt.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                 |
| wcdc_synchronizer_type    |                                                                                                                                                                                                                                                        |
|                           | <p>Specify the type of synchronizer circuit used in design for CDC checks.<br/>Valid values are dff or mux or &lt;module name&gt;.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> dff mux</p> <p><b>Edit:</b> Yes</p>                               |
| write_lec_dft_constraints |                                                                                                                                                                                                                                                        |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>This attribute is a TCL dict and holds the LEC dft pin constraints from the last LEC run by Genus. It is passed forward through write_design. The write_do_lec command will automatically include this constraints in the do file.</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                    |
| write_lec_directory_naming_style |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                  | <p>The directory name where 'write_do_lec' will write verification files when the 'write_lec_files' attribute is 'true'. The directory will be created if it does not already exist, and will overwrite an existing directory of the same name. A %s in the string is replaced with the design name (e.g. it will overwrite the directory for the same design but not for a different design). A %d in the string is replaced with a unique integer to avoid overwriting any existing directory.</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> fv/invs/%s<br/><b>Edit:</b> Yes</p> |

## flexible\_htree

### Parent Objects

[pin](#), [root](#)

### Definition

flexible htree

| Attribute                                            | Description                                                                                                                                                                                     |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cts_flexible_htree_adjust_sink_grid_for_aspect_ratio |                                                                                                                                                                                                 |
|                                                      | <p>If true adjust the sink grid for the aspect ratio of the sink grid box.<br/>Valid values: boolean</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> true<br/><b>Edit:</b> Yes</p> |
| cts_flexible_htree_auto_sinks                        |                                                                                                                                                                                                 |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>The number of sinks based on the density of clock sinks under the root of the flexible H-tree. It can also be specified per partition hinst and top level. This attribute is not supported for designs containing master/clone partitions.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                            |
| cts_flexible_htree_auto_sinks_rects     |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <p>The automatically computed sink rectangles of the flexible H-tree when auto sinks are specified. This attribute is not supported for designs containing master/clone partitions.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                       |
| cts_flexible_htree_auto_sinks_sink_area |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <p>The initial width and height of automatically computed sink rectangles of the flexible H-tree, which can be increased due to floorplan restrictions. This attribute is not supported for designs containing master/clone partitions. The attribute is mandatory if auto_sinks attribute is specified.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| cts_flexible_htree_final_cell           |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <p>The library cell to use for the H-tree sinks.</p> <p>Valid values: string</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                             |
| cts_flexible_htree_hv_balance           |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <p>Specifies whether horizontal and vertical wires can only be balanced against other wires of the same orientation (true), or whether any wire can be balanced against any other wire (false).</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes</p>                                                                          |
| cts_flexible_htree_image_directory      |                                                                                                                                                                                                                                                                                                                                                                                                       |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Name of the directory to which images generated by the H-tree synthesis algorithm are written.</p> <p>Valid values: string</p> <p>Color coding of images:</p> <p>White Unobstructed edges of the synthesis grid</p> <p>Red Grid points that are blocked for trunk cell placement in all modules</p> <p>Orange Grid points that are blocked for final cell placement in all modules</p> <p>Red orange Grid points that are blocked for trunk and final cell placement in all modules</p> <p>Yellow circle The grid point of the source</p> <p>Yellow points Candidate grid points of H-tree sinks</p> <p>Yellow rectangle The sink area containing target grid point candidates of H-sinks, adjusted to the synthesis grid</p> <p>Brown If specified, the sink grid box adjusted to the synthesis grid</p> <p>Green/blue The edges of the synthesized H-tree</p> <p>Purple H-tree repeaters</p> <p>Cyan Partition boundaries in hierarchical designs</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_flexible_htree_inverting           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                        | <p>Specifies whether the flexible H-tree will invert its input.</p> <p>Valid values: boolean</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| cts_flexible_htree_layer_density       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                        | <p>The assumed layer density used to compute the parasitics for timing estimates of H-tree nets during H-tree synthesis based on the non default rule for top nets.</p> <p>Valid values: Any float in the range 0 to 1.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| cts_flexible_htree_max_driver_distance |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                        | <p>If specified, ignore DRVs and use the given value as the maximum length of the nets connecting H-tree drivers.</p> <p>Valid values: float</p> <p><b>Type:</b> string</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cts_flexible_htree_max_root_distance |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      | <p>If specified, ignore DRVs and use the given value as the maximum length of the net connecting the root and the first driver of the H-tree. This value overrides the attribute cts_max_driver_distance of this net and can only be specified if -max_driver_distance is also specified.</p> <p>Valid values: float</p> <p><b>Type:</b> string</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p>                                                                                                                            |
| cts_flexible_htree_mode              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      | <p>The driver insertion mode for the H-tree.</p> <p>If set to 'drv', the algorithm inserts drivers to avoid DRVs. It also minimizes the insertion delay of the H-tree if the power weight is less than 1.0.</p> <p>If set to 'distance', the attributes 'cts_max_driver_distance' and 'cts_max_root_distance' determine the maximum net lengths allowed. Transitions and delays are not computed in this case.</p> <p>Valid values: drv distance</p> <p><b>Type:</b> string</p> <p><b>Default:</b> drv</p> <p><b>Edit:</b> Yes</p> |
| cts_flexible_htree_omit_symmetry     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>Controls the omission of symmetry features to balance a flexible H-tree.</p> <p>Symmetry drivers are drivers that are added to balance pin capacitances at branch points. For example, if buffer pairs are inserted at branch points, one of these buffers may not drive any fanout and is inserted to match other buffer pairs at the same level of the flexible H-tree. Omitting symmetry drivers can reduce the power of the H-tree but increase its skew. Drivers that are added to terminate symmetry branches (see below) do not count as symmetry drivers and are never omitted.</p> <p>Similarly, symmetry branches are branches that are needed to balance the wire load at branch points. They are added to match other branchpoints at the same level of the flexible H-tree. Symmetry branches are always terminated with drivers, which are not counted as symmetry drivers.</p> <p>By default, symmetry branches and drivers are added.</p> <p>Possible values for this attribute:</p> <p>false Add symmetry branches and drivers</p> <p>true Omit both symmetry drivers and branches</p> <p>drivers Omit symmetry drivers</p> <p>branches Omit symmetry branches</p> <p>{drivers branches} Omit both symmetry drivers and branches</p> <p>{ } Add symmetry branches and drivers</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> |
| cts_flexible_htree_partition_boundary_polarity  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                 | <p>Specifies the polarity of the clock signal at partition entries with regard to the source of the flexible H-tree. Set this attribute to 'ignore' if the polarity at partition entries is not constrained, which can reduce the number of driver levels and the latency of the flexible H-tree. Note that -partition_boundary_polarity does not affect partition polarity constraints if the -partition_groups setting is used, which by default assumes that the polarity is unconstrained.</p> <p>Valid values: non_inverting inverting ignore</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> non_inverting</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cts_flexible_htree_partition_convergence_points |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                 | <p>For hierarchical designs controls how many partition convergence points are allowed. (default single)</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> single</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| cts_flexible_htree_partition_groups             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>The groups in which partition are clustered in channelless designs. Nested lists imply allowed crossings between groups. Each group has zero or one input port and each partition must only be specified once.</p> <p>Optionally, a maximum pre-route net length from the boundary of a partition group can be specified. This argument must be specified as the second parameter of a nested partition group.</p> <p>Optionally, the next argument specifies the clock phase when entering the partition group in relation to the root pin of the H-tree. Allowed values are 'inverting' and 'non-inverting'. If no value is specified, the clock phase is unconstrained. The clock phase is unconstrained when crossing the boundaries of partitions within the same group. All specified clock phases must be either inverting or non-inverting.</p> <p>Example:<br/> <pre>{{A} {{C D} 50 non_inverting {{E F}} {{G}}}}</pre> </p> <ul style="list-style-type: none"> <li>- The H-tree starts in partition A and descends into group C/D.</li> <li>- From partition group C/D, the tree descends into groups E/F and G.</li> <li>- Any clustering of sinks inside the C/D and E/F groups are allowed, potentially crossing internal partition boundaries several times.</li> <li>- Partition A has no clock input port and one clock output port</li> <li>- Partition group C/D has one clock input port and two clock output ports</li> <li>- Partition group E/F and G have one clock input port and no clock output port</li> <li>- The maximum net length from the entry point into partition group C/D is 50um (pre-route)</li> <li>- The clock phase is non-inverting when entering partition group C/D. The clock phase is unconstrained when entering other partition groups.</li> </ul> <p>Valid values: string<br/> <b>Type:</b> string<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| cts_flexible_htree_power_weight |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                 | <p>The power versus insertion delay trade-off. The value specifies the weight that is put on the optimization of power related attributes , in particular the number of repeaters in the H-tree, during the synthesis of a flexible H-tree.</p> <p>Valid values: Any float in the range 0 to 1.</p> <p><b>Type:</b> string<br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| cts_flexible_htree_root         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>The pin under which the H-tree is created. This pin must be part of a clock tree at the time of synthesis.</p> <p>Valid values: pin</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| cts_flexible_htree_sink_grid     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                  | <p>Specifies the columns and rows of a grid of H-tree sinks, where the columns and rows may be swapped if the cts_flexible_htree_adjust_sink_grid_for_aspect_ratio attribute is true.</p> <p>If using synthesize_flexible_htrees -partition the sink grids may be specified for each partition. If only some partitions have specified sink grids, then for the other partitions the sink grid will be inferred from the floorplan sink grid. To specify a sink grid for the top-level outside partitions, the design name is used. The cts_flexible_htree_adjust_sink_grid_for_aspect_ratio attribute is not applied to partition-based sink grids.</p> <p>Example:</p> <p>For an H-Tree spanning partitions A, B, C and the top_cell a sink grid of { 8 8 { hinst:A 2 1 } { hinst:B 2 2 } { design:top 8 2 } } means that partition A will have 2x1 grid, partition B will have a 2x2 grid, the top-level will have an 8x2 grid and the grid for partition C will be inferred from its size and the 8x8 floorplan grid.</p> <p>To achieve a low global skew of the flexible H-tree and avoid balancing induced detours, the pitch between H-tree sinks in all partitions and in the top level should be similar. This can be achieved by aligning the number of columns and rows with the dimensions of the partitions and adjusting the bounding box of the sink grids using the cts_flexible_htree_sink_grid_box attribute. It may also be beneficial to exclude isolated H-tree sinks by reducing the sink grid box of a partition if the sinks are not critical for timing purposes.</p> <p>Valid values: {[columns rows] [{partition_hinst design columns rows}]...}</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_flexible_htree_sink_grid_box |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | <p>The box describing the area that the grid of H-tree sinks should cover. This attribute only has an effect if the <code>cts_flexible_htree_sink_grid</code> attribute of the flexible H-tree is set.</p> <p>If using <code>synthesize_flexible_htrees</code> -partition the sink grid boxes may be specified for each partition. The tool will use the intersection of the specified partition box with the bounding box of the partition area. If a floorplan box is used, the box for partitions without a specified box will be the intersection of the floorplan box and the partition boundary. To specify a sink grid box for the top-level, the design name is used.</p> <p>Example:<br/>For an H-Tree spanning partitions A, B, C and the top-level a sink grid of { { hinst:A 0 600 500 800 } { hinst:B 600 700 1000 1000 } { design:top 0 0 1000 500 } } means that the sink grid for partition A will cover the box {0 600 500 800}, partition B will cover {600 700 1000 1000}, top-level will cover {0 0 1000 500} and the grid for partition C will cover the whole partition.</p> <p>Valid values: {[xmin ymin xmax ymax] [{partition_hinst design xmin ymin xmax ymax}]...}</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p> |
| cts_flexible_htree_sink_grid_exclusion_areas |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                              | <p>Boxes describing zones that should not be covered by the grid of H-tree sinks. This attribute only has an effect if the <code>cts_flexible_htree_sink_grid</code> attribute of the flexible H-tree is set.</p> <p>Valid values: list {xmin ymin xmax ymax}</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| cts_flexible_htree_sink_grid_sink_area       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                              | <p>The approximate size of the rectangle describing valid locations for final cells (given by -final_cell) per H-tree sink in the grid. This attribute only has an effect if the <code>cts_flexible_htree_sink_grid</code> attribute of the flexible H-tree is set.</p> <p>Valid values: {width height}</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cts_flexible_htree_sink_instance_prefix      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                              | <p>Prefix used for instance names of final cells (given by -final_cell). The name of the cell will be &lt;prefix&gt;_&lt;htree_name&gt;_&lt;id&gt;, where id is a running index.</p> <p>Valid values: string</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                            |                                                                                                                                                                                                                                                                    |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cts_flexible_htree_sinks                   |                                                                                                                                                                                                                                                                    |
|                                            | <p>Specifies H-tree sinks as approximate rectangular areas for locations of final cells (given by -final_cell) or pins to wire to.<br/>Valid values: list {pin   {xmin ymin xmax ymax}}</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p> |
| cts_flexible_htree_stop_at_sdc_clock_roots |                                                                                                                                                                                                                                                                    |
|                                            | <p>If specified, stop searching for parts of the clock tree through SDC clock roots when defining generated clock trees for H-tree sinks.</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p>                                              |
| cts_flexible_htree_trunk_cell              |                                                                                                                                                                                                                                                                    |
|                                            | <p>The library cell to use inside the H-tree.<br/>Valid values: string</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                  |
| cts_route_type_flexible_htree              |                                                                                                                                                                                                                                                                    |
|                                            | <p>Route type to use for flexible h-trees.</p> <p><b>Type:</b> string<br/><b>Default:</b> default<br/><b>Edit:</b> Yes</p>                                                                                                                                         |
| cts_route_type_flexible_htree_backside     |                                                                                                                                                                                                                                                                    |
|                                            | <p>Route type to use for backside flexible h-trees.</p> <p><b>Type:</b> string<br/><b>Default:</b> default<br/><b>Edit:</b> Yes</p>                                                                                                                                |
| insts                                      |                                                                                                                                                                                                                                                                    |
|                                            | <p>instances in the clock tree</p> <p><b>Type:</b> obj(inst)*<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                       |
| name                                       |                                                                                                                                                                                                                                                                    |

|          |                                                                                                                                                                                                               |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | name of flexible_htree<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                        |
| nets     |                                                                                                                                                                                                               |
|          | nets in the clock tree<br><b>Type:</b> <a href="#">obj(net)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                     |
| obj_type |                                                                                                                                                                                                               |
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">flexible_htree</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

## flow

### Parent Objects

[root](#)

### Definition

flow

| Attribute      | Description                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------|
| end_steps      |                                                                                                             |
|                | List of steps for the flow<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| feature_values |                                                                                                             |



|             |                                                                                                                                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>Feature values for instances of this flow</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                |
| features    |                                                                                                                                                                                                                                                                              |
|             | <p>Features defined for this flow</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                           |
| name        |                                                                                                                                                                                                                                                                              |
|             | <p>The name of the flow</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                     |
| obj_type    |                                                                                                                                                                                                                                                                              |
|             | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">flow</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                         |
| owner       |                                                                                                                                                                                                                                                                              |
|             | <p>Owner of this flow</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                       |
| run_count   |                                                                                                                                                                                                                                                                              |
|             | <p>An integer indicating how many times the flow has been run. The value is initially zero, so it may be used as a Boolean indicating that the flow has been run at all.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p> |
| skip_metric |                                                                                                                                                                                                                                                                              |
|             | <p>Do not create a metric snapshot for the flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                            |

|              |                                                                                                                              |
|--------------|------------------------------------------------------------------------------------------------------------------------------|
| start_steps  |                                                                                                                              |
|              | List of steps for the flow<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                  |
| steps        |                                                                                                                              |
|              | List of steps for the flow<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                  |
| tool         |                                                                                                                              |
|              | The tool to run this flow<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                   |
| tool_options |                                                                                                                              |
|              | The tool options for the tool for this flow<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes |

## flow\_step

### Parent Objects

[root](#)

### Definition

flow step

| Attribute | Description |
|-----------|-------------|
| begin_tcl |             |

|             |                                                                                                                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>The Tcl body to run at the start of the step</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                |
| body_tcl    |                                                                                                                                                                                                                 |
|             | <p>The Tcl body of the step. The initial value for this attribute is provided by the create_flow_step command.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| categories  |                                                                                                                                                                                                                 |
|             | <p>Metric categories to calculate for this flow step.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                          |
| check_tcl   |                                                                                                                                                                                                                 |
|             | <p>A block of Tcl to run to perform step-specific static checks.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                               |
| description |                                                                                                                                                                                                                 |
|             | <p>Purpose for this flow step.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                 |
| end_steps   |                                                                                                                                                                                                                 |
|             | <p>A list of flows or flow_steps that will be run at the end of this step.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                     |
| end_tcl     |                                                                                                                                                                                                                 |
|             | <p>The Tcl body to run at the end of the step.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                 |

|                     |                                                                                                                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| exclude_time_metric |                                                                                                                                                                                                          |
|                     | Do not include cpu and wall time of this step in any parent steps<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                      |
| feature_values      |                                                                                                                                                                                                          |
|                     | Feature values for instances for this flow step<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                         |
| features            |                                                                                                                                                                                                          |
|                     | Features defined for this flow step<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                     |
| name                |                                                                                                                                                                                                          |
|                     | The name of the step<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                    |
| obj_type            |                                                                                                                                                                                                          |
|                     | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">flow_step</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| owner               |                                                                                                                                                                                                          |
|                     | Owner of this step.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                     |
| run_count           |                                                                                                                                                                                                          |

|             |                                                                                                                                                                                                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>An integer indicating how many times the step has been run. The value is initially zero, so it may be used as a Boolean indicating that the step has been run at all.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> no_value<br/> <b>Edit:</b> Yes</p> |
| skip_db     |                                                                                                                                                                                                                                                                          |
|             | <p>Do not auto-save a db at the end of this step</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes</p>                                                                                                                            |
| skip_metric |                                                                                                                                                                                                                                                                          |
|             | <p>Do not auto-save a db at the end of this step</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                           |
| start_steps |                                                                                                                                                                                                                                                                          |
|             | <p>A list of flows or flow_steps that will be run at the start of this step.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                |
| status      |                                                                                                                                                                                                                                                                          |
|             | <p>Flow step status</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                         |

## foreign\_cell

### Parent Objects

[base\\_cell](#), [root](#)

### Definition

LEF MACRO FOREIGN information

| Attribute | Description                                                                                                                                                                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name      |                                                                                                                                                                                                                                                           |
|           | Name of the foreign cell (can be the same as the master if only a shift/offset is required)<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                               |
| obj_type  |                                                                                                                                                                                                                                                           |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">foreign_cell</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                               |
| orient    |                                                                                                                                                                                                                                                           |
|           | Orientation of the foreign cell (non-R0 values are only allowed if the cell name is different from the master cell)<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> r0 r90 r180 r270 mx mx90 my my90<br><b>Default:</b> ""<br><b>Edit:</b> No |
| point     |                                                                                                                                                                                                                                                           |
|           | Location (offset) of the foreign cell<br><b>Type:</b> <a href="#">point</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                      |

# gcell

## Parent Objects

[design](#), [root](#)

## Definition

A global-routing cell. It only exists after global routing has been run, and includes the number of routing tracks available (supply), and the tracks used (demand) for each gcell.

| Attribute            | Description                                                                                                                                                                                                                                                |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name            |                                                                                                                                                                                                                                                            |
|                      | Object name derived from index value<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                       |
| demand               |                                                                                                                                                                                                                                                            |
|                      | Number of routing tracks used in the preferred routing direction for the layer. If no layer index is given, the value is not useful, so -1 is returned.<br><b>Type:</b> int<br><b>Allowed -index values:</b> layer<br><b>Default:</b> 0<br><b>Edit:</b> No |
| horizontal_demand    |                                                                                                                                                                                                                                                            |
|                      | Number of routing tracks used in horizontal direction for all routing layers.<br><b>Type:</b> int<br><b>Default:</b> 0<br><b>Edit:</b> No                                                                                                                  |
| horizontal_remaining |                                                                                                                                                                                                                                                            |
|                      | Number of routing tracks remaining in horizontal direction for all routing layers.<br><b>Type:</b> int<br><b>Default:</b> 0<br><b>Edit:</b> No                                                                                                             |
| horizontal_supply    |                                                                                                                                                                                                                                                            |
|                      | Number of routing tracks available in horizontal direction for all routing layers.<br><b>Type:</b> int<br><b>Default:</b> 0<br><b>Edit:</b> No                                                                                                             |
| index                |                                                                                                                                                                                                                                                            |
|                      | The X and Y index values into the gcell array. The bottom, left gcell has index = {0 0}<br><b>Type:</b> int*<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                      |
| name                 |                                                                                                                                                                                                                                                            |

|                    |                                                                                                                                                                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | Object name derived from index value<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                 |
| obj_type           |                                                                                                                                                                                                                                                                                                                      |
|                    | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">gcell</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                 |
| rect               |                                                                                                                                                                                                                                                                                                                      |
|                    | Physical dimensions {{llx lly} {urx ury}}<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                              |
| remaining          |                                                                                                                                                                                                                                                                                                                      |
|                    | Number of routing tracks remaining in the preferred routing direction for the layer. A negative value indicates overflow. If no layer index is given, the value is not useful, so -1 is returned.<br><b>Type:</b> <a href="#">int</a><br><b>Allowed -index values:</b> layer<br><b>Default:</b> 0<br><b>Edit:</b> No |
| supply             |                                                                                                                                                                                                                                                                                                                      |
|                    | Number of routing tracks available in the preferred routing direction for the layer. If no layer index is given, the value is not useful, so -1 is returned.<br><b>Type:</b> <a href="#">int</a><br><b>Allowed -index values:</b> layer<br><b>Default:</b> 0<br><b>Edit:</b> No                                      |
| vertical_demand    |                                                                                                                                                                                                                                                                                                                      |
|                    | Number of routing tracks used in vertical direction for all routing layers.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> No                                                                                                                                                              |
| vertical_remaining |                                                                                                                                                                                                                                                                                                                      |



|                 |                                                                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | Number of routing tracks remaining in vertical direction for all routing layers.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> No |
| vertical_supply |                                                                                                                                                              |
|                 | Number of routing tracks available in vertical direction for all routing layers.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> No |

## gcell\_grid

### Parent Objects

[design](#), [root](#)

### Definition

Equivalent to DEF GCELLGRID statements

| Attribute | Description                                                                                                                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| direction |                                                                                                                                                                                                                                    |
|           | Specifies the location and direction of the first grid defined. x indicates vertical lines; y indicates horizontal lines.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> y x<br><b>Default:</b> ""<br><b>Edit:</b> No |
| num_grids |                                                                                                                                                                                                                                    |
|           | Specifies the number of grid lines to create (number of rows or columns is numGrids-1)<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                |
| obj_type  |                                                                                                                                                                                                                                    |

|       |                                                                                                                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">gcell_grid</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| start |                                                                                                                                                                                                           |
|       | Specifies the coordinate of the first line<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                 |
| step  |                                                                                                                                                                                                           |
|       | Specifies the spacing between the grids<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                    |

## group

### Parent Objects

[hinst](#), [inst](#), [design](#), [power\\_domain](#), [root](#),

### Definition

group of hinsts, insts, or groups

| Attribute | Description                                                                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| area      |                                                                                                                                                         |
|           | Area of the group as defined by the LEF MACRO SIZE or OVERLAP information<br><b>Type:</b> <a href="#">area</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| base_name |                                                                                                                                                         |

|                      |                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Name of group<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                         |
| constraint_type      |                                                                                                                                                                                                                                                                                                                                                                                    |
|                      | <p>Constraint type for the group<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> guide region fence cluster none<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                 |
| density              |                                                                                                                                                                                                                                                                                                                                                                                    |
|                      | <p>Group density (legal range: 0.0-1.0)<br/> <b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> no_value<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                           |
| exclusive_group_gap  |                                                                                                                                                                                                                                                                                                                                                                                    |
|                      | <p>This is the gap should be maintained between exclusive_groups (per safety island groups). The value is measured in microns. It can be set by command create_exclusive_groups -gap. It is only valid when the group's constraint_type is region or fence.<br/> <b>Type:</b> <a href="#">coord</a><br/> <b>Default:</b> 5<br/> <b>Edit:</b> No</p>                                |
| exclusive_group_type |                                                                                                                                                                                                                                                                                                                                                                                    |
|                      | <p>This attribute specifies the logical type of this group. It describes the logical relationship of the group relative to other groups which can be used to implement safety islands in automotive application designs.<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none master checker mutually_exclusive<br/> <b>Default:</b> none<br/> <b>Edit:</b> No</p> |
| exclusive_groups     |                                                                                                                                                                                                                                                                                                                                                                                    |
|                      | <p>This attribute specifies a list of groups that are exclusive of this group. It can be used to implement safety islands in automotive application designs.<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                             |

|              |                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_floating  |                                                                                                                                                                                                                                                                                                                                                                                  |
|              | <p>Only affects groups with .constraint_type = fence or region. If true, the global placer can move the fence or region. The .rects value must also be set, and is currently restricted to a single rect. The global placer will not change the size of the rect, but may move it.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p> |
| members      |                                                                                                                                                                                                                                                                                                                                                                                  |
|              | <p>List of group's members</p> <p><b>Type:</b> <a href="#">obj(hinst)*</a> <a href="#">obj(inst)*</a> <a href="#">obj(group)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                               |
| name         |                                                                                                                                                                                                                                                                                                                                                                                  |
|              | <p>Name of group</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                     |
| obj_type     |                                                                                                                                                                                                                                                                                                                                                                                  |
|              | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">group</a>)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                |
| parent       |                                                                                                                                                                                                                                                                                                                                                                                  |
|              | <p>The parent group if is a sub-group</p> <p><b>Type:</b> <a href="#">obj(group)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                            |
| power_domain |                                                                                                                                                                                                                                                                                                                                                                                  |
|              | <p>The Power Domain (if group is a power domain)</p> <p><b>Type:</b> <a href="#">obj(power_domain)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                          |
| rects        |                                                                                                                                                                                                                                                                                                                                                                                  |

|                             |                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>List of non-overlapping rectangles that defines the shape of the group</p> <p><b>Type:</b> <a href="#">rect*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                         |
| safety_dcls_isolate_clock   |                                                                                                                                                                                                                                                                                                                                                                                   |
|                             | <p>The type of isolation required for the clock signals driving this dcls (dual core lock step) group.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none split_net split_network</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p>                                                                                    |
| safety_dcls_isolate_inputs  |                                                                                                                                                                                                                                                                                                                                                                                   |
|                             | <p>The type of isolation required for all the input signals driving this dcls (dual core lock step) group.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none split_net</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p>                                                                                              |
| safety_dcls_isolate_is_halo |                                                                                                                                                                                                                                                                                                                                                                                   |
|                             | <p>When true, this group is an isolation halo for a dcls (dual core lock step) group. The isolation cells on the input/output nets for the dcls-group will be placed inside this halo around the dcls-group boundary. The halo-group will be the parent-group of the dcls-group.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p> |
| safety_dcls_isolate_outputs |                                                                                                                                                                                                                                                                                                                                                                                   |
|                             | <p>The type of isolation required for all the output signals driven by this dcls (dual core lock step) group.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none split_net</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p>                                                                                           |
| safety_dcls_isolate_reset   |                                                                                                                                                                                                                                                                                                                                                                                   |

|                                 |                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>The type of isolation required for the reset signals driving this dcls (dual core lock step) group.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none split_network split_net</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p>                                |
| safety_dcls_isolate_scan_enable |                                                                                                                                                                                                                                                                                                                               |
|                                 | <p>The type of isolation required for the scan_enable signals driving this dcls (dual core lock step) group.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none split_network split_net</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p>                          |
| safety_dcls_isolate_use_halo    |                                                                                                                                                                                                                                                                                                                               |
|                                 | <p>When true, the isolation cells on the input/output nets for this dcls (dual core lock step) group will be placed inside a halo around the dcls-group boundary. The halo-group will be the parent-group of the dcls-group.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p> |
| safety_dcls_route_use_halo      |                                                                                                                                                                                                                                                                                                                               |
|                                 | <p>When true, the internal nets for this dcls (dual core lock step) group will be routing inside a halo around the dcls-group boundary.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                      |
| safety_dclst_compare_group      |                                                                                                                                                                                                                                                                                                                               |
|                                 | <p>The compare group of this dclst (dual core lock step temporal). Checks for errors by comparing the outputs of primary &amp; shadow.</p> <p><b>Type:</b> <a href="#">obj(group)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                    |
| safety_dclst_delay_cycles       |                                                                                                                                                                                                                                                                                                                               |

|                                 |                                                                                                                                                                                                                                                                                          |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>The number of clock-cycles the shadow is delayed compared to the primary in dclst (dual core lock step temporal).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                     |
| safety_dclst_group_type         |                                                                                                                                                                                                                                                                                          |
|                                 | <p>The function of this group in a dclst (dual core lock step temporal).</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none primary shadow compare input_delay output_delay</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p> |
| safety_dclst_input_delay_group  |                                                                                                                                                                                                                                                                                          |
|                                 | <p>The input_delay group of this dclst (dual core lock step temporal). Delays the inputs to the shadow.</p> <p><b>Type:</b> <a href="#">obj(group)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                              |
| safety_dclst_output_delay_group |                                                                                                                                                                                                                                                                                          |
|                                 | <p>The output_delay group of this dclst (dual core lock step temporal). Delays the outputs from the primary to the compare.</p> <p><b>Type:</b> <a href="#">obj(group)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                          |
| safety_dclst_primary_group      |                                                                                                                                                                                                                                                                                          |
|                                 | <p>The primary group of this dclst (dual core lock step temporal). Performs a core function of the design, and drives the rest of the design logic.</p> <p><b>Type:</b> <a href="#">obj(group)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                  |
| safety_dclst_shadow_group       |                                                                                                                                                                                                                                                                                          |
|                                 | <p>The shadow group of this dclst (dual core lock step temporal). Clone of the primary.</p> <p><b>Type:</b> <a href="#">obj(group)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                              |
| safety_failure_mode             |                                                                                                                                                                                                                                                                                          |

|                       |                                                                                                                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | The name of the failure-mode to which this group belongs, as defined in the USF (Unified Safety Format) file.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| safety_mechanism      |                                                                                                                                                                                               |
|                       | The name of the safety-mechanism applied to this group, as defined in the USF (Unified Safety Format) file.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No   |
| safety_mechanism_type |                                                                                                                                                                                               |
|                       | The type of safety-mechanism applied to this group, as defined in the USF (Unified Safety Format) file.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No       |

## gui\_line

### Parent Objects

[design](#), [root](#)

### Definition

A line that can be displayed on the GUI, and is not output to DEF.

| Attribute      | Description                                                                                                                                                      |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| arrow          |                                                                                                                                                                  |
|                | Draw arrow in the middle of the line with direction from start point to end point.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| gui_layer_name |                                                                                                                                                                  |



|             |                                                                                                                                                                                                                                                                                            |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | Normally a GUI-only layer name that is not a tech-file layer. If the name is the same as a tech-file layer, it will be drawn on the GUI with other shapes on that layer, but it will not be output to DEF.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| name        |                                                                                                                                                                                                                                                                                            |
|             | GUI line name.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                                                                            |
| obj_type    |                                                                                                                                                                                                                                                                                            |
|             | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">gui_line</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                    |
| pixel_width |                                                                                                                                                                                                                                                                                            |
|             | Number of pixels in the border of the object.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                 |
| points      |                                                                                                                                                                                                                                                                                            |
|             | 2 points for line<br><b>Type:</b> <a href="#">point</a> *<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                         |

## gui\_polygon

### Parent Objects

[design](#), [root](#)

### Definition

The polygon shape that can be displayed on the GUI, and is not output to DEF.

| Attribute      | Description                                                                                                                                                                                                                                                                                               |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gui_layer_name |                                                                                                                                                                                                                                                                                                           |
|                | <p>Normally a GUI-only layer name that is not a tech-file layer. If the name is the same as a tech-file layer, it will be drawn on the GUI with other shapes on that layer, but it will not be output to DEF.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| name           |                                                                                                                                                                                                                                                                                                           |
|                | <p>GUI polygon name.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                         |
| obj_type       |                                                                                                                                                                                                                                                                                                           |
|                | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">gui_polygon</a>)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                   |
| pixel_width    |                                                                                                                                                                                                                                                                                                           |
|                | <p>Number of pixels in the border of the object.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                 |
| points         |                                                                                                                                                                                                                                                                                                           |
|                | <p>point list for poly</p> <p><b>Type:</b> <a href="#">point</a>*<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                        |

# gui\_rect

## Parent Objects

[design](#), [root](#)

## Definition

A rect that can be displayed on the GUI, and is not output to DEF.

| Attribute      | Description                                                                                                                                                                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gui_layer_name |                                                                                                                                                                                                                                                                                            |
|                | Normally a GUI-only layer name that is not a tech-file layer. If the name is the same as a tech-file layer, it will be drawn on the GUI with other shapes on that layer, but it will not be output to DEF.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| name           |                                                                                                                                                                                                                                                                                            |
|                | GUI rect name.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                                                                            |
| obj_type       |                                                                                                                                                                                                                                                                                            |
|                | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">gui_rect</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                    |
| pixel_width    |                                                                                                                                                                                                                                                                                            |
|                | Number of pixels in the border of the object.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                 |
| rect           |                                                                                                                                                                                                                                                                                            |

|  |                                                                                                                          |
|--|--------------------------------------------------------------------------------------------------------------------------|
|  | Rectangle that defines the gui_rect shape.<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
|--|--------------------------------------------------------------------------------------------------------------------------|

## gui\_text

### Parent Objects

[design](#), [root](#)

### Definition

A text that can be displayed on the GUI, and is not output to DEF.

| Attribute      | Description                                                                                    |
|----------------|------------------------------------------------------------------------------------------------|
| gui_layer_name |                                                                                                |
|                | GUI layer name<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| height         |                                                                                                |
|                | Text height<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No     |
| label          |                                                                                                |
|                | Text string<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No    |
| location       |                                                                                                |

|          |                                                                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Text location (lower left)<br><b>Type:</b> <a href="#">point</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                               |
| obj_type |                                                                                                                                                                                                         |
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">gui_text</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| orient   |                                                                                                                                                                                                         |
|          | Text Orientation<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> r0 r90 r180 r270 mx mx90 my my90<br><b>Default:</b> ""<br><b>Edit:</b> No                                                  |

## hinst

### Parent Objects

[violation](#), [group](#), [hport\\_bus](#), [hpin](#), [hnet](#), [actual\\_scan\\_segment](#), [partition](#), [design](#), [bump](#), [root](#), [boundary](#), [hport](#), [module](#), [inst](#), [scan\\_segment](#), [pin](#), [hpin\\_bus](#)

### Definition

Hierarchical instance

| Attribute            | Description                                                                                                                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apply_booth_encoding |                                                                                                                                                                                                                                                      |
|                      | Apply booth encoding by using user specified method (for a specified instance).<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> nonbooth auto_bitwidth auto_togglerate manual inherited<br><b>Default:</b> inherited<br><b>Edit:</b> Yes |
| area                 |                                                                                                                                                                                                                                                      |

|                          |                                                                                                                                                                                                                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>The area of all the insts contained inside this hinst and below it.</p> <p><b>Type:</b> <a href="#">area</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                       |
| base_name                |                                                                                                                                                                                                                                                                                                                            |
|                          | <p>The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                        |
| bbox                     |                                                                                                                                                                                                                                                                                                                            |
|                          | <p>Bounding box of all the insts inside this hinst.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                          |
| block_power_intent_file  |                                                                                                                                                                                                                                                                                                                            |
|                          | <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                               |
| boundary                 |                                                                                                                                                                                                                                                                                                                            |
|                          | <p>The placement boundary (e.g. fence, region, group, cluster) for this hinst if it exists.</p> <p><b>Type:</b> <a href="#">obj(boundary)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                         |
| clock_gating_test_signal |                                                                                                                                                                                                                                                                                                                            |
|                          | <p>Set specified test signal to connect up test logic in clock-gating circuitry. Specify use_shift_enable to derive test logic from shift enable of the gated flops, or use_none to disable test connection to the hinst.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| clock_gating_xor_status  |                                                                                                                                                                                                                                                                                                                            |

|                                         |                                                                                                                                                                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Set clock-gating xor status.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> enable_updated inserted unset</p> <p><b>Default:</b> unset</p> <p><b>Edit:</b> No</p>                                    |
| dft_abstract_dont_scan                  |                                                                                                                                                                                                                              |
|                                         | <p>Exclude this instance from scan-synthesis as an abstract segment has been defined on it.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                 |
| dft_dont_scan                           |                                                                                                                                                                                                                              |
|                                         | <p>Exclude this design from scan-synthesis {inherited   true   false}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p> |
| dft_exclude_flop_from_los_pipeline      |                                                                                                                                                                                                                              |
|                                         | <p>Avoid insertion of launch-off-shift pipelined logic to flop's shift enable.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                             |
| dft_force_blackbox_for_atpg             |                                                                                                                                                                                                                              |
|                                         | <p>Determines is the instance is blackbox for ATPG.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none true false</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p>                              |
| dft_hier_instance_for_dedicated_wrapper |                                                                                                                                                                                                                              |
|                                         | <p>The hier instance for dedicated wrapper cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                          |
| dft_is_blackbox_for_atpg                |                                                                                                                                                                                                                              |

|                              |                                                                                                                                                                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | Blackbox for ATPG.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> No                                                                                                                                                   |
| dft_is_los_pipeline_flop     |                                                                                                                                                                                                                                                    |
|                              | Determines if this flop is launch-off-shift pipeline flop.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                          |
| dft_part_of_segment          |                                                                                                                                                                                                                                                    |
|                              | The type of segment this instance belongs to {abstract   fixed   floating   preserved   shift_re<br><b>Type:</b> <code>enum</code><br><b>Enum Values:</b> fixed floating preserve abstract shift_register<br><b>Default:</b> ""<br><b>Edit:</b> No |
| dft_partition                |                                                                                                                                                                                                                                                    |
|                              | Determine the DFT partition of the specified instance/hierarchical instance.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                          |
| dft_scan_chain               |                                                                                                                                                                                                                                                    |
|                              | The actual scan chain this instance belongs to.<br><b>Type:</b> <code>obj(scan_chain) obj(scan_segment) obj(actual_scan_segment) obj(actual_scan_ch</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                |
| dft_scan_chain_in_multi_mode |                                                                                                                                                                                                                                                    |
|                              | The actual scan chains this instance belongs to in multi mode stitching.<br><b>Type:</b> <code>obj(scan_chain)* obj(scan_segment)* obj(actual_scan_segment)* obj(actual_scan_</code><br><b>Default:</b> ""<br><b>Edit:</b> No                      |
| dont_retime                  |                                                                                                                                                                                                                                                    |
|                              | Do not retime this instance.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                        |



## dont\_touch

This attribute defines the user preservation status of the hinst during optimization. Setting this attribute will set the dont\_touch attribute on the parent module of this hinst and all hinsts of same module. This setting will apply to all insts within the hinst unless overridden at a lower hinst or on the inst object itself. The dont\_touch\_effective attribute on each child inst and hinst return the resolved value.

Use 'help inst dont\_touch' to see the enum value definitions but note that map\_size\_ok is not allowed at the instance level.

**Type:** enum

**Enum Values:** none false true delete\_ok const\_prop\_size\_delete\_ok const\_prop\_delete\_size\_delete\_ok size\_ok size\_same\_height\_ok size\_same\_footprint\_ok

**Default:** none

**Edit:** Yes

## dont\_touch\_effective

This attribute defines the effective preservation status of this hinst during optimization from the dont\_touch\_sources values. If the partition source is 'true' or the ilm source is 'true', then the effective value for the hinst is 'true'. Otherwise, the user value has precedence. If the user value is false, then the parent value is returned.

See help on inst dont\_touch attribute for details on possible values.

**Type:** enum

**Enum Values:** none false true delete\_ok const\_prop\_size\_delete\_ok const\_prop\_delete\_size\_delete\_ok size\_ok size\_same\_height\_ok size\_same\_footprint\_ok

**Default:** false

**Edit:** No

## dont\_touch\_file

Specifies the sourced file to set dont\_touch.

**Type:** string

**Default:** ""

**Edit:** Yes

## dont\_touch\_hports

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>This attribute defines the user preservation status for the hports of this hinst during optimization.</p> <p><b>Supported values:</b></p> <p>none: Unconstrained</p> <p>false: Can add/remove ports</p> <p>true: Cannot add/remove ports</p> <p>delete_ok: Can delete ports (if they have no fanout)</p> <p>add_ok: We cannot delete, or change the polarity or any hport but can add or duplicate hports</p> <p>invert_ok: We cannot delete, duplicate, or add but can change the polarity of any hport</p> <p>add_invert_ok: We cannot delete any hport but can add, duplicate, and change the polarity</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> none false true delete_ok add_ok invert_ok add_invert_ok</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> |
| dont_touch_sources       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                          | <p>Dictionary of {source &lt;value&gt;} pairs contributing to the dont_touch_effective attribute for this hinst:</p> <p>{user &lt;value&gt;} {parent &lt;value&gt;} {read_only_effective &lt;value&gt;}</p> <p>user # This hinst dont_touch value</p> <p>parent # The effective dont_touch value from an hinst above (the closest hinst above that is not false)</p> <p>read_only_effective # True when this hinst is read_only because it is an ilm or set_module or set it or a parent partition to read_only. Local read_only value overrides parent read_only value</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                           |
| dont_use_cells           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                          | <p>List of cell names (wildcards supported) to disallow for this hinst during optimization. Setting this attribute on the module of the hinst (setting on the hinst is just for convenience). If a cell is added to this list that is already in the .use_cells list, it will be removed from the .use_cells list so that the lists are non-overlapping.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                         |
| dont_use_cells_effective |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>The resolved list of all cell names to disallow during optimization for this hinst, based on the dont_use and the dont_use_cells and use_cells attributes of this hinst or the closest parent with a non-empty list. The precedence is: union of the use_cells of this hinst (or closest parent if empty), then dont_use_cells of this hinst (or closest parent if empty), then the library dont_use setting. When there are multiple hinsts that share the same module, the dont_touch_effective is calculated for the master hinst and the other (clone) hinsts inherit.</p> <p>A cell C is in the don't use cell effective list for a hinst H if and only if.</p> <p>(1) Walk from the hinst H up in the parent hierarchy (including H) to find the closest parent where C is referred to in use cell list or don't use cell list.</p> <p>(2) If such a hinst is found and if it occurs as use cell in hinst then it is not in the effective don't use list. If it occurs in the don't_use cell list of hinst then it is in the effective don't use list.</p> <p>(3) If such a hinst is not found the library don't use determines if the cell is in the effective don't use list or not.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| dp_sharing                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                 | <p>To control the datapath sharing on a datapath instance in 'synthesize -to_generic' (advanced/none/inherited).</p> <p><b>Type:</b> enum<br/><b>Enum Values:</b> advanced none inherited<br/><b>Default:</b> inherited<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| dpopt_instance_constant_input_based_speculation |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                 | <p>Controls constant input based datapath speculation of datapath module in 'syn_generic'.</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| escaped_name                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                 | <p>The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax: bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a is a scalar.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| exceptions                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                     |                                                                                                                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>Reports the path exceptions<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                |
| fcu_instruction_set |                                                                                                                                                                                                                                                |
|                     | <p>Attribute to define different repair specific instructions for the different instances of the same on which fuse control unit has been inserted.<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| group               |                                                                                                                                                                                                                                                |
|                     | <p>The parent group<br/> <b>Type:</b> <a href="#">obj(group)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                              |
| hdl_proc_name       |                                                                                                                                                                                                                                                |
|                     | <p>Name of the process (VHDL) or always block (Verilog) from which the instance is inferred.<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                        |
| hdl_vname           |                                                                                                                                                                                                                                                |
|                     | <p>The name for this instance as specified in the input HDL file (Verilog or VHDL).<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                  |
| hinsts              |                                                                                                                                                                                                                                                |
|                     | <p>List of all the hinsts at the current level and below this hinst<br/> <b>Type:</b> <a href="#">obj(hinst)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                             |
| hnets               |                                                                                                                                                                                                                                                |
|                     | <p>List of all the hnets at the current level and below this hinst<br/> <b>Type:</b> <a href="#">obj(hnet)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                               |

|                         |                                                                                                                                                                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hpin_busses             |                                                                                                                                                                                                                                          |
|                         | <p>List of all the hpin_bus at the current level and below this hinst</p> <p><b>Type:</b> <a href="#">obj(hpin_bus)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                            |
| hpins                   |                                                                                                                                                                                                                                          |
|                         | <p>List of all the hpins at the current level and below this hinst</p> <p><b>Type:</b> <a href="#">obj(hpin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                   |
| hport_busses            |                                                                                                                                                                                                                                          |
|                         | <p>List of all the hport_bus of this hinst.</p> <p><b>Type:</b> <a href="#">obj(hport_bus)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                     |
| hports                  |                                                                                                                                                                                                                                          |
|                         | <p>List of all the hports of this hinst (e.g. from inside the hinst looking up at the hinst boundary)</p> <p><b>Type:</b> <a href="#">obj(hport)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                               |
| ilm_inst                |                                                                                                                                                                                                                                          |
|                         | <p>Specifies the inst object of this ilm when it is under unflatten view. This attribute is only valid the ilm is under flatten view.</p> <p><b>Type:</b> <a href="#">obj(inst)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| instance_internal_power |                                                                                                                                                                                                                                          |
|                         | <p>The instance's internal power.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                 |
| instance_leakage_power  |                                                                                                                                                                                                                                          |

|                      |                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>The instance's leakage power.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                |
| insts                |                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>List of all the insts at the current level and below this hinst</p> <p><b>Type:</b> <a href="#">obj(inst)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                 |
| is_ilm               |                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>This attribute is true if the hinst is an ILM. It will affect the read_only_effective and dont_touch_effective of this hinst and all hinsts within it, as well as the dont_touch_effective insts within it. It cannot be overridden by other hinst or inst values.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p> |
| is_partition_defined |                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Describes if there is any partition definitions on the hinst.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                         |
| local_hinsts         |                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>List of hinsts in the current level of this hinst (e.g. from inside the hinst looking down at the hinst level)</p> <p><b>Type:</b> <a href="#">obj(hinst)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                 |
| local_hnets          |                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>List of hnets in the current level of this hinst (e.g. from inside the hinst looking down at the hinst level)</p> <p><b>Type:</b> <a href="#">obj(hnet)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                   |
| local_hpins          |                                                                                                                                                                                                                                                                                                                                                                        |

|                                   |                                                                                                                                                                                                                              |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>List of hpins in the current level of this hinst (e.g. from inside the hinst looking down at the hinsts at this level)</p> <p><b>Type:</b> <code>obj(hpin)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| local_insts                       |                                                                                                                                                                                                                              |
|                                   | <p>List of insts in the current level of this hinst (e.g. from inside the hinst looking down at the insts at this level)</p> <p><b>Type:</b> <code>obj(inst)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| local_pins                        |                                                                                                                                                                                                                              |
|                                   | <p>List of pins in the current level of this hinst (e.g. from inside the hinst looking down at the pins at this level)</p> <p><b>Type:</b> <code>obj(pin)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| lp_dynamic_analysis_scope         |                                                                                                                                                                                                                              |
|                                   | <p>Set VCD analysis scope attribute on an instance in the design hierarchy. Valid values for the attribute are true or false.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> |
| merge_combinational_hier_instance |                                                                                                                                                                                                                              |
|                                   | <p>Allow the combinational hierarchical instance to be merged.</p> <p><b>Type:</b> <code>enum</code></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>        |
| module                            |                                                                                                                                                                                                                              |
|                                   | <p>The module cell that corresponds to the hinst</p> <p><b>Type:</b> <code>obj(module)</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                           |
| name                              |                                                                                                                                                                                                                              |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>The same as .escaped_name without any \ escape chars. This name is commonly used to : problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so flattening hierarchical names (e.g. with ungroup) does not require changing the names in T scripts.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| num_exceptions           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                          | <p>Reports the number of path exceptions</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                  |
| obj_type                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                          | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">hinst</a>)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                         |
| orient                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                          | <p>Orientation of the partition.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> r0 r90 r180 r270 mx mx90 my my90<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                     |
| original_name            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                          | <p>Original rtl names of single-bit flops.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                   |
| original_names_for_split |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                          | <p>Returns the original name of single-bit flops merged.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                     |
| parent                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



|                        |                                                                                                                                                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>The parent of the hinst, which is either another hinst, or the design object.</p> <p><b>Type:</b> <a href="#">obj(hinst)</a> <a href="#">obj(design)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                   |
| part_power_intent_file |                                                                                                                                                                                                                                                                    |
|                        | <p>The path of file which will contain final power intent for this block.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                         |
| partition              |                                                                                                                                                                                                                                                                    |
|                        | <p>If the hinst is a partition/blackbox, this partition will contain the pin constraints and related d partition/physical blackbox. Otherwise it is empty.</p> <p><b>Type:</b> <a href="#">obj(partition)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pmbist_instruction_set |                                                                                                                                                                                                                                                                    |
|                        | <p>Attribute to define different programmable MBIST instructions for the different instances of the block on which programmable MBIST has been inserted.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>          |
| power_domain           |                                                                                                                                                                                                                                                                    |
|                        | <p>Power domain of the hierarchical inst if in a group</p> <p><b>Type:</b> <a href="#">obj(power_domain)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                  |
| power_dynamic          |                                                                                                                                                                                                                                                                    |
|                        | <p>The sum of the power_dynamic value for all the insts inside this hinst. A value of no_value is treated as 0.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                    |
| power_internal         |                                                                                                                                                                                                                                                                    |

|                                      |                                                                                                                                                                                                                       |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>The sum of the power_internal value for all the insts inside this hinst. A value of no_value is as 0.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                              |
| power_leakage                        |                                                                                                                                                                                                                       |
|                                      | <p>The sum of the power_leakage value for all the insts inside this hinst. A value of no_value is as 0.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                               |
| power_switching                      |                                                                                                                                                                                                                       |
|                                      | <p>The sum of the power_switching value for all the insts inside this hinst. A value of no_value is treated as 0.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                     |
| power_toggle_rate                    |                                                                                                                                                                                                                       |
|                                      | <p>The average of the power_toggle_rate value for all the insts inside this &lt;hinst/design&gt;. A value of no_value is treated as 0</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| power_total                          |                                                                                                                                                                                                                       |
|                                      | <p>The sum of the power_total value for all the insts inside this hinst. A value of no_value is treated as 0.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                         |
| propagate_constant_from_timing_model |                                                                                                                                                                                                                       |
|                                      | <p>Allow constant propagation from specific timing model instance.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>          |
| read_only                            |                                                                                                                                                                                                                       |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>This attribute is true if the hinst is a read_only. This is set by the set_module_view command. If true, this hinst cannot be optimized and cells inside will not be moved. This attribute will affect dont_touch_effective and place_status_effective attributes on all insts and hinsts within this hinst. This attribute cannot be overridden by other hinst or inst values.</p> <p><b>Supported values:</b></p> <p>false: This hinst is allowed to be optimized</p> <p>true: The hinst read_only due to partition or ILM status</p> <p>none: No constraint.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> none false true</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p> |
| read_only_effective       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                           | <p>This attribute defines the read_only status of this hinst. This can be true when the local .read_only attribute is true or if a parent .read_only attribute is true. The hinst cannot be optimized and cells inside will not be moved.</p> <p><b>Supported values:</b></p> <p>false: This hinst is allowed to be optimized</p> <p>true: The hinst read_only due to local or parent read_only partition</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                    |
| ref_name                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                           | <p>ref_name</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| retime_original_registers |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                           | <p>Original registers before retiming.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| rtls_hdl_instantiated     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                           | <p>Returns true if the instance was present in the user input.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| single_bit_original_name |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>Returns the hdl name of the original single bits.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                   |
| trace_retime             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>Trace this instance through retiming, use the attribute 'retime_original_registers' to get the r the original instance.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                            |
| ungroup_allowed          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>Controls whether this hierarchical instance should be (manually or automatically) ungrouped allows ungrouping on all modules/hinsts. false: disallows ungrouping on all RTL modules/hinsts but still allows ungrouping of tool-generated modules/hinsts (identified by hdl_instantiated=strict_no: disallows ungrouping on all modules/hinsts.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> false true strict_no</p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>      |
| unresolved               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>Returns true if the module is not from input netlist.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                  |
| use_cells                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>List of base_cell names to allow for this hinst during optimization. Setting on an hinst sets the attribute on the module of the hinst (setting on the hinst is just for convenience). All lib_cells base_cell will be allowed. If a cell is added to this list that is already in the .dont_use_cells list it will be removed from the .dont_use_cells list so that the lists are non-overlapping.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |

# hnet

## Parent Objects

[hinst](#), [design](#), [hpin](#), [pin](#), [net](#), [root](#), [hport](#), [pg\\_pin](#), [port](#)

## Definition

Hierarchical net

| Attribute          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                    | <p>The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                        |
| connections        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                    | <p>Returns a list of connections of the hnet (hpins and hports).</p> <p><b>Type:</b> <a href="#">obj(pin)*</a> <a href="#">obj(hpin)*</a> <a href="#">obj(port)*</a> <a href="#">obj(hport)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                      |
| constant           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                    | <p>Returns if this hnet has a constant logic value in the Verilog (supply0/supply1 or assigned 1'b0/1'b1). If it is no_constant, and it is connected to a net, you must check the net .constant value to see if the net is constant for other reasons. A supply0/1 hnet will cause its net to have .use = power or ground.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> 0 1 no_constant<br/> <b>Default:</b> no_constant<br/> <b>Edit:</b> No</p> |
| dft_constant_value |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                    | <p>Constant net value in test mode {logic_0   logic_1   logic_z   no_value}.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                               |
| dont_touch         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>This attribute defines the preservation status of an hnet during optimization. Setting this will preserve all connections on the hnet at the level of hierarchy where the hnet exists (i.e. will stop at the hpins and hports connected to this hnet).</p> <p><b>Supported values:</b><br/> false: Unconstrained<br/> true: Cannot touch<br/> delete_ok: Can delete (if net has no sinks)</p> <p><b>Type:</b> enum<br/> <b>Enum Values:</b> false true delete_ok const_prop_delete_ok<br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p> |
| dont_touch_file    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | <p>Specifies the sourced file to set dont_touch.</p> <p><b>Type:</b> string<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| dont_touch_sources |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | <p>Dictionary of {source value} pairs contributing to the dont_touch_effective attribute for this object :</p> <p>{user &lt;value&gt;} {dont_touch_network &lt;value&gt;} {parent &lt;value&gt;}</p> <p>user # The user dont_touch value<br/> dont_touch_network # The net is in the fanout cone of a set_dont_touch_network constraint<br/> parent # The effective dont_touch value from an hinst above</p> <p><b>Type:</b> string<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                          |
| drivers            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | <p>local to this hinst: pins, hpins that are output or bider, and hports. Ports that are input or bidir</p> <p><b>Type:</b> obj(pin)* obj(hpin)* obj(port)* obj(hport)*<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                      |
| escaped_name       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax, so bus-bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So i1/i2 is a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a[0\] is a scalar.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                   |
| hinst                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>The parent hinst for this hnet</p> <p><b>Type:</b> <a href="#">obj(hinst)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                    |
| loads                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>local to this hinst: pins, hpins that are input or bider, and hports. Ports that are output or bidir</p> <p><b>Type:</b> <a href="#">obj(pin)* obj(hpin)* obj(port)* obj(hport)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                             |
| lp_asserted_probability |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>Probability of signal to be high on net as specified by user.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> no_value<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                  |
| lp_asserted_toggle_rate |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>Signal toggle rate assigned to net by user.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> no_value<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                    |
| name                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>The same as .escaped_name without any \ escape chars. This name is commonly used to avoid problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so that flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |

|                        |                                                                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| net                    |                                                                                                                                                                                                     |
|                        | canonical (flat) net associated with the hierarchical net<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">net</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                      |
| num_drivers            |                                                                                                                                                                                                     |
|                        | number of local drivers<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                |
| num_loads              |                                                                                                                                                                                                     |
|                        | number of local loads<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                  |
| obj_type               |                                                                                                                                                                                                     |
|                        | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">hnet</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| timing_constant        |                                                                                                                                                                                                     |
|                        | Reports constant value on net<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                                  |
| timing_constant_source |                                                                                                                                                                                                     |
|                        | Reports source of constant value on net<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                        |



# hpin

## Parent Objects

[test\\_clock](#), [clock](#), [port](#), [violation](#), [arc](#), [hinst](#), [scan\\_chain](#), [hnet](#), [actual\\_scan\\_segment](#), [dft\\_configuration\\_mode](#), [design](#), [partition](#), [root](#), [hport](#), [test\\_signal](#), [inst](#), [scan\\_segment](#), [actual\\_scan\\_chain](#), [hpin\\_bus](#)

## Definition

Hierarchical instance pin

| Attribute                              | Description                                                                                                                                                                                                                |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name                              |                                                                                                                                                                                                                            |
|                                        | The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                           |
| boundary_feedthru                      |                                                                                                                                                                                                                            |
|                                        | Captures feedthrough created during boundary optimization and used during mapping file generation.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                        |
| boundary_optimize_constant_hpins       |                                                                                                                                                                                                                            |
|                                        | Allow constant propagation through hierarchical boundary pins.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> 1 0 true false inherited<br><b>Default:</b> inherited<br><b>Edit:</b> Yes                       |
| boundary_optimize_equal_opposite_hpins |                                                                                                                                                                                                                            |
|                                        | Allow connection of hierarchical boundary pins through equal/opposite boundary pins.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> 1 0 true false inherited<br><b>Default:</b> inherited<br><b>Edit:</b> Yes |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| boundary_optimize_feedthrough_hpins |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                     | <p>Allow routing of feedthrough boundary pins around a hierarchical instance.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                               |
| boundary_optimize_invert_hpins      |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                     | <p>Allow inversion of hierarchical boundary pins.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                           |
| bus                                 |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                     | <p>The bus for this hpin.</p> <p><b>Type:</b> <a href="#">obj(hpin_bus)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                      |
| clock_gating_exclude_signal         |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                     | <p>Exclude this pin from clock gating enable functions.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                             |
| constant                            |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                     | <p>Returns if this hpin has a constant logic value due to a 1'b0 or 1'b1 asserted directly on the the Verilog. If it is no_constant, and it is connected to a net, you must check the net .constar to see if the net is constant for other reasons.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 0 1 no_constant</p> <p><b>Default:</b> no_constant</p> <p><b>Edit:</b> No</p> |
| cts_isolated                        |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                     | <p>Is this hpin isolated</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                             |
| cts_isolated_effective              |                                                                                                                                                                                                                                                                                                                                                                                                       |

|                              |                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | Is this hpin isolated<br><b>Type:</b> <code>bool</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                                                                                                                                                                                                                              |
| depth                        |                                                                                                                                                                                                                                                                                                                                            |
|                              | Depth constraint of the hpin in microns. This is only valid for partition hpins (see <code>.partition_p</code> affected by pin assignment commands, and returns <code>""</code> if never set. The <code>edit_pin</code> command c used to set it.<br><b>Type:</b> <code>coord</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No |
| dft_constant_value           |                                                                                                                                                                                                                                                                                                                                            |
|                              | Constant pin value in test mode { <code>logic_0</code>   <code>logic_1</code>   <code>logic_z</code>   <code>no_value</code> }.<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                                                                                                                  |
| dft_controllable             |                                                                                                                                                                                                                                                                                                                                            |
|                              | In test mode the output pin can take either the <code>non_inverting</code> or <code>inverting</code> value of the input p<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> Yes                                                                                                                       |
| dft_dedicated_wrapper_reason |                                                                                                                                                                                                                                                                                                                                            |
|                              | Specifies the reason behind putting dedicated wrapper.<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> Yes                                                                                                                                                                                          |
| dft_driven_by_clock          |                                                                                                                                                                                                                                                                                                                                            |
|                              | If pin is driven by clock.<br><b>Type:</b> <code>enum</code><br><b>Enum Values:</b> <code>false</code> <code>true</code><br><b>Default:</b> <code>false</code><br><b>Edit:</b> Yes                                                                                                                                                         |
| dft_partition                |                                                                                                                                                                                                                                                                                                                                            |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Determine the DFT partition of the specified port/pin/hierarchical pin.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| direction            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                      | <p>Direction of pin.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> internal in out inout</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| dont_touch           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                      | <p>The preservation status of an hpin during optimization. A preserved hpin means the logical of the hpin must be preserved to maintain a simulation or test-point hpin in the netlist. However, name does not need to be preserved.</p> <p><b>Supported values:</b></p> <p>false: Unconstrained</p> <p>true: Cannot add/remove ports</p> <p>delete_ok: Can delete ports (if they have no fanout)</p> <p>add_ok: We cannot delete, or change the polarity or any hport but can add or duplicate hport</p> <p>invert_ok: We cannot delete, duplicate, or add but can change the polarity of any hport</p> <p>add_invert_ok: We cannot delete any hport but can add, duplicate, and change the polarity</p> <p>none: No user setting; will inherit from the module/hinst</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> none false true delete_ok add_ok invert_ok add_invert_ok const_prop_del</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> |
| dont_touch_effective |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                      | <p>This attribute defines the effective preservation status of an hpin during optimization based on .dont_touch_sources. If the local hpin .dont_touch is not "none", it will be used. Otherwise, .dont_touch_effective is used if not "none" or "false". If not, then the hinst .dont_touch_hport is used.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> none false true delete_ok add_ok invert_ok add_invert_ok const_prop_del</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| dont_touch_file      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>Specifies the sourced file to set dont_touch.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| dont_touch_sources |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                    | <p>Dictionary of {source value&gt;} pairs contributing to the dont_touch_effective attribute for this {user &lt;value&gt;} {power_intent &lt;value&gt;} {hinst_dont_touch_hports &lt;value&gt;} {hinst_dont_touch_effective}.</p> <p>user # This pins dont_touch value</p> <p>power_intent # Value set by committing UPF/CPF</p> <p>dft # This &lt;obj&gt; is part of design-for-test logic constraints</p> <p>hinst_dont_touch_effective # The hinst.dont_touch_effective value</p> <p>hinst_dont_touch_hports # The hinst.dont_touch_hports value</p> <p><b>Type:</b> string</p> <p><b>Default:</b> {user none} {power_intent false}</p> <p><b>Edit:</b> No</p> |
| edge               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                    | <p>If this hpin is for a partition, or inst that is a black-box, and the hpin is assigned, the edge value indicates along which edge of the boundary polygon the hpin is assigned. The edge number is from the lowest Y, then left-most X vertex, starting with 0, and then counting clock-wise. See set_pin_constraint command document for a figure showing the edge numbering. If the hpin is not assigned, or not for a partition or black-box inst, the value of -1 is returned.</p> <p><b>Type:</b> int</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                |
| escaped_name       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                    | <p>The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax: bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a is a scalar.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                |
| exceptions         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                       |                                                                                                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>Reports the path exceptions</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                      |
| hdl_vname             |                                                                                                                                                                                                              |
|                       | <p>The name for this hpin as specified in the input HDL file (Verilog or VHDL).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                              |
| hinst                 |                                                                                                                                                                                                              |
|                       | <p>The hinst for this hpin.</p> <p><b>Type:</b> <a href="#">obj(hinst)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                              |
| hnet                  |                                                                                                                                                                                                              |
|                       | <p>The hnet connected to this hpin (above the hinst for this hpin).</p> <p><b>Type:</b> <a href="#">obj(hnet)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                       |
| hport                 |                                                                                                                                                                                                              |
|                       | <p>The hport is the internal view of this hpin from inside the hinst.</p> <p><b>Type:</b> <a href="#">obj(hport)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| is_booth_encoded      |                                                                                                                                                                                                              |
|                       | <p>Booth encoded setting assigned to input port by user.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                   |
| is_boundary_duplicate |                                                                                                                                                                                                              |
|                       | <p>Captures duplicated ports after boundary optimizations and attributes is used during mapping generation.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |

|                            |                                                                                                                                                                                                                                                                                                      |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_boundary_feedback       |                                                                                                                                                                                                                                                                                                      |
|                            | <p>Captures feedback ports after boundary optimizations and attributes is used during mapping generation.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> Yes</p>                                                                                 |
| is_boundary_feedthru       |                                                                                                                                                                                                                                                                                                      |
|                            | <p>Captures feedthrough created during boundary optimization and used during mapping file generation.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> <code>false</code></p> <p><b>Edit:</b> Yes</p>                                                                                    |
| is_boundary_phase_inverted |                                                                                                                                                                                                                                                                                                      |
|                            | <p>Captures phase inversion during boundary optimization and attribute is used during mapping generation.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> <code>false</code></p> <p><b>Edit:</b> Yes</p>                                                                                |
| is_initial_phase_inverted  |                                                                                                                                                                                                                                                                                                      |
|                            | <p>This attribute is set during init_design and holds the initial value of the .is_phase_inverted attribute for the pin</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                      |
| is_phase_inverted          |                                                                                                                                                                                                                                                                                                      |
|                            | <p>Identifies the hpin for which phase is inverted during optimizations.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> <code>false</code></p> <p><b>Edit:</b> Yes</p>                                                                                                                 |
| layer                      |                                                                                                                                                                                                                                                                                                      |
|                            | <p>Returns the layer if this hpin is also for a partition (see .partition_pin), otherwise it returns <code>""</code>. edit_pin command can be used to change a partition hpin .layer.</p> <p><b>Type:</b> <code>obj(layer)</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> Yes</p> |
| location                   |                                                                                                                                                                                                                                                                                                      |

|                                |                                                                                                                                                                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Returns the location if this hpin is also for a partition (see .partition_pin) or physical hierarchy, otherwise it returns "". The edit_pin command can be used to change a partition hpin location.</p> <p><b>Type:</b> point<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| lp_asserted_probability        |                                                                                                                                                                                                                                                                                  |
|                                | <p>Probability of signal to be high on port/subport/pin as specified by user.</p> <p><b>Type:</b> double<br/><b>Default:</b> no_value<br/><b>Edit:</b> Yes</p>                                                                                                                   |
| lp_asserted_toggle_rate        |                                                                                                                                                                                                                                                                                  |
|                                | <p>Signal toggle rate assigned to pin by user.</p> <p><b>Type:</b> double<br/><b>Default:</b> no_value<br/><b>Edit:</b> Yes</p>                                                                                                                                                  |
| lp_system_asserted_probability |                                                                                                                                                                                                                                                                                  |
|                                | <p>System/Internal asserted probability.</p> <p><b>Type:</b> double<br/><b>Default:</b> no_value<br/><b>Edit:</b> Yes</p>                                                                                                                                                        |
| lp_system_asserted_toggle_rate |                                                                                                                                                                                                                                                                                  |
|                                | <p>System/Internal asserted toggle rate.</p> <p><b>Type:</b> double<br/><b>Default:</b> no_value<br/><b>Edit:</b> Yes</p>                                                                                                                                                        |
| lssd_master_clock              |                                                                                                                                                                                                                                                                                  |
|                                | <p>Defines the master clock port related to this slave clock port.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                    |
| name                           |                                                                                                                                                                                                                                                                                  |



|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>The same as .escaped_name without any \ escape chars. This name is commonly used to avoid problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| net            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                | <p>The canonical (flat) net attached to this hpin.</p> <p><b>Type:</b> <a href="#">obj(net)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                |
| num_exceptions |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                | <p>Reports the number of path exceptions</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                        |
| obj_type       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum (hpin)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                |
| original_name  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                | <p>Stores hdl_name on output pin</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                   |
| partition_pin  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>If this hpin is also for an uncommitted partition (the hinst for this hpin has a .partition object) returns the corresponding base_pin of the partition object, otherwise it returns "". The hinst .hpins and .hports) for an uncommitted partition carry the logical netlist connectivity, while the partition object carries the physical pin data (.base_pins) and pin constraints for pin assignment commands. Some of partition_pin physical and constraint data are also available directly from hpin (see .layer, .location, .width, .depth, .side, .place_status).</p> <p><b>Type:</b> <a href="#">obj(base_pin)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| place_status                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                    | <p>Returns the place_status if this hpin is also for a partition (see .partition_pin), otherwise it returns the edit_pin command can be used to set it.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> unplaced placed fixed cover<br/> <b>Default:</b> unplaced<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                    |
| prune_unused_logic                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                    | <p>Allow pruning of logic driving a hierarchical pin.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| safety_dcls_isolate_pin            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                    | <p>The type of isolation required for the signal driving this pin inside a dcls (dual core lock step).</p> <p><b>Supported values:</b><br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none split_network split_net<br/> <b>Default:</b> none<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                           |
| safety_dclst_error_signal_endpoint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                    | <p>This object is the endpoint for a safety-mechanism's error-signal.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| safety_dmr_error_signal_endpoint   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                    | <p>This object is the endpoint for a safety-mechanism's error-signal.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                     |                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| safety_parity_error_signal_endpoint |                                                                                                                                                                                                                                                                                                                                                                  |
|                                     | <p>This object is the endpoint for a safety-mechanism's error-signal.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                            |
| side                                |                                                                                                                                                                                                                                                                                                                                                                  |
|                                     | <p>Side constraint of the hpin. This is only valid for partition hpins (see .partition_pin) affected by pin assignment commands, and returns "" if never set. The edit_pin command can be used to set it.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> north west south east up none</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| timing_constant                     |                                                                                                                                                                                                                                                                                                                                                                  |
|                                     | <p>Reports constant value on pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                        |
| timing_constant_source              |                                                                                                                                                                                                                                                                                                                                                                  |
|                                     | <p>Reports source of constant value on pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                              |
| width                               |                                                                                                                                                                                                                                                                                                                                                                  |
|                                     | <p>Width constraint of the hpin in microns. This is only valid for partition hpins (see .partition_pin) affected by pin assignment commands, and returns "" if never set. The edit_pin command can be used to set it.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                             |
| wrapper_control                     |                                                                                                                                                                                                                                                                                                                                                                  |
|                                     | <p>Specifies as wrapper control signal.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                        |

|                 |                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| wrapper_segment |                                                                                                                                                                                                               |
|                 | Specifies the wrapper segments associated with the pin.<br><b>Type:</b> <code>obj(scan_chain)* obj(scan_segment)* obj(actual_scan_segment)* obj(actual_scan_</code><br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| wrapper_type    |                                                                                                                                                                                                               |
|                 | Specifies the type of wrapper segment associated with the pin.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                  |

## hport

### Parent Objects

`test_clock`, `clock`, `port`, `violation`, `arc`, `hinst`, `hport_bus`, `hpin`, `hnet`, `scan_chain`, `actual_scan_segment`, `dft_configuration_mode`, `root`, `test_signal`, `inst`, `scan_segment`, `actual_scan_chain`

### Definition

hierarchical terminal

| Attribute                   | Description                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name                   |                                                                                                                                                   |
|                             | The name without the leading hinst hierarchy (e.g. p3 for h1/h2/p3).<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> No |
| bus                         |                                                                                                                                                   |
|                             | The bus for this hport.<br><b>Type:</b> <code>obj(hport_bus)</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                      |
| clock_gating_exclude_signal |                                                                                                                                                   |

|                              |                                                                                                                                                           |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | Exclude this pin from clock gating enable functions.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> Yes                       |
| dft_dedicated_wrapper_reason |                                                                                                                                                           |
|                              | Specifies the reason behind putting dedicated wrapper.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> Yes                      |
| dft_driven_by_clock          |                                                                                                                                                           |
|                              | If pin is driven by clock.<br><b>Type:</b> <code>enum</code><br><b>Enum Values:</b> false true<br><b>Default:</b> false<br><b>Edit:</b> Yes               |
| direction                    |                                                                                                                                                           |
|                              | Direction of pin.<br><b>Type:</b> <code>enum</code><br><b>Enum Values:</b> internal in out inout<br><b>Default:</b> ""<br><b>Edit:</b> No                 |
| escaped_name                 |                                                                                                                                                           |
|                              | The escaped name of the hport.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                               |
| hdl_vname                    |                                                                                                                                                           |
|                              | The name for this port as specified in the input HDL file (Verilog or VHDL).<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> No |
| hinst                        |                                                                                                                                                           |
|                              | The hinst for this hport.<br><b>Type:</b> <code>obj(hinst)</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                |

|                                |                                                                                                                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hnet                           |                                                                                                                                                                                      |
|                                | <p>The hnet connected to this hport (inside the hinst of this hport).</p> <p><b>Type:</b> <a href="#">obj(hnet)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>             |
| hpin                           |                                                                                                                                                                                      |
|                                | <p>The hpin is the external view from outside the hinst for this hport.</p> <p><b>Type:</b> <a href="#">obj(hpin)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| is_booth_encoded               |                                                                                                                                                                                      |
|                                | <p>Booth encoded setting assigned to input port by user.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                           |
| lp_asserted_probability        |                                                                                                                                                                                      |
|                                | <p>Probability of signal to be high on port/subport/pin as specified by user.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p> |
| lp_asserted_toggle_rate        |                                                                                                                                                                                      |
|                                | <p>Signal toggle rate assigned to pin by user.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                |
| lp_system_asserted_probability |                                                                                                                                                                                      |
|                                | <p>System/Internal asserted probability.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                      |
| lp_system_asserted_toggle_rate |                                                                                                                                                                                      |
|                                | <p>System/Internal asserted toggle rate.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                      |

|                   |                                                                                                                                                                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lssd_master_clock |                                                                                                                                                                                                                                     |
|                   | <p>Defines the master clock port related to this slave clock port.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                 |
| name              |                                                                                                                                                                                                                                     |
|                   | <p>The fully qualified path name of the hport (e.g. h1/h2/p3).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                      |
| obj_type          |                                                                                                                                                                                                                                     |
|                   | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">hport</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>               |
| original_name     |                                                                                                                                                                                                                                     |
|                   | <p>Stores port_vname on port bits.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                 |
| wrapper_control   |                                                                                                                                                                                                                                     |
|                   | <p>Specifies as wrapper control signal.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                           |
| wrapper_segment   |                                                                                                                                                                                                                                     |
|                   | <p>Specifies the wrapper segments associated with the pin.</p> <p><b>Type:</b> <a href="#">obj(scan_chain)* obj(scan_segment)* obj(actual_scan_segment)* obj(actual_scan_</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| wrapper_type      |                                                                                                                                                                                                                                     |
|                   | <p>Specifies the type of wrapper segment associated with the pin.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                  |

# inst

## Parent Objects

[flexible\\_htree](#), [marker](#), [io\\_constraint](#), [place\\_blockage](#), [violation](#), [group](#), [hinst](#), [sdp](#), [actual\\_scan\\_segment](#), [inst\\_physical\\_pin](#), [partition](#), [design](#), [clock\\_tree](#), [root](#), [pg\\_pin](#), [route\\_blockage](#), [scan\\_segment](#), [pin](#), [inst\\_obs\\_shape](#)

## Definition

instance

| Attribute                         | Description                                                                                                                                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow_seq_in_out_phase_opto       |                                                                                                                                                                                                                          |
|                                   | Allow phase inversion on sequential instance.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes                                                                                           |
| aocv_bootstrap_fall_derate_factor |                                                                                                                                                                                                                          |
|                                   | Returns aocv coefficient aocv_bootstrap_fall_derate_factor in bootstrap mode on type cell<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| aocv_bootstrap_rise_derate_factor |                                                                                                                                                                                                                          |
|                                   | Returns aocv coefficient aocv_bootstrap_rise_derate_factor in bootstrap mode on type cell<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| apply_booth_encoding              |                                                                                                                                                                                                                          |



|                |                                                                                                                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>Apply booth encoding by using user specified method (for a specified instance).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> nonbooth auto_bitwidth auto_togglerate manual inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p> |
| arcs           |                                                                                                                                                                                                                                                                             |
|                | <p>Returns a list of arc objects that are associated with this inst</p> <p><b>Type:</b> <a href="#">obj(arc)*</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                   |
| area           |                                                                                                                                                                                                                                                                             |
|                | <p>area of inst</p> <p><b>Type:</b> <a href="#">area</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                               |
| base_cell      |                                                                                                                                                                                                                                                                             |
|                | <p>cell of inst</p> <p><b>Type:</b> <a href="#">obj(base_cell)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                     |
| base_name      |                                                                                                                                                                                                                                                                             |
|                | <p>The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                         |
| bbox           |                                                                                                                                                                                                                                                                             |
|                | <p>Bounding box of the overlap rects that define the placement area used by this inst.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                        |
| bottom_padding |                                                                                                                                                                                                                                                                             |
|                | <p>Value of bottom padding for the instance.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                   |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| clock_gating_exclude         |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                              | <p>Exclude this instance from clock-gating.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                         |
| clock_gating_integrated_cell |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                              | <p>clock_gating_integrated_cell</p> <p><b>Type:</b> <code>string</code></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                    |
| clock_gating_method          |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                              | <p>The clock gating method used.</p> <p><b>Type:</b> <code>enum</code></p> <p><b>Enum Values:</b> combine common_enable_extraction feedback other sena sena_and_feed ungated</p> <p><b>Default:</b> ungated</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                             |
| clock_gating_test_signal     |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                              | <p>Set specified test signal to connect up test logic in clock-gating circuitry. Specify use_shift_ to derive test logic from shift enable of the gated flops, or use_none to disable test connecti the inst.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                        |
| clock_gating_xor_status      |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                              | <p>Set clock-gating xor status.</p> <p><b>Type:</b> <code>enum</code></p> <p><b>Enum Values:</b> enable_updated inserted unset</p> <p><b>Default:</b> unset</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                             |
| cts_cannot_clone_reasons     |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                              | <p>A list of reasons to explain why the inst could not be cloned.</p> <p>This attribute will not be set if attributes cts_clone_clock_gates/cts_clone_clock_logic are</p> <p>The attribute returns a list of reasons why the inst cannot be cloned.</p> <p>The reasons why a inst cannot be cloned can be divided into three categories:</p> <ol style="list-style-type: none"> <li>1. The inst is marked as dont_touch</li> </ol> |

2. Reasons specific to the clock node

3. General reasons

The possible dont\_touch reasons are:

dont\_touch.add\_port\_driver Set if inst was added above an output port or below an input port  
attribute cts\_add\_port\_driver

dont\_touch.add\_driver\_cell Set if inst was added below the root via attribute cts\_add\_driver

dont\_touch.clock\_root Set during clock tree extraction if identified as having the root pin

dont\_touch.clock\_sink Set on inst if any pin is a clock sink in a clock tree

dont\_touch.clock\_tree\_generator\_path Set if nodes and wires are in clock tree generator

dont\_touch.clock\_wire Set on inst if clock input wire is user dont\_touch

dont\_touch.clockgate\_no\_power\_domain Set if clock gate / clock logic is a clocknode in a clock gate / clock logic power domain

dont\_touch.drives\_multi\_driver\_net Set if inst drives wires with other drivers

dont\_touch.external\_skew\_group\_pin Set if a pin on the inst is a skew group sink, source or  
ignore pin for a skew group created by the user

dont\_touch.flexible\_htree Set if inst was added to a flexible H-tree

dont\_touch.internal\_skew\_group\_pin Set if a pin on the inst is a skew group sink, source or  
pin for a skew group created by CCOpt

dont\_touch.neg\_edge\_clock\_gate Set if a clock gate gates the falling edge

dont\_touch.non\_flop\_clock\_gating Set during CTS on insts where cloning them could cause  
non-equivalence failure in LEC, including clock gates whose transitive fanout contains flop  
triggered by the opposite clock edge, macros, or latches

dont\_touch.non\_standard\_inputs\_clock\_gate Set if inst is a clock gate with 'non-standard'  
to avoid disconnecting them

dont\_touch.observability\_clock\_gate Set if inst is a clock gate with an observability output

dont\_touch.output\_wire Set if inst drives a net that is dont\_touch

dont\_touch.placer.lock Set if inst is user locked or locked by DEF

dont\_touch.power\_management Set if inst is a power management inst

dont\_touch.prevent\_assign Set if inst is necessary to prevent an assign

dont\_touch.sdc Set if constrained by SDC timing

dont\_touch.sdc\_path\_group Set if there is an SDC path group start / endpoint on one of the  
this inst

dont\_touch.sub\_block Set if inst is in dont\_touch module

dont\_touch.unmergeable\_composite\_clock\_gate Set if some of the clock gate is dont\_touch

dont\_touch.user Set by user

Contact Cadence Support if any of the following reasons are listed:

dont\_touch.cannot\_understand\_clock\_gate Set if clock gate not recognized

dont\_touch.composite\_clock\_gate\_enable\_test\_or\_gate

The possible reasons specific to the clock node are:

PowerManagement Set if inst is a power management inst

PowerManagementInconsistency Set if a disconnection of the inst from the netlist would cause  
power management inconsistency

PreservedUMPB Set if clock node is a User Module Port Bit which is preserved

Contact Cadence Support if any of the following reasons are listed:

ClockGatesAlways Set if inst is clock gate, considered always dont\_touch

ClockLogicAlways Set if inst is clock logic, considered always dont\_touch

ModelsRoot Set if the clock node is the clock root  
 NoRoot Set if the clock node has no root  
 RootIgnored Set if the clock root is ignored  
 SpineCell Set if cell is a clock spine inst and is marked as dont\_touch  
 The possible general reasons are:  
 DrivesAcrossPowerDomains The inst drives across power domain boundaries  
 NodeHasSGConstraints The inst has user mode skew group constraints in default mode  
 ClockDriverCannotCloneInverter  
 ClockDriverCreatedByBuffLongNets Set if the inst was created by 'Buffering long nets'.  
 ClockDriverInverterCloningDisabled The inst is an inverter and cloning inverters is disabled  
 UnclonableClockSource Set if the inst carries the -source of an SDC clock  
 UnclonableIODelayReference Set if the inst is associated with an I/O timing constraint  
 UnclonablePLLReference Set if the inst is associated with a PLL constraint  
 UnclonablePLLFeedback Set if the inst is analyzed as a PLL feedback  
 UnclonablePLLOutput Set if the inst is analyzed as a PLL output  
 UnsupportedNodeType Set if the inst is not a supported type. Supported types are: clock gate, clock logic and clock driver.  
 SingleOrZeroFanoutNode Set if the inst has fewer than two fanout.  
 Contact Cadence Support if any of the following reasons are listed:  
 NoCellInst  
 ClockLogicMultiOutputCell  
 IgnoredTree  
 IsUnclonable  
 DiscreteClockGate  
 NotTreeViewNode  
 OutputPinCellPinNull  
 ClockDriverNeedToKeep  
 Valid values: list string  
**Type:** string  
**Default:** ""  
**Edit:** Yes

#### cts\_cannot\_merge\_reasons

A list of reasons to explain why the inst could not be merged.  
 The attribute returns a list of reasons why the inst cannot be merged. These attributes are set on clock gates and clock logic instances.  
 For clock gates, the merge is driven by the enable function. If two clock gates share the same enable function, potentially they may be merged together. The reasons explains why potentially mergeable clock gates could not in fact be merged.  
 For clock logic, the merge is driven by the logical expression, or logic function, at the clock output pin. If two clock logic express the same logic function, at least potentially they may be merged. The reasons lists why the potential merge could not be realized.  
 User configuration disabled merging:

ClockGateMergingDisabledOnTree Merging disabled: attribute cts\_declone\_clock\_gates  
ClockLogicMergingDisabledOnTree Merging disabled: attribute cts\_declone\_clock\_logic  
IsDontTouch The instance cannot be merged as it is marked dont\_touch  
Attributes of a clock node rendered it unmergeable:  
GloballyUnique The clock node has no potential candidate with which it might merge  
UniqueUnderParent There are potential merge candidates, but not on the same clock net  
HasSkewGroupConstraints The instance carries a skew group source, sink or ignore pin  
UndrivenEnablePins The clock gate has a floating enable input  
UndrivenRetentionPins The clock gate has a floating retention input  
UndrivenTestPins The clock gate has a floating test enable input  
DifferentSplitOnInputs The multi-input clock logic has different skew groups at its clock in  
SiblingsNotMergeable Could not merge because all other candidates were unmergeable  
Otherwise mergeable clock nodes may have incompatibilities that prevent merging:  
DifferentSkewGroupForInputs Two clock nodes have different skew groups at their clock in  
DifferentSkewGroupForOutputs Two clock nodes have different skew groups at their clock  
DifferentCellFamilies Two clock nodes belong to incompatible cell families  
DifferentNumberOfInputPins Two clock nodes have different numbers of input pins  
DifferentNonStandardInputs Two clock gates have non-standard\* inputs which are non-ec  
MismatchingNonClockInputs Two clock nodes have non-clock inputs which receive non-equivalent signals  
MismatchingClockInputs Two clock nodes have clock inputs which receive non-equivalent  
MismatchingPowerDomains Two clock nodes are in incompatible power domains  
NotLogicalClones Two clock nodes have outputs which are not logically equivalent  
DifferentNumberOfParents Two clock nodes have different numbers of clock-carrying inputs  
DifferentOutputPowerContext Two clock nodes have outputs with different power domain/voltage  
DifferentInputPowerContext Two clock nodes have inputs with different power domain/voltage  
IncompatibleRestrictedRegions Two clock nodes have incompatible restricted regions  
DifferentAnnotations Two clock nodes have different annotations on its pins  
(\*) Non-standard inputs on a clock gate are inputs besides the clock, enable, test enable and retention inputs.  
Valid values: list string  
**Type:** string  
**Default:** ""  
**Edit:** Yes

cts\_effective\_clock\_halo\_x

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>This read only attribute shows the x component of the clock halo The units are um.<br/>The following attributes can be used to assign x direction clock halos within CCOpt:</p> <ul style="list-style-type: none"><li>cts_cell_halo_x</li><li>cts_cell_density</li><li>cts_cell_halo_sites</li></ul> <p>See also:</p> <ul style="list-style-type: none"><li>.cts_effective_clock_halo_x_source</li><li>.cts_effective_clock_halo_y_source</li><li>.cts_effective_clock_halo_y</li></ul> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| cts_effective_clock_halo_x_source |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                   | <p>This read only attribute shows which attribute defines the x component of the clock halo.<br/>The following attributes can be used to assign x direction clock halos within CCOpt:</p> <ul style="list-style-type: none"><li>cts_cell_halo_x</li><li>cts_cell_density</li><li>cts_cell_halo_sites</li></ul> <p>See also:</p> <ul style="list-style-type: none"><li>.cts_effective_clock_halo_x</li><li>.cts_effective_clock_halo_y</li><li>.cts_effective_clock_halo_y_source</li></ul> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| cts_effective_clock_halo_y        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>This read only attribute shows the x component of the clock halo. The units are um.<br/>The following attributes can be used to assign y direction clock halos within CCOpt:</p> <p>cts_cell_halo_y<br/>cts_adjacent_rows_legal<br/>cts_cell_halo_rows</p> <p>See also:</p> <p>.cts_effective_clock_halo_x_source<br/>.cts_effective_clock_halo_y_source<br/>.cts_effective_clock_halo_x</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| cts_effective_clock_halo_y_source |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                   | <p>This read only attribute shows which attribute defines the y component of the clock halo.<br/>The following attributes can be used to assign y direction clock halos within CCOpt:</p> <p>cts_cell_halo_y<br/>cts_adjacent_rows_legal<br/>cts_cell_halo_rows</p> <p>See also:</p> <p>.cts_effective_clock_halo_x<br/>.cts_effective_clock_halo_y<br/>.cts_effective_clock_halo_x_source</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>  |
| cts_node_type                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                   | <p>node type of instance within cts graph</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                   |
| cts_original_names                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Specifies for a clockgate or clocklogic which has been merged or is a clone a list of names from the original netlist which are equivalent to the clockgate/clocklogic. For example:<br/>If A and B are merged to form C then original_names for C is { A B }.<br/>If D_clone is a clone of D then original_names for D_clone is { D }.<br/>If E is a clone of C then original_names for E is { A B } (remembering C was a merger of A<br/>Valid values: list string<br/><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p> |
| cts_preserve_from_deletion |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                            | <p>Do not allow this instance to be deleted. Default false. Valid values are true or false.<br/><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                            |
| delta_temperature          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                            | <p>temperature of the inst<br/><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| dft_abstract_dont_scan     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                            | <p>Exclude this instance from scan-synthesis as an abstract segment has been defined on it.<br/><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                             |
| dft_custom_se              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                            | <p>Shift Enable for flop.<br/><b>Type:</b> obj(test_signal)<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| dft_dont_scan              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                            | <p>Exclude this design from scan-synthesis {inherited   true   false}.<br/><b>Type:</b> enum<br/><b>Enum Values:</b> false true inherited<br/><b>Default:</b> inherited<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                |



|                                    |                                                                                                                                                                                 |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dft_exclude_flop_from_los_pipeline |                                                                                                                                                                                 |
|                                    | <p>Avoid insertion of launch-off-shift pipelined logic to flop's shift enable.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                |
| dft_exclude_from_shift_register    |                                                                                                                                                                                 |
|                                    | <p>Determines if flop should be excluded from shift register.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                 |
| dft_exempt_from_system_clock_check |                                                                                                                                                                                 |
|                                    | <p>Exempt flop from system clock check.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                       |
| dft_force_blackbox_for_atpg        |                                                                                                                                                                                 |
|                                    | <p>Determines is the instance is blackbox for ATPG.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> none true false</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> |
| dft_is_blackbox_for_atpg           |                                                                                                                                                                                 |
|                                    | <p>Blackbox for ATPG.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                          |
| dft_is_los_pipeline_flop           |                                                                                                                                                                                 |
|                                    | <p>Determines if this flop is launch-off-shift pipeline flop.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                 |
| dft_is_testpoint                   |                                                                                                                                                                                 |

|                              |                                                                                                                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | Attribute for flop which indicates if it is a testpoint flop or not.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                           |
| dft_mapped                   |                                                                                                                                                                                                                                                                                  |
|                              | This flop is mapped to scan for DFT purposes.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> No                                                                                                                                                   |
| dft_part_of_segment          |                                                                                                                                                                                                                                                                                  |
|                              | The type of segment this instance belongs to {abstract   fixed   floating   preserved   shift_re<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> fixed floating preserve abstract shift_register<br><b>Default:</b> ""<br><b>Edit:</b> No                            |
| dft_partition                |                                                                                                                                                                                                                                                                                  |
|                              | Determine the DFT partition of the specified instance/hierarchical instance.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                     |
| dft_scan_chain               |                                                                                                                                                                                                                                                                                  |
|                              | The actual scan chain this instance belongs to.<br><b>Type:</b> <a href="#">obj(scan_chain)</a> <a href="#">obj(scan_segment)</a> <a href="#">obj(actual_scan_segment)</a> <a href="#">obj(actual_scan_ch</a><br><b>Default:</b> ""<br><b>Edit:</b> No                           |
| dft_scan_chain_in_multi_mode |                                                                                                                                                                                                                                                                                  |
|                              | The actual scan chains this instance belongs to in multi mode stitching.<br><b>Type:</b> <a href="#">obj(scan_chain)*</a> <a href="#">obj(scan_segment)*</a> <a href="#">obj(actual_scan_segment)*</a> <a href="#">obj(actual_scan_</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| dft_status                   |                                                                                                                                                                                                                                                                                  |
|                              | DFT-rule-check status as analyzed by 'check_dft_rules' {Passes DFT rules   Fails DFT rules<br>Abstract Segment Dont scan   Dont scan   Misc. non scan}.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                          |

|                              |                                                                                                                                                                                                                                                                                                                    |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dft_test_clock               |                                                                                                                                                                                                                                                                                                                    |
|                              | Test clock as analyzed by 'check_dft_rules'.<br><b>Type:</b> obj(test_clock)<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                              |
| dft_test_clock_edge          |                                                                                                                                                                                                                                                                                                                    |
|                              | Test clock edge as analyzed by 'check_dft_rules'.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                  |
| dft_test_clock_source        |                                                                                                                                                                                                                                                                                                                    |
|                              | Test clock source as analyzed by 'check_dft_rules'.<br><b>Type:</b> obj(pin) obj(hpin) obj(port) obj(hport)<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                               |
| dft_test_clock_waveform_edge |                                                                                                                                                                                                                                                                                                                    |
|                              | Test clock waveform leading or trailing edge as analyzed by 'check_dft_rules'.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                     |
| dft_violation                |                                                                                                                                                                                                                                                                                                                    |
|                              | DFT rule violation as analyzed by 'check_dft_rules'.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                               |
| dont_merge_multibit          |                                                                                                                                                                                                                                                                                                                    |
|                              | This attribute is denotes whether the instance can be merged during multibit optimization. 1 enabled only when the root attribute 'use_multibit_cells' is set to true.<br><b>Supported values:</b><br>true: Cannot merge<br>false: Can be merged<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes |
| dont_retime                  |                                                                                                                                                                                                                                                                                                                    |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Do not retime this instance.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| dont_split_multibit  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                      | <p>This attribute is denotes whether the instance can be split (unmerged) during multibit optim<br/>This is enabled only when the root attribute 'use_multibit_cells' is set to true.</p> <p><b>Supported values:</b></p> <ul style="list-style-type: none"> <li>true: Cannot split</li> <li>false: Can be split</li> </ul> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| dont_touch           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                      | <p>This attribute defines the user preservation status of an instance during optimization.</p> <p><b>Supported values:</b></p> <ul style="list-style-type: none"> <li>none (default): Unconstrained</li> <li>false: Can be mapped, sized, deleted, and constants can be propagated through it</li> <li>const_prop_size_delete_ok: Can be resized or deleted and constants can be propagated through it</li> <li>const_prop_delete_ok: Can be deleted and constants can be propagated through it</li> <li>size_delete_ok: Can be resized or deleted if no fanout</li> <li>delete_ok: Can be deleted if it has no fanout, but cannot be resized</li> <li>size_ok: Can only be resized</li> <li>size_same_height_ok: Can only be resized to a cell of the same height</li> <li>size_same_footprint_ok: Can only be resized to a cell of the same footprint that has exact same pin shapes</li> <li>map_size_ok: Can be mapped or sized (but not deleted). Applies only to sequential insta cannot be applied on the module or hinst object.</li> <li>true: Cannot be touched</li> </ul> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none false true delete_ok const_prop_size_delete_ok const_prop_delete_size_delete_ok size_ok size_same_height_ok size_same_footprint_ok map_size_ok</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> |
| dont_touch_effective |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>This attribute defines the effective (most pessimistic) preservation status of an instance during optimization based on the 'sources'.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none false true delete_ok const_prop_size_delete_ok const_prop_delete_size_delete_ok size_ok size_same_height_ok size_same_footprint_ok map_size_ok</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| dont_touch_file    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | <p>Specifies the sourced file to set dont_touch.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| dont_touch_pins    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none false true delete_ok const_prop_size_delete_ok const_prop_delete_size_delete_ok size_ok size_same_height_ok size_same_footprint_ok map_size_ok</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| dont_touch_sources |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | <p>Dictionary of {source &lt;value&gt;} pairs contributing to the dont_touch_effective attribute for this {user &lt;value&gt;} {lib &lt;value&gt;} {parent &lt;value&gt;} {scan &lt;value&gt;} {dft &lt;value&gt;} {place_status &lt;value&gt;}</p> <p>user # This inst dont_touch value</p> <p>lib # The base_cell dont_touch value</p> <p>parent # The effective dont_touch value from an hinst above (the closest hinst above that is not false)</p> <p>scan # Is this inst part of a scan-chain (value is either size_ok or none). There is an additional attribute that holds this information.</p> <p>dft # This &lt;obj&gt; is part of design-for-test logic constraints.</p> <p>place_status # The dont_touch come from placement status of an instance during placement optimization</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| dont_use_qbar_pin  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                                 |                                                                                                                                                                                                                                            |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>Do not use inverted output of sequential instance.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                    |
| dpopt_instance_constant_input_based_speculation |                                                                                                                                                                                                                                            |
|                                                 | <p>Controls constant input based datapath speculation of datapath module in 'syn_generic'.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                               |
| early_cell_check_derate_factor                  |                                                                                                                                                                                                                                            |
|                                                 | <p>Returns the derating factor on timing check values for early paths (e.g Hold checks)</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_cell_check_mean_derate_factor             |                                                                                                                                                                                                                                            |
|                                                 | <p>Reports early mean derate factor for timing check arc of the specified cell.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| early_cell_check_sigma_derate_factor            |                                                                                                                                                                                                                                            |
|                                                 | <p>Reports early sigma derate factor for timing check arc of the specified cell.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| early_clk_cell_derate_factor                    |                                                                                                                                                                                                                                            |
|                                                 | <p>Returns the derating factor for instances on early clock paths</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| early_clk_cell_mean_derate_factor               |                                                                                                                                                                                                                                            |

|                                             |                                                                                                                                                                                                                                           |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>Reports early mean timing derate factor for clock line</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                        |
| early_clk_cell_sigma_derate_factor          |                                                                                                                                                                                                                                           |
|                                             | <p>Reports early sigma timing derate factor for clock line</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| early_clock_check_incremental_derate_factor |                                                                                                                                                                                                                                           |
|                                             | <p>Reports incremental derate factor specified with -early -clock for check arcs</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_data_cell_derate_factor               |                                                                                                                                                                                                                                           |
|                                             | <p>Returns the derating factor for instances on early data paths</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| early_data_cell_mean_derate_factor          |                                                                                                                                                                                                                                           |
|                                             | <p>Reports mean derate factor specified with -early -data</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                        |
| early_data_cell_sigma_derate_factor         |                                                                                                                                                                                                                                           |
|                                             | <p>Reports sigma derate factor specified with -early -data</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| early_data_check_incremental_derate_factor  |                                                                                                                                                                                                                                           |

|                                           |                                                                                                                                                                                                                                          |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>Reports incremental derate factor specified with -early -data for check arcs</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_fall_cell_check_derate_factor       |                                                                                                                                                                                                                                          |
|                                           | <p>Returns the early cell check derating factor for falling arrivals</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                  |
| early_fall_cell_check_mean_derate_factor  |                                                                                                                                                                                                                                          |
|                                           | <p>Reports mean derate factor specified with -early -fall for check arcs</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| early_fall_cell_check_sigma_derate_factor |                                                                                                                                                                                                                                          |
|                                           | <p>Reports sigma derate factor specified with -early -fall for check arcs</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| early_fall_clk_cell_derate_factor         |                                                                                                                                                                                                                                          |
|                                           | <p>Returns the early derating factor for falling delays through clock tree instances</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| early_fall_clk_cell_mean_derate_factor    |                                                                                                                                                                                                                                          |
|                                           | <p>Reports mean derate factor specified with -early -fall for clock logic</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| early_fall_clk_cell_sigma_derate_factor   |                                                                                                                                                                                                                                          |



|                                            |                                                                                                                                                                                                                          |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Reports sigma derate factor specified with -early -fall for clock logic</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| early_fall_clock_incremental_derate_factor |                                                                                                                                                                                                                          |
|                                            | <p>Reports incremental derate factor specified with -early -fall -clock</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>         |
| early_fall_data_cell_derate_factor         |                                                                                                                                                                                                                          |
|                                            | <p>Returns the early derating factor for falling delays through data path instances</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| early_fall_data_cell_mean_derate_factor    |                                                                                                                                                                                                                          |
|                                            | <p>Reports mean derate factor specified with -early -fall -data</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                 |
| early_fall_data_cell_sigma_derate_factor   |                                                                                                                                                                                                                          |
|                                            | <p>Reports sigma derate factor specified with -early -rise -data for check arcs</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_fall_data_incremental_derate_factor  |                                                                                                                                                                                                                          |
|                                            | <p>Reports incremental derate factor specified with -early --fall -data</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>         |
| early_rise_cell_check_derate_factor        |                                                                                                                                                                                                                          |

|                                            |                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Returns the early cell check derating factor for rising arrivals</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>               |
| early_rise_cell_check_mean_derate_factor   |                                                                                                                                                                                                                      |
|                                            | <p>Reports mean derate factor specified with -early -rise for check arcs</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| early_rise_cell_check_sigma_derate_factor  |                                                                                                                                                                                                                      |
|                                            | <p>Reports sigma derate factor specified with -early -rise for check arcs</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| early_rise_clk_cell_derate_factor          |                                                                                                                                                                                                                      |
|                                            | <p>Returns the early derating factor for rise delays through clock tree instances</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_rise_clk_cell_mean_derate_factor     |                                                                                                                                                                                                                      |
|                                            | <p>Reports mean derate factor specified with -early -rise -clock</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>            |
| early_rise_clk_cell_sigma_derate_factor    |                                                                                                                                                                                                                      |
|                                            | <p>Reports sigma derate factor specified with -early -rise -clock</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| early_rise_clock_incremental_derate_factor |                                                                                                                                                                                                                      |

|                                           |                                                                                                                                                                                                                       |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>Reports incremental derate factor specified with -early -rise -clock</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| early_rise_data_cell_derate_factor        |                                                                                                                                                                                                                       |
|                                           | <p>Returns the early derating factor for rising delays through data path instances</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_rise_data_cell_mean_derate_factor   |                                                                                                                                                                                                                       |
|                                           | <p>Reports mean derate factor specified with -early -rise -data</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>              |
| early_rise_data_cell_sigma_derate_factor  |                                                                                                                                                                                                                       |
|                                           | <p>Reports sigma derate factor specified with -early -rise -data</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>             |
| early_rise_data_incremental_derate_factor |                                                                                                                                                                                                                       |
|                                           | <p>Reports incremental derate factor specified with -early -rise -data</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| eco_opt_use_cells                         |                                                                                                                                                                                                                       |
|                                           | <p>Specifies the cells to be used by eco opt engine for replacing the inst.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                          |
| escaped_name                              |                                                                                                                                                                                                                       |

|                     |                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax: bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a scalar.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| exceptions          |                                                                                                                                                                                                                                                                                                                                                                                         |
|                     | <p>Reports the path exceptions</p> <p><b>Type:</b> string<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                          |
| fcu_instruction_set |                                                                                                                                                                                                                                                                                                                                                                                         |
|                     | <p>Attribute to define different repair specific instructions for the different instances of the same on which fuse control unit has been inserted.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                          |
| group               |                                                                                                                                                                                                                                                                                                                                                                                         |
|                     | <p>The floorplan group that contains this inst. The instances in a group will be placed close together.</p> <p><b>Type:</b> obj(group)<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                   |
| has_aging           |                                                                                                                                                                                                                                                                                                                                                                                         |
|                     | <p>Returns true if the associated library cell is recognized has aging information</p> <p><b>Type:</b> bool<br/><b>Allowed -index values:</b> analysis_view<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                              |
| has_bti_aging       |                                                                                                                                                                                                                                                                                                                                                                                         |
|                     | <p>Returns true if the associated library cell is recognized has BTI aging information</p> <p><b>Type:</b> bool<br/><b>Allowed -index values:</b> analysis_view<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                          |
| has_hci_aging       |                                                                                                                                                                                                                                                                                                                                                                                         |

|                    |                                                                                                                                                                                                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | Returns true if the associated library cell is recognized has HCI aging information<br><b>Type:</b> <code>bool</code><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                       |
| has_lvf            |                                                                                                                                                                                                                                                                                     |
|                    | Returns true if library arc(s) of libcell has LVF information<br><b>Type:</b> <code>bool</code><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                             |
| has_sdc_assertions |                                                                                                                                                                                                                                                                                     |
|                    | Reports true if SDC assertions are asserted<br><b>Type:</b> <code>bool</code><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                         |
| hdl_name           |                                                                                                                                                                                                                                                                                     |
|                    | Same name attribute on inst from genus pass on the value to innovus attribute . to be used hdl name of Hard Macro/BBOX during 1 PAss LEC flow .<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                       |
| hdl_proc_name      |                                                                                                                                                                                                                                                                                     |
|                    | Name of the process (VHDL) or always block (Verilog) from which the instance is inferred.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                             |
| hdl_vname          |                                                                                                                                                                                                                                                                                     |
|                    | The name for this instance as specified in the input HDL file (Verilog or VHDL). For flip-flop instances, hdl_vname is the name of the variable for which the flip-flop or latch instance wa inferred.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> No |
| hierarchical_level |                                                                                                                                                                                                                                                                                     |

|                         |                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | Returns hierarchical level for an Instance<br><b>Type:</b> <a href="#">int</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| hierarchical_name       |                                                                                                                                                                              |
|                         | hierarchical_name<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                       |
| hinst                   |                                                                                                                                                                              |
|                         | The hinst of the inst, which is a hinst for an instance within hierarchy.<br><b>Type:</b> <a href="#">obj(hinst)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                |
| instance_internal_power |                                                                                                                                                                              |
|                         | The instance's internal power.<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> no_value<br><b>Edit:</b> Yes                                                        |
| instance_leakage_power  |                                                                                                                                                                              |
|                         | The instance's leakage power.<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> no_value<br><b>Edit:</b> Yes                                                         |
| instance_model          |                                                                                                                                                                              |
|                         | Reports the noise instance model<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No        |
| inverted_phase          |                                                                                                                                                                              |

|                       |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>The sequential instance was implemented using a inverted phase.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                        |
| is_always_on          |                                                                                                                                                                                                                                                                                                                                                                        |
|                       | <p>Returns true if the associated library cell is recognized as always on type cell</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                         |
| is_black_box          |                                                                                                                                                                                                                                                                                                                                                                        |
|                       | <p>Returns a value of true if the cell is a black box cell.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                 |
| is_buffer             |                                                                                                                                                                                                                                                                                                                                                                        |
|                       | <p>This inst is a buffer.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                    |
| is_clock_gating_check |                                                                                                                                                                                                                                                                                                                                                                        |
|                       | <p>Returns true if a clock gating check is performed by this instance. The clock gating check m inferred from logic in netlist, asserted by SDC constraints, or may be due to this instance be integrated clock gating cell</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| is_clock_network_inst |                                                                                                                                                                                                                                                                                                                                                                        |
|                       | <p>Reports true if specified inst is part of clock network</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                          |
| is_combinational      |                                                                                                                                                                                                                                                                                                                                                                        |

|                        |                                                                                                                                                                                                                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Returns a value of true if the instance is a combinational cell (not a sequential cell).</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| is_disable_timing      |                                                                                                                                                                                                                                                        |
|                        | <p>Returns true if set_disable_timing has been set on this cell instance</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                              |
| is_fall_edge_triggered |                                                                                                                                                                                                                                                        |
|                        | <p>Returns a value of true if the instance is triggered by the falling edge of the clock.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| is_fixed_mask          |                                                                                                                                                                                                                                                        |
|                        | <p>Indicates the base_cell of the inst has FIXEDMASK keyword in LEF, so mask-shifting is not allowed, except for the layers with LAYERMASKSHIFT keyword.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_flop                |                                                                                                                                                                                                                                                        |
|                        | <p>Returns true if the library cell is recognized as flip-flop/register type cell</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                        |
| is_genus_clock_gate    |                                                                                                                                                                                                                                                        |
|                        | <p>Set by the tool in the iSpatial flow to indicate that this integrated clock gating instance was created by Genus clock gating</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                     |
| is_hierarchical        |                                                                                                                                                                                                                                                        |



|                            |                                                                                                                                                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>is_hierarchical</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                         |
| is_inside_ilm              |                                                                                                                                                                                                                                             |
|                            | <p>This attribute denotes whether the inst is the child of a parent hinst the has an ILM specifiec</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                            |
| is_integrated_clock_gating |                                                                                                                                                                                                                                             |
|                            | <p>Returns true if this instance's library cell has the Liberty clock_gating_integrated_cell set to</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| is_interface_timing        |                                                                                                                                                                                                                                             |
|                            | <p>Returns a value of true if this instance's library cell has the Liberty interface_timing attribute true</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_inverter                |                                                                                                                                                                                                                                             |
|                            | <p>This inst is an inverter.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                  |
| is_isolation               |                                                                                                                                                                                                                                             |
|                            | <p>This inst is an isolation.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                 |
| is_isolation_cell          |                                                                                                                                                                                                                                             |

|                      |                                                                                                                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | Returns true if the associated library cell is recognized as isolation type cell<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                       |
| is_jtag              |                                                                                                                                                                                                                                     |
|                      | The inst is a JTAG element.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                       |
| is_latch             |                                                                                                                                                                                                                                     |
|                      | Returns true if the associated library cell is recognized as latch type cell<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                          |
| is_legal             |                                                                                                                                                                                                                                     |
|                      | Specifies the legal status that was seen by the last place_detail or check_place.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                     |
| is_level_shifter     |                                                                                                                                                                                                                                     |
|                      | Returns true if the associated library cell is recognized as level shifter type cell<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                   |
| is_macro             |                                                                                                                                                                                                                                     |
|                      | Returns a value of true if the associated library cell has the Liberty attribute is_macro_cell is true<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| is_master_slave_flop |                                                                                                                                                                                                                                     |

|                             |                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | Returns true if the Liberty clock_on_also attribute is specified for the associated library cell<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                |
| is_master_slave_lssd_flop   |                                                                                                                                                                                                                                                                                                                                               |
|                             | Returns true if this cell has been recognized as a master/slave LSSD device<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                     |
| is_memory                   |                                                                                                                                                                                                                                                                                                                                               |
|                             | Returns true or false depending on whether the associated library cell of the inst is recognized as a Liberty memory cell. Cells which include a Liberty memory group definition are recognized as memory cells.<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| is_msv                      |                                                                                                                                                                                                                                                                                                                                               |
|                             | Returns true if object is a multiple supply voltage cell<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                   |
| is_mux                      |                                                                                                                                                                                                                                                                                                                                               |
|                             | Returns true if object is a mux cell<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                       |
| is_negative_level_sensitive |                                                                                                                                                                                                                                                                                                                                               |
|                             | Returns a value of true if the associated library cell is identified as a negative level-sensitive cell.<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                         |
| is_pad                      |                                                                                                                                                                                                                                                                                                                                               |

|                             |                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>Returns a value of true if the associated library cell's Liberty is_pad attribute is set to true</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| is_physical                 |                                                                                                                                                                                                                                        |
|                             | <p>is physical only of inst</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                              |
| is_place_halo               |                                                                                                                                                                                                                                        |
|                             | <p>Indicates that the instance has a placement halo.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                 |
| is_positive_level_sensitive |                                                                                                                                                                                                                                        |
|                             | <p>Returns a value of true if the associated library cell is identified as a positive level-sensitive</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_power_switch             |                                                                                                                                                                                                                                        |
|                             | <p>Returns true if the associated library cell is recognized as power switch type cell</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                |
| is_retention                |                                                                                                                                                                                                                                        |
|                             | <p>Returns true if the associated library cell is recognized as retention type cell</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                   |
| is_rise_edge_triggered      |                                                                                                                                                                                                                                        |

|                         |                                                                                                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Returns a value of true if the instance is triggered by the rising edge of the clock.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                |
| is_sequential           |                                                                                                                                                                                                                                                                |
|                         | <p>Returns a value of true if the library cell is a latch or flip-flop, or if the cell has sequential time checks.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>            |
| is_spare                |                                                                                                                                                                                                                                                                |
|                         | <p>The inst is a spare instance. These are used by post-mask ECO flows.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                      |
| is_synchronizer         |                                                                                                                                                                                                                                                                |
|                         | <p>Instance is a synchronizer.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                                                                |
| is_synthesis_clock_gate |                                                                                                                                                                                                                                                                |
|                         | <p>Attribute stored on clock gate instances to indicate if the clock gate was added by the tool or manually.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                 |
| is_tristate             |                                                                                                                                                                                                                                                                |
|                         | <p>Returns a value of true if the associated library cell definition includes the Liberty attribute three_state set to true .</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_unmapped             |                                                                                                                                                                                                                                                                |

|                                     |                                                                                                                                                                                                                                            |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Returns true if object is not mapped in netlist</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                  |
| is_user_clock_gate                  |                                                                                                                                                                                                                                            |
|                                     | <p>To indicate user clock gate.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                           |
| late_cell_check_derate_factor       |                                                                                                                                                                                                                                            |
|                                     | <p>Returns the derating factor on timing check values for late paths (e.g Setup checks)</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| late_cell_check_mean_derate_factor  |                                                                                                                                                                                                                                            |
|                                     | <p>Returns mean derate factor for -late for check arcs</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                            |
| late_cell_check_sigma_derate_factor |                                                                                                                                                                                                                                            |
|                                     | <p>Returns sigma derate factor for -late for check arcs</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                           |
| late_clk_cell_derate_factor         |                                                                                                                                                                                                                                            |
|                                     | <p>Returns the derating factor for instances on late clock paths</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                        |
| late_clk_cell_mean_derate_factor    |                                                                                                                                                                                                                                            |

|                                            |                                                                                                                                                                             |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | Returns mean derate factor for -late -clock<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No            |
| late_clk_cell_sigma_derate_factor          |                                                                                                                                                                             |
|                                            | Returns sigma derate factor for -late -clock<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No           |
| late_clock_check_incremental_derate_factor |                                                                                                                                                                             |
|                                            | Returns incremental derate factor for -late -clock<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No     |
| late_data_cell_derate_factor               |                                                                                                                                                                             |
|                                            | Returns the derating factor for instances on late data paths<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| late_data_cell_mean_derate_factor          |                                                                                                                                                                             |
|                                            | Returns mean derate factor for -late -data<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No             |
| late_data_cell_sigma_derate_factor         |                                                                                                                                                                             |
|                                            | Returns sigma derate factor for -late -data<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No            |
| late_data_check_incremental_derate_factor  |                                                                                                                                                                             |

|                                          |                                                                                                                                                                                                 |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | Returns incremental derate factor for -late -data<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                          |
| late_fall_cell_check_derate_factor       |                                                                                                                                                                                                 |
|                                          | Returns the late cell check derating factor for falling arrivals<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                 |
| late_fall_cell_check_mean_derate_factor  |                                                                                                                                                                                                 |
|                                          | Returns mean derate factor for -late -fall for check arcs<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                  |
| late_fall_cell_check_sigma_derate_factor |                                                                                                                                                                                                 |
|                                          | Returns sigma derate factor for -late -fall for check arcs<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                 |
| late_fall_clk_cell_derate_factor         |                                                                                                                                                                                                 |
|                                          | Returns the late derating factor for falling delays through clock tree instances<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| late_fall_clk_cell_mean_derate_factor    |                                                                                                                                                                                                 |
|                                          | Returns mean derate factor for -late -fall -clock<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                          |
| late_fall_clk_cell_sigma_derate_factor   |                                                                                                                                                                                                 |



|                                           |                                                                                                                                                                                                                       |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>Returns sigma derate factor for -late -fall -clock</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                        |
| late_fall_clock_incremental_derate_factor |                                                                                                                                                                                                                       |
|                                           | <p>Returns incremental derate factor for -late -fall -clock</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                  |
| late_fall_data_cell_derate_factor         |                                                                                                                                                                                                                       |
|                                           | <p>Returns the late derating factor for falling delays through data path instances</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| late_fall_data_cell_mean_derate_factor    |                                                                                                                                                                                                                       |
|                                           | <p>Returns mean derate factor for -late -fall -data</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                          |
| late_fall_data_cell_sigma_derate_factor   |                                                                                                                                                                                                                       |
|                                           | <p>Returns sigma derate factor for -late -fall -data</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                         |
| late_fall_data_incremental_derate_factor  |                                                                                                                                                                                                                       |
|                                           | <p>Returns incremental derate factor for -late -fall -data</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                   |
| late_rise_cell_check_derate_factor        |                                                                                                                                                                                                                       |

|                                           |                                                                                                                                                                                              |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | Returns the late cell check derating factor for rising arrivals<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No               |
| late_rise_cell_check_mean_derate_factor   |                                                                                                                                                                                              |
|                                           | Returns mean derate factor for -late -rise for check arcs<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No               |
| late_rise_cell_check_sigma_derate_factor  |                                                                                                                                                                                              |
|                                           | Returns sigma derate factor for -late -rise for check arcs<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No              |
| late_rise_clk_cell_derate_factor          |                                                                                                                                                                                              |
|                                           | Returns the late derating factor for rise delays through clock tree instances<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| late_rise_clk_cell_mean_derate_factor     |                                                                                                                                                                                              |
|                                           | Returns mean derate factor for -late -rise -clock<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                       |
| late_rise_clk_cell_sigma_derate_factor    |                                                                                                                                                                                              |
|                                           | Returns sigma derate factor for -late -rise -clock<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                      |
| late_rise_clock_incremental_derate_factor |                                                                                                                                                                                              |

|                                          |                                                                                                                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | Returns incremental derate factor for -late -rise -clock<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                 |
| late_rise_data_cell_derate_factor        |                                                                                                                                                                                               |
|                                          | Returns the late derating factor for rising delays through data path instances<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| late_rise_data_cell_mean_derate_factor   |                                                                                                                                                                                               |
|                                          | Returns mean derate factor for -late -rise -data<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                         |
| late_rise_data_cell_sigma_derate_factor  |                                                                                                                                                                                               |
|                                          | Returns sigma derate factor for -late -rise -data<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                        |
| late_rise_data_incremental_derate_factor |                                                                                                                                                                                               |
|                                          | Returns incremental derate factor for -late -rise -data<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                  |
| Ide_distance_left                        |                                                                                                                                                                                               |
|                                          | The COPDE spacing to the left side of the inst.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                               |
| Ide_distance_right                       |                                                                                                                                                                                               |

|                      |                                                                                                                                                                                                                                                |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>The COPDE spacing to the right side of the inst.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                            |
| leakage_power        |                                                                                                                                                                                                                                                |
|                      | <p>Returns the leakage power of the instance from library cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                              |
| left_padding         |                                                                                                                                                                                                                                                |
|                      | <p>Value of left padding for the instance.</p> <p><b>Type:</b> int</p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                                                        |
| level_shifter_type   |                                                                                                                                                                                                                                                |
|                      | <p>Returns the Level Shifter type for associated library cell. The supported values are LH, HL &amp; HL_LH.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| lib_binding_info_max |                                                                                                                                                                                                                                                |
|                      | <p>Returns max voltage library binding information</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                    |
| lib_binding_info_min |                                                                                                                                                                                                                                                |
|                      | <p>Returns min voltage library binding information</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                    |
| lib_cell             |                                                                                                                                                                                                                                                |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Returns the bind library cell</p> <p><b>Type:</b> <a href="#">obj(lib_cell)</a>*</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                        |
| lib_cells                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                            | <p>Returns a list of library cell objects associated with this instance. You can use -index to refine list to lib_cells associated with a specific view</p> <p><b>Type:</b> <a href="#">obj(lib_cell)</a>*</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                 |
| litho_halo                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                            | <p>A litho_halo on an inst forces parallelrouting away from the block boundary to meet lithographic DRC rules on a few routing layers but allows perpendicular access to pins. It is only allowed if LITHOMACROHALO values exists for some routing layers. See the LEF manual for figures and more details of this DRC rule. It cannot be added to standard cells (cells with a CLASS COF SITE).</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> |
| location                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                            | <p>The location of the inst. It is always the lower-left corner of the inst bounding-box, independent of the inst orientation.</p> <p><b>Type:</b> <a href="#">point</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                     |
| lp_dynamic_analysis_scope  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                            | <p>Set VCD analysis scope attribute on an instance in the design hierarchy. Valid values for the attribute are true or false.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                    |
| map_to_multibit_bank_label |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Defines a bank label for a sequential instance. Instances with the same label can be considered multibit merging.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> default_bank</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                    |
| map_to_multibit_register          |                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                   | <p>Enables predefined multibit cell merging for the sequential instance and limits the multibit register of this instance to the specified multibit library cells. Since this attribute enables predefined cell inferencing, it implies a forced type of mapping and is therefore not QoR-driven.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| mask_shift                        |                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                   | <p>Digit encoded value indicates the mask shifting for the instance contents (0 = unshifted, for shift cases refer to the DEF COMP + MASKSHIFT documentation).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                    |
| merge_combinational_hier_instance |                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                   | <p>Allow the combinational hierarchical instance to be merged.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                           |
| merge_multibit                    |                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                   | <p>Infer multibit cell for this instance.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                        |
| multibit_allow_async_phase_map    |                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                   | <p>Allow interchanging async pins for multibit mapping.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                  |
| multibit_aware_seq_mapping        |                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | Mapper prefers flops which have corresponding multibit variants.<br><b>Type:</b> <code>enum</code><br><b>Enum Values:</b> 1 0 true false auto<br><b>Default:</b> auto<br><b>Edit:</b> Yes                                                                                                                                                                                                                                                                                        |
| multibit_flop_rejection_reason |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                | Reports the reason why the instance could not be merged into a multibit cell.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                                                                                                                                                                                                      |
| multibit_rejection_reason      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                | Reports rejection reason for not creating multibit.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                                                                                                                                                 |
| multibit_rejection_reason_id   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                | Reports the genus rejection reason category why the instance could not be merged into a r<br>cell in map.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                                                                                                                                                                          |
| name                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                | The same as .escaped_name without any \ escape chars. This name is commonly used to<br>problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC<br>commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 s<br>flattening hierarchical names (e.g. with ungroup) does not require changing the names in T<br>scripts.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> No |
| num_exceptions                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                | Reports the number of path exceptions<br><b>Type:</b> <code>int</code><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                                                                                                             |
| obj_type                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                |                                                                                                                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">inst</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| optimize_constant_0_seq        |                                                                                                                                                                                                                         |
|                                | <p>Allow constant 0 propagation through sequential instances.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p> |
| optimize_constant_1_seq        |                                                                                                                                                                                                                         |
|                                | <p>Allow constant 1 propagation through sequential instances.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p> |
| optimize_constant_feedback_seq |                                                                                                                                                                                                                         |
|                                | <p>Allow constant propagation through feedback sequential instances.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                   |
| optimize_merge_seq             |                                                                                                                                                                                                                         |
|                                | <p>Allow merging of equivalent sequential instances.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>          |
| orient                         |                                                                                                                                                                                                                         |
|                                | <p>Instance placement orientation.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> r0 r90 r180 r270 mx mx90 my my90</p> <p><b>Default:</b> r0</p> <p><b>Edit:</b> Yes</p>                           |
| original_name                  |                                                                                                                                                                                                                         |



|                          |                                                                                                                                                                                                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | Original rtl names of single-bit flops.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                          |
| original_names_for_split |                                                                                                                                                                                                                                                   |
|                          | Original names of single-bit flops merged.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                       |
| overlap_rects            |                                                                                                                                                                                                                                                   |
|                          | List of rectangles that define the shape of instance<br><b>Type:</b> <a href="#">rect*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                               |
| parent                   |                                                                                                                                                                                                                                                   |
|                          | The parent of the inst, which is either a hinst for an instance within hierarchy or the design c<br>an instance at the top level.<br><b>Type:</b> <a href="#">obj(design)</a> <a href="#">obj(hinst)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| parent_cell              |                                                                                                                                                                                                                                                   |
|                          | Returns the enclosing parent hinst<br><b>Type:</b> <a href="#">obj(hinst)*</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                      |
| partition                |                                                                                                                                                                                                                                                   |
|                          | If this inst is a physical black_box, this partition will carry the pin constraints and related dat<br>blackbox. Otherwise it is empty.<br><b>Type:</b> <a href="#">obj(partition)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                   |
| peak_current             |                                                                                                                                                                                                                                                   |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>Peak current value of the inst</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                             |
| pg_pins           |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                   | <p>List of pg_pins for this inst. These are declared as power or ground pins in the .lib or LEF/O library, and separated from the signal pins of the inst.</p> <p><b>Type:</b> <a href="#">obj(pg_pin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                              |
| pin_count         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                   | <p>pin_count</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                            |
| pins              |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                   | <p>The pins of this inst.</p> <p><b>Type:</b> <a href="#">obj(pin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                  |
| place_halo_bbox   |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                   | <p>Bounding box of the inst placement halo. For a rectilinear block this might include area that covered by the place halo.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                     |
| place_halo_bottom |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                   | <p>Specifies an extra halo of space along the bottom edges of the inst that should not be used placement. This area can still be used later during optimization and clock tree creation to add repeaters. It can only be used on blocks and cannot be added to standard-cells. See the place attributes on the base_cell to add extra space to standard-cells.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| place_halo_left   |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>Specifies an extra halo of space along the left edges of the inst that should not be used during placement. This area can still be used later during optimization and clock tree creation to add repeaters. It can only be used on blocks and cannot be added to standard-cells. See the place attributes on the base_cell to add extra space to standard-cells.</p> <p><b>Type:</b> coord<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>  |
| place_halo_polygon |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | <p>Specifies the polygon shape of the inst placement halo. When the inst is unplaced, it preter inst is placed at coordinate origin and use it's base_cell to calculate polygon. For Example, if base_cell's size is X by Y, then the polygon of the unplaced inst is {0 0 . . . .}. Based on that place_halo_polygon is calculated.</p> <p><b>Type:</b> point<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                 |
| place_halo_right   |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | <p>Specifies an extra halo of space along the right edges of the inst that should not be used during placement. This area can still be used later during optimization and clock tree creation to add repeaters. It can only be used on blocks and cannot be added to standard-cells. See the place attributes on the base_cell to add extra space to standard-cells.</p> <p><b>Type:</b> coord<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| place_halo_top     |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | <p>Specifies an extra halo of space along the top edges of the inst that should not be used during placement. This area can still be used later during optimization and clock tree creation to add repeaters. It can only be used on blocks and cannot be added to standard-cells. See the place attributes on the base_cell to add extra space to standard-cells.</p> <p><b>Type:</b> coord<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>   |
| place_status       |                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>This attribute is the placement status of an instance during placement and optimization. The will look at both place_status and place_status_cts and use the more restrictive value. When attribute is explicitly set by the user, it will reset the place_status_cts attribute so the user in precedence.</p> <p><b>Supported values:</b><br/> unplaced: Unconstrained<br/> placed: Is placed but can be moved<br/> soft_fixed: Is fixed, but can move a short distance to legalize it<br/> fixed: Is fixed and can only be moved by the user, will not be moved by placement or optimization commands.<br/> cover: Is fixed and cannot be moved by any commands, you must change the place_status to move it.</p> <p><b>Type:</b> enum<br/> <b>Enum Values:</b> unplaced placed fixed cover soft_fixed<br/> <b>Default:</b> unplaced<br/> <b>Edit:</b> Yes</p> |
| place_status_cts       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                        | <p>This attribute is the CTS placement status of an instance during placement and optimization. The placer will look at both place_status and place_status_cts and use the more restrictive value.</p> <p><b>Supported values:</b><br/> unset: Unconstrained<br/> soft_fixed: For future use<br/> fixed: Cannot be moved</p> <p><b>Type:</b> enum<br/> <b>Enum Values:</b> unset fixed soft_fixed<br/> <b>Default:</b> unset<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                              |
| place_status_effective |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                        | <p>This attribute is the effective placement status for the instance. It is the worst case of the instance's place_status, instance place_status_cts, and hinst parent read only attribute (closest hinst that is not set to none). See place_status for the description of possible values.</p> <p><b>Type:</b> enum<br/> <b>Enum Values:</b> unplaced placed fixed cover soft_fixed<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| pmbist_instruction_set |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                        | <p>Attribute to define different programmable MBIST instructions for the different instances of the block on which programmable MBIST has been inserted.</p> <p><b>Type:</b> string<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                        |                                                                                                                                                                                                                   |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_domain           |                                                                                                                                                                                                                   |
|                        | <p>The power_domain for the inst.</p> <p><b>Type:</b> obj(power_domain)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                      |
| power_dynamic          |                                                                                                                                                                                                                   |
|                        | <p>The switching + internal power of this inst computed by report_power.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                          |
| power_frequency_domain |                                                                                                                                                                                                                   |
|                        | <p>signifies Frequency of the inst with which it is constrained, sdc/twf file has clock frequency r information for the instances</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| power_internal         |                                                                                                                                                                                                                   |
|                        | <p>The internal power of this inst computed by report_power.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                      |
| power_internal_density |                                                                                                                                                                                                                   |
|                        | <p>Internal power density of the inst which is inst power internal / inst area</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                    |
| power_leakage          |                                                                                                                                                                                                                   |
|                        | <p>The leakage power of this inst computed by report_power.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                       |
| power_leakage_density  |                                                                                                                                                                                                                   |

|                           |                                                                                                                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | Leakage power density of the inst which is inst power leakage / inst area<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                       |
| power_loading_capacitance |                                                                                                                                                                                                                                 |
|                           | Loading capacitance of the inst, it signifies the output net cap + pin cap of the next instance connecting to.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                  |
| power_ref_clock           |                                                                                                                                                                                                                                 |
|                           | The reference clock for toggle and activity calculations. By default this is the fastest clock for clock domains but can be modified by some power commands.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No    |
| power_switch_type         |                                                                                                                                                                                                                                 |
|                           | Returns the switch cell type associated with library cell. The supported values are coarse_grain and fine_grain.<br><b>Type:</b> string<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| power_switching           |                                                                                                                                                                                                                                 |
|                           | The switching power of the nets driven by this inst computed by report_power.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                   |
| power_switching_density   |                                                                                                                                                                                                                                 |
|                           | Switching power density of the inst which is power switching / inst area<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                        |
| power_toggle_rate         |                                                                                                                                                                                                                                 |

|                                      |                                                                                                                                                                                                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>The average number of toggles read from VCD,TCF,SAIF, etc. or from propagation that occurs second on all the pins of this inst. no_value is returned if it cannot be computed.</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| power_total                          |                                                                                                                                                                                                                                                             |
|                                      | <p>The switching + internal + leakage power of this inst computed by report_power.</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                |
| power_total_density                  |                                                                                                                                                                                                                                                             |
|                                      | <p>Total power density of the inst which is inst power total / inst area</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                          |
| propagate_constant_from_timing_model |                                                                                                                                                                                                                                                             |
|                                      | <p>Allow constant propagation from specific timing model instance.</p> <p><b>Type:</b> enum<br/><b>Enum Values:</b> 1 0 true false inherited<br/><b>Default:</b> inherited<br/><b>Edit:</b> Yes</p>                                                         |
| rail_domain_voltage_drop             |                                                                                                                                                                                                                                                             |
|                                      | <p>Worst Effective drop (VDD-VSS) of the inst in elapse or timing/switching window. Values are loaded in GUI after user loads the voltus state directory</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                          |
| rail_ground_voltage_drop             |                                                                                                                                                                                                                                                             |
|                                      | <p>Ground bounce for ground rail of the inst, Values are loaded in GUI after user loads the voltus state directory</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                |
| rail_power_voltage_drop              |                                                                                                                                                                                                                                                             |

|                           |                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Voltage drop for power rail of the inst, Values are loaded in GUI after user loads the voltus state directory</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                         |
| rail_reff                 |                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p>signifies equivalent resistance for the instance pin from the voltage source, values are loaded in GUI after user load the effective resistance state directory.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                      |
| rail_rlrp                 |                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p>signifies least resistance path value of the inst from a voltage source to a pin of the instance, values are loaded in GUI after user load the voltus state directory. It is enabled when enable_rlrp option of set_rail_analysis_config is turned on</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| ref_lib_cell_name         |                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p>ref_lib_cell_name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                            |
| ref_name                  |                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p>ref_name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                     |
| retention_cell            |                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p>Returns the retention_cell derived from Liberty</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                    |
| retime_original_registers |                                                                                                                                                                                                                                                                                                                                                          |



|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | Original registers before retiming.<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| right_padding           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                         | Value of right padding for the instance.<br><b>Type:</b> <code>int</code><br><b>Default:</b> 0<br><b>Edit:</b> Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| route_halo_bottom       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                         | Specifies the size of a routing halo around the inst on the bottom side. A value of 0 means there is no route_halo. It is used to keep routes away from the block edges to reduce cross-coupling of wires inside the block. A route_halo can only be added to blocks and is not allowed on standard cells. The router will only route through the halo to reach pins on the boundary of the block. The value applies to all sides of the block and only positive values are allowed.<br><b>Type:</b> <code>coord</code><br><b>Default:</b> no_value<br><b>Edit:</b> Yes |
| route_halo_bottom_layer |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                         | The bottom layer for the routing halo.<br><b>Type:</b> <code>obj(layer)</code><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| route_halo_left         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                         | Specifies the size of a routing halo around the inst on the left side. A value of 0 means there is no route_halo. It is used to keep routes away from the block edges to reduce cross-coupling of wires inside the block. A route_halo can only be added to blocks and is not allowed on standard cells. The router will only route through the halo to reach pins on the boundary of the block. The value applies to all sides of the block and only positive values are allowed.<br><b>Type:</b> <code>coord</code><br><b>Default:</b> no_value<br><b>Edit:</b> Yes   |
| route_halo_polygon      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                         | Specifies the polygon shape of the inst routing halo.<br><b>Type:</b> <code>point</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_halo_right      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                       | <p>Specifies the size of a routing halo around the inst on the right side. A value of 0 means the route_halo. It is used to keep routes away from the block edges to reduce cross-coupling to inside the block. A route_halo can only be added to blocks and is not allowed on standard-. The router will only route through the halo to reach pins on the boundary of the block. The v applies to all sides of the block and only positive values are allowed.</p> <p><b>Type:</b> coord<br/> <b>Default:</b> no_value<br/> <b>Edit:</b> Yes</p> |
| route_halo_top        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                       | <p>Specifies the size of a routing halo around the inst on the top side. A value of 0 means there route_halo. It is used to keep routes away from the block edges to reduce cross-coupling to inside the block. A route_halo can only be added to blocks and is not allowed on standard-. The router will only route through the halo to reach pins on the boundary of the block. The v applies to all sides of the block and only positive values are allowed.</p> <p><b>Type:</b> coord<br/> <b>Default:</b> no_value<br/> <b>Edit:</b> Yes</p> |
| route_halo_top_layer  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                       | <p>The top layer of the routing halo.</p> <p><b>Type:</b> obj(layer)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                            |
| rtls_hdl_instantiated |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                       | <p>Returns true if the instance was present in the user input.</p> <p><b>Type:</b> bool<br/> <b>Default:</b> false<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                       |
| safety_dcls_cell_type |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>The type of cell added to support the implementation of a dcls (dual core lock step) safety mechanism.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none isolate_split_net_group_input isolate_split_net_group_output isolate_split_network_start isolate_split_network_branch isolate_split_network_group_inp common_cause_failure isolate_split_network_last_combo isolate_split_network_stop</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p> |
| safety_dcls_isolate_groups               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                          | <p>The dcls (dual core lock step) groups isolated by this inst.</p> <p><b>Type:</b> <a href="#">obj(group)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                             |
| safety_dcls_isolate_is_group_input_cell  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                          | <p>If true, then this inst is an isolation cell on an input net for a dcls (dual core lock step) group</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                          |
| safety_dcls_isolate_is_group_output_cell |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                          | <p>If true, then this inst is an isolation cell on an output net for a dcls (dual core lock step) group</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                         |
| safety_dcls_isolate_signal_type          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                          | <p>The type of signal isolated by this inst for a dcls (dual core lock step) group.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none clock reset scan_enable pin input output</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                     |
| safety_dmr_cell_type                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                       |                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>The type of cell added to support the implementation of a dmr (dual modular redundancy) s mechanism.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none isolate_split_net isolate_split_network_start isolate_split_network_br isolate_split_network_last_combo</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p> |
| safety_dmr_parent     |                                                                                                                                                                                                                                                                                                                                                                                 |
|                       | <p>The dmr (dual modular redundancy) parent-flop of this object.</p> <p><b>Type:</b> <a href="#">obj(pin)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                              |
| safety_failure_mode   |                                                                                                                                                                                                                                                                                                                                                                                 |
|                       | <p>The name of the failure-mode to which this inst belongs, as defined in the USF (Unified Saf Format) file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                    |
| safety_mechanism      |                                                                                                                                                                                                                                                                                                                                                                                 |
|                       | <p>The name of the safety-mechanism applied to this inst, as defined in the USF (Unified Safe Format) file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                     |
| safety_mechanism_type |                                                                                                                                                                                                                                                                                                                                                                                 |
|                       | <p>The type of safety-mechanism applied to this inst, as defined in the USF (Unified Safety For</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                 |
| safety_tmr_cell_type  |                                                                                                                                                                                                                                                                                                                                                                                 |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>The type of cell added to support the implementation of a tmr (triple modular redundancy) s mechanism.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none isolate_split_net isolate_split_network_start isolate_split_network_br isolate_split_network_last_combo</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p> |
| safety_tmr_clones                |                                                                                                                                                                                                                                                                                                                                                                                   |
|                                  | <p>The tmr (triple modular redundancy) clone flops of this parent flop. This attribute is only stor tmr parent-flops.</p> <p><b>Type:</b> <a href="#">obj(inst)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                         |
| safety_tmr_errors                |                                                                                                                                                                                                                                                                                                                                                                                   |
|                                  | <p>The tmr (triple modular redundancy) error-detection-logic of this parent-flop-bit.</p> <p><b>Type:</b> <a href="#">obj(inst)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                         |
| safety_tmr_parent                |                                                                                                                                                                                                                                                                                                                                                                                   |
|                                  | <p>The tmr (triple modular redundancy) parent flop of this inst. The tmr safety-mechanism is ini applied to the parent flop, adding clone flops and voting logic.</p> <p><b>Type:</b> <a href="#">obj(inst)</a> <a href="#">obj(pin)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                     |
| safety_tmr_voters                |                                                                                                                                                                                                                                                                                                                                                                                   |
|                                  | <p>The tmr (triple modular redundancy) voting logic of this parent flop. This attribute is only sto tmr parent-flops.</p> <p><b>Type:</b> <a href="#">obj(inst)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                         |
| safety_tmr_well_tap_left_padding |                                                                                                                                                                                                                                                                                                                                                                                   |
|                                  | <p>The padding used to reserve space for well-tap cells to be placed next to the left edges of tr modular redundancy) flops</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> No</p>                                                                                                                                                           |

|                                   |                                                                                                                                                                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| safety_tmr_well_tap_right_padding |                                                                                                                                                                                                                                                                           |
|                                   | <p>The padding used to reserve space for well-tap cells to be placed next to the right edges of (triple modular redundancy) flops.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> No</p>                                                |
| sdp                               |                                                                                                                                                                                                                                                                           |
|                                   | <p>The parent sdp group which the instance belongs to.</p> <p><b>Type:</b> <a href="#">obj(sdp)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                      |
| single_bit_original_name          |                                                                                                                                                                                                                                                                           |
|                                   | <p>Stores the hdl name of the original single bits.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                          |
| std_cell_main_rail_name           |                                                                                                                                                                                                                                                                           |
|                                   | <p>Returns the rail name associated with pg pin for which std_cell_main_rail is enabled.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                     |
| timing_model_type                 |                                                                                                                                                                                                                                                                           |
|                                   | <p>Returns the Liberty model type for a given cell or instance. The supported values are abstract, extracted, and qtm.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| top_padding                       |                                                                                                                                                                                                                                                                           |
|                                   | <p>Value of top padding for the instance.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                        |
| trace_retime                      |                                                                                                                                                                                                                                                                           |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | Trace this instance through retiming, use the attribute 'retime_original_registers' to get the original instance.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                                                                                                                                                                 |
| transition_density           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                              | Transition density of the inst<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                                                                                                                                                                      |
| unused_bit_count_in_multibit |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                              | Tells the number of unused bits in multibit cell.<br><b>Type:</b> int<br><b>Default:</b> 0<br><b>Edit:</b> Yes                                                                                                                                                                                                                                                                                                                                                      |
| use_cells_only               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                              | This attribute is the list of allowable cells that this instance can be resized to. Wildcards are supported. If set, any parent (hinst) .dont_use_cells_effective values are ignored. This attrit applies to operations on mapped designs. The .dont_touch_effective values size_same_height and size_same_footprint_ok will filter this list further based on height and footprint, respectively.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| x_coordinate_max             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                              | x_coordinate_max<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                                                                                                                               |
| x_coordinate_min             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                              | x_coordinate_min<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                                                                                                                               |
| y_coordinate_max             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                  |                                                                                                                                                       |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | y_coordinate_max<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| y_coordinate_min |                                                                                                                                                       |
|                  | y_coordinate_min<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |

## inst\_obs

### Parent Objects

### Definition

obstruction shape on an instance

| Attribute | Description                                                                                                                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| obj_type  |                                                                                                                                                                                                         |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">inst_obs</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

## inst\_obs\_shape

### Parent Objects



## Definition

This corresponds to one of the .base\_cell.obs\_layer\_shapes.shapes (LEF OBS shapes) for a given inst. It carries a link to the 'inst' object, that a 'shape' object does not have. This allows GUI operations that need both the shape and the inst object together to select, highlight or query the shape. It can only be accessed by selecting the object with the GUI.

| Attribute        | Description                                                                                                                                                                                                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| inst             |                                                                                                                                                                                                                                                                            |
|                  | <p>The inst of this OBS shape<br/> <b>Type:</b> <a href="#">obj(inst)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                 |
| is_ignore_pg_net |                                                                                                                                                                                                                                                                            |
|                  | <p>Indicates that Power/Ground routing is ignored when checking for DRC violations (including shorts) involving the current shape (equivalent to LEF MACRO OBS LAYER EXCEPTPGNET).<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| layer            |                                                                                                                                                                                                                                                                            |
|                  | <p>The layer of the obstruction<br/> <b>Type:</b> <a href="#">obj(layer)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                              |
| mask             |                                                                                                                                                                                                                                                                            |
|                  | <p>Indicates mask number for multiple mask layer usage. Refer to layer's .num_mask attribute for legal range, 0 indicates uncolored.<br/> <b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                |
| obj_type         |                                                                                                                                                                                                                                                                            |
|                  | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br/> <b>Type:</b> <a href="#">enum (inst_obs_shape)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                   |

|         |                                                                                                                                                                                                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| polygon |                                                                                                                                                                                                                           |
|         | <p>Points of the polygon (if type = polygon) in design coordinates (e.g. base_cell values are transformed for this inst).</p> <p><b>Type:</b> <a href="#">point</a>*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| rect    |                                                                                                                                                                                                                           |
|         | <p>Box of the shape (if type = rect) in design coordinates (e.g. base_cell values are transformed for this inst).</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| spacing |                                                                                                                                                                                                                           |
|         | <p>LEF OBS SPACING value, no_value if not specified in LEF.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                |
| type    |                                                                                                                                                                                                                           |
|         | <p>Type of shape (rect, polygon)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> rect polygon</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                    |

## io\_constraint

### Parent Objects

[design](#), [root](#)

### Definition

IO object for block (term) or chip design (inst) constraints

| Attribute | Description |
|-----------|-------------|
|-----------|-------------|

|              |                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| area         |                                                                                                                                                                                  |
|              | <p>Area of the instance as defined by the LEF MACRO SIZE or OVERLAP information</p> <p><b>Type:</b> <a href="#">area</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| bbox         |                                                                                                                                                                                  |
|              | <p>Bounding box of the inst overlap rects. This is only correct if type = inst.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>   |
| indent       |                                                                                                                                                                                  |
|              | <p>Indent of instance from the boundary of the design</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                      |
| inst         |                                                                                                                                                                                  |
|              | <p>The IO instance (null/0x0 for block design case)</p> <p><b>Type:</b> <a href="#">obj(inst)</a> <a href="#">obj(bump)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_assigned  |                                                                                                                                                                                  |
|              | <p>Indicates that the IO location has been set by the IO placer</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                |
| is_corner    |                                                                                                                                                                                  |
|              | <p>Indicates that the IO refers to a corner cell</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                               |
| is_gap_fixed |                                                                                                                                                                                  |
|              | <p>Indicates that the IO spacing constraint is applied</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                         |

|                 |                                                                                                                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_ground       |                                                                                                                                                                                                                               |
|                 | <p>Indicates that the IO is a Ground</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                            |
| is_offset_fixed |                                                                                                                                                                                                                               |
|                 | <p>Indicates that the IO offset constraint is applied</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                       |
| is_power        |                                                                                                                                                                                                                               |
|                 | <p>Indicates that the IO is a Power</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                             |
| name            |                                                                                                                                                                                                                               |
|                 | <p>IO name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                    |
| obj_type        |                                                                                                                                                                                                                               |
|                 | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">io_constraint</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| offset          |                                                                                                                                                                                                                               |
|                 | <p>IO relative location from left (North &amp; South) or bottom (East &amp; West) die edge</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                              |
| order           |                                                                                                                                                                                                                               |
|                 | <p>IO order per side. Order is left-to-right (North &amp; South) or bottom-to-top (East &amp; West)</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                             |

|         |                                                                                                                                                                                                                                                                    |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| row     |                                                                                                                                                                                                                                                                    |
|         | <p>IO row/ring number (0 is outermost IO row)</p> <p><b>Type:</b> <code>int</code></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                                                            |
| side    |                                                                                                                                                                                                                                                                    |
|         | <p>Side constraint of IO</p> <p><b>Type:</b> <code>enum</code></p> <p><b>Enum Values:</b> north west south east none</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                         |
| spacing |                                                                                                                                                                                                                                                                    |
|         | <p>Spacing between IO and previous IO (left or below)</p> <p><b>Type:</b> <code>coord</code></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                           |
| term    |                                                                                                                                                                                                                                                                    |
|         | <p>The IO terminal (only used when 'type = term')</p> <p><b>Type:</b> <code>obj(port)</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                 |
| type    |                                                                                                                                                                                                                                                                    |
|         | <p>Type of IO (endSpace indicates from last IO to corner, inst indicates real instance case, obs indicates obstruction between IOs, term indicates block design case)</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

# layer

## Parent Objects

marker, bump\_pin, layer\_shape, text, port, via\_def, pin\_blockage, ndr\_eol\_spacing, special\_wire, pg\_base\_pin, hpin, what\_if\_wire, inst\_physical\_pin, what\_if\_via, trim\_grid, base\_pin, partition, track\_pattern, design, tap\_layer\_distance, tech, pin\_guide, root, layer\_rule, virtual\_wire, port\_shape, via\_def\_rule, route\_blockage, patch\_wire, antenna\_data, inst, wire, bus\_guide, net, pin, interposer\_route\_rule, route\_type, bump\_grid, inst\_obs\_shape

## Definition

A layer from the LEF or OA technology file.

| Attribute          | Description                                                                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| antenna_models     |                                                                                                                                                                                                   |
|                    | The list of antenna model objects for the Oxides of the layer<br><b>Type:</b> <code>obj(antenna_model)*</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                           |
| area               |                                                                                                                                                                                                   |
|                    | Layer minimum area from LEF/OpenAccess. If AREA rule is not specified in LEF, the value of -1 in dbu will be returned.<br><b>Type:</b> <code>area</code><br><b>Default:</b> ""<br><b>Edit:</b> No |
| backside           |                                                                                                                                                                                                   |
|                    | Indicates that the layer is a backside (underside of the die) layer.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                   |
| backside_cut_index |                                                                                                                                                                                                   |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>An index for BACKSIDE cut layers. It is 0 for the TSV cut-layer, -1 for the BACKSIDE cut layer between the first (.backside_route_index = -1) and second (.backside_route_index = -2) backside routing layers below the substrate, -2 for the BACKSIDE cut layer between the second and third backside routing layers, etc. It is 1000 for all other layers.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                           |
| backside_route_index |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                      | <p>An index for BACKSIDE routing layers. It is -1 for the first backside routing layer closest (just below) the substrate (e.g. the last LEF TYPE ROUTING layer with BACKSIDE property in the technology file), -2 for the second backside routing layer below the substrate, etc. It is 1000 for any other layer.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                        |
| base_name            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                      | <p>Layer name from LEF/OpenAccess technology file definition.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                          |
| cut_index            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                      | <p>An index into the cut layers used by vias in the routing. It is 1 for the cut-layer above the first routing layer, 2 for the cut-layer above the second routing layer, etc. It is 0 for the cut-layer just below the first routing layer. The cut_index is -1 for any other layer, including LEF cut layers with TYPE MIMCAP, TSV or BACKSIDE. In OpenAccess, the cut layers are determined by the LEFDefaultRouteSpec validVias list.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| density_step_x       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                      | <p>Layer density window step from LEF/OpenAccess</p> <p><b>Type:</b> <a href="#">coord</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                        |
| density_step_y       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                     |                                                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | Layer density window step from LEF/OpenAccess<br><b>Type:</b> coord<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                 |
| density_window_x    |                                                                                                                                                                              |
|                     | Layer density window length from LEF/OpenAccess<br><b>Type:</b> coord<br><b>Default:</b> ""<br><b>Edit:</b> No                                                               |
| density_window_y    |                                                                                                                                                                              |
|                     | Layer density window width from LEF/OpenAccess<br><b>Type:</b> coord<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                |
| direction           |                                                                                                                                                                              |
|                     | Layer pref. direction from LEF/OpenAccess<br><b>Type:</b> enum<br><b>Enum Values:</b> horizontal vertical unassigned diag45 diag135<br><b>Default:</b> ""<br><b>Edit:</b> No |
| eol_track_rules     |                                                                                                                                                                              |
|                     | PITCH and TRACKDISTANCE are stored in LEF while offset is stored in floorplan PLE grid.<br><b>Type:</b> string*<br><b>Default:</b> ""<br><b>Edit:</b> No                     |
| fill_active_spacing |                                                                                                                                                                              |
|                     | Layer fill minimum spacing from LEF/OpenAccess<br><b>Type:</b> coord<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                |
| fill_gap_spacing    |                                                                                                                                                                              |
|                     | Layer fill to fill spacing from LEF/OpenAccess<br><b>Type:</b> coord<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                |



|                 |                                                                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_trim_virtual |                                                                                                                                                          |
|                 | <p>Indicates that the layer is a trim metal virtual layer.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| max_density     |                                                                                                                                                          |
|                 | <p>Layer fill maximum density from LEF/OpenAccess</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| max_width       |                                                                                                                                                          |
|                 | <p>Layer maximum wire width from LEF/OpenAccess</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| mfg_grid        |                                                                                                                                                          |
|                 | <p>Manufacturing grid</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                     |
| min_density     |                                                                                                                                                          |
|                 | <p>Layer fill minimum density from LEF/OpenAccess</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| min_spacing     |                                                                                                                                                          |
|                 | <p>Layer minimum spacing from LEF/OpenAccess</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>              |
| min_width       |                                                                                                                                                          |
|                 | <p>Layer minimum wire width from LEF/OpenAccess</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |

|           |                                                                                                                                                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name      |                                                                                                                                                                                                                       |
|           | <p>Layer name from LEF/OpenAccess technology file definition.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                         |
| num_masks |                                                                                                                                                                                                                       |
|           | <p>Indicates how many masks will be used for the layer (1 = single mask, 2 = double-patterning, 3 = triple-patterning)</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| obj_type  |                                                                                                                                                                                                                       |
|           | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">layer</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| offset_x  |                                                                                                                                                                                                                       |
|           | <p>Layer offset X from LEF/OpenAccess</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                  |
| offset_y  |                                                                                                                                                                                                                       |
|           | <p>Layer offset Y from LEF/OpenAccess</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                  |
| pitch_x   |                                                                                                                                                                                                                       |
|           | <p>Layer wire pitch X from LEF/OpenAccess</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                              |
| pitch_y   |                                                                                                                                                                                                                       |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>Layer wire pitch Y from LEF/OpenAccess</p> <p><b>Type:</b> coord</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| route_index    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                | <p>An index into the routing layers. It is 1 for the first LEF TYPE ROUTING layer in the technology file, 2 for the second routing layer, etc. It is 0 for the layer just below the first routing layer (e.g a poly layer or non-routing metal layer that may have pin shapes in some cells). The route_index is -1 for any non-routing layers, or LEF routing layers with LEF TYPE MIMCAP or TYPE BACKSIDE. In OpenAccess, the routing layers are determined by the LEFDefaultRouteSpec validLayers list.</p> <p><b>Type:</b> int</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                      |
| spacing_tables |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                | <p>A list of all the LEF spacing-table rules for this layer in the LEF SPACINGTABLE syntax. All SPACINGTABLE and LEF58_SPACINGTABLE property values are returned.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| tech           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                | <p>The tech for the layer.</p> <p><b>Type:</b> obj(tech)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| type           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                | <p>The layer type (routing, cut, etc.) from the LEF LAYER TYPE statements or OA tech for this layer.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> invalid masterslice cut overlap routing implant tsv mimcap passivation poly_routing nwell pwell stacked_die cut_mimcap above_die_edge below_die_edge diffusion ignore trim_poly trim_metal region cut_region routing_region trim_metal_region middle_eol high_resistance_routing high_resistance_cut tsv_metal pad_metal stacked_mimcap well_distance well_distance_boundary_tap pad_metal_cut special_cut copde rc_blockage trim_diffusion abut_logic abut_filler cut_pg_rail probe substract</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| width          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>Layer wire width from LEF/OpenAccess</p> <p><b>Type:</b> coord</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                              |
| wrong_way_min_width                   |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                       | <p>min width in the non-preferred direction (from LEF LAYER MINWIDTH WRONGDIRECTION). A value of 0 indicates that there is no special value assigned for a wrong direction behavior.</p> <p><b>Type:</b> coord</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                 |
| wrong_way_spacing                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                       | <p>min spacing in the non-preferred direction (from LEF LAYER SPACING WRONGDIRECTION). A value of 0 indicates that there is no special value assigned for a wrong direction behavior.</p> <p><b>Type:</b> coord</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                |
| wrong_way_width                       |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                       | <p>min width in the non-preferred direction (from LEF LAYER WIDTH WRONGDIRECTION). A value of 0 indicates that there is no special value assigned for a wrong direction behavior.</p> <p><b>Type:</b> coord</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                    |
| wsp_oa_width_spacing_pattern          |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                       | <p>A list of Tcl dict style parameters that match the OA widthSpacingPattern parameters like this: {name &lt;string&gt; is_from_lib &lt;bool&gt; offset &lt;coord&gt;}. The 'name' and 'offset' match the corresponding OA parameters, and 'is_from_lib &lt;bool&gt;' is true if it is from the OA tech graph and false if it is from the cellview.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| wsp_oa_width_spacing_snap_pattern_def |                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>A list of Tcl dict style parameters that match the OA widthSpacingSnapPatternDef parameters like this: {name &lt;string&gt; is_from_lib &lt;bool&gt; offset &lt;coord&gt; offset_reference &lt;string&gt; wire_type &lt;name&gt; purpose &lt;name&gt;}. The 'name', 'offset', 'wire_type', and 'purpose' match the corresponding OA parameters. 'offset_reference' indicates whether the 'offset' is from the lower-left corner of the boundary or the origin. 'is_from_lib &lt;bool&gt;' is true if it is from the OA tech graph and false if it is from the cellview.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| wsp_offset        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                   | <p>Offset from lower-left corner of the core box to the first track. For a horizontal routing layer track, this is a Y offset.</p> <p><b>Type:</b> <a href="#">coord</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| wsp_pattern       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                   | <p>This is set by add_tracks or by reading an OA design with widthSpacingPatterns defined. It is a list of {width pitch repeat} values that created the tracks. For example, {1.5 2.5 1} {1.0 2.0 3} means a track of 1.5 width, 2.5 track-to-track pitch repeated 1 time, and then a track with width 1.0, track-to-track pitch 2.0 repeated 3 times, then the full pattern repeats. If this is not an SADP layer, there is no width assigned to the track, and a width value of 0.0 is returned. For a horizontal routing layer, the width and pitch are Y values.</p> <p><b>Type:</b> <a href="#">string*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>      |
| wsp_pattern_masks |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                   | <p>A list of mask values for each track in the wsp_pattern after the pattern repeat sections are expanded.</p> <p><b>Type:</b> <a href="#">int*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## layer\_rule

### Parent Objects

[route\\_rule](#)

## Definition

### Layer Rule

| Attribute | Description                                                                                                                                                                            |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| layer     |                                                                                                                                                                                        |
|           | The layer of layer rule<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                            |
| obj_type  |                                                                                                                                                                                        |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum(layer_rule)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| spacing   |                                                                                                                                                                                        |
|           | min spacing (from NONDEFAULTRULE LAYER SPACING, LEF LAYER SPACING or SPACINGTABLE)<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No                      |
| width     |                                                                                                                                                                                        |
|           | wire width (from NONDEFAULTRULE LAYER WIDTH or LEF LAYER WIDTH)<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                         |

## layer\_shape

### Parent Objects

[base\\_cell](#), [physical\\_pin](#)

## Definition

## layer\_shape

| Attribute        | Description                                                                                                                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_ignore_pg_net |                                                                                                                                                                                                                                                                                                        |
|                  | <p>Indicates that Power/Ground routing is ignored when checking for DRC violations (including shorts) involving the current shape (equivalent to LEF MACRO OBS LAYER EXCEPTPGNET)}</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                           |
| layer            |                                                                                                                                                                                                                                                                                                        |
|                  | <p>The layer of blockage</p> <p><b>Type:</b> <a href="#">obj(layer)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                               |
| obj_type         |                                                                                                                                                                                                                                                                                                        |
|                  | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum (layer_shape)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                |
| obs_type         |                                                                                                                                                                                                                                                                                                        |
|                  | <p>The type of this LEF OBS shape corresponding to the LEF MACRO OBS REAL, ABSTRACT, NOROUTE keywords. The type should always be real for any LEF PIN shapes.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> abstract real noroute<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| shapes           |                                                                                                                                                                                                                                                                                                        |
|                  | <p>List of shapes that define the blockage area</p> <p><b>Type:</b> <a href="#">obj(shape)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                       |
| spacing          |                                                                                                                                                                                                                                                                                                        |

|                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LEF OBS SPACING equivalent min_spacing value, no_value if not specified in LEF.<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|

## lef\_set

### Parent Objects

### Definition

A lef set is a group of lef files used by design(s).

| Attribute | Description                                                                                                                                                                                            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| obj_type  |                                                                                                                                                                                                        |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">lef_set</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

## lib\_arc

### Parent Objects

[lib\\_cell](#), [arc](#), [root](#), [lib\\_pin](#)

### Definition

cte library timing arc

| Attribute   | Description |
|-------------|-------------|
| aocv_weight |             |



|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <p>Returns the AOCV stage weight for this library arc. By default, all library cells and library arcs have default stage weight of 1.0. The aocv_weight property is specified as a user-defined library attribute in the Liberty timing library explicitly - or, by asserting it via command. You can use -index to return a value for a specific view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| base_name     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|               | <p>Returns the simple name for this library timing arc in the form: inputPin_outputPin_uniqueIntegerSuffix</p> <p><b>Type:</b> string<br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                              |
| from_lib_pin  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|               | <p>Returns a lib_pin object for beginning pin of this timing arc. You can use -index to return a value for a specific view.</p> <p><b>Type:</b> obj(lib_pin)*<br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                      |
| full_name     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|               | <p>Returns the unique name for this library arc in the form: librarySet/libraryName/libraryCell/inputPin_outputPin_uniqueIntegerSuffix. You can use -index to return a value for a specific view.</p> <p><b>Type:</b> string<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                        |
| has_bti_aging |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|               | <p>Returns true if library arc(s) has BTI aging information</p> <p><b>Type:</b> bool<br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                               |
| has_ccs_noise |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|               |                                                                                                                                                                                                                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <p>Returns true if cell has ccs noise data.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                             |
| has_hci_aging |                                                                                                                                                                                                                                                                                                                                |
|               | <p>Returns true if library arc(s) has HCI aging information</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                   |
| has_mis_fall  |                                                                                                                                                                                                                                                                                                                                |
|               | <p>Return library attribute has_mis for fall output transition.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                               |
| has_mis_rise  |                                                                                                                                                                                                                                                                                                                                |
|               | <p>Return library attribute has_mis for rise output transition.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                               |
| has_socv      |                                                                                                                                                                                                                                                                                                                                |
|               | <p>Returns true if library arc(s) has SOCV information</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                        |
| is_disabled   |                                                                                                                                                                                                                                                                                                                                |
|               | <p>Returns a value of true if this library timing arc has been explicitly disabled by the user via a set_disable_timing constraint. You can use -index to return a value for a specific view.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

|                          |                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lib_cell                 |                                                                                                                                                                                                                                                                                                                                  |
|                          | <p>Returns the parent library cell object for this library arc. You can use -index to return a value for a specific view.</p> <p><b>Type:</b> <a href="#">obj(lib_cell)</a>*</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                             |
| mode                     |                                                                                                                                                                                                                                                                                                                                  |
|                          | <p>If this timing arc is defined as part of a Liberty mode group, this attribute will return the name of the group. You can use -index to return a value for a specific view.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| name                     |                                                                                                                                                                                                                                                                                                                                  |
|                          | <p>Returns an identifier for this library arc in the form:<br/>libraryName/libraryCell/inputPin_outputPin_uniqueIntegerSuffix</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                    |
| obj_type                 |                                                                                                                                                                                                                                                                                                                                  |
|                          | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">lib_arc</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                          |
| output_signal_level_high |                                                                                                                                                                                                                                                                                                                                  |
|                          | <p>Returns the upper/higher limit of partial voltage swing value that is defined for a library cell output and inout pins.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                              |
| output_signal_level_low  |                                                                                                                                                                                                                                                                                                                                  |

|             |                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>Returns the lower limit of partial voltage swing value that is defined for a library cell output and inout pins.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                            |
| sdf_cond    |                                                                                                                                                                                                                                                                                                                                                                         |
|             | <p>Returns the value of the Liberty sdf_cond attribute if specified for this arc. You can use -index to return a value for a specific view.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                          |
| sense       |                                                                                                                                                                                                                                                                                                                                                                         |
|             | <p>Returns the Liberty timing_sense value associated with this arc. This value is one of: positive_unate, negative_unate, or non_unate. You can use -index to return a value for a specific view.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                    |
| timing_type |                                                                                                                                                                                                                                                                                                                                                                         |
|             | <p>Returns the Liberty timing_type associated with this arc. You can consult the Liberty documentation for the list of possible values for this attribute. You can use -index to return a value for a specific view.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| to_lib_pin  |                                                                                                                                                                                                                                                                                                                                                                         |
|             | <p>Returns a lib_pin object for terminating pin of this timing arc. You can use -index to return a value for a specific view.</p> <p><b>Type:</b> <a href="#">obj(lib_pin)*</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                 |
| when        |                                                                                                                                                                                                                                                                                                                                                                         |

|                     |                                                                                                                                                                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | Returns the value of the Liberty 'when' attribute if specified for this arc. You can use -index to return a value for a specific view.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No       |
| when_end            |                                                                                                                                                                                                                                                                             |
|                     | Returns the value of the Liberty 'when_end' attribute if specified for this arc. You can use -index to return a value for a specific view.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No   |
| when_library_string |                                                                                                                                                                                                                                                                             |
|                     | Returns the exact value of the Liberty 'when' attribute if specified for this arc. You can use -index to return a value for a specific view.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| when_start          |                                                                                                                                                                                                                                                                             |
|                     | Returns the value of the Liberty 'when_start' attribute if specified for this arc. You can use -index to return a value for a specific view.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |

## lib\_cell

### Parent Objects

[seq\\_function](#), [inst](#), [base\\_cell](#), [lib\\_arc](#), [root](#), [library](#), [lib\\_pin](#)

### Definition

cte lib cell

| Attribute                         | Description                                                                                                                                                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| aocv_bootstrap_fall_derate_factor |                                                                                                                                                                                                                                                     |
|                                   | Returns aocv coefficient aocv_bootstrap_fall_derate_factor in bootstrap mode on the lib_cell<br><b>Type:</b> float<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                                          |
| aocv_bootstrap_rise_derate_factor |                                                                                                                                                                                                                                                     |
|                                   | Returns aocv coefficient aocv_bootstrap_rise_derate_factor in bootstrap mode on the lib_cell<br><b>Type:</b> float<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                                          |
| aocv_weight                       |                                                                                                                                                                                                                                                     |
|                                   | Returns the AOCV stage weight specified for the library cell either explicitly in the library or as a user-defined library attribute<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| area                              |                                                                                                                                                                                                                                                     |
|                                   | Returns the area of the library cell as specified by the Liberty timing library<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                                                      |
| async_clear_pins                  |                                                                                                                                                                                                                                                     |
|                                   | Returns a list of lib_pins that have both the is_async and is_clear attributes set to true<br><b>Type:</b> obj(lib_pin)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                   |
| async_preset_pins                 |                                                                                                                                                                                                                                                     |

|                              |                                                                                                                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Returns a list of lib_pins that have both the is_async and is_clear attributes set to true</p> <p><b>Type:</b> obj(lib_pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                |
| base_cell                    |                                                                                                                                                                                                                                     |
|                              | <p>Returns a pointer the base_cell object associated with this library cell. You can use chaining to reach the attributes of the base_cell</p> <p><b>Type:</b> obj(base_cell)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| base_name                    |                                                                                                                                                                                                                                     |
|                              | <p>The base_name of the cell.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                       |
| bit_width                    |                                                                                                                                                                                                                                     |
|                              | <p>Returns the bit width of the cell</p> <p><b>Type:</b> int</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                   |
| cell_footprint               |                                                                                                                                                                                                                                     |
|                              | <p>Returns the library cell footprint.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                     |
| clock_gating_integrated_cell |                                                                                                                                                                                                                                     |
|                              | <p>clock_gating_integrated_cell</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                  |
| clock_pins                   |                                                                                                                                                                                                                                     |

|                                             |                                                                                                                                                                                                                                                                         |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | Returns a list of lib_pins which have the attribute is_clock equal to true<br><b>Type:</b> obj(lib_pin)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                       |
| data_pins                                   |                                                                                                                                                                                                                                                                         |
|                                             | Returns a list of lib_pins which have the attribute is_data equal to true<br><b>Type:</b> obj(lib_pin)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                        |
| dft_mapper_usable                           |                                                                                                                                                                                                                                                                         |
|                                             | The lib_cell is mapper usable.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                                                        |
| dft_timing_model                            |                                                                                                                                                                                                                                                                         |
|                                             | libcell has timing model<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                                                              |
| early_clock_check_incremental_derate_factor |                                                                                                                                                                                                                                                                         |
|                                             | Returns the early clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| early_data_check_incremental_derate_factor  |                                                                                                                                                                                                                                                                         |
|                                             | Returns the early data path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| early_fall_clock_incremental_derate_factor  |                                                                                                                                                                                                                                                                         |



|                                            |                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Returns the early clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -fall parameter is specified.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_fall_data_incremental_derate_factor  |                                                                                                                                                                                                                                                                                                                                        |
|                                            | <p>Returns the early data path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -fall parameter is specified.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| early_fall_mean_slew_derate_factor         |                                                                                                                                                                                                                                                                                                                                        |
|                                            | <p>Returns early fall mean transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                 |
| early_fall_sigma_slew_derate_factor        |                                                                                                                                                                                                                                                                                                                                        |
|                                            | <p>Returns early fall sigma transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                |
| early_fall_slew_derate_factor              |                                                                                                                                                                                                                                                                                                                                        |
|                                            | <p>Returns early fall transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                      |
| early_rise_clock_incremental_derate_factor |                                                                                                                                                                                                                                                                                                                                        |

|                                           |                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>Returns the early clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -rise parameter is specified.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_rise_data_incremental_derate_factor |                                                                                                                                                                                                                                                                                                                                        |
|                                           | <p>Returns the early data path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -rise parameter is specified.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| early_rise_mean_slew_derate_factor        |                                                                                                                                                                                                                                                                                                                                        |
|                                           | <p>Returns early rise mean transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                 |
| early_rise_sigma_slew_derate_factor       |                                                                                                                                                                                                                                                                                                                                        |
|                                           | <p>Returns early rise sigma transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                |
| early_rise_slew_derate_factor             |                                                                                                                                                                                                                                                                                                                                        |
|                                           | <p>Returns early rise transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                      |
| fall_input_switching_derate_factor        |                                                                                                                                                                                                                                                                                                                                        |

|               |                                                                                                                                                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <p>Returns user or library defined value for controlling the arrival sensitivity window for simultaneous rising inputs to this cell type</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| ground_pins   |                                                                                                                                                                                                                                                                             |
|               | <p>Returns a collection of the ground pins of a library cell.</p> <p><b>Type:</b> obj(lib_pin)*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                               |
| has_aging     |                                                                                                                                                                                                                                                                             |
|               | <p>Returns true if library cell has aging information</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                      |
| has_bti_aging |                                                                                                                                                                                                                                                                             |
|               | <p>Returns true if library cell has BTI aging information</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                  |
| has_ccs_noise |                                                                                                                                                                                                                                                                             |
|               | <p>Returns true if cell has ccs noise data.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                          |
| has_hci_aging |                                                                                                                                                                                                                                                                             |
|               | <p>Returns true if library cell has HCI aging information</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                  |
| has_lvf       |                                                                                                                                                                                                                                                                             |

|                   |                                                                                                                                                                                                                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>Returns true if library arc(s) of libcell has LVF information</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                |
| has_mis           |                                                                                                                                                                                                                                  |
|                   | <p>Returns true if library cell has multi-instance switching data.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| has_socv          |                                                                                                                                                                                                                                  |
|                   | <p>Returns true if library cell has socv data.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                            |
| has_spatial       |                                                                                                                                                                                                                                  |
|                   | <p>Returns a value of true if a spatial derate exists for a library cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| hierarchical_name |                                                                                                                                                                                                                                  |
|                   | <p>hierarchical_name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                          |
| is_always_on      |                                                                                                                                                                                                                                  |
|                   | <p>Returns a value of true if the lib_cell is identified as always on type</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| is_black_box      |                                                                                                                                                                                                                                  |

|                        |                                                                                                                                                                                                                                  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Returns a value of true if the cell is a black box cell.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                     |
| is_buffer              |                                                                                                                                                                                                                                  |
|                        | <p>Returns a value of true if the lib_cell is identified as a buffer type</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| is_combinational       |                                                                                                                                                                                                                                  |
|                        | <p>Returns a value of true if the library cell is a combinational cell (not a sequential cell).</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_disable_timing      |                                                                                                                                                                                                                                  |
|                        | <p>Returns true if set_disable_timing has been set on this library cell</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                         |
| is_dont_touch          |                                                                                                                                                                                                                                  |
|                        | <p>Returns a value of true if a set_dont_touch constraint exists for the library cell.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| is_exist               |                                                                                                                                                                                                                                  |
|                        | <p>Returns true if library cell present in view.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                |
| is_fall_edge_triggered |                                                                                                                                                                                                                                  |

|                            |                                                                                                                                                                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Returns a value of true if the library cell is triggered by the falling edge of the clock.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| is_flop                    |                                                                                                                                                                                                                                      |
|                            | <p>Returns true if the library cell is recognized as flip-flop/register type cell</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                      |
| is_generic                 |                                                                                                                                                                                                                                      |
|                            | <p>This cell is Synthesis Generic cell.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                |
| is_instantiated            |                                                                                                                                                                                                                                      |
|                            | <p>Returns a value of true if the library cell is instantiated.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                     |
| is_integrated_clock_gating |                                                                                                                                                                                                                                      |
|                            | <p>Returns true if the library cell has the Liberty clock_gating_integrated_cell set to true</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| is_interface_timing        |                                                                                                                                                                                                                                      |
|                            | <p>Returns a value of true if a library cell has the Liberty interface_timing attribute set to true</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_inverter                |                                                                                                                                                                                                                                      |

|                           |                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Returns a value of true if the lib_cell is identified as an inverter type</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| is_isolation_cell         |                                                                                                                                                                                                                     |
|                           | <p>Returns a value of true if the lib_cell is identified as an isolation cell type</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_latch                  |                                                                                                                                                                                                                     |
|                           | <p>Returns true if the library cell is recognized as latch type cell</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                  |
| is_level_shifter          |                                                                                                                                                                                                                     |
|                           | <p>Returns a value of true if the lib_cell is identified as a level shifter type</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| is_macro                  |                                                                                                                                                                                                                     |
|                           | <p>Returns a value of true if the Liberty attribute is_macro_cell is set true</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| is_master_slave_flop      |                                                                                                                                                                                                                     |
|                           | <p>Returns true if the lib_cell is recognized as a master/slave flip-flop</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                             |
| is_master_slave_lssd_flop |                                                                                                                                                                                                                     |

|                             |                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>Returns true if this cell has been recognized as a master/slave LSSD device</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                             |
| is_memory                   |                                                                                                                                                                                                                                                                                                                                          |
|                             | <p>Returns true or false depending on whether the associated library cell is recognized as a Liberty memory cell. Cells which include a Liberty memory group definition are recognized as memory cells.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_msv                      |                                                                                                                                                                                                                                                                                                                                          |
|                             | <p>Returns true if cell has multi-supply voltages.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                |
| is_negative_level_sensitive |                                                                                                                                                                                                                                                                                                                                          |
|                             | <p>Returns a value of true if the library cell is identified as a negative level-sensitive latch.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                       |
| is_pad                      |                                                                                                                                                                                                                                                                                                                                          |
|                             | <p>Returns a value of true if Liberty attribute is_pad is set to true</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                   |
| is_pll                      |                                                                                                                                                                                                                                                                                                                                          |
|                             | <p>Returns a value of true if the Liberty library is_pll_cell attribute is set to true for this library cell.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                           |



|                             |                                                                                                                                                                                                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_positive_level_sensitive |                                                                                                                                                                                                                                                                       |
|                             | <p>Returns a value of true if the library cell is identified as a positive level-sensitive latch.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                    |
| is_power_switch             |                                                                                                                                                                                                                                                                       |
|                             | <p>Returns a value of true if the lib_cell is identified as power switch type</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                        |
| is_retention                |                                                                                                                                                                                                                                                                       |
|                             | <p>Returns a value of true if the lib_cell is identified as an retention cell type</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                   |
| is_rise_edge_triggered      |                                                                                                                                                                                                                                                                       |
|                             | <p>Returns a value of true if the library cell is triggered by the rising edge of the clock.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                         |
| is_sequential               |                                                                                                                                                                                                                                                                       |
|                             | <p>Returns a value of true if the library cell is a latch or flip-flop, or if the cell has sequential timing checks.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_synchronizer             |                                                                                                                                                                                                                                                                       |
|                             | <p>The lib_cell or base_cell is a synchronizer. Library cell having back to back connected flops is considered as synchronizer.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                      |

|                                            |                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_tristate                                |                                                                                                                                                                                                                                                                                               |
|                                            | <p>Returns a value of true if the library cell definition includes the Liberty attribute three_state set to true.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                            |
| is_unmapped                                |                                                                                                                                                                                                                                                                                               |
|                                            | <p>Returns true if library cell is unmapped.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                           |
| is_usable                                  |                                                                                                                                                                                                                                                                                               |
|                                            | <p>The lib_cell can be inferred during mapping/optimization.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> true</p> <p><b>Edit:</b> No</p>                                                                                                                                                  |
| latch_enable_pins                          |                                                                                                                                                                                                                                                                                               |
|                                            | <p>Returns a list of lib_pin objects which function as the enable/gate pin of a latch</p> <p><b>Type:</b> obj(lib_pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                  |
| late_clock_check_incremental_derate_factor |                                                                                                                                                                                                                                                                                               |
|                                            | <p>Returns the late clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| late_data_check_incremental_derate_factor  |                                                                                                                                                                                                                                                                                               |
|                                            | <p>Returns the late data path instances' incremental derating factor specified through the set_timing_derate -incremental_adjust parameter.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>          |

|                                            |                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| late_fall_clock_incremental_derate_factor  |                                                                                                                                                                                                                                                                                                                                       |
|                                            | <p>Returns the late clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -fall parameter is specified.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| late_fall_clock_period_derate_factor       |                                                                                                                                                                                                                                                                                                                                       |
|                                            | <p>Returns clock period derate factor for fall transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                          |
| late_fall_clock_period_mean_derate_factor  |                                                                                                                                                                                                                                                                                                                                       |
|                                            | <p>Returns mean clock period derate factor for fall transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                     |
| late_fall_clock_period_sigma_derate_factor |                                                                                                                                                                                                                                                                                                                                       |
|                                            | <p>Returns sigma clock period derate factor for fall transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                    |
| late_fall_data_incremental_derate_factor   |                                                                                                                                                                                                                                                                                                                                       |
|                                            | <p>Returns the late data path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -fall parameter is specified.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| late_fall_mean_slew_derate_factor          |                                                                                                                                                                                                                                                                                                                                       |

|                                           |                                                                                                                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>Returns late fall mean transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>             |
| late_fall_pulse_width_derate_factor       |                                                                                                                                                                                                                   |
|                                           | <p>Returns pulse width derate factor for fall transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| late_fall_pulse_width_mean_derate_factor  |                                                                                                                                                                                                                   |
|                                           | <p>Returns mean pulse width derate factor for fall transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| late_fall_pulse_width_sigma_derate_factor |                                                                                                                                                                                                                   |
|                                           | <p>Returns sigma pulse width derate factor for fall transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| late_fall_sigma_slew_derate_factor        |                                                                                                                                                                                                                   |
|                                           | <p>Returns late fall sigma transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>            |
| late_fall_slew_derate_factor              |                                                                                                                                                                                                                   |
|                                           | <p>Returns late fall transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                  |
| late_rise_clock_incremental_derate_factor |                                                                                                                                                                                                                   |

|                                            |                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Returns the late clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -rise parameter is specified.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| late_rise_clock_period_derate_factor       |                                                                                                                                                                                                                                                                                                                                       |
|                                            | <p>Returns clock period derate factor for rise transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                          |
| late_rise_clock_period_mean_derate_factor  |                                                                                                                                                                                                                                                                                                                                       |
|                                            | <p>Returns mean clock period derate factor for rise transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                     |
| late_rise_clock_period_sigma_derate_factor |                                                                                                                                                                                                                                                                                                                                       |
|                                            | <p>Returns sigma clock period derate factor for rise transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                    |
| late_rise_data_incremental_derate_factor   |                                                                                                                                                                                                                                                                                                                                       |
|                                            | <p>Returns the late data path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -rise parameter is specified.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| late_rise_mean_slew_derate_factor          |                                                                                                                                                                                                                                                                                                                                       |

|                                           |                                                                                                                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>Returns late rise mean transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>             |
| late_rise_pulse_width_derate_factor       |                                                                                                                                                                                                                   |
|                                           | <p>Returns pulse width derate factor for rise transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| late_rise_pulse_width_mean_derate_factor  |                                                                                                                                                                                                                   |
|                                           | <p>Returns mean pulse width derate factor for rise transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| late_rise_pulse_width_sigma_derate_factor |                                                                                                                                                                                                                   |
|                                           | <p>Returns sigma pulse width derate factor for rise transition on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| late_rise_sigma_slew_derate_factor        |                                                                                                                                                                                                                   |
|                                           | <p>Returns late rise sigma transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>            |
| late_rise_slew_derate_factor              |                                                                                                                                                                                                                   |
|                                           | <p>Returns late rise transition derate factor on the lib cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                  |
| leakage                                   |                                                                                                                                                                                                                   |

|                      |                                                                                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Returns a cell_leakage_power property for a lib_cell</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                            |
| leakage_power        |                                                                                                                                                                                                                                       |
|                      | <p>Returns the leakage power of the library cell as specified by the Liberty timing library</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| leakage_scale_factor |                                                                                                                                                                                                                                       |
|                      | <p>The scaling factor for lib_cell's leakage_power.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p>                                                                                                 |
| level_shifter_type   |                                                                                                                                                                                                                                       |
|                      | <p>Returns the Level Shifter type for a given lib_cell. The supported values are LH, HL and HL_LH.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| lib_arcs             |                                                                                                                                                                                                                                       |
|                      | <p>Returns a list of lib_arc objects contained with the lib_cell</p> <p><b>Type:</b> obj(lib_arc)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                               |
| lib_pins             |                                                                                                                                                                                                                                       |
|                      | <p>Returns a list of lib_pin objects for the lib_cell</p> <p><b>Type:</b> obj(lib_pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                          |
| library              |                                                                                                                                                                                                                                       |

|                              |                                                                                                                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | Returns the parent library object for the lib_cell<br><b>Type:</b> <a href="#">obj(library)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                           |
| master_physical_variant_cell |                                                                                                                                                                                                     |
|                              | Returns the master physical variant cell of the library cell.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| name                         |                                                                                                                                                                                                     |
|                              | Returns the unique name of the library cell in the form lib_cell:librarySetName/libraryName/libraryCellName<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No         |
| obj_type                     |                                                                                                                                                                                                     |
|                              | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (lib_cell)</a><br><b>Default:</b> ""<br><b>Edit:</b> No               |
| pg_lib_pins                  |                                                                                                                                                                                                     |
|                              | Returns a list of the power and ground lib_pin objects of the lib_cell<br><b>Type:</b> <a href="#">obj(lib_pin)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                       |
| physical_variant_cells       |                                                                                                                                                                                                     |
|                              | Returns the list of physical variant cells of the master cell.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| pin_count                    |                                                                                                                                                                                                     |



|                                    |                                                                                                                                                                                                                                                                                          |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Returns the number of pins of the cell.</p> <p><b>Type:</b> <code>int</code></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                            |
| power_pins                         |                                                                                                                                                                                                                                                                                          |
|                                    | <p>Returns the collection of power pins of the cell.</p> <p><b>Type:</b> <code>obj(lib_pin)*</code></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                        |
| power_switch_type                  |                                                                                                                                                                                                                                                                                          |
|                                    | <p>Returns the switch cell type associated with library cell. The supported values are coarse_grain, fine_grain.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                         |
| receiver_capacitance_type          |                                                                                                                                                                                                                                                                                          |
|                                    | <p>Returns input cap group being used for delay calculations, C1C2 or C1CN</p> <p><b>Type:</b> <code>string</code></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                               |
| retention_cell                     |                                                                                                                                                                                                                                                                                          |
|                                    | <p>Returns the retention_cell derived from Liberty</p> <p><b>Type:</b> <code>string</code></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                       |
| rise_input_switching_derate_factor |                                                                                                                                                                                                                                                                                          |
|                                    | <p>Returns user or library defined value for controlling the arrival sensitivity window for simultaneous rising inputs to this cell type</p> <p><b>Type:</b> <code>double</code></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

|                         |                                                                                                                                                                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| seq_function            |                                                                                                                                                                                                                                             |
|                         | <p>Returns a list of seq_function associated with that library cell object.</p> <p><b>Type:</b> <a href="#">obj(seq_function)</a>*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                     |
| signal_pins             |                                                                                                                                                                                                                                             |
|                         | <p>Returns the collection of signal pins</p> <p><b>Type:</b> <a href="#">obj(lib_pin)</a>*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| std_cell_main_rail_name |                                                                                                                                                                                                                                             |
|                         | <p>Returns the rail name associated with pg pin for which std_cell_main_rail is enabled.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| sync_clear              |                                                                                                                                                                                                                                             |
|                         | <p>Returns the collection of sync clear pins</p> <p><b>Type:</b> <a href="#">obj(lib_pin)</a>*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                |
| sync_enable             |                                                                                                                                                                                                                                             |
|                         | <p>Returns the collection of sync enable pins</p> <p><b>Type:</b> <a href="#">obj(lib_pin)</a>*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                               |
| sync_preset             |                                                                                                                                                                                                                                             |
|                         | <p>Returns the collection of sync preset pins</p> <p><b>Type:</b> <a href="#">obj(lib_pin)</a>*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                               |
| test_scan_enable        |                                                                                                                                                                                                                                             |

|                           |                                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | Returns the collection of test_scan_enable pins<br><b>Type:</b> <a href="#">obj(lib_pin)*</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| test_scan_enable_inverted |                                                                                                                                                                                                      |
|                           | Returns the collection of test_scan_enable_inverted pins<br><b>Type:</b> <a href="#">obj(lib_pin)*</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| test_scan_in              |                                                                                                                                                                                                      |
|                           | Returns the collection of test_scan_in pins<br><b>Type:</b> <a href="#">obj(lib_pin)*</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No              |
| test_scan_in_inverted     |                                                                                                                                                                                                      |
|                           | Returns the collection of test_scan_in_inverted pins<br><b>Type:</b> <a href="#">obj(lib_pin)*</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No     |
| test_scan_out             |                                                                                                                                                                                                      |
|                           | Returns the collection of test_scan_out pins<br><b>Type:</b> <a href="#">obj(lib_pin)*</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No             |
| test_scan_out_inverted    |                                                                                                                                                                                                      |
|                           | Returns the collection of test_scan_out_inverted pins<br><b>Type:</b> <a href="#">obj(lib_pin)*</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No    |
| threshold_voltage_group   |                                                                                                                                                                                                      |

|                   |                                                                                                                                                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | Returns name of a group/category – that is specified based on the cell's threshold voltage (high or low) characteristics.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| timing_model_type |                                                                                                                                                                                                                                                                |
|                   | Returns the Liberty model type for a given cell or instance. The supported values are abstracted , extracted , and qtm .<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No        |

## lib\_pin

### Parent Objects

[lib\\_cell](#), [pin](#), [lib\\_arc](#), [root](#), [port](#)

### Definition

cte lib pin

| Attribute          | Description                                                                                                                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all_q_pin_of_d_pin |                                                                                                                                                                                                                 |
|                    | Returns the collection of all_q_pin_of_d_pin pins<br><b>Type:</b> <a href="#">obj(lib_pin)*</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                   |
| base_name          |                                                                                                                                                                                                                 |
|                    | Returns the simple name of this library pin. The base_name of BUFFERX1/A is 'A'.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |

|                      |                                                                                                                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_pin             |                                                                                                                                                                                                                   |
|                      | <p>Returns the related base_pin object of this library pin. Use chaining to access the base_pin</p> <p><b>Type:</b> <a href="#">obj(base_pin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| capacitance          |                                                                                                                                                                                                                   |
|                      | <p>Returns the capacitance for the library pin.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                |
| capacitance_max_fall |                                                                                                                                                                                                                   |
|                      | <p>Returns the maximum value of the falling capacitance range.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| capacitance_max_rise |                                                                                                                                                                                                                   |
|                      | <p>Returns the maximum value of the rising capacitance range.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| capacitance_min_fall |                                                                                                                                                                                                                   |
|                      | <p>Returns the minimum value of the falling capacitance range.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| capacitance_min_rise |                                                                                                                                                                                                                   |
|                      | <p>Returns the minimum value of the rising capacitance range.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| cell_name            |                                                                                                                                                                                                                   |

|                       |                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>Returns the name of the lib_cell of the lib_pin mentioned.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                      |
| clock_polarity        |                                                                                                                                                                                                                                                                                                                                                       |
|                       | <p>Returns the clock polarity for the library pin.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                 |
| direction             |                                                                                                                                                                                                                                                                                                                                                       |
|                       | <p>Returns the direction of the library pin: in , out , inout or internal. Internal is for a lib_pin that in the GUI, or Verilog, but may appear in timing reports.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> internal in out inout</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| drive_resistance_fall |                                                                                                                                                                                                                                                                                                                                                       |
|                       | <p>Specifies the driving resistance of the cell for falling transitions</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                            |
| drive_resistance_rise |                                                                                                                                                                                                                                                                                                                                                       |
|                       | <p>Specifies the driving resistance of the cell for falling transitions</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                            |
| fanout_load           |                                                                                                                                                                                                                                                                                                                                                       |
|                       | <p>Returns the fanout load value of the library pin.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                               |

|                         |                                                                                                                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| from_lib_arcs           |                                                                                                                                                                                                                                                    |
|                         | <p>Returns the list of lib_arc objects that begin from this library pin</p> <p><b>Type:</b> <a href="#">obj(lib_arc)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                     |
| full_name               |                                                                                                                                                                                                                                                    |
|                         | <p>Returns a unique reference for the library cell of the form: libset/lib_name/cell_name/pin_n</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| function                |                                                                                                                                                                                                                                                    |
|                         | <p>Returns the function property for a lib pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                  |
| hierarchical_name       |                                                                                                                                                                                                                                                    |
|                         | <p>Returns the hierarchical name of the library pin.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                      |
| input_signal_level      |                                                                                                                                                                                                                                                    |
|                         | <p>Returns the rail name associated with input library pin.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                     |
| input_signal_level_high |                                                                                                                                                                                                                                                    |
|                         | <p>Returns partial high voltage swing of an input pin.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                          |
| input_signal_level_low  |                                                                                                                                                                                                                                                    |

|                            |                                                                                                                                                                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Returns partial low voltage swing of an input pin.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                |
| input_signal_level_voltage |                                                                                                                                                                                                                                                         |
|                            | <p>Returns rail voltage of signal which is driving the input pin.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| is_always_on               |                                                                                                                                                                                                                                                         |
|                            | <p>Returns a value of true if the Liberty library is_always_on attribute is set to true for this libra</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_analog                  |                                                                                                                                                                                                                                                         |
|                            | <p>Returns true if lib pin has analog signal</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                     |
| is_async                   |                                                                                                                                                                                                                                                         |
|                            | <p>Returns a value of true if the library pin is an asynchronous preset pin, or an asynchronous</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| is_clear                   |                                                                                                                                                                                                                                                         |
|                            | <p>Returns a value of true if the pin is an asynchronous clear pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| is_clock                   |                                                                                                                                                                                                                                                         |



|                        |                                                                                                                                                                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Returns a value of true if the library pin definition includes clock:true .</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                    |
| is_clock_gate_clock    |                                                                                                                                                                                                                                                    |
|                        | <p>Returns a value of true if the pin has the Liberty attribute clock_gate_clock_pin set to true</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| is_clock_gate_enable   |                                                                                                                                                                                                                                                    |
|                        | <p>Returns a value of true if the pin has the Liberty attribute clock_gate_enable_pin set to true</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_clock_gating_clock  |                                                                                                                                                                                                                                                    |
|                        | <p>Returns a value true if the pin is defined as a clock pin of a clock gating cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| is_clock_gating_enable |                                                                                                                                                                                                                                                    |
|                        | <p>Returns a value of true if the pin corresponds to the enable pin of a clock gating cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_clock_gating_test   |                                                                                                                                                                                                                                                    |
|                        | <p>Returns a value of true if the pin corresponds to the test pin of a clock gating cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| is_clock_pin           |                                                                                                                                                                                                                                                    |

|                              |                                                                                                                                                                                                                                                                                                |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Returns a value of true if the library pin definition includes timing_type:setup_rising/setup_falling/hold_rising/hold_falling/recovery_rising/recovery_fa</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_data                      |                                                                                                                                                                                                                                                                                                |
|                              | <p>Returns a value of true if the library pin is the data pin of a flip-flop.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                 |
| is_dynamic_clock_gating_data |                                                                                                                                                                                                                                                                                                |
|                              | <p>Returns true if pin is data pin of a dynamic clock gating cell</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                       |
| is_exist                     |                                                                                                                                                                                                                                                                                                |
|                              | <p>Returns a value of true if the library pin exists in view.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                 |
| is_fall_edge_triggered_clock |                                                                                                                                                                                                                                                                                                |
|                              | <p>is_fall_edge_triggered_clock</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                               |
| is_fall_edge_triggered_data  |                                                                                                                                                                                                                                                                                                |
|                              | <p>Returns a value of true if the library pin is the data pin of a falling edge triggered device.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                             |
| is_iq_function               |                                                                                                                                                                                                                                                                                                |

|                          |                                                                                                                                                                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Returns a value of true if the q pin has a function.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                             |
| is_iqn_function          |                                                                                                                                                                                                                                            |
|                          | <p>Returns a value of true if the q bar pin has a function.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                         |
| is_isolated              |                                                                                                                                                                                                                                            |
|                          | <p>Returns a value of true if the Liberty library is_isolated attribute is set to true for this library p</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_isolation_cell_clock  |                                                                                                                                                                                                                                            |
|                          | <p>Returns true if a clock lib pin is part of an isolatoin cell,</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                          |
| is_isolation_cell_data   |                                                                                                                                                                                                                                            |
|                          | <p>Returns a value of true if the Liberty library is_isolation_cell_data attribute is set to true for t</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| is_isolation_cell_enable |                                                                                                                                                                                                                                            |
|                          | <p>Returns a value of true if the Liberty library is_isolation_cell_enable attribute is set to true fr</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| is_level_shifter_data    |                                                                                                                                                                                                                                            |

|                                   |                                                                                                                                                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Returns a value of true if the Liberty library is_level_shifter_data attribute is set to true for th</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_level_shifter_enable           |                                                                                                                                                                                                                                                          |
|                                   | <p>Returns a value of true if the Liberty library is_level_shifter_enable attribute is set to true for</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| is_negative_level_sensitive_clock |                                                                                                                                                                                                                                                          |
|                                   | <p>Returns a value of true if the library pin is an enable pin of an active low level-sensitive dev</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>     |
| is_negative_level_sensitive_data  |                                                                                                                                                                                                                                                          |
|                                   | <p>Returns a value of true if the pin is a data pin of an active low level-sensitive device.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>            |
| is_pad                            |                                                                                                                                                                                                                                                          |
|                                   | <p>Returns the pad value of a pin</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                       |
| is_pll_feedback_pin               |                                                                                                                                                                                                                                                          |
|                                   | <p>Returns a value of true if the Liberty library is_pll_feedback_pin attribute is set to true for thi</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| is_pll_output_pin                 |                                                                                                                                                                                                                                                          |

|                                   |                                                                                                                                                                                                                                                           |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Returns a value of true if the Liberty library is_pll_output_pin attribute is set to true for this li</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_pll_reference_pin              |                                                                                                                                                                                                                                                           |
|                                   | <p>Returns a value of true if the Liberty library is_pll_reference_pin attribute is set to true for th</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| is_positive_level_sensitive_clock |                                                                                                                                                                                                                                                           |
|                                   | <p>Returns a value of true if the library pin is an enable pin of an active high level-sensitive de</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| is_positive_level_sensitive_data  |                                                                                                                                                                                                                                                           |
|                                   | <p>Returns a value of true if the pin is a data pin of an active high level-sensitive device.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>            |
| is_power_switch_enable            |                                                                                                                                                                                                                                                           |
|                                   | <p>Returns a value of true if the Liberty library is_power_switch_enable attribute is set to true f</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| is_preset                         |                                                                                                                                                                                                                                                           |
|                                   | <p>Returns a value of true if the pin is a preset pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                   |
| is_ref_clock_pin                  |                                                                                                                                                                                                                                                           |

|                              |                                                                                                                                                                                                                                                        |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Returns true if object is a reference clock pin</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                              |
| is_retention_cell_enable     |                                                                                                                                                                                                                                                        |
|                              | <p>Returns a value of true if the Liberty library is_retention_cell_enable attribute is set to true f</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_rise_edge_triggered_clock |                                                                                                                                                                                                                                                        |
|                              | <p>Returns a value of true if the library pin is the clock pin of a rising edge triggered device.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>     |
| is_rise_edge_triggered_data  |                                                                                                                                                                                                                                                        |
|                              | <p>Returns a value of true if the library pin is the data pin of a rising edge triggered device.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| is_tristate                  |                                                                                                                                                                                                                                                        |
|                              | <p>Returns a value of true if the library pin definition includes three_state:true .</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                  |
| is_tristate_enable           |                                                                                                                                                                                                                                                        |
|                              | <p>Returns a value of true if the library pin is part of a Liberty three_state_enable logical expre</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| is_tristate_output           |                                                                                                                                                                                                                                                        |

|                            |                                                                                                                                                                                                                                                          |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Returns a value of true if the library pin definition includes three_state:true .</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                    |
| lib_cell                   |                                                                                                                                                                                                                                                          |
|                            | <p>Returns the lib_cell parent object of this lib_pin</p> <p><b>Type:</b> <a href="#">obj(lib_cell)</a>*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                            |
| max_capacitance            |                                                                                                                                                                                                                                                          |
|                            | <p>Returns the maximum capacitance limit for the library pin.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                         |
| max_capacitance_clock_path |                                                                                                                                                                                                                                                          |
|                            | <p>Returns the maximum value of the capacitance for the clock path.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                             |
| max_fanout                 |                                                                                                                                                                                                                                                          |
|                            | <p>Returns the maximum fanout value for the library pin.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                              |
| max_transition             |                                                                                                                                                                                                                                                          |
|                            | <p>Returns the maximum transition time limit specified for a given library pin. If the limit is not s</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| max_transition_clock_path  |                                                                                                                                                                                                                                                          |

|                          |                                                                                                                                                                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Returns the maximum value of the transition for the clock path.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                               |
| max_transition_data_path |                                                                                                                                                                                                                                                                           |
|                          | <p>Returns the maximum value of the transition for the data path.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                |
| min_capacitance          |                                                                                                                                                                                                                                                                           |
|                          | <p>Returns the minimum capacitance limit for the library pin.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                          |
| min_fanout               |                                                                                                                                                                                                                                                                           |
|                          | <p>Returns the minimum fanout value for the library pin.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                               |
| min_transition           |                                                                                                                                                                                                                                                                           |
|                          | <p>Returns the minimum transition time limit specified for a given library pin. If the limit is not specified, the default value is 0.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| name                     |                                                                                                                                                                                                                                                                           |
|                          | <p>Returns the name of the library pin in the form: &lt;library_name&gt;/&lt;cell_name&gt;/&lt;pin_name&gt;</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                            |
| obj_type                 |                                                                                                                                                                                                                                                                           |



|                          |                                                                                                                                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">lib_pin</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| output_signal_level      |                                                                                                                                                                                                                         |
|                          | <p>Returns the rail name associated with output library pin.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>         |
| output_signal_level_high |                                                                                                                                                                                                                         |
|                          | <p>Returns partial high voltage swing of an output pin.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>              |
| output_signal_level_low  |                                                                                                                                                                                                                         |
|                          | <p>Returns partial low voltage swing of an output pin.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>               |
| pin_capacitance          |                                                                                                                                                                                                                         |
|                          | <p>Returns capacitance of library pin.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                         |
| pin_capacitance_max_fall |                                                                                                                                                                                                                         |
|                          | <p>Returns max fall capacitance of lib pin.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                    |
| pin_capacitance_max_rise |                                                                                                                                                                                                                         |

|                                 |                                                                                                                                                                                 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | Returns max rise capacitance of lib pin.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                   |
| pin_capacitance_min_fall        |                                                                                                                                                                                 |
|                                 | Returns min fall capacitance of lib Pin.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                   |
| pin_capacitance_min_rise        |                                                                                                                                                                                 |
|                                 | Returns min rise capacitance of lib pin.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                   |
| related_ground_pin              |                                                                                                                                                                                 |
|                                 | Returns the related ground pin.<br><b>Type:</b> obj(pg_pin)* obj(pg_base_pin)*<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No    |
| related_ground_pin_name         |                                                                                                                                                                                 |
|                                 | related_ground_pin_name<br><b>Type:</b> string<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                                    |
| related_ground_pin_rail_voltage |                                                                                                                                                                                 |
|                                 | Returns rail voltage of associated ground pin for the given pin.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| related_power_pin               |                                                                                                                                                                                 |

|                                  |                                                                                                                                                                                                                                           |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Returns the related power pin.<br/> <b>Type:</b> <code>obj(pg_pin)* obj(pg_base_pin)*</code><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                   |
| related_power_pin_name           |                                                                                                                                                                                                                                           |
|                                  | <p>related_power_pin_name<br/> <b>Type:</b> <code>string</code><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                   |
| related_power_pin_rail_voltage   |                                                                                                                                                                                                                                           |
|                                  | <p>Returns rail voltage of associated power pin for the given pin.<br/> <b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                |
| slew_threshold_percent_fall_high |                                                                                                                                                                                                                                           |
|                                  | <p>Specifies the upper threshold point used to model a falling transition on this pin. This value<br/> <b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| slew_threshold_percent_fall_low  |                                                                                                                                                                                                                                           |
|                                  | <p>Specifies the lower threshold point used to model a falling transition on this pin. This value<br/> <b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| slew_threshold_percent_rise_high |                                                                                                                                                                                                                                           |
|                                  | <p>Specifies the upper threshold point used to model a rising transition on this pin. This value<br/> <b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| slew_threshold_percent_rise_low  |                                                                                                                                                                                                                                           |

|             |                                                                                                                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | Specifies the lower threshold point used to model a rising transition on this pin. This value i<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| to_lib_arcs |                                                                                                                                                                                                                                |
|             | Returns the list of lib_arc objects that terminate at this pin<br><b>Type:</b> <a href="#">obj(lib_arc)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                          |

## library

### Parent Objects

[timing\\_condition](#), [lib\\_cell](#), [library\\_set](#), [root](#)

### Definition

cte lib

| Attribute              | Description                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name              |                                                                                                                                                                       |
|                        | Returns the simple name of the library as defined by the Liberty library() group name<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| cap_scale_in_ff        |                                                                                                                                                                       |
|                        | Returns the capacitance scaling of the library in femtofarads<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> ""<br><b>Edit:</b> No                         |
| default_max_transition |                                                                                                                                                                       |

|                                    |                                                                                                                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Returns the default max transition of the library</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| default_power_rail                 |                                                                                                                                                                                                                      |
|                                    | <p>Returns the power rail defined by the Liberty default_power_rail attribute</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                        |
| default_wireload                   |                                                                                                                                                                                                                      |
|                                    | <p>Returns the wire-load model defined by the Liberty default_wire_load attribute</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| files                              |                                                                                                                                                                                                                      |
|                                    | <p>Returns the list of library files associated with this library</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| has_cells_having_power_ground_pins |                                                                                                                                                                                                                      |
|                                    | <p>Returns a value of true if the library cells in this library have power/ground pin modeling</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                         |
| hierarchical_name                  |                                                                                                                                                                                                                      |
|                                    | <p>hierarchical_name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                              |
| input_threshold_pct_fall           |                                                                                                                                                                                                                      |

|                           |                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Specifies the delay threshold for a falling input signal. This value is specified as a percentage.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| input_threshold_pct_rise  |                                                                                                                                                                                                                                          |
|                           | <p>Specifies the delay threshold for a rising input signal. This value is specified as a percentage.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| is_exist                  |                                                                                                                                                                                                                                          |
|                           | <p>Returns a value of true if library is present in view.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                  |
| leakage_power_scale_in_nw |                                                                                                                                                                                                                                          |
|                           | <p>Returns the leakage power scaling used in this library in units of nanoWatts</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                          |
| lib_cells                 |                                                                                                                                                                                                                                          |
|                           | <p>Returns a list of library cell objects contained within this library</p> <p><b>Type:</b> obj(lib_cell)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                          |
| lib_file_path             |                                                                                                                                                                                                                                          |
|                           | <p>Returns the library file path name.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                          |
| lib_version               |                                                                                                                                                                                                                                          |

|                     |                                                                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>Returns the library version.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                |
| name                |                                                                                                                                                                                                                         |
|                     | <p>Returns an identifier for this library of the form: library_set/library_name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                         |
| nominal_process     |                                                                                                                                                                                                                         |
|                     | <p>Returns the nominal process of the library as specified by the Liberty nom_process attribute</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                         |
| nominal_temperature |                                                                                                                                                                                                                         |
|                     | <p>Returns the nominal temperature of the library as specified by the Liberty nom_temperature attribute</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                 |
| nominal_voltage     |                                                                                                                                                                                                                         |
|                     | <p>Returns the nominal voltage of the library as specified by the Liberty nom_voltage attribute</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                         |
| obj_type            |                                                                                                                                                                                                                         |
|                     | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">library</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| operating_condition |                                                                                                                                                                                                                         |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Returns the default operating conditions from the library</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                  |
| output_threshold_pct_fall     |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                               | <p>Specifies the delay threshold for a falling output signal. This value is specified as a percentage.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                        |
| output_threshold_pct_rise     |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                               | <p>Specifies the delay threshold for a rising output signal. This value is specified as a percentage.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                         |
| power_rails                   |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                               | <p>Returns the power rails defined for this library</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                              |
| slew_derate_from_library      |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                               | <p>Specifies the multiplier used to translate between the transition time range used during characterization and the transition time range used in the timing library tables. A characterization range of 30% and library range of 10% would result in a derate value of 0.5.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slew_lower_threshold_pct_fall |                                                                                                                                                                                                                                                                                                                                                                                                                  |



|                               |                                                                                                                                                                                                                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Specifies the lower threshold point used to model a falling transition on this pin. This value is specified as a percentage.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>              |
| slew_lower_threshold_pct_rise |                                                                                                                                                                                                                                                                                                 |
|                               | <p>Specifies the lower threshold point used to model a rising transition on this pin. This value is specified as a percentage.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>               |
| slew_upper_threshold_pct_fall |                                                                                                                                                                                                                                                                                                 |
|                               | <p>Specifies the upper threshold point used to model a falling transition on this pin.. This value is specified as a percentage.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>             |
| slew_upper_threshold_pct_rise |                                                                                                                                                                                                                                                                                                 |
|                               | <p>Specifies the upper threshold point used to model a rising transition on this pin. This value is specified as a percentage.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>               |
| threshold_voltage_group       |                                                                                                                                                                                                                                                                                                 |
|                               | <p>Returns name of a group/category – that is specified based on the timing library's threshold voltage (high or low) characteristics.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| wireload_models               |                                                                                                                                                                                                                                                                                                 |

|  |                                                                                                                                              |
|--|----------------------------------------------------------------------------------------------------------------------------------------------|
|  | Returns a list of the wireload models defined in the library<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
|--|----------------------------------------------------------------------------------------------------------------------------------------------|

## library\_set

### Parent Objects

[timing\\_condition](#), [root](#)

### Definition

Specifies a group of library files to be treated as a single entity so that higher-level descriptions (delay\_corners) can simply refer to the library configuration by name. All non-physical libraries used by the timing must be part of a library\_set - including Liberty, AOCV, SOCV, and signal integrity library formats. Use the create\_library\_set command to define new library\_sets and the update\_library\_set command to update the attributes of existing library\_sets.

| Attribute     | Description                                                                                                                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| aocv_files    |                                                                                                                                                                                      |
|               | Specifies the list of optional AOCV derating library files associated with the library_set<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No           |
| libraries     |                                                                                                                                                                                      |
|               | Specifies the list of timing library objects that result from the import of the library_files<br><b>Type:</b> <a href="#">obj(library)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| library_files |                                                                                                                                                                                      |
|               | Specifies the list of Liberty timing library files associated with the library_set<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                   |
| name          |                                                                                                                                                                                      |

|            |                                                                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | Provides the name of this library_set object as specified by create_library_set.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                           |
| obj_type   |                                                                                                                                                                                                            |
|            | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">library_set</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| si_files   |                                                                                                                                                                                                            |
|            | Specifies the list of optional signal integrity (.cdb) library files associated with the library_set<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                       |
| socv_files |                                                                                                                                                                                                            |
|            | Specifies the list of optional SOCV variation library files associated with the library_set<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                |

# marker

## Parent Objects

[design](#), [root](#)

## Definition

A violation marker. All markers have a .bbox and .originator set, and optionally .polygon for polygon shapes. Other attributes depend on the marker. Markers created by internal check or report commands will have .originator != external and have a .layer, .type, .subtype, .message, and .message\_id defined by those commands. Markers created with read\_markers or create\_marker will have .originator == external and will have .layer, .user\_type, .user\_subtype, .user\_originator, and .message based on the external input.

| Attribute | Description |
|-----------|-------------|
|-----------|-------------|

|            |                                                                                                                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bbox       |                                                                                                                                                                                                        |
|            | <p>Bounding box of the marker shape.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                     |
| is_false   |                                                                                                                                                                                                        |
|            | <p>Indicates the marker is marked as false violation by user.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                        |
| is_hidden  |                                                                                                                                                                                                        |
|            | <p>Indicates the marker is hidden by user in violation browser.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                      |
| layer      |                                                                                                                                                                                                        |
|            | <p>The layer of marker</p> <p><b>Type:</b> <a href="#">obj(layer)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                             |
| message    |                                                                                                                                                                                                        |
|            | <p>The message attached to the marker</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                  |
| message_id |                                                                                                                                                                                                        |
|            | <p>The message ID used internally for this message</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                        |
| obj_type   |                                                                                                                                                                                                        |
|            | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum (marker)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| objects         |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                 | <p>The objects which caused the DRC. The list may be empty, and is currently limited to at most 2 objects.</p> <p><b>Type:</b> <code>obj(inst)* obj(net)* obj(pin)* obj(port)* obj(bump)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                 |
| originator      |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                 | <p>The originator of the marker from an internal command, or 'external' if the marker is from create_marker or read_markers.</p> <p><b>Type:</b> <code>enum</code></p> <p><b>Enum Values:</b> unknown check external check_place route_litho verify_power_domain check_filler nano_route euler route_design check_pin_assignment pg ccopt check_floorplan check_bus_guide check_safety_mechanism</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| polygon         |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                 | <p>Polygon boundary for the marker if it is not a rectangle. The first point is not repeated as the last point in the list.</p> <p><b>Type:</b> <code>polygon</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                             |
| subtype         |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                 | <p>The marker subtype if it was created be an internal check or report command (when .originator != external).</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> {}</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                           |
| type            |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                 | <p>The marker type if it was created by an internal check or report command (when .originator != external).</p> <p><b>Type:</b> <code>enum</code></p> <p><b>Enum Values:</b> none drc antenna connectivity floorplan overlap density ir_drop xtalk ac_limit mixed_signal electrical place route_litho busplanning msv ccopt safety scan_chain_probe via_cell</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                     |
| user_originator |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | The originator given by the user for external markers from create_marker or read_marker (when .originator == external).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> {}<br><b>Edit:</b> No |
| user_subtype |                                                                                                                                                                                                         |
|              | The subtype given by the user for external markers from create_marker or read_marker (when .originator == external).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> {}<br><b>Edit:</b> No    |
| user_type    |                                                                                                                                                                                                         |
|              | The type given by the user for external markers from create_marker or read_marker (when .originator == external).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> {}<br><b>Edit:</b> No       |

## message

### Parent Objects

[root](#)

### Definition

Formatted error, warning and info messages (e.g. ERROR

| Attribute | Description                                                                                                                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name |                                                                                                                                                                                                              |
|           | The message name (e.g. ABCDEF-123 for the message "ERROR: (ABCDEF-123):...").<br>The DPO name would be "message:ABCDEF-123".<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| count     |                                                                                                                                                                                                              |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | <p>The number of times message has been issued in current session. The count can be reset by <code>set_db \$message .count 0</code> or <code>set_message -id xxx -reset_count</code>.</p> <p><b>Type:</b> <code>int</code><br/> <b>Default:</b> <code>no_value</code><br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                        |
| long_help |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|           | <p>Some messages have additional help information. It is also available with 'man ABCDEF-123' in the DESCRIPTION section.</p> <p>This is also the "DESCRIPTION" part of <code>get_message -long</code>. If the "DESCRIPTION" of a message is "There are no further details for this message", <code>get_db</code> doesn't need to return it, but return an empty string instead.</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p> |
| max_print |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|           | <p>Specifies the maximum number of times the message will be printed to the log file and screen. -1 is unlimited. 0 is suppressed.</p> <p><b>Type:</b> <code>int</code><br/> <b>Default:</b> 20<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                              |
| message   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|           | <p>A brief explanation of the message. It is also available with 'help ABCDEF-123'. It uses printf syntax with %s, %d, etc. for message specific arguments.</p> <p>This is the <code>get_message -short</code> string.</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                           |
| name      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|           | <p>The message name (e.g. ABCDEF-123 for the message "ERROR: (ABCDEF-123):..."). The DPO name would be "message:ABCDEF-123".</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                     |
| obj_type  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|           | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <code>enum</code> (<code>message</code>)<br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                             |

| on_error |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <p>The error action for the specified message. This only applies if the message has severity = error, and is ignored for messages with severity of warning or info. Valid values are:</p> <ul style="list-style-type: none"><li>msg_only: Write out message, with normal command behavior.</li><li>exit: Fatal error. Exit and return to Linux.</li><li>stop_script: If message occurs inside 'source &lt;file&gt;', then the source command stops and returns.</li></ul> <p><b>Type:</b> <a href="#">enum</a><br/><b>Enum Values:</b> exit stop_script msg_only<br/><b>Default:</b> msg_only<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                      |
| severity |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|          | <p>The severity of the message. The severity is "default" until some application writes it out and sets the severity, or the user sets it. Setting it to "default" will undo any previous setting, and then applications will set it if again the next time they write it out. Valid values are:</p> <ul style="list-style-type: none"><li>default: A default severity is set to all messages until it is issued at least once.</li><li>info: An information message. No user action needed.</li><li>warning: A possible problem, but the command can continue.</li><li>error: A problem causing wrong behavior. In some cases, the command cannot continue and stops, and sometimes it can continue to enable debugging or prototyping with errors.</li></ul> <p><b>Type:</b> <a href="#">enum</a><br/><b>Enum Values:</b> default error warning info<br/><b>Default:</b> default<br/><b>Edit:</b> Yes</p> |

## module

### Parent Objects

[hinst](#), [design](#), [root](#)

### Definition

Cell module

| Attribute | Description |
|-----------|-------------|
|-----------|-------------|



|                                        |                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow_ilm_eco                          |                                                                                                                                                                                                                                             |
|                                        | <p>The attribute is only valid when the module is an ILM. If true, optimizer can optimize the ILM boundary interface logic to improve timing</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>  |
| arch_filename                          |                                                                                                                                                                                                                                             |
|                                        | <p>Source file that contains the specified module (Verilog) or architecture (VHDL).</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                            |
| arch_name                              |                                                                                                                                                                                                                                             |
|                                        | <p>The name of the specified module (Verilog) or architecture (VHDL).</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                          |
| base_name                              |                                                                                                                                                                                                                                             |
|                                        | <p>Name of cell</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                 |
| boundary_optimize_constant_hpins       |                                                                                                                                                                                                                                             |
|                                        | <p>Allow constant propagation through hierarchical boundary pins.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> 1 0 true false inherited<br/> <b>Default:</b> inherited<br/> <b>Edit:</b> Yes</p>                       |
| boundary_optimize_equal_opposite_hpins |                                                                                                                                                                                                                                             |
|                                        | <p>Allow connection of hierarchical boundary pins through equal/opposite boundary pins.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> 1 0 true false inherited<br/> <b>Default:</b> inherited<br/> <b>Edit:</b> Yes</p> |
| boundary_optimize_feedthrough_hpins    |                                                                                                                                                                                                                                             |

|                                           |                                                                                                                                                                                                                                         |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>Allow routing of feedthrough boundary pins around a hierarchical instance.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p> |
| boundary_optimize_invert_hpins            |                                                                                                                                                                                                                                         |
|                                           | <p>Allow inversion of hierarchical boundary pins.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                             |
| boundary_opto                             |                                                                                                                                                                                                                                         |
|                                           | <p>Control boundary-optimization on a subdesign.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false strict_no</p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                       |
| clock_gating_auto_path_adjust             |                                                                                                                                                                                                                                         |
|                                           | <p>Controls automatic timing adjustment on clock gate enable paths.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> inherited none fixed variable</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>      |
| clock_gating_auto_path_adjust_fixed_delay |                                                                                                                                                                                                                                         |
|                                           | <p>Fixes the path adjustment automatically added to clock gate enable pins.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                            |
| clock_gating_auto_path_adjust_multiplier  |                                                                                                                                                                                                                                         |
|                                           | <p>Scales the automatic load-dependent path adjustment added to clock gate enable pins.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                          |
| clock_gating_exclude                      |                                                                                                                                                                                                                                         |

|                            |                                                                                                                                                                                                                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Exclude this design from clock-gating.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                  |
| clock_gating_hierarchical  |                                                                                                                                                                                                                                                                                                                           |
|                            | <p>Create hierarchical clock gates in this module.</p> <p><b>Type:</b> <code>enum</code></p> <p><b>Enum Values:</b> auto false</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p>                                                                                                                                    |
| clock_gating_max_flops     |                                                                                                                                                                                                                                                                                                                           |
|                            | <p>Set fanout limit to clock-gating cell.</p> <p><b>Type:</b> <code>int</code></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                |
| clock_gating_min_flops     |                                                                                                                                                                                                                                                                                                                           |
|                            | <p>Set minimum number of flops needed to add a clock-gating cell.</p> <p><b>Type:</b> <code>int</code></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                        |
| clock_gating_module        |                                                                                                                                                                                                                                                                                                                           |
|                            | <p>Module used for clock gating circuitry.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                  |
| clock_gating_test_signal   |                                                                                                                                                                                                                                                                                                                           |
|                            | <p>Set specified test signal to connect up test logic in clock-gating circuitry. Specify use_shift_enable to derive test logic from shift enable of the gated flops, or use_none to disable test connection for the module.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| control_logic_optimization |                                                                                                                                                                                                                                                                                                                           |

|                            |                                                                                                                                                                                                                                                        |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | Controls the effort level for optimization of control logic on this module (basic/advanced/none/inherited).<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> basic advanced none inherited<br><b>Default:</b> inherited<br><b>Edit:</b> Yes |
| delete_unloaded_insts      |                                                                                                                                                                                                                                                        |
|                            | Control deletion of unloaded hierarchical instances.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> 1 0 true false inherited<br><b>Default:</b> inherited<br><b>Edit:</b> Yes                                                             |
| delete_unloaded_seqs       |                                                                                                                                                                                                                                                        |
|                            | Control deletion of unloaded sequential instances.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> 1 0 true false inherited<br><b>Default:</b> inherited<br><b>Edit:</b> Yes                                                               |
| design                     |                                                                                                                                                                                                                                                        |
|                            | The design that contains this module.<br><b>Type:</b> <a href="#">obj(design)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                            |
| design_verification_effort |                                                                                                                                                                                                                                                        |
|                            | Specifies design verification effort level for LEC. Can be used to disable problematic optimizations in bulk.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> inherited<br><b>Edit:</b> Yes                                                  |
| dft_dont_scan              |                                                                                                                                                                                                                                                        |
|                            | Exclude this design from scan-synthesis {inherited   true   false}.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> false true inherited<br><b>Default:</b> inherited<br><b>Edit:</b> Yes                                                  |
| dont_touch                 |                                                                                                                                                                                                                                                        |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>This attributes defines the user preservation status of the module during optimization. Setting this attribute will set the dont_touch attribute on all hinsts of the same module. This setting will apply to all insts within the hinst unless overridden at a lower level hinst or on the inst object itself. The dont_touch_effective attribute on each child inst and hinst will return the resolved value.</p> <p>Use 'help inst dont_touch' to see the enum value definitions.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> none false true delete_ok const_prop_size_delete_ok const_prop_delete_ok size_delete_ok size_ok size_same_height_ok size_same_footprint_ok</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p>                                    |
| dont_touch_file   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                   | <p>Specifies the sourced file to set dont_touch.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| dont_touch_hports |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                   | <p>This attribute defines the user preservation status for the module object hports during optimization.</p> <p><b>Supported values:</b></p> <p>none: Unconstrained</p> <p>false: Can add/remove ports</p> <p>true: Cannot add/remove ports</p> <p>delete_ok: Can delete ports (if they have no fanout)</p> <p>add_ok: We cannot delete, or change the polarity or any hport but can add or duplicate hports</p> <p>invert_ok: We cannot delete, duplicate, or add but can change the polarity of any hport</p> <p>add_invert_ok: We cannot delete any hport but can add, duplicate, and change the polarity</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> none false true delete_ok add_ok invert_ok add_invert_ok</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> |
| dont_use_cells    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>List of cell names (wildcards supported) to disallow for this module during optimization. Setting this applies to all hinsts sharing the module. Overrides any library dont_use values</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                    |
| dont_use_cells_effective |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>The resolved list of all cell names to disallow during optimization for hinsts of this module, based on the library dont_use and the dont_use_cells and use_cells attributes of this module or the closest parent hinst with a non-empty list. The precedence is: use_cells of this hinst (or closest parent if empty), then dont_use_cells of this hinst (or closest parent if empty), then the library dont_use setting.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| dp_csa                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Controls CSA operation on design/subdesign in 'synthesize -to_generic' (basic/none/inherited).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic none inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                       |
| dp_csa_factorize         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Controls datapath csa_factorize on a design in 'syn_generic'.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic none inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                        |
| dp_opt_power_opt         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Controls datapath power optimization flow on a given subdesign.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                  |
| dp_rename                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                     |                                                                                                                                                                                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>Attribute to control the datapath rewriting optimization level on a subdesign (basic/advanced/none/inherited).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic advanced none inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p> |
| dp_sharing          |                                                                                                                                                                                                                                                                                  |
|                     | <p>To control the datapath sharing on a design/subdesign in 'synthesize -to_generic' (advanced/none/inherited).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> advanced none inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>         |
| dp_speculation      |                                                                                                                                                                                                                                                                                  |
|                     | <p>Controls datapath speculation on a design in 'syn_generic'.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic none inherited</p> <p><b>Default:</b> inherited</p> <p><b>Edit:</b> Yes</p>                                                             |
| embedded_script     |                                                                                                                                                                                                                                                                                  |
|                     | <p>Stores the SDC command script embedded in the source file that contains the module (Verilog) or entity (VHDL) specification for this design.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                 |
| entity_filename     |                                                                                                                                                                                                                                                                                  |
|                     | <p>Source file that contains the specified module (Verilog) or entity (VHDL).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                   |
| entity_name         |                                                                                                                                                                                                                                                                                  |
|                     | <p>Name of the specified module (Verilog) or entity (VHDL).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                      |
| has_dp_rewrite_gint |                                                                                                                                                                                                                                                                                  |

|                         |                                                                                                                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>This design/subdesign has rewritten instance.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> No</p>                                                                |
| has_dp_shared_gint      |                                                                                                                                                                                                              |
|                         | <p>This design/subdesign has shared instance.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> No</p>                                                                   |
| hdl_all_filelist        |                                                                                                                                                                                                              |
|                         | <p>Attribute to get all the files along with the included hdl files and package files for a design or subdesign.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| hdl_config_name         |                                                                                                                                                                                                              |
|                         | <p>Name of the configuration used to build this module.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                         |
| hdl_cw_list             |                                                                                                                                                                                                              |
|                         | <p>List of CW components used by a design.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                      |
| hdl_elab_command_params |                                                                                                                                                                                                              |
|                         | <p>List of {parameter value} pair that are passed to 'elaborate -parameters' command.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                            |
| hdl_filelist            |                                                                                                                                                                                                              |
|                         | <p>List of HDL files read prior to the elaboration of a design.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                 |



|                           |                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hdl_parameters            |                                                                                                                                                                                                                                                                  |
|                           | <p>List of {parameter value cmdset is_default} used to 'elaborate' a design. 'cmdset' is '1' only if the parameter is set using the 'elaborate -parameters' command.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| hdl_pipeline_comp         |                                                                                                                                                                                                                                                                  |
|                           | <p>Specifies whether the design contains a \$pipeline component.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> No</p>                                                                                                    |
| hdl_user_name             |                                                                                                                                                                                                                                                                  |
|                           | <p>Name of the module (Verilog) or entity (VHDL) for this design.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                   |
| hdl_vdp_list              |                                                                                                                                                                                                                                                                  |
|                           | <p>Information about datapath function primitives used in a module.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                  |
| hinsts                    |                                                                                                                                                                                                                                                                  |
|                           | <p>The hinsts that are derived from (instantiate) this module.</p> <p><b>Type:</b> <a href="#">obj(hinst)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                  |
| incrsyn_force_not_replace |                                                                                                                                                                                                                                                                  |
|                           | <p>Disable cache replacement, and force resynthesis of this module during incremental run.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                         |
| is_ilm                    |                                                                                                                                                                                                                                                                  |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>This attribute is true if the module is a ILM. This attribute will affect the read_only_effective and dont_touch_effective attribute on all insts and hinsts within the hinsts of this module. It cannot be overridden by other hinst or inst values.</p> <p><b>Supported values:</b><br/>false: This module is not an ILM<br/>true: This module is an ILM</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> No</p> |
| is_retimed                     |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                | <p>Indicates that the design has been retimed.</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                               |
| library_name                   |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                | <p>Name of the (logical) library in which module (Verilog) or entity (VHDL) reside.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                            |
| logical_hier                   |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                | <p>Indicates if the module is a logical hierarchy object, that is, it corresponds to a user-defined entity (in VHDL) or user-defined module (in Verilog).</p> <p><b>Type:</b> bool<br/><b>Default:</b> true<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                     |
| minimize_uniquify              |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                | <p>Try to avoid uniquification of specific subdesign.</p> <p><b>Type:</b> enum<br/><b>Enum Values:</b> 1 0 true false inherited<br/><b>Default:</b> inherited<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                   |
| multibit_allow_async_phase_map |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                | <p>Allow interchanging async pins for multibit mapping.</p> <p><b>Type:</b> enum<br/><b>Enum Values:</b> 1 0 true false inherited<br/><b>Default:</b> inherited<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                 |

|                                        |                                                                                                                                                                                                       |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name                                   |                                                                                                                                                                                                       |
|                                        | Name of cell<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                          |
| obj_type                               |                                                                                                                                                                                                       |
|                                        | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">module</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| opt_retime_move_mux_loop_with_register |                                                                                                                                                                                                       |
|                                        | Move the feedback mux loop along with the register during retiming. Module level control.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes                            |
| pbs_mim_keep_folded                    |                                                                                                                                                                                                       |
|                                        | Keep the module folded in MiM flow.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                 |
| retime                                 |                                                                                                                                                                                                       |
|                                        | Retime this design during synthesis.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                |
| retime_hard_region                     |                                                                                                                                                                                                       |
|                                        | Do not move flops across the boundaries of this subdesigns.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                         |
| retime_period_percentage               |                                                                                                                                                                                                       |

|                        |                                                                                                                                                                                                                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | Adjust target clock period in retiming.<br><b>Type:</b> double<br><b>Default:</b> 1.0<br><b>Edit:</b> Yes                                                                                                                                                                                                            |
| retime_ssw_sync_enable |                                                                                                                                                                                                                                                                                                                      |
|                        | Directs retiming to use SAT-sweep to find equivalence classes for synchronous enable signals. Module level control.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                |
| speed_grade            |                                                                                                                                                                                                                                                                                                                      |
|                        | Effective speed_grade for a subdesign.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                               |
| sub_arch               |                                                                                                                                                                                                                                                                                                                      |
|                        | Datapath sub architecture type.<br><b>Type:</b> enum<br><b>Enum Values:</b> no_value non_booth booth radix8 ao barrel<br><b>Default:</b> no_value<br><b>Edit:</b> Yes                                                                                                                                                |
| ungroup_ok             |                                                                                                                                                                                                                                                                                                                      |
|                        | To allow ungroup for this sub-design.<br><b>Type:</b> enum<br><b>Enum Values:</b> false true strict_no<br><b>Default:</b> true<br><b>Edit:</b> Yes                                                                                                                                                                   |
| use_cells              |                                                                                                                                                                                                                                                                                                                      |
|                        | List of cell names to allow for this module during optimization. Setting this applies to all hinsts sharing the module All lib_cells of each base_cell will be allowed. Overrides cells in the dont_use_cells list and any library dont_use values.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| user_speed_grade       |                                                                                                                                                                                                                                                                                                                      |

|               |                                                                                                                                                                                                                                                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | User defined speed grade.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                                                                                                                                                  |
| user_sub_arch |                                                                                                                                                                                                                                                                                                                                                                             |
|               | User defined sub_architecture. Valid values 'booth', 'nonbooth', 'radix8', and 'radix8_fast' apply to multipliers. Valid values 'ao' and 'barrel' apply to shifters.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> no_value non_booth booth radix8 ao barrel barrel_radix4 barrel_radix8 barrel_radix16 radix8_fast<br><b>Default:</b> ""<br><b>Edit:</b> Yes |

## net

### Parent Objects

[flexible\\_htree](#), [marker](#), [bus](#), [resistor](#), [port](#), [special\\_wire](#), [hpin](#), [hnet](#), [what\\_if\\_wire](#), [via](#), [what\\_if\\_via](#), [timing\\_point](#), [design](#), [clock\\_tree](#), [net\\_group](#), [timing\\_path](#), [bump](#), [root](#), [power\\_domain](#), [virtual\\_wire](#), [port\\_shape](#), [pg\\_pin](#), [patch\\_wire](#), [wire](#), [pin](#), [interposer\\_route\\_rule](#), [special\\_via](#), [route\\_type](#)

### Definition

canonical net

| Attribute                 | Description                                                                                                                                                                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| annotated_capacitance_max |                                                                                                                                                                                                                                                                            |
|                           | Returns maximum annotated capacitance of the net for late path analysis. You can use -index to return the derate factor for specific views.<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| annotated_capacitance_min |                                                                                                                                                                                                                                                                            |

|                          |                                                                                                                                                                                                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Returns the minimum annotated capacitance of the net for early path analysis. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| annotated_resistance_max |                                                                                                                                                                                                                                                                                        |
|                          | <p>Returns the maximum annotated resistance of the net for late path analysis. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| annotated_resistance_min |                                                                                                                                                                                                                                                                                        |
|                          | <p>Returns the minimum annotated resistance of the net for early path analysis. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| arcs                     |                                                                                                                                                                                                                                                                                        |
|                          | <p>Returns a list of arc objects that are associated with this net.</p> <p><b>Type:</b> obj(arc)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                 |
| attackers                |                                                                                                                                                                                                                                                                                        |
|                          | <p>Returns the RC value.</p> <p><b>Type:</b> obj(net)*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                   |
| attackers_rc             |                                                                                                                                                                                                                                                                                        |
|                          | <p>Returns the RC value.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                            |

|                               |                                                                                                                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name                     |                                                                                                                                                                                                                                                                                        |
|                               | <p>The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                        |
| bbox                          |                                                                                                                                                                                                                                                                                        |
|                               | <p>Bounding box of all the wires, vias, pins, and ports of the net. Does not include special wires or special vias.</p> <p><b>Type:</b> <a href="#">rect</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                          |
| bottom_preferred_shield_layer |                                                                                                                                                                                                                                                                                        |
|                               | <p>The bottom preferred layer on which to add shield on for the net.</p> <p><b>Type:</b> <a href="#">obj(layer)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                  |
| bus                           |                                                                                                                                                                                                                                                                                        |
|                               | <p>The bus for this net.</p> <p><b>Type:</b> <a href="#">obj(bus)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                 |
| capacitance_max               |                                                                                                                                                                                                                                                                                        |
|                               | <p>Returns the total capacitance of the net used for late path analysis. You can use -index to return the capacitance for specific views.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| capacitance_min               |                                                                                                                                                                                                                                                                                        |
|                               | <p>Returns the total capacitance of the net for early path analysis. You can use -index to return the capacitance for specific views.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>     |
| connections                   |                                                                                                                                                                                                                                                                                        |

|                          |                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Returns a list of connections of the net (pins and ports).</p> <p><b>Type:</b> <code>obj(pin)* obj(port)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                |
| constant                 |                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>Returns if this net has a 1'b0 or 1'b1 on any hpin or hnet, or a supply0/supply1 for any hnet. It does not include timing set_case_analysis assertions, or propagation through cells, or the effects of a constant driver on the net.</p> <p><b>Type:</b> <code>enum</code></p> <p><b>Enum Values:</b> 0 1 no_constant</p> <p><b>Default:</b> no_constant</p> <p><b>Edit:</b> No</p> |
| coupling_capacitance_max |                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>Returns the maximum value of coupling capacitance of a net over all the views.</p> <p><b>Type:</b> <code>double</code></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                       |
| coupling_capacitance_min |                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>Returns the minimum value of coupling capacitance of a net over all the views.</p> <p><b>Type:</b> <code>double</code></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                       |
| cts_ideal_net            |                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>If set the clock tree timing engine will consider this net as ideal.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                  |
| cts_net_type             |                                                                                                                                                                                                                                                                                                                                                                                         |



|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>For a net, return the CTS net type.<br/>Will be one of:</p> <ul style="list-style-type: none"> <li>. "top" - A clock net that has fanout above cts_top_fanout_transitive_count.</li> <li>. "trunk" - A CTS clock net that is not a "top" net nor a "leaf" net.</li> <li>. "leaf" - A CTS clock net that has fanout to only clock sinks.</li> <li>. "unknown" - A CTS clock net that has not been analyzed.</li> <li>. "" - Not a CTS clock net.</li> </ul> <p>See also the get_clock_tree_nets command.<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| delta_delay_max      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                      | <p>Return the delta delay of max corner<br/><b>Type:</b> <a href="#">double</a><br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                              |
| delta_delay_max_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                      | <p>Returns the delta fall delay of max corner.<br/><b>Type:</b> <a href="#">double</a><br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                       |
| delta_delay_max_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                      | <p>Returns the delta rise delay for max corner.<br/><b>Type:</b> <a href="#">double</a><br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                      |
| delta_delay_min      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                      | <p>Returns the delta delay for min corner.<br/><b>Type:</b> <a href="#">double</a><br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                           |
| delta_delay_min_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Returns the delta fall delay for min corner.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                               |
| delta_delay_min_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Returns the delta rise delay for min corner.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                               |
| dont_touch           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>This attribute defines the preservation status of an net during optimization. Setting this attribute will preserve all connections on this net. When set, this overrides any setting on hnets of this net. Also, note that the .use attribute for the net can also cause the net to be preserved.</p> <p><b>Supported values:</b></p> <p>false: Unconstrained</p> <p>true: Cannot touch</p> <p>delete_ok: Can delete (if they have no sinks)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true delete_ok const_prop_delete_ok</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> |
| dont_touch_sources   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Dictionary of {source value} pairs contributing to the dont_touch_effective attribute for this object :</p> <p>{user &lt;value&gt;} {dont_touch_network &lt;value&gt;} {parent &lt;value&gt;}</p> <p>user # The user dont_touch value</p> <p>dont_touch_network # The net is in the fanout cone of a set_dont_touch_network constraint</p> <p>parent # The effective dont_touch value from an hinst above</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                           |
| driver_pins          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>These are the drivers of the net that are pins (no port drivers are returned). Pins that have direction = out or inout are considered drivers.</p> <p><b>Type:</b> <code>obj(pin)*</code><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                    |
| driver_ports                           |                                                                                                                                                                                                                                                                                                  |
|                                        | <p>These are the drivers of the net that are ports (no pin drivers are returned). Ports that have direction = in or inout are considered drivers.</p> <p><b>Type:</b> <code>obj(port)*</code><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                   |
| drivers                                |                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Ports and pins that drive the net. These are pins that have direction = out or inout, and ports that have direction = in or inout.</p> <p><b>Type:</b> <code>obj(pin)* obj(port)*</code><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                     |
| duty_cycle                             |                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Returns the duty cycle.</p> <p><b>Type:</b> <code>string</code><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                             |
| early_fall_clk_net_delta_derate_factor |                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Returns the early derating factor for falling SI/delta delays on data path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| early_fall_clk_net_derate_factor       |                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Returns the early derating factor for falling static delays on clock path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |

|                                         |                                                                                                                                                                                                                                                                                            |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| early_fall_data_net_delta_derate_factor |                                                                                                                                                                                                                                                                                            |
|                                         | <p>Returns the early derating factor for falling SI/delta delays on clock path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| early_fall_data_net_derate_factor       |                                                                                                                                                                                                                                                                                            |
|                                         | <p>Returns the early derating factor for falling static delays on data path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| early_rise_clk_net_delta_derate_factor  |                                                                                                                                                                                                                                                                                            |
|                                         | <p>Returns the early derating factor for rising SI/delta delays on data path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| early_rise_clk_net_derate_factor        |                                                                                                                                                                                                                                                                                            |
|                                         | <p>Returns the early derating factor for rising static delays on clock path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| early_rise_data_net_delta_derate_factor |                                                                                                                                                                                                                                                                                            |
|                                         | <p>Returns the early derating factor for rising SI/delta delays on clock path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| early_rise_data_net_derate_factor       |                                                                                                                                                                                                                                                                                            |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Returns the early derating factor for rising static delays on data path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                 |
| eco_opt_add_cells       |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                         | <p>Specifies the cells to be used by eco opt engine for inserting buffer/inverter on the net.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                         |
| escaped_name            |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                         | <p>The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax, so bus-bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So i1/i2 is a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a[0\] is a scalar.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| frequency               |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                         | <p>Reports the frequency of the net.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                   |
| has_detailed_parasitics |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                         | <p>Returns a value of true if the net, or one of its parts, has detailed parasitics associated with it from either RC extraction or SPEF annotation.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                  |
| has_valid_parasitics    |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                         | <p>Returns true if the net has complete RC info from SPEF.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                            |

|                         |                                                                                                                                                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hierarchical_name       |                                                                                                                                                                                                                                    |
|                         | <p>hierarchical_name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |
| hnets                   |                                                                                                                                                                                                                                    |
|                         | <p>List of hNets which make up this net.</p> <p><b>Type:</b> <a href="#">obj(hnet)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                       |
| is_clock                |                                                                                                                                                                                                                                    |
|                         | <p>Indicates that net is a clock according to timing constraints and tracing. It is only valid if the timing-graph has been created.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_clock_source_latency |                                                                                                                                                                                                                                    |
|                         | <p>Returns true if object is part of source latency network</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                 |
| is_custom_route         |                                                                                                                                                                                                                                    |
|                         | <p>Indicates that the net has Virtuoso/OA custom routing constraints that NR does not support, so NR should not route this net.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| is_dirty                |                                                                                                                                                                                                                                    |
|                         | <p>Indicates that the net is dirty.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                              |
| is_early_global_routed  |                                                                                                                                                                                                                                    |

|                 |                                                                                                                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <p>Indicate that the net is from early global route</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                           |
| is_edit         |                                                                                                                                                                                                                                                                                                                 |
|                 | <p>Indicate that the wire/via on the net has been modified. This attribute only works when "set_db edit_wire_create_is_edit_flag 1". Please look up for more details by "help edit_wire_create_is_edit_flag".</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> |
| is_external     |                                                                                                                                                                                                                                                                                                                 |
|                 | <p>Indicates that net is connected to a to a top-level port.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                      |
| is_fixed_bump   |                                                                                                                                                                                                                                                                                                                 |
|                 | <p>Indicates that net is fixed to bump connection.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                |
| is_ground       |                                                                                                                                                                                                                                                                                                                 |
|                 | <p>Indicate that the net is ground net</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                            |
| is_hierarchical |                                                                                                                                                                                                                                                                                                                 |
|                 | <p>Returns true if net is a part of hierarchichal module</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                 |
| is_ideal        |                                                                                                                                                                                                                                                                                                                 |

|                                       |                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>Returns true if the net is part of an ideal network due to the assertion or propagation of the set_ideal_network constraint. You can use -index to return the is_ideal value for specific views.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_ilm                                |                                                                                                                                                                                                                                                                                                                                      |
|                                       | <p>Specify if a top level net is connecting to ILM (Interface Logic Model) modules and some or all terms (either drive or sink) of the net are inside ILM module.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                      |
| is_partial_parallel_route_net         |                                                                                                                                                                                                                                                                                                                                      |
|                                       | <p>Returns true if the net is a partial parallel routed net.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                           |
| is_physical_only                      |                                                                                                                                                                                                                                                                                                                                      |
|                                       | <p>Indicates that the net is a physical only net that does not get written to logical Verilog netlist.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                 |
| is_power                              |                                                                                                                                                                                                                                                                                                                                      |
|                                       | <p>Indicate that the net is power net</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                  |
| is_voltage_asserted                   |                                                                                                                                                                                                                                                                                                                                      |
|                                       | <p>Returns whether a pg net voltage is inferred by tool or user asserted.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                     |
| late_fall_clk_net_delta_derate_factor |                                                                                                                                                                                                                                                                                                                                      |



|                                        |                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Returns the late derating factor for falling SI/delta delays on data path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| late_fall_clk_net_derate_factor        |                                                                                                                                                                                                                                                                                           |
|                                        | <p>Returns the late derating factor for falling static delays on clock path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| late_fall_data_net_delta_derate_factor |                                                                                                                                                                                                                                                                                           |
|                                        | <p>Returns the late derating factor for falling SI/delta delays on clock path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| late_fall_data_net_derate_factor       |                                                                                                                                                                                                                                                                                           |
|                                        | <p>Returns the late derating factor for falling static delays on data path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| late_rise_clk_net_delta_derate_factor  |                                                                                                                                                                                                                                                                                           |
|                                        | <p>Returns the late derating factor for rising SI/delta delays on data path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| late_rise_clk_net_derate_factor        |                                                                                                                                                                                                                                                                                           |

|                                        |                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Returns the late derating factor for rising static delays on clock path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| late_rise_data_net_delta_derate_factor |                                                                                                                                                                                                                                                                                          |
|                                        | <p>Returns the late derating factor for rising SI/delta delays on clock path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| late_rise_data_net_derate_factor       |                                                                                                                                                                                                                                                                                          |
|                                        | <p>Returns the late derating factor for rising static delays on data path nets. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| load_pins                              |                                                                                                                                                                                                                                                                                          |
|                                        | <p>These are the loads of the net that are pins (no port loads are returned). Pins that have direction = in or inout are considered loads.</p> <p><b>Type:</b> obj(pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                            |
| load_ports                             |                                                                                                                                                                                                                                                                                          |
|                                        | <p>These are the loads of the net that are ports (no pin loads are returned). Ports that have direction = out or inout are considered loads.</p> <p><b>Type:</b> obj(port)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                         |
| loads                                  |                                                                                                                                                                                                                                                                                          |
|                                        | <p>Ports and pins that are loads for the net. These are pins that have direction = in or inout, and ports that have direction = out or inout.</p> <p><b>Type:</b> obj(pin)* obj(port)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                              |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lumped_capacitance_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                        | <p>Returns the total lumped capacitance of the net used for late path analysis. You can use -index to return the capacitance for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                      |
| lumped_capacitance_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                        | <p>Returns the total lumped capacitance of the net for early path analysis. You can use -index to return the capacitance for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                          |
| lumped_resistance_max  |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                        | <p>Returns the maximum resistance of the net for late path analysis. You can use -index to return the value for a specific view</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                       |
| lumped_resistance_min  |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                        | <p>Returns the maximum resistance of the net for late path analysis. You can use -index to return the value for a specific view</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                       |
| max_voltage            |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                        | <p>Max voltage for the net, determined by the max net voltage of all active hold views. Net voltage is determined using the following procedure: voltage from net's driver pin, associated power from CPF related_power_pins command or Liberty related_pg_pin attribute or LEF SUPPLYSENSITIVITY statement; power domain operating voltage; or default system voltage.</p> <p><b>Type:</b> voltage</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| min_voltage            |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <p>Min voltage for the net, determined by the min net voltage of all active setup views. Net voltage is determined using the following procedure: voltage from net's driver pin, associated power from CPF related _power_pins command or Liberty related _pg_pin attribute or LEF SUPPLYSENSITIVITY statement; power domain operating voltage; or default system voltage.</p> <p><b>Type:</b> voltage<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                              |
| name            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                 | <p>The same as .escaped_name without any \ escape chars. This name is commonly used to avoid problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so that flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| num_connections |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                 | <p>Number of connections to the net (number of pins + number of ports)</p> <p><b>Type:</b> int<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                      |
| num_driver_pins |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                 | <p>Returns the number of driver pins of the net.</p> <p><b>Type:</b> int<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                      |
| num_drivers     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                 | <p>Number of drivers for the net. These are pins that have direction = out or inout, and ports that have direction = in or inout.</p> <p><b>Type:</b> int<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                           |
| num_load_pins   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                          |                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Returns the number of load pins of the net.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                          |
| num_loads                |                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Number of loads for the net. These are pins that have direction = in or inout, and ports that have direction = out or inout.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                  |
| obj_type                 |                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">net</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                           |
| patch_wires              |                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>List of patch_wire</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">patch_wire</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                               |
| pin_capacitance_max      |                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Returns the portion of the net's total capacitance which comes the library max pin capacitance values. This is used for late path delay calculation. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pin_capacitance_max_fall |                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>pin_capacitance_max_fall</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                |
| pin_capacitance_max_rise |                                                                                                                                                                                                                                                                                                                                                                               |

|                          |                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>pin_capacitance_max_rise</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                 |
| pin_capacitance_min      |                                                                                                                                                                                                                                                                                                                                                                |
|                          | <p>Returns the portion of the net's total capacitance which comes the library min pin capacitance values. This is used for early path delay calculation. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pin_capacitance_min_fall |                                                                                                                                                                                                                                                                                                                                                                |
|                          | <p>pin_capacitance_min_fall</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                 |
| pin_capacitance_min_rise |                                                                                                                                                                                                                                                                                                                                                                |
|                          | <p>pin_capacitance_min_rise</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                 |
| pins                     |                                                                                                                                                                                                                                                                                                                                                                |
|                          | <p>Returns all pins on the net.</p> <p><b>Type:</b> obj(pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                             |
| power_duty_cycle         |                                                                                                                                                                                                                                                                                                                                                                |
|                          | <p>.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                           |
| power_ref_clock          |                                                                                                                                                                                                                                                                                                                                                                |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>The static probability of the signal to stay high during one clock cycle. It is a value between 0.0 and 1.0. no_value is returned if it cannot be computed.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                               |
| power_switching          |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                          | <p>The switching power of this net computed by report_power.</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                 |
| power_toggle_per_clock   |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                          | <p>The number of toggles per clock cycle. It is computed from power_toggle_rate * &lt;period of power_ref_clock&gt;. It can be as high as 2.0 for a clock, and is between 0 and 1 for a signal net. If there is more than one output pin, then it is the average of all the output pin power_toggle_rates. no_value is returned if no value can be computed.</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| power_toggle_rate        |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                          | <p>The average number of toggles read from VCD, TCF, SAIF, etc. or from propagation that occur in 1 second on the net.</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                       |
| power_toggle_rate_max    |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                          | <p>The average number of toggles read from VCD, TCF, SAIF, etc. or from propagation that occur in 1 second on the net.</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                       |
| power_toggle_rate_source |                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>The source of the power_toggle_rate value. The enum values mean: asserted: the value is user-specified by direct assertion or from a switching activity file. clock: the value is derived from the clock waveform. computed: the value is computed by propagating internal switching activity. constant: the value is driven by a constant value (e.g. from set_case_analysis or tie-offs). default: the value is not user-specified, but determined from the root power_default_toggle_rate attribute.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| probe_marker       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                    | <p>the probe marker port_shape of this net</p> <p><b>Type:</b> obj(port_shape)<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| probe_pad          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                    | <p>the probe pad port_shape of this net</p> <p><b>Type:</b> obj(port_shape)<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| resistance_max     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                    | <p>Returns the maximum resistance of the net for late path analysis. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> resistance<br/><b>Allowed -index values:</b> analysis_view<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                       |
| resistance_min     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                    | <p>Returns the minimum resistance of the net for early path analysis. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> resistance<br/><b>Allowed -index values:</b> analysis_view<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                      |
| route_avoid_detour |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Avoids detours of roughly more than a few gcell grids on the specified nets. This attribute affects global routing only. It's recommended that you use caution with this attribute, as it adds congestion to the design.</p> <p><b>Type:</b> <code>bool</code><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| route_bottom_preferred_layer      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                   | <p>Specifies the preferred lowest routing layer. This attribute is a soft limit; that is, the router might use a layer below the specified layer if necessary to complete routing. You can set route_preferred_layer_effort to determine how strict the router needs to honor this limit. If the top routing layer is equal to or greater the 4th routing layer (layer with .route_index = 4), and no bottom preferred routing layer is specified for clock nets, the router automatically sets the route_bottom_preferred_layer for clock nets to the 3rd routing layer (layer with .route_index = 3), to prevent the clock nets from blocking other nets.</p> <p><b>Type:</b> <code>obj(layer)</code><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                |
| route_coaxial_shield_interval     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                   | <p>Specifies the interval of crossties for the shield wires by routing track. A single value applies to all layers, or you can use a list of &lt;layer&gt; &lt;value&gt; pairs for individual layer controls. The default of 0 means there are no gaps between cross-ties. For example, '2' means 2 empty tracks between cross-ties on all layers, while 'metal2 2 metal3 4' means metal2 has 2 empty track and, metal3 have 4 empty tracks, and the default of 0 for all the other layers. The &lt;layer&gt; args can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal2 2 layer:metal3 4}.</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p> |
| route_coaxial_shield_layers       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                   | <p>Specifies the bottom and top routing layer for coaxial shields added around a shielded net. For example {metal4 metal6} means no coaxial shield shapes are added below metal4 or above metal6. The &lt;layer&gt; names can be input as a DPO name "layer:metal3", or a layer name "metal3", or as a layer index "3". The output is always in a DPO name format like {layer:metal3 layer:metal6}.</p> <p><b>Type:</b> <code>obj(layer)*</code><br/> <b>Default:</b> {}<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                       |
| route_coaxial_shield_layers_limit |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Specifies the bottom and top routing layer limit for routing that should be shielded by coaxial shields. For example, {metal3 metal6} means the routing wires on layers metal3 to metal6 should have coaxial shields added around them. The &lt;layer&gt; names can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal3 layer:metal6}.</p> <p><b>Type:</b> <a href="#">obj(layer)</a>*</p> <p><b>Default:</b> {}</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                        |
| route_em_rule             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p>Specifies a route_rule which is used for fixing EM issues. It is used to add a wider route_rule for the routing at output driver pins for a distance given by route_em_rule_distance.</p> <p><b>Type:</b> <a href="#">obj(route_rule)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| route_em_rule_distance    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p>Specifies the routing distance in microns that routing at output driver pins should use the route_em_rule rather than the normal net route_rule to avoid EM problems.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| route_em_width_rule       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p>The layer width and distance of routing at output driver pins needed to fix EM issues. The format is {&lt;layer&gt; &lt;width&gt; &lt;dist&gt; ...}. For example {metal2 0.4 5.0 metal3 0.5 6.0}. The router will choose a route_rule that meets the &lt;width&gt; required for each layer and use it for at least the &lt;dist&gt; from the output pin on each layer before changing back to the normal route_rule for the net. This attribute cannot be set at the same time as route_em_rule as only one parameter can be used to specify the route_rule to use. The &lt;layer&gt; names can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal2 0.4 5.0 layer:metal3 0.5 6.0}.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> {}</p> <p><b>Edit:</b> Yes</p> |
| route_hard_spacing_length |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>Specifies the parallel run length allowed on each layer to relax a route_rule with a hard spacing constraint (e.g. DEF NONDEFAULTRULE with SPACING constraint) into a soft constraint (e.g. only spacing needed by DRC rules are enforced rather than extra space from the route_rule). The soft constraint is allowed if the parallel run length between any wire on this net, and another wire is smaller than the specified value. For example, {metal3 2.0 metal4 3.0} would use soft-spacing for any metal3 wire on this net that does not have parallel lengths of more than 2.0 microns with other metal3 wires, and metal4 wires for 3.0 microns. The &lt;layer&gt; names can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal3 2.0 layer:metal4 3.0}.</p> <p><b>Type:</b> string<br/><b>Default:</b> {}<br/><b>Edit:</b> Yes</p> |
| route_initial_bottom_preferred_layer        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                             | <p>The initial lowest layer set by set_route_attributes or set_db. Opt will use it as a reference to set bottom_preferred_layer to any value equal or greater than initial_bottom_preferred_layer and smaller than top_preferred_layer for better performance.</p> <p><b>Type:</b> obj(layer)<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| route_interposer_interlayer_shielding_rules |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                             | <p>Returns the interposer inter-layer shielding rules of the net.</p> <p><b>Type:</b> obj(interposer_route_rule)*<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| route_interposer_properties                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                             | <p>This attributes returns interposer routing related properties as a dict.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| route_interposer_samelayer_shielding_rules  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                             | <p>Returns the interposer same-layer shielding rules of the net.</p> <p><b>Type:</b> obj(interposer_route_rule)*<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| route_interposer_shielding_mode             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Returns the shielding mode of the net. The valid modes are GSG, GSSG, GxSG:n.</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                             |
| route_interposer_trunk_rules |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                              | <p>Returns the interposer trunk rules of the net.</p> <p><b>Type:</b> <code>obj(interposer_route_rule)*</code><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                       |
| route_mask                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                              | <p>Masks constraint setting for the net. Legal values are from 0 to the mask number for the layers. A value of 0 indicates that there is no mask constraint on the net. A non-zero value indicates that all wiring on the net should be created on the specified mask. Layers that do not support the specified value will be treated as unconstrained.</p> <p><b>Type:</b> <code>int</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                         |
| route_mask_layer_range       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                              | <p>Specifies the bottom and top layer range to which the mask constraint should be applied. For example, {metal3 metal5} means that the mask constraint should be applied on layers metal3 to metal5. The &lt;layer&gt; names can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal3 layer:metal5}.</p> <p><b>Type:</b> <code>obj(layer)*</code><br/> <b>Default:</b> {}<br/> <b>Edit:</b> Yes</p> |
| route_max_fanout             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                              | <p>Specifies the maximum fanout value for each "steiner tree" for trunk routing (e.g. .route_pattern = trunk). If not set to 0, this value overrides the root attribute route_design_trunk_with_cluster_target_size value for this net.</p> <p><b>Type:</b> <code>int</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                         |
| route_min_stack_via_layer    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Specifies the lowest stack_via layer for output pins attached to this net. It indicates that stack_vias should be added up to the specified layer. If a pin already has a pre-defined custom stack_via, and the upper most layer of the via is still below the specified layer, additional stack_vias should be added up to the specified layer. Then regular routing would start from the specified layer.</p> <p><b>Type:</b> obj(layer)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                    |
| route_multi_cut_via_effort         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                    | <p>Specifies how much effort the router should use to add multi-cut vias. Higher settings will result in more multi-cut vias at the expense of more congestion and higher total routing length.</p> <p><b>Type:</b> enum<br/> <b>Enum Values:</b> low medium high<br/> <b>Default:</b> low<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                               |
| route_one_side_spacing_layer_range |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                    | <p>Specifies the bottom to top layer range that a one_side_spacing constraint should be applied on. This constraint means that extra spacing from the route_rule (e.g. LEF or DEF NONDEFAULTRULE SPACING values with or without HARDSPACING) is reserved on only one side. For example, {metal3 metal5} means only one side of the routing wires will get extra spacing for layers metal3 to metal5.</p> <p><b>Type:</b> obj(layer)*<br/> <b>Default:</b> {}<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                             |
| route_pattern                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                    | <p>Specifies the routing pattern.</p> <ul style="list-style-type: none"> <li>steiner: Routes all pins with a single Steiner tree.</li> <li>trunk: Routes pins in clusters to the nearest special_route trunk on the net. Trunk routing will cluster pins into separate Steiner trees and connects each tree to the nearest special_route trunk.</li> </ul> <p>By default the max number of pins in any single Steiner tree is controlled by the root attribute route_design_trunk_with_cluster_target_size, unless this net has the route_max_fanout attribute set to override it.</p> <p><b>Type:</b> enum<br/> <b>Enum Values:</b> steiner trunk fishbone<br/> <b>Default:</b> steiner<br/> <b>Edit:</b> Yes</p> |
| route_preferred_extra_space_tracks |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Specifies the preferred extra spacing tracks for this net. It adds extra spacing in units of "routing tracks" (which vary in width on different layers). Use this attribute to give critical nets extra space to reduce coupling. Setting it to 1 gives the routing 1 extra routing pitch of spacing (e.g. one empty routing track on both sides of each wire). This parameter creates a soft rule and, if the design is congested, might not give the extra space to the net. If you specify this attribute, Early Global Route also gives additional space to the specified nets.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p> |
| route_preferred_layer_effort |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                              | <p>Determines how much effort the router uses to meet the preferred layer limits for this net. Use this attribute with route_top_preferred_layer and route_bottom_preferred_layer attributes. Higher values will make the router try to obey the preferred routing layer range more strongly at the expense of more congestion and longer total routing length.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> low medium high<br/> <b>Default:</b> low<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                            |
| route_rule                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                              | <p>The route_rule (e.g. LEF or DEF NONDEFAULTRULE) to use for this net. Use this attribute for critical nets, such as clock nets, that might need to have wider wires or wider spacing than other nets.</p> <p><b>Type:</b> <a href="#">obj(route_rule)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                   |
| route_rule_effort            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                              | <p>Specifies the route_rule effort to use for this net.</p> <p>auto: Follows the route_rule .is_hard_spacing setting (e.g. LEF NONDEFAULTRULE HARDSPACING).<br/> hard: Makes the route_rule hard even if .is_hard_spacing == false for the route_rule.<br/> soft: Makes the route_rule soft even if .is_hard_spacing == true for the route_rule.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> auto hard soft<br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                           |
| route_shield_nets            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Specifies one or two PG nets to use as shields for critical nets. You should route these PG nets before routing this net so the shields can be connected to the PG mesh properly when the signal router runs. Normally only one shield net is used for shielding both sides. You can also specify two route_shield_nets (one for each side), but you cannot control which side is used if two nets are given. The shields are added as special_wires on each side of the signal wires with .shield_net pointing at the net that is shielded. The signal wires should be shielded for the full length, except near pins.</p> <p><b>Type:</b> obj(net)*<br/> <b>Default:</b> {}<br/> <b>Edit:</b> Yes</p> |
| route_shield_side         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                           | <p>By default both sides of a shielded signal wire have special_wire shields. If this value is set to 1, then only one side will be shielded, which will improve congestion at the expense of less effective shielding.</p> <p><b>Type:</b> enum<br/> <b>Enum Values:</b> 1 2<br/> <b>Default:</b> 2<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                             |
| route_si_post_repair      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                           | <p>Repairs crosstalk violations on this net. This includes changing topology, layer assignments, spacing and location to avoid coupling to other nets. This attribute only has an effect if the root attribute route_design_with_si_driven is true.</p> <p><b>Type:</b> bool<br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                          |
| route_skip                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                           | <p>Stops the router from routing this net. When set true for a net, the router treats the net as if you had marked it + FIXED in the DEF file and it becomes an obstruction for other nets. If the net is partially routed, the router does not complete the routing. This attribute also applies to Early Global Route.</p> <p><b>Type:</b> bool<br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                     |
| route_skip_antenna_repair |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Prevents the router from repairing process antenna violations on this net by either layer-hopping to higher layers or by adding diodes. This attribute is useful on nets that will have antenna violations repaired at the next level of hierarchy or with some other method.</p> <p><b>Type:</b> <code>bool</code><br/> <b>Default:</b> <code>false</code><br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                              |
| route_stack_via_distance  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                           | <p>Specifies the maximum distance in microns that stack_vias can be added from any pin that requires a stack_via. By default, the value of 0 means stack_vias must be directly on top of the pin.</p> <p><b>Type:</b> <code>double</code><br/> <b>Default:</b> <code>0.0</code><br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                             |
| route_stripe_layer_range  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                           | <p>This is used for nets that have pre-routed special_wires (e.g. clock trunks, or PG mesh for always-on PG nets, etc.) to limit the signal routing to connect to special_wires in the given layer range. The root attribute route_design_tie_net_to_shape controls which special_wire shapes (stripe, ring, etc.) can be used for connections.</p> <p><b>Type:</b> <code>obj(layer)*</code><br/> <b>Default:</b> <code>{}</code><br/> <b>Edit:</b> Yes</p>                                                                                                                           |
| route_top_preferred_layer |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                           | <p>Specifies the preferred highest layer for routing this net. This attribute is a soft limit; that is, the router might use a layer above the specified layer if necessary to complete routing. You can set route_preferred_layer_effort to determine how strictly the router should honor this limit. Use this attribute to improve timing by restricting the routing layers available for critical nets, while simultaneously routing other nets on all routing layers.</p> <p><b>Type:</b> <code>obj(layer)</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> Yes</p> |
| route_weight              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Specifies a relative weight for routing nets. In each switch box, the router routes nets with the highest weight first, then the next highest weight, and so on. Specify a value higher than 2 to ensure a net is routed before other nets. A switch box is a collection of global routing cells used during detailed routing. When you specify a net weight and route a net, the router does not route the entire net according to the weight. It routes the nets in each switch box with the highest weight first, then the next highest weight, and so on. Net weight.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 2<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| shield_tap_instance_insertion_effort |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | <p>different route type for clock shielding, including high frequency(high) and low frequency(standard) that main difference is what ground tie cell instance/via used to achieve shield segments electrical connection.</p> <p>None: default<br/> Standard: use existing standard cell vss m0 pins as much as possible, create a new instance of a ground tie cell under the shield route if cannot find an existing vss m0 pin within a user controlled search distance of a required connection, the new cell vss pins can have a "max fanout" of 2 shield nets.<br/> High: must create ground tie cell instances, and user can specify the physical distance of these new ground tie cells along the clock segments. new instance m0 vss pin to the clock shield segments will be a via stack of single vias on ever layer to the shield nets. It is acceptable for the router to create a n-1 metal shape to connect adjacent vss shield segments together and then route this single vss shape down to m0 ground tie pin. Shield segments may be electrically connected between adjacent route layers through vias. If this is done, there shall be no Manhattan distance between ground tie cells that is larger than the user provided value. Newly inserted ground tie cells cannot be shared between clocks.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none standard high<br/> <b>Default:</b> none<br/> <b>Edit:</b> Yes</p> |
| special_vias                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | <p>List of special vias.</p> <p><b>Type:</b> <a href="#">obj(special_via)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| special_wires                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                            |                                                                                                                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>List of special_wires (DEF SPECIALNETS equivalent)<br/> <b>Type:</b> <a href="#">obj(special_wire)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                       |
| timing_constant            |                                                                                                                                                                                                                                   |
|                            | <p>Reports constant value on net<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                 |
| timing_constant_source     |                                                                                                                                                                                                                                   |
|                            | <p>Reports source of constant value on net<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                       |
| toggle_rate                |                                                                                                                                                                                                                                   |
|                            | <p>Returns the toggle rate.<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                            |
| top_preferred_shield_layer |                                                                                                                                                                                                                                   |
|                            | <p>The top preferred layer on which to add shield on for the net.<br/> <b>Type:</b> <a href="#">obj(layer)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                  |
| total_capacitance_max_fall |                                                                                                                                                                                                                                   |
|                            | <p>Returns the total falling capacitance of the net used for late path analysis.<br/> <b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| total_capacitance_max_rise |                                                                                                                                                                                                                                   |

|                                   |                                                                                                                                                                                                                                                                              |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Returns the total rising capacitance of the net used for late path analysis.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                     |
| total_capacitance_min_fall        |                                                                                                                                                                                                                                                                              |
|                                   | <p>Returns the total falling capacitance of the net used for early path analysis.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                   |
| total_capacitance_min_rise        |                                                                                                                                                                                                                                                                              |
|                                   | <p>Returns the total rising capacitance of the net used for early path analysis.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                    |
| total_lumped_capacitance_max_fall |                                                                                                                                                                                                                                                                              |
|                                   | <p>Returns the total capacitance of the net used for late path analysis. You can use -index to return the capacitance for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| total_lumped_capacitance_max_rise |                                                                                                                                                                                                                                                                              |
|                                   | <p>Returns the total capacitance of the net used for late path analysis. You can use -index to return the capacitance for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| total_lumped_capacitance_min_fall |                                                                                                                                                                                                                                                                              |
|                                   | <p>Returns the total capacitance of the net for early path analysis. You can use -index to return the capacitance for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>     |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| total_lumped_capacitance_min_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                   | <p>Returns the total capacitance of the net for early path analysis. You can use -index to return the capacitance for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| use                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                   | <p>Indicates how this net is used. The enum values correspond to DEF and OpenAccess enum names. Note that the .use value is not used by any timing analysis commands because they only rely on SDC, .lib and Verilog data.</p> <p>You can change the .use value between signal, clock and analog, but you cannot change any power or ground values (you must delete and create a new net instead).</p> <p>signal: By default, nets have this value.</p> <p>clock: This value is set by clock-tree creation commands like ccopt_design. The 'real' clock status from SDC and .lib files is found in .is_clock after report_timing is run. So the .use 'clock' value will not match the .is_clock value unless you have run ccopt_design on this clock net.</p> <p>analog: This marks the net as an analog net. This can be used with some route_design controls to skip these nets or handle them in a special way.</p> <p>ground: A ground net.</p> <p>power: A power net.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> signal analog clock ground power scan tieoff</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| vias                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                   | <p>List of viaInsts.</p> <p><b>Type:</b> obj(via)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| virtual_wires                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                   | <p>List of virtual_wire</p> <p><b>Type:</b> obj(virtual_wire)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| vl_vh_analysis_status             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                           |                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Reports the glitch analysis status for net</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                |
| vlu_vho_analysis_status   |                                                                                                                                                                                                                                                                                                                                                                                       |
|                           | <p>Reports the glitch analysis status for net</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                |
| what_if_vias              |                                                                                                                                                                                                                                                                                                                                                                                       |
|                           | <p>List of 'what if' vias.</p> <p><b>Type:</b> <a href="#">obj(what_if_via)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                 |
| what_if_wires             |                                                                                                                                                                                                                                                                                                                                                                                       |
|                           | <p>List of 'what if' wires</p> <p><b>Type:</b> <a href="#">obj(what_if_wire)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                |
| wire_capacitance_max      |                                                                                                                                                                                                                                                                                                                                                                                       |
|                           | <p>Returns the portion of the net's total capacitance which comes extracted or annotated wire capacitance values. This is used for late path delay calculation. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| wire_capacitance_max_fall |                                                                                                                                                                                                                                                                                                                                                                                       |
|                           | <p>Returns the wire capacitance of the net for the maximum value of the falling capacitance range.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                 |
| wire_capacitance_max_rise |                                                                                                                                                                                                                                                                                                                                                                                       |

|                                  |                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Returns the wire capacitance of the net for the maximum value of the rising capacitance range.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                   |
| wire_capacitance_min             |                                                                                                                                                                                                                                                                                                                                                                        |
|                                  | <p>Returns the portion of the net's total capacitance which comes extracted or annotated wire capacitance values. This is used for early path delay calculation. You can use -index to return the derate factor for specific views.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| wire_capacitance_min_fall        |                                                                                                                                                                                                                                                                                                                                                                        |
|                                  | <p>Returns the wire capacitance of the net for the minimum value of the falling capacitance range.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                  |
| wire_capacitance_min_rise        |                                                                                                                                                                                                                                                                                                                                                                        |
|                                  | <p>Returns the wire capacitance of the net for the minimum value of the rising capacitance range.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                   |
| wire_lumped_capacitance_max      |                                                                                                                                                                                                                                                                                                                                                                        |
|                                  | <p>Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                     |
| wire_lumped_capacitance_max_fall |                                                                                                                                                                                                                                                                                                                                                                        |

|                                  |                                                                                                                                                                                                                                                                    |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Returns the lumped wire capacitance of the net for the maximum value of the falling capacitance range.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| wire_lumped_capacitance_max_rise |                                                                                                                                                                                                                                                                    |
|                                  | <p>Returns the lumped wire capacitance of the net for the maximum value of the rising capacitance range.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                        |
| wire_lumped_capacitance_min      |                                                                                                                                                                                                                                                                    |
|                                  | <p>Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| wire_lumped_capacitance_min_fall |                                                                                                                                                                                                                                                                    |
|                                  | <p>Returns the lumped wire capacitance of the net for the minimum value of the falling capacitance range.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| wire_lumped_capacitance_min_rise |                                                                                                                                                                                                                                                                    |
|                                  | <p>Returns the lumped wire capacitance of the net for the minimum value of the rising capacitance range.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                        |
| wires                            |                                                                                                                                                                                                                                                                    |

|  |                                                                                                   |
|--|---------------------------------------------------------------------------------------------------|
|  | List of wires<br><b>Type:</b> <a href="#">obj(wire)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
|--|---------------------------------------------------------------------------------------------------|

## net\_group

### Parent Objects

[design](#), [bus\\_guide](#), [pin\\_guide](#), [root](#)

### Definition

Net Group

| Attribute       | Description                                                                                                                                                                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| exclude_net     |                                                                                                                                                                                                                                                                                                                  |
|                 | Indicates group net exclude type(all_layer, same_layer, inclusive, all_layer_in_guided_area)<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> all_layer_in_guided_area all_layer same_layer inclusive<br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                       |
| is_compact      |                                                                                                                                                                                                                                                                                                                  |
|                 | Indicate whether the nets in the group are assigned tightly together. By default, nets which are part of a group and associated to a guide can be spread inside the guide based on the area available and alignment to targets<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes |
| is_guided       |                                                                                                                                                                                                                                                                                                                  |
|                 | Indicates if net group is guided<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                                                                               |
| is_mixed_signal |                                                                                                                                                                                                                                                                                                                  |



|                   |                                                                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | Returns if the net group is a mixed signal group<br><b>Type:</b> <code>bool</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                                                         |
| is_optimize_order |                                                                                                                                                                                                  |
|                   | Indicates whether net order will be optimized<br><b>Type:</b> <code>bool</code><br><b>Default:</b> <code>false</code><br><b>Edit:</b> Yes                                                        |
| is_spread         |                                                                                                                                                                                                  |
|                   | Indicates whether member nets distributed evenly<br><b>Type:</b> <code>bool</code><br><b>Default:</b> <code>false</code><br><b>Edit:</b> Yes                                                     |
| keep_out_space    |                                                                                                                                                                                                  |
|                   | Minimum spacing with pin of foreign nets (Unit: track)<br><b>Type:</b> <code>int</code><br><b>Default:</b> <code>0</code><br><b>Edit:</b> Yes                                                    |
| name              |                                                                                                                                                                                                  |
|                   | Group name<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                                                                                             |
| nets              |                                                                                                                                                                                                  |
|                   | List of member nets to this net group<br><b>Type:</b> <code>obj(net)*</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                                                               |
| obj_type          |                                                                                                                                                                                                  |
|                   | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <code>enum (net_group)</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No |
| spacing           |                                                                                                                                                                                                  |

|  |                                                                                                                                          |
|--|------------------------------------------------------------------------------------------------------------------------------------------|
|  | Specifies minimum spacing between the pins of the net group<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
|--|------------------------------------------------------------------------------------------------------------------------------------------|

## obj\_type

### Parent Objects

[attribute](#), [root](#)

### Definition

| Attribute                      | Description                                                                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| accept_user_defined_attributes |                                                                                                                                                 |
|                                | Specifies whether new attributes can be added by user.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No            |
| attributes                     |                                                                                                                                                 |
|                                | list of attribute for this obj_type<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">attribute</a> )*<br><b>Default:</b> ""<br><b>Edit:</b> No |
| help                           |                                                                                                                                                 |
|                                | Specifies the help text for the object<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                          |
| name                           |                                                                                                                                                 |
|                                | obj_type name like inst, lib_cell, etc.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                         |

| obj_type |                                                                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">obj_type</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

## opcond

### Parent Objects

[timing\\_condition](#), [root](#)

### Definition

An opcond represents a named operating condition which is defined by specific process, voltage, and temperature values. Opcond objects may preexist within a Liberty library or "virtual" opconds may be created using the create\_opcond command. Virtual opconds can also be modified by using the update\_opcond command.

| Attribute  | Description                                                                                                                                                                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name  |                                                                                                                                                                                                                                                  |
|            | The base name of the opcond without any leading library_set or library names (e.g. op1 for libset1/lib1/op1).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                    |
| is_virtual |                                                                                                                                                                                                                                                  |
|            | Indicates that this opcond object is a virtual opcond created by the create_opcond command vs. an opcond that is created by reading a Liberty timing library file.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| name       |                                                                                                                                                                                                                                                  |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>The full name of the opcond. The opcond name from a Liberty file includes the library_set and library names (e.g. libset1/lib1/op1) while the full name from create_opcond is just the opcond name (e.g. op1).</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                             |
| obj_type    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|             | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">opcond</a>)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| process     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|             | <p>Specifies the Process value of the opcond</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| temperature |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|             | <p>Specifies the Temperature value of the opcond in Celsius.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| tree_type   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|             | <p>The Liberty tree_type model. It is an enum with these choices: none means there is no tree_type value given; best_case_tree models each load pin is close to the driver, so wire capacitance is incurred, but wire resistance is ignored; balanced_tree models when all load pins are on equal branches, so each load pin has an equal portion of the total wire cap and resistance; worst_case_tree models when the load pin is at the extreme end of the wire, so each load pin incurs the full wire capacitance and resistance</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none best_case_tree balanced_tree worst_case_tree<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| voltage     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|             | <p>Specifies the Voltage value of the opcond in units of the library it came from.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

# package\_component

## Parent Objects

[design](#)

## Definition

pkg component

| Attribute  | Description                                                                                                                                                                                                      |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cell_name  |                                                                                                                                                                                                                  |
|            | The name of other chip cell in the package<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                       |
| inst_name  |                                                                                                                                                                                                                  |
|            | The name of other chips in the package<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                           |
| obj_type   |                                                                                                                                                                                                                  |
|            | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">package_component</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| pt         |                                                                                                                                                                                                                  |
|            | Location of the other chips in the package<br><b>Type:</b> <a href="#">point</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                        |
| ref_design |                                                                                                                                                                                                                  |

|      |                                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------|
|      | Another identifying name of the chip in the package<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| size |                                                                                                                                     |
|      | Size of the cell<br><b>Type:</b> <a href="#">point</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                     |

## package\_object

### Parent Objects

[design](#), [root](#)

### Definition

pkg object

| Attribute    | Description                                                                                                                                |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| die_net_name |                                                                                                                                            |
|              | the name of die net which this package object connected to<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| layer_name   |                                                                                                                                            |
|              | The package layer that this package object belongs to<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No      |
| name         |                                                                                                                                            |

|                  |                                                                                                                                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <p>The name of this package object</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                             |
| obj_type         |                                                                                                                                                                                                                                |
|                  | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">package_object</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| package_net_name |                                                                                                                                                                                                                                |
|                  | <p>the name of package net which this package object connected to</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                              |
| pt               |                                                                                                                                                                                                                                |
|                  | <p>Location of the package object</p> <p><b>Type:</b> <a href="#">point</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                               |
| size             |                                                                                                                                                                                                                                |
|                  | <p>Size of the package object</p> <p><b>Type:</b> <a href="#">point</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                   |
| type             |                                                                                                                                                                                                                                |
|                  | <p>the type of this package object</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> bga_ball bond_finger bond_wire route flight_line component other</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |

# partition

## Parent Objects

hinst, inst, design, pin\_group, pin\_guide, root

## Definition

Partition Object

| Attribute      | Description                                                                                                                                                                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_pins      |                                                                                                                                                                                                                                                                                                                     |
|                | <p>List of partition pins. It is valid only if partition is in non-committed state, i.e. partition module is instantiated as hinst. Otherwise it is set to NULL.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">base_pin</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                            |
| clones         |                                                                                                                                                                                                                                                                                                                     |
|                | <p>List of clone inst/hinsts. The obj_type returned will be inst if the partition is committed or a blackbox, otherwise the obj_type will be hinst.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">inst</a>)* <a href="#">obj</a>(<a href="#">hinst</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| core_to_bottom |                                                                                                                                                                                                                                                                                                                     |
|                | <p>Spacing between the partition boundary and core design area of the partition module</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                        |
| core_to_left   |                                                                                                                                                                                                                                                                                                                     |
|                | <p>Spacing between the partition boundary and core design area of the partition module</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                        |
| core_to_right  |                                                                                                                                                                                                                                                                                                                     |



|                  |                                                                                                                                                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | Spacing between the partition boundary and core design area of the partition module<br><b>Type:</b> coord<br><b>Default:</b> no_value<br><b>Edit:</b> Yes                                                                 |
| core_to_top      |                                                                                                                                                                                                                           |
|                  | Spacing between the partition boundary and core design area of the partition module<br><b>Type:</b> coord<br><b>Default:</b> no_value<br><b>Edit:</b> Yes                                                                 |
| hpins            |                                                                                                                                                                                                                           |
|                  | Hierarchical instance pin of the partition. It is only valid if the partition is uncommitted and still an hinst.<br><b>Type:</b> obj(hpin)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                      |
| is_black_box     |                                                                                                                                                                                                                           |
|                  | Specifies if partition is a blackbox<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                        |
| is_committed     |                                                                                                                                                                                                                           |
|                  | Specifies if partition is committed<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                         |
| master           |                                                                                                                                                                                                                           |
|                  | The master inst/hInst. The obj_type returned will be inst if the partition is committed or a blackbox, otherwise the objType will be hInst.<br><b>Type:</b> obj(inst) obj(hinst)<br><b>Default:</b> ""<br><b>Edit:</b> No |
| min_pitch_bottom |                                                                                                                                                                                                                           |
|                  | Specifies the pin pitch (in tracks)<br><b>Type:</b> int<br><b>Default:</b> 0<br><b>Edit:</b> Yes                                                                                                                          |

|                   |                                                                                                                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| min_pitch_left    |                                                                                                                                                                                                                           |
|                   | <p>Specifies the pin pitch (in tracks)</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                       |
| min_pitch_right   |                                                                                                                                                                                                                           |
|                   | <p>Specifies the pin pitch (in tracks)</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                       |
| min_pitch_top     |                                                                                                                                                                                                                           |
|                   | <p>Specifies the pin pitch (in tracks)</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                       |
| name              |                                                                                                                                                                                                                           |
|                   | <p>name of the partition</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                  |
| obj_type          |                                                                                                                                                                                                                           |
|                   | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">partition</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pin_bottom_layers |                                                                                                                                                                                                                           |
|                   | <p>List of bottom layers to use for the partition</p> <p><b>Type:</b> <a href="#">obj(layer)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                   |
| pin_left_layers   |                                                                                                                                                                                                                           |
|                   | <p>List of left layers to use for the partition</p> <p><b>Type:</b> <a href="#">obj(layer)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                     |

|                      |                                                                                                                                                                                                                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pin_right_layers     |                                                                                                                                                                                                                                                                                                          |
|                      | <p>List of right layers to use for the partition</p> <p><b>Type:</b> <a href="#">obj(layer)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                   |
| pin_to_corner_tracks |                                                                                                                                                                                                                                                                                                          |
|                      | <p>Minimum number of routing tracks between pins and each corner. It is a list of int values, where the first value is for the lower-left corner, and the remaining corners are listed in clockwise order.</p> <p><b>Type:</b> <a href="#">int*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pin_top_layers       |                                                                                                                                                                                                                                                                                                          |
|                      | <p>List of top layers to use for the partition</p> <p><b>Type:</b> <a href="#">obj(layer)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                     |
| place_halo_bottom    |                                                                                                                                                                                                                                                                                                          |
|                      | <p>Specifies extra spacing around the partition that should not be used for placement</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                              |
| place_halo_left      |                                                                                                                                                                                                                                                                                                          |
|                      | <p>Specifies extra spacing around the partition that should not be used for placement</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                              |
| place_halo_right     |                                                                                                                                                                                                                                                                                                          |
|                      | <p>Specifies extra spacing around the partition that should not be used for placement</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                              |
| place_halo_top       |                                                                                                                                                                                                                                                                                                          |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Specifies extra spacing around the partition that should not be used for placement</p> <p><b>Type:</b> coord</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| rail_width              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                         | <p>Specifies the cell rail width</p> <p><b>Type:</b> coord</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| read_only               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                         | <p>This attributes is set by set_module_view to identify if a partition is read only or not. It means the partition cannot be optimized, and cells inside will not be moved. Setting this attribute will set the dont_touch_effective attribute on all insts and hinsts within the partition unless overridden at a lower level partition. It cannot be overridden by other hinst or inst values.</p> <p><b>Supported values:</b></p> <p>false: This partition is allowed to be optimized, even if a partition or design above has read_only = true.</p> <p>true: The partition is read_only.</p> <p>none: No constraint. This partition inherits any read_only from a partition or design above.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> none false true</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p> |
| reserved_layers         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                         | <p>List of metal layers which are used for routing in the partition and generating partition pins. Any metal layers that are not specified, usually the top-most metal layers, are allowed to route over the partition</p> <p><b>Type:</b> obj(layer)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| route_halo_bottom_layer |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                         | <p>The bottom partition layer for which routing halo will be created</p> <p><b>Type:</b> obj(layer)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| route_halo_to_boundary  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                      |                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Specifies routing halo around the partition (honored by signal router). Positive values indicate the halo is outside the partition. Negative values indicate the halo is inside of the boundary of the partition and will be pushed into the partition when the partition is committed.</p> <p><b>Type:</b> coord<br/><b>Default:</b> no_value<br/><b>Edit:</b> Yes</p> |
| route_halo_top_layer |                                                                                                                                                                                                                                                                                                                                                                            |
|                      | <p>The top partition layer for which routing halo will be created</p> <p><b>Type:</b> obj(layer)<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                           |

## patch\_wire

### Parent Objects

net

### Definition

DEF NETS RECT wire

| Attribute      | Description                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------------------|
| has_trim_metal |                                                                                                                                  |
|                | <p>Indicate the path wire has trim metal or not</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>          |
| is_pin_probe   |                                                                                                                                  |
|                | <p>Returns true if the patch wire is pin probe patch.</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> No</p> |
| layer          |                                                                                                                                  |

|            |                                                                                                                                                                                                                                  |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <p>The layer of patch</p> <p><b>Type:</b> <a href="#">obj(layer)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                        |
| location   |                                                                                                                                                                                                                                  |
|            | <p>Reference point to symbolic location</p> <p><b>Type:</b> <a href="#">point</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                           |
| mask       |                                                                                                                                                                                                                                  |
|            | <p>Indicates mask number for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> |
| net        |                                                                                                                                                                                                                                  |
|            | <p>The net that the patch belongs to</p> <p><b>Type:</b> <a href="#">obj(net)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                           |
| obj_type   |                                                                                                                                                                                                                                  |
|            | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum(patch_wire)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                        |
| rect       |                                                                                                                                                                                                                                  |
|            | <p>Rectangle that defines the patch_wire shape</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                     |
| route_rule |                                                                                                                                                                                                                                  |
|            | <p>The non-default rule corresponding to the wire, wires with the default routing rule will return NULL (0x0).</p> <p><b>Type:</b> <a href="#">obj(route_rule)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>          |

|                  |                                                                                                                                                                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| status           |                                                                                                                                                                                                                                                                                                 |
|                  | Wiring status (equivalent to DEF NETS regular wiring status)<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> unknown routed fixed cover noshield<br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                          |
| trim_metal_color |                                                                                                                                                                                                                                                                                                 |
|                  | Color of this patch wire trim metal<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                |
| trim_metal_rect  |                                                                                                                                                                                                                                                                                                 |
|                  | The trim_metal rect if the patch_wire has a trim_metal shape attached. This only occurs for some advanced node layers that use self-aligned patterning. {0 0 0 0} is returned if there is no trim_metal attached.<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> No |

## path\_group

### Parent Objects

[timing\\_path](#)

### Definition

cte path group

| Attribute | Description                                                                                                 |
|-----------|-------------------------------------------------------------------------------------------------------------|
| name      |                                                                                                             |
|           | The name of the path_group.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |

|          |                                                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| obj_type |                                                                                                                                                                                                           |
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">path_group</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

## pg\_base\_pin

### Parent Objects

[base\\_cell](#), [pin](#), [tech](#), [root](#), [lib\\_pin](#), [pg\\_pin](#)

### Definition

A power or ground pin for a base\_cell.

| Attribute | Description                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_cell |                                                                                                                                                          |
|           | The base_cell of this pg_base_pin.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">base_cell</a> )*<br><b>Default:</b> ""<br><b>Edit:</b> No           |
| base_name |                                                                                                                                                          |
|           | The base name of this pg_base_pin without the base_cell name (e.g. vdd).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| direction |                                                                                                                                                          |



|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>The pin direction from .lib if available, otherwise from LEF/OA. It can be in, out, inout or internal. Internal means it is an internal pin from a .lib file for the timing model, and is not part of the netlist. If there is no .lib for this cell, then the direction comes from the LEF PIN DIRECTION or equivalent OA oaTermType. The LEF DIRECTION values (and equivalent oaTermType values) are mapped this way: INPUT = in, OUTPUT = out, OUTPUT TRISTATE = out, INOUT = inout, FEEDTHRU = inout, and the oaTermType unknown = inout.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> in out inout internal<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| effective_stack_via_rule |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                          | <p>The stack_via_rule that is expected, but not required, to be used by the router for connecting this pg_base_pin. The actual stack via rule used (if any) may be effected by other pg_pin and pg_base_pin attributes, or by choices made by the software (optimization, clock tree synthesis, the router etc.)</p> <p><b>Type:</b> <a href="#">obj(stack_via_rule)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                 |
| is_always_on             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                          | <p>This pg_base_pin is an always on power pin. This attribute can be set by liberty files.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| name                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                          | <p>The name of this pg_base_pin with the base_cell name (e.g. and2/vdd).</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| obj_type                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                          | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum (pg_base_pin)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| pg_type                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>The type of the power or ground pin from liberty data. PG pins with no liberty entry will have "invalid".</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> primary_power primary_ground backup_power backup_ground internal_power internal_ground pwell nwell deep_pwell deep_nwell invalid</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                           |
| physical_direction           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                              | <p>Direction of this pin.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> input output inout unknown feedthrough tristate</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                               |
| physical_pins                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                              | <p>Physical pins for the pg_base_pin. One logical pg_base_pin can have multiple physical_pins (equivalent to separate LEF or DEF PORT statements for one LEF or DEF PIN). Each physical_pin normally has one shape, but may have multiple shapes that are all strongly connected to each other. By default, at least one shape of each physical_pin should be connected to the power-mesh.</p> <p><b>Type:</b> <a href="#">obj(physical_pin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                           |
| route_bottom_preferred_layer |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                              | <p>Specifies the preferred lowest routing layer. This attribute is a soft limit; that is, the router might use a layer below the specified layer if necessary to complete routing. You can set route_preferred_layer_effort to determine how strict the router needs to honor this limit. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint).</p> <p><b>Type:</b> <a href="#">obj(layer)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| route_max_fanout             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Specifies the maximum fanout value for each "steiner tree" for trunk routing (e.g. route_pattern = trunk). This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint).</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                  |
| route_pattern                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                              | <p>Specifies the routing pattern. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint).</p> <p>    steiner: Routes all pins with a single Steiner tree.<br/>         trunk: Routes pins in clusters to the nearest special_route trunk on the net.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> steiner trunk<br/> <b>Default:</b> steiner<br/> <b>Edit:</b> Yes</p>                                                                                                                                                     |
| route_preferred_layer_effort |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                              | <p>Determines how much effort the router uses to meet the preferred layer limits. Use this attribute with route_top_preferred_layer and route_bottom_preferred_layer attributes. Higher values will make the router try to obey the preferred routing layer range more strongly at the expense of more congestion and longer total routing length. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint).</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> low medium high<br/> <b>Default:</b> low<br/> <b>Edit:</b> Yes</p> |
| route_rule                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                              | <p>The route_rule (e.g. LEF or DEF NONDEFAULTRULE) to use. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint).</p> <p><b>Type:</b> <a href="#">obj(route_rule)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                        |
| route_stripe_layer_range     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Limits the signal routing to connect to special_wires in the given layer range. Exactly two layers should be given for the bottom and top layers in the range. For example, {metal3 metal5} means the router will only connect to special_wires on layers metal3 to metal5. The &lt;layer&gt; names can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal3 layer:metal5}. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint).</p> <p><b>Type:</b> obj(layer)*<br/> <b>Default:</b> {}<br/> <b>Edit:</b> Yes</p> |
| route_top_preferred_layer |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>Specifies the preferred highest layer for routing. This attribute is a soft limit; that is, the router might use a layer above the specified layer if necessary to complete routing. You can set route_preferred_layer_effort to determine how strictly the router should honor this limit. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint).</p> <p><b>Type:</b> obj(layer)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                   |
| stack_via_required        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>Sets whether stack vias are required for this pg_base_pin.</p> <p><b>Type:</b> bool<br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| stack_via_rule_list       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>List of stack_via_rule that are valid choices for connecting this pg_base_pin. If the list is empty, no stack via is allowed.</p> <p><b>Type:</b> obj(stack_via_rule)*<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| stack_via_rule_required   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <p>Specifies whether a stack via is required when connecting to this pg_base_pin. If set to 0, a stack via is optional. If set to 1, the min one of the stack_via_rule from stack_via_rules must be used to generate a stack via even if a design rule violation occurs. If set to 2, the min one the stack_via_rule from stack_via_rules is used to generate a stack via if a design rule violation does not occur. If set to 3, tool enforces multiple tries from large to min one of the stack_via_rule from stack_via_rules to generate a stack via even if a design rule violation occurs. If set to 4, tool performs multiple tries from large to min one the stack_via_rule from stack_via_rules to generate a stack via if a design rule violation does not occur.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> 0 1 2 3 4<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| taper_rule |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|            | <p>The taper route_rule for the pg_pin. By default, if tapered routing is needed to access the pin, the default route_rule will be used unless this attribute is set (see LEF MACRO PIN TAPERRULE).</p> <p><b>Type:</b> <a href="#">obj(route_rule)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| tech       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|            | <p>The tech for this pg_base_pin.</p> <p><b>Type:</b> <a href="#">obj(tech)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| tied_to    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|            | <p>If this pg_base_pin is a substrate bias-pin (e.g. pg_type is nwell or pwell), and it is internally tied-to a "master" PG pin, this specifies the PG pin name (e.g. an nwell pin might have tied_to = vdd if it is internally connected to the vdd pin).</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| use        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

Indicates how this pg\_base\_pin is used from the LEF USE value or OA equivalent. Only power or ground enum values make sense in normal usage, but if there are conflicts between Liberty and LEF/OA definitions, you may have pg\_base\_pins that are marked in LEF/OA with the other values.

**Type:** [enum](#)

**Enum Values:** signal analog power ground clock

**Default:** ""

**Edit:** No

## pg\_pin

### Parent Objects

[inst](#), [design](#), [pin](#), [root](#), [lib\\_pin](#)

### Definition

Power or ground pin information

| Attribute                | Description                                                                                                                                                                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name                |                                                                                                                                                                                                                                                                                       |
|                          | The base name for this pg_pin without the base_cell name (e.g. vdd).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                  |
| effective_stack_via_rule |                                                                                                                                                                                                                                                                                       |
|                          | The stack_via_rule that is expected to be used by the router for connecting to this pg_pin. This value is derived from the other pg_pin and pg_base_pin attributes related to stack via.<br><b>Type:</b> <a href="#">obj(stack_via_rule)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| escaped_name             |                                                                                                                                                                                                                                                                                       |
|                          | The escaped name of this pg_pin.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                      |

|                  |                                                                                                                                                                                                       |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hnet             |                                                                                                                                                                                                       |
|                  | <p>The hierarchical net connected to this pg_pin.</p> <p><b>Type:</b> <a href="#">obj(hnet)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                  |
| inst             |                                                                                                                                                                                                       |
|                  | <p>The inst containing this pg_pin.</p> <p><b>Type:</b> <a href="#">obj(inst)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                |
| name             |                                                                                                                                                                                                       |
|                  | <p>The pg_pin name including the inst path (e.g. i1/i2/vdd)</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                           |
| net              |                                                                                                                                                                                                       |
|                  | <p>The net connected to this pg_pin.</p> <p><b>Type:</b> <a href="#">obj(net)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                |
| obj_type         |                                                                                                                                                                                                       |
|                  | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum(pg_pin)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pg_base_pin      |                                                                                                                                                                                                       |
|                  | <p>The corresponding pg_base_pin on the base_cell for this pg_pin.</p> <p><b>Type:</b> <a href="#">obj(pg_base_pin)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                          |
| route_max_fanout |                                                                                                                                                                                                       |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Specifies the maximum fanout value for each "steiner tree" for trunk routing (e.g. route_pattern = trunk). Normally, every pg_pin instantiated by the same pg_base_pin honors the same max fanout. So user only needs to set it on the pg_base_pin. For the few cases where pg_pin has different requirement than the pg_base_pin, NanoRoute will honor the one on pg_pin. This is intended for usage on always-on power pins routed by the signal router rather than the power router, It is most commonly used during late ECO routing to meet EM and IR limits by forcing an isolated pg_pin into its own "cluster" by setting the value to 1.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| stack_via_rule          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | <p>This value serves as a pg_pin-specific override to use this stack_via_rule when connecting to this pg_pin. If set, it must match one of the elements from the stack_via_rule_list attribute of the corresponding pg_base_pin. The interpretation of this value depends on this pg_pin's stack_via_rule_required attribute.</p> <p>stack_via_rule == {} &amp;&amp; stack_via_rule_required == false: stack_via_rule does not affect the choice.</p> <p>stack_via_rule == stackrule1 &amp;&amp; stack_via_rule_required == false: Router will prefer stackrule1, but it may select another (or possibly none) if necessary to avoid design rule violations.</p> <p>stack_via_rule == {} &amp;&amp; stack_via_rule_required == true: The required stack_via_rule is empty, so the router will not use any stack_via_rule for connecting to this pg_pin (even ignoring stack_via_rule_required on the pg_base_pin if necessary).</p> <p>stack_via_rule == stackrule1 &amp;&amp; stack_via_rule_required == true: Router will use stackrule1, even if it leads to design rule violations.</p> <p><b>Type:</b> <a href="#">obj(stack_via_rule)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| stack_via_rule_required |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | <p>Specifies whether the router must use this pg_pin's stack_via_rule value. If false, the stack_via_rule value will be preferred by the router. If true, the stack_via_rule value is required by the router. See the stack_via_rule attribute for more details.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



# physical\_pin

## Parent Objects

[pg\\_base\\_pin](#), [base\\_pin](#), [port](#)

## Definition

physical pin

| Attribute    | Description                                                                                                                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| class        |                                                                                                                                                                                                             |
|              | Physical pin class<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> undefined none core bump<br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                           |
| layer_shapes |                                                                                                                                                                                                             |
|              | List of layer_shapes that define the terminal pin geometries.<br><b>Type:</b> <a href="#">obj(layer_shape)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                    |
| obj_type     |                                                                                                                                                                                                             |
|              | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">physical_pin</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| port_number  |                                                                                                                                                                                                             |
|              | The port number for geometries under the physical_pin.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                         |
| shape_vias   |                                                                                                                                                                                                             |

List of shape\_vias that define the terminal pin geometries.

**Type:** [obj\(shape\\_via\)\\*](#)

**Default:** ""

**Edit:** No

## pin

### Parent Objects

[marker](#), [test\\_clock](#), [clock](#), [violation](#), [port](#), [def\\_scan\\_chain](#), [skew\\_group](#), [arc](#), [hinst](#), [scan\\_chain](#), [hnet](#), [actual\\_scan\\_segment](#), [timing\\_point](#), [dft\\_configuration\\_mode](#), [design](#), [bus\\_sink\\_group](#), [clock\\_tree](#), [timing\\_path](#), [root](#), [test\\_signal](#), [inst](#), [scan\\_segment](#), [net](#), [actual\\_scan\\_chain](#)

### Definition

Instance terminal

| Attribute                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| actual_latency_early_fall_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                               | Returns the computed early falling clock latency to this clock pin. If the analysis mode is set case/worst-case, the value will be the Setup capture latency. When operating in OCV mode max qualification is ignored. If there are multiple arrival times from different clock phases - t conservative latency value is reported. You can use the -index function to return latencies v respect to a specific clock. When operating in concurrent MMMC mode, the worst latency a views is reported. You can use the -index function to filter latencies based on view.<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| actual_latency_early_fall_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Returns the computed early falling clock latency to this clock pin. If the analysis mode is set case/worst-case, the value will be the Hold launch latency. When operating in OCV mode, 1 qualification is ignored. If there are multiple arrival times from different clock phases - the m conservative latency value is reported. You can use the -index function to return latencies v respect to a specific clock. When operating in concurrent MMMC mode, the worst latency a views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| actual_latency_early_rise_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                               | <p>Returns the computed early rising clock latency to this clock pin. If the analysis mode is set case/worst-case, the value will be the Setup capture latency. When operating in OCV mode max qualification is ignored. If there are multiple arrival times from different clock phases - t conservative latency value is reported. You can use the -index function to return latencies v respect to a specific clock. When operating in concurrent MMMC mode, the worst latency a views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>   |
| actual_latency_early_rise_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                               | <p>Returns the computed early rising clock latency to this clock pin. If the analysis mode is set case/worst-case, the value will be the Hold launch latency. When operating in OCV mode, 1 qualification is ignored. If there are multiple arrival times from different clock phases - the m conservative latency value is reported. You can use the -index function to return latencies v respect to a specific clock. When operating in concurrent MMMC mode, the worst latency a views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| actual_latency_late_fall_max  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Returns the computed late falling clock latency to this clock pin. If the analysis mode is set to case/worst-case, the value will be the Setup launch latency. When operating in OCV mode, max qualification is ignored. If there are multiple arrival times from different clock phases - the conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| actual_latency_late_fall_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                              | <p>Returns the computed late falling clock latency to this clock pin. If the analysis mode is set to case/worst-case, the value will be the Hold capture latency. When operating in OCV mode, max qualification is ignored. If there are multiple arrival times from different clock phases - the conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| actual_latency_late_rise_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                              | <p>Returns the computed late rising clock latency to this clock pin. If the analysis mode is set to case/worst-case, the value will be the Setup launch latency. When operating in OCV mode, max qualification is ignored. If there are multiple arrival times from different clock phases - the conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| actual_latency_late_rise_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Returns the computed late rising clock latency to this clock pin. If the analysis mode is set to case/worst-case, the value will be the Hold capture latency. When operating in OCV mode, qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| adaptive_ipba_clock_period_threshold |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                      | <p>Reports percentage clock period threshold used for filtering violation during adaptive IPBA</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| arrival_clocks                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                      | <p>Reports clocks that correspond to the arrival times</p> <p><b>Type:</b> obj(clock)*<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| arrival_level                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                      | <p>Reports arrival level of the pin</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| arrival_max                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                      | <p>Reports maximum arrival time</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| arrival_max_fall                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>Returns the latest falling arrival time to the specified pin across all concurrent MMMC views can use -index to determine the worst arrival for a specific view. You can also use -index to latest arrival time related to a specific clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                            |
| arrival_max_rise      |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                       | <p>Returns the latest rising arrival time to the specified pin across all concurrent MMMC views can use -index to determine the worst arrival for a specific view. You can also use -index to latest arrival time related to a specific clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                             |
| arrival_mean_max_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                       | <p>In SOCV analysis mode, this returns the mean value of the latest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| arrival_mean_max_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                       | <p>In SOCV analysis mode, this returns the mean value of the latest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| arrival_mean_min_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>In SOCV analysis mode, this returns the mean value of the earliest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| arrival_mean_min_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                       | <p>In SOCV analysis mode, this returns the mean value of the earliest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| arrival_min           |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                       | <p>Reports minimum arrival time</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                   |
| arrival_min_fall      |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                       | <p>Returns the earliest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                  |
| arrival_min_rise      |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                       | <p>Returns the earliest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                   |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| arrival_sigma_max_fall    | <p>In SOCV analysis mode, this returns the variation component of the latest falling arrival time specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>               |
| arrival_sigma_max_rise    | <p>In SOCV analysis mode, this returns the variation component of the latest rising arrival time specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                |
| arrival_sigma_min_fall    | <p>In SOCV analysis mode, this returns the variation component of the earliest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| arrival_sigma_min_rise    | <p>In SOCV analysis mode, this returns the mean value of the earliest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>           |
| arrival_skewness_max_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>In SOCV analysis mode, this returns the skewness value of the latest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>        |
| arrival_skewness_max_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p>In SOCV analysis mode, this returns the skewness value of the latest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>         |
| arrival_skewness_min_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p>In SOCV analysis mode, this returns the skewness value of the earliest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| arrival_skewness_min_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p>In SOCV analysis mode, this returns the skewness value of the earliest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| arrival_window            |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                      |                                                                                                                                                                                                                                                                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Returns a list of the earliest and latest, rising and falling arrival times per clock phase at the across all concurrent MMMC views. You can use -index to make the arrival times specific to view</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| attacker_fall_slew   |                                                                                                                                                                                                                                                                                                                                                          |
|                      | <p>Reports fall slew of attacker seen for net connected to this pin</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                   |
| attacker_rise_slew   |                                                                                                                                                                                                                                                                                                                                                          |
|                      | <p>Reports rise slew of attacker seen for net connected to this pin</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                   |
| base_name            |                                                                                                                                                                                                                                                                                                                                                          |
|                      | <p>The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                          |
| base_pin             |                                                                                                                                                                                                                                                                                                                                                          |
|                      | <p>pointer to equivalent base pin</p> <p><b>Type:</b> <a href="#">obj(base_pin)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                     |
| capacitance_max      |                                                                                                                                                                                                                                                                                                                                                          |
|                      | <p>Reports maximum capacitance of pin</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                 |
| capacitance_max_fall |                                                                                                                                                                                                                                                                                                                                                          |

|                      |                                                                                                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Returns the maximum value of the falling capacitance range of the corresponding library pin can use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| capacitance_max_rise |                                                                                                                                                                                                                                                                                                           |
|                      | <p>Returns the maximum value of the rising capacitance range of the corresponding library pin can use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| capacitance_min      |                                                                                                                                                                                                                                                                                                           |
|                      | <p>Reports minimum capacitance of pin</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                            |
| capacitance_min_fall |                                                                                                                                                                                                                                                                                                           |
|                      | <p>Returns the minimum value of the falling capacitance range of the corresponding library pin can use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| capacitance_min_rise |                                                                                                                                                                                                                                                                                                           |
|                      | <p>Returns the minimum value of the rising capacitance range of the corresponding library pin can use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| ccopt_schedule       |                                                                                                                                                                                                                                                                                                           |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>Specifies the restrictions, if any, on the scope of useful skew scheduling for this pin. Settings 'timing_only', 'off', and 'on' override other sources of per-pin useful skew restrictions, such as the 'opt_skew_no_boundary' attribute. Allowed values are:</p> <p>auto Neither restrictions nor overrides are to be applied to the pin (default).</p> <p>on Specifies no restrictions on sink adjustment scope, overriding other sources of per-pin useful skew restrictions such as the 'opt_skew_no_boundary' setting.</p> <p>timing_only Restricts the sink to useful skew adjustments for timing if necessary.</p> <p>off Useful skew must avoid adjusting the schedule of this pin, even if this degrades timing.</p> <p>Valid values: auto off on timing_only</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> |
| cell                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                             | <p>Reports cell of pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| cell_name                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                             | <p>Returns the name of the instance of the instance pin mentioned.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| clock_gating_exclude_signal |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                             | <p>Exclude this pin from clock gating enable functions.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| clock_pulse_width_high      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                       |                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>return pulse_width_high for arrival clocks<br/> <b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                       |
| clock_pulse_width_low |                                                                                                                                                                                                                                                                                                                                                                      |
|                       | <p>return pulse_width_low for arrival clocks<br/> <b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                        |
| clock_sources         |                                                                                                                                                                                                                                                                                                                                                                      |
|                       | <p>Reports the collection of clock roots of clocks associated with pin<br/> <b>Type:</b> obj(pin)*<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                           |
| clocks                |                                                                                                                                                                                                                                                                                                                                                                      |
|                       | <p>Returns a list of all of the clock objects arriving at the pin. You can use -index to return the v from a specific analysis view.<br/> <b>Type:</b> obj(clock)*<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                           |
| constant              |                                                                                                                                                                                                                                                                                                                                                                      |
|                       | <p>Returns if this pin has a constant logic value of 1'b0 or 1'b1 directly on the pin in the Verilog no_constant, and it is connected to a net, you must also check the net .constant value to see if the net driving the pin is constant.<br/> <b>Type:</b> enum<br/> <b>Enum Values:</b> 0 1 no_constant<br/> <b>Default:</b> no_constant<br/> <b>Edit:</b> No</p> |
| constant_value        |                                                                                                                                                                                                                                                                                                                                                                      |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Returns a constant value of 0 or 1 if logic state has been asserted or propagated to this pin. Use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                   |
| cts_add_pin_driver       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>Specifies a cell type or pair of cell types, so that a cell inst can to be added above an input or below an output pin.</p> <p>The pin is specified by the argument pin.</p> <p>No more than two cell types may be specified. A warning will be emitted if no AON cells are specified.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                              |
| cts_add_port_driver      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>Specifies a cell type or pair of cell types, so that a cell inst can to be added above an output or below an input port.</p> <p>The port is specified by the argument pin.</p> <p>If the pin specified is not a design IO pin or it is not in the clock network then CCOpt will emit a warning and will not add cell insts at that position. No more than two cell types may be specified. A warning will be emitted if no AON cells are specified.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_annotated_delay_to   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | <p>Override any clock tree timing engine computed cell arc or net arc delays to this pin, in a similar manner to SDC set_annotated_delay.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> delay_corner</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                            |
| cts_annotated_transition |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>Override any clock tree timing engine computed transition at this pin, in a similar manner to set_annotated_transition.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> delay_corner</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                    |
| cts_assign_clock_tree |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                       | <p>Can be used to make sure a clock tree sink pin gets driven by a particular clock tree. The clock tree can alternatively be specified as the tree's root pin, in case extraction rennumbers the clock trees defined from a single SDC clock.</p> <p>By default CCOpt automatically selects appropriate pins.</p> <p>Valid values: cts_clock_tree   list of pins</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_case_analysis     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                       | <p>Specifies a constant value to be used as pin signal when analyzing timing within CTS. Only applies to input pins of cells in the clock tree. If set to 'none' (the default), CTS will consider the input as non-constant.</p> <p>Valid values: 0 1 none</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p>                                                                                                      |
| cts_clock_tree        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                       | <p>The clock tree to which this object belongs. Flops do not belong to a clock tree, but their clock pins do.</p> <p>Valid values: cts_clock_tree</p> <p><b>Type:</b> <a href="#">obj(clock_tree)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                         |
| cts_clock_trees       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                       | <p>A list of clock trees the pin is contained within. This includes parents of generated clock trees and all relevant parents when clock trees overlap.</p> <p>Valid values: list cts_clock_tree</p> <p><b>Type:</b> <a href="#">obj(clock_tree)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                          |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cts_effective_mixed_buffer_budget | <p>Allows the user to query the mixed buffer budget at a particular pin.<br/>This is the number of buffers allowed above the given pin, and is the smallest of the remaining budgets from any fanout below this point or the override set at this point if one has been set.</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> auto<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                  |
| cts_effective_routing_preference  | <p>Returns the pin's effective routing preference; one of leaf, trunk, top, none. The value "none" returned for pins that have no preference such as antenna diodes or load cap inputs. The routing preference value for a pin does not solely determine the net_type of the driving net has multiple fanout. The fanout routing preferences for all sinks are taken into account when determining the net_type for the net as a whole.</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                         |
| cts_flexible_htree                | <p>The flexible HTree, if any, associated with the given object.</p> <p><b>Type:</b> <a href="#">obj(flexible_htree)*</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| cts_is_sdc_clock_root             | <p>Specifies whether the given pin is the root (source pin) of an SDC clock.<br/>The create_clock_tree_spec populates this attribute with the location of the SDC clock root (source) pins.<br/>This attribute controls the behavior of clock tree definition commands create_clock_tree and create_generated_clock_tree, when the -stop_at_sdc_clock_roots argument is specified. In such a case, pins and ports for which this attribute is true will be treated as being SDC clock root pins.<br/>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p> |
| cts_isolated_effective            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>This is the isolated value CCOpt will use for the given pin.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| cts_net_unbufferable_reasons |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                              | <p>This attribute contains a list of reasons why CCOpt was not able to buffer the clock net attached to the specified pin.</p> <p>Valid values: string</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| cts_node_type                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                              | <p>node type of pin within cts graph</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| cts_pin_capacitance_sources  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                              | <p>List of the places the capacitance could be retrieved from.</p> <p>Requesting this attribute for a pin not in the clock tree will result in an error.</p> <p>The possible places are:</p> <ul style="list-style-type: none"> <li>- library - CTS will retrieve capacitance values from the library for one or more timing_corner combinations.</li> <li>- cts_capacitance_override - CTS will retrieve capacitance values from the cts_capacitance_override attribute for one or more timing_corner/event combinations.</li> <li>- cts_blackbox_default_load_base_pin - CTS will retrieve capacitance values from the cts_blackbox_default_load_base_pin attribute for one or more timing_corner/event combinations.</li> </ul> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| cts_pin_insertion_delay      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>The amount of insertion delay under this pin. Clock tree synthesis will attempt to make the insertion delay to this pin less than that to other sinks in the same skew group by this amount if a positive value is set. A negative value should be used if you would like the insertion delay to this pin to be greater than that to other sinks. The value 'auto' means there is no insertion delay offset for the pin.</p> <p>This attribute exposes the 'total' pin insertion delay. For further discussion on this, consult the documentation for attribute <code>cts_pin_insertion_delay_source</code></p> <p>Valid values: double   ignore   min   auto</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> delay_corner</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> |
| cts_pin_insertion_delay_sources |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

The amount of insertion delay under this pin, broken down by the source of the pin insertion delay (PID).  
The value of this attribute is a Tcl dictionary of PID values, keyed on the source of that PID. This provides an 'exploded' view into the total pin insertion delay that is present, categorizing the various contributions by their origin.  
Valid Tcl dict keys:  
user Asserted by user Tcl scripting or restored from an older DB.  
sdc Extracted from SDC set\_clock\_latency by spec creation.  
sink\_type Asserted to reflect the impact of the cts\_sink\_type attribute.  
ilm Records the mean clock latency of the ILM clock network that is rooted at this pin.  
useful\_skew A useful skew generated by pre-CTS optimization.  
incremental An incremental offset, used to specify useful skews to incremental CTS.  
total The formal sum of the above contributions. This value is the 'overall' pin insertion delay accessible via the cts\_pin\_insertion\_delay attribute.  
Valid Tcl dict values: double | ignore | min | auto  
Reading the attribute returns a dict value that exposes the current state of the PID, recording all the PID sources and the associated PID value of each one. Sources whose value are auto are omitted. If there are no non-auto sources, the returned dict is empty. For convenience, the total is always included in the dict if it is non-auto.  
Writing a dict value to the attribute will update each of the specified sources with the specified PID value. The total PID value will then be updated. Sources in the dict that are assigned a value of 'auto' will be reset. Sources omitted from the dict are left unchanged from their current value: assigning an empty dict is a no-op.  
Resetting the attribute will clear all the sources and the total back, resetting them to the default value of 'auto', thereby completely removing the PID from the pin.  
Valid values: A Tcl dictionary of PID values, keyed on PID source.  
**Type:** [string](#)  
**Allowed -index values:** delay\_corner  
**Default:** ""  
**Edit:** Yes

cts\_pin\_route\_type\_propagation

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Describes how <code>cts_pin_route_type</code> should be applied to the net driven by the specified pin. The possible values are:</p> <ul style="list-style-type: none"> <li>- none - it applies to net driven by the pin and only that net, irrespective of whether the net buffered or not.</li> <li>- net - it applies to all nets generated by buffering the pre-CTS net driven by pin (default).</li> </ul> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> net<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                      |
| cts_routing_trunk_override |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                            | <p>Prefer trunk routing rules for this pin. Only applies to clock tree sinks.</p> <p>Valid values: true false</p> <p><b>Type:</b> <code>bool</code><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cts_sink_type              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                            | <p>The type of sink this pin represents.</p> <p>Valid values are as follows:</p> <ul style="list-style-type: none"> <li>auto The sink type will be automatically determined by CCOpt.</li> <li>through Through pin. Trace the clock tree through this pin.</li> <li>stop Stop pin. When defining clock trees, CCOpt stops searching for parts of the clock tree at stop pins.</li> <li>ignore Ignore pin. CCOpt stops searching for parts of the clock tree at ignore pins and it does not attempt to balance the insertion delay of ignore pins.</li> <li>exclude Exclude pin. Exclude this pin from the clock tree.</li> </ul> <p>Valid values: auto through stop ignore exclude</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes</p> |
| cts_sink_type_effective    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                            | <p>Indicates how CCOpt will treat a given pin, taking into account both its <code>cts_sink_type_implicit</code> and any <code>cts_sink_type</code> settings.</p> <p>Setting a non-default value for the <code>cts_sink_type</code> attribute will override the <code>cts_sink_type_implicit</code> attribute.</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> auto<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                       |
| cts_sink_type_implicit     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Indicates the type of sink that CCOpt classified this pin as. Note that the cts_sink_type attribute override these internal classifications.</p> <p>Possible values are:</p> <ul style="list-style-type: none"> <li>exclude Indicates that this pin is a sink which represents a non-clock pin.</li> <li>ignore Indicates that CCOpt has determined not to search for more clock tree through this pin. Additionally, this pin will not be balanced.</li> <li>stop Indicates that CCOpt has determined not to search for more clock tree through this pin.</li> </ul> <p>An empty value for this attribute indicates either that this pin is either not a sink, or that it is that is not implicitly exclude or ignore or stop.</p> <p><b>Type:</b> string<br/><b>Default:</b> exclude<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| cts_sink_type_reasons  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                        | <p>The reasons why this pin has the given cts_sink_type.</p> <p>This attribute is configured by the create_clock_tree_spec command in order to record the reasons that the cts_sink_type attribute has been configured for the given pin.</p> <p>This attribute is a list of values. Valid values are as follows:</p> <ul style="list-style-type: none"> <li>auto The cts_sink_type attribute is set to 'auto'</li> <li>implicit The pin is an implicit sink (flop/latch)</li> <li>user The user has set the cts_sink_type attribute</li> <li>design_io This pin is a design I/O</li> <li>multiple_outputs This pin is on an instance with multiple outputs, and the cts_spec_config_trace_through_to attribute has not been set</li> <li>set_disable_timing SDC set_disable_timing stops the clock at this pin</li> <li>set_case_analysis SDC set_case_analysis stops the clock at this pin</li> <li>generated_clock_tree This pin is the generator input to an SDC generated clock</li> <li>no_sdc_clock The SDC clock is stopped at this pin for other reasons</li> <li>ilm The create_clock_tree_spec command has detected an ILM below this pin</li> </ul> <p>Valid values: auto implicit user design_io multiple_outputs set_disable_timing set_case_analysis generated_clock_tree no_sdc_clock ilm</p> <p><b>Type:</b> string<br/><b>Default:</b> auto<br/><b>Edit:</b> Yes</p> |
| cts_skew_groups_active |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Returns the list of active skew groups for this pin.<br/>For sink pins, this attribute lists both skew groups that pass through this pin and skew groups for which this sink is an endpoint.<br/>For non-sink pins, shows skew groups that pass through this pin.<br/>Valid values: list skew_groups<br/><b>Type:</b> <a href="#">obj(skew_group)*</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                        |
| cts_skew_groups_active_sink       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                   | <p>Returns the list of skew groups for which this pin is an active sink.<br/>For sink pins, this attribute lists the skew groups for which this sink is an endpoint. Skew groups that pass through this pin are not included.<br/>For non-sink pins, this attribute always returns null.<br/>Valid values: list skew_groups<br/><b>Type:</b> <a href="#">obj(skew_group)*</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                    |
| cts_skew_groups_constraining      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                   | <p>The list of delay-constraining skew groups which are active at this pin.<br/>This attribute reports similar data to<br/>cts_skew_groups_active. The only difference is that<br/>the reporting-only skew groups are not included in this attribute's value.<br/>For sink pins, this attribute lists both delay-constraining skew<br/>groups that pass through this pin and delay-constraining skew groups for which<br/>this sink is an endpoint.<br/>For non-sink pins, shows delay-constraining skew groups that pass through this pin.<br/>Valid values: list skew_groups<br/><b>Type:</b> <a href="#">obj(skew_group)*</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| cts_skew_groups_constraining_sink |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>The list of delay-constraining skew groups for which this pin is an active sink. This attribute lists the delay-constraining skew groups for which this pin is an active sink. This attribute reports similar data to <code>cts_skew_groups_active_sink</code>. The only difference is that the reporting-only (constrains none) skew groups are not included in this attribute's value.</p> <p>For sink pins, this attribute lists the delay-constraining skew groups for which this sink is an endpoint. Skew groups that pass through this pin are not included.</p> <p>For non-sink pins, this attribute always returns null.</p> <p>Valid values: list skew_groups</p> <p><b>Type:</b> <code>obj(skew_group)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| cts_skew_groups_ignore           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                  | <p>The list of skew groups for which paths through this pin are ignored.</p> <p>Valid values: list skew_groups</p> <p><b>Type:</b> <code>obj(skew_group)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| cts_skew_groups_sink             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                  | <p>The list of skew groups for which this pin is a sink.</p> <p>Valid values: list skew_groups</p> <p><b>Type:</b> <code>obj(skew_group)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| cts_skew_groups_source_pin       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                  | <p>The list of skew groups for which this clock tree or pin is specified as a source.</p> <p>Valid values: list skew_groups</p> <p><b>Type:</b> <code>obj(skew_group)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| cts_spec_config_trace_through_to |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>Clock tree definition will, by default, not continue through certain types of cell arc (for instance, the clock to Q arc in a DFF). This attribute allows you to override this default behavior, permitting the clock tree to trace through such a cell.</p> <p>The attribute should be configured on the input pin at which the clock arrives. The value of the attribute specifies the output pin to which the clock should propagate. The specified output pin must be another pin on the same instance. The output pin may be specified either by its fully qualified name (i.e. inclusive of the instance name), or else simply by its local (cell-relative) name.</p> <p>There must be a pre-existing (library-defined) chain of one or more delay arcs that connect the input and output pins together. It is not possible to use trace_through_to to synthesize delay arcs.</p> <p>If multiple input pins are annotated on a given instance, the value of trace_through_to at each of those pins must select the same output pin: i.e. the configuration must identify a single clock output for the instance. If multiple clock outputs are required then trace_through_to should not be used: instead define a generated clock tree at each of the clock-carrying outputs.</p> <p>If the configuration of trace_through_to settings for a given instance does not meet these requirements, a warning will be issued and the settings for that instance will be ignored.</p> <p>Note: if both trace_through_to and library_trace_through_to are applicable at a given netlist instance pin, the trace_through_to value takes precedence.</p> <p>Valid values: pin</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p> |
| cts_top_fanout_count_override   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                 | <p>The number of clock sinks this sink counts for when applying the top routing rules.</p> <p>Note that this attribute is only valid for sink pins, and it returns auto for non-sink pins.</p> <p>For a sink pin, a non-auto value means that this sink is counted as though it were multiple sinks, for the purposes of determining which nets should have top routing. An auto value for a sink pin means that the sink counts as a single sink.</p> <p>Valid values: integer &gt; 0</p> <p><b>Type:</b> string<br/><b>Default:</b> auto<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| cts_top_fanout_transitive_count |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>The number of clock sinks in the transitive fanout of the pin as counted for applying the top rules.</p> <p>This attribute is very similar to the <code>cts_transitive_fanout</code> attribute but counts sink fanout using <code>cts_top_fanout_count_override</code> attribute instead of always counting sinks as a single item fanout.</p> <p>Requesting this attribute for a pin not in the clock tree will result in an error.</p> <p>Valid values: integer</p> <p><b>Type:</b> <code>int</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> No</p> |
| <code>cts_transitive_fanout</code>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>The number of clock sinks in the transitive fanout of the pin, within the clock tree.</p> <p>Requesting this attribute for a pin not in the clock tree will result in an error.</p> <p>Valid values: int</p> <p><b>Type:</b> <code>int</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                          |
| <code>cts_virtual_delay_early_fall</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>The amount of virtual delay that has been applied under this pin.</p> <p>Valid values: double</p> <p><b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> <code>delay_corner</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                    |
| <code>cts_virtual_delay_early_rise</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>The amount of virtual delay that has been applied under this pin.</p> <p>Valid values: double</p> <p><b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> <code>delay_corner</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                    |
| <code>cts_virtual_delay_late_fall</code>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>The amount of virtual delay that has been applied under this pin.</p> <p>Valid values: double</p> <p><b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> <code>delay_corner</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                    |

|                             |                                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cts_virtual_delay_late_rise |                                                                                                                                                                                                                                   |
|                             | <p>The amount of virtual delay that has been applied under this pin.<br/>Valid values: double<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> delay_corner<br/><b>Default:</b> 0<br/><b>Edit:</b> Yes</p>               |
| delay_max_fall              |                                                                                                                                                                                                                                   |
|                             | <p>Returns the maximum falling delay. You can use -index to return the value from a specific a view.<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| delay_max_rise              |                                                                                                                                                                                                                                   |
|                             | <p>Returns the maximum rising delay. You can use -index to return the value from a specific a view.<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>  |
| delay_min_fall              |                                                                                                                                                                                                                                   |
|                             | <p>Returns the minimum falling delay. You can use -index to return the value from a specific a view.<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| delay_min_rise              |                                                                                                                                                                                                                                   |
|                             | <p>Returns the minimum rising delay. You can use -index to return the value from a specific ar view.<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| dft_constant_value          |                                                                                                                                                                                                                                   |

|                              |                                                                                                                                                                                                                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | Constant pin value in test mode {logic_0   logic_1   logic_z   no_value}.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                            |
| dft_controllable             |                                                                                                                                                                                                                                                                                                      |
|                              | In test mode the output pin can take either the non_inverting or inverting value of the input.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                      |
| dft_dedicated_wrapper_reason |                                                                                                                                                                                                                                                                                                      |
|                              | Specifies the reason behind putting dedicated wrapper.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                                                              |
| dft_driven_by_clock          |                                                                                                                                                                                                                                                                                                      |
|                              | If pin is driven by clock.<br><b>Type:</b> enum<br><b>Enum Values:</b> false true<br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                                                       |
| dft_partition                |                                                                                                                                                                                                                                                                                                      |
|                              | Determine the DFT partition of the specified port/pin/hierarchical pin.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                              |
| direction                    |                                                                                                                                                                                                                                                                                                      |
|                              | Pin's direction from the corresponding base_pin. It can be in, out, inout or internal. Internal is an internal pin from a .lib file for the timing model, and is not part of the netlist.<br><b>Type:</b> enum<br><b>Enum Values:</b> internal in out inout<br><b>Default:</b> ""<br><b>Edit:</b> No |
| dont_invert_phase            |                                                                                                                                                                                                                                                                                                      |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| dont_touch                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                               | <p>The preservation status of a pin during optimization. A preserved pin means the logical function of the pin must be preserved to maintain a simulation or test-point pin in the netlist. However, the pin name does not need to be preserved.</p> <p><b>Supported values:</b></p> <ul style="list-style-type: none"> <li>false: Unconstrained</li> <li>true: Cannot add/remove ports</li> <li>delete_ok: Can delete ports (if they have no fanout)</li> <li>invert_ok: We cannot delete, duplicate, or add but can change the polarity</li> <li>none: No user setting; will inherit from the module/hinst</li> </ul> <p><b>Type:</b> <code>enum</code><br/> <b>Enum Values:</b> none false true delete_ok invert_ok const_prop_delete_ok<br/> <b>Default:</b> none<br/> <b>Edit:</b> Yes</p> |
| dont_touch_file               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                               | <p>Specifies the sourced file to set dont_touch.</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| driver_cell                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                               | <p>Reports driver cell of pin</p> <p><b>Type:</b> <code>obj(inst)*</code><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| driver_cell_name              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                               | <p>Reports driver cell name of pin</p> <p><b>Type:</b> <code>string</code><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| driver_ground_pin_voltage_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Reports maximum ground pin voltage of driver cell</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                        |
| driver_ground_pin_voltage_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                               | <p>Reports minimum ground pin voltage of driver cell</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                        |
| driver_power_pin_voltage_max  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                               | <p>Reports maximum power pin voltage of driver cell</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                         |
| driver_power_pin_voltage_min  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                               | <p>Reports minimum power pin voltage of driver cell</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                         |
| driver_ref_lib_cell_name      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                               | <p>Reports library cell name of driver cell of the pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                      |
| edge                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                               | <p>If this pin is for a partition, or inst that is a black-box, and the pin is assigned, the edge value indicates along which edge of the boundary polygon the pin is assigned. The edge number from the lowest Y, then left-most X vertex, starting with 0, and then counting clock-wise. See set_pin_constraint command document for a figure showing the edge numbering. If the pin assigned, or not for a partition or black-box inst, the value of -1 is returned.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| effective_stack_via_rule |                                                                                                                                                                                                                                                                                                                                                                                                          |
|                          | <p>The stack_via_rule that is expected to be used by the router for connecting to this pin. This derived from the other pin and base_pin attributes related to stack via.</p> <p><b>Type:</b> <a href="#">obj(stack_via_rule)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                       |
| escaped_name             |                                                                                                                                                                                                                                                                                                                                                                                                          |
|                          | <p>The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a scalar.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| exceptions               |                                                                                                                                                                                                                                                                                                                                                                                                          |
|                          | <p>Reports the path exceptions</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                        |
| fanin                    |                                                                                                                                                                                                                                                                                                                                                                                                          |
|                          | <p>fanin</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                 |
| fanout                   |                                                                                                                                                                                                                                                                                                                                                                                                          |
|                          | <p>fanout</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                |
| fanout_load              |                                                                                                                                                                                                                                                                                                                                                                                                          |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Returns the fanout load for the pin. You can use -index to return the value for a specific view</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                     |
| fastest_clock_name   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                      | <p>Reports the clock name with maximum frequency associated with the pin</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                               |
| fastest_clock_period |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                      | <p>Reports clock period of maximum frequency clock associated with the pin</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                             |
| from_arcs            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                      | <p>Returns a list of arc objects for which this pin is the starting pin of the timing arc.</p> <p><b>Type:</b> obj(arc)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                   |
| hdl_name             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                      | <p>This is the original RTL name for this pin. It is used to map RTL simulation results with RTL to the current netlist for switching activity analysis. It is only maintained properly on the output of sequential cells. Optimization will copy this name during any multi-bit merge or splitting transforms to the equivalent pin, but not on output pins of combinational cells. The is_phase_inverted attribute will be flipped if the phase is inverted.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| hierarchical_level   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                      | <p>Returns hierarchical level for an Instance Pin.</p> <p><b>Type:</b> int</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                        |

|                         |                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hnet                    |                                                                                                                                                                                                                                                                                                                                    |
|                         | <p>pointer to the hnet connected to the pin</p> <p><b>Type:</b> <a href="#">obj(hnet)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                     |
| hold_uncertainty        |                                                                                                                                                                                                                                                                                                                                    |
|                         | <p>Returns the most conservative uncertainty of all possible uncertainty assertions associated pin. You can use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| holding_resistance_high |                                                                                                                                                                                                                                                                                                                                    |
|                         | <p>Reports high holding resistance of pin</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                 |
| holding_resistance_low  |                                                                                                                                                                                                                                                                                                                                    |
|                         | <p>Reports low holding resistance of pin</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                  |
| imm_parent_cell_id      |                                                                                                                                                                                                                                                                                                                                    |
|                         | <p>Reports numeric identifier assigned to parent cell during inter module matrix analysis</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                    |
| initial_name            |                                                                                                                                                                                                                                                                                                                                    |
|                         | <p>This will be used for verification of initial input netlist (post-synthesis) to any other netlist generated during Innovus flow.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                               |



|                            |                                                                                                                                                                                                                            |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input_signal_level         |                                                                                                                                                                                                                            |
|                            | <p>Returns the rail name associated with input pin.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>               |
| input_signal_level_voltage |                                                                                                                                                                                                                            |
|                            | <p>Returns rail voltage of signal which is driving the input pin.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| inst                       |                                                                                                                                                                                                                            |
|                            | <p>pointer to the inst containing the pin</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">inst</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                               |
| interpolation_info_max     |                                                                                                                                                                                                                            |
|                            | <p>Reports interpolation information related to max corner</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| interpolation_info_min     |                                                                                                                                                                                                                            |
|                            | <p>Reports interpolation information related to min corner</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| is_always_on               |                                                                                                                                                                                                                            |
|                            | <p>Returns a value of true if the pin is always on type pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| is_async                   |                                                                                                                                                                                                                            |

|                        |                                                                                                                                                                                                                                                                                                                |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Returns a value of true if the pin is an asynchronous clear or preset pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                           |
| is_booth_encoded       |                                                                                                                                                                                                                                                                                                                |
|                        | <p>Booth encoded setting assigned to input port by user.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                     |
| is_clear               |                                                                                                                                                                                                                                                                                                                |
|                        | <p>Returns a value of true if the pin is an asynchronous clear pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                     |
| is_clock               |                                                                                                                                                                                                                                                                                                                |
|                        | <p>Returns a value of true if the pin has the Liberty pin attribute: clock .</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                            |
| is_clock_gating        |                                                                                                                                                                                                                                                                                                                |
|                        | <p>Returns a value of true if the pin is defined as a pin of a clock gating cell. You can use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_clock_gating_clock  |                                                                                                                                                                                                                                                                                                                |
|                        | <p>Returns a value of true if the pin is defined as a clock pin of a clock gating cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                 |
| is_clock_gating_enable |                                                                                                                                                                                                                                                                                                                |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Returns a value of true if the pin corresponds to the enable pin of a clock gating cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                    |
| is_clock_network_pin    |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                         | <p>Reports true if specified pin is part of clock network</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                    |
| is_clock_source_latency |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                         | <p>Returns true if object is part of latency network</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                           |
| is_clock_source_pin     |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                         | <p>Returns true if object is a clock source pin</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                |
| is_clock_used_as_clock  |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                         | <p>Returns a value of true if the pin lies in the clock network and at least one of the clocks arriving at the pin is used as a clock in the downstream network of the pin. You can use -index to return a value for a specific analysis view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_clock_used_as_data   |                                                                                                                                                                                                                                                                                                                                                                                                       |

|                          |                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Returns a value of true if the pin lies in the clock source path and at least one of the clocks on the pin is used as data in the downstream network of the pin. You can use -index to return value for a specific analysis view.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_constant_in_all_views |                                                                                                                                                                                                                                                                                                                                                                                          |
|                          | <p>Returns true if logic state has been asserted or propagated to this pin in all the views</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                          |
| is_constant_in_any_view  |                                                                                                                                                                                                                                                                                                                                                                                          |
|                          | <p>Returns true if logic state has been asserted or propagated to this pin in atleast one view.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                      |
| is_constrained           |                                                                                                                                                                                                                                                                                                                                                                                          |
|                          | <p>Returns true if specified has a constraint applied</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                |
| is_constrained_endpoint  |                                                                                                                                                                                                                                                                                                                                                                                          |
|                          | <p>Reports true if pin is a constrained endpoint</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                     |
| is_data                  |                                                                                                                                                                                                                                                                                                                                                                                          |
|                          | <p>Returns a value of true if the pin is a data pin (that is, is not a clock pin).</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                   |

|                              |                                                                                                                                                                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_disable_timing            |                                                                                                                                                                                                                                                                               |
|                              | <p>Returns a value of true if the pin's timing has been disabled. You can use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_dynamic_clock_gating_data |                                                                                                                                                                                                                                                                               |
|                              | <p>Returns true if pin is data pin of a dynamic clock gating cell</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                      |
| is_endpoint                  |                                                                                                                                                                                                                                                                               |
|                              | <p>Returns true if specified pin is an endpoint</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                        |
| is_existing_cte_pin          |                                                                                                                                                                                                                                                                               |
|                              | <p>Returns true if pin exists as part of timing graph during analysis</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                  |
| is_fall_edge_triggered_clock |                                                                                                                                                                                                                                                                               |
|                              | <p>Returns a value of true if the pin is a clock pin of a flop, and is triggered by the falling edge of the clock.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                     |
| is_fall_edge_triggered_data  |                                                                                                                                                                                                                                                                               |

|                         |                                                                                                                                                                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Returns a value of true if the pin corresponds to the data pin of a fall edge triggered device.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>     |
| is_hier_crossing_needed |                                                                                                                                                                                                                                                               |
|                         | <p>Returns true if needs hierarchical crossing arcs to be built</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                        |
| is_hierarchical         |                                                                                                                                                                                                                                                               |
|                         | <p>is_hierarchical</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                     |
| is_inside_partition     |                                                                                                                                                                                                                                                               |
|                         | <p>Returns whether pin is in partition or not.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                         |
| is_inter_die_feedthru   |                                                                                                                                                                                                                                                               |
|                         | <p>Returns true if pin is a freethru inter-die pin</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                     |
| is_internal_disable     |                                                                                                                                                                                                                                                               |
|                         | <p>Returns the internal disabled assertion on endpoints. Such endpoints will have no timing c on them.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_isolation_cell_clock |                                                                                                                                                                                                                                                               |

|                          |                                                                                                                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | Returns true if a clock pin is part of an isolatoin cell,<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No              |
| is_isolation_cell_data   |                                                                                                                                                                                                           |
|                          | Returns a value of true if the pin is isolation data pin type.<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No         |
| is_isolation_cell_enable |                                                                                                                                                                                                           |
|                          | Returns a value of true if the pin is isolation enable pin type.<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No       |
| is_latency_network_pin   |                                                                                                                                                                                                           |
|                          | Returns a value of true if the pin has latency phase propogated to it.<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| is_level_shifter_data    |                                                                                                                                                                                                           |
|                          | Returns a value of true if the pin is level shifter data pin type.<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No     |
| is_level_shifter_enable  |                                                                                                                                                                                                           |
|                          | Returns a value of true if the pin is level shifter enable pin type.<br><b>Type:</b> <a href="#">bool</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No   |
| is_marked                |                                                                                                                                                                                                           |

|                                   |                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Returns true if pin is marked</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                              |
| is_multiple_clock_fanin_point     |                                                                                                                                                                                                                                                                                                      |
|                                   | <p>Returns a value of true if multiple clock phases converge at the pin. You can use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_mux_select_pin                 |                                                                                                                                                                                                                                                                                                      |
|                                   | <p>Returns the is_mux_select_pin property for a pin. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                              |
| is_negative_level_sensitive_clock |                                                                                                                                                                                                                                                                                                      |
|                                   | <p>Returns a value of true if the library pin is an enable pin of an active low level-sensitive device.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                       |
| is_negative_level_sensitive_data  |                                                                                                                                                                                                                                                                                                      |
|                                   | <p>Returns a value of true if the pin is a data pin of an active low level-sensitive device.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                  |
| is_owner                          |                                                                                                                                                                                                                                                                                                      |
|                                   | <p>Returns true if client is the owner for pin in distributed STA.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                            |



|                                   |                                                                                                                                                                                                                                                            |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_pba_begin_point_candidate      |                                                                                                                                                                                                                                                            |
|                                   | <p>Returns a value of true if the pin is PBA begin point candidate.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                 |
| is_phase_inverted                 |                                                                                                                                                                                                                                                            |
|                                   | <p>Identifies the pin for which phase is inverted during optimizations.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                  |
| is_pin                            |                                                                                                                                                                                                                                                            |
|                                   | <p>Returns true if object is a pin</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                  |
| is_port                           |                                                                                                                                                                                                                                                            |
|                                   | <p>Returns true if pin is a port</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                    |
| is_positive_level_sensitive_clock |                                                                                                                                                                                                                                                            |
|                                   | <p>Returns a value of true if the library pin is an enable pin of an active high level-sensitive de</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_positive_level_sensitive_data  |                                                                                                                                                                                                                                                            |
|                                   | <p>Returns a value of true if the pin is a data pin of an active high level-sensitive device.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| is_power_mode_disabled            |                                                                                                                                                                                                                                                            |

|                              |                                                                                                                                                                                                                                                                                                                       |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Returns true if power mode is disabled</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                            |
| is_preset                    |                                                                                                                                                                                                                                                                                                                       |
|                              | <p>Returns a value of true if the pin is a preset pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                         |
| is_propagated_clock          |                                                                                                                                                                                                                                                                                                                       |
|                              | <p>Returns a value of true if there is an explicit set_propagated_clock assertion at the pin. You use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_ref_clock_pin             |                                                                                                                                                                                                                                                                                                                       |
|                              | <p>Returns true if object is a reference clock pin</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                             |
| is_rise_edge_triggered_clock |                                                                                                                                                                                                                                                                                                                       |
|                              | <p>Returns a value of true if the pin is a clock pin of a flop, and is triggered by the rising edge clock.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                     |
| is_rise_edge_triggered_data  |                                                                                                                                                                                                                                                                                                                       |
|                              | <p>Returns a value of true if the pin corresponds to the data pin of a rise edge-triggered device</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                              |

|                         |                                                                                                                                                                                                                           |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_sequential_duplicate |                                                                                                                                                                                                                           |
|                         | <p>Identifies the Pin which is cloned or duplicated.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                     |
| is_sequential_merged    |                                                                                                                                                                                                                           |
|                         | <p>Keep track of Pins after merge or split during optimization.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                          |
| is_sequential_retimed   |                                                                                                                                                                                                                           |
|                         | <p>is retimed of pin</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                    |
| is_shared               |                                                                                                                                                                                                                           |
|                         | <p>Returns true if pin is shared across clients in distributed STA</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_special              |                                                                                                                                                                                                                           |
|                         | <p>pin is belong DEF SPECIALNETS section</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                |
| is_timing_ignored       |                                                                                                                                                                                                                           |
|                         | <p>Returns true if pin is ignored fur timing analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>              |
| is_tristate             |                                                                                                                                                                                                                           |

|                             |                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>Returns a value of true if the pin has the three_state attribute in the Liberty timing library.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                         |
| is_tristate_enable          |                                                                                                                                                                                                                                                                                                                                                                                   |
|                             | <p>Returns a value of true if the pin is the source pin of timing arcs with either the three_state_ or three_state_disable attribute in the Liberty timing library.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                        |
| is_tristate_output          |                                                                                                                                                                                                                                                                                                                                                                                   |
|                             | <p>Returns a value of true if the pin corresponds to a three-state output pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                             |
| is_valid_for_reports        |                                                                                                                                                                                                                                                                                                                                                                                   |
|                             | <p>Returns the pin is valid for reporting.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                 |
| is_valid_pin_for_imm_report |                                                                                                                                                                                                                                                                                                                                                                                   |
|                             | <p>Returns true if pin is classified for inter-module-matrix analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                      |
| latch_time_given            |                                                                                                                                                                                                                                                                                                                                                                                   |
|                             | <p>In latch-based analysis, this returns the value by which the arrival time on this pin is adjusted account for time borrowed at the previous stage. You can use -index to return the value for specific analysis view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

|                      |                                                                                                                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| layer                |                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Layer of the pin. For pins with more than one shape, it is the layer of the first shape found in This is also the same shape used for the .location value. If the pin has no shapes or vias de then {} is returned.</p> <p><b>Type:</b> <a href="#">obj(layer)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| lib_binding_info_max |                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Reports the type of bindinf, i.e. Match/Interpolation/Extrapolation/UserSpecified</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                |
| lib_binding_info_min |                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Reports the type of bindinf, i.e. Match/Interpolation/Extrapolation/UserSpecified</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                |
| lib_pins             |                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Returns a list of library pin (lib_pin) objects which are associated with this pin.</p> <p><b>Type:</b> <a href="#">obj(lib_pin)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                              |
| local_slack_max      |                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Reports maximum local slack of pin</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                               |
| local_slack_max_fall |                                                                                                                                                                                                                                                                                                                        |
|                      | <p>Reports maximum fall local slack of pin</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                          |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| local_slack_max_rise    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>Reports maximum rise local slack of pin</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                  |
| local_slack_min         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>Reports minimum local slack of pin</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                       |
| local_slack_min_fall    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>Reports minimum fall local slack of pin</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                  |
| local_slack_min_rise    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>Reports minimum rise local slack of pin</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                  |
| location                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | <p>Location of the pin. The returned point will overlap some part of the pin shape. For pins with than one shape, the point overlaps the first shape found in the DB. This is also the same sh used for the .layer value. If the pin has no shapes or vias defined, then the inst .location is r It is first initialized when the LEF is read in, but the location might be changed by the router with the routing tracks and default routing via when it first runs.</p> <p><b>Type:</b> point</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| lp_asserted_probability |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                |                                                                                                                                                                                                                                                                     |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Probability of signal to be high on port/subport/pin as specified by user.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                |
| lp_asserted_toggle_rate        |                                                                                                                                                                                                                                                                     |
|                                | <p>Signal toggle rate assigned to pin by user.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                               |
| lp_system_asserted_probability |                                                                                                                                                                                                                                                                     |
|                                | <p>System/Internal asserted probability.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                                     |
| lp_system_asserted_toggle_rate |                                                                                                                                                                                                                                                                     |
|                                | <p>System/Internal asserted toggle rate.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                                     |
| lssd_master_clock              |                                                                                                                                                                                                                                                                     |
|                                | <p>Defines the master clock port related to this slave clock port.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                 |
| max_capacitance                |                                                                                                                                                                                                                                                                     |
|                                | <p>Returns the maximum capacitance limit for the pin. You can use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| max_cpvr_early                 |                                                                                                                                                                                                                                                                     |

|                    |                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>Reports maximum early CPPR</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                       |
| max_cppr_early     |                                                                                                                                                                                                                                                                                                                                                                                              |
|                    | <p>Reports maximum late CPPR</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                        |
| max_cppr_late      |                                                                                                                                                                                                                                                                                                                                                                                              |
|                    | <p>Returns the maximum fanout load that the pin can drive. This value is set using set_max_fanout the default_max_fanout library attribute. You can use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| max_fanout         |                                                                                                                                                                                                                                                                                                                                                                                              |
|                    | <p>In latch-based analysis, returns the maximum time that a path arriving at this latch input can borrow from the next stage - as specified by set_max_time_borrow. You can use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>            |
| max_time_borrow    |                                                                                                                                                                                                                                                                                                                                                                                              |
|                    | <p>In latch-based analysis, returns the maximum time that a path arriving at this latch input can borrow from the next stage - as specified by set_max_time_borrow -interface. You can use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| max_time_borrow_io |                                                                                                                                                                                                                                                                                                                                                                                              |
|                    | <p>In latch-based analysis, returns the maximum time that a path arriving at this latch input can borrow from the next stage - as specified by set_max_time_borrow -interface. You can use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| max_transition     |                                                                                                                                                                                                                                                                                                                                                                                              |



|                       |                                                                                                                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>Returns the maximum transition time limit specified for the pin. You can use -index to return value for a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                   |
| min_capacitance       |                                                                                                                                                                                                                                                                                                 |
|                       | <p>Returns the minimum capacitance limit for the pin. You can use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                             |
| min_fanout            |                                                                                                                                                                                                                                                                                                 |
|                       | <p>Returns the minimum fanout design rule limit of the corresponding library pin. You can use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| min_transition        |                                                                                                                                                                                                                                                                                                 |
|                       | <p>Returns the minimum transition time limit specified for the pin. You can use -index to return value for a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                   |
| multibit_aware_mapped |                                                                                                                                                                                                                                                                                                 |
|                       | <p>Set when the sequential instance is mapped for multibit.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                   |
| name                  |                                                                                                                                                                                                                                                                                                 |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>The same as .escaped_name without any \ escape chars. This name is commonly used to solve problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| net                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>pointer to the net connected to the pin</p> <p><b>Type:</b> obj(net)<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                        |
| net_name                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>net_name</p> <p><b>Type:</b> string<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                   |
| network_latency_fall_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Returns the maximum fall insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                       |
| network_latency_fall_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Returns the minimum fall insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific analysis view and/or clock.</p> <p><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                          |
| network_latency_rise_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                               |                                                                                                                                                                                                                                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Returns the maximum rise insertion delay specified by an explicit set_clock_latency at the p can use -index to return the value for a specific analysis view and/or clock</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| network_latency_rise_min      |                                                                                                                                                                                                                                                                                                                       |
|                               | <p>Returns the minimum rise insertion delay specified by an explicit set_clock_latency at the p can use -index to return the value for a specific analysis view and/or clock</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| noise_victim                  |                                                                                                                                                                                                                                                                                                                       |
|                               | <p>Reports associated victim noise object</p> <p><b>Type:</b> obj(pin)*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                 |
| num_exceptions                |                                                                                                                                                                                                                                                                                                                       |
|                               | <p>Reports the number of path exceptions</p> <p><b>Type:</b> int</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                        |
| num_phases                    |                                                                                                                                                                                                                                                                                                                       |
|                               | <p>Reports total number of phases on the pin</p> <p><b>Type:</b> int</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                    |
| num_violating_phase_count_max |                                                                                                                                                                                                                                                                                                                       |
|                               | <p>Reports count of maximum violating phases on pin</p> <p><b>Type:</b> int</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                             |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| num_violating_phase_count_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                               | <p>Reports count of minimum violating phases on pin</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                             |
| obj_type                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                               | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">pin</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                   |
| opt_skew_max_allowed_delay    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                               | <p>Specifies in nanoseconds the maximum permitted amount of insertion delay change that the software can make at this pin, when performing useful skew operations. When not specified given pin (value "no_value"), the tool will apply the globally configured skewing limits, for example as specified by attribute opt_skew_max_allowed_delay. This attribute only takes effect when configured on the load of a clock net.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p> |
| output_signal_level           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                               | <p>Returns the rail name associated with output pin.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                         |
| parent_cell                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                               | <p>Reports the parent cell of the cell to which specified pin belongs</p> <p><b>Type:</b> <a href="#">obj(hinst)</a>*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                   |
| pin_capacitance_max           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                          |                                                                                                                                                                                                     |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Reports maximum capacitance of pin</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| pin_capacitance_max_fall |                                                                                                                                                                                                     |
|                          | <p>Reports maximum fall capacitance of pin</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pin_capacitance_max_rise |                                                                                                                                                                                                     |
|                          | <p>Reports maximum rise capacitance of pin</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pin_capacitance_min      |                                                                                                                                                                                                     |
|                          | <p>Reports minimum capacitance of pin</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| pin_capacitance_min_fall |                                                                                                                                                                                                     |
|                          | <p>Reports minimum fall capacitance of pin</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pin_capacitance_min_rise |                                                                                                                                                                                                     |
|                          | <p>Reports minimum rise capacitance of pin</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pin_direction            |                                                                                                                                                                                                     |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Reports direction of pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                          |
| power_domain                       |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                    | <p>The parent power domain of the pin (equivalent to Design Browser effPD)</p> <p><b>Type:</b> <a href="#">obj(power_domain)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                         |
| power_rail_voltage_inout_input_max |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                    | <p>Reports power rail voltage for input part of bidi pins/ports for maximum operating condition. use -index to return the value for a specific analysis view</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                         |
| power_rail_voltage_inout_input_min |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                    | <p>Reports power rail voltage for input part of bidi pins/ports for minimum operating condition. use -index to return the value for a specific analysis view</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                         |
| power_rail_voltage_max             |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                    | <p>Reports power rail voltage for maximum operating condition for bidirectional, input and output. For bidirectional pins, the power rail voltage for output signal is reported. You can use -index to return the value for a specific analysis view</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| power_rail_voltage_min             |                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                |                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Reports power rail voltage for minimum operating condition for bidirectional, input and output. For bidirectional pins, the power rail voltage for output signal is reported. You can use -index to return the value for a specific analysis view</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| receiver_ccc_type              |                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Reports receiver ccc type of pin</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                  |
| related_ground_pin             |                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Reports associated ground pin of specified pin</p> <p><b>Type:</b> obj(pg_pin)* obj(pg_base_pin)*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                            |
| related_ground_pin_voltage_max |                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Reports maximum voltage of related ground pin of specified pin</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                    |
| related_ground_pin_voltage_min |                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Reports minimum voltage of related ground pin of specified pin</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                    |
| related_power_pin              |                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Reports related power pin of specified pin</p> <p><b>Type:</b> obj(pg_pin)* obj(pg_base_pin)*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                |

|                                    |                                                                                                                                                                                                                                                 |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| related_power_pin_voltage_max      |                                                                                                                                                                                                                                                 |
|                                    | <p>Reports maximum voltage of related power pin of specified pin</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| related_power_pin_voltage_min      |                                                                                                                                                                                                                                                 |
|                                    | <p>Reports minimum voltage of related power pin of specified pin</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| required_level                     |                                                                                                                                                                                                                                                 |
|                                    | <p>Reports required level of pin in timing graph</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                       |
| safety_dclst_error_signal_endpoint |                                                                                                                                                                                                                                                 |
|                                    | <p>This object is the endpoint for a safety-mechanism's error-signal.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                           |
| safety_dmr_clone                   |                                                                                                                                                                                                                                                 |
|                                    | <p>The dmr (dual modular redundancy) clone-flop of this parent-flop. This attribute is only stored on dmr parent-flops.</p> <p><b>Type:</b> <a href="#">obj(pin)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| safety_dmr_clone_inv               |                                                                                                                                                                                                                                                 |
|                                    | <p>The dmr (dual modular redundancy) inverter driving the clone-flop of this parent-flop. This attribute is only stored on dmr parent-flops.</p> <p><b>Type:</b> <a href="#">obj(inst)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| safety_dmr_error_signal_endpoint   |                                                                                                                                                                                                                                                 |



|                       |                                                                                                                                                                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>This object is the endpoint for a safety-mechanism's error-signal.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                             |
| safety_dmr_errors     |                                                                                                                                                                                                                                   |
|                       | <p>The dmr (dual modular redundancy) error-detection-logic of this parent-flop. This attribute is stored on dmr parent-flops.</p> <p><b>Type:</b> <a href="#">obj(inst)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| safety_dmr_parent     |                                                                                                                                                                                                                                   |
|                       | <p>The dmr (dual modular redundancy) parent-flop of this object.</p> <p><b>Type:</b> <a href="#">obj(pin)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                |
| safety_failure_mode   |                                                                                                                                                                                                                                   |
|                       | <p>The name of the failure-mode to which this pin belongs, as defined in the USF (Unified Safety Format) file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                    |
| safety_mechanism      |                                                                                                                                                                                                                                   |
|                       | <p>The name of the safety-mechanism applied to this pin, as defined in the USF (Unified Safety Format) file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                      |
| safety_mechanism_type |                                                                                                                                                                                                                                   |
|                       | <p>The type of safety-mechanism applied to this pin, as defined in the USF (Unified Safety Format) file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                          |
| safety_parity_bit     |                                                                                                                                                                                                                                   |

|                                     |                                                                                                                                                                                                                                                                          |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>The parity-bit of the parity-group containing this flop.</p> <p><b>Type:</b> <a href="#">obj(pin)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                            |
| safety_parity_error_signal_endpoint |                                                                                                                                                                                                                                                                          |
|                                     | <p>This object is the endpoint for a safety-mechanism's error-signal.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                    |
| safety_parity_rejection_reason      |                                                                                                                                                                                                                                                                          |
|                                     | <p>The reason why safety-mechanism insertion failed for this object.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                     |
| safety_ser_type                     |                                                                                                                                                                                                                                                                          |
|                                     | <p>The type of ser (soft error resilience) required for those objects with an ser safety mechanism.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> hard soft</p> <p><b>Default:</b> hard</p> <p><b>Edit:</b> No</p> |
| safety_tmr_clones                   |                                                                                                                                                                                                                                                                          |
|                                     | <p>The tmr (triple modular redundancy) clone-flop-bits of this parent-flop-bit.</p> <p><b>Type:</b> <a href="#">obj(pin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                       |
| safety_tmr_errors                   |                                                                                                                                                                                                                                                                          |
|                                     | <p>The tmr (triple modular redundancy) error-detection-logic of this parent-flop-bit.</p> <p><b>Type:</b> <a href="#">obj(inst)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                |
| safety_tmr_parent                   |                                                                                                                                                                                                                                                                          |
|                                     | <p>safety The tmr (triple modular redundancy) parent-flop-bit of this object.</p> <p><b>Type:</b> <a href="#">obj(pin)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                          |

|                                   |                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| safety_tmr_voters                 |                                                                                                                                                                                                                                                                                                                                  |
|                                   | <p>The tmr (triple modular redundancy) voting-logic of this parent-flop-bit.</p> <p><b>Type:</b> <a href="#">obj(inst)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                 |
| safety_tmr_well_tap_left_padding  |                                                                                                                                                                                                                                                                                                                                  |
|                                   | <p>The padding used to reserve space for well-tap cells to be placed next to the left edges of triple modular redundancy) flops.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> No</p>                                                                                                     |
| safety_tmr_well_tap_right_padding |                                                                                                                                                                                                                                                                                                                                  |
|                                   | <p>The padding used to reserve space for well-tap cells to be placed next to the right edges of triple modular redundancy) flops.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> No</p>                                                                                                    |
| setup_uncertainty                 |                                                                                                                                                                                                                                                                                                                                  |
|                                   | <p>Returns the most conservative uncertainty of all possible uncertainty assertions associated pin. You can use -index to return the value for a specific analysis view</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| si_slew_max                       |                                                                                                                                                                                                                                                                                                                                  |
|                                   | <p>Reports the maximum slew considering incremental slew from signal integrity analysis</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                 |
| si_slew_max_fall                  |                                                                                                                                                                                                                                                                                                                                  |
|                                   | <p>Reports the maximum fall slew considering incremental slew from signal integrity analysis</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                            |

|                       |                                                                                                                                                                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| si_slew_max_rise      |                                                                                                                                                                                                                                            |
|                       | <p>Reports the maximum rise slew considering incremental slew from signal integrity analysis</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| si_slew_mean_max_fall |                                                                                                                                                                                                                                            |
|                       | <p>Reports the maximum fall mean slew considering incremental slew from signal integrity analysis</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| si_slew_mean_max_rise |                                                                                                                                                                                                                                            |
|                       | <p>Reports the maximum rise mean slew considering incremental slew from signal integrity analysis</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| si_slew_mean_min_fall |                                                                                                                                                                                                                                            |
|                       | <p>Reports the minimum fall mean slew considering incremental slew from signal integrity analysis</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| si_slew_mean_min_rise |                                                                                                                                                                                                                                            |
|                       | <p>Reports the minimum rise mean slew considering incremental slew from signal integrity analysis</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| si_slew_min           |                                                                                                                                                                                                                                            |
|                       | <p>Reports the minimum slew considering incremental slew from signal integrity analysis</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |

|                  |                                                                                                                                                                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| si_slew_min_fall |                                                                                                                                                                                                                                                                                                                                 |
|                  | <p>Reports the minimum fall slew considering incremental slew from signal integrity analysis</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                           |
| si_slew_min_rise |                                                                                                                                                                                                                                                                                                                                 |
|                  | <p>Reports the minimum rise slew considering incremental slew from signal integrity analysis</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                           |
| si_victim        |                                                                                                                                                                                                                                                                                                                                 |
|                  | <p>Reports the victim object from signal integrity analysis</p> <p><b>Type:</b> <a href="#">obj(pin)*</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                         |
| skip_assertion   |                                                                                                                                                                                                                                                                                                                                 |
|                  | <p>Do not assert stimulus activity.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                           |
| slack_max        |                                                                                                                                                                                                                                                                                                                                 |
|                  | <p>Returns the worst slack across all concurrent MMMC views for Setup-style late data path of You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slack_max_edge   |                                                                                                                                                                                                                                                                                                                                 |
|                  | <p>Returns the data edge of the path responsible for the slack_max value. You can use -index return the value for a specific view and/or clock.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                        |

|                     |                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| slack_max_fall      | <p>Returns the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                              |
| slack_max_rise      | <p>Returns the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                               |
| slack_mean_max      | <p>In SOCV analysis, this returns the mean component of the worst slack across all concurrent views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>              |
| slack_mean_max_fall | <p>In SOCV analysis, this returns the mean component of the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| slack_mean_max_rise | <p>In SOCV analysis, this returns the mean component of the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |

|                     |                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| slack_mean_min      |                                                                                                                                                                                                                                                                                                                                                                     |
|                     | <p>In SOCV analysis, this returns the mean component of the worst slack across all concurrent views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>              |
| slack_mean_min_fall |                                                                                                                                                                                                                                                                                                                                                                     |
|                     | <p>In SOCV analysis, this returns the mean component of the worst falling slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| slack_mean_min_rise |                                                                                                                                                                                                                                                                                                                                                                     |
|                     | <p>In SOCV analysis, this returns the mean component of the worst rising slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| slack_min           |                                                                                                                                                                                                                                                                                                                                                                     |
|                     | <p>Returns the worst slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                      |
| slack_min_edge      |                                                                                                                                                                                                                                                                                                                                                                     |
|                     | <p>Returns the data edge of the path responsible for the slack_min value. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> string<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                               |

|                      |                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| slack_min_fall       | <p>Returns the worst falling slack across all concurrent MMMC views for Hold-style early data checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                    |
| slack_min_rise       | <p>Returns the worst rising slack across all concurrent MMMC views for Hold-style early data checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                     |
| slack_sigma_max      | <p>In SOCV analysis, this returns the variation component of the worst slack across all concurrent MMMC views for Setup-style late data path checks.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                              |
| slack_sigma_max_fall | <p>In SOCV analysis, this returns the variation component of the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| slack_sigma_max_rise | <p>In SOCV analysis, this returns the variation component of the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| slack_sigma_min      |                                                                                                                                                                                                                                                                                                                                                                      |



|                         |                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>In SOCV analysis, this returns the variation component of the worst slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>         |
| slack_sigma_min_fall    |                                                                                                                                                                                                                                                                                                                                                                                |
|                         | <p>In SOCV analysis, this returns the variation component of the worst falling slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slack_sigma_min_rise    |                                                                                                                                                                                                                                                                                                                                                                                |
|                         | <p>In SOCV analysis, this returns the variation component of the worst rising slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| slack_skewness_max_fall |                                                                                                                                                                                                                                                                                                                                                                                |
|                         | <p>In SOCV analysis, this returns the skewness component of the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| slack_skewness_max_rise |                                                                                                                                                                                                                                                                                                                                                                                |
|                         | <p>In SOCV analysis, this returns the skewness component of the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |

|                         |                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| slack_skewness_min_fall |                                                                                                                                                                                                                                                                                                                                                                     |
|                         | <p>In SOCV analysis, this returns the skewness component of the worst falling slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| slack_skewness_min_rise |                                                                                                                                                                                                                                                                                                                                                                     |
|                         | <p>In SOCV analysis, this returns the skewness component of the worst rising slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| slew_max                |                                                                                                                                                                                                                                                                                                                                                                     |
|                         | <p>Reports the maximum base slew of pin</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                          |
| slew_max_fall           |                                                                                                                                                                                                                                                                                                                                                                     |
|                         | <p>Returns the maximum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                           |
| slew_max_rise           |                                                                                                                                                                                                                                                                                                                                                                     |
|                         | <p>Returns the maximum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                            |
| slew_mean_max_fall      |                                                                                                                                                                                                                                                                                                                                                                     |

|                    |                                                                                                                                                                                                                                                                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>In SOCV mode, returns the mean component of the maximum slew time for falling transition across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slew_mean_max_rise |                                                                                                                                                                                                                                                                                                                                    |
|                    | <p>In SOCV mode, returns the mean component of the maximum slew time for rising transition all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>         |
| slew_mean_min_fall |                                                                                                                                                                                                                                                                                                                                    |
|                    | <p>In SOCV mode, returns the mean component of the minimum slew time for falling transition all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| slew_mean_min_rise |                                                                                                                                                                                                                                                                                                                                    |
|                    | <p>In SOCV mode, returns the mean component of the minimum slew time for rising transitions all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| slew_min           |                                                                                                                                                                                                                                                                                                                                    |
|                    | <p>Reports minimum base slew of pin</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                       |
| slew_min_fall      |                                                                                                                                                                                                                                                                                                                                    |

|                     |                                                                                                                                                                                                                                                                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>Returns the minimum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                          |
| slew_min_rise       |                                                                                                                                                                                                                                                                                                                                                          |
|                     | <p>Returns the minimum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                           |
| slew_sigma_max_fall |                                                                                                                                                                                                                                                                                                                                                          |
|                     | <p>In SOCV mode, returns the variation component of the maximum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slew_sigma_max_rise |                                                                                                                                                                                                                                                                                                                                                          |
|                     | <p>In SOCV mode, returns the variation component of the maximum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| slew_sigma_min_fall |                                                                                                                                                                                                                                                                                                                                                          |
|                     | <p>In SOCV mode, returns the variation component of the minimum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slew_sigma_min_rise |                                                                                                                                                                                                                                                                                                                                                          |

|                               |                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>In SOCV mode, returns the variation component of the minimum slew time for rising transition across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slew_skewness_max_fall        |                                                                                                                                                                                                                                                                                                                                        |
|                               | <p>In SOCV mode, returns the skewness component of the maximum slew time for falling transition across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slew_skewness_max_rise        |                                                                                                                                                                                                                                                                                                                                        |
|                               | <p>In SOCV mode, returns the skewness component of the maximum slew time for rising transition across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| slew_skewness_min_fall        |                                                                                                                                                                                                                                                                                                                                        |
|                               | <p>In SOCV mode, returns the skewness component of the minimum slew time for falling transition across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slew_skewness_min_rise        |                                                                                                                                                                                                                                                                                                                                        |
|                               | <p>In SOCV mode, returns the skewness component of the minimum slew time for rising transition across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| source_latency_early_fall_max |                                                                                                                                                                                                                                                                                                                                        |

|                               |                                                                                                                                                                                                                                                                                                                            |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Returns the maximum early fall source insertion delay specified by an explicit set_clock_lat the pin.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                          |
| source_latency_early_fall_min |                                                                                                                                                                                                                                                                                                                            |
|                               | <p>Returns the minimum early fall source insertion delay specified by an explicit set_clock_lat the pin. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_latency_early_rise_max |                                                                                                                                                                                                                                                                                                                            |
|                               | <p>Returns the maximum early rise source insertion delay specified by an explicit set_clock_lat the pin. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_latency_early_rise_min |                                                                                                                                                                                                                                                                                                                            |
|                               | <p>Returns the minimum early rise source insertion delay specified by an explicit set_clock_lat the pin. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_latency_late_fall_max  |                                                                                                                                                                                                                                                                                                                            |
|                               | <p>Returns the maximum late fall source insertion delay specified by an explicit set_clock_late the pin. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_latency_late_fall_min  |                                                                                                                                                                                                                                                                                                                            |

|                              |                                                                                                                                                                                                                                                                                                                            |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Returns the minimum late fall source insertion delay specified by an explicit set_clock_late the pin. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_latency_late_rise_max |                                                                                                                                                                                                                                                                                                                            |
|                              | <p>Returns the maximum late rise source insertion delay specified by an explicit set_clock_late the pin. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_latency_late_rise_min |                                                                                                                                                                                                                                                                                                                            |
|                              | <p>Returns the minimum late rise source insertion delay specified by an explicit set_clock_late the pin. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_of_clocks             |                                                                                                                                                                                                                                                                                                                            |
|                              | <p>Reports source pins of clocks associated with this pin</p> <p><b>Type:</b> obj(pin)*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                      |
| stack_via_rule               |                                                                                                                                                                                                                                                                                                                            |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>This value serves as a pin-specific override to use this stack_via_rule when connecting to t<br/>If set, it must match one of the elements from the stack_via_rule_list attribute of the correspo<br/>base_pin. The interpretation of this value depends on this pin's stack_via_rule_required att<br/>stack_via_rule == {} &amp;&amp; stack_via_rule_required == false: stack_via_rule does not affect th<br/>choice.<br/>stack_via_rule == stackrule1 &amp;&amp; stack_via_rule_required == false: Router will prefer stack<br/>but it may select another (or possibly none) if necessary to avoid design rule violations.<br/>stack_via_rule == {} &amp;&amp; stack_via_rule_required == true: The required stack_via_rule is ei<br/>the router will not use any stack_via_rule for connecting to this pin (even ignoring<br/>stack_via_rule_required on the base_pin if necessary).<br/>stack_via_rule == stackrule1 &amp;&amp; stack_via_rule_required == true: Router will use stackrul<br/>if it leads to design rule violations.<br/><b>Type:</b> <a href="#">obj(stack_via_rule)</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p> |
| stack_via_rule_required    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                            | <p>Specifies whether the router must use this pin's stack_via_rule value. If false, the stack_via_<br/>value will be preferred by the router, if true the stack_via_rule value is required by the route<br/>the stack_via_rule attribute for more details.<br/><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| stolen_slack               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                            | <p>Reports stolen (borrowed) slack for latch objects<br/><b>Type:</b> <a href="#">double</a><br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| timing_case_computed_value |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                            | <p>Returns the case computed value<br/><b>Type:</b> <a href="#">bool</a><br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| timing_case_logic_value    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



|                        |                                                                                                                                                                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Returns the case logic value</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                  |
| timing_constant        |                                                                                                                                                                                                                                                                         |
|                        | <p>Reports constant value on pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                               |
| timing_constant_source |                                                                                                                                                                                                                                                                         |
|                        | <p>Reports source of constant value on pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                     |
| to_arcs                |                                                                                                                                                                                                                                                                         |
|                        | <p>Returns a list of arc objects where the current pin is the termination point of the arc</p> <p><b>Type:</b> <a href="#">obj(arc)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                           |
| unused_bit_in_multibit |                                                                                                                                                                                                                                                                         |
|                        | <p>Returns a value of true if the pin is unused in the multibit cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                 |
| user_constant_value    |                                                                                                                                                                                                                                                                         |
|                        | <p>Returns constant values from netlist or constraints. You can use -index to return the value for specific view.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| waveform_max_fall      |                                                                                                                                                                                                                                                                         |

|                   |                                                                                                                                                                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>Reports maximum fall waveform on pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| waveform_max_rise |                                                                                                                                                                                                                                     |
|                   | <p>Reports maximum rise waveform on pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| waveform_min_fall |                                                                                                                                                                                                                                     |
|                   | <p>Reports minimum fall waveform on pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| waveform_min_rise |                                                                                                                                                                                                                                     |
|                   | <p>Reports minimum rise waveform on pin</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| wrapper_control   |                                                                                                                                                                                                                                     |
|                   | <p>Specifies as wrapper control signal.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                           |
| wrapper_segment   |                                                                                                                                                                                                                                     |
|                   | <p>Specifies the wrapper segments associated with the pin.</p> <p><b>Type:</b> <a href="#">obj(scan_chain)* obj(scan_segment)* obj(actual_scan_segment)* obj(actual_scan_</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| wrapper_type      |                                                                                                                                                                                                                                     |

|              |                                                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | Specifies the type of wrapper segment associated with the pin.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                   |
| x_coordinate |                                                                                                                                                                   |
|              | Reports X co-ordinate of pin<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| y_coordinate |                                                                                                                                                                   |
|              | Reports Y co-ordinate of pin<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |

## pin\_blockage

### Parent Objects

[design](#), [root](#)

### Definition

Pin blockage objects

| Attribute     | Description                                                                                                                                                                                                                                                                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| excluded_pins |                                                                                                                                                                                                                                                                                                                                      |
|               | Returns the excluded pins for pin_blockage. The pin list is set by create_pin_blockage - exclude_pin_patterns. If the .included_pins & .excluded_pins were both empty, the pin blockage would be applied to all pins on the cell.<br><b>Type:</b> <a href="#">obj(port)* obj(base_pin)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| included_pins |                                                                                                                                                                                                                                                                                                                                      |

|          |                                                                                                                                                                                                                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Returns the included pins for pin_blockage. The pin list is set by create_pin_blockage - include_pin_patterns. If the .included_pins & .excluded_pins were both empty, the pin blockage would be applied to all pins on the cell.<br><b>Type:</b> <a href="#">obj(port)* obj(base_pin)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| layer    |                                                                                                                                                                                                                                                                                                                                      |
|          | The layer that is being blocked<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                  |
| name     |                                                                                                                                                                                                                                                                                                                                      |
|          | Partition pin blockage name<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                          |
| obj_type |                                                                                                                                                                                                                                                                                                                                      |
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (pin_blockage)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                            |
| rect     |                                                                                                                                                                                                                                                                                                                                      |
|          | Rectangle that define the pin_blockage shape.<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                          |

## pin\_group

### Parent Objects

[design](#), [pin\\_guide](#), [root](#)

### Definition

## Pin group

| Attribute      | Description                                                                                                                                                                                                                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_pins      |                                                                                                                                                                                                                                                                                                                                  |
|                | <p>List of base pins or ports associated with the group.</p> <p><b>Type:</b> <code>obj(port)* obj(base_pin)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                         |
| exclude_pin    |                                                                                                                                                                                                                                                                                                                                  |
|                | <p>Indicates group pin exclude type(all_layer, same_layer, inclusive, all_layer_in_guided_area)</p> <p><b>Type:</b> <code>enum</code></p> <p><b>Enum Values:</b> all_layer_in_guided_area all_layer same_layer inclusive</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                   |
| is_compact     |                                                                                                                                                                                                                                                                                                                                  |
|                | <p>Indicate whether the pins in the group are assigned tightly together. By default, pins which are part of a group and associated to a guide can be spread inside the guide based on the area available and alignment to targets</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> |
| is_guided      |                                                                                                                                                                                                                                                                                                                                  |
|                | <p>Indicates if pin group is guided</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                   |
| is_spread      |                                                                                                                                                                                                                                                                                                                                  |
|                | <p>Indicates whether member pins distributed evenly</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                   |
| keep_out_space |                                                                                                                                                                                                                                                                                                                                  |

|                |                                                                                                                                                                                                                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>Minimum spacing with foreign pins (Unit: track)</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                          |
| name           |                                                                                                                                                                                                                                                                                                                                          |
|                | <p>Name of pin group</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                     |
| obj_type       |                                                                                                                                                                                                                                                                                                                                          |
|                | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">pin_group</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                |
| optimize_order |                                                                                                                                                                                                                                                                                                                                          |
|                | <p>Name of pin group</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                       |
| parent         |                                                                                                                                                                                                                                                                                                                                          |
|                | <p>The parent of the pin_group. It can be a design, a partition (if not committed yet), or base_cell for a black_box.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">design</a>) <a href="#">obj</a>(<a href="#">partition</a>) <a href="#">obj</a>(<a href="#">base_cell</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pin_spacing    |                                                                                                                                                                                                                                                                                                                                          |
|                | <p>Minimum spacing between adjacent pins (Unit: track)</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                      |

# pin\_guide

## Parent Objects

design, root

## Definition

Pin guide

| Attribute      | Description                                                                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| area           |                                                                                                                                                             |
|                | Area of the pin guide as defined by the LEF MACRO SIZE or OVERLAP information<br><b>Type:</b> <a href="#">area</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| layer_priority |                                                                                                                                                             |
|                | layer priority<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                |
| layers         |                                                                                                                                                             |
|                | layers where pin_guide is present<br><b>Type:</b> <a href="#">obj(layer)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                      |
| name           |                                                                                                                                                             |
|                | Name of pin guide<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                           |
| net_group      |                                                                                                                                                             |

|           |                                                                                                                                                                                                                                                                                                                                |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | net_group, if pin_guide is based on net_group otherwise NULL<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">net_group</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                        |
| obj_type  |                                                                                                                                                                                                                                                                                                                                |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">pin_guide</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                       |
| parent    |                                                                                                                                                                                                                                                                                                                                |
|           | The parent of the pin_guide. It can be a design, a partition (if not committed yet), or base_cell for a black_box.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">design</a> ) <a href="#">obj</a> ( <a href="#">partition</a> ) <a href="#">obj</a> ( <a href="#">base_cell</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| pin_group |                                                                                                                                                                                                                                                                                                                                |
|           | pin_group, if pin_guide is based on pin_group otherwise NULL<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">pin_group</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                        |
| rects     |                                                                                                                                                                                                                                                                                                                                |
|           | The rects that define the location of the pin_guide.<br><b>Type:</b> <a href="#">rect</a> *<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                           |

## place\_blockage

### Parent Objects

[design](#), [root](#)

### Definition

Placement blockage(hard, soft, partial).



| Attribute   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|             | <p>Name of placement blockage</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                 |
| density     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|             | <p>The max placement density percent allowed inside this place_blockage. It must be in the range of 5 to 100, in steps of 5. It is only valid if the type = partial or soft. For example, a partial placement percentage of 75 percent means that up to 75 percent of placement density is allowed in the area. If the type is not partial or soft, a value of 0 is returned.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| inst        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|             | <p>The instance that the placement blockage is associated with (equivalent to DEF BLOCKAGES + COMPONENT)</p> <p><b>Type:</b> obj(inst)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                    |
| is_no_flop  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|             | <p>Flip-flops and latches cannot be placed inside this place_blockage. Only has an effect if type = partial.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                 |
| is_pushdown |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|             | <p>This place_blockage has been pushed down from a higher level in the design hierarchy.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                     |
| name        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|             | <p>Name of placement blockage</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                 |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| obj_type |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|          | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">place_blockage</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                        |
| rects    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|          | <p>List of non-overlapping rectangles that defines the shape of place_blockage</p> <p><b>Type:</b> <a href="#">rect</a>*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| shapes   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|          | <p>List of shapes of this placement blockage</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">shape</a>)* <a href="#">obj</a>(<a href="#">density_shape</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                 |
| type     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|          | <p>The type of blockage. hard = no cells allowed, macro_only = standard-cells are allowed but blocks are not allowed, partial = allow cells until reach the .density limit, soft = most logic cells are not allowed, but 'repeater' cells like inverters, buffers, level-shifters, isolation cells, and clock-gating cells are allowed. See the set_selective_blockage_gate command for other methods to control what is allowed inside a soft place_blockage.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> macro_only hard soft partial</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |

## port

### Parent Objects

[marker](#), [bump\\_pin](#), [test\\_clock](#), [clock](#), [io\\_constraint](#), [violation](#), [def\\_scan\\_chain](#), [skew\\_group](#), [pin\\_blockage](#), [scan\\_chain](#), [hnet](#), [actual\\_scan\\_segment](#), [timing\\_point](#), [dft\\_configuration\\_mode](#), [bus\\_sink\\_group](#), [design](#), [clock\\_tree](#), [pin\\_group](#), [timing\\_path](#), [root](#), [bump](#), [test\\_signal](#), [port\\_shape](#), [inst](#), [port\\_bus](#), [scan\\_segment](#), [net](#), [actual\\_scan\\_chain](#)

## Definition

External logical ports of the design. See pg\_ports for power/ground ports

| Attribute                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| actual_latency_early_fall_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                               | <p>Returns the computed early falling clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Setup capture latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| actual_latency_early_fall_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                               | <p>Returns the computed early falling clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Hold launch latency. When operating in OCV mode, the min qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>   |
| actual_latency_early_rise_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                               | <p>Returns the computed early rising clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Setup capture latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| actual_latency_early_rise_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Returns the computed early rising clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Hold launch latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| actual_latency_early_fall_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                               | <p>Returns the computed late falling clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Setup launch latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| actual_latency_early_fall_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                               | <p>Returns the computed late falling clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Hold capture latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| actual_latency_early_rise_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Returns the computed late rising clock latency to this clock port. If the analysis mode is set to case/worst-case, the value will be the Setup launch latency. When operating in OCV mode, max qualification is ignored. If there are multiple arrival times from different clock phases - the conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| actual_latency_late_rise_min   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Returns the computed late rising clock latency to this clock port. If the analysis mode is set to case/worst-case, the value will be the Hold capture latency. When operating in OCV mode, max qualification is ignored. If there are multiple arrival times from different clock phases - the conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| antenna_data                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>List of antenna data for the terminal</p> <p><b>Type:</b> obj(antenna_data)*<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| aocv_early_input_stage_weight  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Returns the number of external path stages to consider in AOCV early analysis of paths from the input port.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| aocv_early_output_stage_weight |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Returns the number of external path stages to consider in AOCV early analysis of paths to the output port.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| aocv_late_input_stage_weight  |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>Returns the number of external path stages to consider in AOCV late analysis of paths from port.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                            |
| aocv_late_output_stage_weight |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>Returns the number of external path stages to consider in AOCV late analysis of paths to th</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                 |
| arrival_max_fall              |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>Returns the latest falling arrival time to the specified port across all concurrent MMMC view can use -index to determine the worst arrival for a specific view. You can also use -index to latest arrival time related to a specific clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                |
| arrival_max_rise              |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>Returns the latest rising arrival time to the specified port across all concurrent MMMC views can use -index to determine the worst arrival for a specific view. You can also use -index to latest arrival time related to a specific clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                |
| arrival_mean_max_fall         |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>In SOCV analysis mode, this returns the mean value of the latest falling arrival time to the s port across all concurrent MMMC views. You can use -index to determine the latest arrival f specific view. You can also use -index to get the worst arrival time related to a specific clock</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| arrival_mean_max_rise         |                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>In SOCV analysis mode, this returns the mean value of the latest rising arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the latest arrival time for a specific view. You can also use -index to get the worst arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>         |
| arrival_mean_min_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                       | <p>In SOCV analysis mode, this returns the mean value of the earliest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the earliest arrival time for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| arrival_mean_min_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                       | <p>In SOCV analysis mode, this returns the mean value of the earliest rising arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the earliest arrival time for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| arrival_min_fall      |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                       | <p>Returns the earliest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                       |
| arrival_min_rise      |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Returns the earliest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                              |
| arrival_sigma_max_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                        | <p>In SOCV analysis mode, this returns the variation component of the latest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| arrival_sigma_max_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                        | <p>In SOCV analysis mode, this returns the variation component of the latest rising arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>         |
| arrival_sigma_min_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                        | <p>In SOCV analysis mode, this returns the variation component of the earliest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| arrival_sigma_min_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>In SOCV analysis mode, this returns the mean value of the earliest rising arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| arrival_window |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                | <p>Returns a list of the earliest and latest, rising and falling arrival times per clock phase at the across all concurrent MMMC views. You can use -index to make the arrival times specific to view</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                             |
| base_name      |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                | <p>The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                        |
| bumps          |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                | <p>Specifies the bumps and bump_pins assigned to this port.</p> <p><b>Type:</b> obj(bump)* obj(bump_pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                             |
| bus            |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                | <p>The bus of this port.</p> <p><b>Type:</b> obj(port_bus)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                            |
| ccopt_schedule |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>Specifies the restrictions, if any, on the scope of useful skew scheduling for this pin. Settings 'timing_only', 'off', and 'on' override other sources of per-pin useful skew restrictions, such as the 'opt_skew_no_boundary' attribute. Allowed values are:</p> <p>auto Neither restrictions nor overrides are to be applied to the pin (default).</p> <p>on Specifies no restrictions on sink adjustment scope, overriding other sources of per-pin useful skew restrictions such as the 'opt_skew_no_boundary' setting.</p> <p>timing_only Restricts the sink to useful skew adjustments for timing if necessary.</p> <p>off Useful skew must avoid adjusting the schedule of this pin, even if this degrades timing.</p> <p>Valid values: auto off on timing_only</p> <p><b>Type:</b> string</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> |
| clock_gating_exclude_signal |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                             | <p>Exclude this pin from clock gating enable functions.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| clocks                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                             | <p>Returns a list of all of the clock objects arriving at the port. You can use -index to return the from a specific analysis view.</p> <p><b>Type:</b> obj(clock)*</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| constant_value              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                             | <p>Returns a constant value of 0 or 1 if logic state has been asserted or propagated to this pin. use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> int</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| context_constant_value      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Returns a context_constant value of 0 or 1 if logic state has been asserted or propagated to i<br/>You can use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                       |
| cts_add_pin_driver       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                          | <p>Specifies a cell type or pair of cell types, so that a cell inst can to be added above an input p<br/>below an output pin.<br/> The pin is specified by the argument pin.<br/> No more than two cell types may be specified. A warning will be emitted if no AON cells ar<br/>specified.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                          |
| cts_add_port_driver      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                          | <p>Specifies a cell type or pair of cell types, so that a cell inst can to be added above an output<br/>below an input port.<br/> The port is specified by the argument pin.<br/> If the pin specified is not a design IO pin or it is not in the clock network then CCOpt will en<br/>warning<br/> and will not add cell insts at that position. No more than two cell types may be specified. A<br/>will be emitted if no AON cells are specified.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| cts_annotated_delay_to   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                          | <p>Override any clock tree timing engine computed cell arc or net arc delays to this pin, in a sin<br/>manner to SDC set_annotated_delay.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> delay_corner<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                            |
| cts_annotated_transition |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>Override any clock tree timing engine computed transition at this pin, in a similar manner to set_annotated_transition.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> delay_corner</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                    |
| cts_assign_clock_tree |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                       | <p>Can be used to make sure a clock tree sink pin gets driven by a particular clock tree. The clock tree can alternatively be specified as the tree's root pin, in case extraction rennumbers the clock trees defined from a single SDC clock.</p> <p>By default CCOpt automatically selects appropriate pins.</p> <p>Valid values: cts_clock_tree   list of pins</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_case_analysis     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                       | <p>Specifies a constant value to be used as pin signal when analyzing timing within CTS. Only applies to input pins of cells in the clock tree. If set to 'none' (the default), CTS will consider the input as non-constant.</p> <p>Valid values: 0 1 none</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p>                                                                                                      |
| cts_clock_tree        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                       | <p>The clock tree to which this object belongs. Flops do not belong to a clock tree, but their clock pins do.</p> <p>Valid values: cts_clock_tree</p> <p><b>Type:</b> <a href="#">obj(clock_tree)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                         |
| cts_clock_trees       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                       | <p>A list of clock trees the pin is contained within. This includes parents of generated clock trees and all relevant parents when clock trees overlap.</p> <p>Valid values: list cts_clock_tree</p> <p><b>Type:</b> <a href="#">obj(clock_tree)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                          |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cts_effective_mixed_buffer_budget | <p>Allows the user to query the mixed buffer budget at a particular pin.<br/>This is the number of buffers allowed above the given pin, and is the smallest of the remaining budgets from any fanout below this point or the override set at this point if one has been set.</p> <p><b>Type:</b> string<br/><b>Default:</b> auto<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                  |
| cts_effective_routing_preference  | <p>Returns the pin's effective routing preference; one of leaf, trunk, top, none. The value "none" returned for pins that have no preference such as antenna diodes or load cap inputs. The routing preference value for a pin does not solely determine the net_type of the driving net has multiple fanout. The fanout routing preferences for all sinks are taken into account when determining the net_type for the net as a whole.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                         |
| cts_is_sdc_clock_root             | <p>Specifies whether the given pin is the root (source pin) of an SDC clock.<br/>The create_clock_tree_spec populates this attribute with the location of the SDC clock root (source) pins.<br/>This attribute controls the behavior of clock tree definition commands create_clock_tree and create_generated_clock_tree, when the -stop_at_sdc_clock_roots argument is specified. In such a case, pins and ports for which this attribute is true will be treated as being SDC clock root pins.<br/>Valid values: true false</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p> |
| cts_isolated_effective            | <p>This is the isolated value CCOpt will use for the given pin.</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| cts_net_unbufferable_reasons      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>This attribute contains a list of reasons why CCOpt was not able to buffer the clock net attached to the specified pin.<br/>Valid values: string<br/><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| cts_node_type                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>node type of port within cts graph<br/><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| cts_pin_capacitance_sources     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>List of the places the capacitance could be retrieved from.<br/>Requesting this attribute for a pin not in the clock tree will result in an error.<br/>The possible places are:<br/>- library - CTS will retrieve capacitance values from the library for one or more timing_corner combinations.<br/>- cts_capacitance_override - CTS will retrieve capacitance values from the cts_capacitance_override attribute for one or more timing_corner/event combinations.<br/>- cts_blackbox_default_load_base_pin - CTS will retrieve capacitance values from the cts_blackbox_default_load_base_pin attribute for one or more timing_corner/event combinations.<br/><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                |
| cts_pin_insertion_delay         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>The amount of insertion delay under this pin. Clock tree synthesis will attempt to make the insertion delay to this pin less than that to other sinks in the same skew group by this amount if a positive value is set. A negative value should be used if you would like the insertion delay to this pin to be greater than that to other sinks. The value 'auto' means there is no insertion delay offset for the pin.<br/>This attribute exposes the 'total' pin insertion delay. For further discussion on this, consult the documentation for attribute cts_pin_insertion_delay_source<br/>Valid values: double   ignore   min   auto<br/><b>Type:</b> string<br/><b>Allowed -index values:</b> delay_corner<br/><b>Default:</b> auto<br/><b>Edit:</b> Yes</p> |
| cts_pin_insertion_delay_sources |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>The amount of insertion delay under this pin, broken down by the source of the pin insertion delay (PID).</p> <p>The value of this attribute is a Tcl dictionary of PID values, keyed on the source of that PID. This provides an 'exploded' view into the total pin insertion delay that is present, categorizing the various contributions by their origin.</p> <p>Valid Tcl dict keys:</p> <ul style="list-style-type: none"> <li>user Asserted by user Tcl scripting or restored from an older DB.</li> <li>sdv Extracted from SDC set_clock_latency by spec creation.</li> <li>sink_type Asserted to reflect the impact of the cts_sink_type attribute.</li> <li>ilm Records the mean clock latency of the ILM clock network that is rooted at this pin.</li> <li>useful_skew A useful skew generated by pre-CTS optimization.</li> <li>incremental An incremental offset, used to specify useful skews to incremental CTS.</li> <li>total The formal sum of the above contributions. This value is the 'overall' pin insertion delay accessible via the cts_pin_insertion_delay attribute.</li> </ul> <p>Valid Tcl dict values: double   ignore   min   auto</p> <p>Reading the attribute returns a dict value that exposes the current state of the PID, recording all the PID sources and the associated PID value of each one. Sources whose value are auto are omitted. If there are no non-auto sources, the returned dict is empty. For convenience, the total is always included in the dict if it is non-auto.</p> <p>Writing a dict value to the attribute will update each of the specified sources with the specified PID value. The total PID value will then be updated. Sources in the dict that are assigned a value of 'auto' will be reset. Sources omitted from the dict are left unchanged from their current value: assigning an empty dict is a no-op.</p> <p>Resetting the attribute will clear all the sources and the total back, resetting them to the default value of 'auto', thereby completely removing the PID from the pin.</p> <p>Valid values: A Tcl dictionary of PID values, keyed on PID source.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> delay_corner</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| cts_pin_route_type_propagation |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                | <p>Describes how cts_pin_route_type should be applied to the net driven by the specified pin.</p> <p>The possible values are:</p> <ul style="list-style-type: none"> <li>- none - it applies to net driven by the pin and only that net, irrespective of whether the net buffered or not.</li> <li>- net - it applies to all nets generated by buffering the pre-CTS net driven by pin (default).</li> </ul> <p><b>Type:</b> string</p> <p><b>Default:</b> net</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cts_routing_trunk_override |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                            | <p>Prefer trunk routing rules for this pin. Only applies to clock tree sinks.<br/>Valid values: true false</p> <p><b>Type:</b> bool<br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| cts_sink_type              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                            | <p>The type of sink this pin represents.<br/>Valid values are as follows:<br/>auto The sink type will be automatically determined by CCOpt.<br/>through Through pin. Trace the clock tree through this pin.<br/>stop Stop pin. When defining clock trees, CCOpt stops searching for parts of the clock tree at stop pins.<br/>ignore Ignore pin. CCOpt stops searching for parts of the clock tree at ignore pins and it does not attempt to balance the insertion delay of ignore pins.<br/>exclude Exclude pin. Exclude this pin from the clock tree.<br/>Valid values: auto through stop ignore exclude</p> <p><b>Type:</b> string<br/><b>Default:</b> auto<br/><b>Edit:</b> Yes</p> |
| cts_sink_type_effective    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                            | <p>Indicates how CCOpt will treat a given pin, taking into account both its cts_sink_type_implicit and any cts_sink_type settings.<br/>Setting a non-default value for the cts_sink_type attribute will override the cts_sink_type_implicit attribute.</p> <p><b>Type:</b> string<br/><b>Default:</b> auto<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                      |
| cts_sink_type_implicit     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Indicates the type of sink that CCOpt classified this pin as. Note that the <code>cts_sink_type</code> attribute override these internal classifications.</p> <p>Possible values are:</p> <ul style="list-style-type: none"> <li><code>exclude</code> Indicates that this pin is a sink which represents a non-clock pin.</li> <li><code>ignore</code> Indicates that CCOpt has determined not to search for more clock tree through this pin. Additionally, this pin will not be balanced.</li> <li><code>stop</code> Indicates that CCOpt has determined not to search for more clock tree through this pin.</li> </ul> <p>An empty value for this attribute indicates either that this pin is either not a sink, or that it is that is not implicitly <code>exclude</code> or <code>ignore</code> or <code>stop</code>.</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> <code>exclude</code><br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>cts_sink_type_reasons</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                               | <p>The reasons why this pin has the given <code>cts_sink_type</code>.</p> <p>This attribute is configured by the <code>create_clock_tree_spec</code> command in order to record the reasons that the <code>cts_sink_type</code> attribute has been configured for the given pin.</p> <p>This attribute is a list of values. Valid values are as follows:</p> <ul style="list-style-type: none"> <li><code>auto</code> The <code>cts_sink_type</code> attribute is set to 'auto'</li> <li><code>implicit</code> The pin is an implicit sink (flop/latch)</li> <li><code>user</code> The user has set the <code>cts_sink_type</code> attribute</li> <li><code>design_io</code> This pin is a design I/O</li> <li><code>multiple_outputs</code> This pin is on an instance with multiple outputs, and the <code>cts_spec_config_trace_through_to</code> attribute has not been set</li> <li><code>set_disable_timing</code> SDC <code>set_disable_timing</code> stops the clock at this pin</li> <li><code>set_case_analysis</code> SDC <code>set_case_analysis</code> stops the clock at this pin</li> <li><code>generated_clock_tree</code> This pin is the generator input to an SDC generated clock</li> <li><code>no_sdc_clock</code> The SDC clock is stopped at this pin for other reasons</li> <li><code>ilm</code> The <code>create_clock_tree_spec</code> command has detected an ILM below this pin</li> </ul> <p>Valid values: <code>auto</code> <code>implicit</code> <code>user</code> <code>design_io</code> <code>multiple_outputs</code> <code>set_disable_timing</code> <code>set_case_analysis</code> <code>generated_clock_tree</code> <code>no_sdc_clock</code> <code>ilm</code></p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> <code>auto</code><br/> <b>Edit:</b> Yes</p> |
| <b>cts_skew_groups_active</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Returns the list of active skew groups for this pin.<br/>For sink pins, this attribute lists both skew groups that pass through this pin and skew groups for which this sink is an endpoint.<br/>For non-sink pins, shows skew groups that pass through this pin.<br/>Valid values: list skew_groups<br/><b>Type:</b> obj(skew_group)*<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                        |
| cts_skew_groups_active_sink       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                   | <p>Returns the list of skew groups for which this pin is an active sink.<br/>For sink pins, this attribute lists the skew groups for which this sink is an endpoint. Skew groups that pass through this pin are not included.<br/>For non-sink pins, this attribute always returns null.<br/>Valid values: list skew_groups<br/><b>Type:</b> obj(skew_group)*<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                    |
| cts_skew_groups_constraining      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                   | <p>The list of delay-constraining skew groups which are active at this pin.<br/>This attribute reports similar data to cts_skew_groups_active. The only difference is that the reporting-only skew groups are not included in this attribute's value.<br/>For sink pins, this attribute lists both delay-constraining skew groups that pass through this pin and delay-constraining skew groups for which this sink is an endpoint.<br/>For non-sink pins, shows delay-constraining skew groups that pass through this pin.<br/>Valid values: list skew_groups<br/><b>Type:</b> obj(skew_group)*<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| cts_skew_groups_constraining_sink |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>The list of delay-constraining skew groups for which this pin is an active sink. This attribute lists the delay-constraining skew groups for which this pin is an active sink. This attribute reports similar data to <code>cts_skew_groups_active_sink</code>. The only difference is that the reporting-only (constrains none) skew groups are not included in this attribute's value.</p> <p>For sink pins, this attribute lists the delay-constraining skew groups for which this sink is an endpoint. Skew groups that pass through this pin are not included.</p> <p>For non-sink pins, this attribute always returns null.</p> <p>Valid values: list skew_groups</p> <p><b>Type:</b> <code>obj(skew_group)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| cts_skew_groups_ignore           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                  | <p>The list of skew groups for which paths through this pin are ignored.</p> <p>Valid values: list skew_groups</p> <p><b>Type:</b> <code>obj(skew_group)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| cts_skew_groups_sink             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                  | <p>The list of skew groups for which this pin is a sink.</p> <p>Valid values: list skew_groups</p> <p><b>Type:</b> <code>obj(skew_group)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| cts_skew_groups_source_pin       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                  | <p>The list of skew groups for which this clock tree or pin is specified as a source.</p> <p>Valid values: list skew_groups</p> <p><b>Type:</b> <code>obj(skew_group)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| cts_spec_config_trace_through_to |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>Clock tree definition will, by default, not continue through certain types of cell arc (for instance, the clock to Q arc in a DFF). This attribute allows you to override this default behavior, permitting the clock tree to trace through such a cell.</p> <p>The attribute should be configured on the input pin at which the clock arrives. The value of the attribute specifies the output pin to which the clock should propagate. The specified output pin must be another pin on the same instance. The output pin may be specified either by its fully qualified name (i.e. inclusive of the instance name), or else simply by its local (cell-relative) name.</p> <p>There must be a pre-existing (library-defined) chain of one or more delay arcs that connect the input and output pins together. It is not possible to use trace_through_to to synthesize delay arcs.</p> <p>If multiple input pins are annotated on a given instance, the value of trace_through_to at each of those pins must select the same output pin: i.e. the configuration must identify a single clock output for the instance. If multiple clock outputs are required then trace_through_to should not be used: instead define a generated clock tree at each of the clock-carrying outputs.</p> <p>If the configuration of trace_through_to settings for a given instance does not meet these requirements, a warning will be issued and the settings for that instance will be ignored.</p> <p>Note: if both trace_through_to and library_trace_through_to are applicable at a given netlist instance pin, the trace_through_to value takes precedence.</p> <p>Valid values: pin</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| cts_top_fanout_count_override   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                 | <p>The number of clock sinks this sink counts for when applying the top routing rules.</p> <p>Note that this attribute is only valid for sink pins, and it returns auto for non-sink pins.</p> <p>For a sink pin, a non-auto value means that this sink is counted as though it were multiple sinks, for the purposes of determining which nets should have top routing. An auto value for a sink pin means that the sink counts as a single sink.</p> <p>Valid values: integer &gt; 0</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| cts_top_fanout_transitive_count |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>The number of clock sinks in the transitive fanout of the pin as counted for applying the top rules.</p> <p>This attribute is very similar to the <code>cts_transitive_fanout</code> attribute but counts sink fanout using <code>cts_top_fanout_count_override</code> attribute instead of always counting sinks as a single item fanout.</p> <p>Requesting this attribute for a pin not in the clock tree will result in an error.</p> <p>Valid values: integer</p> <p><b>Type:</b> <code>int</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> No</p> |
| <code>cts_transitive_fanout</code>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>The number of clock sinks in the transitive fanout of the pin, within the clock tree.</p> <p>Requesting this attribute for a pin not in the clock tree will result in an error.</p> <p>Valid values: int</p> <p><b>Type:</b> <code>int</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                          |
| <code>cts_virtual_delay_early_fall</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>The amount of virtual delay that has been applied under this pin.</p> <p>Valid values: double</p> <p><b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> <code>delay_corner</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                    |
| <code>cts_virtual_delay_early_rise</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>The amount of virtual delay that has been applied under this pin.</p> <p>Valid values: double</p> <p><b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> <code>delay_corner</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                    |
| <code>cts_virtual_delay_late_fall</code>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>The amount of virtual delay that has been applied under this pin.</p> <p>Valid values: double</p> <p><b>Type:</b> <code>double</code><br/> <b>Allowed -index values:</b> <code>delay_corner</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                    |

|                             |                                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cts_virtual_delay_late_rise |                                                                                                                                                                                                                                   |
|                             | <p>The amount of virtual delay that has been applied under this pin.<br/>Valid values: double<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> delay_corner<br/><b>Default:</b> 0<br/><b>Edit:</b> Yes</p>               |
| delay_max_fall              |                                                                                                                                                                                                                                   |
|                             | <p>Returns the maximum falling delay. You can use -index to return the value from a specific a view.<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| delay_max_rise              |                                                                                                                                                                                                                                   |
|                             | <p>Returns the maximum rising delay. You can use -index to return the value from a specific a view.<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>  |
| delay_min_fall              |                                                                                                                                                                                                                                   |
|                             | <p>Returns the minimum falling delay. You can use -index to return the value from a specific a view.<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| delay_min_rise              |                                                                                                                                                                                                                                   |
|                             | <p>Returns the minimum rising delay. You can use -index to return the value from a specific ar view.<br/><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| depth                       |                                                                                                                                                                                                                                   |

|                              |                                                                                                                                                                                                                                                                                 |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Depth constraint of the port in microns. Pin assignment commands will create a pin shape for port that extends 'depth' microns inside the design edge. The edit_pin command can be used to edit it.</p> <p><b>Type:</b> coord<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| design                       |                                                                                                                                                                                                                                                                                 |
|                              | <p>The design of this port</p> <p><b>Type:</b> obj(design)<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                       |
| dft_dedicated_wrapper_reason |                                                                                                                                                                                                                                                                                 |
|                              | <p>Specifies the reason behind putting dedicated wrapper.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                            |
| dft_driven_by_clock          |                                                                                                                                                                                                                                                                                 |
|                              | <p>If pin is driven by clock.</p> <p><b>Type:</b> enum<br/><b>Enum Values:</b> false true<br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p>                                                                                                                                    |
| dft_enable_hookup_pin        |                                                                                                                                                                                                                                                                                 |
|                              | <p>Hookup pin of shift enable port.</p> <p><b>Type:</b> obj(pin) obj(hpin) obj(port) obj(hport)<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                 |
| dft_enable_hookup_polarity   |                                                                                                                                                                                                                                                                                 |
|                              | <p>Hookup polarity of shift enable port.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                             |
| dft_partition                |                                                                                                                                                                                                                                                                                 |

|                           |                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Determine the DFT partition of the specified port/pin/hierarchical pin.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                              |
| dft_sdi_output_hookup_pin |                                                                                                                                                                                                                                                                                                         |
|                           | <p>Input hookup pin of a bidirectional port.</p> <p><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                          |
| dft_sdo_input_hookup_pin  |                                                                                                                                                                                                                                                                                                         |
|                           | <p>Output hookup pin of a bidirectional port.</p> <p><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                         |
| direction                 |                                                                                                                                                                                                                                                                                                         |
|                           | <p>Direction of the port from the Verilog netlist.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> in out inout</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                |
| drive_resistance_fall_max |                                                                                                                                                                                                                                                                                                         |
|                           | <p>Returns the falling linear drive resistance at the port for late timing paths. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| drive_resistance_fall_min |                                                                                                                                                                                                                                                                                                         |
|                           | <p>Returns the falling linear drive resistance at the port for early timing paths. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| drive_resistance_rise_max |                                                                                                                                                                                                                                                                                                         |



|                           |                                                                                                                                                                                                                                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Returns the rising linear drive resistance at the port for late timing paths.. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| drive_resistance_rise_min |                                                                                                                                                                                                                                                                                        |
|                           | <p>Returns the rising linear drive resistance at the port for early timing paths. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| driver_from_pin_fall_max  |                                                                                                                                                                                                                                                                                        |
|                           | <p>Returns the driving cell input pin specified for max falling delays at the port. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> obj(lib_pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                 |
| driver_from_pin_fall_min  |                                                                                                                                                                                                                                                                                        |
|                           | <p>Returns the driving cell input pin specified for min falling delays at the port. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> obj(lib_pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                 |
| driver_from_pin_rise_max  |                                                                                                                                                                                                                                                                                        |
|                           | <p>Returns the driving cell input pin specified for max rising delays at the port. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> obj(lib_pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                  |
| driver_from_pin_rise_min  |                                                                                                                                                                                                                                                                                        |
|                           | <p>Returns the driving cell input pin specified for min rising delays at the port. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> obj(lib_pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                  |

|                                    |                                                                                                                                                                                                                                                                                           |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| driver_ignore_drc                  |                                                                                                                                                                                                                                                                                           |
|                                    | <p>driver_ignore_drc</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                               |
| driver_input_slew_fall_to_fall_max |                                                                                                                                                                                                                                                                                           |
|                                    | <p>Returns the driver input pin slew used for the max fall-to-fall delay. You can use -index to re value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| driver_input_slew_fall_to_fall_min |                                                                                                                                                                                                                                                                                           |
|                                    | <p>Returns the driver input pin slew used for the min fall-to-fall delay. You can use -index to re value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| driver_input_slew_fall_to_rise_max |                                                                                                                                                                                                                                                                                           |
|                                    | <p>Returns the driver input pin slew used for the max fall-to-rise delay. You can use -index to r the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| driver_input_slew_fall_to_rise_min |                                                                                                                                                                                                                                                                                           |
|                                    | <p>Returns the driver input pin slew used for the min fall-to-rise delay. You can use -index to re value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| driver_input_slew_rise_to_fall_max |                                                                                                                                                                                                                                                                                           |

|                                    |                                                                                                                                                                                                                                                                                            |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Returns the driver input pin slew used for the max rise-to-fall delay. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>             |
| driver_input_slew_rise_to_fall_min |                                                                                                                                                                                                                                                                                            |
|                                    | <p>Returns the driver input pin slew used for the min rise-to-fall delay. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>             |
| driver_input_slew_rise_to_rise_max |                                                                                                                                                                                                                                                                                            |
|                                    | <p>Returns the driver input pin slew used for the max rise-to-rise delay. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>             |
| driver_input_slew_rise_to_rise_min |                                                                                                                                                                                                                                                                                            |
|                                    | <p>Returns the driver input pin slew used for the min rise-to-rise delay. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>             |
| driver_pin_fall_max                |                                                                                                                                                                                                                                                                                            |
|                                    | <p>Returns the driving cell output pin specified for max falling delays at the port. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> obj(lib_pin)*</p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| driver_pin_fall_min                |                                                                                                                                                                                                                                                                                            |

|                                |                                                                                                                                                                                                                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Returns the driving cell output pin specified for min falling delays at the port. You can use -i return the value for a specific view.</p> <p><b>Type:</b> <code>obj(lib_pin)*</code></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>     |
| driver_pin_rise_max            |                                                                                                                                                                                                                                                                                                      |
|                                | <p>Returns the driving cell output pin specified for max rising delays at the port. You can use -i return the value for a specific view.</p> <p><b>Type:</b> <code>obj(lib_pin)*</code></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| driver_pin_rise_min            |                                                                                                                                                                                                                                                                                                      |
|                                | <p>Returns the driving cell output pin specified for min rising delays at the port. You can use -i return the value for a specific view.</p> <p><b>Type:</b> <code>obj(lib_pin)*</code></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| driving_cell_fall_max          |                                                                                                                                                                                                                                                                                                      |
|                                | <p>Returns the name of the library cell used to compute max falling delays at the port. You can index to return the value for a specific view.</p> <p><b>Type:</b> <code>double</code></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| driving_cell_fall_min          |                                                                                                                                                                                                                                                                                                      |
|                                | <p>Returns the name of the library cell used to compute min falling delays at the port. You can index to return the value for a specific view.</p> <p><b>Type:</b> <code>double</code></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| driving_cell_from_pin_fall_max |                                                                                                                                                                                                                                                                                                      |

|                                |                                                                                                                                                                                                                                                                                    |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Returns the driving cell input pin specified for max falling delays at the port. You can use -in return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| driving_cell_from_pin_fall_min |                                                                                                                                                                                                                                                                                    |
|                                | <p>Returns the driving cell input pin specified for min falling delays at the port. You can use -in return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| driving_cell_from_pin_rise_max |                                                                                                                                                                                                                                                                                    |
|                                | <p>Returns the driving cell input pin specified for max rising delays at the port. You can use -in return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| driving_cell_from_pin_rise_min |                                                                                                                                                                                                                                                                                    |
|                                | <p>Returns the driving cell input pin specified for min rising delays at the port. You can use -in return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| driving_cell_library_fall_max  |                                                                                                                                                                                                                                                                                    |
|                                | <p>Returns the driving cell library specified for max falling delays at the port. You can use -in return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| driving_cell_library_fall_min  |                                                                                                                                                                                                                                                                                    |

|                               |                                                                                                                                                                                                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Returns the driving cell library specified for min falling delays at the port. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| driving_cell_library_rise_max |                                                                                                                                                                                                                                                                                           |
|                               | <p>Returns the driving cell library specified for max rising delays at the port. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>     |
| driving_cell_library_rise_min |                                                                                                                                                                                                                                                                                           |
|                               | <p>Returns the driving cell library specified for min rising delays at the port. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>     |
| driving_cell_pin_fall_max     |                                                                                                                                                                                                                                                                                           |
|                               | <p>Returns the driving cell output pin specified for max falling delays at the port. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| driving_cell_pin_fall_min     |                                                                                                                                                                                                                                                                                           |
|                               | <p>Returns the driving cell output pin specified for min falling delays at the port. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| driving_cell_pin_rise_max     |                                                                                                                                                                                                                                                                                           |

|                           |                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Returns the driving cell output pin specified for max rising delays at the port. You can use -i return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                            |
| driving_cell_pin_rise_min |                                                                                                                                                                                                                                                                                                                                                                                              |
|                           | <p>Returns the driving cell output pin specified for min rising delays at the port. You can use -i return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                            |
| driving_cell_rise_max     |                                                                                                                                                                                                                                                                                                                                                                                              |
|                           | <p>Returns the name of the library cell used to compute max rising delays at the port. You can index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                       |
| driving_cell_rise_min     |                                                                                                                                                                                                                                                                                                                                                                                              |
|                           | <p>Returns the name of the library cell used to compute max rising delays at the port. You can index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                       |
| escaped_name              |                                                                                                                                                                                                                                                                                                                                                                                              |
|                           | <p>The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a scalar.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| exceptions                |                                                                                                                                                                                                                                                                                                                                                                                              |

|                          |                                                                                                                                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Reports the path exceptions</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                  |
| external_capacitance_max |                                                                                                                                                                                                                                                                                                                          |
|                          | <p>Returns the total max external loading on the port from all sources - including set_load ass or wireload models. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| external_capacitance_min |                                                                                                                                                                                                                                                                                                                          |
|                          | <p>Returns the total min external loading on the port from all sources - including set_load asse wireload models. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| external_fanout_load     |                                                                                                                                                                                                                                                                                                                          |
|                          | <p>Returns the external fanout load specified by set_fanout_load. You can use -index to return value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                       |
| external_wireload_model  |                                                                                                                                                                                                                                                                                                                          |
|                          | <p>Returns the wireload model used to model external connections to the port. You can use -i return the value for a specific view.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                       |
| gpin_verilog_sim_type    |                                                                                                                                                                                                                                                                                                                          |
|                          | <p>To identify if the net is wand or wor type.</p> <p><b>Type:</b> int</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                       |



|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hdl_name            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     | <p>This is the original RTL name for this port. It is used to map RTL simulation results with RTL to the current netlist for switching activity analysis. It is only maintained properly on the output of sequential cells. Optimization will copy this name during any multi-bit merge or splitting transforms to the equivalent port, but not on output ports of combinational cells. The is_phase_inverted attribute will be flipped if the phase is inverted.</p> <p><b>Type:</b> string<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| hnet                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     | <p>hnet of this port</p> <p><b>Type:</b> obj(hnet)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| hold_uncertainty    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     | <p>Returns the most conservative uncertainty of all possible uncertainty Hold assertions associated with the port. You can use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                    |
| input_slew_max_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     | <p>Returns the slowest falling transition on the port as specified by set_input_transition. You can use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                           |
| input_slew_max_rise |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     | <p>Returns the slowest rising transition on the port as specified by set_input_transition. You can use -index to return the value from a specific analysis view.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                            |
| input_slew_min_fall |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                         |                                                                                                                                                                                                                                                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Returns the fastest falling transition on the port as specified by set_input_transition. You can index to return the value from a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| input_slew_min_rise     |                                                                                                                                                                                                                                                                                                       |
|                         | <p>Returns the fastest rising transition on the port as specified by set_input_transition. You can index to return the value from a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| is_booth_encoded        |                                                                                                                                                                                                                                                                                                       |
|                         | <p>Booth encoded setting assigned to input port by user.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                            |
| is_clock_network_port   |                                                                                                                                                                                                                                                                                                       |
|                         | <p>Reports true if specified port is part of clock network</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                            |
| is_clock_source_latency |                                                                                                                                                                                                                                                                                                       |
|                         | <p>Returns true if object is part of latency network</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                           |
| is_clock_source_pin     |                                                                                                                                                                                                                                                                                                       |
|                         | <p>Returns true if object is a clock source port</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                               |
| is_clock_used_as_clock  |                                                                                                                                                                                                                                                                                                       |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Returns a value of true if the port lies in the clock network and at least one of the clocks arriving at the pin is used as a clock in the downstream network of the pin. You can use -index to return a value for a specific analysis view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_clock_used_as_data    |                                                                                                                                                                                                                                                                                                                                                                                                        |
|                          | <p>Returns a value of true if the port lies in the clock source path and at least one of the clocks on the pin is used as data in the downstream network of the pin. You can use -index to return a value for a specific analysis view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>         |
| is_constant_in_all_views |                                                                                                                                                                                                                                                                                                                                                                                                        |
|                          | <p>Returns true if logic state has been asserted or propagated to this pin in all the views</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                     |
| is_constant_in_any_view  |                                                                                                                                                                                                                                                                                                                                                                                                        |
|                          | <p>Returns true if logic state has been asserted or propagated to this pin in at least one view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                |
| is_disable_timing        |                                                                                                                                                                                                                                                                                                                                                                                                        |
|                          | <p>Returns a value of true if the port's timing has been disabled. You can use -index to return a value for a specific analysis view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                           |
| is_ideal_driver          |                                                                                                                                                                                                                                                                                                                                                                                                        |

|                        |                                                                                                                                                                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Returns true if the port is an ideal driver due to the set_ideal_latency or set_ideal_transition constraint.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                   |
| is_ideal_network       |                                                                                                                                                                                                                                                                 |
|                        | <p>Returns true if the port is part of an ideal network due to the set_ideal_network constraint. You use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_inside_partition    |                                                                                                                                                                                                                                                                 |
|                        | <p>Returns whether port is inside partition or not.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |
| is_internal_disable    |                                                                                                                                                                                                                                                                 |
|                        | <p>Return is_internal_disable property for a pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                        |
| is_owner               |                                                                                                                                                                                                                                                                 |
|                        | <p>Returns true if client is the owner for pin in distributed STA.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                       |
| is_power_mode_disabled |                                                                                                                                                                                                                                                                 |
|                        | <p>Returns true if power mode is disabled</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                      |
| is_propagated_clock    |                                                                                                                                                                                                                                                                 |

|                         |                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Returns a value of true if there is an explicit set_propagated_clock assertion at the port. You use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                           |
| is_shared               |                                                                                                                                                                                                                                                                                                                                                  |
|                         | <p>Returns true if pin is shared across clients in distributed STA.</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                       |
| is_special              |                                                                                                                                                                                                                                                                                                                                                  |
|                         | <p>Indicates that the port is Special(not set for base_cell terms)</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                             |
| is_valid_for_reports    |                                                                                                                                                                                                                                                                                                                                                  |
|                         | <p>Return whether a pin is valid for reporting or not</p> <p><b>Type:</b> bool</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                     |
| layer                   |                                                                                                                                                                                                                                                                                                                                                  |
|                         | <p>Layer of the port. For ports with more than one shape, it is the layer of the first shape (which same shape used for the .location value). If there are no shapes for the port, {} is returned.</p> <p><b>Type:</b> obj(layer)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                           |
| location                |                                                                                                                                                                                                                                                                                                                                                  |
|                         | <p>Location of port. For port with only one shape, it is the center of the port shape edge abutting design boundary. The edit_pin command can be used to set it. For ports that have more than one shape, the location will overlap the first shape of the port.</p> <p><b>Type:</b> point</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| lp_asserted_probability |                                                                                                                                                                                                                                                                                                                                                  |

|                                |                                                                                                                                                                      |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Probability of signal to be high on port/subport/pin as specified by user.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p> |
| lp_asserted_toggle_rate        |                                                                                                                                                                      |
|                                | <p>Signal toggle rate assigned to pin by user.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                |
| lp_system_asserted_probability |                                                                                                                                                                      |
|                                | <p>System/Internal asserted probability.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                      |
| lp_system_asserted_toggle_rate |                                                                                                                                                                      |
|                                | <p>System/Internal asserted toggle rate.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                      |
| lssd_master_clock              |                                                                                                                                                                      |
|                                | <p>Defines the master clock port related to this slave clock port.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                  |
| max_capacitance                |                                                                                                                                                                      |
|                                | <p>max_capacitance</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>          |
| max_fanout                     |                                                                                                                                                                      |

|                 |                                                                                                                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <p>Returns the maximum fanout load that the port can drive. . You can use -index to return the for a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>          |
| max_transition  |                                                                                                                                                                                                                                                                                |
|                 | <p>Returns the maximum transition time limit specified for the port. You can use -index to return value for a specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| min_capacitance |                                                                                                                                                                                                                                                                                |
|                 | <p>Returns the minimum capacitance limit for the port. You can use -index to return the value f specific analysis view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>               |
| min_fanout      |                                                                                                                                                                                                                                                                                |
|                 | <p>Returns the minimum fanout value for the port.</p> <p><b>Type:</b> int</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                 |
| min_transition  |                                                                                                                                                                                                                                                                                |
|                 | <p>Returns the minimum transition time limit specified for the port.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                           |
| name            |                                                                                                                                                                                                                                                                                |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>The same as .escaped_name without any \ escape chars. This name is commonly used to solve problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| net                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>The canonical (flat) net connected to the terminal</p> <p><b>Type:</b> obj(net)<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                             |
| network_latency_fall_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Returns the maximum fall insertion delay specified by an explicit set_clock_latency at the port. Can use -index to return the value for a specific analysis view.</p> <p><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                          |
| network_latency_fall_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Returns the minimum fall insertion delay specified by an explicit set_clock_latency at the port. Can use -index to return the value for a specific analysis view and/or clock.</p> <p><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                             |
| network_latency_rise_max |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                          | <p>Returns the maximum rise insertion delay specified by an explicit set_clock_latency at the port. Can use -index to return the value for a specific analysis view and/or clock</p> <p><b>Type:</b> double<br/><b>Allowed -index values:</b> analysis_view clock<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                              |
| network_latency_rise_min |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>Returns the minimum rise insertion delay specified by an explicit set_clock_latency at the p can use -index to return the value for a specific analysis view and/or clock</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                         |
| num_exceptions |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                | <p>Reports the number of path exceptions</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                |
| obj_type       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">port</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                          |
| physical_pins  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                | <p>List of ports</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">physical_pin</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| pin_edge       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                | <p>If this port is for a design, and the port is assigned (has place_status of placed/fixed/cover), edge value indicates along which edge of the boundary polygon the port is assigned. The number starts from the lowest Y, then left-most X vertex, starting with 0, and then counting clockwise. See the set_pin_constraint command document for a figure showing the edge number the port is not assigned, or not for a design, the value of -1 is returned.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| place_status   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                | <p>Placement Status of terminal (not for base_cell terms)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> unplaced placed fixed cover</p> <p><b>Default:</b> unplaced</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                         |

|                                    |                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_domain                       |                                                                                                                                                                                                                                                                                                                                                                         |
|                                    | <p>The power domain of the terminal (equivalent to Design Browser effPD)</p> <p><b>Type:</b> obj(power_domain)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                     |
| power_rail_voltage_max             |                                                                                                                                                                                                                                                                                                                                                                         |
|                                    | <p>power_rail_voltage_max</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                      |
| power_rail_voltage_min             |                                                                                                                                                                                                                                                                                                                                                                         |
|                                    | <p>power_rail_voltage_min</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                      |
| related_ground_pin                 |                                                                                                                                                                                                                                                                                                                                                                         |
|                                    | <p>Specifies which ground pin drives this signal pin. It must be one of the ground pins defined cell. It can be set by CPF, Liberty, or LEF/OA, with CPF having highest precedence, then Lil then LEF/OA. It is often only set when there is more than one ground pin for the cell.</p> <p><b>Type:</b> obj(port)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| related_power_pin                  |                                                                                                                                                                                                                                                                                                                                                                         |
|                                    | <p>Specifies which power pin drives this signal pin. It must be one of the power pins defined for cell. It can be set by CPF, Liberty, or LEF/OA, with CPF having highest precedence, then Lil then LEF/OA. It is often only set when there is more than one power pin for the cell.</p> <p><b>Type:</b> obj(port)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| safety_dclst_error_signal_endpoint |                                                                                                                                                                                                                                                                                                                                                                         |
|                                    | <p>This object is the endpoint for a safety-mechanism's error-signal.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                   |

|                                     |                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| safety_dmr_error_signal_endpoint    |                                                                                                                                                                                                                                                                                                                                   |
|                                     | <p>This object is the endpoint for a safety-mechanism's error-signal.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                             |
| safety_parity_error_signal_endpoint |                                                                                                                                                                                                                                                                                                                                   |
|                                     | <p>This object is the endpoint for a safety-mechanism's error-signal.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                             |
| setup_uncertainty                   |                                                                                                                                                                                                                                                                                                                                   |
|                                     | <p>Returns the most conservative uncertainty of all possible uncertainty assertions associated port. You can use -index to return the value for a specific analysis view</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| shape                               |                                                                                                                                                                                                                                                                                                                                   |
|                                     | <p>Terminal shape</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none ring abutment feed_through</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                        |
| side                                |                                                                                                                                                                                                                                                                                                                                   |
|                                     | <p>Side constraint of the port. The edit_pin command can be used to set this value.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> north west south east up none</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                        |
| skip_assertion                      |                                                                                                                                                                                                                                                                                                                                   |
|                                     | <p>Do not assert stimulus activity.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                             |
| slack_max                           |                                                                                                                                                                                                                                                                                                                                   |

|                     |                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>Returns the worst slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                         |
| slack_max_edge      |                                                                                                                                                                                                                                                                                                                                                        |
|                     | <p>Returns the edge (rise or fall) of the worst slack-causing path at the specified port in late mode. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> string<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                     |
| slack_max_fall      |                                                                                                                                                                                                                                                                                                                                                        |
|                     | <p>Returns the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                 |
| slack_max_rise      |                                                                                                                                                                                                                                                                                                                                                        |
|                     | <p>Returns the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                  |
| slack_mean_max      |                                                                                                                                                                                                                                                                                                                                                        |
|                     | <p>In SOCV analysis, this returns the mean component of the worst slack across all concurrent views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| slack_mean_max_fall |                                                                                                                                                                                                                                                                                                                                                        |

|                     |                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>In SOCV analysis, this returns the mean component of the worst falling slack across all concurrent MMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| slack_mean_max_rise |                                                                                                                                                                                                                                                                                                                                                                    |
|                     | <p>In SOCV analysis, this returns the mean component of the worst rising slack across all concurrent MMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| slack_mean_min      |                                                                                                                                                                                                                                                                                                                                                                    |
|                     | <p>In SOCV analysis, this returns the mean component of the worst slack across all concurrent views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>             |
| slack_mean_min_fall |                                                                                                                                                                                                                                                                                                                                                                    |
|                     | <p>In SOCV analysis, this returns the mean component of the worst falling slack across all concurrent MMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| slack_mean_min_rise |                                                                                                                                                                                                                                                                                                                                                                    |
|                     | <p>In SOCV analysis, this returns the mean component of the worst rising slack across all concurrent MMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |

|                      |                                                                                                                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| slack_min            |                                                                                                                                                                                                                                                                                               |
|                      | <p>Returns the worst slack across all concurrent MMMC views for hold-style late data path checks.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                    |
| slack_min_edge       |                                                                                                                                                                                                                                                                                               |
|                      | <p>Returns the edge (rise or fall) of the worst slack-causing path at the specified port in early non-secure analysis.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                               |
| slack_min_fall       |                                                                                                                                                                                                                                                                                               |
|                      | <p>Returns the worst hold slack for a falling signal at the port endpoint.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                           |
| slack_min_rise       |                                                                                                                                                                                                                                                                                               |
|                      | <p>Returns the worst hold slack for a rising signal at the port endpoint.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                            |
| slack_sigma_max      |                                                                                                                                                                                                                                                                                               |
|                      | <p>In SOCV analysis, this returns the variation component of the worst slack across all concurrent MMMC views for Setup-style late data path checks.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slack_sigma_max_fall |                                                                                                                                                                                                                                                                                               |

|                      |                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>In SOCV analysis, this returns the variation component of the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| slack_sigma_max_rise |                                                                                                                                                                                                                                                                                                                                                                      |
|                      | <p>In SOCV analysis, this returns the variation component of the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |
| slack_sigma_min      |                                                                                                                                                                                                                                                                                                                                                                      |
|                      | <p>In SOCV analysis, this returns the variation component of the worst slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>     |
| slack_sigma_min_fall |                                                                                                                                                                                                                                                                                                                                                                      |
|                      | <p>In SOCV analysis, this returns the variation component of the worst falling slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| slack_sigma_min_rise |                                                                                                                                                                                                                                                                                                                                                                      |
|                      | <p>In SOCV analysis, this returns the variation component of the worst rising slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return value for a specific view and/or clock.</p> <p><b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>  |

|                    |                                                                                                                                                                                                                                                                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| slew_max_fall      |                                                                                                                                                                                                                                                                                                                                    |
|                    | <p>Returns the maximum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                    |
| slew_max_rise      |                                                                                                                                                                                                                                                                                                                                    |
|                    | <p>Returns the maximum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                     |
| slew_mean_max_fall |                                                                                                                                                                                                                                                                                                                                    |
|                    | <p>In SOCV mode, returns the mean component of the maximum slew time for falling transition across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slew_mean_max_rise |                                                                                                                                                                                                                                                                                                                                    |
|                    | <p>In SOCV mode, returns the mean component of the maximum slew time for rising transition across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| slew_mean_min_fall |                                                                                                                                                                                                                                                                                                                                    |
|                    | <p>In SOCV mode, returns the mean component of the minimum slew time for falling transition across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slew_mean_min_rise |                                                                                                                                                                                                                                                                                                                                    |



|                     |                                                                                                                                                                                                                                                                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>In SOCV mode, returns the mean component of the minimum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| slew_min_fall       |                                                                                                                                                                                                                                                                                                                                                          |
|                     | <p>Returns the minimum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                          |
| slew_min_rise       |                                                                                                                                                                                                                                                                                                                                                          |
|                     | <p>Returns the minimum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                           |
| slew_sigma_max_fall |                                                                                                                                                                                                                                                                                                                                                          |
|                     | <p>In SOCV mode, returns the variation component of the maximum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slew_sigma_max_rise |                                                                                                                                                                                                                                                                                                                                                          |
|                     | <p>In SOCV mode, returns the variation component of the maximum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| slew_sigma_min_fall |                                                                                                                                                                                                                                                                                                                                                          |

|                               |                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>In SOCV mode, returns the variation component of the minimum slew time for falling transition across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| slew_sigma_min_rise           |                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>In SOCV mode, returns the variation component of the minimum slew time for rising transition across all concurrent MMMC views. You can use -index to return the value for a specific view.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| source_latency_early_fall_max |                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>Returns the maximum early fall source insertion delay specified by an explicit set_clock_latency at the port.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                               |
| source_latency_early_fall_min |                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>Returns the minimum early fall source insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| source_latency_early_rise_max |                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>Returns the maximum early rise source insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| source_latency_early_rise_min |                                                                                                                                                                                                                                                                                                                                         |

|                               |                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Returns the minimum early rise source insertion delay specified by an explicit set_clock_lat the port. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| source_latency_early_fall_max |                                                                                                                                                                                                                                                                                                                                             |
|                               | <p>Returns the maximum late fall source insertion delay specified by an explicit set_clock_lat the port. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| source_latency_early_fall_min |                                                                                                                                                                                                                                                                                                                                             |
|                               | <p>Returns the minimum late fall source insertion delay specified by an explicit set_clock_lat the port. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| source_latency_early_rise_max |                                                                                                                                                                                                                                                                                                                                             |
|                               | <p>Returns the maximum late rise source insertion delay specified by an explicit set_clock_lat the port. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| source_latency_early_rise_min |                                                                                                                                                                                                                                                                                                                                             |
|                               | <p>Returns the minimum late rise source insertion delay specified by an explicit set_clock_lat the port. You can use -index to return the value for a specific view and/or clock.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| timing_case_computed_value    |                                                                                                                                                                                                                                                                                                                                             |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Returns the case computed value</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| timing_case_logic_value |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                         | <p>Returns the case logic value</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| timing_constant         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                         | <p>Reports constant value on port</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| timing_constant_source  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                         | <p>Reports source of constant value on port</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| use                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                         | <p>Indicates how this port is used from the DEF PIN +USE value or OA equivalent. The legal values are: analog clock ground power signal. Note that timing analysis does not use these values; the .lib data instead (e.g. is_clock, is_analog, etc.). So the signal, clock or analog values are normally used by applications. The power/ground values are used by many applications. DEF +USE RESET, SCAN or TIEOFF are not supported, so read_def will ignore them, and keep existing DB use value.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> signal clock power ground analog</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| width                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                  |                                                                                                                                                                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | Width constraint of the port in microns. The edit_pin command can be used to set it.<br><b>Type:</b> coord<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                           |
| wire_lumped_capacitance_max      |                                                                                                                                                                                                                                               |
|                                  | Returns the portion of the net's total lumped capacitance which comes extracted or annotat<br>lumped capacitance values.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| wire_lumped_capacitance_max_fall |                                                                                                                                                                                                                                               |
|                                  | Returns the portion of the net's total lumped capacitance which comes extracted or annotat<br>lumped capacitance values.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| wire_lumped_capacitance_max_rise |                                                                                                                                                                                                                                               |
|                                  | Returns the portion of the net's total lumped capacitance which comes extracted or annotat<br>lumped capacitance values.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| wire_lumped_capacitance_min      |                                                                                                                                                                                                                                               |
|                                  | Returns the portion of the net's total lumped capacitance which comes extracted or annotat<br>lumped capacitance values.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| wire_lumped_capacitance_min_fall |                                                                                                                                                                                                                                               |

|                                  |                                                                                                                                                                                                                                                                                   |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Returns the portion of the net's total lumped capacitance which comes extracted or annotat lumped capacitance values.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| wire_lumped_capacitance_min_rise |                                                                                                                                                                                                                                                                                   |
|                                  | <p>Returns the portion of the net's total lumped capacitance which comes extracted or annotat lumped capacitance values.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| wrapper_control                  |                                                                                                                                                                                                                                                                                   |
|                                  | <p>Specifies as wrapper control signal.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                         |
| wrapper_segment                  |                                                                                                                                                                                                                                                                                   |
|                                  | <p>Specifies the wrapper segments associated with the pin.</p> <p><b>Type:</b> <a href="#">obj(scan_chain)* obj(scan_segment)* obj(actual_scan_segment)* obj(actual_scan_</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                               |
| wrapper_type                     |                                                                                                                                                                                                                                                                                   |
|                                  | <p>Specifies the type of wrapper segment associated with the pin.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                |

## port\_shape

### Parent Objects

[design](#), [net](#), [root](#)

## Definition

This corresponds to one of a port's .physical\_pins.layer\_shapes.shapes, but is directly accessible from the design and root objects. It carries a link to the port object, that a layer\_shape object does not have. This allows GUI operations that need both the layer\_shape and the port object together as one object like delete\_obj or wire-editing commands.

| Attribute | Description                                                                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| layer     |                                                                                                                                                                                                                               |
|           | Layer of this port shape<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                 |
| mask      |                                                                                                                                                                                                                               |
|           | The mask number for this port_shape if this layer has multiple masks. 0 means it is uncolored. Refer to layer .num_masks for legal range.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> no_value<br><b>Edit:</b> Yes |
| name      |                                                                                                                                                                                                                               |
|           | Name of the port for this port_shape. Note that all the port_shapes for one port will have the same name.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                     |
| net       |                                                                                                                                                                                                                               |
|           | The net this port shape connects to.<br><b>Type:</b> <a href="#">obj(net)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                        |
| obj_type  |                                                                                                                                                                                                                               |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (port_shape)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                       |
| polygon   |                                                                                                                                                                                                                               |

|      |                                                                                                                                                                                                                                                                              |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | Points of the polygon of the port_shape for type rect or polygon.<br><b>Type:</b> <a href="#">point*</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                           |
| port |                                                                                                                                                                                                                                                                              |
|      | The port this port shape belongs to.<br><b>Type:</b> <a href="#">obj(port)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                      |
| rect |                                                                                                                                                                                                                                                                              |
|      | Bounding box of the port_shape for type rect or polygon.<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                      |
| type |                                                                                                                                                                                                                                                                              |
|      | Type of port_shape (rect polygon). Note if a polygon was used to create the shape, but the specified polygon actually is rectangle, the type will be rect.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> rect polygon<br><b>Default:</b> ""<br><b>Edit:</b> No |

## power\_domain

### Parent Objects

[hinst](#), [inst](#), [design](#), [pin](#), [root](#), [group](#), [port](#)

### Definition

Power domain

| Attribute             | Description |
|-----------------------|-------------|
| available_supply_nets |             |



|                   |                                                                                                                                                                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>Specifies the power nets physically available for this power domain to use for secondary power pin connections.</p> <p><b>Type:</b> <a href="#">obj(net)</a>*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                 |
| base_domains      |                                                                                                                                                                                                                                                       |
|                   | <p>Specifies the base power domains (always-on domains) that supply the power to this switchable power domain through power switch cells.</p> <p><b>Type:</b> <a href="#">obj(power_domain)</a>*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| core_to_bottom    |                                                                                                                                                                                                                                                       |
|                   | <p>Distance between the power domain edge and its core box</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                             |
| core_to_left      |                                                                                                                                                                                                                                                       |
|                   | <p>Distance between the power domain edge and its core box</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                             |
| core_to_right     |                                                                                                                                                                                                                                                       |
|                   | <p>Distance between the power domain edge and its core box</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                             |
| core_to_top       |                                                                                                                                                                                                                                                       |
|                   | <p>Distance between the power domain edge and its core box</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                             |
| default_tech_site |                                                                                                                                                                                                                                                       |
|                   | <p>default tech. site of the domain</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                   |

|                         |                                                                                                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| disjoint_hinst_box_list |                                                                                                                                                                                                                                                                |
|                         | <p>List of hierarchical instances and disjoint boxes defining power domain boundary. It is defined as an hinst and region pair. The listed hinst will be placed inside this domain</p> <p><b>Type:</b> string<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| extend_power_bottom     |                                                                                                                                                                                                                                                                |
|                         | <p>Maximum search distance for power connections</p> <p><b>Type:</b> coord<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                    |
| extend_power_edges      |                                                                                                                                                                                                                                                                |
|                         | <p>List of maximum search distances for power extension connections in clockwise order starting with the vertical edge at the lower-left corner (smallest Y, then smallest X)</p> <p><b>Type:</b> coord*<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>      |
| extend_power_left       |                                                                                                                                                                                                                                                                |
|                         | <p>Maximum search distance for power connections</p> <p><b>Type:</b> coord<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                    |
| extend_power_right      |                                                                                                                                                                                                                                                                |
|                         | <p>Maximum search distance for power connections</p> <p><b>Type:</b> coord<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                    |
| extend_power_top        |                                                                                                                                                                                                                                                                |
|                         | <p>Maximum search distance for power connections</p> <p><b>Type:</b> coord<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                    |
| first_row_site_index    |                                                                                                                                                                                                                                                                |

|            |                                                                                                                                                                                                                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <p>Specifies the first row site index value.</p> <p><b>Type:</b> <code>int</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                  |
| gap_bottom |                                                                                                                                                                                                                                                                                      |
|            | <p>Minimum spacing to other power domains or rows</p> <p><b>Type:</b> <code>coord</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                           |
| gap_edges  |                                                                                                                                                                                                                                                                                      |
|            | <p>List of minimum spacing values to other power domains or rows in clockwise order starting with the vertical edge at the lower-left corner (smallest Y, then smallest X)</p> <p><b>Type:</b> <code>coord*</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p> |
| gap_left   |                                                                                                                                                                                                                                                                                      |
|            | <p>Minimum spacing to other power domains or rows</p> <p><b>Type:</b> <code>coord</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                           |
| gap_right  |                                                                                                                                                                                                                                                                                      |
|            | <p>Minimum spacing to other power domains or rows</p> <p><b>Type:</b> <code>coord</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                           |
| gap_top    |                                                                                                                                                                                                                                                                                      |
|            | <p>Minimum spacing to other power domains or rows</p> <p><b>Type:</b> <code>coord</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                           |
| group      |                                                                                                                                                                                                                                                                                      |
|            | <p>The group of PD</p> <p><b>Type:</b> <code>obj(group)</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                                                                                                                                     |

|               |                                                                                                                                                                                                                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_always_on  |                                                                                                                                                                                                                                                                                                                     |
|               | <p>Indicates that the power domain is always on</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                       |
| is_default    |                                                                                                                                                                                                                                                                                                                     |
|               | <p>The power domain is the default power domain or not.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                               |
| is_internal   |                                                                                                                                                                                                                                                                                                                     |
|               | <p>Indicates that the domain is created by tool for the supply set which is not the primary set of any domains defined in IEEE1801 file. It is created for supply set PVT specification and leaf instance pin's domain assignment.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p> |
| is_logical    |                                                                                                                                                                                                                                                                                                                     |
|               | <p>Indicates that the domain might have a physical constraint</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                                                                                      |
| is_macro_only |                                                                                                                                                                                                                                                                                                                     |
|               | <p>Indicates whether the domain member has only hardmacro members.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                    |
| is_physical   |                                                                                                                                                                                                                                                                                                                     |
|               | <p>Indicates that the domain might have a physical constraint</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                                                                                      |
| is_virtual    |                                                                                                                                                                                                                                                                                                                     |

|                       |                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Specifies the power domain is a virtual domain, meaning that this domain does not have any inst member.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> No |
| isolation_rules       |                                                                                                                                                                                       |
|                       | Get the domain's isolation_rules<br><b>Type:</b> <code>obj(isolation_rule)*</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                           |
| last_row_site_index   |                                                                                                                                                                                       |
|                       | Specifies the last row site index value.<br><b>Type:</b> <code>int</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                    |
| level_shifter_rules   |                                                                                                                                                                                       |
|                       | Get the domain's level_shifter_rules<br><b>Type:</b> <code>obj(level_shifter_rule)*</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                   |
| logical_power_domains |                                                                                                                                                                                       |
|                       | Indicates the domain's associated logical domains.<br><b>Type:</b> <code>string*</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                      |
| name                  |                                                                                                                                                                                       |
|                       | Name of the power domain<br><b>Type:</b> <code>string</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                 |
| nwell_supply_net      |                                                                                                                                                                                       |
|                       | Specifies the nwell bias net for this power_domain to use for nwell pin connections.<br><b>Type:</b> <code>obj(net)</code><br><b>Default:</b> ""<br><b>Edit:</b> No                   |

|                        |                                                                                                                                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| obj_type               |                                                                                                                                                                                                                              |
|                        | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">power_domain</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| physical_power_domain  |                                                                                                                                                                                                                              |
|                        | <p>Indicates the domain's physical domain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                   |
| power_switch_rule_name |                                                                                                                                                                                                                              |
|                        | <p>Indicates the domain's power switch rule name.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                            |
| power_switch_rules     |                                                                                                                                                                                                                              |
|                        | <p>Get the domain's power_switch_rules</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">power_switch_rule</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |
| primary_ground_net     |                                                                                                                                                                                                                              |
|                        | <p>Specifies the main ground rail net of the power domain.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">net</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                 |
| primary_power_net      |                                                                                                                                                                                                                              |
|                        | <p>Specifies the main power rail net of the power domain.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">net</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                  |
| pwell_supply_net       |                                                                                                                                                                                                                              |
|                        | <p>Specifies the pwell bias net for this power_domain to use for pwell pin connections.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">net</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                    |

|                  |                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| retention_rules  |                                                                                                                                                                                                                                                                                                                                                                                            |
|                  | <p>Get the domain's retention_rules</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">retention_rule</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                          |
| row_flip         |                                                                                                                                                                                                                                                                                                                                                                                            |
|                  | <p>Flips the orientation of either the first or second row, then follows the flipped and abutted row pattern. Refer to "update_power_domain -row_flip" for more details.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> noflip second first auto</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p>                                                              |
| row_pattern_site |                                                                                                                                                                                                                                                                                                                                                                                            |
|                  | <p>Specifies the site name of the row pattern for the power doamin. For non-default power domain, the default value is "". For default power domain, the default value is the default row pattern site set by floorplan setting or "" if floorplan setting is not set.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">site</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| row_space_type   |                                                                                                                                                                                                                                                                                                                                                                                            |
|                  | <p>Determines whether the row spacing value applies between no row (0), each row (1), or each pair of rows (2).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 0 1 2</p> <p><b>Default:</b> 2</p> <p><b>Edit:</b> Yes</p>                                                                                                                                             |
| row_spacing      |                                                                                                                                                                                                                                                                                                                                                                                            |
|                  | <p>Specifies the row spacing between each row (1) or each pair of rows (2) as specified in row_space_type.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                 |
| safety_ser_cells |                                                                                                                                                                                                                                                                                                                                                                                            |

|                           |                                                                                                                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>The list of base_cells allowed for those insts inside this power_domain with an ser (soft error resilience) safety mechanism.</p> <p><b>Type:</b> <code>obj(base_cell)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| safety_tmr_well_tap_cells |                                                                                                                                                                                                                                              |
|                           | <p>The well-tap cells to be placed next to the left and right edges of tmr (triple modular redundancy) flops within this power_domain.</p> <p><b>Type:</b> <code>obj(base_cell)*</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| shutoff_condition         |                                                                                                                                                                                                                                              |
|                           | <p>Indicates the power domain's shutoff condition. It is a logic expression with pin or hpin names.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                             |

## preferred\_cell\_stripe

### Parent Objects

### Definition

preferred cell stripe

| Attribute | Description                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------|
| name      |                                                                                                                               |
|           | <p>name of preferred_cell_stripe</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| obj_type  |                                                                                                                               |



The obj\_type name for this object. To see all obj\_type names, go to [Contents](#)  
**Type:** [enum](#) ([preferred\\_cell\\_stripe](#))  
**Default:** ""  
**Edit:** No

## rc\_corner

### Parent Objects

[delay\\_corner](#), [root](#)

### Definition

The rc\_corner object represents a specific interconnect parasitic corner. It contains configuration information for controlling the extraction of the parasitics and possible scaling. Parasitic files are annotated or exported with respect to an rc\_corner object name. The rc\_corner object is typically referenced from one or many higher-level delay\_corner objects. Use the create\_rc\_corner command to define new rc\_corners and the update\_rc\_corner command to update the attributes of existing rc\_corners.

| Attribute      | Description                                                                                                                                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cap_table_file |                                                                                                                                                                                                                                                                         |
|                | Specifies the layer capacitance table file associated with the rc_corner. This is not recommended for designs below 32nm. See the write_cap_table command for more details on the file.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| is_active      |                                                                                                                                                                                                                                                                         |
|                | Indicates that the rc_corner is associated with an analysis_view that is used in the active analysis_view<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                 |
| is_dynamic     |                                                                                                                                                                                                                                                                         |

|                |                                                                                                                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | Indicates that the rc_corner is associated with an analysis_view that is used in the active dynamic analysis_view<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No          |
| is_hold        |                                                                                                                                                                                                          |
|                | Indicates that the rc_corner is associated with an analysis_view that is used in the active hold analysis_view<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No             |
| is_leakage     |                                                                                                                                                                                                          |
|                | Indicates that the rc_corner is associated with an analysis_view that is used in the active leakage analysis_view<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No          |
| is_setup       |                                                                                                                                                                                                          |
|                | Indicates that the rc_corner is associated with an analysis_view that is used in the active setup analysis_view<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No            |
| name           |                                                                                                                                                                                                          |
|                | Provides the name of this rc_corner object as specified by create_rc_corner.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                             |
| obj_type       |                                                                                                                                                                                                          |
|                | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">rc_corner</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| post_route_cap |                                                                                                                                                                                                          |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Specifies scaling factor(s) to be used for capacitances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting.</p> <p><b>Type:</b> double*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                        |
| post_route_clock_cap       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                            | <p>Specifies scaling factor(s) to be used for clock network capacitances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting.</p> <p><b>Type:</b> double*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>          |
| post_route_clock_cross_cap |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                            | <p>Specifies scaling factor(s) to be used for clock network coupling capacitances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting.</p> <p><b>Type:</b> double*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| post_route_clock_res       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                            | <p>Specifies scaling factor(s) to be used for clock network resistances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting.</p> <p><b>Type:</b> double*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| post_route_cross_cap       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>Specifies scaling factor(s) to be used for coupling-capacitances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting.</p> <p><b>Type:</b> double*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| post_route_res      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                     | <p>Specifies scaling factor(s) to be used for resistances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting.</p> <p><b>Type:</b> double*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| pre_route_cap       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                     | <p>Specifies a scaling factor to be used for capacitances in pre_route flow steps. Defaults to 1.0 if not given.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                    |
| pre_route_clock_cap |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                     | <p>Specifies a scaling factor to be used for clock network capacitances in pre_route flow steps. Defaults to 1.0 if not given.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                      |
| pre_route_clock_res |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                     | <p>Specifies a scaling factor to be used for clock network resistance in pre_route flow steps. Defaults to 1.0 if not given.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                        |
| pre_route_res       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Specifies a scaling factor to be used for resistances in pre_route flow steps. Defaults to 1.0 if not given.</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| qrc_tech_file              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                            | <p>Specifies the QRC technology file associated with the rc_corner</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| temperature                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                            | <p>Specifies the temperature, in units of Celsius, to use to derate resistance values for this rc_corner. Use this parameter when you want to override the default temperature specified in the capacitance table or the QRC technology file. By default, the RC extractor uses the temperature of 25 degrees Celsius, unless it is specified by the capacitance table or the QRC technology file. Note, the opcond temperature is not used for RC extraction, the temperature for RC extraction must be set with this attribute if you want to override the techfile or captable settings.</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| thru_substrate_subckt_file |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                            | <p>Specifies sub circuit RC for FTV cells.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| via_variation_file         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                            | <p>Specifies the via layer file so that all the VIA resistance in SPEF could be mapped to have own variation multiplier.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## resistor

### Parent Objects

## Definition

After loading a Voltus IR-drop analysis result, you can select a resistor in the GUI. The pointer of the object is the same as its name, like resistor

| Attribute            | Description                                                                                                                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| capacitance          |                                                                                                                                                                                                                                                                                                                          |
|                      | <p>The capacitance for the two nodes before any effects of set_rail_what_if_capacitance in units of farads.</p> <p><b>Type:</b> double*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                             |
| current              |                                                                                                                                                                                                                                                                                                                          |
|                      | <p>The current of the resistor in units of A. For static rail_analysis it is the average value, for dynamic power analysis it is the rms value.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                          |
| current_direction    |                                                                                                                                                                                                                                                                                                                          |
|                      | <p>The direction of current. no_value is returned for non-orthogonal resistors or if it is not available.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> no_value left right up down</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                           |
| em_jmax_dc_avg_ratio |                                                                                                                                                                                                                                                                                                                          |
|                      | <p>The ratio of the current / em_jmax_dc_avg limit in the Quantus ICT file. This is computed during static rail analysis. The jmax name is used for legacy reasons, but the value is a current value ratio, not a current density ratio.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| em_jmax_dc_rms_ratio |                                                                                                                                                                                                                                                                                                                          |

|                 |                                                                                                                                                                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <p>The ratio of the current / em_jmax_dc_rms limit in the Quantus ICT file. This is computed during dynamic rail analysis. The jmax name is used for legacy reasons, but the value is a current value ratio, not a current density ratio.</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| failure_in_time |                                                                                                                                                                                                                                                                                                                     |
|                 | <p>The failure in time value of the resistor in units of 1e9 hours (e.g. 1.0 means one failure in 1e9 hours of operation).</p> <p><b>Type:</b> double<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                |
| layer           |                                                                                                                                                                                                                                                                                                                     |
|                 | <p>The layer of the resistor. A via resistor will have a via-layer, and a routing layer resistor will have a routing layer.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                               |
| length          |                                                                                                                                                                                                                                                                                                                     |
|                 | <p>The length of the resistor in unit of um.</p> <p><b>Type:</b> coord<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                               |
| location        |                                                                                                                                                                                                                                                                                                                     |
|                 | <p>The x, y location of the two nodes at each end of the resistor.</p> <p><b>Type:</b> point*<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                        |
| name            |                                                                                                                                                                                                                                                                                                                     |
|                 | <p>This is an R followed by an integer to identify the resistor.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                          |
| net             |                                                                                                                                                                                                                                                                                                                     |

|                   |                                                                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>The net of the resistor.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">net</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                            |
| node_names        |                                                                                                                                                                                                                          |
|                   | <p>This is an N followed by an integer to identify the two nodes at each end of the resistor.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                            |
| node_reff         |                                                                                                                                                                                                                          |
|                   | <p>The values are in units of ohms and are valid whenever the node reff db is available in the loaded state directory.</p> <p><b>Type:</b> <a href="#">double</a>*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| node_voltage_drop |                                                                                                                                                                                                                          |
|                   | <p>The IR voltage drop of the two nodes in units of mV.</p> <p><b>Type:</b> <a href="#">double</a>*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                 |
| node_voltages     |                                                                                                                                                                                                                          |
|                   | <p>The IR voltage drop of the two nodes in units of mV.</p> <p><b>Type:</b> <a href="#">double</a>*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                 |
| obj_type          |                                                                                                                                                                                                                          |
|                   | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">resistor</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| resistance        |                                                                                                                                                                                                                          |
|                   | <p>The resistance before any effects from set_rail_what_if_resistance in units of ohms.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                  |



|                     |                                                                                                                                                                                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| switch_net          |                                                                                                                                                                                                                                                 |
|                     | <p>The switched net of the resistor.</p> <p><b>Type:</b> <a href="#">obj(net)</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                |
| what_if_capacitance |                                                                                                                                                                                                                                                 |
|                     | <p>The capacitance after any effects of set_rail_what_if_capacitance. This is the capacitance actually used in the rail analysis in units of farads.</p> <p><b>Type:</b> <a href="#">double*</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| what_if_resistance  |                                                                                                                                                                                                                                                 |
|                     | <p>The resistance after any effects of set_rail_what_if_resistance. This is the resistance actually used in the rail analysis in units of ohms.</p> <p><b>Type:</b> <a href="#">double</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>       |
| width               |                                                                                                                                                                                                                                                 |
|                     | <p>The width of the resistor in unit of um.</p> <p><b>Type:</b> <a href="#">coord</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                            |

## resize\_blockage

### Parent Objects

[design](#), [root](#)

### Definition

resize blkg

| Attribute     | Description |
|---------------|-------------|
| is_resizeable |             |

|          |                                                                                                                                                                                                                                                                                              |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <p>Specifies that the size blockage can be resized, however alignment and the minimum space between the objects in the blockage area will be maintained during floorplan resize.(1 = resizable).</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p> |
| name     |                                                                                                                                                                                                                                                                                              |
|          | <p>Name of the size blockage</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                                      |
| obj_type |                                                                                                                                                                                                                                                                                              |
|          | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">resize_blockage</a>)<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                    |
| rect     |                                                                                                                                                                                                                                                                                              |
|          | <p>Rectangle that defines the <a href="#">resize_blockage</a> shape.</p> <p><b>Type:</b> <a href="#">rect</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                 |

# root

## Parent Objects

## Definition

root

| Attribute                  | Description |
|----------------------------|-------------|
| -enable_sens_based_scaling |             |

|                                                            |                                                                                                                                                                                                       |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            | <b>Type:</b> <a href="#">bool</a><br><b>Edit:</b> Yes                                                                                                                                                 |
| <a href="#">_timing_sdf_nochange_verilog_compatibility</a> |                                                                                                                                                                                                       |
|                                                            | <b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes                                                                                                                         |
| <a href="#">actual_scan_chains</a>                         |                                                                                                                                                                                                       |
|                                                            | List of all actual_scan_chains<br><b>Type:</b> <a href="#">obj(actual_scan_chain)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                       |
| <a href="#">actual_scan_segments</a>                       |                                                                                                                                                                                                       |
|                                                            | List of all actual_scan_segments<br><b>Type:</b> <a href="#">obj(actual_scan_segment)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                   |
| <a href="#">add_dtc_areas</a>                              |                                                                                                                                                                                                       |
|                                                            | specify the areas to insert DTC cells.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_dtc_areas</a>                            |
| <a href="#">add_dtc_dtc_cell</a>                           |                                                                                                                                                                                                       |
|                                                            | specify the DTC cell to be inserted. Only DTC cell is allowed.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_dtc_dtc_cell</a> |
| <a href="#">add_dtc_dtc_max_area</a>                       |                                                                                                                                                                                                       |

|                                        |                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>specify the maximum area of DTC cell pattern, if DTC cell pattern's area greater than the value swap DTC cells to filler cells.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 1e+06<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_dtc_dtc_max_area</a></p>     |
| add_dtc_dtc_max_length                 |                                                                                                                                                                                                                                                                                                    |
|                                        | <p>specify max DTC vertical stack length rule by micron.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 1e+06<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_dtc_dtc_max_length</a></p>                                                                             |
| add_dtc_dtc_max_width                  |                                                                                                                                                                                                                                                                                                    |
|                                        | <p>specify max DTC horizontal stack length rule by micron.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 1e+06<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_dtc_dtc_max_width</a></p>                                                                            |
| add_dtc_dtc_min_area                   |                                                                                                                                                                                                                                                                                                    |
|                                        | <p>specify the minimum area of DTC cell pattern, if DTC cell pattern's area less than the value, DTC cells to filler cells.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_dtc_dtc_min_area</a></p>              |
| add_dtc_dtc_min_enclosed_area          |                                                                                                                                                                                                                                                                                                    |
|                                        | <p>specify the minimum enclosure area of DTC cell pattern, if enclosed area less than the value swap DTC cells to filler cells.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_dtc_dtc_min_enclosed_area</a></p> |
| add_dtc_dtc_to_horizontal_wire_spacing |                                                                                                                                                                                                                                                                                                    |
|                                        | <p>define the spacing between DTC cell and horizontal PG wire.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_dtc_dtc_to_horizontal_wire_spacing</a></p>                                                         |

|                                 |                                                                                                                                                                                                                                                      |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_dtc_dtc_vss_ratio           |                                                                                                                                                                                                                                                      |
|                                 | <p>specify the ratio between CAP unit cell counts and VSS unit cell counts.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 4.0</p> <p><b>Edit:</b> Yes</p>                                                                        |
| add_dtc_filler_cell             |                                                                                                                                                                                                                                                      |
|                                 | <p>specify the filler cell.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_dtc_filler_cell</a></p>                                                            |
| add_dtc_ground_cell             |                                                                                                                                                                                                                                                      |
|                                 | <p>specify the ground cell.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_dtc_ground_cell</a></p>                                                            |
| add_dtc_max_dtc_ground_distance |                                                                                                                                                                                                                                                      |
|                                 | <p>define max spacing between DTC to ground cell in same PIN_VSS group.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1e+06</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_dtc_max_dtc_ground_distance</a></p> |
| add_dtc_min_cell_tsv_spacing    |                                                                                                                                                                                                                                                      |
|                                 | <p>define min spacing between DTC cell boundary and TSV cell boundary.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p>                                                                             |
| add_dtc_min_dtc_gap             |                                                                                                                                                                                                                                                      |
|                                 | <p>specify the min spacing between DTC cells. DTC cell abutment allowed.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_dtc_min_dtc_gap</a></p>              |
| add_dtc_power_cell              |                                                                                                                                                                                                                                                      |

|                                          |                                                                                                                                                                                                                                              |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>specify the power cell.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_dtc_power_cell</a></p>                                                      |
| add_dtc_power_domain                     |                                                                                                                                                                                                                                              |
|                                          | <p>insert DTC cells inside the power domain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_dtc_power_domain</a></p>                                  |
| add_endcaps_antenna_tap_fill_wall_cells  |                                                                                                                                                                                                                                              |
|                                          | <p>N2 spr flow fillwall cells</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_antenna_tap_fill_wall_cells</a></p>                              |
| add_endcaps_avoid_two_sites_cell_abut    |                                                                                                                                                                                                                                              |
|                                          | <p>avoid 2 sites boundary cell abut to incorner cell in N10</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_avoid_two_sites_cell_abut</a></p> |
| add_endcaps_barrier_border_cell_bottom_y |                                                                                                                                                                                                                                              |
|                                          | <p>horizontal bottom border break cell</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_barrier_border_cell_bottom_y</a></p>                    |
| add_endcaps_barrier_border_cell_top_y    |                                                                                                                                                                                                                                              |
|                                          | <p>horizontal top border break cell</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_barrier_border_cell_top_y</a></p>                          |
| add_endcaps_barrier_cell_x               |                                                                                                                                                                                                                                              |

|                                           |                                                                                                                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | vertical break cell<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_barrier_cell_x</a>                                                    |
| add_endcaps_barrier_cell_xy               |                                                                                                                                                                                                                         |
|                                           | horizontal vertical overlap break cell<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_barrier_cell_xy</a>                                |
| add_endcaps_barrier_cell_y                |                                                                                                                                                                                                                         |
|                                           | horizontal break cell<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_barrier_cell_y</a>                                                  |
| add_endcaps_barrier_keepout_from_boundary |                                                                                                                                                                                                                         |
|                                           | break cell x and y Keepout distance from boundary value<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_barrier_keepout_from_boundary</a> |
| add_endcaps_barrier_pitch                 |                                                                                                                                                                                                                         |
|                                           | break cell x and y pitch value<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_barrier_pitch</a>                                          |
| add_endcaps_barrier_termination_bottom_y  |                                                                                                                                                                                                                         |
|                                           | horizontal termination bottom break cell<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_barrier_termination_bottom_y</a>                 |
| add_endcaps_barrier_termination_top_y     |                                                                                                                                                                                                                         |

|                                    |                                                                                                                                                                                                                       |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | horizontal termination top break cell<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_barrier_termination_top_y</a>                     |
| add_endcaps_barrier_termination_x  |                                                                                                                                                                                                                       |
|                                    | vertical termination break cell<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_barrier_termination_x</a>                               |
| add_endcaps_barrier_y_mode         |                                                                                                                                                                                                                       |
|                                    | use mode 1 or 2 for barrier y cell insertion<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> 1 2<br><b>Default:</b> 2<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_barrier_y_mode</a> |
| add_endcaps_bottom_edge            |                                                                                                                                                                                                                       |
|                                    | EndCaps with n-well at the bottom edge<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_bottom_edge</a>                                  |
| add_endcaps_boundary_tap           |                                                                                                                                                                                                                       |
|                                    | New flow to add cap cell and tap cell together<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_boundary_tap</a>                        |
| add_endcaps_boundary_tap_swap_flow |                                                                                                                                                                                                                       |
|                                    | boundary tap swap flow<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_boundary_tap_swap_flow</a>                                      |
| add_endcaps_cells                  |                                                                                                                                                                                                                       |



|                                               |                                                                                                                                                                                                                                        |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | Specify endCaps cell candidates for N10 that has defined in LEF<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_cells</a>                                |
| add_endcaps_create_rows                       |                                                                                                                                                                                                                                        |
|                                               | Create rows for endtap cell techSites which do not have rows in the floor plan<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_create_rows</a>          |
| add_endcaps_double_height_incorner_protrusion |                                                                                                                                                                                                                                        |
|                                               | allow double height in-corner cell to protrude into core area<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_double_height_incorner_protrusion</a>     |
| add_endcaps_enable_shrink_physical_cell_flow  |                                                                                                                                                                                                                                        |
|                                               | Samsung shrink left/right edge cell flow<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_enable_shrink_physical_cell_flow</a>                           |
| add_endcaps_flip_y                            |                                                                                                                                                                                                                                        |
|                                               | Flips the orientation of the endcap instances in Y direction if the site symmetry allows it.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_flip_y</a> |
| add_endcaps_incremental_left_edge             |                                                                                                                                                                                                                                        |
|                                               | EndCap lists with n-well at the left edge<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_incremental_left_edge</a>                                      |
| add_endcaps_incremental_right_edge            |                                                                                                                                                                                                                                        |

|                                         |                                                                                                                                                                                                                                                                      |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>EndCap list with n-well at the right edge</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_incremental_right_edge</a></p>                                            |
| add_endcaps_insert_nppp_wall            |                                                                                                                                                                                                                                                                      |
|                                         | <p>cells will be used for np wall</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_insert_nppp_wall</a></p>                                                             |
| add_endcaps_left_bottom_corner          |                                                                                                                                                                                                                                                                      |
|                                         | <p>EndCap with n-well at the left bottom corner</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_bottom_corner</a></p>                                             |
| add_endcaps_left_bottom_corner_even     |                                                                                                                                                                                                                                                                      |
|                                         | <p>EndCap with n-well at the left bottom corner, even sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_bottom_corner_even</a></p>                            |
| add_endcaps_left_bottom_corner_neighbor |                                                                                                                                                                                                                                                                      |
|                                         | <p>The cell to be abutted with the left side of leftBottomCorner in R0 orientation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_bottom_corner_neighbor</a></p> |
| add_endcaps_left_bottom_corner_odd      |                                                                                                                                                                                                                                                                      |
|                                         | <p>EndCap with n-well at the left bottom corner, odd sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_bottom_corner_odd</a></p>                              |
| add_endcaps_left_bottom_edge            |                                                                                                                                                                                                                                                                      |

|                                       |                                                                                                                                                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>EndCap with n-well at the left bottom edge</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_bottom_edge</a></p>                                              |
| add_endcaps_left_bottom_edge_even     |                                                                                                                                                                                                                                                                   |
|                                       | <p>EndCap with n-well at the left bottom edge, even sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_bottom_edge_even</a></p>                             |
| add_endcaps_left_bottom_edge_neighbor |                                                                                                                                                                                                                                                                   |
|                                       | <p>The cell to be abutted with the right edge of leftBottomEdge in R0 orientation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_bottom_edge_neighbor</a></p> |
| add_endcaps_left_bottom_edge_odd      |                                                                                                                                                                                                                                                                   |
|                                       | <p>EndCap with n-well at the left bottom edge, odd sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_bottom_edge_odd</a></p>                               |
| add_endcaps_left_corner_bottom_border |                                                                                                                                                                                                                                                                   |
|                                       | <p>inner corner's border cell</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_corner_bottom_border</a></p>                                                     |
| add_endcaps_left_corner_top_border    |                                                                                                                                                                                                                                                                   |
|                                       | <p>inner corner's border cell</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_corner_top_border</a></p>                                                        |
| add_endcaps_left_edge                 |                                                                                                                                                                                                                                                                   |

|                                     |                                                                                                                                                                                                                                                    |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>EndCap with n-well at the left edge</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_edge</a></p>                                             |
| add_endcaps_left_edge_bottom_border |                                                                                                                                                                                                                                                    |
|                                     | <p>EndCaps with n-well at the left edge, poly at the bottom boundary</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_edge_bottom_border</a></p> |
| add_endcaps_left_edge_even          |                                                                                                                                                                                                                                                    |
|                                     | <p>EndCap with n-well at the left edge, even sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_edge_even</a></p>                            |
| add_endcaps_left_edge_odd           |                                                                                                                                                                                                                                                    |
|                                     | <p>EndCap with n-well at the left edge, odd sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_edge_odd</a></p>                              |
| add_endcaps_left_edge_top_border    |                                                                                                                                                                                                                                                    |
|                                     | <p>EndCaps with n-well at the left edge, poly at the top boundary</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_edge_top_border</a></p>       |
| add_endcaps_left_top_corner         |                                                                                                                                                                                                                                                    |
|                                     | <p>EndCap with n-well at the left top corner</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_top_corner</a></p>                                 |
| add_endcaps_left_top_corner_even    |                                                                                                                                                                                                                                                    |

|                                      |                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>EndCap with n-well at the left top corner, even sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_top_corner_even</a></p>                            |
| add_endcaps_left_top_corner_neighbor |                                                                                                                                                                                                                                                                |
|                                      | <p>The cell to be abutted with the left side of leftTopCorner in R0 orientation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_top_corner_neighbor</a></p> |
| add_endcaps_left_top_corner_odd      |                                                                                                                                                                                                                                                                |
|                                      | <p>EndCap with n-well at the left top corner, odd sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_top_corner_odd</a></p>                              |
| add_endcaps_left_top_edge            |                                                                                                                                                                                                                                                                |
|                                      | <p>EndCap with n-well at the left top edge</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_top_edge</a></p>                                                 |
| add_endcaps_left_top_edge_even       |                                                                                                                                                                                                                                                                |
|                                      | <p>EndCap with n-well at the left top edge, even sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_top_edge_even</a></p>                                |
| add_endcaps_left_top_edge_neighbor   |                                                                                                                                                                                                                                                                |
|                                      | <p>The cell to be abutted with the right edge of leftTopEdge in R0 orientation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_top_edge_neighbor</a></p>    |
| add_endcaps_left_top_edge_odd        |                                                                                                                                                                                                                                                                |

|                                          |                                                                                                                                                                                                                               |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>EndCap with n-well at the left top edge, odd sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_left_top_edge_odd</a></p> |
| add_endcaps_min_horizontal_channel_width |                                                                                                                                                                                                                               |
|                                          | <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_min_horizontal_channel_width</a></p>                                                    |
| add_endcaps_min_jog_height               |                                                                                                                                                                                                                               |
|                                          | <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_min_jog_height</a></p>                                                                  |
| add_endcaps_min_jog_width                |                                                                                                                                                                                                                               |
|                                          | <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_min_jog_width</a></p>                                                                   |
| add_endcaps_min_vertical_channel_width   |                                                                                                                                                                                                                               |
|                                          | <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_min_vertical_channel_width</a></p>                                                      |
| add_endcaps_nppp_wall_offset             |                                                                                                                                                                                                                               |
|                                          | <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_nppp_wall_offset</a></p>                                                           |
| add_endcaps_nppp_wall_pitch              |                                                                                                                                                                                                                               |

|                                           |                                                                                                                                                                                     |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <b>Type:</b> double<br><b>Default:</b> 0.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_nppp_wall_pitch</a>                                                     |
| add_endcaps_nppp_wall_shift_step          |                                                                                                                                                                                     |
|                                           | <b>Type:</b> double<br><b>Default:</b> 0.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_nppp_wall_shift_step</a>                                                |
| add_endcaps_nppp_wall_to_fillwall_spacing |                                                                                                                                                                                     |
|                                           | <b>Type:</b> double<br><b>Default:</b> 0.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_nppp_wall_to_fillwall_spacing</a>                                       |
| add_endcaps_nppp_wall_to_incorner_spacing |                                                                                                                                                                                     |
|                                           | <b>Type:</b> double<br><b>Default:</b> 0.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_nppp_wall_to_incorner_spacing</a>                                       |
| add_endcaps_prefix                        |                                                                                                                                                                                     |
|                                           | Prefix of EndCap cells<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_prefix</a>                                     |
| add_endcaps_right_bottom_corner           |                                                                                                                                                                                     |
|                                           | EndCap with n-well at the right bottom corner<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_endcaps_right_bottom_corner</a> |
| add_endcaps_right_bottom_corner_even      |                                                                                                                                                                                     |

|                                          |                                                                                                                                                                                                                                                                         |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>EndCap with n-well at the right bottom corner, even sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_bottom_corner_even</a></p>                             |
| add_endcaps_right_bottom_corner_neighbor |                                                                                                                                                                                                                                                                         |
|                                          | <p>The cell to be abutted with the right side of rightBottomCorner in R0 orientation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_bottom_corner_neighbor</a></p> |
| add_endcaps_right_bottom_corner_odd      |                                                                                                                                                                                                                                                                         |
|                                          | <p>EndCap with n-well at the right bottom corner, odd sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_bottom_corner_odd</a></p>                               |
| add_endcaps_right_bottom_edge            |                                                                                                                                                                                                                                                                         |
|                                          | <p>EndCap with n-well at the right bottom edge</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_bottom_edge</a></p>                                                  |
| add_endcaps_right_bottom_edge_even       |                                                                                                                                                                                                                                                                         |
|                                          | <p>EndCap with n-well at the right bottom edge, even sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_bottom_edge_even</a></p>                                 |
| add_endcaps_right_bottom_edge_neighbor   |                                                                                                                                                                                                                                                                         |
|                                          | <p>The cell to be abutted with the left edge of rightBottomEdge in R0 orientation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_bottom_edge_neighbor</a></p>      |
| add_endcaps_right_bottom_edge_odd        |                                                                                                                                                                                                                                                                         |



|                                        |                                                                                                                                                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>EndCap with n-well at the right bottom edge, odd sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_bottom_edge_odd</a></p>                |
| add_endcaps_right_corner_bottom_border |                                                                                                                                                                                                                                                      |
|                                        | <p>inner corner's border cell</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_corner_bottom_border</a></p>                                       |
| add_endcaps_right_corner_top_border    |                                                                                                                                                                                                                                                      |
|                                        | <p>inner corner's border cell</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_corner_top_border</a></p>                                          |
| add_endcaps_right_edge                 |                                                                                                                                                                                                                                                      |
|                                        | <p>EndCap with n-well at the right edge</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_edge</a></p>                                             |
| add_endcaps_right_edge_bottom_border   |                                                                                                                                                                                                                                                      |
|                                        | <p>EndCaps with n-well at the right edge, poly at the bottom boundary</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_edge_bottom_border</a></p> |
| add_endcaps_right_edge_even            |                                                                                                                                                                                                                                                      |
|                                        | <p>EndCap with n-well at the right edge, even sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_edge_even</a></p>                            |
| add_endcaps_right_edge_odd             |                                                                                                                                                                                                                                                      |

|                                       |                                                                                                                                                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>EndCap with n-well at the right edge, odd sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_edge_odd</a></p>                                           |
| add_endcaps_right_edge_top_border     |                                                                                                                                                                                                                                                                   |
|                                       | <p>EndCaps with n-well at the right edge, poly at the top boundary</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_edge_top_border</a></p>                    |
| add_endcaps_right_top_corner          |                                                                                                                                                                                                                                                                   |
|                                       | <p>EndCap with n-well at the right top corner</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_top_corner</a></p>                                              |
| add_endcaps_right_top_corner_even     |                                                                                                                                                                                                                                                                   |
|                                       | <p>EndCap with n-well at the right top corner, even sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_top_corner_even</a></p>                             |
| add_endcaps_right_top_corner_neighbor |                                                                                                                                                                                                                                                                   |
|                                       | <p>The cell to be abutted with the right side of rightTopCorner in R0 orientation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_top_corner_neighbor</a></p> |
| add_endcaps_right_top_corner_odd      |                                                                                                                                                                                                                                                                   |
|                                       | <p>EndCap with n-well at the right top corner, odd sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_top_corner_odd</a></p>                               |
| add_endcaps_right_top_edge            |                                                                                                                                                                                                                                                                   |

|                                     |                                                                                                                                                                                                                                                              |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>EndCap with n-well at the right top edge</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_top_edge</a></p>                                             |
| add_endcaps_right_top_edge_even     |                                                                                                                                                                                                                                                              |
|                                     | <p>EndCap with n-well at the right top edge, even sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_top_edge_even</a></p>                            |
| add_endcaps_right_top_edge_neighbor |                                                                                                                                                                                                                                                              |
|                                     | <p>The cell to be abutted with the left edge of rightTopEdge in R0 orientation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_top_edge_neighbor</a></p> |
| add_endcaps_right_top_edge_odd      |                                                                                                                                                                                                                                                              |
|                                     | <p>EndCap with n-well at the right top edge, odd sites</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_right_top_edge_odd</a></p>                              |
| add_endcaps_top_bottom_edge         |                                                                                                                                                                                                                                                              |
|                                     | <p>EndCaps with n-well at the both top and bottom edges</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_top_bottom_edge</a></p>                                |
| add_endcaps_top_edge                |                                                                                                                                                                                                                                                              |
|                                     | <p>EndCaps with n-well at the top edge</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_top_edge</a></p>                                                        |
| add_endcaps_use_even_odd_sites      |                                                                                                                                                                                                                                                              |

|                                                 |                                                                                                                                                                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>use even or odd poly sites</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none even odd</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_use_even_odd_sites</a></p> |
| add_endcaps_wall_keepout_from_vertical_boundary |                                                                                                                                                                                                                                                 |
|                                                 | <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_wall_keepout_from_vertical_boundary</a></p>                                                          |
| add_endcaps_wall_offset                         |                                                                                                                                                                                                                                                 |
|                                                 | <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_wall_offset</a></p>                                                                                  |
| add_endcaps_wall_pitch                          |                                                                                                                                                                                                                                                 |
|                                                 | <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_wall_pitch</a></p>                                                                                   |
| add_endcaps_wall_shift_step                     |                                                                                                                                                                                                                                                 |
|                                                 | <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_wall_shift_step</a></p>                                                                              |
| add_endcaps_wall_to_convex_corner_spacing       |                                                                                                                                                                                                                                                 |
|                                                 | <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_endcaps_wall_to_convex_corner_spacing</a></p>                                                                |
| add_fillers_avoid_abutment_patterns             |                                                                                                                                                                                                                                                 |

|                                   |                                                                                                                                                                                                                                                                                          |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>illegal abutment patterns</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_avoid_abutment_patterns</a></p>                                                                               |
| add_fillers_cell_name_style       |                                                                                                                                                                                                                                                                                          |
|                                   | <p>Add physical cells into hierarchical modules, or as top level cells (flat)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> hier flat</p> <p><b>Default:</b> hier</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_cell_name_style</a></p> |
| add_fillers_cells                 |                                                                                                                                                                                                                                                                                          |
|                                   | <p>List of filler cells</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_cells</a></p>                                                                                                      |
| add_fillers_check_different_cells |                                                                                                                                                                                                                                                                                          |
|                                   | <p>Check violations between different cell types</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_check_different_cells</a></p>                                                            |
| add_fillers_check_drc             |                                                                                                                                                                                                                                                                                          |
|                                   | <p>DRC checking after filler insertion</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_check_drc</a></p>                                                                                   |
| add_fillers_check_trim_rule       |                                                                                                                                                                                                                                                                                          |
|                                   | <p>check trim-2-trim rule between cell trim and routing trim.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_check_trim_rule</a></p>                                                     |
| add_fillers_create_rows           |                                                                                                                                                                                                                                                                                          |

|                                            |                                                                                                                                                                                                                              |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | Create rows for filler cell techSites which do not have rows in the floor plan<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_create_rows</a> |
| add_fillers_default_diffusion_filler       |                                                                                                                                                                                                                              |
|                                            | List of default diffusion cells<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_default_diffusion_filler</a>                                   |
| add_fillers_distribute_implant_evenly      |                                                                                                                                                                                                                              |
|                                            | distribute implant evenly for fillers<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_distribute_implant_evenly</a>                           |
| add_fillers_double_height_filler_insertion |                                                                                                                                                                                                                              |
|                                            | enable double height filler during filler insertion.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_double_height_filler_insertion</a>       |
| add_fillers_eco_mode                       |                                                                                                                                                                                                                              |
|                                            | Fillers inserted in eco flow<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_eco_mode</a>                                                     |
| add_fillers_enable_restore_filler_flow     |                                                                                                                                                                                                                              |
|                                            | enable restore filler flow.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_enable_restore_filler_flow</a>                                     |
| add_fillers_honor_preroute_as_obs          |                                                                                                                                                                                                                              |

|                                       |                                                                                                                                                                                                                                                         |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>Filler command honors place_detail_preroute_as_obs value</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_honor_preroute_as_obs</a></p>                |
| add_fillers_horizontal_exception_cell |                                                                                                                                                                                                                                                         |
|                                       | <p>List of exceptional cells for horizontal stack checking</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_horizontal_exception_cell</a></p>              |
| add_fillers_horizontal_max_length     |                                                                                                                                                                                                                                                         |
|                                       | <p>specify instance max length (in micron) aligned in horizontal direction</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_horizontal_max_length</a></p> |
| add_fillers_horizontal_repair_cell    |                                                                                                                                                                                                                                                         |
|                                       | <p>List of filler cells to fix horizontal max length violation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_horizontal_repair_cell</a></p>             |
| add_fillers_keep_fixed                |                                                                                                                                                                                                                                                         |
|                                       | <p>'FIXED' fillers are kepted</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_keep_fixed</a></p>                                                         |
| add_fillers_no_single_site_gap        |                                                                                                                                                                                                                                                         |
|                                       | <p>Filler command fit 1-site gap</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_no_single_site_gap</a></p>                                               |
| add_fillers_prefix                    |                                                                                                                                                                                                                                                         |

|                                                     |                                                                                                                                                                                                                                                         |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | Prefix of filler cells<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_prefix</a>                                                                                         |
| add_fillers_preserve_user_order                     |                                                                                                                                                                                                                                                         |
|                                                     | Instruct tool to insert fillers by the user defined order<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_preserve_user_order</a>                                        |
| add_fillers_scheme                                  |                                                                                                                                                                                                                                                         |
|                                                     | Instruct tool to select locationFirst cellFirst scheme<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> locationFirst cellFirst<br><b>Default:</b> locationFirst<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_scheme</a> |
| add_fillers_swap_cell                               |                                                                                                                                                                                                                                                         |
|                                                     | swap cell pairs<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_swap_cell</a>                                                                                             |
| add_fillers_vertical_stack_exception_cell           |                                                                                                                                                                                                                                                         |
|                                                     | List of exceptional cells for vertical stack checking<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_vertical_stack_exception_cell</a>                                   |
| add_fillers_vertical_stack_left_edge_exception_cell |                                                                                                                                                                                                                                                         |
|                                                     | List of cells only check right edge for vertical stack violation<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">add_fillers_vertical_stack_left_edge_exception_cell</a>              |
| add_fillers_vertical_stack_max_length               |                                                                                                                                                                                                                                                         |



|                                                      |                                                                                                                                                                                                                                                                         |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | <p>specify instance max length (in micron) aligned in vertical direction</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_vertical_stack_max_length</a></p>               |
| add_fillers_vertical_stack_repair_cell               |                                                                                                                                                                                                                                                                         |
|                                                      | <p>List of filler cells to fix vertical stack violation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_vertical_stack_repair_cell</a></p>                                |
| add_fillers_vertical_stack_repair_edge               |                                                                                                                                                                                                                                                                         |
|                                                      | <p>specify the edge to use of vertical repair cell</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> left right</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_vertical_stack_repair_edge</a></p> |
| add_fillers_vertical_stack_right_edge_exception_cell |                                                                                                                                                                                                                                                                         |
|                                                      | <p>List of cells only check left edge for vertical stack violation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_vertical_stack_right_edge_exception_cell</a></p>       |
| add_fillers_with_drc                                 |                                                                                                                                                                                                                                                                         |
|                                                      | <p>call second add_fillers without drc checking</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_with_drc</a></p>                                                          |
| add_fillers_y_flip_type                              |                                                                                                                                                                                                                                                                         |
|                                                      | <p>Defines first or last cell with certain cell size should be y-flipped</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_fillers_y_flip_type</a></p>                              |
| add_reinforce_pg_critical_path_slack                 |                                                                                                                                                                                                                                                                         |

|                                                                  |                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                  | <p>Specifies a slack threshold, in nanoseconds, for critical nets during timing-driven reinforce_</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_reinforce_pg_critical_path_slack</a></p>                                               |
| add_reinforce_pg_irdrop_hierarchical_block                       |                                                                                                                                                                                                                                                                                                                                 |
|                                                                  | <p>Specifies hierarchical blocks to fix irdrop.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_reinforce_pg_irdrop_hierarchical_block</a></p>                                                                                            |
| add_reinforce_pg_irdrop_hierarchical_top_rail_analysis_directory |                                                                                                                                                                                                                                                                                                                                 |
|                                                                  | <p>Specifies the directory of rail analysis information of full chip for hierarchical blocks to fix ird</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_reinforce_pg_irdrop_hierarchical_top_rail_analysis_directory</a></p>              |
| add_reinforce_pg_respect_defined_nets                            |                                                                                                                                                                                                                                                                                                                                 |
|                                                                  | <p>Specifies reinforce_pg to respect user defined nets in file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_reinforce_pg_respect_defined_nets</a></p>                                                                                 |
| add_reinforce_pg_respect_routes                                  |                                                                                                                                                                                                                                                                                                                                 |
|                                                                  | <p>Specifies whether to honor pre-existing routes check during stripe generation</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none fixed fixed_and_clock all</p> <p><b>Default:</b> fixed_and_clock</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_reinforce_pg_respect_routes</a></p> |
| add_reinforce_pg_respect_stdcell_geometry                        |                                                                                                                                                                                                                                                                                                                                 |
|                                                                  | <p>Specifies the cell type to be respected in via generation.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none sequential all</p> <p><b>Default:</b> all</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_reinforce_pg_respect_stdcell_geometry</a></p>                                 |

|                                             |                                                                                                                                                                                                                                                 |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_rings_avoid_short                       |                                                                                                                                                                                                                                                 |
|                                             | <p>Specifies whether to avoid short violation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_avoid_short</a></p>                                  |
| add_rings_break_core_ring_io_list           |                                                                                                                                                                                                                                                 |
|                                             | <p>Specifies io instance names for which core ring wires are broken</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_break_core_ring_io_list</a></p> |
| add_rings_continue_on_no_selection          |                                                                                                                                                                                                                                                 |
|                                             | <p>Continues w/o selection</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_continue_on_no_selection</a></p>                                        |
| add_rings_detailed_log                      |                                                                                                                                                                                                                                                 |
|                                             | <p>Outputs detailed log file addring.log</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_detailed_log</a></p>                                      |
| add_rings_extend_block_ring_search_distance |                                                                                                                                                                                                                                                 |
|                                             | <p>Specifies blocking extension search distance</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_extend_block_ring_search_distance</a></p>          |
| add_rings_extend_core_ring_search_distance  |                                                                                                                                                                                                                                                 |
|                                             | <p>Specifies corering extension search distance</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_extend_core_ring_search_distance</a></p>           |

|                                         |                                                                                                                                                                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_rings_extend_merge_with_pre_wires   |                                                                                                                                                                                                                                           |
|                                         | <p>Specifies whether to merge with prerouted wires</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_extend_merge_with_pre_wires</a></p>        |
| add_rings_extend_over_row               |                                                                                                                                                                                                                                           |
|                                         | <p>Specifies whether to extend over row</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_extend_over_row</a></p>                              |
| add_rings_extend_search_nets            |                                                                                                                                                                                                                                           |
|                                         | <p>Specifies search net names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_extend_search_nets</a></p>                                      |
| add_rings_extend_stripe_search_distance |                                                                                                                                                                                                                                           |
|                                         | <p>Specifies stripe extension search distance</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_extend_stripe_search_distance</a></p>          |
| add_rings_gap_width_without_io          |                                                                                                                                                                                                                                           |
|                                         | <p>Specifies the max missing IO Pads area width between IO Pads</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_gap_width_without_io</a></p> |
| add_rings_ignore_rows                   |                                                                                                                                                                                                                                           |
|                                         | <p>Specifies whether to ignore all rows</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_ignore_rows</a></p>                                  |

|                                  |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_rings_max_via_size           |                                                                                                                                                                                                                                                                                                                                                           |
|                                  | <p>the maximum size of a crossover via</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_max_via_size</a></p>                                                                                                                                                   |
| add_rings_orthogonal_only        |                                                                                                                                                                                                                                                                                                                                                           |
|                                  | <p>vias connect to orthogonal targets only</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_orthogonal_only</a></p>                                                                                                                                            |
| add_rings_skip_shared_inner_ring |                                                                                                                                                                                                                                                                                                                                                           |
|                                  | <p>Specifies whether to skip crossing trunks</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> horizontal vertical none</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_skip_shared_inner_ring</a></p>                                                                               |
| add_rings_skip_via_on_pin        |                                                                                                                                                                                                                                                                                                                                                           |
|                                  | <p>prevent vias from being generated on the specified type of pins</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> pad block cover standardcell physicalpin</p> <p><b>Default:</b> standardcell</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_skip_via_on_pin</a></p>                                        |
| add_rings_skip_via_on_wire_shape |                                                                                                                                                                                                                                                                                                                                                           |
|                                  | <p>prevent vias from being generated on the specified wire shapes</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> blockring stripe followpin corewire blockwire iowire padring ring fillwire nos</p> <p><b>Default:</b> noshape</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_skip_via_on_wire_shape</a></p> |
| add_rings_spacing_from_block     |                                                                                                                                                                                                                                                                                                                                                           |

|                                          |                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Specifies the spacing from block</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_spacing_from_block</a></p>                                                                                                                                   |
| add_rings_split_long_via                 |                                                                                                                                                                                                                                                                                                                                               |
|                                          | <p>Split vias longer than &lt;threshold&gt; into smaller vias with specified &lt;step&gt; and bottom/left end &lt;offset&gt; and vertical/horizontal &lt;length&gt;</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 0 0 -1 -1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_split_long_via</a></p> |
| add_rings_stacked_via_bottom_layer       |                                                                                                                                                                                                                                                                                                                                               |
|                                          | <p>lef layer name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> bottomLayer</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_stacked_via_bottom_layer</a></p>                                                                                                                                       |
| add_rings_stacked_via_top_layer          |                                                                                                                                                                                                                                                                                                                                               |
|                                          | <p>lef layer name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> topLayer</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_stacked_via_top_layer</a></p>                                                                                                                                             |
| add_rings_target                         |                                                                                                                                                                                                                                                                                                                                               |
|                                          | <p>Specifies ring target</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> core_ring default first_ring pad_ring stripe</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_target</a></p>                                                                                |
| add_rings_via_using_exact_crossover_size |                                                                                                                                                                                                                                                                                                                                               |
|                                          | <p>generate partial vias of the exact crossover size</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_via_using_exact_crossover_size</a></p>                                                                                                       |

|                                     |                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_rings_wire_center_offset        |                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>offset value is from wire center to domain/core boundary</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_rings_wire_center_offset</a></p>                                                                                      |
| add_route_vias_auto                 |                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>enable auto via generation when design is loaded</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_route_vias_auto</a></p>                                                                                                       |
| add_route_vias_ndr_only             |                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>generate vias for non default rules only</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_route_vias_ndr_only</a></p>                                                                                                           |
| add_stripes_allow_jog               |                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>Specifies the allowed jog type.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none padcore_ring block_ring</p> <p><b>Default:</b> padcore_ring block_ring</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_allow_jog</a></p>                                            |
| add_stripes_allow_non_preferred_dir |                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>Specifies whether to allow non-preferred direction to jog to target</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none blocking corering padring stripe</p> <p><b>Default:</b> padring</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_allow_non_preferred_dir</a></p> |
| add_stripes_area                    |                                                                                                                                                                                                                                                                                                                          |

|                                      |                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Specify the rectangular areas or rectilinear areas or a mix of them, the correct value should ly ux uy&gt;   &lt;x1 y1 x2 y2 x3 y3 ...&gt;}</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_stripes_area</a></p>                                                  |
| add_stripes_blocks_without_same_net  |                                                                                                                                                                                                                                                                                                                                                      |
|                                      | <p>Specifies the stripe break point.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none block_ring selected_block unassigned_bump overlap_ringpin outer_outside_ringmacro blocks_without_same_net<br/> <b>Default:</b> none<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_stripes_blocks_without_same_net</a></p> |
| add_stripes_break_at                 |                                                                                                                                                                                                                                                                                                                                                      |
|                                      | <p>Specifies the stripe break point.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none block_ring selected_block unassigned_bump overlap_ringpin outer_outside_ringmacro blocks_without_same_net<br/> <b>Default:</b> none<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_stripes_break_at</a></p>                |
| add_stripes_continue_on_no_selection |                                                                                                                                                                                                                                                                                                                                                      |
|                                      | <p>Specifies whether to continue w/o selection</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_stripes_continue_on_no_selection</a></p>                                                                                                                             |
| add_stripes_detailed_log             |                                                                                                                                                                                                                                                                                                                                                      |
|                                      | <p>Specifies whether to output detailed log file addStripe.log</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_stripes_detailed_log</a></p>                                                                                                                         |
| add_stripes_domain_offset_from_core  |                                                                                                                                                                                                                                                                                                                                                      |



|                                            |                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Specifies whether to use global offset for power domain stripes</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_domain_offset_from_core</a></p>                                                    |
| add_stripes_extend_to_closest_target       |                                                                                                                                                                                                                                                                                                      |
|                                            | <p>Specifies the stripe to specified target</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> ring stripe same_dir_stripe area_boundary none</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_extend_to_closest_target</a></p> |
| add_stripes_extend_to_first_ring           |                                                                                                                                                                                                                                                                                                      |
|                                            | <p>Specifies whether to extend stripe antennas to closest ring</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_extend_to_first_ring</a></p>                                                           |
| add_stripes_ignore_block_check             |                                                                                                                                                                                                                                                                                                      |
|                                            | <p>Specifies whether to ignore block check</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_ignore_block_check</a></p>                                                                                 |
| add_stripes_ignore_block_ring_during_break |                                                                                                                                                                                                                                                                                                      |
|                                            | <p>Specifies whether to break stripe outside block</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_ignore_block_ring_during_break</a></p>                                                             |
| add_stripes_ignore_drc                     |                                                                                                                                                                                                                                                                                                      |
|                                            | <p>Specifies whether to ignore DRC check during stripe generation, it is always used for proto flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_ignore_drc</a></p>                                |

|                                        |                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_stripes_ignore_non_default_domains |                                                                                                                                                                                                                                                                                                                                                            |
|                                        | <p>Specifies whether to break at non-default domain</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_ignore_non_default_domains</a></p>                                                                                                                      |
| add_stripes_in_cell_allow_shift        |                                                                                                                                                                                                                                                                                                                                                            |
|                                        | <p>Specifies to allow the created pillar inside cell can be shifted when meeting DRC</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_in_cell_allow_shift</a></p>                                                                                            |
| add_stripes_in_cell_only               |                                                                                                                                                                                                                                                                                                                                                            |
|                                        | <p>Specifies whether to generate stripes only inside selected or specified cell</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_in_cell_only</a></p>                                                                                                        |
| add_stripes_keep_pitch_after_snap      |                                                                                                                                                                                                                                                                                                                                                            |
|                                        | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_keep_pitch_after_snap</a></p>                                                                                                                                                                                   |
| add_stripes_mask_color_balance         |                                                                                                                                                                                                                                                                                                                                                            |
|                                        | <p>same_color would correspond to interlace with groups and alternate_color would correspond to interlace within group</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> same_color alternate_color none</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_mask_color_balance</a></p> |
| add_stripes_max_extension_distance     |                                                                                                                                                                                                                                                                                                                                                            |

|                                          |                                                                                                                                                                                                                                                                                              |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Specifies maximum extension distance(default=max integer) when -extend_to_closest_target specified.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 2.14748e+09</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_max_extension_distance</a></p> |
| add_stripes_max_via_size                 |                                                                                                                                                                                                                                                                                              |
|                                          | <p>the maximum size of a crossover via, the correct value should be {shape width% height% target_penetration%}</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_max_via_size</a></p>            |
| add_stripes_merge_with_all_layers        |                                                                                                                                                                                                                                                                                              |
|                                          | <p>Specifies whether to merge a stripe on all layers</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_merge_with_all_layers</a></p>                                                            |
| add_stripes_mesh_via                     |                                                                                                                                                                                                                                                                                              |
|                                          | <p>use mesh via to reduce memory</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_mesh_via</a></p>                                                                                             |
| add_stripes_offset_from_core             |                                                                                                                                                                                                                                                                                              |
|                                          | <p>Specifies whether to offset first or last stripe from the core area</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_offset_from_core</a></p>                                                |
| add_stripes_opt_stripe_for_routing_track |                                                                                                                                                                                                                                                                                              |
|                                          | <p>move stripe location to preserve routing resource</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> shift none</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_opt_stripe_for_routing_track</a></p>                |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_stripes_orthogonal_offset                 |                                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>Specifies the offsets for edges orthogonally, the correct value should be {{{edge1 orthogonal_offset1} {edge2 edge3 orthogonal_offset2} {edge4 edge5 ... orthogonal_offset3} [all orthogonal_offset]}}</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> {none}</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_orthogonal_offset</a></p> |
| add_stripes_orthogonal_only                   |                                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>vias connect to orthogonal targets only</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_orthogonal_only</a></p>                                                                                                                                                                      |
| add_stripes_over_row_extension                |                                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>Specifies whether to extend the stripe to cover followpin at the row end</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_over_row_extension</a></p>                                                                                                                                 |
| add_stripes_partial_set_through_domain        |                                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>Specifies whether to go over power domain if domain contains the specified net</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_partial_set_through_domain</a></p>                                                                                                                   |
| add_stripes_remove_floating_stapling          |                                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>remove floating stapling stripe if there is no via connection</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_remove_floating_stapling</a></p>                                                                                                                                       |
| add_stripes_remove_floating_stripe_over_block |                                                                                                                                                                                                                                                                                                                                                                                       |

|                                       |                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>Specifies whether to remove stripe fragments start and end inside same block or ring macro</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_remove_floating_stripe_over_block</a></p>                                        |
| add_stripes_remove_stripe_under_ring  |                                                                                                                                                                                                                                                                                                                              |
|                                       | <p>Specifies whether to remove those segments under rings.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_remove_stripe_under_ring</a></p>                                                                                   |
| add_stripes_respect_routes            |                                                                                                                                                                                                                                                                                                                              |
|                                       | <p>Specifies whether to honor pre-existing routes check during stripe generation</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none   fixed fixed_and_clock all</p> <p><b>Default:</b> fixed_and_clock</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_respect_routes</a></p> |
| add_stripes_route_over_rows_only      |                                                                                                                                                                                                                                                                                                                              |
|                                       | <p>Specifies whether to route stripe over rows within the boundary</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_route_over_rows_only</a></p>                                                                               |
| add_stripes_rows_without_stripes_only |                                                                                                                                                                                                                                                                                                                              |
|                                       | <p>Specifies whether to generate stripes over rows w/o stripes</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_rows_without_stripes_only</a></p>                                                                              |
| add_stripes_same_size_stack_vias      |                                                                                                                                                                                                                                                                                                                              |
|                                       | <p>generate same-sized stacked vias</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_same_size_stack_vias</a></p>                                                                                                              |
| add_stripes_skip_via_on_pin           |                                                                                                                                                                                                                                                                                                                              |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>prevent vias from being generated on the specified type of pins</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> pad block cover standardcell physicalpin</p> <p><b>Default:</b> standardcell</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_skip_via_on_pin</a></p>                                                                               |
| add_stripes_skip_via_on_wire_shape |                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                    | <p>prevent vias from being generated on the specified wire shapes</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> blockring stripe followpin corewire blockwire iowire padding ring fillwire nos</p> <p><b>Default:</b> noshape</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_skip_via_on_wire_shape</a></p>                                        |
| add_stripes_spacing_from_block     |                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                    | <p>Specifies the spacing from block</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_spacing_from_block</a></p>                                                                                                                                                                                      |
| add_stripes_spacing_type           |                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                    | <p>Specify the type of intra spacing for add_stripes -spacing</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> edge_to_edge center_to_center</p> <p><b>Default:</b> edge_to_edge</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_spacing_type</a></p>                                                                                                  |
| add_stripes_split_long_via         |                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                    | <p>Split vias longer than &lt;threshold&gt; into smaller vias with specified &lt;step&gt; and bottom/left en &lt;offset&gt; and vertical/horizontal &lt;length&gt;, the correct value should be {threshold step offset</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 0 0 -1 -1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_split_long_via</a></p> |
| add_stripes_split_vias             |                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                                |                                                                                                                                                                                                                                                             |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>create multiple partial vias</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_split_vias</a></p>                                                           |
| add_stripes_split_wire_spacing                 |                                                                                                                                                                                                                                                             |
|                                                | <p>Specifies the spacing between each split wire</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_split_wire_spacing</a></p>                                  |
| add_stripes_split_wire_weight                  |                                                                                                                                                                                                                                                             |
|                                                | <p>Deletes fewest IO wires to accommodate approximately n of 10 split stripe wires</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_split_wire_weight</a></p> |
| add_stripes_split_wire_width                   |                                                                                                                                                                                                                                                             |
|                                                | <p>Specifies the width of each split wire</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_split_wire_width</a></p>                                           |
| add_stripes_stacked_via_bottom_layer           |                                                                                                                                                                                                                                                             |
|                                                | <p>lef layer name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> bottomLayer</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_stacked_via_bottom_layer</a></p>                                                   |
| add_stripes_stacked_via_top_layer              |                                                                                                                                                                                                                                                             |
|                                                | <p>lef layer name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> topLayer</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_stacked_via_top_layer</a></p>                                                         |
| add_stripes_stapling_extend_to_minimum_spacing |                                                                                                                                                                                                                                                             |

|                                        |                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Specified to automatically extend the adjacent stapling as long as enough to satisfy the end minimum spacing with DRC free</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_stapling_extend_to_minimum_spacing</a></p> |
| add_stripes_stapling_nets_style        |                                                                                                                                                                                                                                                                                                                         |
|                                        | <p>Specify the stapling style if multiple nets are specified</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> end_to_end side_to_side side_to_side_full_nets</p> <p><b>Default:</b> end_to_end</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_stapling_nets_style</a></p>  |
| add_stripes_stapling_shift             |                                                                                                                                                                                                                                                                                                                         |
|                                        | <p>shift stapling stripe to avoid the drc automatically</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_stapling_shift</a></p>                                                                                            |
| add_stripes_stop_at_closest_target     |                                                                                                                                                                                                                                                                                                                         |
|                                        | <p>Specifies the stripe to specified target when over_pins is enabled.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none block_ring core_ring stripe</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_stop_at_closest_target</a></p>         |
| add_stripes_stop_at_last_wire_for_area |                                                                                                                                                                                                                                                                                                                         |
|                                        | <p>Specifies whether to trim back stripe antennas to the closest wire or pin of the same net</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_stop_at_last_wire_for_area</a></p>                                          |
| add_stripes_stripe_min_length          |                                                                                                                                                                                                                                                                                                                         |
|                                        | <p>Specifies the minimum stripe length</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> stripe_width</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_stripe_min_length</a></p>                                                                                                |



|                                         |                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_stripes_stripe_min_width            |                                                                                                                                                                                                                                                                                                                               |
|                                         | <p>Specifies the min width of remaining stripe after being trimmed</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_stripe_min_width</a></p>                                                                                    |
| add_stripes_switch_cell_name            |                                                                                                                                                                                                                                                                                                                               |
|                                         | <p>Specifies the switch cells to be treated as rings</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_switch_cell_name</a></p>                                                                                                 |
| add_stripes_switch_layer_overlap_length |                                                                                                                                                                                                                                                                                                                               |
|                                         | <p>Specifies the overlap distance used for the via between a stripe layer and the layer being s to</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_switch_layer_overlap_length</a></p>                                         |
| add_stripes_trim_antenna_back_to_shape  |                                                                                                                                                                                                                                                                                                                               |
|                                         | <p>Specifies the distance between the end of antenna and blocked target</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none block_ring core_ring pad_ring stripe</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_trim_antenna_back_to_shape</a></p> |
| add_stripes_trim_antenna_max_distance   |                                                                                                                                                                                                                                                                                                                               |
|                                         | <p>Specifies the distance between the end of antenna and blocked target</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_stripes_trim_antenna_max_distance</a></p>                                                                      |
| add_stripes_trim_stripe                 |                                                                                                                                                                                                                                                                                                                               |

|                                            |                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Specifies if stripe is trimmed by the specified shape.<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none design_boundary core_boundary<br/> <b>Default:</b> none<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_stripes_trim_stripe</a></p> |
| add_stripes_use_exact_spacing              |                                                                                                                                                                                                                                                                                 |
|                                            | <p>Specifies whether to use exact spacing<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_stripes_use_exact_spacing</a></p>                                                                      |
| add_stripes_use_point_to_point_route       |                                                                                                                                                                                                                                                                                 |
|                                            | <p>Specifies whether to use exact spacing<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_stripes_use_point_to_point_route</a></p>                                                               |
| add_stripes_use_stripe_width               |                                                                                                                                                                                                                                                                                 |
|                                            | <p>Specifies whether to use stripe width<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_stripes_use_stripe_width</a></p>                                                                         |
| add_stripes_via_using_exact_crossover_size |                                                                                                                                                                                                                                                                                 |
|                                            | <p>generate partial vias of the exact crossover size<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_stripes_via_using_exact_crossover_size</a></p>                                               |
| add_stripes_width_file                     |                                                                                                                                                                                                                                                                                 |
|                                            | <p>Specify the values of recommended width in the file name<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">add_stripes_width_file</a></p>                                                            |
| add_target_pg_allow_weak_connect           |                                                                                                                                                                                                                                                                                 |

|                                      |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Decreate specified pillar number to avoid drc</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_target_pg_allow_weak_connect</a></p>                                                                                                   |
| add_target_pg_max_extension_distance |                                                                                                                                                                                                                                                                                                                                |
|                                      | <p>Specify the max distance that target PG pin can extend to target layer</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 2.14748e+09</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_target_pg_max_extension_distance</a></p>                                                              |
| add_target_pg_pins                   |                                                                                                                                                                                                                                                                                                                                |
|                                      | <p>Specify the cell pins to be connected</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_target_pg_pins</a></p>                                                                                                                          |
| add_target_pg_pins_group_distance    |                                                                                                                                                                                                                                                                                                                                |
|                                      | <p>Specify the side_to_side distance of pins to be treated as a group during connecting target pins</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_target_pg_pins_group_distance</a></p>                                              |
| add_target_pg_respect_routes         |                                                                                                                                                                                                                                                                                                                                |
|                                      | <p>Specifies whether to honor pre-existing routes check during stripe generation</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none   fixed fixed_and_clock all</p> <p><b>Default:</b> fixed_and_clock</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_target_pg_respect_routes</a></p> |
| add_target_pg_share_resource         |                                                                                                                                                                                                                                                                                                                                |
|                                      | <p>New added PG connect to the existing PG at middle layer</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_target_pg_share_resource</a></p>                                                                                              |

|                              |                                                                                                                                                                                                                                               |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_tieoffs_cells            |                                                                                                                                                                                                                                               |
|                              | <p>Set tie cell lists for set_db command</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_tieoffs_cells</a></p>                                          |
| add_tieoffs_create_hports    |                                                                                                                                                                                                                                               |
|                              | <p>Add tie-cells in different hierarchy than tie-off pins when possible</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_tieoffs_create_hports</a></p>  |
| add_tieoffs_honor_dont_touch |                                                                                                                                                                                                                                               |
|                              | <p>honor dont_touch attribute on nets, insts, cells and modules</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_tieoffs_honor_dont_touch</a></p>       |
| add_tieoffs_honor_dont_use   |                                                                                                                                                                                                                                               |
|                              | <p>honor dont_use attribute on cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_tieoffs_honor_dont_use</a></p>                                   |
| add_tieoffs_max_distance     |                                                                                                                                                                                                                                               |
|                              | <p>distance between tie-cell and tie-pins should be less than given value</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_tieoffs_max_distance</a></p> |
| add_tieoffs_max_fanout       |                                                                                                                                                                                                                                               |
|                              | <p>the number of tie-pins a tie-net can drive. A '0' means no-limit</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_tieoffs_max_fanout</a></p>              |

|                                          |                                                                                                                                                                                                                                                                        |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_tieoffs_module_prevention            |                                                                                                                                                                                                                                                                        |
|                                          | <p>add tie cell for module instantiation port</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_tieoffs_module_prevention</a></p>                                                 |
| add_tieoffs_prefix                       |                                                                                                                                                                                                                                                                        |
|                                          | <p>Prefix of tie cell instances</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_tieoffs_prefix</a></p>                                                                           |
| add_tieoffs_report_hports                |                                                                                                                                                                                                                                                                        |
|                                          | <p>Report created ports when add_tieoffs_create_hports is set</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_tieoffs_report_hports</a></p>                                     |
| add_well_taps_abut_boundary_tap_distance |                                                                                                                                                                                                                                                                        |
|                                          | <p>Specifies boundary cells and their tap distance when they are vertically abutted</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_abut_boundary_tap_distance</a></p> |
| add_well_taps_antenna_tap_break_cell     |                                                                                                                                                                                                                                                                        |
|                                          | <p>Specifies cells can break antenna tap</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_antenna_tap_break_cell</a></p>                                                |
| add_well_taps_antenna_tap_cell           |                                                                                                                                                                                                                                                                        |
|                                          | <p>Specifies normal antenna tap cell</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_antenna_tap_cell</a></p>                                                          |

|                                               |                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_well_taps_antenna_tap_left_cell           |                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>Specifies antenna tap cell for left edge</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_antenna_tap_left_cell</a></p>                                                                                               |
| add_well_taps_antenna_tap_pitch               |                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>Specifies the pitch of antenna tap cell</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_antenna_tap_pitch</a></p>                                                                                                   |
| add_well_taps_antenna_tap_right_cell          |                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>Specifies antenna tap cell for right edge</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_antenna_tap_right_cell</a></p>                                                                                             |
| add_well_taps_antenna_tap_right_top_edge_cell |                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>Specifies antenna tap cell for right top edge</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_antenna_tap_right_top_edge_cell</a></p>                                                                                |
| add_well_taps_avoid_abutment                  |                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>Specifies whether the inserted taps should avoid horizontal and vertical abutment</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_avoid_abutment</a></p>                                                            |
| add_well_taps_avoid_vertical_well_abutment    |                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>Specifies which type of well-tap cells to be forbidden from vertical abutment</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none ntap ptap all_tap</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_avoid_vertical_well_abutment</a></p> |

|                                       |                                                                                                                                                                                                                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_well_taps_block_boundary_only     |                                                                                                                                                                                                                                                       |
|                                       | <p>Specifies whether inserting termination cells on the core boundary</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_block_boundary_only</a></p>    |
| add_well_taps_bottom_tap_cell         |                                                                                                                                                                                                                                                       |
|                                       | <p>Cell to be used as bottom tap for add_endcaps boundary cell insertion flow</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_bottom_tap_cell</a></p> |
| add_well_taps_bottom_termination_cell |                                                                                                                                                                                                                                                       |
|                                       | <p>Specifies the list of bottom termination cells to be used</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_bottom_termination_cell</a></p>          |
| add_well_taps_cell                    |                                                                                                                                                                                                                                                       |
|                                       | <p>Cell to be used as well_taps</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_cell</a></p>                                                          |
| add_well_taps_channel_offset          |                                                                                                                                                                                                                                                       |
|                                       | <p>Specifies the offset value between taps and blockage in channel</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_channel_offset</a></p>            |
| add_well_taps_check_channel           |                                                                                                                                                                                                                                                       |
|                                       | <p>Specifies whether checking blockage channel</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_check_channel</a></p>                                 |

|                                              |                                                                                                                                                                                                                                                                       |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_well_taps_column_cells                   |                                                                                                                                                                                                                                                                       |
|                                              | <p>Specifies the list of cells which can be replaced by termination cell</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_column_cells</a></p>                         |
| add_well_taps_create_rows                    |                                                                                                                                                                                                                                                                       |
|                                              | <p>Create rows for welltap cell techSites which do not have rows in the floor plan</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_create_rows</a></p>               |
| add_well_taps_disable_check_zone_at_boundary |                                                                                                                                                                                                                                                                       |
|                                              | <p>disable check zone at bounadry</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none vdd vss both</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_disable_check_zone_at_boundary</a></p> |
| add_well_taps_in_row_offset                  |                                                                                                                                                                                                                                                                       |
|                                              | <p>Specifies the distance between the first wellTap in a row and the start of the row</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_in_row_offset</a></p>          |
| add_well_taps_insert_cells                   |                                                                                                                                                                                                                                                                       |
|                                              | <p>Insert well-taps based on cell, layer and inertval</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_insert_cells</a></p>                                            |
| add_well_taps_replace_intersection_cell      |                                                                                                                                                                                                                                                                       |
|                                              | <p>Cell to be used for replacing intersection break cell</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                             |



|                                         |                                                                                                                                                                                                                                                                |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_well_taps_rule                      |                                                                                                                                                                                                                                                                |
|                                         | <p>Specifies the well latch-up rule</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_rule</a></p>                                                              |
| add_well_taps_site_offset               |                                                                                                                                                                                                                                                                |
|                                         | <p>Specifies the offset value in number of sites that vertical taps need to be aware of</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_site_offset</a></p>        |
| add_well_taps_tap_function_cells        |                                                                                                                                                                                                                                                                |
|                                         | <p>Specifies the list of tap function cells to be used</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_tap_function_cells</a></p>                              |
| add_well_taps_tap_termination_alignment |                                                                                                                                                                                                                                                                |
|                                         | <p>Specifies whether to align well-taps and termination cells</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_tap_termination_alignment</a></p>               |
| add_well_taps_termination_align         |                                                                                                                                                                                                                                                                |
|                                         | <p>Specifies the termination cell align</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> left right center</p> <p><b>Default:</b> left</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_termination_align</a></p> |
| add_well_taps_termination_cells         |                                                                                                                                                                                                                                                                |
|                                         | <p>Specifies the list of termination cells to be used</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_termination_cells</a></p>                                |

|                                         |                                                                                                                                                                                                                                                               |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_well_taps_top_tap_cell              |                                                                                                                                                                                                                                                               |
|                                         | <p>Cell to be used as top tap for add_endcaps boundary cell insertion flow</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_top_tap_cell</a></p>               |
| add_well_taps_top_termination_cell      |                                                                                                                                                                                                                                                               |
|                                         | <p>Specifies the list of top termination cells to be used</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_top_termination_cell</a></p>                        |
| add_well_taps_vertical_boundary_spacing |                                                                                                                                                                                                                                                               |
|                                         | <p>Specifies the spacing between center of tap cell and vertical boundary</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_vertical_boundary_spacing</a></p> |
| add_well_taps_well_cut_cells            |                                                                                                                                                                                                                                                               |
|                                         | <p>Specifies the list of cells with the well cuts</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">add_well_taps_well_cut_cells</a></p>                                      |
| ai_server_url                           |                                                                                                                                                                                                                                                               |
|                                         | <p>the server address which ai_db_query will connect to</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                      |
| analysis_views                          |                                                                                                                                                                                                                                                               |
|                                         | <p>Returns the information about the analysis views in the design.</p> <p><b>Type:</b> <a href="#">obj(analysis_view)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                               |
| aocv_library                            |                                                                                                                                                                                                                                                               |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | <p>To specify AOCV library.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">aocv_library</a></p>                                                                                                                                                                                                                                    |
| apply_booth_encoding                             |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                  | <p>Apply booth encoding by using user specified method (for all instances).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> nonbooth auto_bitwidth auto_togglerate manual inherited</p> <p><b>Default:</b> auto_bitwidth</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">apply_booth_encoding</a></p>                                                                                |
| assemble_design_generic_hier                     |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                  | <p>Read DEF for generic hierarchy.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assemble_design_generic_hier</a></p>                                                                                                                                                                                                            |
| assign_pins_advanced_node_rule_support           |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                  | <p>If you use this parameter, all the pin related command will use design object shapes for determining pin location which will give accurate result but will have larger runtime.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_advanced_node_rule_support</a></p>                                                   |
| assign_pins_allow_non_ndr_net_pins_on_ndr_tracks |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                  | <p>A true value allows general pins to be placed on ndr tracks if available. Alternately, if the value is set to false, the ndr-track rule is followed strictly and the net's pins are placed on matching ndr tracks only.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_allow_non_ndr_net_pins_on_ndr_tracks</a></p> |
| assign_pins_allow_unconnected_in_abutted_edge    |                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>If you use this parameter, unconnected partition pin can be placed in abutted edge</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">assign_pins_allow_unconnected_in_abutted_edge</a></p>                                                                                                                                                                                                                                                             |
| assign_pins_blocked_boundary_macro_distance |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                             | <p>This option blocks automatic pin assignment, from putting pins on the part of partition boundary facing block on layers with OBS, in the places where channel width between partition boundary and block is less than the value specified. For channel width below specified value, block a on boundary facing this channel on layers with OBS.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 1.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">assign_pins_blocked_boundary_macro_distance</a></p> |
| assign_pins_bottom_layer                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                             | <p>Specifies the minimum metal layer name for partition/block pin assignment</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> 2<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">assign_pins_bottom_layer</a></p>                                                                                                                                                                                                                                                                                            |
| assign_pins_edit_in_batch                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                             | <p>To enable/disable pin-editing in batch mode.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">assign_pins_edit_in_batch</a></p>                                                                                                                                                                                                                                                                                                                      |
| assign_pins_force_abutment_with_fixed       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                             | <p>For fixed abutted pins's connected pin, ignore soft constraints checks</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">assign_pins_force_abutment_with_fixed</a></p>                                                                                                                                                                                                                                                                                 |
| assign_pins_inside_pin_search_distance      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                             | <p>To control the search distance for legalizing the internal pins</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 10<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">assign_pins_inside_pin_search_distance</a></p>                                                                                                                                                                                                                                                                                          |

|                                                         |                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| assign_pins_max_channel_width_as_abutted                | <p>If you set this parameter value then all channel less than equal to specified value will be considered as abutted and abutment rules will be applied</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_max_channel_width_as_abutted</a></p> |
| assign_pins_max_distance_pair                           | <p>Ignores moving corresponding connected pins of 2-pin-connection nets where distance (in microns) between the pins is equal or greater than distance specified</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 50.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_max_distance_pair</a></p>  |
| assign_pins_off_stripe                                  | <p>Prevents the generation of pins on metal layers with respect to the power and ground stripe</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> below all none</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_off_stripe</a></p>                                   |
| assign_pins_pin_to_stripe_distance                      | <p>Layer specific, keep distance (in microns) from pin to stripe</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_pin_to_stripe_distance</a></p>                                                                                               |
| assign_pins_pin_to_via_distance_non_preferred_direction | <p>Layer specific, keep distance (in microns) from pin to via-stack</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_pin_to_via_distance_non_preferred_direction</a></p>                                                                       |
| assign_pins_pin_to_via_distance_preferred_direction     |                                                                                                                                                                                                                                                                                                                                             |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Layer specific, keep distance (in microns) from pin to via-stack from the partition fence boundary</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_pin_to_via_distance_preferred_direction</a></p>                                                                                                                                                                                                                                                                                                               |
| assign_pins_promoted_macro_bottom_layer        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                | <p>Specifies the minimum metal layer name for promoting macro pins</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 31</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_promoted_macro_bottom_layer</a></p>                                                                                                                                                                                                                                                                                                                                                              |
| assign_pins_promoted_macro_top_layer           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                | <p>Specifies the maximum metal layer name for promoting macro pins</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 31</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_promoted_macro_top_layer</a></p>                                                                                                                                                                                                                                                                                                                                                                 |
| assign_pins_restricted_boundary_macro_distance |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                | <p>This option blocks automatic pin assignment, from putting pins on the part of partition boundary facing block on layers with OBS, in the places where channel width between partition boundary and block is less than the value specified. For channel width below specified value: OBS at the layer, channel length and channel width considered to allow suitable number of pins or boundary facing this channel.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 10.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_restricted_boundary_macro_distance</a></p> |
| assign_pins_strict_abutment                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                | <p>To relax abutment violations, for placing multi partition pin of a net and non neighbor pins on abutted edges</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_strict_abutment</a></p>                                                                                                                                                                                                                                                                                                                            |
| assign_pins_top_layer                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                            |                                                                                                                                                                                                                                                                          |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Specifies the maximum metal layer name for partition/block pin assignment</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 31</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_top_layer</a></p>                             |
| assign_pins_use_common_color_engine_for_io |                                                                                                                                                                                                                                                                          |
|                                            | <p>assignIoPins command will use coloring engine api to assign mask-color to pins.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">assign_pins_use_common_color_engine_for_io</a></p> |
| attributes                                 |                                                                                                                                                                                                                                                                          |
|                                            | <p>Short-cut for [get_db obj_types .attributes]</p> <p><b>Type:</b> <a href="#">obj(attribute)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                 |
| auto_file_dir                              |                                                                                                                                                                                                                                                                          |
|                                            | <p>Represents the top-level directory to store files/sub-directories 'auto-generated' by the tool.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ./</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">auto_file_dir</a></p>               |
| auto_file_prefix                           |                                                                                                                                                                                                                                                                          |
|                                            | <p>Represents the prefix to be applied to all files/sub-directories 'auto-generated' by the tool.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">auto_file_prefix</a></p>             |
| auto_partition                             |                                                                                                                                                                                                                                                                          |
|                                            | <p>Create temporary hierarchies based on connectivity to guide optimization.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">auto_partition</a></p>                                    |
| auto_ungroup                               |                                                                                                                                                                                                                                                                          |

|                                 |                                                                                                                                                                                                                                                                            |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>Automatically ungroup user hierarchies to improve optimization.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none area timing both</p> <p><b>Default:</b> both</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">auto_ungroup</a></p> |
| auto_ungroup_max_threshold      |                                                                                                                                                                                                                                                                            |
|                                 | <p>To allow ungroup for the instance according to max threshold.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">auto_ungroup_max_threshold</a></p>                                          |
| auto_voltage_scaling            |                                                                                                                                                                                                                                                                            |
|                                 | <p>To enable library scaling when libraries with different voltages mentioned in single list.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                             |
| avoid_tied_inputs               |                                                                                                                                                                                                                                                                            |
|                                 | <p>Avoid creation of tied input pins of mapped gates in mapping if possible.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">avoid_tied_inputs</a></p>                                  |
| bank_based_multibit_inferencing |                                                                                                                                                                                                                                                                            |
|                                 | <p>Removes map to multibit register constraint in predefined multibit.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">bank_based_multibit_inferencing</a></p>                          |
| base_cells                      |                                                                                                                                                                                                                                                                            |
|                                 | <p>All the base_cells from .lib, Verilog stubs, LEF or OA.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">base_cell</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                        |
| base_pins                       |                                                                                                                                                                                                                                                                            |



|                                        |                                                                                                                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>All the base_pins in all the base_cells.</p> <p><b>Type:</b> <a href="#">obj(base_pin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                      |
| bit_blasted_port_style                 |                                                                                                                                                                                                                                                                          |
|                                        | <p>the default bus bit blasted port name as defined by this tcl global</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> %s_%d</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">bit_blasted_port_style</a></p>                               |
| boundaries                             |                                                                                                                                                                                                                                                                          |
|                                        | <p>Short-cut to all the boundary objects (fence, region, etc.) in the design.</p> <p><b>Type:</b> <a href="#">obj(boundary)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                    |
| boundary_optimize_constant_hpins       |                                                                                                                                                                                                                                                                          |
|                                        | <p>Allow constant propagation through hierarchical boundary pins.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">boundary_optimize_constant_hpins</a></p>                             |
| boundary_optimize_equal_opposite_hpins |                                                                                                                                                                                                                                                                          |
|                                        | <p>Allow connection of hierarchical boundary pins through equal/opposite boundary pins.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">boundary_optimize_equal_opposite_hpins</a></p> |
| boundary_optimize_feedthrough_hpins    |                                                                                                                                                                                                                                                                          |
|                                        | <p>Allow routing of feedthrough boundary pins around a hierarchical instance.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">boundary_optimize_feedthrough_hpins</a></p>              |
| boundary_optimize_invert_hpins         |                                                                                                                                                                                                                                                                          |

|                                            |                                                                                                                                                                                                                                                                                         |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Allow inversion of hierarchical boundary pins.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">boundary_optimize_invert_hpins</a></p>                                                             |
| boundary_optimize_invert_hpins_rename_nets |                                                                                                                                                                                                                                                                                         |
|                                            | <p>Rename nets driven by inverted hierarchical boundary output pins.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">boundary_optimize_invert_hpins_rename_nets</a></p>                               |
| budget_abutted                             |                                                                                                                                                                                                                                                                                         |
|                                            | <p>Assumes all partitions are abutted</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_abutted</a></p>                                                                                         |
| budget_accumulated                         |                                                                                                                                                                                                                                                                                         |
|                                            | <p>Write both interface and flat constraints for nested partitions under if {budget_use_accumul 1} {} else {} conditions.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_accumulated</a></p> |
| budget_boundary_condition_template_path    |                                                                                                                                                                                                                                                                                         |
|                                            | <p>Directory path containing model files to be used with option useBoundaryCondition templa</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_boundary_condition_template_path</a></p>           |
| budget_buffer_delay_lib_cell               |                                                                                                                                                                                                                                                                                         |
|                                            | <p>user defined buffer which should be used for virtual buffer adjustment</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_buffer_delay_lib_cell</a></p>                                        |
| budget_buffer_delay_selection_effort       |                                                                                                                                                                                                                                                                                         |

|                             |                                                                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | controls the tool effort for calculating the buffer delay adjustment internally<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> low high none<br><b>Default:</b> none<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_buffer_delay_selection_effort</a> |
| budget_ccd                  |                                                                                                                                                                                                                                                                                 |
|                             | Generates scripts for verification of time budgeting top and block constraints against pre-bu chip constraints using the CCD<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_ccd</a>                  |
| budget_constant_model       |                                                                                                                                                                                                                                                                                 |
|                             | Specifies the type of timing model created<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_constant_model</a>                                                                                          |
| budget_distribute_mmmc      |                                                                                                                                                                                                                                                                                 |
|                             | Distribute multiple analysis views over multiple sessions<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_distribute_mmmc</a>                                                                         |
| budget_dont_write_wire_load |                                                                                                                                                                                                                                                                                 |
|                             | write set_load with -wire_load information<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_dont_write_wire_load</a>                                                                                   |
| budget_drive_constraint     |                                                                                                                                                                                                                                                                                 |
|                             | Driving boundary constraint to be written on block input port<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> drive drive_cell input_transition<br><b>Default:</b> drive_cell<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_drive_constraint</a>      |

|                                      |                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| budget_fix_top_level_paths           |                                                                                                                                                                                                                                                                                                                                                                            |
|                                      | <p>Slack value for which the top-level timing budget is fixed, and remaining budget is proportionally distributed only for partitions</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none all negative_only positive_only</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_fix_top_level_paths</a></p> |
| budget_handle_complex_sdc            |                                                                                                                                                                                                                                                                                                                                                                            |
|                                      | <p>handle sdc files with complex constructs and regular expressions in push down</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_handle_complex_sdc</a></p>                                                                                                                      |
| budget_honor_port_boundary_condition |                                                                                                                                                                                                                                                                                                                                                                            |
|                                      | <p>Push down chip IO constraints to block boundary pins which are directly connected to chip</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_honor_port_boundary_condition</a></p>                                                                                                |
| budget_honor_port_delays             |                                                                                                                                                                                                                                                                                                                                                                            |
|                                      | <p>Push down chip IO SID SOD values to block IO if the two are directly connected and abutting</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_honor_port_delays</a></p>                                                                                                          |
| budget_honor_report_timing_format    |                                                                                                                                                                                                                                                                                                                                                                            |
|                                      | <p>honor set_global report_timing_format in justification reports</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_honor_report_timing_format</a></p>                                                                                                                              |
| budget_ignore_dont_touch             |                                                                                                                                                                                                                                                                                                                                                                            |

|                                      |                                                                                                                                                                                                                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Specifies the handling of don't touch objects</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_ignore_dont_touch</a></p>                                                        |
| budget_include_latency               |                                                                                                                                                                                                                                                                             |
|                                      | <p>Specifies whether the clock latency is included in the set_input_delay and set_output_delay constraints</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_include_latency</a></p> |
| budget_include_wire_loads_in_lib     |                                                                                                                                                                                                                                                                             |
|                                      | <p>Include the wire loads in library model pin capacitance</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_include_wire_loads_in_lib</a></p>                                      |
| budget_input_load                    |                                                                                                                                                                                                                                                                             |
|                                      | <p>write set_load for input pins</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_input_load</a></p>                                                                               |
| budget_justify                       |                                                                                                                                                                                                                                                                             |
|                                      | <p>controls the type of justification</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> delay exception lib_arc boundary_condition all</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_justify</a></p>      |
| budget_keep_pin_list_for_block_ports |                                                                                                                                                                                                                                                                             |
|                                      | <p>Preserve all fan-in and fan-out lists</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_keep_pin_list_for_block_ports</a></p>                                                     |

|                                       |                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| budget_latency_on_clocks              |                                                                                                                                                                                                                                                                                                                   |
|                                       | <p>write set_clock_latency on clocks</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_latency_on_clocks</a></p>                                                                                                           |
| budget_local_latency                  |                                                                                                                                                                                                                                                                                                                   |
|                                       | <p>Write user specific set_clock_latency constraints, provided through set_partition_user_constraints_file command, in partition constraint files</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_local_latency</a></p> |
| budget_local_uncertainty              |                                                                                                                                                                                                                                                                                                                   |
|                                       | <p>Use local uncertainty specified through set_partition_user_constraints_file command for partition constraints</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_local_uncertainty</a></p>                              |
| budget_make_negative_input_delay_zero |                                                                                                                                                                                                                                                                                                                   |
|                                       | <p>To be backward compatible where negative set input delays were made zero</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_make_negative_input_delay_zero</a></p>                                                       |
| budget_make_reduced_timing_graph      |                                                                                                                                                                                                                                                                                                                   |
|                                       | <p>Enable running ART before Budgeting.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_make_reduced_timing_graph</a></p>                                                                                               |
| budget_master_clone                   |                                                                                                                                                                                                                                                                                                                   |

|                                               |                                                                                                                                                                                                                                                                |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | controls the tool behaviour for master clones<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> master_only merged unique_view_per_hinst<br><b>Default:</b> master_only<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_master_clone</a> |
| budget_no_false_paths_for_unconstrained_ports |                                                                                                                                                                                                                                                                |
|                                               | do not create false paths for unconstrained ports<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_no_false_paths_for_unconstrained_ports</a>                                         |
| budget_no_hold_view                           |                                                                                                                                                                                                                                                                |
|                                               | Disable hold view budgeting<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_no_hold_view</a>                                                                                         |
| budget_no_setup_view                          |                                                                                                                                                                                                                                                                |
|                                               | Disable setup view budgeting<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_no_setup_view</a>                                                                                       |
| budget_report_dir                             |                                                                                                                                                                                                                                                                |
|                                               | Alternate directory name for justify reports when -justify is given with create_timing_budget<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> budget_justify<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_report_dir</a>              |
| budget_report_negative_slack_on_ports         |                                                                                                                                                                                                                                                                |
|                                               | save warnings for ports having slack less than the specified value<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> -1e+06<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">budget_report_negative_slack_on_ports</a>                             |
| budget_report_or_update_budget                |                                                                                                                                                                                                                                                                |

|                                          |                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Enable the collection of data to use the update_timing_budget command</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_report_or_update_budget</a></p>                                                                     |
| budget_sdc_contents                      |                                                                                                                                                                                                                                                                                                                        |
|                                          | <p>option to control SDC contents</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> full push_down_only boundary_constraints_only no_boundary_condition no_boundary_constraints</p> <p><b>Default:</b> full</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_sdc_contents</a></p> |
| budget_stage_based_fanout_driving_factor |                                                                                                                                                                                                                                                                                                                        |
|                                          | <p>Fanout count of the pin, above which we will add buffer stages in Timing Graph</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> -1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_stage_based_fanout_driving_factor</a></p>                                                      |
| budget_stage_based_partition_multiplier  |                                                                                                                                                                                                                                                                                                                        |
|                                          | <p>List of partition hinsts with its corresponding multiplication factor</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_stage_based_partition_multiplier</a></p>                                                             |
| budget_stage_based_weight                |                                                                                                                                                                                                                                                                                                                        |
|                                          | <p>List of weights given for {Sequential Combinational Buffer} cells</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 1 1 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_stage_based_weight</a></p>                                                                            |
| budget_use_boundary_condition            |                                                                                                                                                                                                                                                                                                                        |
|                                          | <p>Mechanism used for finding buffers to be used as driver/load</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> optimized template actual empty</p> <p><b>Default:</b> template</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_use_boundary_condition</a></p>                 |



|                                                 |                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| budget_virtual_clock                            |                                                                                                                                                                                                                                                                                       |
|                                                 | <p>Create additional virtual clocks for budgeting</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_virtual_clock</a></p>                                                                      |
| budget_virtual_opt_engine                       |                                                                                                                                                                                                                                                                                       |
|                                                 | <p>Specifies the virtual optimization engine to be used</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none early_timing_engine proto</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_virtual_opt_engine</a></p> |
| budget_write_constraints_for_clock_output_ports |                                                                                                                                                                                                                                                                                       |
|                                                 | <p>Write constraints for clock output ports</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_write_constraints_for_clock_output_ports</a></p>                                                |
| budget_write_false_path_for_hold                |                                                                                                                                                                                                                                                                                       |
|                                                 | <p>write false path -hold for all I/Os in setup budgeting</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_write_false_path_for_hold</a></p>                                                 |
| budget_write_latency_per_clock                  |                                                                                                                                                                                                                                                                                       |
|                                                 | <p>write latency per clock instead of per port</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">budget_write_latency_per_clock</a></p>                                                              |
| bump_grids                                      |                                                                                                                                                                                                                                                                                       |
|                                                 | <p>List of bump_grid object.</p> <p><b>Type:</b> <a href="#">obj(bump_grid)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                 |

|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bumps                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                       | <p>Short-cut for [get_db current_design .bumps]</p> <p><b>Type:</b> <a href="#">obj(bump)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| bus_sink_groups                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                       | <p>A group of sinks (loads) that some floorplan and routing commands use to control adding b and routing for a bus. See 'help *bus_sink*' for a list of commands related to this object.</p> <p><b>Type:</b> <a href="#">obj(bus_sink_group)*</a></p> <p><b>Default:</b> {}</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                  |
| busses                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                       | <p>Short-cut for [get_db current_design .busses]</p> <p><b>Type:</b> <a href="#">obj(bus)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| categories                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                       | <p>available categories of root attributes</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| ccopt_auto_limit_insertion_delay_factor               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                       | <p>CCOpt attempts to keep the insertion delays of each clock tree a fixed multiple of the longest insertion delay that would result from a global skew approach. This multiple can be modified by design timing and the presence of other clock trees, but will start at a fixed fraction above the global skew insertion delay. This attribute specifies that fixed fraction.</p> <p>Valid values: real</p> <p>See also:</p> <p>.ccopt_auto_limit_insertion_delay_factor_skew_group</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.5</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ccopt_auto_limit_insertion_delay_factor</a></p> |
| check_ac_limit_additional_individual_violation_report |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                                    |                                                                                                                                                                                                                                                                              |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | Report calculated duty ratio in addition with applied duty ratio in report file.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">check_ac_limit_additional_individual_violation_report</a>                |
| check_ac_limit_avg_recovery                        |                                                                                                                                                                                                                                                                              |
|                                                    | override QRC tech em_recover factor for all layers used in lavg limits.<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 1.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">check_ac_limit_avg_recovery</a>                                                   |
| check_ac_limit_check_thermal_aware_em              |                                                                                                                                                                                                                                                                              |
|                                                    | check thermal aware EM<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                                                     |
| check_ac_limit_current_file                        |                                                                                                                                                                                                                                                                              |
|                                                    | Specify current file to provide currents for cell/instance's pin.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">check_ac_limit_current_file</a>                                                          |
| check_ac_limit_current_scale_factor                |                                                                                                                                                                                                                                                                              |
|                                                    | Scale Signal EM current, syntax: -current_scale_factor { [rms <rms_value>] [peak <peak_value>] [avg <avg_value>] }<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">check_ac_limit_current_scale_factor</a> |
| check_ac_limit_current_scale_table                 |                                                                                                                                                                                                                                                                              |
|                                                    | Specifies the layer-based scale table for current<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">check_ac_limit_current_scale_table</a>                                                                   |
| check_ac_limit_default_freq_for_unconstrained_nets |                                                                                                                                                                                                                                                                              |

|                                             |                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>Specifies the frequency for EM calculation when a net has no defined frequency or a defined frequency of 0Hz in the design. Unit: Hz.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1e+06</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_default_freq_for_unconstrained_nets</a></p> |
| check_ac_limit_delta_temperature            |                                                                                                                                                                                                                                                                                                                                          |
|                                             | <p>maximum change in temperature allowed in units of Celsius. Used in the QRC tech for RM!</p> <p>Default=5.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 5.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_delta_temperature</a></p>                                                 |
| check_ac_limit_delta_temperature_layer_list |                                                                                                                                                                                                                                                                                                                                          |
|                                             | <p>Specifies layer based delta temperature for EM RMS current limit analysis</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_delta_temperature_layer_list</a></p>                                                                       |
| check_ac_limit_detailed                     |                                                                                                                                                                                                                                                                                                                                          |
|                                             | <p>Generates a detailed report containing information for all signal nets, including those with violations</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_detailed</a></p>                                                            |
| check_ac_limit_effort_level                 |                                                                                                                                                                                                                                                                                                                                          |
|                                             | <p>Enum_list legal values are: low, medium, high. Default: low.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> low</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_effort_level</a></p>                                                          |
| check_ac_limit_em_cdf_percentage            |                                                                                                                                                                                                                                                                                                                                          |

|                                          |                                                                                                                                                                                                                                                                                        |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | Specify em cumulative distribution factor (cdf) percentage.<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 1e-09<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">check_ac_limit_em_cdf_percentage</a>                                                                  |
| check_ac_limit_em_limit_scale_factor     |                                                                                                                                                                                                                                                                                        |
|                                          | Scale Signal EM limit, syntax: -em_limit_scale_factor { [rms <rms_value>] [peak <peak_val<br>[avg <avg_value>]}<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">check_ac_limit_em_limit_scale_factor</a>             |
| check_ac_limit_em_limit_scale_table      |                                                                                                                                                                                                                                                                                        |
|                                          | Specifies the layer-based scale table for em_limit<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">check_ac_limit_em_limit_scale_table</a>                                                                           |
| check_ac_limit_em_res_width              |                                                                                                                                                                                                                                                                                        |
|                                          | Specifies width used to get EM limit. enum_list legal values are: drawn, silicon. Default: dra<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> drawn silicon<br><b>Default:</b> drawn<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">check_ac_limit_em_res_width</a> |
| check_ac_limit_em_temperature            |                                                                                                                                                                                                                                                                                        |
|                                          | Specifies the temperature used to lookup the temperature scaling factor. By default, no scal<br>done. Unit: Celsius.<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> -1.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">check_ac_limit_em_temperature</a>             |
| check_ac_limit_em_temperature_layer_list |                                                                                                                                                                                                                                                                                        |
|                                          | Specifies layer based EM temperature for EM AVG current limit analysis<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">check_ac_limit_em_temperature_layer_list</a>                                                  |

|                                     |                                                                                                                                                                                                                                                                            |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| check_ac_limit_em_threshold         |                                                                                                                                                                                                                                                                            |
|                                     | <p>The value is the ratio of signal current (avg/peak/rms) to the respective EM limit. Default: 1.0</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_em_threshold</a></p> |
| check_ac_limit_enable_seb           |                                                                                                                                                                                                                                                                            |
|                                     | <p>Enable SEB calculation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_enable_seb</a></p>                                                                             |
| check_ac_limit_env_temperature      |                                                                                                                                                                                                                                                                            |
|                                     | <p>environment temperature. Unit: Celsius.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_env_temperature</a></p>                                                      |
| check_ac_limit_extraction_tech_file |                                                                                                                                                                                                                                                                            |
|                                     | <p>Specify qrcTechFile name.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_extraction_tech_file</a></p>                                                                 |
| check_ac_limit_force_hold_view      |                                                                                                                                                                                                                                                                            |
|                                     | <p>Turn this on to force hold view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_force_hold_view</a></p>                                                              |
| check_ac_limit_handle_pin_obs_via   |                                                                                                                                                                                                                                                                            |
|                                     | <p>Turn on to handle pin obs via.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_handle_pin_obs_via</a></p>                                                             |

|                                            |                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| check_ac_limit_ict_em_models               |                                                                                                                                                                                                                                                                                                                                    |
|                                            | <p>Specify ict EM file to provide EM rule if EM rule not exists in qrcTechFile.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_ict_em_models</a></p>                                                                             |
| check_ac_limit_ircx_models                 |                                                                                                                                                                                                                                                                                                                                    |
|                                            | <p>Specify ircx model file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_ircx_models</a></p>                                                                                                                                   |
| check_ac_limit_lifetime                    |                                                                                                                                                                                                                                                                                                                                    |
|                                            | <p>Specifies the hours of operation used to lookup the lifetime scaling factor. By default, no scaling is done. Unit: hour.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 87600.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_lifetime</a></p>                                 |
| check_ac_limit_lifetime_rating_factor_file |                                                                                                                                                                                                                                                                                                                                    |
|                                            | <p>Specify lifetime rating factor file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                           |
| check_ac_limit_max_error                   |                                                                                                                                                                                                                                                                                                                                    |
|                                            | <p>Specifies the maximum number of errors to report.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 10000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_max_error</a></p>                                                                                                            |
| check_ac_limit_method                      |                                                                                                                                                                                                                                                                                                                                    |
|                                            | <p>Specifies the current waveform calculation method, check one or more of Irms, Ipeak, and Iavg limits. enum_list legal values are: rms, peak, avg. Default: rms.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_method</a></p> |

|                                    |                                                                                                                                                                                                                                                                                        |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| check_ac_limit_min_peak_duty_ratio |                                                                                                                                                                                                                                                                                        |
|                                    | <p>Do not check lpeak for nets with duty ratio below &lt;duty_ratio&gt;. default = 0.05, range 0.0 to 1</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_min_peak_duty_ratio</a></p> |
| check_ac_limit_min_peak_freq       |                                                                                                                                                                                                                                                                                        |
|                                    | <p>Do not check lpeak for nets with frequency below &lt;freq&gt; in units of hertz. default = 1e6. Uni</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1e+06</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_min_peak_freq</a></p>       |
| check_ac_limit_net_file            |                                                                                                                                                                                                                                                                                        |
|                                    | <p>Specifies from file the net to check in signalEM analysis</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_net_file</a></p>                                                         |
| check_ac_limit_nets                |                                                                                                                                                                                                                                                                                        |
|                                    | <p>Specifies whether to check named or selected nets</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_nets</a></p>                                                                     |
| check_ac_limit_out_file            |                                                                                                                                                                                                                                                                                        |
|                                    | <p>Specifies the report file for the violation data</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_out_file</a></p>                                                                  |
| check_ac_limit_peak_td_method      |                                                                                                                                                                                                                                                                                        |



|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | <p>Specifies Td calculation method for lpeak check. enum_list legal values are: effective_width_from_integration, effective_half_peak_width, max_equivalent_dc_peak, sum_half_peak_width. Default: effective_width_from_integration.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> effective_width_from_integration effective_half_peak_width max_equivalent_dc_peak sum_half_peak_width</p> <p><b>Default:</b> effective_width_from_integration</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_peak_td_method</a></p> |
| check_ac_limit_read_detail_delta_temperature_file |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                   | <p>Specify detail delta temperature file for thermal aware checking.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_read_detail_delta_temperature_file</a></p>                                                                                                                                                                                                                                                                                                          |
| check_ac_limit_report_db                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                   | <p>Enable writing a db file to store data for GUI display.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_report_db</a></p>                                                                                                                                                                                                                                                                                                                                            |
| check_ac_limit_report_unannotated_shapes          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                   | <p>Report shapes without resistance back-annotated.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| check_ac_limit_seb_lifetime                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                   | <p>Specifies the hours of operation. By default, it's 5 years. Unit : hour.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 43800.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_seb_lifetime</a></p>                                                                                                                                                                                                                                                                                                                    |
| check_ac_limit_seb_table                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                   | <p>Specify seb table file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_seb_table</a></p>                                                                                                                                                                                                                                                                                                                                                                             |

|                                    |                                                                                                                                                                                                                                                                                             |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| check_ac_limit_seb_temperature     |                                                                                                                                                                                                                                                                                             |
|                                    | <p>Specifies the temperature of operation. By default, it's 110c. Unit: Celsius.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -273.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_seb_temperature</a></p>                               |
| check_ac_limit_selected            |                                                                                                                                                                                                                                                                                             |
|                                    | <p>Specifies whether to check named or selected nets</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_selected</a></p>                                                                     |
| check_ac_limit_simulation_net      |                                                                                                                                                                                                                                                                                             |
|                                    | <p>Specifies a list of nets to use simulation mode</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_simulation_net</a></p>                                                                  |
| check_ac_limit_simulation_net_file |                                                                                                                                                                                                                                                                                             |
|                                    | <p>Specifies a file contained a list of nets to use simulation mode</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_simulation_net_file</a></p>                                            |
| check_ac_limit_skip_category_mode  |                                                                                                                                                                                                                                                                                             |
|                                    | <p>Specify the mode that under which condition a net should be skipped and which category a skipped net belongs to.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_skip_category_mode</a></p> |
| check_ac_limit_skip_net            |                                                                                                                                                                                                                                                                                             |
|                                    | <p>Specifies the net to be skipped in signalEM analysis</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_skip_net</a></p>                                                                   |

|                                |                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| check_ac_limit_skip_net_file   |                                                                                                                                                                                                                                                                                                                                                           |
|                                | <p>Specifies the net to be skipped in signalEM analysis</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_skip_net_file</a></p>                                                                                                                            |
| check_ac_limit_toggle          |                                                                                                                                                                                                                                                                                                                                                           |
|                                | <p>Specifies the toggle rate for signal nets</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_toggle</a></p>                                                                                                                                             |
| check_ac_limit_use_db_freq     |                                                                                                                                                                                                                                                                                                                                                           |
|                                | <p>Uses the database frequency value as the effective frequency per net</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_use_db_freq</a></p>                                                                                                             |
| check_ac_limit_use_qrc_tech    |                                                                                                                                                                                                                                                                                                                                                           |
|                                | <p>Turn this on to force lrms checks to use the QRC tech file rather than the LEF tech. If either lavg is also checked, then all checks will use the QRC tech file, including lrms.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_use_qrc_tech</a></p> |
| check_ac_limit_use_rms_delta_t |                                                                                                                                                                                                                                                                                                                                                           |
|                                | <p>Turn on to use rms delta temperature.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_use_rms_delta_t</a></p>                                                                                                                                        |
| check_ac_limit_view            |                                                                                                                                                                                                                                                                                                                                                           |
|                                | <p>Specifies the name of the view</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_ac_limit_view</a></p>                                                                                                                                                           |

|                                     |                                                                                                                                                                                                                                                                            |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| check_design_across_hierarchy       |                                                                                                                                                                                                                                                                            |
|                                     | <p>Enables checking of unloaded, undriven, multi-driven and constant pins across design hierarchy.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_design_across_hierarchy</a></p> |
| check_design_report_async_pins      |                                                                                                                                                                                                                                                                            |
|                                     | <p>Includes async pins in netlist design checks.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_design_report_async_pins</a></p>                                                  |
| check_design_report_scan_pins       |                                                                                                                                                                                                                                                                            |
|                                     | <p>Includes scan pins in netlist design checks.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_design_report_scan_pins</a></p>                                                    |
| check_design_threshold_fanout       |                                                                                                                                                                                                                                                                            |
|                                     | <p>Upper threshold for netlist design checks.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 10000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_design_threshold_fanout</a></p>                                                      |
| check_drc_area                      |                                                                                                                                                                                                                                                                            |
|                                     | <p>Specifies the coordinates of the area to verify</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> 0.0 0.0 0.0 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_area</a></p>                                                     |
| check_drc_check_illegal_trim_shapes |                                                                                                                                                                                                                                                                            |
|                                     | <p>check illegal trim shapes.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_check_illegal_trim_shapes</a></p>                                                               |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| check_drc_check_only                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                     | <p>Specifies what kind of shapes to be checked.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> all regular special selected_net selected cell default</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_check_only</a></p>                                                                                                                                                                                                                                                                             |
| check_drc_check_reverse             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                     | <p>report CONVEXCORNER REVERSE violation.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_check_reverse</a></p>                                                                                                                                                                                                                                                                                                                                                                    |
| check_drc_check_routing_halo        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                     | <p>Check routing halo.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_check_routing_halo</a></p>                                                                                                                                                                                                                                                                                                                                                                                  |
| check_drc_check_routing_halo_corner |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                     | <p>Check routing halo in the corner</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_check_routing_halo_corner</a></p>                                                                                                                                                                                                                                                                                                                                                              |
| check_drc_disable_rules             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                     | <p>Disable the rule that from the provided rule list. the rule list as follow: (jog2jog_spacing   eol_spacing cut_spacing min_cut enclosure color min_step protrusion min_area out_of_die off_manufacturing_grid off_routing_track)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> jog2jog_spacing eol_spacing cut_spacing min_cut enclosure color min_step protrusion min_area out_of_die off_manufacturing_grid off_routing_track</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_disable_rules</a></p> |
| check_drc_enable_post_passive_fill  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                       |                                                                                                                                                                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>Check full fill trim.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_enable_post_passive_fill</a></p>                 |
| check_drc_exclude_pg_net              |                                                                                                                                                                                                                        |
|                                       | <p>Excludes the checking of power and ground nets.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_exclude_pg_net</a></p> |
| check_drc_ignore_cell_blockage        |                                                                                                                                                                                                                        |
|                                       | <p>Ignore geometries of cell blockage.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_ignore_cell_blockage</a></p>       |
| check_drc_ignore_fill_wire            |                                                                                                                                                                                                                        |
|                                       | <p>Ignore metalfill wires when check drc.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_ignore_fill_wire</a></p>        |
| check_drc_ignore_non_rectangle_shapes |                                                                                                                                                                                                                        |
|                                       | <p>ignore non-rectangle shapes.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_ignore_non_rectangle_shapes</a></p>       |
| check_drc_ignore_trial_route          |                                                                                                                                                                                                                        |
|                                       | <p>Ignore trial route when check drc.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_ignore_trial_route</a></p>          |
| check_drc_inside_via_def              |                                                                                                                                                                                                                        |

|                              |                                                                                                                                                                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Check the cut spacing from the same via</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_inside_via_def</a></p>                                                               |
| check_drc_layer_range        |                                                                                                                                                                                                                                                                             |
|                              | <p>Checks between the range of metal layers including cut layers in between</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_layer_range</a></p>                                 |
| check_drc_limit              |                                                                                                                                                                                                                                                                             |
|                              | <p>Specifies the maximum number of errors to report</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_limit</a></p>                                                                |
| check_drc_max_wrong_way_halo |                                                                                                                                                                                                                                                                             |
|                              | <p>Specifies the maximum wrong way halo</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_max_wrong_way_halo</a></p>                                                            |
| check_drc_ndr_spacing        |                                                                                                                                                                                                                                                                             |
|                              | <p>Specifies what kind of non-default rule spacing to be checked</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false auto</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_ndr_spacing</a></p> |
| check_drc_quiet              |                                                                                                                                                                                                                                                                             |
|                              | <p>Don't print sub-area message in logv file.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                             |
| check_drc_report             |                                                                                                                                                                                                                                                                             |

|                                            |                                                                                                                                                                                                                                                                        |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Specifies the report file that contains the violations information</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_report</a></p>                                       |
| check_drc_short_only                       |                                                                                                                                                                                                                                                                        |
|                                            | <p>Exclude all DRC check except short.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_short_only</a></p>                                                                 |
| check_drc_trim_length                      |                                                                                                                                                                                                                                                                        |
|                                            | <p>Consider max length in the trim shape layer.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_trim_length</a></p>                                                       |
| check_drc_uncolored                        |                                                                                                                                                                                                                                                                        |
|                                            | <p>report color change violation when the dpt layer shape is free color.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_uncolored</a></p>                                |
| check_drc_use_min_spacing_on_block_obs     |                                                                                                                                                                                                                                                                        |
|                                            | <p>Use min spacing to check for block OBS.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false auto</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">check_drc_use_min_spacing_on_block_obs</a></p> |
| clock_gating_auto_control                  |                                                                                                                                                                                                                                                                        |
|                                            | <p>Set automatically generated control port to improve testability.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none scan_enable test_mode</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p>                                             |
| clock_gating_connect_test_clock_gate_types |                                                                                                                                                                                                                                                                        |



|                                                                  |                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                  | <p>Set the types of clock gates that will be considered for test connection.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> all user synthesis</p> <p><b>Default:</b> synthesis</p> <p><b>Edit:</b> Yes</p>                                                                              |
| clock_gating_connect_test_consider_constant_enabled_as_connected |                                                                                                                                                                                                                                                                                                               |
|                                                                  | <p>Will clock gating test connection consider a constant enabled clock gate as already connected.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">clock_gating_connect_test_consider_constant_enabled_as_connected</a></p> |
| clock_gating_coverage_effort                                     |                                                                                                                                                                                                                                                                                                               |
|                                                                  | <p>Effort level for clock gating enable discovery. Possible values are standard and high.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> standard high</p> <p><b>Default:</b> standard</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">clock_gating_coverage_effort</a></p> |
| clock_gating_exceptions_aware                                    |                                                                                                                                                                                                                                                                                                               |
|                                                                  | <p>Check timing exceptions and copy onto flops during clock-gating.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">clock_gating_exceptions_aware</a></p>                                                                   |
| clock_gating_prefix                                              |                                                                                                                                                                                                                                                                                                               |
|                                                                  | <p>Set name prefix for clock-gating objects.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">clock_gating_prefix</a></p>                                                                                                    |
| clock_tree_source_groups                                         |                                                                                                                                                                                                                                                                                                               |
|                                                                  | <p>list of clock_tree_source_group</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">clock_tree_source_group</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">clock_tree_source_groups</a></p>                                                                   |
| clock_trees                                                      |                                                                                                                                                                                                                                                                                                               |

|                                                  |                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | list of clock_tree<br><b>Type:</b> <a href="#">obj(clock_tree)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No<br><b>Reference:</b> <a href="#">clock_trees</a>                                                                                                                                                                         |
| clocks                                           |                                                                                                                                                                                                                                                                                                                                       |
|                                                  | clocks<br><b>Type:</b> <a href="#">obj(clock)*</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                            |
| clp_consider_errors_as_warnings                  |                                                                                                                                                                                                                                                                                                                                       |
|                                                  | Specify the CLP messages whose severity should be reduced to 'Warning'.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                                                              |
| clp_enable_1801_hierarchical_bbox                |                                                                                                                                                                                                                                                                                                                                       |
|                                                  | To enable hierarchical bbox flow for 1801 based design.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">clp_enable_1801_hierarchical_bbox</a>                                                                                                                      |
| clp_ignore_level_shifter_high_to_low             |                                                                                                                                                                                                                                                                                                                                       |
|                                                  | Specify if high to low level shifter checks should be skipped.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                                                                      |
| cmd_file                                         |                                                                                                                                                                                                                                                                                                                                       |
|                                                  | The cmd file name of the program. It has each command run in the current session. It can be changed during the session to direct cmd file output to a different file. Type 'man log_file' for details.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">cmd_file</a> |
| combinational_sequential_merge_message_threshold |                                                                                                                                                                                                                                                                                                                                       |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>Merging message is printed when hier instance has cells more than this threshold value.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 10</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">combinational_sequential_merge_message_threshold</a></p>                                                                                                                                                                                                                                                                        |
| constraint_modes                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                 | <p>Returns the information about the constraint modes in the design.</p> <p><b>Type:</b> <a href="#">obj(constraint_mode)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                       |
| control_logic_optimization                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                 | <p>Controls the effort level for optimization of control logic (basic/advanced/none).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic advanced none</p> <p><b>Default:</b> basic</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">control_logic_optimization</a></p>                                                                                                                                                                                                                                                |
| cts_add_port_driver_max_distance_from_port      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                 | <p>This attribute specifies the allowable distance, in um if no units suffix is specified, that a port is allowed to be placed from the port being buffered. If the driver cannot be added close enough to the port it will not be added at all. This attribute, if set, overrides any value set in the attribute cts_add_port_driver_max_distance_from_port_rows.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_add_port_driver_max_distance_from_port</a></p> |
| cts_add_port_driver_max_distance_from_port_rows |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                 | <p>This attribute specifies the allowable distance, in standard cell row heights, that a port drive is allowed to be placed from the port being buffered. If the driver cannot be added close enough to the port it will not be added at all. The attribute cts_add_port_driver_max_distance_from_port, if set, overrides this value.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 50</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_add_port_driver_max_distance_from_port_rows</a></p>                              |
| cts_adjacent_rows_legal                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | <p>Defines the clock halo in the y direction. If set to true the y direction clock halo is zero and clock instances are allowed in adjacent rows.</p> <p>The following attributes can be used to assign y direction clock halos within CCOpt:</p> <ul style="list-style-type: none"> <li>cts_cell_halo_y</li> <li>cts_adjacent_rows_legal</li> <li>cts_cell_halo_rows</li> </ul> <p>Only one of these attributes is used to determine the clock halo in the y direction. The following rules determine which:</p> <ul style="list-style-type: none"> <li>. If cts_cell_halo_y is set to a non-auto value, then this defines the y direction clock halo. The attributes cts_adjacent_rows_legal and cts_cell_halo_rows have no effect.</li> <li>. If cts_cell_halo_y is set to auto and cts_adjacent_rows_legal is set to a non-auto value then cts_adjacent_rows_legal defines the clock halo in the y direction. The attribute cts_cell_halo_rows has no effect.</li> <li>. If both cts_cell_halo_y and cts_adjacent_rows_legal are set to auto then cts_cell_halo_rows defines the clock halo in the y direction.</li> </ul> <p>Valid values: true false</p> <p>See also:</p> <ul style="list-style-type: none"> <li>. cts_cell_halo_y</li> <li>. cts_cell_halo_rows</li> </ul> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_adjacent_rows_legal</a></p> |
| cts_allow_non_std_clock_gate_inputs               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                   | <p>If this attribute is set, CTS will allow the use of clock gates with non-standard pins. CCOpt considers the following pin types to be standard: clock pins, enable pins, test enable pins, retention pins and power gating pins. Before starting CTS CCOpt will emit a warning, indicating which pin(s) it considers non-standard.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_allow_non_std_clock_gate_inputs</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| cts_allow_post_route_sizing_stack_via_rule_update |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Whether to allow post-route resizing to make cell swaps which require stack via rules to change a clock_tree pin. Enabling this may cause disruption to routed nets, i.e. ECO routing will have more to repair.</p> <p>Valid values: resize_allow_svr_change resize_dont_allow_svr_change</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> resize_dont_allow_svr_change</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_allow_post_route_sizing_stack_via_rule_update</a></p>                                                                                                                                                                                                            |
| cts_balance_mode                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                      | <p>Replace CCOpt mode setting cts_opt_type {full   cluster   trial}. If not full, causes CCOpt and CTS to halt before final completion of the clock tree to facilitate clock tree inspection.</p> <p>The possible values for this attribute are as follows:</p> <ul style="list-style-type: none"> <li>- full - default value, a full CTS is performed.</li> <li>- cluster - a cluster-only CTS is performed. The clock tree has no balancing delay applied.</li> <li>- trial - The clock has only virtual (numeric annotation) balancing delays applied.</li> </ul> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> full</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_balance_mode</a></p> |
| cts_blackbox_default_driver_base_pin |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                      | <p>Base pin that will be used for all timing and capacitance modeling for clock roots with a source at a blackbox output.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_blackbox_default_driver_base_pin</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| cts_blackbox_default_load_base_pin   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                      | <p>Base pin that will be used for all timing and capacitance modeling for clock sinks at blackbox inputs. Any cts_capacitance_override attribute will be used in preference to this.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_blackbox_default_load_base_pin</a></p>                                                                                                                                                                                                                                                                                                                                                             |
| cts_buffer_cells                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>Specifies the buffer cells for CTS. If none are specified CCOpt will choose buffers from the libraries.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>Different buffer cells may be specified for any combination of clock tree and power domain.</p> <p>To use different buffers for each net type set the <code>cts_buffer_cells_top</code> and <code>cts_buffer_cells_bottom</code> attributes.</p> <p>Some examples follow:</p> <p>To specify buffer cells for all clock trees and all power domains:</p> <pre>set_db cts_buffer_cells {bufAX* bufBX*}</pre> <p>To specify buffer cells for a particular clock tree and all power domains:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_buffer_cells {bufX20 bufX18}</pre> <p>To specify buffer cells for a particular clock tree and power domain:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_buffer_cells -index {power_domain &lt;pd&gt;} {bufX12 bufX8}</pre> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_buffer_cells</a></p> |
| cts_buffer_cells_leaf |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                       | <p>Specifies the buffer cells available for CTS to use on leaf nets. If none are specified CCOpt will use the same buffers as on trunk nets (as specified in the <code>cts_buffer_cells</code> attribute).</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>Different leaf buffer cells may be specified for any combination of clock tree and power domain.</p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_buffer_cells_leaf</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cts_buffer_cells_top  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Specifies the buffers cells available for CTS to use on top nets. If none are specified CCOpt use the same buffers as on trunk nets<br/>(as specified in the <code>cts_buffer_cells</code> attribute).<br/>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.<br/>If set explicitly, CCOpt will ignore any <code>dont_use</code> settings for the cells specified.<br/>Different top buffer cells may be specified for any combination of clock tree and power domain.<br/>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Allowed -index values:</b> <code>power_domain</code><br/><b>Default:</b> <code>""</code><br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_buffer_cells_top</a></p> |
| <code>cts_cell_density</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Specifies the clock halo in the x direction.  
Any x direction clock halo defined by this attribute is proportional to the cell width.  
The constant of proportionality is defined by the attribute value. For example:  
. If `cts_cell_density` = 0.25 then the x direction clock halo equals 3 \* cell width.  
. If `cts_cell_density` = 0.5 then the x direction clock halo equals cell width.  
. If `cts_cell_density` = 0.75 then the x direction clock halo equals cell width / 3.  
. If `cts_cell_density` = 1 then the x direction clock halo equals zero.  
This attribute can specify the x direction clock halo for all clock trees via:  
`set_db cts_cell_density 0.8`  
This attribute can specify the x direction clock halo per-clock tree via:  
`set_db clock_tree:<clk> .cts_cell_density 0.9`  
The following attributes can be used to assign x direction clock halos within CCOpt:  
`cts_cell_halo_x`  
`cts_cell_density`  
`cts_cell_halo_sites`  
Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which:  
. If `cts_cell_halo_x` is set to a non-auto value, then this defines the x direction clock halo. The attributes `cts_cell_density` and `cts_cell_halo_sites` have no effect.  
. If `cts_cell_halo_x` is set to auto and `cts_cell_density` is set to a non-auto value then `cts_cell_density` defines the clock halo in the x direction. The attribute `cts_cell_halo_sites` has no effect.  
. If both `cts_cell_halo_x` and `cts_cell_density` are set to auto then `cts_cell_halo_sites` defines the clock halo in the x direction.  
Valid values: 0.01 to 1  
See also:  
. `cts_cell_halo_x`  
. `cts_cell_halo_sites`  
**Type:** [string](#)  
**Default:** 0.75  
**Edit:** Yes  
**Reference:** [cts\\_cell\\_density](#)

`cts_cell_halo_mode`



Specifies how clock halos are used to determine the minimum legal separation between a clock instances. There are two possible modes 'max' and 'sum'. When set to 'max' the minimum legal separation is the larger of the two clock halos. When set to 'sum' the minimum legal separation is the sum of the two clock halos.

Valid values: max sum

See also:

- .cts\_cell\_halo\_x
- .cts\_cell\_halo\_y
- .cts\_cell\_density
- .cts\_adjacent\_rows\_legal
- .cts\_cell\_halo\_sites
- .cts\_cell\_halo\_rows
- .cts\_effective\_clock\_halo\_x
- .cts\_effective\_clock\_halo\_y
- .cts\_effective\_clock\_halo\_x\_source
- .cts\_effective\_clock\_halo\_y\_source

**Type:** [string](#)

**Default:** max

**Edit:** Yes

**Reference:** [cts\\_cell\\_halo\\_mode](#)

---

cts\_cell\_halo\_rows

---

Specifies the clock halo in the y direction in rows for all clock cells.  
The following attributes can be used to assign y direction clock halos within CCOpt:  
cts\_cell\_halo\_y  
cts\_adjacent\_rows\_legal  
cts\_cell\_halo\_rows  
Only one of these attributes is used to determine the clock halo in the y direction. The following rules determine which:  
    . If cts\_cell\_halo\_y is set to a non-auto value, then this defines the y direction clock halo. The attributes cts\_adjacent\_rows\_legal and cts\_cell\_halo\_rows have no effect.  
    . If cts\_cell\_halo\_y is set to auto and cts\_adjacent\_rows\_legal is set to a non-auto value then cts\_adjacent\_rows\_legal defines the clock halo in the y direction. The attribute cts\_cell\_halo\_rows has no effect.  
    . If both cts\_cell\_halo\_y and cts\_adjacent\_rows\_legal are set to auto then cts\_cell\_halo\_rows defines the clock halo in the y direction.  
See also:  
    . cts\_cell\_halo\_y  
    . cts\_adjacent\_rows\_legal  
**Type:** [int](#)  
**Default:** 1  
**Edit:** Yes  
**Reference:** [cts\\_cell\\_halo\\_rows](#)

cts\_cell\_halo\_sites

Specifies the clock halo in the x direction in sites for all clock cells.  
The following attributes can be used to assign x direction clock halos within CCOpt:  
cts\_cell\_halo\_x  
cts\_cell\_density  
cts\_cell\_halo\_sites  
Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which:  
. If cts\_cell\_halo\_x is set to a non-auto value, then this defines the x direction clock halo. The attributes cts\_cell\_density and cts\_cell\_halo\_sites have no effect.  
. If cts\_cell\_halo\_x is set to auto and cts\_cell\_density is set to a non-auto value then cts\_cell\_density defines the clock halo in the x direction. The attribute cts\_cell\_halo\_sites has no effect.  
. If both cts\_cell\_halo\_x and cts\_cell\_density are set to auto then cts\_cell\_halo\_sites defines the clock halo in the x direction.  
See also:  
. cts\_cell\_halo\_x  
. cts\_cell\_density  
**Type:** [int](#)  
**Default:** 4  
**Edit:** Yes  
**Reference:** [cts\\_cell\\_halo\\_sites](#)

---

cts\_cell\_halo\_x

---

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <p>Specifies the clock halo distance in the x direction. The default value of this attribute is auto. The following attributes can be used to assign x direction clock halos within CCOpt:</p> <ul style="list-style-type: none"><li>cts_cell_halo_x</li><li>cts_cell_density</li><li>cts_cell_halo_sites</li></ul> <p>Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which:</p> <ul style="list-style-type: none"><li>. If cts_cell_halo_x is set to a non-auto value, then this defines the x direction clock halo. The attributes cts_cell_density and cts_cell_halo_sites have no effect.</li><li>. If cts_cell_halo_x is set to auto and cts_cell_density is set to a non-auto value then cts_cell_density defines the clock halo in the x direction. The attribute cts_cell_halo_sites has no effect.</li><li>. If both cts_cell_halo_x and cts_cell_density are set to auto then cts_cell_halo_sites defines the clock halo in the x direction.</li></ul> <p>See also:</p> <ul style="list-style-type: none"><li>. cts_cell_density</li><li>. cts_cell_halo_sites</li></ul> <p><b>Type:</b> <a href="#">string</a><br/><b>Allowed -index values:</b> power_domain<br/><b>Default:</b> auto<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_cell_halo_x</a></p> |
| cts_cell_halo_y |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Specifies the clock halo distance in the y direction. The default value of this attribute is auto. The following attributes can be used to assign y direction clock halos within CCOpt:</p> <ul style="list-style-type: none"> <li>cts_cell_halo_y</li> <li>cts_adjacent_rows_legal</li> <li>cts_cell_halo_rows</li> </ul> <p>Only one of these attributes is used to determine the clock halo in the y direction. The following rules determine which:</p> <ul style="list-style-type: none"> <li>. If cts_cell_halo_y is set to a non-auto value, then this defines the y direction clock halo. The attributes cts_adjacent_rows_legal and cts_cell_halo_rows have no effect.</li> <li>. If cts_cell_halo_y is set to auto and cts_adjacent_rows_legal is set to a non-auto value then cts_adjacent_rows_legal defines the clock halo in the y direction. The attribute cts_cell_halo_rows has no effect.</li> <li>. If both cts_cell_halo_y and cts_adjacent_rows_legal are set to auto then cts_cell_halo_rows defines the clock halo in the y direction.</li> </ul> <p>See also:</p> <ul style="list-style-type: none"> <li>. cts_adjacent_rows_legal</li> <li>. cts_cell_halo_rows</li> </ul> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> power_domain<br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_cell_halo_y</a></p> |
| cts_clock_gate_movement_limit |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                               | <p>Each clock gate is restricted to a Manhattan ball centered on its original location with CTS factor. The radius of the ball is a multiple of the clock gate height. This controls the default value of that multiple.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> 10<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_clock_gate_movement_limit</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| cts_clock_gating_cells        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Specifies the clock gates for CTS. If none are specified CCOpt will choose clock gates from libraries.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>A list of lists can be used to specify separate clock gate sub-lists. All CCOpt transforms will only use clock gates within the clock gate sub-list they are initially part of. If an incoming clock gate is part of any clock gate sub-list, it will be resized to the first clock gate sub-list.</p> <p>Different clock gates may be specified for any combination of clock tree and power domain. Some examples follow:</p> <p>To specify clock gates for all clock trees and all power domains:<br/> <code>set_db cts_clock_gating_cells {cgAX* cgBX*}</code></p> <p>To specify clock gates for a particular clock tree and all power domains:<br/> <code>set_db clock_tree:&lt;clk&gt; .cts_clock_gating_cells {cgX20 cgX18}</code></p> <p>To specify clock gates for a particular clock tree and power domain:<br/> <code>set_db clock_tree:&lt;clk&gt; .cts_clock_gating_cells -index {power_domain &lt;pd&gt;} {cgX12 cgX18}</code></p> <p>To specify separate clock gate sub-lists:<br/> <code>set_db .cts_clock_gating_cells {{cgX12A1 cgX8A1} {cgX12A2 cgX8A2}}</code></p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names or lists to specify separate clock gate sub-lists</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> power_domain<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_clock_gating_cells</a></p> |
| cts_clock_source_cells                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                         | <p>Specifies the cells available for CTS to size clock sources if the cts_size_clock_sources attribute is set to true. If none are specified the tool will choose cells from the libraries.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, the tool will ignore any dont_use settings for the cells specified.</p> <p>Different cells may be specified for clock trees or power domains. Only clock sources that are buffers, inverters, logic and clock gating cells with a single output can be resized.</p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> power_domain<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_clock_source_cells</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| cts_clock_tree_source_group_clock_trees |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | <p>A list of the clock trees relevant to this source group.<br/>Valid values: list <code>cts_clock_trees</code></p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_clock_tree_source_group_clock_trees</a></p>                                                                                                                                                                       |
| cts_clock_tree_source_input_max_transition_time      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                      | <p>The slew which will be assumed at the input of the root driver.<br/>Valid values: double</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Allowed -index values:</b> <code>delay_corner</code><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_clock_tree_source_input_max_transition_time</a></p>                                                                                                                       |
| cts_cloning_effort                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                      | <p>Specifies the amount of cloning during clustering. By default the cloning effort is high.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> high<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_cloning_effort</a></p>                                                                                                                                                                                                             |
| cts_cloning_inst_name_suffix                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                      | <p>If set, the suffix will be used generally for all insts cloned by CTS, else the suffix will default "clone".</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> clone<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_cloning_inst_name_suffix</a></p>                                                                                                                                                                               |
| cts_cloning_inst_name_suffix_source_group_assignment |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                      | <p>Controls the suffix used by multi-tap cloning. If set to a non-empty string, that string will be used when naming cloned instances created by multi-tap cloning. Otherwise, the suffix will be controlled by the attribute <code>cts_cloning_inst_name_suffix</code>.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_cloning_inst_name_suffix_source_group_assignment</a></p> |
| cts_declone_clock_gates                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>This attribute controls clock gate decloning (merging) in both CTS-only and full clock_opt_design (or ECF) flows.</p> <p>Valid values:</p> <ul style="list-style-type: none"><li>false - decloning is disabled in all flows</li><li>cts_only - decloning is enabled for CTS-only flows</li><li>ccopt_only - decloning is enabled for full flows</li><li>true - decloning is enabled for all flows</li></ul> <p>See also attribute cts_declone_clock_logic.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ccopt_only</p> <p><b>Edit:</b> Yes</p>  |
| cts_declone_clock_logic |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                         | <p>This attribute controls clock logic decloning (merging) in both CTS-only and full clock_opt_design (or ECF) flows.</p> <p>Valid values:</p> <ul style="list-style-type: none"><li>false - decloning is disabled in all flows</li><li>cts_only - decloning is enabled for CTS-only flows</li><li>ccopt_only - decloning is enabled for full flows</li><li>true - decloning is enabled for all flows</li></ul> <p>See also attribute cts_declone_clock_gates.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ccopt_only</p> <p><b>Edit:</b> Yes</p> |
| cts_delay_cells         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Specifies the delay cells available for CTS. If none are specified CCOpt will not use delay cells. Setting this attribute to the string 'auto' means that CCOpt will choose delay cells from the library to use.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>Different delay cells may be specified for any combination of clock tree and power domain omitting those arguments a global setting can be applied.</p> <p>Some examples follow:</p> <p>To specify delay cells for all clock trees and power domains:<br/>set_db cts_delay_cells {delayAX* delayBX*}</p> <p>To specify delay cells for a particular clock tree and all power domains:<br/>set_db clock_tree:&lt;clk&gt; .cts_delay_cells {delayX1 delayX2}</p> <p>To specify delay cells for a particular clock tree and power domain:<br/>set_db clock_tree:&lt;clk&gt; .cts_delay_cells -index {power_domain &lt;pd&gt;} {delayX2 delayX3}</p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names string 'auto'</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_delay_cells</a></p> |
| cts_detailed_cell_warnings    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                               | <p>If set to true, CCOpt outputs detailed cell warning diagnostics when it encounters issues with library cell selection, power domains and/or signal levels.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_detailed_cell_warnings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| cts_exit_if_no_placeable_area |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                               | <p>Specifies that CTS should exit if the design has zero placeable area.</p> <p>Setting this attribute may be useful to temporarily work-around problems with row definition and/or blockages causing placeable area to be zero.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_exit_if_no_placeable_area</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cts_exit_if_skew_target_over_constrained              | <p>Specifies that CCOpt should exit as current skew targets are likely to lead to poor results. By default, CCOpt computes a minimum skew target which should not lead to excessive buffering, and does not allow any skew target to be set lower than this minimum. This attribute overrides that check, and allows any skew target to be used.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_exit_if_skew_target_over_constrained</a></p> |
| cts_exit_if_stage_delay_sigma_target_over_constrained | <p>If set, allows any max stage delay sigma target to be set. If the target is too low, it is likely to runtime problems.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_exit_if_stage_delay_sigma_target_over_constrained</a></p>                                                                                                                                                                                                                                           |
| cts_fix_clock_sinks                                   | <p>If set to true, we will DEF lock clock tree sinks after routing in addition to any clock node locked (fixed).</p> <p>If set to soft, we will DEF lock clock tree sinks after routing in addition to any clock node soft_locking (softFixed).</p> <p>Valid values: true false soft</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_fix_clock_sinks</a></p>                                                                                                               |
| cts_flexible_htree_allow_mixed_sink_polarity          | <p>If set to true, the tool is allowed to create flexible H-trees with mixed sink polarities for sink and final cells, where the polarity is honored for the final cell (default false).</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_flexible_htree_allow_mixed_sink_polarity</a></p>                                                                                                                                                                                    |
| cts_flexible_htree_enable_metrics_reporting           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | <p>A three valued enum comprising: false, post_route, true. Default is post_route. The values true/false enable/disable all reporting, while post_route enables only the report at the end of flexible H-Tree synthesis.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> post_route<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_flexible_htree_enable_metrics_reporting</a></p>                                                                                                                                                                             |
| cts_flexible_htree_placement_legalization_effort         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                          | <p>The legalization effort for finding placement unblocked points on the synthesis grid for flexible trees. High placement legalization effort can avoid having to relax placement constraints when implementing H-trees but may lead to increased runtime.</p> <p>Valid values: low, high (default low)</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> low<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_flexible_htree_placement_legalization_effort</a></p>                                                                                               |
| cts_force_target_max_transition_time                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                          | <p>Specifies that CCOpt should allow any transition target to be used, even if it is likely to lead to runtime problems. By default, CCOpt computes a minimum transition target which should not lead to runtime problems, and does not allow any transition target set lower than this minimum. This attribute overrides that check, and allows any transition target to be used.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_force_target_max_transition_time</a></p> |
| cts_ignore_problematic_skew_as_result_of_dont_touch_nets |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                          | <p>If set, sinks directly connected to nets that are causing unfixable skew problems will be ignored during skew balancing.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_ignore_problematic_skew_as_result_of_dont_touch_nets</a></p>                                                                                                                                                                                                                                                                    |
| cts_inst_name_prefix                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>The name prefix of instances created by CTS. The default value is "CTS". The default names of instances are CTS_.*.</p> <p>Valid values: string</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> CTS</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_inst_name_prefix</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| cts_inverter_cells      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                         | <p>Specifies the inverter cells available for CTS. If none are specified CCOpt will choose inverter from the libraries.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>Different inverter cells may be specified for any combination of clock tree and power domain.</p> <p>To use different inverters for each net type set the cts_inverter_cells_top and cts_inverter_cells_leaf attributes .</p> <p>Some examples follow:</p> <p>To specify inverter cells for all clock trees and all power domains:</p> <pre>set_db cts_inverter_cells {invAX* invBX*}</pre> <p>To specify inverter cells for a particular clock tree and all power domains:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_inverter_cells {invX20 invX18}</pre> <p>To specify inverter cells for a particular clock tree and power domain:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_inverter_cells -index {power_domain &lt;pd&gt;} {invX12 invX8}</pre> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_inverter_cells</a></p> |
| cts_inverter_cells_leaf |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Specifies the inverter cells available for CTS to use on leaf nets. If none are specified CCOpt will use the same inverters as on trunk nets (as specified in the <code>cts_inverter_cells</code> attribute).</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any <code>dont_use</code> settings for the cells specified.</p> <p>Different leaf inverter cells may be specified for any combination of clock tree and power domain.</p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> <code>power_domain</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_inverter_cells_leaf</a></p> |
| <code>cts_inverter_cells_top</code>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                         | <p>Specifies the inverter cells available for CTS to use on top nets. If none are specified CCOpt will use the same inverters as on trunk nets (as specified in the <code>cts_inverter_cells</code> attribute).</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any <code>dont_use</code> settings for the cells specified.</p> <p>Different top inverter cells may be specified for any combination of clock tree and power domain.</p> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> <code>power_domain</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_inverter_cells_top</a></p>    |
| <code>cts_load_capacitance_cells</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                         | <p>Specifies the load capacitance cells available for CTS. CTS will use cells from this collection for load capacitance optimizations.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> <code>power_domain</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_load_capacitance_cells</a></p>                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>cts_logic_cells</code>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Specifies the clock logics for CTS. If none are specified CCOpt will choose clock logics from libraries.</p> <p>Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names.</p> <p>If set explicitly, CCOpt will ignore any dont_use settings for the cells specified.</p> <p>Different logic cells may be specified for any combination of clock tree and power domain.</p> <p>Some examples follow:</p> <p>To specify logic cells for all clock trees and all power domains:</p> <pre>set_db cts_logic_cells {and* mux*}</pre> <p>To specify logic cells for a particular clock tree and all power domains:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_logic_cells {andX20 andX18}</pre> <p>To specify logic cells for a particular clock tree and power domain:</p> <pre>set_db clock_tree:&lt;clk&gt; .cts_logic_cells -index {power_domain &lt;pd&gt;} {andX12 andX8}</pre> <p>Valid values: a list of library cell names, or a list of patterns to expand to library cell names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> power_domain</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_logic_cells</a></p> |
| cts_manage_power_intent_violations |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                    | <p>If this attribute is set, the CTS algorithm will work around power management illegalities in the clock tree, as opposed to failing with an error when it encounters them. This allows the clock tree to be synthesized, but any power management illegalities will remain in the exported design.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_manage_power_intent_violations</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| cts_max_fanout                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                    | <p>The maximum fanout at any point in the clock tree.</p> <p>Valid values: integer ranged between 2 and 1000 inclusive</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_max_fanout</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| cts_max_instance_height            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>When selecting cells for use, CTS will filter the selection to remove all buffers, inverters and clock gates taller than this many row heights. This helps quickly eliminate very large cells that are unlikely to be useful.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 3<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_max_instance_height</a></p>                                                                                                                                                                                                                                                                                                               |
| cts_max_source_to_sink_net_length      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                        | <p>The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe. This constraint can be applied to either a pin, a clock tree, or a net type. By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length.</p> <p>Valid values: double</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> top auto trunk auto leaf auto<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_max_source_to_sink_net_length</a></p> |
| cts_max_source_to_sink_net_length_leaf |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                        | <p>The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe. This constraint can be applied to either a pin, a clock tree, or a net type. By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length.</p> <p>Valid values: double</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_max_source_to_sink_net_length_leaf</a></p>                     |
| cts_max_source_to_sink_net_length_top  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe.<br/>This constraint can be applied to either a pin, a clock tree, or a net type.<br/>By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length.<br/>Valid values: double<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> auto<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_max_source_to_sink_net_length_top</a></p>   |
| cts_max_source_to_sink_net_length_trunk |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <p>The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe.<br/>This constraint can be applied to either a pin, a clock tree, or a net type.<br/>By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length.<br/>Valid values: double<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> auto<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_max_source_to_sink_net_length_trunk</a></p> |
| cts_mixed_fanout_net_type               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <p>Controls how CCOpt considers nets that have fanout consisting partially but not entirely of sinks.<br/>By default, having any sinks (eg. DFFs) will make the net be considered leaf, but when set to trunk, a net has to drive only sinks to be considered leaf. For example, a net driving a clock gate and a DFF would no longer count as a leaf net.<br/>Valid values: leaf trunk<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> leaf<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_mixed_fanout_net_type</a></p>                                                                                                                                     |
| cts_move_clock_gates                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |



|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>If this attribute is set, the CTS algorithm will move clock gates that appear in the clock tree. If this attribute may cause the clock tree to have a lower insertion delay, but might break data timing. During optimization, this is not a problem, because the timing will be automatically recovered during the optimization process. If this attribute is false, CTS permits small moves of ICGs for legalization.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_move_clock_gates</a></p>                                                                                                                                                                                                                                                                                            |
| cts_move_logic                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                            | <p>If this attribute is set, the CTS algorithm will move logic that appears in the clock tree. "Logic" does not include clock gates, buffers, and inverters in a clock tree, which are always moved unless they are locked, or clock generators that are above the root of the clock tree. Usually, this will affect multiplexers used for selecting one of a number of clocks, or for switching between a test clock and the main clock. Setting this attribute may cause the clock tree to have a lower insertion delay, but might break datapath timing. During optimization, this is not a problem, because the timing will be automatically recovered during the optimization process.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_move_logic</a></p> |
| cts_net_name_prefix                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                            | <p>The name prefix of nets created by CTS. The default value is "CTS". The default names of nets are CTS, CTS_1, CTS_2... ..</p> <p>Valid values: string</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> CTS<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_net_name_prefix</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| cts_pin_insertion_delay_histogram_bin_size |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>This attribute specifies the preferred size of the histogram bins that should be used when reporting the distribution of pin insertion delays. The bin size is specified in time units and must be greater than zero. A value of auto (the default) leaves it to the tool to calculate a bin size. This attribute affects pin insertion delay histograms that are logged during various flow commands. It also determines the preferred histogram bin size that is used by the <code>report_pin_insertion_delays</code> command when an explicit <code>-bin_size</code> argument has not been specified.</p> <p>Valid values: auto, time value in library units</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_pin_insertion_delay_histogram_bin_size</a></p> |
| cts_post_route_enable                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                         | <p>Enable post-conditioning optimization after clock nets are routed.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_post_route_enable</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| cts_post_route_enable_routing_eco       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                         | <p>If set to false, post-conditioning will skip its ECO-routing step.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_post_route_enable_routing_eco</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| cts_post_route_repair_drv               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                         | <p>If set to false, post-conditioning will skip its DRV-fixing step.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_post_route_repair_drv</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| cts_post_route_repair_skew_by_buffering |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | <p>If set to true, post-conditioning will attempt skew-fixing using rebuffering.<br/>Valid values: true false<br/><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_post_route_repair_skew_by_buffering</a></p>                                                                                                                                                                                                                                                            |
| cts_primary_delay_corner                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                     | <p>This specifies the delay corner in which clock tree balancing applies the slew and insertion delay targets. If more than one timing corner is defined, this must be set before running CCOpt. By default, this is set to the delay corner contained in the first member of the list of setup views provided to the set_analysis_view command.<br/>Valid values: corner name, or empty<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_primary_delay_corner</a></p> |
| cts_primary_reporting_skew_groups                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                     | <p>The primary skew groups used for reporting.<br/>By default, the value is specified as auto that automatically takes the skew group with max number of sinks<br/>as the primary reporting skew group. For invalid values, the default (auto) will be consider<br/>Valid values: a list of existing skew group names, auto or none<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> auto<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_primary_reporting_skew_groups</a></p>                                           |
| cts_primary_reporting_skew_groups_log_min_max_sinks |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                     | <p>If set to on, the sinks with the shortest and longest paths in each primary reporting skew gro<br/>be logged. If set to logv, they will be logged only to the logv file.<br/>Valid values: on, off, logv<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> logv<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_primary_reporting_skew_groups_log_min_max_sinks</a></p>                                                                                                                                                 |
| cts_repair_congestion                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>If set, place_fix_congestion is called during CTS. This can reduce congestion and ease late routing issues.<br/>Valid values: true or false<br/><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> true<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_repair_congestion</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| cts_repair_drv_by_buffering              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                          | <p>If set, the standalone clock_post_route_repair command called after route_design and the clock_post_route_repair called inside opt_design -post_route will fix clock DRVs by adding buffers, otherwise the tool will not add buffers to fix DRVs.<br/>Valid values: true false<br/><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_repair_drv_by_buffering</a></p>                                                                                                                                                                                                                                                                                                                                                              |
| cts_report_skew_groups_only_with_targets |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                          | <p>The skew groups report (run using the report_skew_groups command) displays insertion delay, skew, and min/max path information for different combinations of skew group, timing corner, and early/late path.<br/>Set the cts_report_skew_groups_only_with_targets attribute to false (the default) to report skew group/timing corner/path combinations regardless of whether a skew target has been set.<br/>Set the cts_report_skew_groups_only_with_targets attribute to true to report only on skew group/timing corner/path combinations where a skew target has been set (either explicitly or using the 'auto' setting)<br/><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_report_skew_groups_only_with_targets</a></p> |
| cts_route_clock_tree_nets                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                          | <p>Perform detailed routing, during the final implementation clock routing phase.<br/>Valid values: true or false<br/><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> true<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_route_clock_tree_nets</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| cts_route_type_flexible_htree            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Route type to use for flexible h-trees.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_route_type_flexible_htree</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| cts_route_type_flexible_htree_backside |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Route type to use for backside flexible h-trees.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_route_type_flexible_htree_backside</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| cts_route_type_leaf                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Specifies the route type. Setting this attribute binds an existing user-defined route_type to one or more types of clock tree nets. Binding a route_type to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that route_type.</p> <p>In the most common usage, the route_type is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional -net_type argument. Omitting the -net_type argument causes the route_type to be bound to all three types of clock tree nets. The optional -clock_tree &lt;pattern&gt; argument limits the binding to the clock trees whose name matches &lt;pattern&gt;. Omitting the -clock_tree argument causes the binding to apply to all clock trees.</p> <p>For a route_type to be used in CTS, it must be bound to at least one net type. If net_type is bound to any route_type, a default route_type will be created for that net type at the start of the CTS run.</p> <p>Valid values: names of route_types created with create_route_type</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_route_type_leaf</a></p> |
| cts_route_type_top                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Specifies the route type. Setting this attribute binds an existing user-defined route_type to one or more types of clock tree nets. Binding a route_type to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that route_type.</p> <p>In the most common usage, the route_type is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional -net_type argument. Omitting the -net_type argument causes the route_type to be bound to all three types of clock tree nets. The optional -clock_tree &lt;pattern&gt; argument limits the binding to the clock trees whose name matches &lt;pattern&gt;. Omitting the -clock_tree argument causes the binding to apply to all clock trees.</p> <p>For a route_type to be used in CTS, it must be bound to at least one net type. If a net type is bound to any route_type, a default route_type will be created for that net type at the start of the CTS run.</p> <p>Valid values: names of route_types created with create_route_type</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> default<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_route_type_top</a></p>   |
| cts_route_type_trunk                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                     | <p>Specifies the route type. Setting this attribute binds an existing user-defined route_type to one or more types of clock tree nets. Binding a route_type to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that route_type.</p> <p>In the most common usage, the route_type is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional -net_type argument. Omitting the -net_type argument causes the route_type to be bound to all three types of clock tree nets. The optional -clock_tree &lt;pattern&gt; argument limits the binding to the clock trees whose name matches &lt;pattern&gt;. Omitting the -clock_tree argument causes the binding to apply to all clock trees.</p> <p>For a route_type to be used in CTS, it must be bound to at least one net type. If a net type is bound to any route_type, a default route_type will be created for that net type at the start of the CTS run.</p> <p>Valid values: names of route_types created with create_route_type</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> default<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_route_type_trunk</a></p> |
| cts_routing_preferred_layer_effort  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                     | <p>Control preferred layer effort for clock routing.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> standard<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_routing_preferred_layer_effort</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| cts_safety_tmr_isolation_route_type |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Sets a route type for nets driven by TMR buffers/inverters. These should be short single fan and this route type is expected to be easily routable (eg. minimal shielding, spacing require</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> default<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_safety_tmr_isolation_route_type</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| cts_size_clock_gates   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                        | <p>When set to true (the default), the CTS algorithm sizes clock gates that appear in the clock t<br/> Setting this attribute may cause the clock tree to have a lower insertion delay, but might cha<br/> the cell types of logic gates in the clock tree, which in turn may require them to be moved sl<br/> find a legal location for the new cell.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_size_clock_gates</a></p>                                                                                                                                                                                                                                                                                                                                                                                                      |
| cts_size_clock_sources |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                        | <p>When set to true, CTS will try to size the clock source. Only clock sources that are buffers, ir<br/> logic and clock gating cells with a single output will be sized. The cells available for CTS to<br/> clock sources can be set specified using the attribute 'cts_clock_source_cells'.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_size_clock_sources</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                           |
| cts_size_logic         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                        | <p>When set to true (the default), the CTS algorithm sizes logic that appears in<br/> the clock tree. "Logic" does not include clock gates, buffers, and inverters<br/> in a clock tree, which are always sized unless they are locked, or clock<br/> generators that are above the root of the clock tree. Usually, this affects<br/> multiplexers used for selecting one of a number of clocks, or for switching<br/> between a test clock and the main clock. Setting this attribute may cause the<br/> clock tree to have a lower insertion delay, but might change the cell types of<br/> logic gates in the clock tree, which in turn may require them to be moved<br/> slightly to find a legal location for the new cell.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_size_logic</a></p> |

#### cts\_skew\_group\_report\_columns

A Tcl list of columns to include in skew group reports produced by the skew group report. You can use this attribute to specify the columns you would like to include in the skew group report, and the order in which the columns should appear.

Valid values:

capacitance  
distance  
event  
fanout  
increment  
lib\_cell  
location  
name  
slew  
status  
time

**Type:** [string](#)

**Default:** name lib\_cell event increment time slew capacitance location distance fanout status voltage

**Edit:** Yes

**Reference:** [cts\\_skew\\_group\\_report\\_columns](#)

#### cts\_skew\_group\_report\_histogram\_bin\_size



When set to a numeric value, that numeric value will be used as the histogram range size (in library units). For example, if the library time units are set to 1 nanosecond, a value of 0.010 for `report_skew_groups_histogram_bin_size` will result in histogram ranges of 10 picoseconds.

When set to auto, the size of the histogram ranges are dependent on the skew targets that are set. If a skew target is set for a given half corner and skew group combination, then the histogram range size will be 10% of the skew target for that half corner and skew target combination. If no skew target is set for a half corner and skew group combination but a skew target is set for the primary half corner and skew group combination, then the histogram range size will be 10% of the skew target for the primary half corner and skew group combination.

In the event that no skew targets are set and `report_skew_groups_histogram_bin_size` is set to auto, a default value of 10 picoseconds will be used for the histogram range size.

Valid values: auto | string

**Type:** [string](#)

**Default:** auto

**Edit:** Yes

**Reference:** [cts\\_skew\\_group\\_report\\_histogram\\_bin\\_size](#)

---

`cts_spec_config_base_pin_trace_through_to`

---

Clock tree definition will, by default, not continue through certain types of cell arc (for instance, the clock to Q arc in a DFF). This attribute allows you to override this default behavior, permitting the clock tree to trace through all instances of such a cell.

This attribute serves the same function as `trace_through_to`, except that here the clock path is specified at the level of the library cell.

The attribute should be configured on the input library pin at which the clock will arrive. The value of the attribute specifies the output library pin to which the clock should propagate. The specified output pin must be another pin on the same library cell. The output pin may be specified either by its fully qualified name (i.e. inclusive of the cell name), or else simply by its local (cell-relative) name.

There must be a pre-existing (library-defined) chain of one or more delay arcs that connect the input and output pins together. It is not possible to use `library_trace_through_to` to synthesize delay arcs.

If multiple input pins are annotated on a given library cell, the value of `library_trace_through_to` at each of those pins must select the same output pin: i.e. the configuration must identify a single clock output for the cell. If multiple clock outputs are necessary then `library_trace_through_to` should not be used: instead for each instance of the library cell, define a generated clock tree at each of the clock-carrying outputs.

If the configuration of `library_trace_through_to` settings for a given library cell does not meet these requirements, a warning will be issued and the settings for that cell will be ignored.

All instances of the library cell will be affected by this setting. If both `trace_through_to` and `library_trace_through_to` are applicable at a given instance pin, the `trace_through_to` value will take precedence.

Valid values: `base_pin`

**Type:** [string](#)  
**Default:** ""  
**Edit:** Yes  
**Reference:** [cts\\_spec\\_config\\_base\\_pin\\_trace\\_through\\_to](#)

---

`cts_spec_config_create_clock_tree_source_groups`

---

|                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | <p>Causes create_clock_tree_spec to set up a clock tree source group for SDC clocks with multiple source pins. If this attribute is set to 'active', create_clock_tree_spec defines one tree for each source pin and then uses the create_clock_tree_source_group command to collect those clock trees together, so that CTS can distribute sinks between the clock trees.</p> <p>If this attribute is set to 'inactive' or 'true', the source groups are also created during create_clock_tree_spec, but they will be defined as inactive.</p> <p>Only active source groups will take part in sink allocation. If this attribute is set to 'false', no groups are extracted from SDC multiple source pin clocks.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_spec_config_create_clock_tree_source_groups</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| cts_spec_config_create_generator_skew_groups             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                          | <p>This attribute will cause the create_clock_tree_spec command to create skew groups for sequential generators and their adjacent registers. Such skew groups will be specified with same highest rank so that they can be balanced from the other normal skew groups that share some sinks of them. The adjacent registers of a generator are registers that have a datapath path to talk with the generator directly. When this attribute is set to true, one skew group will be created per sequential generator instance, master clock and generated clock tree triple. The resulting skew groups will by default be named in the pattern:</p> <p><code>_clock_gen_&lt;master_clock_name&gt;_&lt;generator_local_name&gt;<br/> &lt;_optional_number&gt;/&lt;constraint_mode_name&gt;.</code></p> <p>For example, for a pair of generators, with the same local name "reg_clkgen", CCOpt created generated clock trees from the same master clock named "fclk" in a constraint mode named "fclk", the skew groups emitted into the clock tree specification file would be named:</p> <p><code>_clock_gen_fclk_reg_clkgen_1/func<br/> _clock_gen_fclk_reg_clkgen_2/func</code></p> <p>The prefix for the names of such skew groups is controlled by the cts_spec_config_create_generator_skew_groups_name_prefix CCOpt attribute and default "_clock_gen" (the start underscore is used to group such skew groups at the end of any skew group listing ordered by name).</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_spec_config_create_generator_skew_groups</a></p> |
| cts_spec_config_create_generator_skew_groups_name_prefix |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                   | <p>This attribute controls the skew group name prefix used for skew groups generated to balance generator flops with their adjacent flops. Default is "_clock_gen". The default has a start underscore at the beginning to cause listings of skew groups ordered by name to collect similar skew groups together at the end of a list.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> _clock_gen<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_spec_config_create_generator_skew_groups_name_prefix</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| cts_spec_config_create_reporting_only_skew_groups                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                   | <p>Controls whether spec creation will synthesize reporting-only skew groups. A reporting-only skew group is a skew group whose cts_skew_group_constrains attribute is 'none'. Such a skew group imposes no clock balancing constraint and will not be considered by CTS. The typical example is a generated clock that is synchronous to its master clock. The relationship between the two clocks implies that the sinks of both clocks should be balanced together. To model this, a clock spec created for such a design will have the skew group corresponding to the master clock span the domain of both the master clock and also the generated clock. The skew group of the generated clock is then a subset of the master clock skew group: as such it is a redundant constraint, and may be safely omitted. If synthesized, such a skew group will be created as a reporting-only skew group. When this attribute is set to false, reporting-only skew groups are completely omitted from the generated spec. When set to true, reporting-only skew groups are included in the generated spec, but are marked with the cts_skew_group_constrains attribute set to 'none'. Either way, they impose no balancing constraint.</p> <p>Valid values: true false</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_spec_config_create_reporting_only_skew_groups</a></p> |
| cts_spec_config_trace_multi_output_cells_with_single_clock_output |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

This attribute controls whether and how `create_clock_tree_spec` will extract through cells that have multiple outputs but where only one of those outputs has a clock emanating from it; call these types of instance multi-output single-clock instances.

If the feature is disabled, spec creation will not trace through multi-output single-clock instances. The resulting clock spec will specify a generated clock tree on the clock-carrying outputs of every multi-output instance. Clock-carrying inputs that are connected to each clock output by way of timing arcs will be generator inputs to the corresponding generated clock tree. The presence of these generated clock trees may impact CTS operations on the instance such as sizing, moving, and cloning.

If the feature is enabled, the resulting clock spec will include `cts_spec_config_trace_through` settings for multi-output single-clock instances. These settings will allow the instance to be treated as a clock instance, most commonly as a clock logic. In this case there will be no need for the generated clock tree on the clock-carrying output. The treatment of instances with multiple clock-carrying outputs is unchanged: on such an instance each clock-carrying output will take a generated clock tree, as before.

The feature may be enabled conditionally, such that it is only applied to multi-output single-clock instances which are combinational. In this case multi-output single-clock instances which are sequential will continue to take a generated clock tree on the clock output.

The possible values for this attribute are as follows:

- false - The feature is completely disabled.
- true - The feature is enabled for all cell types; both combinational and sequential.
- combinational - The feature is enabled for combinational cells, but disabled for sequential.

Valid values: false true combinational

**Type:** [string](#)  
**Default:** combinational  
**Edit:** Yes  
**Reference:** [cts\\_spec\\_config\\_trace\\_multi\\_output\\_cells\\_with\\_single\\_clock\\_output](#)

`cts_target_max_capacitance`

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto   double</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> delay_corner<br/> <b>Default:</b> top auto trunk auto leaf auto<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_target_max_capacitance</a></p> |
| cts_target_max_capacitance_leaf |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                 | <p>The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto   double</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> delay_corner<br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_target_max_capacitance_leaf</a></p>                     |
| cts_target_max_capacitance_top  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto   double</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> delay_corner<br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_target_max_capacitance_top</a></p>   |
| cts_target_max_capacitance_trunk |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                  | <p>The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto   double</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Allowed -index values:</b> delay_corner<br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">cts_target_max_capacitance_trunk</a></p> |
| cts_target_max_stage_delay_sigma |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>The max per-stage SOCV delay sigma target to use for CTS.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> delay_corner</p> <p><b>Default:</b> ignore</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_target_max_stage_delay_sigma</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| cts_target_max_transition_time      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                     | <p>The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner.</p> <p>Valid values: default   auto   ignore   double</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> delay_corner power_domain</p> <p><b>Default:</b> top default trunk default leaf default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_target_max_transition_time</a></p> |
| cts_target_max_transition_time_leaf |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                     | <p>The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner.</p> <p>Valid values: default   auto   ignore   double</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> delay_corner power_domain</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_target_max_transition_time_leaf</a></p>                           |
| cts_target_max_transition_time_top  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner.</p> <p>Valid values: default   auto   ignore   double</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> delay_corner power_domain</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_target_max_transition_time_top</a></p>   |
| cts_target_max_transition_time_trunk |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                      | <p>The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner.</p> <p>Valid values: default   auto   ignore   double</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> delay_corner power_domain</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_target_max_transition_time_trunk</a></p> |
| cts_target_skew                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                 | <p>This specifies the target skew for clock tree balancing. This may be set to a numeric value, or one of 'auto', 'ignore' or 'default'.</p> <p>If set to 'auto' this indicates that an appropriate skew target should be computed.</p> <p>If set to 'ignore' this indicates that skew should not be balanced for this corner/path combination.</p> <p>If unspecified then the value of this attribute is 'default'.</p> <p>If the value of the attribute is 'default' the target skew for late delays in the primary delay corner is interpreted as 'auto' and as 'ignore' otherwise.</p> <p>Valid values: default   auto   ignore   double</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Allowed -index values:</b> delay_corner</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_target_skew</a></p>                                                                                                                                                                                                            |
| cts_timing_connectivity_based_skew_groups                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                                 | <p>SDC false path assertions may render a generated SDC clock asynchronous to its master clock. Under such a condition, the sinks of the generated clock need not be balanced against the sinks of the master clock: there are no timing paths between the two clocks that can be affected by inter-clock skew. This balancing 'relaxation' is realized by adjusting the skew groups that are generated for the two clocks.</p> <p>This attribute specifies which SDC assertions are considered when deciding whether a generated clock is asynchronous to its master. Valid values for this attribute are as follows:</p> <p>off In this mode, every generated SDC clock is treated as being synchronous to its master SDC clock.</p> <p>clock_false_path Only Clock/clock set_false_path and set_clock_group assertions are considered.</p> <p>Valid values: off clock_false_path</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> off</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cts_timing_connectivity_based_skew_groups</a></p> |
| cts_timing_connectivity_based_skew_groups_balance_master_clocks |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

This attribute has no effect if `cts_timing_connectivity_based_skew_groups` is set to 'off'.  
With timing connectivity based skew groups enabled, this attribute controls how CCOpt will address the synchrony of disjoint (i.e. non-overlapping) master (i.e. non-generated) SDC clocks.  
When this attribute is set to false, all disjoint master clocks are assumed to be mutually asynchronous. The skew groups generated by `create_clock_tree_spec` will not constrain the sinks of two such clocks to be balanced together.  
When this attribute is set to true, the synchrony of disjoint master clocks is determined by consulting the SDC assertions in accordance with the value of attribute `cts_timing_connectivity_based_skew_groups`.  
Additional skew groups will be generated by `create_clock_tree_spec`, such that the sinks of each pair of synchronous disjoint master clocks will be balanced together by CTS.  
Valid values true false  
**Type:** `bool`  
**Default:** false  
**Edit:** Yes  
**Reference:** `cts_timing_connectivity_based_skew_groups_balance_master_clocks`

`cts_timing_connectivity_info`

This attribute is populated by clock spec creation when timing connectivity based skew groups are enabled. Configuration of this attribute will be written to the generated spec. It documents the clock/clock balancing relationships that were determined for the production of timing connectivity based skew groups. This attribute is documenting only.

If timing connectivity based skew groups are not enabled, this attribute is not populated by spec creation.

The value of the attribute is a set of nested Tcl dictionaries, four levels in total. The top-level dictionary indicates the information category. Currently there is only one such category: 'clock\_relationships'. The second level is keyed on constrained mode name; the lowest level dictionaries are both keyed on SDC clock name. These keys taken together identify two SDC clocks in a given constraint mode. The leaf value specifies the resolved balancing relationships between the two SDC clocks. It an enum with the following members:

- direct - The two clocks are determined to belong to the same clock group and must be balanced together.
- indirect - The two clocks do not belong to the same clock group; however by transitive closure they must balance together.

Clocks in a direct or indirect balancing relationship will share one or more skew groups so as to ensure that their sinks are balanced together by CTS. Missing entries should be taken as 'need not balance'; i.e. the pair of SDC clocks have neither a direct nor indirect balancing relationship.

The indirect balancing relationship can be explored in more detail using this same Tcl dictionary to probe out the transitive closure over the direct balancing relationships that link two indirect balance clocks together. For example if clock A directly balances with clock B only, and clock B directly balances with clock C only; then clock A indirectly balances with clock C. Below is an example dict value, corresponding to clock groups {clkA clkB} {clkB clkC} in constraint mode cm1, and clock groups {clkP clkQ} {clkR clkS} in constraint mode cm2.

```
{clock_relationships {cm1 {clkA {clkB direct clkC indirect} clkB {clkA direct clkC direct} clkC indirect clkB direct}} cm2 {clkP {clkQ direct} clkQ {clkP direct} clkR {clkS direct} clkS {clkR di
```

Notice that although they are in separate clock groups, clkA and clkC nevertheless indirectly balance. This is due to fact that clkB is present in both clock groups. Note also that we omit information about the relationship between clkP and clkS: they are 'need not balance'.

**Type:** [string](#)  
**Default:** ""  
**Edit:** Yes  
**Reference:** [cts\\_timing\\_connectivity\\_info](#)

cts\_top\_fanout\_threshold

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Minimum number of transitive fanout in the clock tree for a net to be routed as a top net. Nets with at least this many sinks in their transitive fanout in the clock tree will have the special routing rules applied to them.<br/>Valid values: integer<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> unset<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_top_fanout_threshold</a></p>                                                                                                                                                                                                                                                               |
| cts_update_clock_latency                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                            | <p>Determine whether to update IO latencies within ccopt_design.<br/>Valid values: true false<br/><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> true<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_update_clock_latency</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| cts_use_inverters                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                            | <p>Specifies whether clock tree synthesis should prefer to use inverters rather than buffers when balancing the clock tree. If set to true, CTS will use inverters for the clock tree balancing process. If set to false, CTS will use the minimum number of levels of inverters required to maintain logical correctness. If set to auto (the default) CTS will build a tree that is built from buffers. A very small number of inverters may be inserted for surgical balancing optimizations.<br/>Valid values: auto true false<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> auto<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_use_inverters</a></p> |
| cts_use_receiver_model_capacitance_for_drv |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                            | <p>If auto, CCOpt will use receiver model capacitance for drv if global variable timing_report_use_receiver_model_capacitance is true.<br/>If true, CCOpt will use receiver model capacitance for drv even if timing_report_use_receiver_model_capacitance is false.<br/>If false, CCOpt will not use receiver model capacitance for drv.<br/>Valid values: auto true false<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">cts_use_receiver_model_capacitance_for_drv</a></p>                                                                                                                              |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| current_design                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>current design</p> <p><b>Type:</b> <a href="#">obj(design)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                 |
| current_tech                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>The tech for the current_design. This attribute is used to make it easier to type tech DPO na without the leading tech name. For example, if the current_tech = tech:tech1, then both "get layer:tech1/metal1" and "get_db layer:metal1" return the same layer object. current_tech ca directly modified—you must change current_design instead.</p> <p><b>Type:</b> <a href="#">obj(tech)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| cwd_setup_file                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>Developer setup file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">cwd_setup_file</a></p>                                                                                                                                                                                                                                                                      |
| dataflow_hinsts                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>specify hinst to be placed</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                         |
| def_scan_chains                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>Short-cut for [get_db current_design .def_scan_chains].</p> <p><b>Type:</b> <a href="#">obj(def_scan_chain)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                               |
| delay_corners                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>Returns the information about the delay corners in the design.</p> <p><b>Type:</b> <a href="#">obj(delay_corner)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                          |
| delaycal_advanced_accuracy_mode |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                         |                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>advanced accuracy level</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> medium high ultra</p> <p><b>Default:</b> medium</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                 |
| delaycal_advanced_node_pin_cap_settings |                                                                                                                                                                                                                                                                                                                                            |
|                                         | <p>Enable advanced node pin cap settings.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_advanced_node_pin_cap_settings</a></p>                                                                                                               |
| delaycal_advanced_pin_cap_mode          |                                                                                                                                                                                                                                                                                                                                            |
|                                         | <p>Enable advanced receiver pin cap mode for base delay.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 0 1 2</p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_advanced_pin_cap_mode</a></p>                                                                            |
| delaycal_combine_mmmc                   |                                                                                                                                                                                                                                                                                                                                            |
|                                         | <p>Specifies whether delay calculation runs are to be combined for delay calculation simulation.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none early_late early_late_corner</p> <p><b>Default:</b> early_late_corner</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_combine_mmmc</a></p> |
| delaycal_default_net_delay              |                                                                                                                                                                                                                                                                                                                                            |
|                                         | <p>set default net delay.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 1000ps</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_default_net_delay</a></p>                                                                                                                                         |
| delaycal_default_net_load               |                                                                                                                                                                                                                                                                                                                                            |
|                                         | <p>set default net load.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 0.5pf</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_default_net_load</a></p>                                                                                                                                            |

|                                       |                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| delaycal_degrade_slew_on_early_nets   |                                                                                                                                                                                                                                                                                                 |
|                                       | <p>Controls interconnect slew degradation for early paths.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_degrade_slew_on_early_nets</a></p>                                                        |
| delaycal_disable_positive_slack_shift |                                                                                                                                                                                                                                                                                                 |
|                                       | <p>disable positive slack shift caused by BEOL variation.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_disable_positive_slack_shift</a></p>                                                      |
| delaycal_disable_randomcell_in_socv   |                                                                                                                                                                                                                                                                                                 |
|                                       | <p>disable randomcell sensitivity in socv</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_disable_randomcell_in_socv</a></p>                                                                        |
| delaycal_early_irdrop_data_type       |                                                                                                                                                                                                                                                                                                 |
|                                       | <p>IRDrop EIV DB EivMethod for early corner</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> best_average worst best average worst_average</p> <p><b>Default:</b> worst</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_early_irdrop_data_type</a></p> |
| delaycal_enable_accurate_lle_analysis |                                                                                                                                                                                                                                                                                                 |
|                                       | <p>enable accurate lle analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_enable_accurate_lle_analysis</a></p>                                                                                |
| delaycal_enable_corner_based_beol     |                                                                                                                                                                                                                                                                                                 |
|                                       | <p>Enable Anchor corner based BEOL analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_enable_corner_based_beol</a></p>                                                                        |



|                                                      |                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| delaycal_enable_high_fanout                          |                                                                                                                                                                                                                                                                                        |
|                                                      | <p>Enables the default net delay which will be annotated on high fanout nets</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_enable_high_fanout</a></p>                                    |
| delaycal_enable_input_slew_sensitivity_on_constraint |                                                                                                                                                                                                                                                                                        |
|                                                      | <p>Enable input slew ensitivity on constraint</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_enable_input_slew_sensitivity_on_constraint</a></p>                                          |
| delaycal_enable_quiet_receivers_for_hold             |                                                                                                                                                                                                                                                                                        |
|                                                      | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_enable_quiet_receivers_for_hold</a></p>                                                                                                        |
| delaycal_enable_seperate_wire_var_sigma_multiplier   |                                                                                                                                                                                                                                                                                        |
|                                                      | <p>Allow seperate reporting sigma multiplier for wire variations. To be used along with option - wire_var_reporting_sigma_multiplier</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                 |
| delaycal_enable_si                                   |                                                                                                                                                                                                                                                                                        |
|                                                      | <p>Enables SIAware delay calculation that also includes cross-talk induced delays.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_enable_si</a></p> |
| delaycal_enable_wire_load_model                      |                                                                                                                                                                                                                                                                                        |
|                                                      | <p>Enables support of Liberty wire-load models and related SDC commands.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_enable_wire_load_model</a></p>                                         |

|                                                     |                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| delaycal_equivalent_waveform_model                  |                                                                                                                                                                                                                                                                                              |
|                                                     | <p>Controls the equivalent waveform model to be used</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none no_propagation propagation</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_equivalent_waveform_model</a></p> |
| delaycal_equivalent_waveform_model_for_timing_check |                                                                                                                                                                                                                                                                                              |
|                                                     | <p>Enable EWM for timing check delay.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_equivalent_waveform_model_for_timing_check</a></p>                                                         |
| delaycal_equivalent_waveform_type                   |                                                                                                                                                                                                                                                                                              |
|                                                     | <p>Equivalent waveform model type to be used</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> moments simulation</p> <p><b>Default:</b> simulation</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_equivalent_waveform_type</a></p>                 |
| delaycal_honor_slew_propagate_constraint            |                                                                                                                                                                                                                                                                                              |
|                                                     | <p>Determines whether to propagate the slew from the disabled timing arcs to the output pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_honor_slew_propagate_constraint</a></p>              |
| delaycal_ignore_net_load                            |                                                                                                                                                                                                                                                                                              |
|                                                     | <p>Uses zero-cap, zero-resistance wire-load model for estimating delays.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_ignore_net_load</a></p>                                                 |
| delaycal_input_transition_delay                     |                                                                                                                                                                                                                                                                                              |

|                                     |                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>set default input transition time.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 0ps</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_input_transition_delay</a></p>                                                                                |
| delaycal_irdrop_data_type           |                                                                                                                                                                                                                                                                                                 |
|                                     | <p>IRDrop EIV DB EivMethod</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> best_average worst best average worst_average</p> <p><b>Default:</b> worst</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_irdrop_data_type</a></p>                        |
| delaycal_irdrop_window_based        |                                                                                                                                                                                                                                                                                                 |
|                                     | <p>Fetch window based EIV values from EIV DB</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none late early both</p> <p><b>Default:</b> both</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_irdrop_window_based</a></p>                             |
| delaycal_late_irdrop_data_type      |                                                                                                                                                                                                                                                                                                 |
|                                     | <p>IRDrop EIV DB EivMethod for late corner</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> best_average worst best average worst_average</p> <p><b>Default:</b> worst</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_late_irdrop_data_type</a></p>   |
| delaycal_library_interpolation_mode |                                                                                                                                                                                                                                                                                                 |
|                                     | <p>library scaling mode for interpolation flow</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> linear non_linear non_linear_advanced</p> <p><b>Default:</b> linear</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_library_interpolation_mode</a></p> |
| delaycal_ptr_effort                 |                                                                                                                                                                                                                                                                                                 |

|                                             |                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>PTR analysis effort. Should be between 0 and 1: 0 - lowest accuracy and runtime, 1 - highest accuracy and runtime</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.5</p> <p><b>Edit:</b> Yes</p>                                                                                  |
| delaycal_report_out_bound                   |                                                                                                                                                                                                                                                                                                         |
|                                             | <p>Generates a report that contains a list of index values (input transition) in the delay tables that are beyond the index range.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_report_out_bound</a></p> |
| delaycal_separate_params_for_via            |                                                                                                                                                                                                                                                                                                         |
|                                             | <p>Create separate parameters for Via variation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_separate_params_for_via</a></p>                                                                             |
| delaycal_signoff_alignment_settings         |                                                                                                                                                                                                                                                                                                         |
|                                             | <p>Turn on/off signoff alignment settings</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_signoff_alignment_settings</a></p>                                                                                |
| delaycal_skip_slew_merge_from_disabled_path |                                                                                                                                                                                                                                                                                                         |
|                                             | <p>Enable slew merge from disabled path</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_skip_slew_merge_from_disabled_path</a></p>                                                                          |
| delaycal_slew_out_bound_limit_high          |                                                                                                                                                                                                                                                                                                         |
|                                             | <p>Limits the maximum slew used in the analysis to the specified value.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 3.40282e+38</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_slew_out_bound_limit_high</a></p>                                           |
| delaycal_slew_out_bound_limit_low           |                                                                                                                                                                                                                                                                                                         |

|                                      |                                                                                                                                                                                                                                                                                  |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Limits the minimum slew used in the analysis to the specified value.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.5</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_slew_out_bound_limit_low</a></p>                             |
| delaycal_socv_accuracy_mode          |                                                                                                                                                                                                                                                                                  |
|                                      | <p>level-based-accuracy-effort in AAE for delays and slews</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high ultra</p> <p><b>Default:</b> low</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_socv_accuracy_mode</a></p> |
| delaycal_socv_lvf_mode               |                                                                                                                                                                                                                                                                                  |
|                                      | <p>Controls interpretation of LVF data.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> moments early_late</p> <p><b>Default:</b> early_late</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_socv_lvf_mode</a></p>                     |
| delaycal_socv_machine_learning_level |                                                                                                                                                                                                                                                                                  |
|                                      | <p>Enable socv machine learning mode</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_socv_machine_learning_level</a></p>                                                                  |
| delaycal_socv_use_lvf_tables         |                                                                                                                                                                                                                                                                                  |
|                                      | <p>Controls which type of variations are considered during analysis.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> all</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_socv_use_lvf_tables</a></p>                                     |
| delaycal_support_output_pin_cap      |                                                                                                                                                                                                                                                                                  |
|                                      | <p>useOutputPinCap</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_support_output_pin_cap</a></p>                                                                                     |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| delaycal_support_wire_load_model            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                             | <p>Enables support of Liberty wire-load models and related SDC commands.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_support_wire_load_model</a></p>                                                                                                                                                                                                                                          |
| delaycal_timing_create_clock_use_ideal_slew |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                             | <p>When set to true, the software detects created clocks and uses zero input slew. Other slew , including those set using set_annotated_transition command, will be ignored. When set to the software honors set_annotated_transition command settings. By default, this global var set to false.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_timing_create_clock_use_ideal_slew</a></p> |
| delaycal_use_default_delay_limit            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                             | <p>set threshold to apply the default delay</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_use_default_delay_limit</a></p>                                                                                                                                                                                                                                                                    |
| delaycal_via_corner_sigma_multiplier        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                             | <p>Provide scaling multiplier for corner scaling</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_via_corner_sigma_multiplier</a></p>                                                                                                                                                                                                                                                          |
| delaycal_via_correlation_type               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                             | <p>How to handle inter-layer correlation. accepted values corner</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> corner</p> <p><b>Default:</b> corner</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_via_correlation_type</a></p>                                                                                                                                                                                                             |
| delaycal_via_reset                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                                               |                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | <p>reset via variation modes to default value</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_via_reset</a></p>                                                                                                                             |
| delaycal_wire_corner_sigma_multiplier         |                                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>Provide scaling multiplier for corner scaling</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_wire_corner_sigma_multiplier</a></p>                                                                                                        |
| delaycal_wire_correlation_type                |                                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>How to handle inter-layer correlation. accepted values statistical binomial corner</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> statistical correlated binomial corner</p> <p><b>Default:</b> statistical</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_wire_correlation_type</a></p> |
| delaycal_wire_layer_avg_rpt                   |                                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>Enable layer averaging reporting( true/false )</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_wire_layer_avg_rpt</a></p>                                                                                                                |
| delaycal_wire_rc_correlation_factor           |                                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>same layer R&amp;C correlation. accepted values ( 1   -1 )</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_wire_rc_correlation_factor</a></p>                                                                                                 |
| delaycal_wire_reset                           |                                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>reset wire variation modes to default value</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_wire_reset</a></p>                                                                                                                           |
| delaycal_wire_use_nsigma_coupling_capacitance |                                                                                                                                                                                                                                                                                                                                         |

|                                                |                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Use mean+nSigma coupling caps( true/false )</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_wire_use_nsigma_coupling_capacitance</a></p>                                                             |
| delaycal_wire_use_nsigma_slew_for_timing_check |                                                                                                                                                                                                                                                                                                     |
|                                                | <p>Use Wire Var aware slews for timing check lookup( true/false )</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_wire_use_nsigma_slew_for_timing_check</a></p>                                         |
| delaycal_wire_var_reporting_sigma_multiplier   |                                                                                                                                                                                                                                                                                                     |
|                                                | <p>Deterministic sigma multiplier for reporting beol variations</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_wire_var_reporting_sigma_multiplier</a></p>                                             |
| delaycal_wire_variation_correlation_mode       |                                                                                                                                                                                                                                                                                                     |
|                                                | <p>Specify wire variatio inter layer correlation</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> statistical binomial</p> <p><b>Default:</b> statistical</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_wire_variation_correlation_mode</a></p>          |
| delaycal_wire_variation_rc_correlation         |                                                                                                                                                                                                                                                                                                     |
|                                                | <p>Same layer RC correlation for wire variation</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_wire_variation_rc_correlation</a></p>                                                                   |
| delaycal_wire_wire_cell_variation_correlation  |                                                                                                                                                                                                                                                                                                     |
|                                                | <p>Whether to RSS or add the random-cell variation and wire variation for arrivals and slack( true/false )</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delaycal_wire_wire_cell_variation_correlation</a></p> |



|                                      |                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| delete_flops_on_preserved_net        |                                                                                                                                                                                                                                                                |
|                                      | <p>Allows to delete the flops connected to a preserved net.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delete_flops_on_preserved_net</a></p>                            |
| delete_hier_insts_on_preserved_net   |                                                                                                                                                                                                                                                                |
|                                      | <p>Allows to delete the hierarchical instances connected to a preserved net.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delete_hier_insts_on_preserved_net</a></p>      |
| delete_unloaded_insts                |                                                                                                                                                                                                                                                                |
|                                      | <p>Control deletion of unloaded hierarchical instances.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delete_unloaded_insts</a></p>                                        |
| delete_unloaded_seqs                 |                                                                                                                                                                                                                                                                |
|                                      | <p>Control deletion of unloaded sequential instances.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">delete_unloaded_seqs</a></p>                                           |
| design_backside_bottom_routing_layer |                                                                                                                                                                                                                                                                |
|                                      | <p>Specifies the highest backside lef layer name for global and detail routing.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_backside_bottom_routing_layer</a></p> |
| design_backside_top_routing_layer    |                                                                                                                                                                                                                                                                |
|                                      | <p>Specifies the lowest backside lef layer name for global and detail routing.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_backside_top_routing_layer</a></p>     |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| design_bottom_routing_layer |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                             | <p>Specifies the lowest LEF layer name or layer number for global and detail routing. Layer number from the LEF layer sequence. For example, 2 is equivalent to the second routing layer defined in the LEF.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">design_bottom_routing_layer</a></p>                                    |
| design_compressed_pg_db     |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                             | <p>Specifies whether the compressed pg feature is enabled. When it's true, read_db will automatically compress the PG data when load the design. You can use "get_db current_design .is_pg_compressed" to check if PG in this design is compressed.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">design_compressed_pg_db</a></p> |
| design_cong_effort          |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                             | <p>Specify congestion effort level</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> low medium high auto<br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">design_cong_effort</a></p>                                                                                                                                                                         |
| design_dual_rail_via_pitch  |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                             | <p>Min stacked-via pitch for dual std-cell power-rails on first and third routing layers. Space specified is the min stacked-via pitch for dual std-cell power-rails on first and third routing layers.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">design_dual_rail_via_pitch</a></p>                                          |
| design_early_clock_flow     |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                             | <p>Enable early clock flow</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">design_early_clock_flow</a></p>                                                                                                                                                                                                                         |
| design_early_pba_mode       |                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                  |                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | Specify early PBA mode<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> none high<br><b>Default:</b> none<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">design_early_pba_mode</a>                                                                                                                                        |
| design_express_route             |                                                                                                                                                                                                                                                                                                                                            |
|                                  | With "-flow_effort express", should routing-stage be forced.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">design_express_route</a>                                                                                                                                   |
| design_flow_effort               |                                                                                                                                                                                                                                                                                                                                            |
|                                  | Specify flow effort level<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> standard extreme<br><b>Default:</b> standard<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">design_flow_effort</a>                                                                                                                             |
| design_high_freq_interposer_flow |                                                                                                                                                                                                                                                                                                                                            |
|                                  | Specifies the flow is a high frequency interposer flow. When set to true, set_integration_route_constraint will auto turn off the options which are not suitable for inte routing.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">design_high_freq_interposer_flow</a> |
| design_ideal_hold_fixing         |                                                                                                                                                                                                                                                                                                                                            |
|                                  | Enable ideal hold fixing flow<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">design_ideal_hold_fixing</a>                                                                                                                                                              |
| design_ignore_followpin_vias     |                                                                                                                                                                                                                                                                                                                                            |

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>if true, ignore followpin vias during detailed placement, optimization and routing.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_ignore_followpin_vias</a></p>                                                                                                                                                                                            |
| design_merge_trim_shapes                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>Specify the setting for trim shapes merge. layers, specify the trim layer list which need merge shapes. gap, the trim shapes when end to end gap is less the value. max_trim_num, specify max trim shapes number are merged. masks, specify the trim shapes color need to be merged</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_merge_trim_shapes</a></p> |
| design_optimization_density_screen_margin |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>Increase a margin on top of original density screen for optimization</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_optimization_density_screen_margin</a></p>                                                                                                                                                                                              |
| design_pessimistic_mode                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>Enable pessimistic mode</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_pessimistic_mode</a></p>                                                                                                                                                                                                                       |
| design_power_effort                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                           | <p>Specify power effort level</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none low high</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_power_effort</a></p>                                                                                                                                                                                                                      |
| design_slack_weighting_method             |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Specify slack weighting to be used in implementation flow</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> unity_weighting view_based_weighting</p> <p><b>Default:</b> unity_weighting</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_slack_weighting_method</a></p>                                                                                           |
| design_tech_node           |                                                                                                                                                                                                                                                                                                                                                                                                           |
|                            | <p>set design tech node</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> N22 N12 N10 N7 N7Plus N6 N5 N5PlusPlus N4 N3 N3E N2 A16 A14 A12 S10 SF8 SF7 SF5 SF4 SF3 SF2 SF1.4 G7 G5 ICF I7 I5 P1222 P1276 P1277 P1278 P1278. P1280 20A 18A C12 C7 R2 unspecified</p> <p><b>Default:</b> unspecified</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_tech_node</a></p> |
| design_top_routing_layer   |                                                                                                                                                                                                                                                                                                                                                                                                           |
|                            | <p>Specifies the highest LEF layer name or layer number for global and detail routing. Layer n is from the LEF layer sequence. For example, 2 is equivalent to the second routing layer de LEF.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_top_routing_layer</a></p>                                        |
| design_trim_grid_group     |                                                                                                                                                                                                                                                                                                                                                                                                           |
|                            | <p>Specifies the GROUP name of which set of trim metal grid to be used for placement and routing. See the LEF documentation on the TRIMMETALTRACK keyword for more details.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_trim_grid_group</a></p>                                                              |
| design_verification_effort |                                                                                                                                                                                                                                                                                                                                                                                                           |
|                            | <p>Specifies the default design verification effort level for LEC. Can be used to disable problem optimizations in bulk. Used for any module for which the module-level attribute value is set value of 'inherited'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> high</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">design_verification_effort</a></p>               |

|                                                        |                                                                                                                                                                                                                                                                |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| designs                                                |                                                                                                                                                                                                                                                                |
|                                                        | <p>All the designs</p> <p><b>Type:</b> <a href="#">obj(design)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                       |
| dft_1500_hierarchical_parent_child_wrapping            |                                                                                                                                                                                                                                                                |
|                                                        | <p>Wrap parent in presence of wrapped child hierarchy.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                       |
| dft_1500_hierarchical_softcore_wrapping                |                                                                                                                                                                                                                                                                |
|                                                        | <p>Ignore wrapped soft-core while wrapping the top level ports.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                              |
| dft_1500_optimize_wrapper_insertion_bw_two_hierarchies |                                                                                                                                                                                                                                                                |
|                                                        | <p>Optimizing insertion of wrapper cells in between two hierarchies.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                         |
| dft_add_mux_on_pre_connected_ctl_si                    |                                                                                                                                                                                                                                                                |
|                                                        | <p>Adds a MUX when SI pin of a CTL Segment is pre-connected to a functional pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_add_mux_on_pre_connected_ctl_si</a></p> |
| dft_allow_dwc_in_top                                   |                                                                                                                                                                                                                                                                |
|                                                        | <p>Do not print segment and instance info in report_scan_registers.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_allow_dwc_in_top</a></p>                            |
| dft_apply_sdc_constraints                              |                                                                                                                                                                                                                                                                |

|                                    |                                                                                                                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Enables generation and application of SDC constraints for DFT constructs.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                   |
| dft_atpg_executable                |                                                                                                                                                                                                                                                  |
|                                    | <p>Specifies Encounter Test executable called from Genus.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_atpg_executable</a></p>                          |
| dft_atpg_setup_status              |                                                                                                                                                                                                                                                  |
|                                    | <p>This will tell the status of prepare for test point analysis command.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">dft_atpg_setup_status</a></p>          |
| dft_auto_create_chains_as_internal |                                                                                                                                                                                                                                                  |
|                                    | <p>Auto-create scan chains as internal.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_auto_create_chains_as_internal</a></p>                            |
| dft_auto_identify_shift_register   |                                                                                                                                                                                                                                                  |
|                                    | <p>Automatically identifies functional shift register segments.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_auto_identify_shift_register</a></p>      |
| dft_boundary_cell_module_prefix    |                                                                                                                                                                                                                                                  |
|                                    | <p>Specifies the prefix to be added to the boundary scan cell modules.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_boundary_cell_module_prefix</a></p> |
| dft_boundary_scan_timing_mode_name |                                                                                                                                                                                                                                                  |

|                                   |                                                                                                                                                                                                                                                    |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | DFT boundary scan timing mode name (default: DFT_BOUNDARY_SCAN_MODE).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> DFT_BOUNDARY_SCAN_MODE<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_boundary_scan_timing_mode_name</a> |
| dft_capture_11496_reciever_output |                                                                                                                                                                                                                                                    |
|                                   | Avoids inserting 11496 bsr capture logic on receiver output if founds false.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_capture_11496_reciever_output</a>               |
| dft_capture_timing_mode_name      |                                                                                                                                                                                                                                                    |
|                                   | DFT capture timing mode name (default: DFT_CAPTURE_MODE).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> DFT_CAPTURE_MODE<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_capture_timing_mode_name</a>                         |
| dft_check_cfg_mode_aware          |                                                                                                                                                                                                                                                    |
|                                   | Allow check_dft_rules to check segments for each configuration mode.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_check_cfg_mode_aware</a>                               |
| dft_clock_waveform_divide_fall    |                                                                                                                                                                                                                                                    |
|                                   | Default falling edge denominator for test clocks.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 100<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_clock_waveform_divide_fall</a>                                               |
| dft_clock_waveform_divide_period  |                                                                                                                                                                                                                                                    |
|                                   | Default period denominator for test clocks.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 1<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_clock_waveform_divide_period</a>                                                     |
| dft_clock_waveform_divide_rise    |                                                                                                                                                                                                                                                    |



|                                             |                                                                                                                                                                                                                                        |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>Default rising edge denominator for test clocks.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_clock_waveform_divide_rise</a></p>             |
| dft_clock_waveform_fall                     |                                                                                                                                                                                                                                        |
|                                             | <p>Default falling edge numerator (fraction of period) for test clocks.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 90</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_clock_waveform_fall</a></p> |
| dft_clock_waveform_period                   |                                                                                                                                                                                                                                        |
|                                             | <p>Default period for test clocks (in picoseconds).</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 50000.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_clock_waveform_period</a></p>           |
| dft_clock_waveform_rise                     |                                                                                                                                                                                                                                        |
|                                             | <p>Default rising edge numerator (fraction of period) for test clocks.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 50</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_clock_waveform_rise</a></p>  |
| dft_compression_2d_aspect_ratio             |                                                                                                                                                                                                                                        |
|                                             | <p>Specifies the aspect ratio of the 2D compression grid to insert.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                             |
| dft_compression_2d_decomp_pipeline_distance |                                                                                                                                                                                                                                        |
|                                             | <p>Specifies the number of 2D compression grid cells between two decompressor pipelines.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                  |
| dft_compression_2d_grid_max_x               |                                                                                                                                                                                                                                        |

|                                             |                                                                                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | Specifies the x-coordinate to the left of which the 2D compression grid will be inserted.<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>no_value</code><br><b>Edit:</b> Yes  |
| dft_compression_2d_grid_max_y               |                                                                                                                                                                                             |
|                                             | Specifies the y-coordinate below which the 2D compression grid will be inserted.<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>no_value</code><br><b>Edit:</b> Yes           |
| dft_compression_2d_grid_min_x               |                                                                                                                                                                                             |
|                                             | Specifies the x-coordinate to the right of which the 2D compression grid will be inserted.<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>no_value</code><br><b>Edit:</b> Yes |
| dft_compression_2d_grid_min_y               |                                                                                                                                                                                             |
|                                             | Specifies the y-coordinate above which the 2D compression grid will be inserted.<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>no_value</code><br><b>Edit:</b> Yes           |
| dft_compression_auto_create                 |                                                                                                                                                                                             |
|                                             | Specifies to auto-create any required compression test signals that are missing.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> <code>false</code><br><b>Edit:</b> Yes                |
| dft_compression_channel_length              |                                                                                                                                                                                             |
|                                             | Specifies the compression channel length to insert.<br><b>Type:</b> <code>int</code><br><b>Default:</b> <code>0</code><br><b>Edit:</b> Yes                                                  |
| dft_compression_comp_pipeline_max_xor_depth |                                                                                                                                                                                             |
|                                             | Specifies the maximum XOR depth between two compressor pipelines.<br><b>Type:</b> <code>int</code><br><b>Default:</b> <code>0</code><br><b>Edit:</b> Yes                                    |
| dft_compression_compressor_type             |                                                                                                                                                                                             |

|                                               |                                                                                                                                                                          |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | <p>Specifies the compressor type to insert (xor   misr).<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> xor<br/><b>Edit:</b> Yes</p>                        |
| dft_compression_decomp_pipeline_max_xor_depth |                                                                                                                                                                          |
|                                               | <p>Specifies the maximum XOR depth between two decompressor pipelines.<br/><b>Type:</b> <a href="#">int</a><br/><b>Default:</b> 0<br/><b>Edit:</b> Yes</p>               |
| dft_compression_decompressor_type             |                                                                                                                                                                          |
|                                               | <p>Specifies the decompressor type to insert (elastic   xor).<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> elastic<br/><b>Edit:</b> Yes</p>               |
| dft_compression_elastic_parallel_load_support |                                                                                                                                                                          |
|                                               | <p>Specify to enable parallel loading of Elastic decompressor.<br/><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p>                  |
| dft_compression_elasticity_ratio              |                                                                                                                                                                          |
|                                               | <p>Specifies the SmartScan deserializer ratio to insert for Elastic decompression.<br/><b>Type:</b> <a href="#">int</a><br/><b>Default:</b> 1<br/><b>Edit:</b> Yes</p>   |
| dft_compression_extest_decompressor_type      |                                                                                                                                                                          |
|                                               | <p>Specifies the extest decompressor type to insert (xor   elastic).<br/><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> xor<br/><b>Edit:</b> Yes</p>            |
| dft_compression_fullscan_support              |                                                                                                                                                                          |
|                                               | <p>Specifies to insert fullscan muxing to concatenate the compression channels.<br/><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes</p> |
| dft_compression_lp_gating_sharing_ratio       |                                                                                                                                                                          |

|                                                   |                                                                                                                                                                    |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | Specifies how many compression channels are to share a common low power gating regis<br><b>Type:</b> <code>int</code><br><b>Default:</b> 1<br><b>Edit:</b> Yes     |
| dft_compression_lp_gating_support                 |                                                                                                                                                                    |
|                                                   | Specifies to insert low power compression channel gating.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> Yes                           |
| dft_compression_mask_and_lp_gating_unload_support |                                                                                                                                                                    |
|                                                   | Specifies to build unload paths for mask and low power gating registers.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> true<br><b>Edit:</b> Yes             |
| dft_compression_mask_sharing_ratio                |                                                                                                                                                                    |
|                                                   | Specifies how many compression channels are to share a common mask register.<br><b>Type:</b> <code>int</code><br><b>Default:</b> 1<br><b>Edit:</b> Yes             |
| dft_compression_mask_support                      |                                                                                                                                                                    |
|                                                   | Specifies to insert compression (wide1) masking.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> Yes                                    |
| dft_compression_masken_pipeline_depth             |                                                                                                                                                                    |
|                                                   | Specifies the number of pipeline stages to insert on the compression mask enable signal.<br><b>Type:</b> <code>int</code><br><b>Default:</b> 0<br><b>Edit:</b> Yes |
| dft_compression_num_extest_scanin                 |                                                                                                                                                                    |
|                                                   | Specifies the number of scan inputs to use for extest compression.<br><b>Type:</b> <code>int</code><br><b>Default:</b> 0<br><b>Edit:</b> Yes                       |
| dft_compression_num_extest_scanout                |                                                                                                                                                                    |

|                                       |                                                                                                                                                                                                                    |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | Specifies the number of scan outputs to use for extest compression.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes                                                                   |
| dft_compression_num_scanin            |                                                                                                                                                                                                                    |
|                                       | Specifies the number of scan inputs to auto-create for compression.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_compression_num_scanin</a>   |
| dft_compression_num_scanout           |                                                                                                                                                                                                                    |
|                                       | Specifies the number of scan outputs to auto-create for compression.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_compression_num_scanout</a> |
| dft_compression_opcg_unload_support   |                                                                                                                                                                                                                    |
|                                       | Specifies to build unload paths for OPCG programming registers.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes                                                                   |
| dft_compression_post_2d_sdc_file      |                                                                                                                                                                                                                    |
|                                       | Specifies the SDC file to read in after 2D compression is inserted.<br><b>Type:</b> <a href="#">in_file</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                              |
| dft_compression_post_2d_sdc_mode_name |                                                                                                                                                                                                                    |
|                                       | Specifies the name of the timing mode to which the specified SDC file will be applied.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                            |
| dft_compression_ratio                 |                                                                                                                                                                                                                    |
|                                       | Specifies the compression ratio to insert.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes                                                                                            |

|                                          |                                                                                                                                                                                                                                      |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dft_compression_scanin_pipeline_depth    |                                                                                                                                                                                                                                      |
|                                          | <p>Specifies the number of pipeline stages to insert on every compression scan input.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                       |
| dft_compression_scanout_pipeline_depth   |                                                                                                                                                                                                                                      |
|                                          | <p>Specifies the number of pipeline stages to insert on every compression scan output.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                      |
| dft_compression_serial_load_support      |                                                                                                                                                                                                                                      |
|                                          | <p>Specifies to enable loading/unloading the Elastic decompressor, the MISR, the mask register the low power gating register serially.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p> |
| dft_configuration_modes                  |                                                                                                                                                                                                                                      |
|                                          | <p>List of all dft_configuration_modes</p> <p><b>Type:</b> <a href="#">obj(dft_configuration_mode)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                             |
| dft_cross_lp_cells_for_lockup_clk_driver |                                                                                                                                                                                                                                      |
|                                          | <p>Cross low power cells to find lockup clock driver.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                  |
| dft_disable_wrapper_inside_non_scan_elem |                                                                                                                                                                                                                                      |
|                                          | <p>Disable wrapper from being inserted inside a dft_dont_scan hierarchy.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes</p>                                                                |
| dft_dont_merge_multibit_lockup           |                                                                                                                                                                                                                                      |

|                                                          |                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | <p>Sets dont merge multibit on lockups inserted by connect_scan_chains, add_lockup_eleme</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_dont_merge_multibit_lockup</a></p>                                                      |
| dft_dont_wrap_if_shared_threshold_exceed                 |                                                                                                                                                                                                                                                                                                                        |
|                                                          | <p>Stop port wrapping if shared threshold exceed.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                    |
| dft_enable_flop_placement_in_test_point_insertion        |                                                                                                                                                                                                                                                                                                                        |
|                                                          | <p>Enable physical placement of testpoint flop during testpoint insertion. Requires MOD400 lic</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_enable_flop_placement_in_test_point_insertion</a></p>                            |
| dft_enable_wir_function_check                            |                                                                                                                                                                                                                                                                                                                        |
|                                                          | <p>Only test signals with certain functions are allowed in the WIR. Disabling this check may res</p> <p>invalid or unsupported configurations.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_enable_wir_function_check</a></p> |
| dft_exclude_internal_flops_from_shared_wrapper_threshold |                                                                                                                                                                                                                                                                                                                        |
|                                                          | <p>Exclude the internal flops from shared wrapper threshold calculation.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                             |
| dft_exclude_tdrc_fail_seg                                |                                                                                                                                                                                                                                                                                                                        |
|                                                          | <p>Exclude abstract segments failing tdrc checks while connect scan chains.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_exclude_tdrc_fail_seg</a></p>                                                                       |
| dft_extended_scandef                                     |                                                                                                                                                                                                                                                                                                                        |

|                                     |                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | Produce extended scanDEF output to encompass almost all the scan bits in a design.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_extended_scandef</a>                                                                            |
| dft_fence_slow_speed_domains        |                                                                                                                                                                                                                                                                                                          |
|                                     | Fence slow speed domains.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_fence_slow_speed_domains</a>                                                                                                                            |
| dft_generate_atpg_no_testpoint_file |                                                                                                                                                                                                                                                                                                          |
|                                     | Generates a file for Modus that specifies where testpoints cannot be inserted { auto   auto_plus_user   user_only   none }.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> auto<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_generate_atpg_no_testpoint_file</a>                  |
| dft_generate_atpg_testpoint_ctlfile |                                                                                                                                                                                                                                                                                                          |
|                                     | Generates a file for Modus that specifies which blocks needs to be considered as fully controllable/observable during testpoint analysis { false   true }.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                             |
| dft_get_balanced_chains_assignment  |                                                                                                                                                                                                                                                                                                          |
|                                     | Controls even distribution of balanced scan chains among all scan groups, when no max length scan chain is defined.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                    |
| dft_identify_internal_test_clocks   |                                                                                                                                                                                                                                                                                                          |
|                                     | Identify internal test clocks as output pins of multi input gates {false   true   no_cgic_hier}.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> false true no_cgic_hier<br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_identify_internal_test_clocks</a> |



|                                                    |                                                                                                                                                                                                                                                                              |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dft_identify_non_boundary_shift_registers          |                                                                                                                                                                                                                                                                              |
|                                                    | <p>Determines whether identify_shift_register_scan_segments will exclude boundary registers. can be converted to shared IEEE 1500 wrappers from shift register identification.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p> |
| dft_identify_shared_wrapper_cells                  |                                                                                                                                                                                                                                                                              |
|                                                    | <p>Process generic design to identify flops to be used as shared wrapper cells.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                |
| dft_identify_test_signals                          |                                                                                                                                                                                                                                                                              |
|                                                    | <p>Specifies whether TDRC should auto identify test signals. The default value is true.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">dft_identify_test_signals</a></p>                        |
| dft_identify_top_level_test_clocks                 |                                                                                                                                                                                                                                                                              |
|                                                    | <p>Specifies whether TDRC should auto identify top level test clocks. The default value is true.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">dft_identify_top_level_test_clocks</a></p>      |
| dft_identify_xsource_violations_from_timing_models |                                                                                                                                                                                                                                                                              |
|                                                    | <p>Identifies x-source violations from output pins of timing models.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">dft_identify_xsource_violations_from_timing_models</a></p>                 |
| dft_ignore_dont_scan_for_shared_wrapper_processing |                                                                                                                                                                                                                                                                              |
|                                                    | <p>Exclude non scan flops while marking wrapper cells.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                         |
| dft_ignore_non_scan_for_wrapper_processing         |                                                                                                                                                                                                                                                                              |

|                                                   |                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | Ignore non scan logic for wrapper insertion.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                                                                   |
| dft_ignore_segment_mode_name                      |                                                                                                                                                                                                                                                                                                                  |
|                                                   | Ignore scan segment mode name values during scan chain connection.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_ignore_segment_mode_name</a>                                                                                            |
| dft_include_controllable_pins_in_abstract_model   |                                                                                                                                                                                                                                                                                                                  |
|                                                   | Enable writing out controllable pins in abstract model {allmodes   none   test_setup}.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> none test_setup allmodes<br><b>Default:</b> allmodes<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_include_controllable_pins_in_abstract_model</a> |
| dft_include_test_signal_outputs_in_abstract_model |                                                                                                                                                                                                                                                                                                                  |
|                                                   | Enable writing out test signal outputs in abstract model.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_include_test_signal_outputs_in_abstract_model</a>                                                                                |
| dft_insert_dedicated_inside_sink_hierarchy        |                                                                                                                                                                                                                                                                                                                  |
|                                                   | Insert dedicated wrapper inside sink hierarchy.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                                                                |
| dft_is_add_los_scannable                          |                                                                                                                                                                                                                                                                                                                  |
|                                                   | Add scan_flop when scannable is true for add_los.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">dft_is_add_los_scannable</a>                                                                                                                |
| dft_jtag_instance_name                            |                                                                                                                                                                                                                                                                                                                  |

|                                    |                                                                                                                                                                                                                                                                           |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Specifies the name of the the JTAG instance.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> JTAG_MODULE</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_jtag_instance_name</a></p>                                                 |
| dft_jtag_module_name               |                                                                                                                                                                                                                                                                           |
|                                    | <p>Specifies the name of the the JTAG module.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> JTAG_MACRO</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_jtag_module_name</a></p>                                                      |
| dft_lbist_capture_timing_mode_name |                                                                                                                                                                                                                                                                           |
|                                    | <p>DFT LBIST capture timing mode name (default: DFT_LBIST_CAPTURE_MODE).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> DFT_LBIST_CAPTURE_MODE</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_lbist_capture_timing_mode_name</a></p> |
| dft_lbist_shift_timing_mode_name   |                                                                                                                                                                                                                                                                           |
|                                    | <p>DFT LBIST shift timing mode name (default: DFT_LBIST_SHIFT_MODE).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> DFT_LBIST_SHIFT_MODE</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_lbist_shift_timing_mode_name</a></p>         |
| dft_license_scheme                 |                                                                                                                                                                                                                                                                           |
|                                    | <p>Dft tool selection {Modus   Encounter_Test}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> Modus Encounter_Test</p> <p><b>Default:</b> Modus</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_license_scheme</a></p>             |
| dft_lockup_name_prefix             |                                                                                                                                                                                                                                                                           |
|                                    | <p>Prefix to lockup latch and flip-flop instances.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> DFT_</p> <p><b>Edit:</b> Yes</p>                                                                                                                     |
| dft_modedef_internal               |                                                                                                                                                                                                                                                                           |

|                                     |                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Use internal modedef file for build testmode when writing out the ATPG scripts.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_modedef_internal</a></p>                                                                     |
| dft_modus_version                   |                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>Set the Modus version.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 22.11</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_modus_version</a></p>                                                                                                                               |
| dft_opcg_block_input_to_flop_paths  |                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>Propagate DFT clock information from primary inputs.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_opcg_block_input_to_flop_paths</a></p>                                                                                  |
| dft_opcg_domain_blocking            |                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>Enable OPCG domain blocking.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_opcg_domain_blocking</a></p>                                                                                                                    |
| dft_opcg_timing_mode_name           |                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>DFT OPCG timing mode name (default: DFT_OPCG_MODE).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> DFT_OPCG_MODE</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_opcg_timing_mode_name</a></p>                                                                                  |
| dft_optimize_chain_wirelength_level |                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>Level of optimization needed for physical scan flow, chain wirelength. Appropriate values for attribute are low , medium or high.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> medium</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_optimize_chain_wirelength_level</a></p> |
| dft_physical_aware_test_points      |                                                                                                                                                                                                                                                                                                                        |

|                                       |                                                                                                                                                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>Enable physically aware test point sharing.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_physical_aware_test_points</a></p>                                           |
| dft_physical_pd_aware_scan_connection |                                                                                                                                                                                                                                                                   |
|                                       | <p>Allow physically aware power domain scan connection.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_physical_pd_aware_scan_connection</a></p>                          |
| dft_pmbist_jtag_timing_mode_name      |                                                                                                                                                                                                                                                                   |
|                                       | <p>DFT PMBIST jtag timing mode name (default: DFT_PMBIST_JTAG_MODE).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> DFT_PMBIST_JTAG_MODE</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_pmbist_jtag_timing_mode_name</a></p> |
| dft_pmbist_mda_timing_mode_name       |                                                                                                                                                                                                                                                                   |
|                                       | <p>DFT PMBIST mda timing mode name (default: DFT_PMBIST_MDA_MODE).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> DFT_PMBIST_MDA_MODE</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_pmbist_mda_timing_mode_name</a></p>     |
| dft_power_aware_lockup_clock_driver   |                                                                                                                                                                                                                                                                   |
|                                       | <p>Enable power aware lockup clock driver.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_power_aware_lockup_clock_driver</a></p>                                         |
| dft_power_aware_wrapper_insertion     |                                                                                                                                                                                                                                                                   |
|                                       | <p>Power aware wrapper insertion.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                                                                                |
| dft_prefix                            |                                                                                                                                                                                                                                                                   |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | <p>Prefix to DFT-generated object names.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> DFT_</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_prefix</a></p>                                                                                                                                                                                                                         |
| dft_process_multibit_for_shared_wrapper          |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                  | <p>Process multibit cells for shared wrapper cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                   |
| dft_propagate_test_signals_from_hookup_pins_only |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                  | <p>Specifies whether TDRC should propagate test signals values from both the pin and hookup just the hookup pin. The default is false, indicating that test signals values are propagated from both the pin and hookup pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_propagate_test_signals_from_hookup_pins_only</a></p> |
| dft_report_empty_test_clocks                     |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                  | <p>Have dft report commands report information on test clocks that do not drive any registers.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_report_empty_test_clocks</a></p>                                                                                                                                                  |
| dft_report_scan_register_quiet                   |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                  | <p>Do not print segment and instance info in report_scan_registers.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_report_scan_register_quiet</a></p>                                                                                                                                                                           |
| dft_retiming_connections_local_clock_hookup      |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                  | <p>Use local hookup for retiming clock connections</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                     |
| dft_retiming_connections_local_se_hookup         |                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                                            |                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            | <p>Use local hookup for retiming SE connections</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                                                                |
| dft_retiming_head_pipeline_length                          |                                                                                                                                                                                                                                                                 |
|                                                            | <p>Number of pipeline retiming flops to be inserted at chain tail.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                                 |
| dft_retiming_logic_neg_head_chain                          |                                                                                                                                                                                                                                                                 |
|                                                            | <p>Specifies the logic to be inserted at head of negative head chain. Possible Values : 'pos_flop' 'pos_flop neg_flop' 'none' (default : none).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p>              |
| dft_retiming_logic_neg_tail_chain                          |                                                                                                                                                                                                                                                                 |
|                                                            | <p>Specifies the logic to be inserted at tail of negative tail chain. Possible Values : 'pos_flop' 'pos_flop neg_latch' 'neg_latch' 'none' (default : none).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> |
| dft_retiming_logic_pos_head_chain                          |                                                                                                                                                                                                                                                                 |
|                                                            | <p>Specifies the logic to be inserted at head of positive head chain. Possible Values : 'pos_flop' 'pos_flop neg_flop' 'none' (default : none).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p>              |
| dft_retiming_logic_pos_tail_chain                          |                                                                                                                                                                                                                                                                 |
|                                                            | <p>Specifies the logic to be inserted at tail of positive tail chain. Possible Values : 'pos_flop' 'pos_flop neg_latch' 'neg_latch' 'none' (default : none).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> |
| dft_retiming_skip_tail_pipeline_latch_local_hier_placement |                                                                                                                                                                                                                                                                 |

|                                             |                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>Skip local hierarchical placement of tail pipeline lockup latch.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                 |
| dft_retiming_tail_pipeline_length           |                                                                                                                                                                                                                                                                                                                       |
|                                             | <p>Number of pipeline retiming flops to be inserted at chain tail.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                       |
| dft_rtl_insertion                           |                                                                                                                                                                                                                                                                                                                       |
|                                             | <p>Check whether DFT inserted RTL generation is requested or not.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_rtl_insertion</a></p>                                                                                        |
| dft_run_test_point_analysis_for_compression |                                                                                                                                                                                                                                                                                                                       |
|                                             | <p>Run modus directly from genus for testpoint analysis during write_dft_compression_test_po command. Requires MOD400 license.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_run_test_point_analysis_for_compression</a></p> |
| dft_run_test_point_analysis_for_lbist       |                                                                                                                                                                                                                                                                                                                       |
|                                             | <p>Run modus directly from genus for testpoint analysis during write_dft_lbist_test_points com Requires MOD400 license.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_run_test_point_analysis_for_lbist</a></p>              |
| dft_scan_power_domain_crossing_lockup_latch |                                                                                                                                                                                                                                                                                                                       |
|                                             | <p>When attribute is set, connect_scan_chains adds lockup latch at power domain crossing w not skew safe.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_scan_power_domain_crossing_lockup_latch</a></p>                      |
| dft_scan_stitch                             |                                                                                                                                                                                                                                                                                                                       |



|                                        |                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>DFT scan stitch {default   advanced  advanced_v2}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> default advanced advanced_v2</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p>                                                                                                         |
| dft_scan_style                         |                                                                                                                                                                                                                                                                                                                           |
|                                        | <p>DFT scan style {muxed_scan   clocked_lssd_scan}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> muxed_scan clocked_lssd_scan</p> <p><b>Default:</b> muxed_scan</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_scan_style</a></p>                                                |
| dft_scanbit_waveform_analysis          |                                                                                                                                                                                                                                                                                                                           |
|                                        | <p>Applies additional analysis if test clock waveform data is available to determine if a non-scan element in a chain is a scan bit or a lockup.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_scanbit_waveform_analysis</a></p> |
| dft_shared_common_logic_threshold      |                                                                                                                                                                                                                                                                                                                           |
|                                        | <p>Specifies the threshold for common logic to be included in shared threshold calculation.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 2</p> <p><b>Edit:</b> Yes</p>                                                                                                                                  |
| dft_shared_wrapper_through             |                                                                                                                                                                                                                                                                                                                           |
|                                        | <p>Trace through the logic to identify the functional flop in the fanin or fanout of pin location for sharing.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                           |
| dft_shift_register_identification_mode |                                                                                                                                                                                                                                                                                                                           |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Identify shift registers under mode setup {logical_only   testmode_only   test_setup   test_setup_shiftenable   testmode_shiftenable}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> logical_only testmode_only test_setup testmode_shiftenable test_setup_shiftenable</p> <p><b>Default:</b> test_setup_shiftenable</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_shift_register_identification_mode</a></p> |
| dft_shift_register_max_length      |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                    | <p>Maximum sequential length of the shift register for auto-identification.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> inf</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_shift_register_max_length</a></p>                                                                                                                                                                                                         |
| dft_shift_register_min_length      |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                    | <p>Minimum sequential length of the shift register for auto-identification.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 2</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_shift_register_min_length</a></p>                                                                                                                                                                                                           |
| dft_shift_register_with_mbc1       |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                    | <p>Support shift register with MBC1.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_shift_register_with_mbc1</a></p>                                                                                                                                                                                                                                              |
| dft_shift_timing_mode_name         |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                    | <p>DFT shift timing mode name (default: DFT_SHIFT_MODE).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> DFT_SHIFT_MODE</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_shift_timing_mode_name</a></p>                                                                                                                                                                                                                 |
| dft_tap_lockup_clock_from_adjacent |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                    | <p>Connect lock up element clock to adjacent element clock pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dft_tap_lockup_clock_from_adjacent</a></p>                                                                                                                                                                                                             |

|                                                                    |                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dft_tpi_sharing_scope                                              |                                                                                                                                                                                                                                                                                                                                    |
|                                                                    | <p>Controls the scope of sharing for the inserted test points. The valid options are within_hier_within_power_domains_only. across_power_domains.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> within_power_domains_only<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">dft_tpi_sharing_scope</a></p> |
| dft_true_time_flow                                                 |                                                                                                                                                                                                                                                                                                                                    |
|                                                                    | <p>Enables Encounter Test True Time Flow when writing out the ATPG scripts using 'write_et</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">dft_true_time_flow</a></p>                                                                                 |
| dft_use_abstract_segment_instance_power_domain_for_scan_connection |                                                                                                                                                                                                                                                                                                                                    |
|                                                                    | <p>Use abstract segment instance power domain for scan connection.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">dft_use_abstract_segment_instance_power_domain_for_scan_connection</a></p>                                                         |
| dft_use_dft_prefix_for_retiming_element_names                      |                                                                                                                                                                                                                                                                                                                                    |
|                                                                    | <p>Use dft_prefix value for naming of retiming element instances.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                    |
| dft_use_multibit_register_width_for_threshold                      |                                                                                                                                                                                                                                                                                                                                    |
|                                                                    | <p>Specifies whether multibit cell width to be used as threshold.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                    |
| dft_use_wck_as_default_wrapper_clock                               |                                                                                                                                                                                                                                                                                                                                    |
|                                                                    | <p>Use fanin/fanout clock if found else use wck clock if present.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">dft_use_wck_as_default_wrapper_clock</a></p>                                                                                        |
| disable_when_checks                                                |                                                                                                                                                                                                                                                                                                                                    |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | <p>Disable the processing of long when condition having bus operands.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">disable_when_checks</a></p>                                                                                                                                                                                                                                                           |
| distributed_mmmc_disable_reports_auto_redirection |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                   | <p>setting 'true' will preserve path specified in report command. User need to specify unique path for each view to avoid all clients writing to same report file. With default setting (false), distributed mmmc run automatically updates report paths to a view unique directory structure.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">distributed_mmmc_disable_reports_auto_redirection</a></p> |
| double_cell_search_pattern                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                   | <p>In case LEF is not provided, a search pattern is needed to identify which cells are double connected.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">double_cell_search_pattern</a></p>                                                                                                                                                                                                                 |
| dp_analytical_opt                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                   | <p>Specifies the effort level for global datapath optimization.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> standard extreme</p> <p><b>Default:</b> standard</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_analytical_opt</a></p>                                                                                                                                                                                                                   |
| dp_area_mode                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                   | <p>When true, enables datapath optimizations that focus on improving area results.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_area_mode</a></p>                                                                                                                                                                                                                                                    |
| dp_csa                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                                 |                                                                                                                                                                                                                                                                     |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>Controls CSA operation in 'synthesize -to_generic' (basic/none).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic none</p> <p><b>Default:</b> basic</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_csa</a></p>         |
| dp_csa_factorize                                |                                                                                                                                                                                                                                                                     |
|                                                 | <p>Controls datapath csa_factorize in 'syn_generic'.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic none</p> <p><b>Default:</b> basic</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_csa_factorize</a></p>              |
| dp_opt_power_opt                                |                                                                                                                                                                                                                                                                     |
|                                                 | <p>Controls datapath power optimization flow.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_opt_power_opt</a></p>                                                           |
| dp_opt_power_opt_scaling                        |                                                                                                                                                                                                                                                                     |
|                                                 | <p>Scaling for power used within datapath power optimization flow.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_opt_power_opt_scaling</a></p>                              |
| dp_opt_toggle_skew_threshold_for_booth_encoding |                                                                                                                                                                                                                                                                     |
|                                                 | <p>Set an int value for the skew threshold when in the auto toggle rate mode.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_opt_toggle_skew_threshold_for_booth_encoding</a></p> |
| dp_rename                                       |                                                                                                                                                                                                                                                                     |

|                           |                                                                                                                                                                                                                                                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Controls the datapath rewriting optimization level in 'synthesize -to_generic' (basic/advanced/none).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic advanced none</p> <p><b>Default:</b> basic</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_rename</a></p> |
| dp_sharing                |                                                                                                                                                                                                                                                                                                              |
|                           | <p>To control the datapath sharing in 'synthesize -to_generic' (advanced/none).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> advanced none inherited</p> <p><b>Default:</b> advanced</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_sharing</a></p>                  |
| dp_speculation            |                                                                                                                                                                                                                                                                                                              |
|                           | <p>Controls datapath speculation in 'syn_generic'.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic none</p> <p><b>Default:</b> basic</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_speculation</a></p>                                                           |
| dp_ungroup_during_syn_map |                                                                                                                                                                                                                                                                                                              |
|                           | <p>Allow ungrouping of datapath and ChipWare hierarchies during syn_map.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_ungroup_during_syn_map</a></p>                                                                 |
| dp_ungroup_separator      |                                                                                                                                                                                                                                                                                                              |
|                           | <p>Separator for instance names of datapath components in ungrouped hierarchies.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">dp_ungroup_separator</a></p>                                                                                        |
| dpopt_balanced_csa        |                                                                                                                                                                                                                                                                                                              |
|                           | <p>Enable additional rebalancing within datapath optimization and balanced CSA tree builder</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                                                                 |

|                        |                                                                                                                                                                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dpopt_combined_costing |                                                                                                                                                                                                                                                                      |
|                        | <p>Enable combined costing within datapath optimization.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                                                            |
| drc_first              |                                                                                                                                                                                                                                                                      |
|                        | <p>Use DRC costs first.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">drc_first</a></p>                                                                                         |
| drc_max_cap_first      |                                                                                                                                                                                                                                                                      |
|                        | <p>Use max_capacitance costs first.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">drc_max_cap_first</a></p>                                                                     |
| drc_max_fanout_first   |                                                                                                                                                                                                                                                                      |
|                        | <p>Use max_fanout costs first.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">drc_max_fanout_first</a></p>                                                                       |
| drc_max_trans_first    |                                                                                                                                                                                                                                                                      |
|                        | <p>Use max_transition costs first.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">drc_max_trans_first</a></p>                                                                    |
| eco_add_ports          |                                                                                                                                                                                                                                                                      |
|                        | <p>Determines whether to add hierarchical module ports during buffering. It only works for max ECO commands.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">eco_add_ports</a></p> |

|                                              |                                                                                                                                                                                                                                                              |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eco_batch_mode                               |                                                                                                                                                                                                                                                              |
|                                              | Enter/Exit batch mode<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">eco_batch_mode</a>                                                                                                  |
| eco_check_logical_equivalence                |                                                                                                                                                                                                                                                              |
|                                              | Do logical-equivalence checking<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">eco_check_logical_equivalence</a>                                                                          |
| eco_delay_calculation_effort                 |                                                                                                                                                                                                                                                              |
|                                              | Delay calculation effort level { low   medium   high } used in timing update<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> high<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">eco_delay_calculation_effort</a>                            |
| eco_disable_change_net_name_for_flat_netlist |                                                                                                                                                                                                                                                              |
|                                              | Specifies whether nets would be renamed in eco report of eco_compare_netlist for flat netlist<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">eco_disable_change_net_name_for_flat_netlist</a> |
| eco_disable_constraints_loading_from_clients |                                                                                                                                                                                                                                                              |
|                                              | Disable Loading Constraints from Clients<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                   |
| eco_disable_derates_loading_from_clients     |                                                                                                                                                                                                                                                              |
|                                              | Disable Loading Derates from Clients<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                                                       |
| eco_disable_parasitic_loading_from_clients   |                                                                                                                                                                                                                                                              |



|                                               |                                                                                                                                                                                          |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | Disable Loading Parasitics from Clients<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                |
| eco_disable_power_format_loading_from_clients |                                                                                                                                                                                          |
|                                               | Disable Loading Power Format from Clients<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes                                                               |
| eco_hierarchical_module_list                  |                                                                                                                                                                                          |
|                                               | set hierarchical module list<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                            |
| eco_honor_dont_touch                          |                                                                                                                                                                                          |
|                                               | Check dont_touch on instances, cells and nets<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">eco_honor_dont_touch</a> |
| eco_honor_dont_use                            |                                                                                                                                                                                          |
|                                               | Skip library cells marked dont_use<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">eco_honor_dont_use</a>              |
| eco_honor_fixed_status                        |                                                                                                                                                                                          |
|                                               | Honor fixed (placement) instances<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">eco_honor_fixed_status</a>           |
| eco_honor_fixed_wires                         |                                                                                                                                                                                          |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Honor fixed wires of nets</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">eco_honor_fixed_wires</a></p>                                                                                                                                                                                                                                                                                                                                                                  |
| eco_honor_power_intent     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                            | <p>Perform MSV checks during ECO: Do not allow resize of regular cell with always-on cell (and versa), do not change cells belonging to different power domains, do not allow adding buffer to power domain it doesn't belong to, do not buffer a cross power domain net, do not allow deleting an always-on buffer, a level-shifter, or a buffer which will cause redundant isolation cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">eco_honor_power_intent</a></p> |
| eco_inherit_net_attribute  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                            | <p>New net inherits attributes of buffered net</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">eco_inherit_net_attribute</a></p>                                                                                                                                                                                                                                                                                                                                            |
| eco_prefix                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                            | <p>Prefix name to be used by ECO commands</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ECO</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">eco_prefix</a></p>                                                                                                                                                                                                                                                                                                                                                               |
| eco_preserve_hpin_function |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                            | <p>Determines whether to preserve logical functions at hierarchical module ports. It only works with manual ECO commands.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">eco_preserve_hpin_function</a></p>                                                                                                                                                                                                                                                               |
| eco_refine_place           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                           |                                                                                                                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Refine placement after ECO</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">eco_refine_place</a></p>                                              |
| eco_select_partition_list |                                                                                                                                                                                                                                       |
|                           | <p>set partition module list</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                         |
| eco_spread_inverter       |                                                                                                                                                                                                                                       |
|                           | <p>Spread inverters on buffered net</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">eco_spread_inverter</a></p>                                     |
| eco_update_only_layers    |                                                                                                                                                                                                                                       |
|                           | <p>Modify routes only on metal layers in this range (both inclusive)</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">eco_update_only_layers</a></p> |
| eco_update_timing         |                                                                                                                                                                                                                                       |
|                           | <p>Update timing results after ECO</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">eco_update_timing</a></p>                                        |
| edit_wire_align           |                                                                                                                                                                                                                                       |
|                           | <p>Specifies whether to support align objects.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_align</a></p>                              |
| edit_wire_allow_45_degree |                                                                                                                                                                                                                                       |

|                                      |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Specifies whether to allow creation of a 45-degree wire.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_allow_45_degree</a></p>                                                                                               |
| edit_wire_arrow_incremental          |                                                                                                                                                                                                                                                                                                                                |
|                                      | <p>Specifies the step increments in microns to move wires with the arrow keys.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_arrow_incremental</a></p>                                                                          |
| edit_wire_assign_multi_pattern_color |                                                                                                                                                                                                                                                                                                                                |
|                                      | <p>Specifies how to assign the mask color on the DPT layer of the wire segment to be created.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto mask1 mask2 mask3</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_assign_multi_pattern_color</a></p> |
| edit_wire_auto_assign_net            |                                                                                                                                                                                                                                                                                                                                |
|                                      | <p>Auto assign the net from touched objects to edit wires/vias. Only enable this behavior for sp edit function.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> add copy move</p> <p><b>Default:</b> copy</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_auto_assign_net</a></p>   |
| edit_wire_auto_distribute_bus        |                                                                                                                                                                                                                                                                                                                                |
|                                      | <p>Auto distribute a group of wires, and set same spacing based on option -bus_total_width_v or -bus_total_width_horizontal.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_auto_distribute_bus</a></p>                          |
| edit_wire_auto_split_bus             |                                                                                                                                                                                                                                                                                                                                |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Splits buses automatically to avoid DRC violations.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_auto_split_bus</a></p>                                                                                                                                                                                        |
| edit_wire_bus_honor_start_parameters |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      | <p>Specifies that the wire width and spacing of the start points (pins) should be honored when drawing routes for a bus.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_bus_honor_start_parameters</a></p>                                                                                                          |
| edit_wire_bus_honor_width_setting    |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      | <p>Specifies whether the tool should adjust the wire width when snapping bus wires to pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_bus_honor_width_setting</a></p>                                                                                                                                           |
| edit_wire_bus_total_width_horizontal |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      | <p>This option can specify the total width of horizontal bus. So when tool drawing a bus, tool auto spacing as: <math>\text{wire spacing} = (\text{total bus width} - \text{total wire width}) / (\text{wire number} - 1)</math>.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_bus_total_width_horizontal</a></p> |
| edit_wire_bus_total_width_vertical   |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      | <p>This option can specify the total width of vertical bus. So when tool drawing a bus, tool auto spacing as: <math>\text{wire spacing} = (\text{total bus width} - \text{total wire width}) / (\text{wire number} - 1)</math>.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_bus_total_width_vertical</a></p>     |
| edit_wire_change_order_at_turn       |                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                        |                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Specifies whether the order of the wires changes or stays the same when the wire makes a degree turn. (Reverse Order   Keep Order)</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> keep_order</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_change_order_at_turn</a></p> |
| edit_wire_check_design_boundary        |                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Specifies whether stop user to stretch the wire in the forbidden area which defined by edit_wire_pull_back_distance.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_check_design_boundary</a></p>                     |
| edit_wire_close_polygons               |                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Specifies whether to close a special route structure toward itself.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_close_polygons</a></p>                                                                             |
| edit_wire_color_align_with_track       |                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Specifies whether the wire color is to be changed to match the track color during a move or operation.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_color_align_with_track</a></p>                                  |
| edit_wire_connect_pin                  |                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Specifies whether or not generate a via on the pin for specified pin type.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> in out inout nodir</p> <p><b>Default:</b> in out inout nodir</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_connect_pin</a></p>              |
| edit_wire_connect_with_specified_layer |                                                                                                                                                                                                                                                                                                                        |

|                                 |                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>Specifies whether or not the layer of connecting wires should be changed as per the edit_wire_layer_horizontal or edit_wire_layer_vertical setting when moving wires.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_connect_with_specified_layer</a></p> |
| edit_wire_create_crossover_vias |                                                                                                                                                                                                                                                                                                                                                            |
|                                 | <p>Specifies whether the software creates a via when you draw a wire that crosses a wire or pi same net that is on a different layer.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_create_crossover_vias</a></p>                                            |
| edit_wire_create_is_edit_flag   |                                                                                                                                                                                                                                                                                                                                                            |
|                                 | <p>Specifies whether to create is_edit flag for wire edit</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_create_is_edit_flag</a></p>                                                                                                                         |
| edit_wire_create_via_on_pin     |                                                                                                                                                                                                                                                                                                                                                            |
|                                 | <p>Specifies whether the software creates vias at pins.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_create_via_on_pin</a></p>                                                                                                                              |
| edit_wire_cut_class             |                                                                                                                                                                                                                                                                                                                                                            |
|                                 | <p>Specifies the cut class name of the generated via when editing wires.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_cut_class</a></p>                                                                                                                     |
| edit_wire_cut_wire_overlap      |                                                                                                                                                                                                                                                                                                                                                            |
|                                 | <p>Specifies whether or not overlaps are created while cutting specified type of wires.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> regular special</p> <p><b>Default:</b> regular</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_cut_wire_overlap</a></p>                                                 |

|                                           |                                                                                                                                                                                                                                                                               |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| edit_wire_debug_file                      |                                                                                                                                                                                                                                                                               |
|                                           | <p>Specifies the report file name.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_debug_file</a></p>                                                                             |
| edit_wire_delete_pin_with_wire            |                                                                                                                                                                                                                                                                               |
|                                           | <p>Delete Pin with Wire</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_delete_pin_with_wire</a></p>                                                                              |
| edit_wire_delete_tsv                      |                                                                                                                                                                                                                                                                               |
|                                           | <p>Specifies whether to delete the items on TSV layer or not.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_delete_tsv</a></p>                                                 |
| edit_wire_delete_wire_via_through_layers  |                                                                                                                                                                                                                                                                               |
|                                           | <p>Delete Wire Via in deep through mode</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_delete_wire_via_through_layers</a></p>                                                   |
| edit_wire_die_name                        |                                                                                                                                                                                                                                                                               |
|                                           | <p>Specifies the die name of adding wires and vias.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                          |
| edit_wire_display_wire_length_with_cursor |                                                                                                                                                                                                                                                                               |
|                                           | <p>Specifies whether display the total/current wire length with mouse when editing wires.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_display_wire_length_with_cursor</a></p> |
| edit_wire_draw_shield                     |                                                                                                                                                                                                                                                                               |



|                                       |                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>Specifies whether draw a shield net only.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_draw_shield</a></p>                                                                                                                                                            |
| edit_wire_draw_with_group_centerline  |                                                                                                                                                                                                                                                                                                                                                                          |
|                                       | <p>Turn on this option, tool should draw bus along the center-line of whole wires.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_draw_with_group_centerline</a></p>                                                                                                       |
| edit_wire_drawing_wire                |                                                                                                                                                                                                                                                                                                                                                                          |
|                                       | <p>Specifies which of the nets used with the -nets parameter corresponding to the mouse point location when adding an array of wires.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_drawing_wire</a></p>                                                                       |
| edit_wire_drc_aware_cross_metal       |                                                                                                                                                                                                                                                                                                                                                                          |
|                                       | <p>Specifies whether to stretch or move wires back to the closest location available before a vi occurs with the crossing metal. This option only work if "setEditMode -stop_at_drc" set to 1.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_drc_aware_cross_metal</a></p> |
| edit_wire_drc_on                      |                                                                                                                                                                                                                                                                                                                                                                          |
|                                       | <p>Specifies whether to check DRC rules during wire editing.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_drc_on</a></p>                                                                                                                                                  |
| edit_wire_drc_use_non_default_spacing |                                                                                                                                                                                                                                                                                                                                                                          |
|                                       | <p>Specifies whether non-default spacing defined in the Non-Default Rule of the editing net should be used for spacing check during wire editing.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_drc_use_non_default_spacing</a></p>                                       |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| edit_wire_extend_wires            | <p>Specifies whether, after completing a signal or power route, the specified boundary of wire segment extends and connects to the first logical target</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> start end start_cell_boundary end_cell_boundary</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_extend_wires</a></p> |
| edit_wire_final_check_with_verify | <p>Specifies whether to do final check with verify</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_final_check_with_verify</a></p>                                                                                                                                                                   |
| edit_wire_ignore_drc              | <p>Specifies which violation type should be ignored during wire editing.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> conn_antenna max_via_stack min_cut min_enclosed_area min_step protr</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_ignore_drc</a></p>                                                              |
| edit_wire_jog_connect_layer       | <p>Specifies the number of layers for connecting wires above the layers for moving wires. If se parameter of 'set_db edit_wire_jog_connect_layer' to 1, disable this option.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_jog_connect_layer</a></p>                                                    |
| edit_wire_keep_floating_via       | <p>Keep floating vias when user deleting wires.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_keep_floating_via</a></p>                                                                                                                                                                            |
| edit_wire_keep_status             |                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                  |                                                                                                                                                                                                                                                  |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Specifies whether to keep object status during editing.<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">edit_wire_keep_status</a></p>                               |
| edit_wire_keep_via               |                                                                                                                                                                                                                                                  |
|                                  | <p>Specifies whether keep original via in wire move or change width.<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">edit_wire_keep_via</a></p>                       |
| edit_wire_lateral_movement_range |                                                                                                                                                                                                                                                  |
|                                  | <p>Specified the max lateral movement distance to avoid DRC when add a wire.<br/> <b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">edit_wire_lateral_movement_range</a></p> |
| edit_wire_layer                  |                                                                                                                                                                                                                                                  |
|                                  | <p>Specifies the layer for patch wires or polygon wires.<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> M1<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">edit_wire_layer</a></p>                                       |
| edit_wire_layer_horizontal       |                                                                                                                                                                                                                                                  |
|                                  | <p>Specifies the layer for horizontal wires.<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> M1<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">edit_wire_layer_horizontal</a></p>                                        |
| edit_wire_layer_max              |                                                                                                                                                                                                                                                  |
|                                  | <p>Specifies the maximum layer for routing.<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> UNINITIALIZED<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">edit_wire_layer_max</a></p>                                     |
| edit_wire_layer_min              |                                                                                                                                                                                                                                                  |

|                              |                                                                                                                                                                                                                                                                    |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Specifies the minimum layer for routing.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> UNINITIALIZED</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_layer_min</a></p>                                               |
| edit_wire_layer_vertical     |                                                                                                                                                                                                                                                                    |
|                              | <p>Specifies the layer for vertical wires.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> M2</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_layer_vertical</a></p>                                                      |
| edit_wire_look_down_layers   |                                                                                                                                                                                                                                                                    |
|                              | <p>Specifies the number of layers below the current layer that added wires will connect to with</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_look_down_layers</a></p> |
| edit_wire_look_up_layers     |                                                                                                                                                                                                                                                                    |
|                              | <p>Specifies the number of layers above the current layer that added wires will connect to with</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_look_up_layers</a></p>   |
| edit_wire_max_pointer_number |                                                                                                                                                                                                                                                                    |
|                              | <p>Specifies the maximum number of pointers to be returned by wire edit commands.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_max_pointer_number</a></p>             |
| edit_wire_mirror_bus_route   |                                                                                                                                                                                                                                                                    |
|                              | <p>Specifies that whether to reverse the net order for bus route.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_mirror_bus_route</a></p>                            |
| edit_wire_nets               |                                                                                                                                                                                                                                                                    |

|                                      |                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Specifies one or more nets for editing.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_nets</a></p>                                                                                                                       |
| edit_wire_no_merge_special_wire      |                                                                                                                                                                                                                                                                                                                           |
|                                      | <p>Specifies whether automatically merge the added special wires with any existing special w</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_no_merge_special_wire</a></p>                                                   |
| edit_wire_only_show_edit_layer       |                                                                                                                                                                                                                                                                                                                           |
|                                      | <p>Turn on this option, when user editing any wires/vias, only show the layer from edit objects, dim other objects.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_only_show_edit_layer</a></p>                              |
| edit_wire_orthogonal_connection_only |                                                                                                                                                                                                                                                                                                                           |
|                                      | <p>Specifies whether to consider wires and pins that are in the same direction as a target for connection. Only special wire is supported.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_orthogonal_connection_only</a></p> |
| edit_wire_outer_shield_spacing       |                                                                                                                                                                                                                                                                                                                           |
|                                      | <p>Specifies a spacing value in microns to be used by the edit_wire_shield low and edit_wire_high parameters.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_outer_shield_spacing</a></p>                                   |
| edit_wire_outer_shield_width         |                                                                                                                                                                                                                                                                                                                           |

|                                          |                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Specifies a width value in microns to be used for outer shield(s) depending on the values of edit_wire_shield_low or edit_wire_shield_high parameters.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">edit_wire_outer_shield_width</a></p> |
| edit_wire_override                       |                                                                                                                                                                                                                                                                                                                             |
|                                          | <p>Specifies whether to use the override specification.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">edit_wire_override</a></p>                                                                                                             |
| edit_wire_partial_overlap_threshold      |                                                                                                                                                                                                                                                                                                                             |
|                                          | <p>Specifies ratio threshold of special via creation on partial overlap wires.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">edit_wire_partial_overlap_threshold</a></p>                                                                          |
| edit_wire_pull_back_distance             |                                                                                                                                                                                                                                                                                                                             |
|                                          | <p>Specify the value for forbidden distance. Default is Auto, it means use the largest min space in the layer.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> auto<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">edit_wire_pull_back_distance</a></p>                                           |
| edit_wire_regular_wire_honor_track_width |                                                                                                                                                                                                                                                                                                                             |
|                                          | <p>Specifies whether the regular wire is to be changed to match the track width.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                              |
| edit_wire_reshape                        |                                                                                                                                                                                                                                                                                                                             |
|                                          | <p>Specifies that when you add new wires, existing redundant wires within the route are automatically removed.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">edit_wire_reshape</a></p>                                                       |

|                                       |                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| edit_wire_return_obj_pointer          |                                                                                                                                                                                                                                                                                                                                                             |
|                                       | <p>Specifies whether return pointers after wire edit operations.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_return_obj_pointer</a></p>                                                                                                                    |
| edit_wire_rule                        |                                                                                                                                                                                                                                                                                                                                                             |
|                                       | <p>Uses the specified LEF rule for regular wires.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_rule</a></p>                                                                                                                                                  |
| edit_wire_search_route_rule_vias_only |                                                                                                                                                                                                                                                                                                                                                             |
|                                       | <p>Specifies that when the Shift + N/P bindkey is used, the tool should circle through NDR vias</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_search_route_rule_vias_only</a></p>                                                                            |
| edit_wire_shape                       |                                                                                                                                                                                                                                                                                                                                                             |
|                                       | <p>Specifies the shape associated with the wire you draw.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> ring stripe followpin iowire corewire blockwire padring blockring fillwire fillw drcfill blockagewire none</p> <p><b>Default:</b> stripe</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_shape</a></p> |
| edit_wire_shield                      |                                                                                                                                                                                                                                                                                                                                                             |
|                                       | <p>Specifies whether to add a minimum width shield wire for specified side.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> high low adjacent</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_shield</a></p>                                                                           |
| edit_wire_shield_look_down_layers     |                                                                                                                                                                                                                                                                                                                                                             |

|                                        |                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Specifies the number of layers below the current layer that an added shield wire can connect using a via.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_shield_look_down_layers</a></p>                                                                        |
| edit_wire_shield_look_up_layers        |                                                                                                                                                                                                                                                                                                                                                               |
|                                        | <p>Specifies the number of layers above the current layer that an added shield wire can connect using a via.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_shield_look_up_layers</a></p>                                                                          |
| edit_wire_shield_shape                 |                                                                                                                                                                                                                                                                                                                                                               |
|                                        | <p>Specifies the shape associated with the shield wire you draw.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> ring stripe followpin iowire corewire blockwire padring blockring fillwire drc_blockagewire none</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_shield_shape</a></p> |
| edit_wire_shielding_nets               |                                                                                                                                                                                                                                                                                                                                                               |
|                                        | <p>Specifies net names for shield wires, usually power or ground names.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_shielding_nets</a></p>                                                                                                                    |
| edit_wire_show_drc_info_for_edit_shape |                                                                                                                                                                                                                                                                                                                                                               |
|                                        | <p>Turn on this option, when user editing any wires/vias and generate a violation, tool can display detail information nearby marker.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_show_drc_info_for_edit_shape</a></p>                                        |
| edit_wire_sibling_look_down_layers     |                                                                                                                                                                                                                                                                                                                                                               |



|                                  |                                                                                                                                                                                                                                                                                        |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Specifies the number of layers below the sibling layers that an added shield wire can connect using a via.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_sibling_look_down_layers</a></p> |
| edit_wire_sibling_look_up_layers |                                                                                                                                                                                                                                                                                        |
|                                  | <p>Specifies the number of layers above the sibling layers that an added shield wire can connect using a via.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_sibling_look_up_layers</a></p>   |
| edit_wire_snap                   |                                                                                                                                                                                                                                                                                        |
|                                  | <p>Specifies whether to enable wire and via snap.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_snap</a></p>                                                                             |
| edit_wire_snap_align_to          |                                                                                                                                                                                                                                                                                        |
|                                  | <p>Specifies how to align with a pin when ending a route.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> center low high auto</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_snap_align_to</a></p>            |
| edit_wire_snap_bus_to_pin        |                                                                                                                                                                                                                                                                                        |
|                                  | <p>Specifies whether to change wire width, spacing and order according to the connected pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_snap_bus_to_pin</a></p>                      |
| edit_wire_snap_end_to            |                                                                                                                                                                                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Specifies whether to snap the end of wire to specified objects.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> track_regular track_special manufacturing_grid</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_snap_end_to</a></p>                                                                               |
| edit_wire_snap_objects_to_track        |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                        | <p>Specifies whether to snap added or moved specified wires to the closest routing track in the preferred direction for the layer automatically.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> special regular patch pin</p> <p><b>Default:</b> regular patch</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_snap_objects_to_track</a></p> |
| edit_wire_snap_to                      |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                        | <p>Specifies whether to snap added or moved wires to specified objects.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> special pg pin row</p> <p><b>Default:</b> special pg pin</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_snap_to</a></p>                                                                                              |
| edit_wire_snap_to_track_honor_color    |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                        | <p>Specifies whether to snap wires to a track with the same color automatically.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_snap_to_track_honor_color</a></p>                                                                                                                           |
| edit_wire_snap_trim_metal_to_trim_grid |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                        | <p>Specify whether snap the trim metal to the closest trim grid during trim metal editing.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_snap_trim_metal_to_trim_grid</a></p>                                                                                                              |
| edit_wire_spacing                      |                                                                                                                                                                                                                                                                                                                                                                                          |

|                              |                                                                                                                                                                                                                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Specifies the distance between wires.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_spacing</a></p>                                                                                      |
| edit_wire_spacing_horizontal |                                                                                                                                                                                                                                                                                           |
|                              | <p>Specifies the distance between horizontal wires.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_spacing_horizontal</a></p>                                                                |
| edit_wire_spacing_vertical   |                                                                                                                                                                                                                                                                                           |
|                              | <p>Specifies the distance between vertical wires.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_spacing_vertical</a></p>                                                                    |
| edit_wire_status             |                                                                                                                                                                                                                                                                                           |
|                              | <p>Specifies the status associated with the wire you draw.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> cover fixed noshield routed shield auto</p> <p><b>Default:</b> fixed</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_status</a></p> |
| edit_wire_stop_at_drc        |                                                                                                                                                                                                                                                                                           |
|                              | <p>Specifies whether the software can cause a DRC violation by moving or stretching a wire.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_stop_at_drc</a></p>                               |
| edit_wire_stretch_end        |                                                                                                                                                                                                                                                                                           |
|                              | <p>Specifies the direction in which to stretch or reduce wires.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> high low</p> <p><b>Default:</b> high</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_stretch_end</a></p>                       |

|                                           |                                                                                                                                                                                                                                                    |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| edit_wire_stretch_with_intersection       |                                                                                                                                                                                                                                                    |
|                                           | Enables wires with intersects to be stretched easily.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">edit_wire_stretch_with_intersection</a>                                   |
| edit_wire_sub_class                       |                                                                                                                                                                                                                                                    |
|                                           | Specifies the subclass name of wires and vias.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">edit_wire_sub_class</a>                                                           |
| edit_wire_turn_at                         |                                                                                                                                                                                                                                                    |
|                                           | Specifies the start location of new wire when do turn.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> center_line wire_edge<br><b>Default:</b> center_line<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">edit_wire_turn_at</a> |
| edit_wire_type                            |                                                                                                                                                                                                                                                    |
|                                           | Specifies the type of creating wire.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> regular special patch<br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">edit_wire_type</a>                               |
| edit_wire_unrestricted_regular_wire_width |                                                                                                                                                                                                                                                    |
|                                           | Specifies whether the regular wire width is unrestricted in edit.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">edit_wire_unrestricted_regular_wire_width</a>                 |
| edit_wire_update_shield_net               |                                                                                                                                                                                                                                                    |

|                                                  |                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | <p>Specifies whether automatically update shield net attribute on the signal net.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_update_shield_net</a></p>                                                   |
| edit_wire_use_fix_via                            |                                                                                                                                                                                                                                                                                                            |
|                                                  | <p>Specifies whether to use fix_via for special via with minStep DRC violation.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_use_fix_via</a></p>                                                           |
| edit_wire_use_interleaving_wire_group            |                                                                                                                                                                                                                                                                                                            |
|                                                  | <p>Alternates the specified wires in the wire group.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_use_interleaving_wire_group</a></p>                                                                      |
| edit_wire_use_wire_group                         |                                                                                                                                                                                                                                                                                                            |
|                                                  | <p>Groups multiple wires from the same net.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_use_wire_group</a></p>                                                                                            |
| edit_wire_use_wire_group_bits                    |                                                                                                                                                                                                                                                                                                            |
|                                                  | <p>Specifies the number of times to replicate a wire in the wire group.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_use_wire_group_bits</a></p>                                                                |
| edit_wire_use_wire_group_reinforcement           |                                                                                                                                                                                                                                                                                                            |
|                                                  | <p>Creates slots in the wire group by creating wires that connect the wires in the wire group, b<br/>orthogonal to them.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_use_wire_group_reinforcement</a></p> |
| edit_wire_use_wire_group_reinforcement_group_via |                                                                                                                                                                                                                                                                                                            |

|                                                |                                                                                                                                                                                                                                                                         |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Specifies that the wire group uses group vias.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_use_wire_group_reinforcement_group_via</a></p>                           |
| edit_wire_use_wire_group_reinforcement_spacing |                                                                                                                                                                                                                                                                         |
|                                                | <p>Specifies the spacing for the orthogonal wires in the wire groups.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_use_wire_group_reinforcement_spacing</a></p>         |
| edit_wire_use_wire_group_reinforcement_width   |                                                                                                                                                                                                                                                                         |
|                                                | <p>Specifies the width of the orthogonal wires in the wire groups.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_use_wire_group_reinforcement_width</a></p>              |
| edit_wire_verbose                              |                                                                                                                                                                                                                                                                         |
|                                                | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_verbose</a></p>                                                                                                                |
| edit_wire_via_allow_geometry_drc               |                                                                                                                                                                                                                                                                         |
|                                                | <p>Check Geometry DRC</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_allow_geometry_drc</a></p>                                                                        |
| edit_wire_via_auto_snap                        |                                                                                                                                                                                                                                                                         |
|                                                | <p>Controls whether to snap vias to the intersection of wires on the same net or to the manufac grid.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_auto_snap</a></p> |
| edit_wire_via_auto_update                      |                                                                                                                                                                                                                                                                         |

|                            |                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Controls whether to replace existing vias.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_auto_update</a></p>                                                                                                                                   |
| edit_wire_via_cell_name    |                                                                                                                                                                                                                                                                                                                                                     |
|                            | <p>Specifies the name of the via cell from the LEF or DEF file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_cell_name</a></p>                                                                                                                   |
| edit_wire_via_columns      |                                                                                                                                                                                                                                                                                                                                                     |
|                            | <p>Specify the rows of via cut.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_columns</a></p>                                                                                                                                                         |
| edit_wire_via_create_by    |                                                                                                                                                                                                                                                                                                                                                     |
|                            | <p>Controls whether to create vias based on the edit_wire_via_cell_name parameter or to create based on the other parameters.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> viacell parameters</p> <p><b>Default:</b> parameters</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_create_by</a></p> |
| edit_wire_via_cut_layer    |                                                                                                                                                                                                                                                                                                                                                     |
|                            | <p>Specifies the via layer name for the via to created or modify.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> V12</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_cut_layer</a></p>                                                                                                                |
| edit_wire_via_exclude_spec |                                                                                                                                                                                                                                                                                                                                                     |
|                            | <p>Specifies the vias to be ignored by wire editor.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_exclude_spec</a></p>                                                                                                                            |

|                                    |                                                                                                                                                                                                                                             |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| edit_wire_via_override_spec        |                                                                                                                                                                                                                                             |
|                                    | <p>Specifies the default vias to be used by wire editor.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_override_spec</a></p>              |
| edit_wire_via_rows                 |                                                                                                                                                                                                                                             |
|                                    | <p>Specify the columns of via cut.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_rows</a></p>                                                 |
| edit_wire_via_scale_height         |                                                                                                                                                                                                                                             |
|                                    | <p>Specify the via height scale.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_scale_height</a></p>                                         |
| edit_wire_via_scale_width          |                                                                                                                                                                                                                                             |
|                                    | <p>Specify the via width scale.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_scale_width</a></p>                                           |
| edit_wire_via_snap_honor_color     |                                                                                                                                                                                                                                             |
|                                    | <p>Specifies whether to snap the same mask automatically</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_snap_honor_color</a></p>           |
| edit_wire_via_snap_to_intersection |                                                                                                                                                                                                                                             |
|                                    | <p>Specifies whether to snap vias to the intersection of wires</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">edit_wire_via_snap_to_intersection</a></p> |



#### edit\_wire\_via\_type

Specifies the type of creating via.  
**Type:** [enum](#)  
**Enum Values:** auto regular special  
**Default:** auto  
**Edit:** Yes  
**Reference:** [edit\\_wire\\_via\\_type](#)

#### edit\_wire\_width

Specifies the width of patch wires.  
**Type:** [double](#)  
**Default:** 0.0  
**Edit:** Yes  
**Reference:** [edit\\_wire\\_width](#)

#### edit\_wire\_width\_horizontal

Specifies the width for horizontal wires.  
**Type:** [double](#)  
**Default:** 0.0  
**Edit:** Yes  
**Reference:** [edit\\_wire\\_width\\_horizontal](#)

#### edit\_wire\_width\_vertical

Specifies the width of vertical wires.  
**Type:** [double](#)  
**Default:** 0.0  
**Edit:** Yes  
**Reference:** [edit\\_wire\\_width\\_vertical](#)

#### edit\_wire\_wire\_override\_spec

Contains the details of the override specification.  
**Type:** [string](#)  
**Default:** ""  
**Edit:** Yes  
**Reference:** [edit\\_wire\\_wire\\_override\\_spec](#)

#### eeq\_variant\_site\_start

|                                     |                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Site variant_id of the first site on rows in core area. This is specified in library LEF58_CELLVARIANTS property by value of STARTVARIANT in 'CELLVARIANTS totalNum [STARTVARIANT num] YFLIPMAP {flippedVariantNum siteVariantNum} ...'</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> no_value<br/> <b>Edit:</b> Yes</p> |
| eeq_variant_y_flip_table            |                                                                                                                                                                                                                                                                                                                                            |
|                                     | <p>The LEF YFLIPMAP table defined in LEF58_CELLVARIANTS property as 'CELLVARIANTS variantTotalNum YFLIPMAP {flippedVariantNum siteVariantNum} ...'</p> <p><b>Type:</b> <a href="#">int*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                               |
| enable_library_pins_sorting_in_mmmc |                                                                                                                                                                                                                                                                                                                                            |
|                                     | <p>Sort library pins in MMMC flow.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">enable_library_pins_sorting_in_mmmc</a></p>                                                                                                                                 |
| enable_odd_via_metal_rect_parameter |                                                                                                                                                                                                                                                                                                                                            |
|                                     | <p>Control if still parameterizes via_def's metal rect if it's odd.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                          |
| enable_strict_percent_control       |                                                                                                                                                                                                                                                                                                                                            |
|                                     | <p>Threshold value to be used as hard limit.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">enable_strict_percent_control</a></p>                                                                                                                            |
| enable_ui_precision                 |                                                                                                                                                                                                                                                                                                                                            |
|                                     | <p>Enable ui precision reporting.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">enable_ui_precision</a></p>                                                                                                                                                 |
| exclusive_group_min_gap             |                                                                                                                                                                                                                                                                                                                                            |

|                                  |                                                                                                                                                                                                                                                                                                  |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>This is the minimum gap should be maintained between exclusive_groups (all exclusive gr<br/>The value is measured in microns. It can be set by command create_exclusive_groups -mi</p> <p><b>Type:</b> <a href="#">coord</a><br/> <b>Default:</b> 5<br/> <b>Edit:</b> No</p>                  |
| extract_keep_fill_wires          |                                                                                                                                                                                                                                                                                                  |
|                                  | <p>Keep metal fill wires for extraction</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">extract_keep_fill_wires</a></p>                                                                                             |
| extract_rc_assume_metal_fill     |                                                                                                                                                                                                                                                                                                  |
|                                  | <p>Assume metal fill</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">extract_rc_assume_metal_fill</a></p>                                                                                                           |
| extract_rc_cap_filter_mode       |                                                                                                                                                                                                                                                                                                  |
|                                  | <p>Coupling Capacitance Filtering Mode</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> relative_only relative_and_coupling relative_or_coupling<br/> <b>Default:</b> relative_only<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">extract_rc_cap_filter_mode</a></p> |
| extract_rc_cerebrus_license_only |                                                                                                                                                                                                                                                                                                  |
|                                  | <p>Fall back to the regular QTS license(s) when Cerebrus is not available</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">extract_rc_cerebrus_license_only</a></p>                                                  |
| extract_rc_compress_rcdb         |                                                                                                                                                                                                                                                                                                  |
|                                  | <p>Create compressed RCDB</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">extract_rc_compress_rcdb</a></p>                                                                                                          |
| extract_rc_coupled               |                                                                                                                                                                                                                                                                                                  |

|                                   |                                                                                                                                                                                                                                                                                             |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Extract coupling cap</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_coupled</a></p>                                                                                                       |
| extract_rc_coupling_cap_threshold |                                                                                                                                                                                                                                                                                             |
|                                   | <p>Coupling C threshold value</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 3.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_coupling_cap_threshold</a></p>                                                                                  |
| extract_rc_def_via_cap            |                                                                                                                                                                                                                                                                                             |
|                                   | <p>Consider via caps for extraction</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_def_via_cap</a></p>                                                                                       |
| extract_rc_effort_level           |                                                                                                                                                                                                                                                                                             |
|                                   | <p>Specifies the effort level of the post-route extraction engine</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high signoff</p> <p><b>Default:</b> undefined</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_effort_level</a></p> |
| extract_rc_engine                 |                                                                                                                                                                                                                                                                                             |
|                                   | <p>Extraction engine</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> pre_route post_route</p> <p><b>Default:</b> pre_route</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_engine</a></p>                                                       |
| extract_rc_extra_cmd_file         |                                                                                                                                                                                                                                                                                             |
|                                   | <p>Extra command file for Quantus QRC</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_extra_cmd_file</a></p>                                                                                   |

|                                                       |                                                                                                                                                                                                                                             |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| extract_rc_hard_block_obs                             |                                                                                                                                                                                                                                             |
|                                                       | Makes obstructions and power/ground pin shapes visible to the extractor for hard blocks or<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">extract_rc_hard_block_obs</a> |
| extract_rc_incremental                                |                                                                                                                                                                                                                                             |
|                                                       | Use incremental signoff mode<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">extract_rc_incremental</a>                                                                  |
| extract_rc_layer_independent                          |                                                                                                                                                                                                                                             |
|                                                       | Enable layer independent extraction<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 1<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">extract_rc_layer_independent</a>                                                          |
| extract_rc_lef_tech_file_map                          |                                                                                                                                                                                                                                             |
|                                                       | LEF-QRCTechfile layer map file<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">extract_rc_lef_tech_file_map</a>                                                           |
| extract_rc_local_cpu                                  |                                                                                                                                                                                                                                             |
|                                                       | <b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">extract_rc_local_cpu</a>                                                                                                         |
| extract_rc_post_route_iquantus_cap_high_accuracy_mode |                                                                                                                                                                                                                                             |
|                                                       | Use for iquantus capacitance high accuracy mode<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">extract_rc_post_route_iquantus_cap_high_accuracy_mode</a>                |

|                                |                                                                                                                                                                                                                                   |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| extract_rc_pvs_fill            |                                                                                                                                                                                                                                   |
|                                | Use to send the PVS fill data attached to the Innovus DB (see set_pvs_fill) to Quantus<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">extract_rc_pvs_fill</a> |
| extract_rc_qrc_cmd_file        |                                                                                                                                                                                                                                   |
|                                | Quantus QRC user command file name<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">extract_rc_qrc_cmd_file</a>                                                  |
| extract_rc_qrc_cmd_type        |                                                                                                                                                                                                                                   |
|                                | Quantus QRC command input type<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> auto partial custom<br><b>Default:</b> auto<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">extract_rc_qrc_cmd_type</a>           |
| extract_rc_qrc_output_mode     |                                                                                                                                                                                                                                   |
|                                | Quantus QRC output mode<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> spef rcdB<br><b>Default:</b> spef<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">extract_rc_qrc_output_mode</a>                         |
| extract_rc_qrc_run_mode        |                                                                                                                                                                                                                                   |
|                                | Quantus QRC MMMC run mode<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> concurrent sequential<br><b>Default:</b> concurrent<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">extract_rc_qrc_run_mode</a>        |
| extract_rc_qrc_stream_map_file |                                                                                                                                                                                                                                   |

|                                         |                                                                                                                                                                                                                                                  |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Quantus stream/oasis file</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_qrc_stream_map_file</a></p>                                            |
| extract_rc_quantus_executable           |                                                                                                                                                                                                                                                  |
|                                         | <p>QRC Executable file path</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_quantus_executable</a></p>                                              |
| extract_rc_relative_cap_threshold       |                                                                                                                                                                                                                                                  |
|                                         | <p>Relative C threshold value</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.03</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_relative_cap_threshold</a></p>                                      |
| extract_rc_shrink_factor                |                                                                                                                                                                                                                                                  |
|                                         | <p>Sets the shrink factor used by the extract_rc command.(Double, default=1)</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_shrink_factor</a></p> |
| extract_rc_signoff_stream_layer_map     |                                                                                                                                                                                                                                                  |
|                                         | <p>QRC stream map file</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_signoff_stream_layer_map</a></p>                                             |
| extract_rc_signoff_stream_layer_map_ccl |                                                                                                                                                                                                                                                  |
|                                         | <p>QRC stream map file in CCL format</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_signoff_stream_layer_map_ccl</a></p>                           |
| extract_rc_total_cap_threshold          |                                                                                                                                                                                                                                                  |

|                                         |                                                                                                                                                                                                                                        |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Total C threshold value</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 5.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_total_cap_threshold</a></p>                                   |
| extract_rc_tquantus_model_file          |                                                                                                                                                                                                                                        |
|                                         | <p>TQuantus model file name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_tquantus_model_file</a></p>                                   |
| extract_rc_tsv_subckt_file              |                                                                                                                                                                                                                                        |
|                                         | <p>TSV subckt file name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_tsv_subckt_file</a></p>                                           |
| extract_rc_turbo_reduce                 |                                                                                                                                                                                                                                        |
|                                         | <p>Reduce XCap (IQuantus)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false auto</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_turbo_reduce</a></p> |
| extract_rc_use_qrc_oa_interface         |                                                                                                                                                                                                                                        |
|                                         | <p>Use OA interface for invoking Quantus QRC extraction</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_use_qrc_oa_interface</a></p>     |
| extract_rc_use_shielding_in_detail_mode |                                                                                                                                                                                                                                        |
|                                         | <p>Considers shielding for a wire segment</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">extract_rc_use_shielding_in_detail_mode</a></p>           |
| extract_rc_via_cap                      |                                                                                                                                                                                                                                        |



|                                                  |                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | <p>Consider via caps for extraction<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">extract_rc_via_cap</a></p>                                                                                                                                                                |
| extract_rc_write_def_options_for_signoff_extract |                                                                                                                                                                                                                                                                                                                                                         |
|                                                  | <p>write_def options<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">extract_rc_write_def_options_for_signoff_extract</a></p>                                                                                                                                                 |
| finish_floorplan_abutted_edge_as_non_row_area    |                                                                                                                                                                                                                                                                                                                                                         |
|                                                  | <p>Specify whether the abutted edge of power domain should be nonRowArea<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">finish_floorplan_abutted_edge_as_non_row_area</a></p>                                                                                                |
| finish_floorplan_active_objs                     |                                                                                                                                                                                                                                                                                                                                                         |
|                                                  | <p>active_object list<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> macro macro_halo core io_pad io_cell fence hard_blockage soft_blockage partial_blockage route_blockage row die partition<br/> <b>Default:</b> macro macro_halo core<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">finish_floorplan_active_objs</a></p> |
| finish_floorplan_add_blockage_direction          |                                                                                                                                                                                                                                                                                                                                                         |
|                                                  | <p>direction<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> x y xy<br/> <b>Default:</b> xy<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">finish_floorplan_add_blockage_direction</a></p>                                                                                                                                    |
| finish_floorplan_add_blockage_fill_option        |                                                                                                                                                                                                                                                                                                                                                         |

|                                  |                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>fill option</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> channel boundary</p> <p><b>Default:</b> channel</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">finish_floorplan_add_blockage_fill_option</a></p>                                                                                                     |
| finish_floorplan_drc_region_objs |                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>drc_region object list</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> macro macro_halo hard_blockage min_gap core_spacing non_row_area r</p> <p><b>Default:</b> macro macro_halo hard_blockage min_gap core_spacing row</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">finish_floorplan_drc_region_objs</a></p> |
| finish_floorplan_override        |                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>override or not</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">finish_floorplan_override</a></p>                                                                                                                                                               |
| fix_min_drcs                     |                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>Fix min_capacitance, min_transition, and min_fanout design rule violations.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">fix_min_drcs</a></p>                                                                                                                |
| flexible_htrees                  |                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>list of flexible_htree</p> <p><b>Type:</b> <a href="#">obj(flexible_htree)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">flexible_htrees</a></p>                                                                                                                                                      |
| flip_chip_allow_layer_change     |                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>Specifies if turning on layer change feature</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_allow_layer_change</a></p>                                                                                                                               |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| flip_chip_allow_routed_bump_edit       |                                                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>By default bumps are allowed to be edited (moved, deleted, reassigned, swapped). If this v is set to FALSE, any bump edit command for a bump with routing connected to it will give an message and fail.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flip_chip_allow_routed_bump_edit</a></p> |
| flip_chip_auto_pairing_file            |                                                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Specifies the output file name to dump out route_flip_chip auto pairing information for bump connection</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flip_chip_auto_pairing_file</a></p>                                                                                                       |
| flip_chip_bottom_layer                 |                                                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Specifies the bottom-most metal layer by layer number or layer name that the software can when routing bumps</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flip_chip_bottom_layer</a></p>                                                                                                       |
| flip_chip_bump_use_octagon_shape       |                                                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>route_flip_chip use circumferential octagon to replace the complex pin shape</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flip_chip_bump_use_octagon_shape</a></p>                                                                                                                            |
| flip_chip_cell_connection_type         |                                                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Specifies cell connection, default is bump center</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> route_width bump_edge bump_center bump_end<br/> <b>Default:</b> bump_center<br/> <b>Edit:</b> Yes</p>                                                                                                                                                     |
| flip_chip_check_bump_access_directions |                                                                                                                                                                                                                                                                                                                                                                                  |

|                                      |                                                                                                                                                                                                                                                                                      |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Specifies whether check the special bump access directions</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_check_bump_access_directions</a></p>                                      |
| flip_chip_compaction                 |                                                                                                                                                                                                                                                                                      |
|                                      | <p>Specifies if turning on compaction routing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_compaction</a></p>                                                                        |
| flip_chip_connect_bump_to_block      |                                                                                                                                                                                                                                                                                      |
|                                      | <p>Connect bump to block pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                      |
| flip_chip_connect_bump_to_bump       |                                                                                                                                                                                                                                                                                      |
|                                      | <p>Connect bump to bump.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_connect_bump_to_bump</a></p>                                                                                   |
| flip_chip_connect_power_cell_to_bump |                                                                                                                                                                                                                                                                                      |
|                                      | <p>Connects all power bumps to the I/O cell pin. It does not connect to a power or ground strip ring</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_connect_power_cell_to_bump</a></p> |
| flip_chip_constraint_file            |                                                                                                                                                                                                                                                                                      |
|                                      | <p>Specifies the file that contains constraints for flip chip routing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_constraint_file</a></p>                                            |
| flip_chip_drop_via_on_all_shapes     |                                                                                                                                                                                                                                                                                      |

|                                          |                                                                                                                                                                                                                                                                           |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>route_flip_chip_drop_via_on_all_geometries</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_drop_via_on_all_shapes</a></p>                                                 |
| flip_chip_drop_via_on_power_mesh         |                                                                                                                                                                                                                                                                           |
|                                          | <p>Specifies the metal layer by layer number or layer name that the software can use to drop v</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_drop_via_on_power_mesh</a></p> |
| flip_chip_extra_config                   |                                                                                                                                                                                                                                                                           |
|                                          | <p>Specifies the name of an extra configuration file for flip chip routing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_extra_config</a></p>                               |
| flip_chip_finger_direction               |                                                                                                                                                                                                                                                                           |
|                                          | <p>Specifies main expansion direction for finger routing</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> n e s w</p> <p><b>Default:</b> n</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_finger_direction</a></p>             |
| flip_chip_finger_max_width               |                                                                                                                                                                                                                                                                           |
|                                          | <p>Specifies the max routing width for one wire segment</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_finger_max_width</a></p>                                             |
| flip_chip_finger_min_width               |                                                                                                                                                                                                                                                                           |
|                                          | <p>Specifies the min routing width for one wire segment</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_finger_min_width</a></p>                                             |
| flip_chip_finger_target_mesh_layer_range |                                                                                                                                                                                                                                                                           |

|                                                |                                                                                                                                                                                                                                                        |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Target mesh layer range for finger routing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_finger_target_mesh_layer_range</a></p>                       |
| flip_chip_honor_bump_connect_target_constraint |                                                                                                                                                                                                                                                        |
|                                                | <p>Specifies if enabling port numbering routing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_honor_bump_connect_target_constraint</a></p>              |
| flip_chip_ignore_pad_type_check                |                                                                                                                                                                                                                                                        |
|                                                | <p>Specifies if supporting CLASS PAD cells in flipchip flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_ignore_pad_type_check</a></p>                  |
| flip_chip_lower_layer_prevent_diagonal_routing |                                                                                                                                                                                                                                                        |
|                                                | <p>Specifies if preventing 45-degree routing for lower layer</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_lower_layer_prevent_diagonal_routing</a></p> |
| flip_chip_lower_layer_route_width              |                                                                                                                                                                                                                                                        |
|                                                | <p>Specifies the value for lower routing layer</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_lower_layer_route_width</a></p>                             |
| flip_chip_merge_nearby_pin                     |                                                                                                                                                                                                                                                        |
|                                                | <p>Merge pins which are close but not touched</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_merge_nearby_pin</a></p>                                    |
| flip_chip_multi_pad_routing_style              |                                                                                                                                                                                                                                                        |

|                               |                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Specifies the routing style for multi-pads routing</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> default serial star</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_multi_pad_routing_style</a></p>                                                                                                           |
| flip_chip_multiple_connection |                                                                                                                                                                                                                                                                                                                                                                                               |
|                               | <p>Specifies routing connections between multiple pads and bumps. Default is no multiple connection</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> multiple_pads_to_bump multiple_bumps_to_pad multiple_bumps_to_multiple_pads default</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_multiple_connection</a></p> |
| flip_chip_pg_mesh_direction   |                                                                                                                                                                                                                                                                                                                                                                                               |
|                               | <p>Specifies the main routing direction of PG mesh</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> default horizontal vertical</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_pg_mesh_direction</a></p>                                                                                                            |
| flip_chip_pg_mesh_main_width  |                                                                                                                                                                                                                                                                                                                                                                                               |
|                               | <p>Specifies the main width of PG mesh</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_pg_mesh_main_width</a></p>                                                                                                                                                                                |
| flip_chip_pg_mesh_max_width   |                                                                                                                                                                                                                                                                                                                                                                                               |
|                               | <p>Specifies the max width of PG mesh</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_pg_mesh_max_width</a></p>                                                                                                                                                                                  |
| flip_chip_pg_mesh_width       |                                                                                                                                                                                                                                                                                                                                                                                               |

|                                              |                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | <p>Specify the exact width of PG mesh. One value for both horizontal and vertical width. And w values, the first value for horizontal width and the second value for vertical width</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flip_chip_pg_mesh_width</a></p> |
| flip_chip_prevent_diagonal_wire_access_bumps |                                                                                                                                                                                                                                                                                                                                                 |
|                                              | <p>Specifies whether use diagonal wire to access bump or not</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flip_chip_prevent_diagonal_wire_access_bumps</a></p>                                                                                                  |
| flip_chip_prevent_via_under_bump             |                                                                                                                                                                                                                                                                                                                                                 |
|                                              | <p>Specifies if generating vias directly under the bump</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flip_chip_prevent_via_under_bump</a></p>                                                                                                                   |
| flip_chip_prevent_via_under_bump_extension   |                                                                                                                                                                                                                                                                                                                                                 |
|                                              | <p>specifies the spacing value around bump to avoid dropping vias</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> default<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flip_chip_prevent_via_under_bump_extension</a></p>                                                                                           |
| flip_chip_route_bump_cluster                 |                                                                                                                                                                                                                                                                                                                                                 |
|                                              | <p>Route bump cluster after bump grouped</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flip_chip_route_bump_cluster</a></p>                                                                                                                                      |
| flip_chip_route_pg_style                     |                                                                                                                                                                                                                                                                                                                                                 |
|                                              | <p>Specifies the routing pattern for route_flip_chip power routing</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none finger mesh<br/> <b>Default:</b> none<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flip_chip_route_pg_style</a></p>                                                                       |



|                                         |                                                                                                                                                                                                                                                                           |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| flip_chip_route_style                   |                                                                                                                                                                                                                                                                           |
|                                         | <p>Specifies routing style</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> manhattan diagonal</p> <p><b>Default:</b> diagonal</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_route_style</a></p>                              |
| flip_chip_route_width                   |                                                                                                                                                                                                                                                                           |
|                                         | <p>Specifies routing width</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_route_width</a></p>                                                                               |
| flip_chip_top_layer                     |                                                                                                                                                                                                                                                                           |
|                                         | <p>Specifies the top-most metal layer by layer number or layer name that the software can use routing bumps</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_top_layer</a></p> |
| flip_chip_via_abut_bump                 |                                                                                                                                                                                                                                                                           |
|                                         | <p>via abut bump only if it is success</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_via_abut_bump</a></p>                                                                 |
| flip_chip_wire_to_pad_honor_min_spacing |                                                                                                                                                                                                                                                                           |
|                                         | <p>Specifies the min spacing between bump/pad with wire</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flip_chip_wire_to_pad_honor_min_spacing</a></p>                                |
| floorplan_auto_sync_master_clone        |                                                                                                                                                                                                                                                                           |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>Specify the objects used to sync-up master to clones.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> bus_guides constraints stdcells macros route_blockages place_blockages</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_auto_sync_master_clone</a></p>                                                                                                                                                                                                                              |
| floorplan_auto_sync_power_domain_extra_row_pattern |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                    | <p>Enable extra row pattern setting sync up between default and non-default power domains</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_auto_sync_power_domain_extra_row_pattern</a></p>                                                                                                                                                                                                                                                                          |
| floorplan_cells_for_extra_sites                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                    | <p>Specify cell names whose related sites will be created.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_cells_for_extra_sites</a></p>                                                                                                                                                                                                                                                                                                                             |
| floorplan_check_types                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                    | <p>Specify the check types for check_floorplan. Default is the basic.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> basic fence odd_even_site_row macro_pin color alignment_following_pin alignment_partition_clone bus_guide_connectivity feed_through partition_in_partition multi_layer_pin power_domain partition place same_length_site narrow_channel block_or unified_vdd_on_bottom tsv exclusive_group all</p> <p><b>Default:</b> basic</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_check_types</a></p> |
| floorplan_cut_off_place_blockage_outside_die       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                    | <p>When creating placement blockages, will cut the part outside of die and snap it by default.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_cut_off_place_blockage_outside_die</a></p>                                                                                                                                                                                                                                                                            |
| floorplan_cut_off_route_blockage_outside_die       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                        |                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>When creating routing blockages around inst/Hinst/Partition, will cut the part outside of die and snap it by default. If this option is set to false, don't cut and snap it.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_cut_off_route_blockage_outside_die</a></p> |
| floorplan_default_blockage_name_prefix |                                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Name prefix for placement blockage and routing blockage.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_default_blockage_name_prefix</a></p>                                                                                                                           |
| floorplan_default_power_domain_site    |                                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Create rows based on default power domain site instead of the design's default site.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_default_power_domain_site</a></p>                                                                                                 |
| floorplan_default_row_pattern_site     |                                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Specify rows pattern.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_default_row_pattern_site</a></p>                                                                                                                                                                  |
| floorplan_default_tech_site            |                                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Specify a site as default technical site. After setting a forced one, the auto-calculating default technical-site functionality will be disabled.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_default_tech_site</a></p>                                             |
| floorplan_enable_rectilinear_design    |                                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Specify whether design boundary can be changed to rectilinear shape.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_enable_rectilinear_design</a></p>                                                                                                                 |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| floorplan_extra_row_pattern         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                     | <p>Specify row pattern site names which will be created.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_extra_row_pattern</a></p>                                                                                                                                                                                                                                                                                            |
| floorplan_extra_sites               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                     | <p>Specify site names which will be created.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_extra_sites</a></p>                                                                                                                                                                                                                                                                                                              |
| floorplan_finfet_inst_grid          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                     | <p>The finfet_inst grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of core box. (e.g. the core rows). The x_pitch is default technical site's width and separates the vertical grid lines. The y_pitch is FinFET pitch as defined in LEF and separates the horizontal lines.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> 0.0 0.0 0.0 0.0</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">floorplan_finfet_inst_grid</a></p>                          |
| floorplan_finfet_manufacturing_grid |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                     | <p>The finfet_manufacturing grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of die box (the design .boundary). The x_pitch is Manufacture Grid value and defined in LEF. The y_pitch is FinFET pitch and defined in LEF.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> 0.0 0.0 0.0 0.0</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">floorplan_finfet_manufacturing_grid</a></p>                                                                     |
| floorplan_finfet_placement_grid     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                     | <p>The finfet_placement grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of die box (the design .boundary). The x_pitch is the pitch of lowest vertical metal layer and separates the vertical grid lines. The y_pitch is FinFET pitch as defined in LEF and separates the horizontal grid lines.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> 0.0 0.0 0.0 0.0</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">floorplan_finfet_placement_grid</a></p> |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| floorplan_first_row_site_index                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                 | <p>Specify the first row site to be created.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_first_row_site_index</a></p>                                                                                                                                                                                                                                                             |
| floorplan_include_io_when_init_area             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                 | <p>Include IO cells when calculating the area of modules.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_include_io_when_init_area</a></p>                                                                                                                                                                                                                                      |
| floorplan_initial_all_compatible_core_site_rows |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                 | <p>If true, initCoreRow will create all SITE with type core.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_initial_all_compatible_core_site_rows</a></p>                                                                                                                                                                                                                       |
| floorplan_inst_grid                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                 | <p>The inst grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of box (e.g. the core rows). The x_pitch is default technical site's width and separates the vertical lines. The y_pitch is default technical site's height and separates the horizontal grid lines.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> 0.0 0.0 0.0 0.0</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">floorplan_inst_grid</a></p> |
| floorplan_keep_rows_when_moving_power_domain    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                 | <p>specify whether to keep all existing rows in disjoint power domain when moving, resizing or reshaping a power domain on GUI or by command.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_keep_rows_when_moving_power_domain</a></p>                                                                                                                                         |
| floorplan_last_row_site_index                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>Specify the last row site to be created.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> -1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_last_row_site_index</a></p>                                                                                                                                                                                                                                                                                                            |
| floorplan_layer_track_grid                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                 | <p>The layer_track grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of die box (the design .boundary). The x_pitch is the pitch of lowest vertical metal layer and separates the vertical grid lines. The y_pitch is the pitch of lowest horizontal metal layer and separates the horizontal grid lines.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> 0.0 0.0 0.0 0.0</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">floorplan_layer_track_grid</a></p> |
| floorplan_manufacturing_grid                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                 | <p>The manufacturing grid is in the format {x_origin y_origin x_pitch y_pitch} from the LEF global MANUFACTURINGGRID statement. The x_origin y_origin is lower left corner of die box(the .boundary). See the layer .mfg_grid values if the mfg_grid varies per layer.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> 0.0 0.0 0.0 0.0</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">floorplan_manufacturing_grid</a></p>                                                                     |
| floorplan_max_io_height                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                 | <p>Use maximum IO height to calculate a die box with IO placement. By default, minimum IO height is used.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_max_io_height</a></p>                                                                                                                                                                                                                                                |
| floorplan_minimum_sites                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                 | <p>Specify the number of minimal sites (N) that check_floorplan will check for the same length rule. When same_length_site is specified in floorplan_check_types, this option is required.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_minimum_sites</a></p>                                                                                                                                                                    |
| floorplan_move_child_constraint_with_constraint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                         |                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Specify child constraints move with parent constraint modules at the same time.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_move_child_constraint_with_constraint</a></p>                                                                             |
| floorplan_move_macros_with_constraint   |                                                                                                                                                                                                                                                                                                                                                           |
|                                         | <p>Specify macros move with constraint modules at the same time.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> guide region fence none all</p> <p><b>Default:</b> fence</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_move_macros_with_constraint</a></p>                                                  |
| floorplan_move_preplaced_std_cell_only  |                                                                                                                                                                                                                                                                                                                                                           |
|                                         | <p>Specify only pre-placed std cells move with constraint modules at the same time.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_move_preplaced_std_cell_only</a></p>                                                                                     |
| floorplan_move_std_cell_with_constraint |                                                                                                                                                                                                                                                                                                                                                           |
|                                         | <p>Specify std cells move with constraint modules at the same time.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> guide region fence none all</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_move_std_cell_with_constraint</a></p>                                              |
| floorplan_narrow_channel_threshold      |                                                                                                                                                                                                                                                                                                                                                           |
|                                         | <p>Reports narrow channels whose width (in micros) is smaller than the specified value. When narrow_channel is specified in floorplan_check_types, this option is required.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_narrow_channel_threshold</a></p> |
| floorplan_no_cut_row                    |                                                                                                                                                                                                                                                                                                                                                           |
|                                         | <p>Specify no need to do cut row before other operations.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_no_cut_row</a></p>                                                                                                                                 |

|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| floorplan_placement_grid                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                       | <p>The placement grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of the design .boundary). The x_pitch is the pitch of lowest vertical metal layer and separates the vertical grid lines. The y_pitch is pitch of lowest horizontal metal layer and separates the horizontal grid lines.</p> <p><b>Type:</b> <a href="#">rect</a><br/> <b>Default:</b> 0.0 0.0 0.0 0.0<br/> <b>Edit:</b> No<br/> <b>Reference:</b> <a href="#">floorplan_placement_grid</a></p> |
| floorplan_power_rail_layer                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                       | <p>Specify the layers for calculating power/ground on bottom attribute of cell or techsite. By default the lowest metal layer is used.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_power_rail_layer</a></p>                                                                                                                                                                                                |
| floorplan_row_height_increment_corner_to_corner       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                       | <p>Specify the offset value for user defined row height constraint with corner to corner edge.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_row_height_increment_corner_to_corner</a></p>                                                                                                                                                                                                                       |
| floorplan_row_height_increment_in_corner_to_corner    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                       | <p>Specify the offset value for user defined row height constraint with incorners to corner edge.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_row_height_increment_in_corner_to_corner</a></p>                                                                                                                                                                                                                 |
| floorplan_row_height_increment_in_corner_to_in_corner |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                       | <p>Specify the offset value for user defined row height constraint with incorners to incorners edge.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_row_height_increment_in_corner_to_in_corner</a></p>                                                                                                                                                                                                           |
| floorplan_row_height_multiple                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



|                                    |                                                                                                                                                                                                                                                       |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | Specify the pitch value for user defined row height constraint.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">floorplan_row_height_multiple</a>                                       |
| floorplan_row_site_height          |                                                                                                                                                                                                                                                       |
|                                    | Specify the odd even row height constraints.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> any odd even computed<br><b>Default:</b> any<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">floorplan_row_site_height</a>              |
| floorplan_row_site_width           |                                                                                                                                                                                                                                                       |
|                                    | Specify the odd even row width constraints.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> any odd even<br><b>Default:</b> any<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">floorplan_row_site_width</a>                         |
| floorplan_skip_violations          |                                                                                                                                                                                                                                                       |
|                                    | Specify message ID then the message will be not reported and shown in violation browser.<br>optional)<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">floorplan_skip_violations</a> |
| floorplan_snap_all_corners_to_grid |                                                                                                                                                                                                                                                       |
|                                    | Specify to snap all corners of object to grid.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">floorplan_snap_all_corners_to_grid</a>                                              |
| floorplan_snap_block_grid          |                                                                                                                                                                                                                                                       |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Specify the block snap rules. It can be set to snap to manufacture grid, instance grid, placer grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is manufacturing_grid. For Finfet design, the default is finfet_manufacturing_grid.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> manufacturing inst placement user_define layer_track finfet_inst finfet_manufacturing finfet_placement</p> <p><b>Default:</b> manufacturing</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_snap_block_grid</a></p> |
| floorplan_snap_constraint_grid |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                | <p>Specify the constraint snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is inst_grid. For Finfet design, the default is finfet_inst_grid.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> manufacturing inst placement user_define layer_track finfet_inst finfet_manufacturing finfet_placement</p> <p><b>Default:</b> inst</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_snap_constraint_grid</a></p>               |
| floorplan_snap_core_grid       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                | <p>Specify the core snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is placement_grid. For Finfet design, the default is finfet_placement_grid.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> manufacturing inst placement user_define layer_track finfet_inst finfet_manufacturing finfet_placement</p> <p><b>Default:</b> placement</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_snap_core_grid</a></p>            |
| floorplan_snap_die_grid        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                | <p>Specify the die snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is placement_grid. For Finfet design, the default is finfet_placement_grid.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> manufacturing inst placement user_define layer_track finfet_inst finfet_manufacturing finfet_placement</p> <p><b>Default:</b> placement</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">floorplan_snap_die_grid</a></p>              |
| floorplan_snap_io_grid         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Specify the IO snap rules. It can be set to snap to manufacture grid, instance grid, placemer user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placeme The default is manufacturing_grid. For Finfet design, the default is finfet_manufacturing_gri</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> manufacturing inst placement user_define layer_track finfet_inst finfet_manufacturing finfet_placement<br/> <b>Default:</b> manufacturing<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_snap_io_grid</a></p> |
| floorplan_snap_place_blockage_grid |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                    | <p>Specify the placement blockage snap rules. The default is inst.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> manufacturing inst user_define<br/> <b>Default:</b> inst<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_snap_place_blockage_grid</a></p>                                                                                                                                                                                                                                                                                                    |
| floorplan_snap_place_blockage_type |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                    | <p>Specify the snapping type.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> closest upsize sequential<br/> <b>Default:</b> closest<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_snap_place_blockage_type</a></p>                                                                                                                                                                                                                                                                                                                                           |
| floorplan_user_define_grid         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                    | <p>Specify the user-define grid. Format is {x_offset y_offset x_pitch y_pitch}. The x_offset y_off: the offset of user-define grid. The x_pitch separates the vertical grid lines. The y_pitch sepa the horizontal grid lines. The values are in microns.</p> <p><b>Type:</b> <a href="#">rect</a><br/> <b>Default:</b> 0.0 0.0 0.0 0.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_user_define_grid</a></p>                                                                                                                                                                  |
| floorplan_vertical_row             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                    | <p>Specifies that the rows in the floorplan are vertical (1) or horizontal (0).</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">floorplan_vertical_row</a></p>                                                                                                                                                                                                                                                                                                                                                               |
| flow_branch                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                 |                                                                                                                                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>Branch being run for a hierarchical flow.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_branch</a></p>                                |
| flow_caller_data                |                                                                                                                                                                                                                                   |
|                                 | <p>Data used by the caller of the tool to identify this flow.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_caller_data</a></p>          |
| flow_db_directory               |                                                                                                                                                                                                                                   |
|                                 | <p>Flow directory where results databases are stored.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> dbs</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_db_directory</a></p>                |
| flow_error_errorinfo            |                                                                                                                                                                                                                                   |
|                                 | <p>Tcl error stack in error database.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_error_errorinfo</a></p>                              |
| flow_error_message              |                                                                                                                                                                                                                                   |
|                                 | <p>Tcl error message in error database.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_error_message</a></p>                              |
| flow_error_write_db             |                                                                                                                                                                                                                                   |
|                                 | <p>Write a database when a flow step results in an error condition.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_error_write_db</a></p> |
| flow_exclude_time_for_init_flow |                                                                                                                                                                                                                                   |

|                     |                                                                                                                                                                                                                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>Marks all snapshots created during the init_flow command as 'exclude_time_metric'. This is user steps added to the init_flow internal plugin locations.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flow_exclude_time_for_init_flow</a></p> |
| flow_exit_when_done |                                                                                                                                                                                                                                                                                                                                 |
|                     | <p>Exit after running final step.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flow_exit_when_done</a></p>                                                                                                                                      |
| flow_feature_values |                                                                                                                                                                                                                                                                                                                                 |
|                     | <p>Global feature settings.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flow_feature_values</a></p>                                                                                                                                             |
| flow_features       |                                                                                                                                                                                                                                                                                                                                 |
|                     | <p>Global feature definitions.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flow_features</a></p>                                                                                                                                                |
| flow_footer_tcl     |                                                                                                                                                                                                                                                                                                                                 |
|                     | <p>TCL script to run when a flow ends.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flow_footer_tcl</a></p>                                                                                                                                      |
| flow_header_tcl     |                                                                                                                                                                                                                                                                                                                                 |
|                     | <p>TCL script to run when a flow starts.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">flow_header_tcl</a></p>                                                                                                                                    |
| flow_hier_path      |                                                                                                                                                                                                                                                                                                                                 |

|                           |                                                                                                                                                                                                                         |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Path of current flow within the flow hierarchy.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_hier_path</a></p>             |
| flow_history              |                                                                                                                                                                                                                         |
|                           | <p>Complete flow run history.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_history</a></p>                                    |
| flow_log_directory        |                                                                                                                                                                                                                         |
|                           | <p>Flow directory where log files are stored.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> logs</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_log_directory</a></p>            |
| flow_log_prefix_generator |                                                                                                                                                                                                                         |
|                           | <p>TCL script to create log filenames in Flow tool.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_log_prefix_generator</a></p> |
| flow_mail_on_error        |                                                                                                                                                                                                                         |
|                           | <p>Email to flow_mail_to if an error is detected.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_mail_on_error</a></p>         |
| flow_mail_to              |                                                                                                                                                                                                                         |
|                           | <p>Email address where results are sent.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_mail_to</a></p>                         |

|                                   |                                                                                                                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| flow_metrics_file                 |                                                                                                                                                                                                                           |
|                                   | Flow metrics file for reporting results.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_metrics_file</a>                                          |
| flow_metrics_snapshot_parent_uuid |                                                                                                                                                                                                                           |
|                                   | The snapshot uuid the results from this flow will be appended to.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_metrics_snapshot_parent_uuid</a> |
| flow_metrics_snapshot_uuid        |                                                                                                                                                                                                                           |
|                                   | The snapshot uuid of the most recent flow step executed.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_metrics_snapshot_uuid</a>                 |
| flow_overwrite_db                 |                                                                                                                                                                                                                           |
|                                   | Enables overwriting databases when saving.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_overwrite_db</a>                                       |
| flow_plugin_canonical_current     |                                                                                                                                                                                                                           |
|                                   | Currently running flow plugin canonical flow step.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                       |
| flow_plugin_names                 |                                                                                                                                                                                                                           |
|                                   | Names of all internal plug-in points.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_plugin_names</a>                                             |

|                        |                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| flow_plugin_steps      |                                                                                                                                                                                                                                                                                                                                                                            |
|                        | List of plug-in steps that have been added through edit_flow.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_plugin_steps</a>                                                                                                                                                                      |
| flow_post_db_overwrite |                                                                                                                                                                                                                                                                                                                                                                            |
|                        | If set within a skip_db flow_step, it allows the user to identify the name and type of database present to future flows as the flow_starting_db. This can be set to no_db to force subsequent to begin without a starting db.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_post_db_overwrite</a> |
| flow_remark            |                                                                                                                                                                                                                                                                                                                                                                            |
|                        | Remark from the last flow yaml file loaded.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_remark</a>                                                                                                                                                                                              |
| flow_report_directory  |                                                                                                                                                                                                                                                                                                                                                                            |
|                        | Flow directory where reports are written.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> reports<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_report_directory</a>                                                                                                                                                                                 |
| flow_run_tag           |                                                                                                                                                                                                                                                                                                                                                                            |
|                        | Tags for this particular flow run.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_run_tag</a>                                                                                                                                                                                                      |
| flow_schedule          |                                                                                                                                                                                                                                                                                                                                                                            |



|                             |                                                                                                                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>Set of flows to run after the current flow completes.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_schedule</a></p>                              |
| flow_starting_db            |                                                                                                                                                                                                                                               |
|                             | <p>Starting database for the flow run.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_starting_db</a></p>                                             |
| flow_startup_directory      |                                                                                                                                                                                                                                               |
|                             | <p>Directory where the tool was started.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> /home/manishk/dbschema/novus/newscript</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_startup_directory</a></p> |
| flow_status_file            |                                                                                                                                                                                                                                               |
|                             | <p>File for flow status.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_status_file</a></p>                                                           |
| flow_step_begin_tcl         |                                                                                                                                                                                                                                               |
|                             | <p>Defines the Tcl to run before running the body of each step.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                              |
| flow_step_canonical_current |                                                                                                                                                                                                                                               |
|                             | <p>Currently running canonical flow step.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_step_canonical_current</a></p>                               |
| flow_step_check_tcl         |                                                                                                                                                                                                                                               |

|                       |                                                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>Procedure to check the steps.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_step_check_tcl</a></p>                   |
| flow_step_current     |                                                                                                                                                                                                                  |
|                       | <p>Currently running flow step.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_step_current</a></p>                      |
| flow_step_end_tcl     |                                                                                                                                                                                                                  |
|                       | <p>Defines the Tcl to run after running the body of each step.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                  |
| flow_step_last        |                                                                                                                                                                                                                  |
|                       | <p>Last flow step to have been completed.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_step_last</a></p>               |
| flow_step_last_msg    |                                                                                                                                                                                                                  |
|                       | <p>Message provided for the last step that was run.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_step_last_msg</a></p> |
| flow_step_last_status |                                                                                                                                                                                                                  |
|                       | <p>Status for the last flow step run.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> not_run</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_step_last_status</a></p>       |
| flow_step_next        |                                                                                                                                                                                                                  |

|                                  |                                                                                                                                                                                                                                  |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Next step to run in the flow.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_step_next</a></p>                                        |
| flow_steps                       |                                                                                                                                                                                                                                  |
|                                  | <p>List of all created flow_steps</p> <p><b>Type:</b> <a href="#">obj(flow_step)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                       |
| flow_summary_tcl                 |                                                                                                                                                                                                                                  |
|                                  | <p>TCL script to run during run_flow summary.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_summary_tcl</a></p>                         |
| flow_template_feature_definition |                                                                                                                                                                                                                                  |
|                                  | <p>Features list and status for the current template.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_template_feature_definition</a></p> |
| flow_template_tools              |                                                                                                                                                                                                                                  |
|                                  | <p>List of tools included in flow templates being run.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_template_tools</a></p>             |
| flow_template_type               |                                                                                                                                                                                                                                  |
|                                  | <p>Type of template being run.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_template_type</a></p>                                      |
| flow_template_version            |                                                                                                                                                                                                                                  |

|                                     |                                                                                                                                                                                                                  |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | Version of the template being run.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_template_version</a>                                   |
| flow_top                            |                                                                                                                                                                                                                  |
|                                     | Flow to run by default.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_top</a>                                                           |
| flow_user_templates                 |                                                                                                                                                                                                                  |
|                                     | Flow user template definitions.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_user_templates</a>                                        |
| flow_verbose                        |                                                                                                                                                                                                                  |
|                                     | Enables printing run information in the logfile.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_verbose</a>                              |
| flow_working_directory              |                                                                                                                                                                                                                  |
|                                     | Flow directory where the flow is being run.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> .<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_working_directory</a>                          |
| flow_write_db_snapshot              |                                                                                                                                                                                                                  |
|                                     | Create a snapshot for write_db on flow steps which save a database.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">flow_write_db_snapshot</a> |
| flow_write_db_snapshot_exclude_time |                                                                                                                                                                                                                  |

|                            |                                                                                                                                                                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Marks all snapshots created for write_db on flow steps which save a database as 'exclude_time_metric'.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_write_db_snapshot_exclude_time</a></p> |
| flow_yamllint_exec         |                                                                                                                                                                                                                                                                                         |
|                            | <p>Yamllint executable command and arguments for checking yaml files.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> yamllint</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flow_yamllint_exec</a></p>                                                |
| flows                      |                                                                                                                                                                                                                                                                                         |
|                            | <p>List of all created flows</p> <p><b>Type:</b> <a href="#">obj(flow)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                        |
| flowtool_exit_timeout      |                                                                                                                                                                                                                                                                                         |
|                            | <p>Maximum amount of time in seconds Flow tool will wait after a tool exits for a completed sta</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 30</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flowtool_exit_timeout</a></p>                            |
| flowtool_extra_arguments   |                                                                                                                                                                                                                                                                                         |
|                            | <p>Extra arguments passed to Flowtool when running in tool mode.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flowtool_extra_arguments</a></p>                                                     |
| flowtool_metrics_qor_excel |                                                                                                                                                                                                                                                                                         |
|                            | <p>Metrics file to write after tools exit.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flowtool_metrics_qor_excel</a></p>                                                                         |
| flowtool_metrics_qor_html  |                                                                                                                                                                                                                                                                                         |

|                                  |                                                                                                                                                                                                                                                                    |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Metrics file to write after tools exit.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flowtool_metrics_qor_html</a></p>                                                     |
| flowtool_metrics_qor_text        |                                                                                                                                                                                                                                                                    |
|                                  | <p>Metrics file to write after tools exit.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flowtool_metrics_qor_text</a></p>                                                     |
| flowtool_metrics_qor_vivid       |                                                                                                                                                                                                                                                                    |
|                                  | <p>Metrics file to write after tools exit.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flowtool_metrics_qor_vivid</a></p>                                                    |
| flowtool_predict_full_names      |                                                                                                                                                                                                                                                                    |
|                                  | <p>Show skip_metric flow name in prediction.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flowtool_predict_full_names</a></p>                                                |
| flowtool_schedule_flow_immediate |                                                                                                                                                                                                                                                                    |
|                                  | <p>Start schedule_flow processes immediately instead of waiting for current tool exit.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flowtool_schedule_flow_immediate</a></p> |
| flowtool_summary_tcl             |                                                                                                                                                                                                                                                                    |
|                                  | <p>Tcl script to run at the end of flowtool.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> puts [report_metric -format text]</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">flowtool_summary_tcl</a></p>                         |
| foreign_cells                    |                                                                                                                                                                                                                                                                    |

|                                                   |                                                                                                                                                                                                                                                                        |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | <p>Short-cut for [get_db base_cells .foreign_cells]</p> <p><b>Type:</b> <a href="#">obj(foreign_cell)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                        |
| formal_verification_constraints                   |                                                                                                                                                                                                                                                                        |
|                                                   | <p>List of all formal_verification_constraints</p> <p><b>Type:</b> <a href="#">obj(formal_verification_constraint)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                           |
| gcell_grids                                       |                                                                                                                                                                                                                                                                        |
|                                                   | <p>Short-cut to all the gcell_grid objects in the design.</p> <p><b>Type:</b> <a href="#">obj(gcell_grid)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                    |
| gcells                                            |                                                                                                                                                                                                                                                                        |
|                                                   | <p>The gcells for this design. The gcells are only created after global route has occurred.</p> <p><b>Type:</b> <a href="#">obj(gcell)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                       |
| generate_special_via_accuracy_effort              |                                                                                                                                                                                                                                                                        |
|                                                   | <p>Specifies viagen effort to generate vias.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> high</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_accuracy_effort</a></p> |
| generate_special_via_add_pin_to_pin_vias          |                                                                                                                                                                                                                                                                        |
|                                                   | <p>Inserts vias to connect pins between cover macros</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_add_pin_to_pin_vias</a></p>                               |
| generate_special_via_align_merged_stack_via_metal |                                                                                                                                                                                                                                                                        |

|                                                |                                                                                                                                                                                                                            |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | Aligns merged stack vias<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_align_merged_stack_via_metal</a>                          |
| generate_special_via_allow_via_expand          |                                                                                                                                                                                                                            |
|                                                | Allows via expansion to meet LEF rules<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_allow_via_expand</a>                         |
| generate_special_via_allow_wire_shape_change   |                                                                                                                                                                                                                            |
|                                                | Allows via to change the metal shape of existing wire<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_allow_wire_shape_change</a>   |
| generate_special_via_area_only                 |                                                                                                                                                                                                                            |
|                                                | adds via only within the specified area<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_area_only</a>                              |
| generate_special_via_bot_enclosure             |                                                                                                                                                                                                                            |
|                                                | Customize bot enclosure<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_bot_enclosure</a>                                           |
| generate_special_via_check_signal_routes       |                                                                                                                                                                                                                            |
|                                                | Honors pre-existing signal routes<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> 0 1 2<br><b>Default:</b> 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_check_signal_routes</a> |
| generate_special_via_create_double_row_cut_via |                                                                                                                                                                                                                            |



|                                             |                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>Expands vias to have an additional cut than original for one-row or one-column vias if poss</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> no_extra_cut add_extra_cut add_extra_cut_and_metal</p> <p><b>Default:</b> no_extra_cut</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_create_double_row_cut_via</a></p> |
| generate_special_via_create_max_row_cut_via |                                                                                                                                                                                                                                                                                                                                                                               |
|                                             | <p>Creates max rows of via cuts</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_create_max_row_cut_via</a></p>                                                                                                                                                        |
| generate_special_via_cut_class_preference   |                                                                                                                                                                                                                                                                                                                                                                               |
|                                             | <p>Specifies via cut class preference</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_cut_class_preference</a></p>                                                                                                                                                |
| generate_special_via_cut_spacing            |                                                                                                                                                                                                                                                                                                                                                                               |
|                                             | <p>Customize cut spacing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_cut_spacing</a></p>                                                                                                                                                                           |
| generate_special_via_disable_via_merge      |                                                                                                                                                                                                                                                                                                                                                                               |
|                                             | <p>Disables via merging</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_disable_via_merge</a></p>                                                                                                                                                                     |
| generate_special_via_enable_check_drc       |                                                                                                                                                                                                                                                                                                                                                                               |
|                                             | <p>Invokes check_drc</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_enable_check_drc</a></p>                                                                                                                                                                         |
| generate_special_via_extend_out_wire_end    |                                                                                                                                                                                                                                                                                                                                                                               |

|                                              |                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | <p>Extends via outside of existing wires end</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_extend_out_wire_end</a></p>                                                                                        |
| generate_special_via_full_cut_via_only       |                                                                                                                                                                                                                                                                                                                        |
|                                              | <p>allows to create full cut via only</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_full_cut_via_only</a></p>                                                                                                |
| generate_special_via_hookup_auto_set_viacell |                                                                                                                                                                                                                                                                                                                        |
|                                              | <p>Hookup auto set viaCell</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_hookup_auto_set_viacell</a></p>                                                                                                     |
| generate_special_via_hookup_contact_max      |                                                                                                                                                                                                                                                                                                                        |
|                                              | <p>Sets the max distance to create hookup via from contact, and the max distance between two hookup via on a continuous M0 PG pin.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 0, 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_hookup_contact_max</a></p> |
| generate_special_via_hookup_contact_pg_track |                                                                                                                                                                                                                                                                                                                        |
|                                              | <p>Sets start point and the pitch of PG track used by hookup contact.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 0, 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_hookup_contact_pg_track</a></p>                                                         |
| generate_special_via_hookup_fixed_grid       |                                                                                                                                                                                                                                                                                                                        |
|                                              | <p>Specifies to add hookup via over the fixed Grid.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_hookup_fixed_grid</a></p>                                                                                  |
| generate_special_via_hookup_min_distance     |                                                                                                                                                                                                                                                                                                                        |

|                                               |                                                                                                                                                                                                                                      |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | min distance between two pg vias<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> min<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_hookup_min_distance</a>                                     |
| generate_special_via_hookup_preplace_fix      |                                                                                                                                                                                                                                      |
|                                               | Enhance hookup via generation runtime<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_hookup_preplace_fix</a>                                     |
| generate_special_via_hookup_rail_pair         |                                                                                                                                                                                                                                      |
|                                               | Create hookup vias over pairing followpins<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_hookup_rail_pair</a>                              |
| generate_special_via_hookup_via_distance      |                                                                                                                                                                                                                                      |
|                                               | Sets via hookup distance and boundary spacing<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> 4pitch 4pitch 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_hookup_via_distance</a>            |
| generate_special_via_hookup_via_rule          |                                                                                                                                                                                                                                      |
|                                               | Specifies the viarule used in specified area<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_hookup_via_rule</a>                              |
| generate_special_via_hookup_via_style         |                                                                                                                                                                                                                                      |
|                                               | Sets via hookup style<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> none loose compact moderate<br><b>Default:</b> none<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">generate_special_via_hookup_via_style</a> |
| generate_special_via_hookup_virtual_trim_grid |                                                                                                                                                                                                                                      |

|                                             |                                                                                                                                                                                                                                                                                    |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>Specifies the virtual trim grid for pg hookup</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_hookup_virtual_trim_grid</a></p>                                           |
| generate_special_via_ignore_design_boundary |                                                                                                                                                                                                                                                                                    |
|                                             | <p>Allow vias to be on design boundary</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_ignore_design_boundary</a></p>                                                      |
| generate_special_via_ignore_drc             |                                                                                                                                                                                                                                                                                    |
|                                             | <p>Ignores DRC</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_ignore_drc</a></p>                                                                                          |
| generate_special_via_ignore_rule_enclosure  |                                                                                                                                                                                                                                                                                    |
|                                             | <p>Ignores enclosures defined in VIARULE GENERATE</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_ignore_rule_enclosure</a></p>                                             |
| generate_special_via_inherit_wire_status    |                                                                                                                                                                                                                                                                                    |
|                                             | <p>Inherits wire status</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_inherit_wire_status</a></p>                                                                        |
| generate_special_via_keep_existing_via      |                                                                                                                                                                                                                                                                                    |
|                                             | <p>Keeps existing vias</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> remove keep_with_new_via keep_with_no_new_via</p> <p><b>Default:</b> remove</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_keep_existing_via</a></p> |
| generate_special_via_keep_fixed_via         |                                                                                                                                                                                                                                                                                    |

|                                                |                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Keeps fixed vias not modifiable</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_keep_fixed_via</a></p>                                                                                                         |
| generate_special_via_mincut_preference         |                                                                                                                                                                                                                                                                                                                          |
|                                                | <p>Specifies min cut preference</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_mincut_preference</a></p>                                                                                                             |
| generate_special_via_opt_cross_via             |                                                                                                                                                                                                                                                                                                                          |
|                                                | <p>Controls the creation of additional via cuts</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_opt_cross_via</a></p>                                                                                            |
| generate_special_via_opt_via_on_routing_track  |                                                                                                                                                                                                                                                                                                                          |
|                                                | <p>Frees up via tracks</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_opt_via_on_routing_track</a></p>                                                                                                          |
| generate_special_via_parameterized_via_only    |                                                                                                                                                                                                                                                                                                                          |
|                                                | <p>generate parameterized vias only, auto sets true for OA flow, false for non-OA flow.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto true false</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_parameterized_via_only</a></p> |
| generate_special_via_partial_overlap_threshold |                                                                                                                                                                                                                                                                                                                          |
|                                                | <p>Specifies ratio threshold of via creation on partial overlap wires</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_partial_overlap_threshold</a></p>                                                          |
| generate_special_via_preferred_vias_only       |                                                                                                                                                                                                                                                                                                                          |

|                                               |                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | <p>Generate only vias set in -viarule_preference</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> use_lef open keep</p> <p><b>Default:</b> use_lef</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_preferred_vias_only</a></p> |
| generate_special_via_prefix                   |                                                                                                                                                                                                                                                                                     |
|                                               | <p>Specifies prefix name for all vias</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_prefix</a></p>                                                                         |
| generate_special_via_reference_boundary       |                                                                                                                                                                                                                                                                                     |
|                                               | <p>Sets reference boundary</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> design core</p> <p><b>Default:</b> design</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_reference_boundary</a></p>                               |
| generate_special_via_respect_stdcell_geometry |                                                                                                                                                                                                                                                                                     |
|                                               | <p>Respects all standard cell patterns</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_respect_stdcell_geometry</a></p>                                                     |
| generate_special_via_respect_wire_ndr         |                                                                                                                                                                                                                                                                                     |
|                                               | <p>Control whether viagen support ndr rules</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_respect_wire_ndr</a></p>                                                        |
| generate_special_via_rule_preference          |                                                                                                                                                                                                                                                                                     |
|                                               | <p>Specifies via rule preference</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_rule_preference</a></p>                                                                |

|                                                 |                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| generate_special_via_set_via_expand_dir         |                                                                                                                                                                                                                                                                                                                 |
|                                                 | <p>Sets via expansion direction</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto horizontal vertical</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_set_via_expand_dir</a></p>                                           |
| generate_special_via_snap_via_center_to_grid    |                                                                                                                                                                                                                                                                                                                 |
|                                                 | <p>Specifies the via center alignment style per routing layer</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_snap_via_center_to_grid</a></p>                                                            |
| generate_special_via_split_long_via_global_grid |                                                                                                                                                                                                                                                                                                                 |
|                                                 | <p>Controls global grids so vias on them are in line</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 0 0 0 0 0 0 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_split_long_via_global_grid</a></p>                                                       |
| generate_special_via_symmetrical_via_only       |                                                                                                                                                                                                                                                                                                                 |
|                                                 | <p>adds symmetrical via only, auto sets true for OA flow, false for non-OA flow.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto true false</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_symmetrical_via_only</a></p> |
| generate_special_via_top_enclosure              |                                                                                                                                                                                                                                                                                                                 |
|                                                 | <p>Customize top enclosure</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_top_enclosure</a></p>                                                                                                         |
| generate_special_via_use_track_offset           |                                                                                                                                                                                                                                                                                                                 |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | <p>Uses track offset</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_use_track_offset</a></p>                                                                                                                                                                                                                                                                                                                                 |
| generate_special_via_use_trim_metal_enclosure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>Force tool to use enclosures with trim layer</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">generate_special_via_use_trim_metal_enclosure</a></p>                                                                                                                                                                                                                                                                                              |
| get_db_display_limit                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>Controls the number of objects that will be displayed when using get_db with an attribute p like '*' rather than single attribute name. If the value is 10, 'get_db insts .*' will only display 1 objects with all their attributes, and will only display the first 10 objects for any attribute that object list (e.g. the first 10 pins for each inst .pins value).</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 10</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">get_db_display_limit</a></p> |
| group_instance_suffix                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>base name of group hinst is concatenation of group module name and suffix defined by this global</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> i</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">group_instance_suffix</a></p>                                                                                                                                                                                                                                                                    |
| groups                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>Short-cut for [get_db current_design .groups]</p> <p><b>Type:</b> <a href="#">obj(group)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                 |
| gui_enabled                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>dummy attribute, always return false in case of innovus</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                           |



|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gui_lines    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|              | <p>Short-cut to all the gui_line objects in the design.</p> <p><b>Type:</b> <a href="#">obj(gui_line)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                           |
| gui_polygons |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|              | <p>Short-cut to all the gui_polygon objects in the design.</p> <p><b>Type:</b> <a href="#">obj(gui_polygon)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                     |
| gui_rects    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|              | <p>Short-cut to all the gui_rect objects in the design.</p> <p><b>Type:</b> <a href="#">obj(gui_rect)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                           |
| gui_shapes   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|              | <p>Short-cut to all the gui_shape objects in the design.</p> <p><b>Type:</b> <a href="#">obj(gui_rect)*</a> <a href="#">obj(gui_line)*</a> <a href="#">obj(gui_polygon)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                         |
| gui_texts    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|              | <p>Short-cut to all the gui_text objects in the design.</p> <p><b>Type:</b> <a href="#">obj(gui_text)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                           |
| gui_verbose  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|              | <p>Controls which GUI actions are logged. The choices are:</p> <ul style="list-style-type: none"> <li>all: all GUI actions;</li> <li>db: GUI actions that affect the DB like gui_select, set_layer_preference, but not display-on actions like gui_zoom, gui_pan;</li> <li>none: no gui_* commands are logged, so the .log and .cmd file are incomplete</li> </ul> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> all db none</p> <p><b>Default:</b> db</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">gui_verbose</a></p> |

|                                              |                                                                                                                                                                                                                                                                                      |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hdl_allow_inout_constant_port_connect        |                                                                                                                                                                                                                                                                                      |
|                                              | <p>When set to true, allows constants to be connected to inout ports.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_allow_inout_constant_port_connect</a></p>                               |
| hdl_allow_instance_name_conflict             |                                                                                                                                                                                                                                                                                      |
|                                              | <p>Allow a Verilog instance and wire to have the same name.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_allow_instance_name_conflict</a></p>                                              |
| hdl_allow_positional_connections_for_pg_inst |                                                                                                                                                                                                                                                                                      |
|                                              | <p>Allows positional port association for instantiation of cell with power and ground pins..</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_allow_positional_connections_for_pg_inst</a></p> |
| hdl_append_generic_ports                     |                                                                                                                                                                                                                                                                                      |
|                                              | <p>When set to false, does not append generic port information to the module name.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_append_generic_ports</a></p>                                |
| hdl_array_naming_style                       |                                                                                                                                                                                                                                                                                      |
|                                              | <p>Format for naming individual bits of array variables.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> %s[%d]</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_array_naming_style</a></p>                                                        |
| hdl_async_set_reset                          |                                                                                                                                                                                                                                                                                      |
|                                              | <p>Specifies that the listed signals be used to drive latch set/reset pins when possible.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_async_set_reset</a></p>                              |

|                                      |                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hdl_auto_async_set_reset             |                                                                                                                                                                                                                                                                                                                       |
|                                      | <p>Forces use of asynchronous set and reset pins, rather than data input pins, to implement asynchronous set and reset operations for latches.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_auto_async_set_reset</a></p>          |
| hdl_auto_sync_set_reset              |                                                                                                                                                                                                                                                                                                                       |
|                                      | <p>When set to true, the set/reset signal will be preserved with the flip-flop during generic optimization.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_auto_sync_set_reset</a></p>                                               |
| hdl_bus_wire_naming_style            |                                                                                                                                                                                                                                                                                                                       |
|                                      | <p>Format for naming individual bits of bus wires.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> %s[%d]<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_bus_wire_naming_style</a></p>                                                                                                  |
| hdl_case_mux_threshold               |                                                                                                                                                                                                                                                                                                                       |
|                                      | <p>Applies to muxes generated for case statements. When the the number of data inputs is less than the threshold, AND/OR logic is generated rather than a mux.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 2<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_case_mux_threshold</a></p> |
| hdl_case_sensitive_instances         |                                                                                                                                                                                                                                                                                                                       |
|                                      | <p>When set to false, allows case insensitive name matching for instances in Verilog whose design exists in VHDL or Verilog files.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> none<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_case_sensitive_instances</a></p>                 |
| hdl_convert_one_bit_vector_to_scalar |                                                                                                                                                                                                                                                                                                                       |

|                                          |                                                                                                                                                                                                                                                       |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>When set to true converts one bit vector to scalar.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_convert_one_bit_vector_to_scalar</a></p>                |
| hdl_convert_onebit_vector_wire_to_scalar |                                                                                                                                                                                                                                                       |
|                                          | <p>Converts a one-bit vector declaration into a scalar.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_convert_onebit_vector_wire_to_scalar</a></p>           |
| hdl_create_label_for_unlabeled_generate  |                                                                                                                                                                                                                                                       |
|                                          | <p>Specifies whether a name is assigned to unnamed generate blocks.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_create_label_for_unlabeled_generate</a></p> |
| hdl_decimal_parameter_name               |                                                                                                                                                                                                                                                       |
|                                          | <p>When set to true it prints out parameters in decimal format.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_decimal_parameter_name</a></p>                 |
| hdl_delete_transparent_latch             |                                                                                                                                                                                                                                                       |
|                                          | <p>Deletes latches that are always enabled.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_delete_transparent_latch</a></p>                                    |
| hdl_early_redundant_logic_removal        |                                                                                                                                                                                                                                                       |
|                                          | <p>Do an early cleanup of redundant steering logic nodes.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_early_redundant_logic_removal</a></p>                 |
| hdl_enable_proc_name                     |                                                                                                                                                                                                                                                       |

|                                      |                                                                                                                                                                                                                                                           |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Enable generation of instance hdl_proc_name attributes.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_enable_proc_name</a></p>                                |
| hdl_enable_real_support              |                                                                                                                                                                                                                                                           |
|                                      | <p>Enable elaboration of constant expressions of real type.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_enable_real_support</a></p>                            |
| hdl_error_on_blackbox                |                                                                                                                                                                                                                                                           |
|                                      | <p>An error is issued if a blackbox is found in a design.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_error_on_blackbox</a></p>                                |
| hdl_error_on_latch                   |                                                                                                                                                                                                                                                           |
|                                      | <p>An error is issued if a latch is inferred for a design.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_error_on_latch</a></p>                                  |
| hdl_error_on_logic_abstract          |                                                                                                                                                                                                                                                           |
|                                      | <p>An error is issued if a logic abstract is found in a design.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_error_on_logic_abstract</a></p>                    |
| hdl_error_on_negedge                 |                                                                                                                                                                                                                                                           |
|                                      | <p>Specifies that an error be issued when a falling edge-triggered flip-flop is inferred.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_error_on_negedge</a></p> |
| hdl_exclude_parameter_in_cell_search |                                                                                                                                                                                                                                                           |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>This attribute applies to situation when a parameterized module instantiation is resolved with Liberty complex cell as opposed to an hdl module. This attribute will contain the list of names of the cells for which the search is done with the name of the module but without the parameter information.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_exclude_parameter_in_cell_search</a></p> |
| hdl_ff_keep_explicit_feedback                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                    | <p>Controls how flip-flop stable states are implemented for feedback assignments that are explicitly specified in the RTL, e.g., D = Q. Explicit feedback is always kept when hdl_ff_keep_feedback is true.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_ff_keep_explicit_feedback</a></p>                                                                                                           |
| hdl_ff_keep_feedback                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                    | <p>Controls how flip-flop stable states are implemented: with a feedback path from the Q output to the D input (true), or with a synchronous enable signal (false).</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_ff_keep_feedback</a></p>                                                                                                                                                           |
| hdl_file_row_column_consider_modules_as_leaf_insts |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                    | <p>Specifies list of primitive modules.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_file_row_column_consider_modules_as_leaf_insts</a></p>                                                                                                                                                                                                                                                          |
| hdl_flatten_complex_port                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                    | <p>To convert complex ports to vector ports.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_flatten_complex_port</a></p>                                                                                                                                                                                                                                                                              |
| hdl_flatten_complex_port_in_bottom_up_flow         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                           |                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | Used for creating one dimensional packed version of complex ports in bottom up flow.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">hdl_flatten_complex_port_in_bottom_up_flow</a>                                                                              |
| hdl_generate_index_style  |                                                                                                                                                                                                                                                                                                                                    |
|                           | Format of 'for generate' block labels.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> %s[%d]<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">hdl_generate_index_style</a>                                                                                                                                          |
| hdl_generate_separator    |                                                                                                                                                                                                                                                                                                                                    |
|                           | String used to separate generate block labels.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> .<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">hdl_generate_separator</a>                                                                                                                                         |
| hdl_ignore_pragma_names   |                                                                                                                                                                                                                                                                                                                                    |
|                           | Specifies pragmas to be ignored.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> coverage<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">hdl_ignore_pragma_names</a>                                                                                                                                               |
| hdl_index_mux_threshold   |                                                                                                                                                                                                                                                                                                                                    |
|                           | Applies to muxes generated for variable index array references, such as 'y = x[i]'. When the number of data inputs is less than the threshold, AND/OR logic is generated rather than a n<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">hdl_index_mux_threshold</a> |
| hdl_inout_assign          |                                                                                                                                                                                                                                                                                                                                    |
|                           | When set to true, interprets Verilog continuous assign statements as being bidirectional.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">hdl_inout_assign</a>                                                                                                  |
| hdl_inout_wand_wor_assign |                                                                                                                                                                                                                                                                                                                                    |

|                                  |                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>When set to true treats assignment to wand/wor net type as bidirectional.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_inout_wand_wor_assign</a></p>                                                                                                                  |
| hdl_instance_array_naming_style  |                                                                                                                                                                                                                                                                                                                                                                    |
|                                  | <p>Format of instance array labels.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> %s[%d]</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_instance_array_naming_style</a></p>                                                                                                                                                  |
| hdl_interface_separator          |                                                                                                                                                                                                                                                                                                                                                                    |
|                                  | <p>String used to separate interface elements.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> _</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_interface_separator</a></p>                                                                                                                                                    |
| hdl_keep_first_module_definition |                                                                                                                                                                                                                                                                                                                                                                    |
|                                  | <p>Keep first seen definition of a module.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_keep_first_module_definition</a></p>                                                                                                                                             |
| hdl_language                     |                                                                                                                                                                                                                                                                                                                                                                    |
|                                  | <p>Specifies the default HDL language mode.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> v2001</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_language</a></p>                                                                                                                                                              |
| hdl_latch_keep_feedback          |                                                                                                                                                                                                                                                                                                                                                                    |
|                                  | <p>Controls how explicitly-specified latch stable states (e.g, q &lt;= q) are implemented: with a feedback path from the Q output to the D input (true), or with a latch enable signal (false).</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_latch_keep_feedback</a></p> |
| hdl_link_from_any_lib            |                                                                                                                                                                                                                                                                                                                                                                    |



|                                  |                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Resolve an instantiation with a 'unique' binding found across all present libs.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_link_from_any_lib</a></p>                                                                                                  |
| hdl_max_loop_limit               |                                                                                                                                                                                                                                                                                                                                                     |
|                                  | <p>Determines the maximum number of iterations for unfolding a loop construct of any type.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 4096</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_max_loop_limit</a></p>                                                                                              |
| hdl_max_map_to_mux_control_width |                                                                                                                                                                                                                                                                                                                                                     |
|                                  | <p>Specifies the maximum multiplexer size, in terms of the width of the control input, that can be generated for HDL 'case' statements marked with the 'map_to_mux' pragma.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 10</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_max_map_to_mux_control_width</a></p> |
| hdl_max_memory_address_range     |                                                                                                                                                                                                                                                                                                                                                     |
|                                  | <p>Sets the maximum addressable memory range to prevent synthesis of large memories.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 4194304</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_max_memory_address_range</a></p>                                                                                       |
| hdl_max_recursion_limit          |                                                                                                                                                                                                                                                                                                                                                     |
|                                  | <p>Sets the maximum number of elaborations for recursive instantiations to prevent possible infinite recursions.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1024</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_max_recursion_limit</a></p>                                                                   |
| hdl_module_definition_resolution |                                                                                                                                                                                                                                                                                                                                                     |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Resolves the definition of module in case of clash to first seen module or to last seen module according to the directory rules. It can take four values viz last, first, first_dir_netlist, first_dir_rtl.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> last</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_module_definition_resolution</a></p> |
| hdl_nc_compatible_module_link     |                                                                                                                                                                                                                                                                                                                                                                                             |
|                                   | <p>When true, implements NC compatible binding rules for linking module/entity instantiations corresponding definitions.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_nc_compatible_module_link</a></p>                                                                                            |
| hdl_new_inout_assign              |                                                                                                                                                                                                                                                                                                                                                                                             |
|                                   | <p>When set to true, interprets Verilog continuous assign statements as being bidirectional.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_new_inout_assign</a></p>                                                                                                                                 |
| hdl_overwrite_command_line_macros |                                                                                                                                                                                                                                                                                                                                                                                             |
|                                   | <p>Enables overwriting of command line macros.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_overwrite_command_line_macros</a></p>                                                                                                                                                                  |
| hdl_parameter_module_name         |                                                                                                                                                                                                                                                                                                                                                                                             |
|                                   | <p>When set to false, skips all parameter values in parameterized module name for top-level design or design specified with 'elaborate' command.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_parameter_module_name</a></p>                                                                        |
| hdl_parameter_naming_style        |                                                                                                                                                                                                                                                                                                                                                                                             |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>Format of the suffix added to original module name per each parameter overwrite.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> <code>_%s%d</code></p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_parameter_naming_style</a></p>                                                                                                                                     |
| hdl_preserve_async_set_reset_priority_logic |                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                             | <p>When true, prevents RC from generating redundant logic when the mapper selects a flip-flop library cell that has the same priority for the asynchronous set and reset operations as is specified in the input HDL.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_preserve_async_set_reset_priority_logic</a></p> |
| hdl_preserve_dangling_output_nets           |                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                             | <p>When true, preserves dangling nets connected to instance output ports.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_preserve_dangling_output_nets</a></p>                                                                                                                                                        |
| hdl_preserve_supply_nets                    |                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                             | <p>Preserve supply nets in user input.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_preserve_supply_nets</a></p>                                                                                                                                                                                                    |
| hdl_preserve_sync_control_logic             |                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                             | <p>Instructs Genus to preserve the sync control logic structure. Setting this attribute can impact QoR.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_preserve_sync_control_logic</a></p>                                                                                                                           |
| hdl_preserve_sync_set_reset                 |                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                             | <p>Instructs Genus to preserve the sync set/reset control logic structure. Setting this attribute can impact QoR.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_preserve_sync_set_reset</a></p>                                                                                                                     |

|                                 |                                                                                                                                                                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hdl_preserve_unused_flop        |                                                                                                                                                                                                                                                                |
|                                 | <p>When set to true, preserves flip-flops that do not affect any outputs.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_preserve_unused_flop</a></p>                  |
| hdl_preserve_unused_latch       |                                                                                                                                                                                                                                                                |
|                                 | <p>When set to true, preserves latches that do not affect any outputs.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_preserve_unused_latch</a></p>                    |
| hdl_preserve_unused_registers   |                                                                                                                                                                                                                                                                |
|                                 | <p>When set to true, preserves flip-flops and latches that do not affect any outputs.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_preserve_unused_registers</a></p> |
| hdl_primitive_input_multibit    |                                                                                                                                                                                                                                                                |
|                                 | <p>Allow a Verilog primitive gate instantiated in RTL to have multi bit input.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_primitive_input_multibit</a></p>         |
| hdl_record_naming_style         |                                                                                                                                                                                                                                                                |
|                                 | <p>Format for naming individual bits of record variables.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> %s[%s]</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_record_naming_style</a></p>                                |
| hdl_register_array_naming_style |                                                                                                                                                                                                                                                                |
|                                 | <p>Format for naming array type registers.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_register_array_naming_style</a></p>                                           |

|                                    |                                                                                                                                                                                                                                                                            |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hdl_register_naming_style          |                                                                                                                                                                                                                                                                            |
|                                    | <p>Format for naming register instances.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> %s_reg%s</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_register_naming_style</a></p>                                                         |
| hdl_register_record_naming_style   |                                                                                                                                                                                                                                                                            |
|                                    | <p>Format for naming record type registers.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_register_record_naming_style</a></p>                                                     |
| hdl_rename_cdn_flop_pins           |                                                                                                                                                                                                                                                                            |
|                                    | <p>Provides the control to change the pin names of internal flops.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_rename_cdn_flop_pins</a></p>                                      |
| hdl_rename_cdn_latch_pins          |                                                                                                                                                                                                                                                                            |
|                                    | <p>Provides the control to change the pin names of internal latches.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_rename_cdn_latch_pins</a></p>                                   |
| hdl_report_case_info               |                                                                                                                                                                                                                                                                            |
|                                    | <p>When set to true the case pragma information will be reported.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_report_case_info</a></p>                                          |
| hdl_resolve_instance_with_lib_cell |                                                                                                                                                                                                                                                                            |
|                                    | <p>Use a technology element rather than a Verilog module or VHDL architecture for any instar</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_resolve_instance_with_lib_cell</a></p> |

|                                                           |                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hdl_resolve_parameterized_instance_with_structural_module |                                                                                                                                                                                                                                                                                                                           |
|                                                           | <p>Uses a module named with a parameterized name for any instance of parameterized Verilog modules or VHDL entities.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_resolve_parameterized_instance_with_structural_module</a></p> |
| hdl_sv_module_wrapper                                     |                                                                                                                                                                                                                                                                                                                           |
|                                                           | <p>Enables generation of System Verilog module wrapper.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_sv_module_wrapper</a></p>                                                                                                  |
| hdl_sync_set_reset                                        |                                                                                                                                                                                                                                                                                                                           |
|                                                           | <p>Specifies that the listed signals be used to drive flip-flop set/reset pins when possible.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_sync_set_reset</a></p>                                                                |
| hdl_track_filename_row_col                                |                                                                                                                                                                                                                                                                                                                           |
|                                                           | <p>Enable/disable HDL file:row:col info tracking on instances.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_track_filename_row_col</a></p>                                                                                      |
| hdl_track_module_elab_memory_and_runtime                  |                                                                                                                                                                                                                                                                                                                           |
|                                                           | <p>Track elaboration memory and runtime for all modules.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_track_module_elab_memory_and_runtime</a></p>                                                                              |
| hdl_unconnected_marking_done                              |                                                                                                                                                                                                                                                                                                                           |
|                                                           | <p>Specifies whether the undriven signals have been connected based on the value of the 'hdl_unconnected_value' attribute.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                              |

|                                            |                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hdl_unconnected_value                      |                                                                                                                                                                                                                                                                                                                                                    |
|                                            | <p>Connects any undriven signal (undriven input in module or cell instantiation/ undriven output undriven signal in a module to the specified value unless the 'none' value is specified.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_unconnected_value</a></p> |
| hdl_use_block_prefix                       |                                                                                                                                                                                                                                                                                                                                                    |
|                                            | <p>Whether block statement labels should be used to prefix instances.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_use_block_prefix</a></p>                                                                                                                     |
| hdl_use_case_generate_prefix               |                                                                                                                                                                                                                                                                                                                                                    |
|                                            | <p>Specifies whether 'case' generate labels should be used to prefix instances.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_use_case_generate_prefix</a></p>                                                                                                   |
| hdl_use_current_dir_before_hdl_search_path |                                                                                                                                                                                                                                                                                                                                                    |
|                                            | <p>Uses current directory before hdl search path.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_use_current_dir_before_hdl_search_path</a></p>                                                                                                                  |
| hdl_use_cw_first                           |                                                                                                                                                                                                                                                                                                                                                    |
|                                            | <p>Uses a ChipWare component rather than a Verilog module or VHDL architecture for any instantiation.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_use_cw_first</a></p>                                                                                        |
| hdl_use_default_parameter_values_in_name   |                                                                                                                                                                                                                                                                                                                                                    |

|                                |                                                                                                                                                                                                                                                                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>When set to true, uses all parameter values in parameterized module name.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_use_default_parameter_values_in_name</a></p>   |
| hdl_use_for_generate_prefix    |                                                                                                                                                                                                                                                                    |
|                                | <p>Specifies whether 'for' generate labels should be used to create instance names.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_use_for_generate_prefix</a></p>          |
| hdl_use_if_generate_prefix     |                                                                                                                                                                                                                                                                    |
|                                | <p>Specifies whether 'if' generate labels should be used to prefix instances.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_use_if_generate_prefix</a></p>                 |
| hdl_use_port_default_value     |                                                                                                                                                                                                                                                                    |
|                                | <p>When true, uses default initial values of VHDL entity input ports.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_use_port_default_value</a></p>                         |
| hdl_verilog_defines            |                                                                                                                                                                                                                                                                    |
|                                | <p>A list of Verilog macro definitions that are passed to read_hdl and read_netlist commands.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> SYNTHESIS</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_verilog_defines</a></p> |
| hdl_vhdl_assign_width_mismatch |                                                                                                                                                                                                                                                                    |
|                                | <p>When set to true, allows arrays with size mismatch to be assigned in VHDL.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hdl_vhdl_assign_width_mismatch</a></p>            |
| hdl_vhdl_case                  |                                                                                                                                                                                                                                                                    |



|                                 |                                                                                                                                                                                                                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>Causes the VHDL analyzer to store VHDL identifiers and operators in lower case, upper case, or the case given in the source file.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> original<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_vhdl_case</a></p> |
| hdl_vhdl_environment            |                                                                                                                                                                                                                                                                                              |
|                                 | <p>Specifies the selection of the predefined arithmetic libraries.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> common<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_vhdl_environment</a></p>                                                              |
| hdl_vhdl_preferred_architecture |                                                                                                                                                                                                                                                                                              |
|                                 | <p>Sets the name of preferred architecture to use with an entity when there are multiple architectures.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_vhdl_preferred_architecture</a></p>                  |
| hdl_vhdl_range_opt              |                                                                                                                                                                                                                                                                                              |
|                                 | <p>When set to false disables range optimization in VHDL.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_vhdl_range_opt</a></p>                                                                            |
| hdl_vhdl_read_version           |                                                                                                                                                                                                                                                                                              |
|                                 | <p>Specifies the VHDL version to be used when reading VHDL designs.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> 1993<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_vhdl_read_version</a></p>                                                              |
| hdl_vhdl_strict_compliance      |                                                                                                                                                                                                                                                                                              |
|                                 | <p>Enforces a more strict interpretation of the VHDL LRM.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">hdl_vhdl_strict_compliance</a></p>                                                                    |
| hier_flow_budgets               |                                                                                                                                                                                                                                                                                              |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>This will set budgeting mode options via setBudgetingMode command for accurate or preliminary budgeting whichever is specified. It has lower precedence than other user specified budget mode settings.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> accurate preliminary none</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hier_flow_budgets</a></p> |
| hinsts               |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>Short-cut for [get_db current_design .hinsts]</p> <p><b>Type:</b> <a href="#">obj(hinst)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                       |
| hnets                |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>Short-cut for [get_db current_design .hnets]</p> <p><b>Type:</b> <a href="#">obj(hnet)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                         |
| honor_valid_location |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>Controls whether to honor the valid location defined on the library cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">honor_valid_location</a></p>                                                                                                                                                                                |
| hpin_busses          |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>Short-cut for [get_db current_design .hpin_busses]</p> <p><b>Type:</b> <a href="#">obj(hpin_bus)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                               |
| hpins                |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>Short-cut for [get_db current_design .hpins]</p> <p><b>Type:</b> <a href="#">obj(hpin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                         |
| hport_busses         |                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                       |                                                                                                                                                                                                                                                  |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>Short-cut for [get_db current_design .hport_busses]</p> <p><b>Type:</b> <a href="#">obj(hport_bus)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                  |
| hports                                |                                                                                                                                                                                                                                                  |
|                                       | <p>Short-cut for [get_db hinsts .hports]</p> <p><b>Type:</b> <a href="#">obj(hport)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                    |
| hybrid_hier_feedthru                  |                                                                                                                                                                                                                                                  |
|                                       | <p>Specifies to enable or disable feed-through insertion</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hybrid_hier_feedthru</a></p>                          |
| hybrid_hier_feedthru_plugin           |                                                                                                                                                                                                                                                  |
|                                       | <p>Specifies the file name having customized feed-through flow from user</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hybrid_hier_feedthru_plugin</a></p>   |
| hybrid_hier_generate_detailed_reports |                                                                                                                                                                                                                                                  |
|                                       | <p>Specifies to enable or disable dumping of pin_assign, feedthru and timing_budget reports</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> pin_assign feedthru budget</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| hybrid_hier_opt_interface_only        |                                                                                                                                                                                                                                                  |
|                                       | <p>Optimize partition interface paths only</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                     |
| hybrid_hier_out_dir                   |                                                                                                                                                                                                                                                  |

|                                                |                                                                                                                                                                                                                                                  |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Specifies directory name for feedthru, budget and pin assignment reports</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ./reports</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hybrid_hier_out_dir</a></p> |
| hybrid_hier_pre_commit_partition_plugin        |                                                                                                                                                                                                                                                  |
|                                                | <p>User file to specify additional settings before commit partition</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                             |
| hybrid_hier_pre_timing_budget_plugin           |                                                                                                                                                                                                                                                  |
|                                                | <p>User file to specify additional budgeting mode settings.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hybrid_hier_pre_timing_budget_plugin</a></p>       |
| hybrid_hier_timing_budget                      |                                                                                                                                                                                                                                                  |
|                                                | <p>Specifies to enable or disable timing budget</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hybrid_hier_timing_budget</a></p>                              |
| hybrid_hier_write_db                           |                                                                                                                                                                                                                                                  |
|                                                | <p>Specifies to save innovus design after intermediate steps</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">hybrid_hier_write_db</a></p>                      |
| ignore_instance_pattern_for_annotation_summary |                                                                                                                                                                                                                                                  |
|                                                | <p>List of leaf instance names to be ignored in the stimulus annotation summary report.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                         |
| ignore_legacy_voltage_scaling                  |                                                                                                                                                                                                                                                  |

|                                        |                                                                                                                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>To ignore trilib format and enable auto voltage scaling method.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                               |
| ignore_max_via_stack_rule              |                                                                                                                                                                                                                                                    |
|                                        | <p>If true, the MaxVIASTACK rule in tlef file is ignored.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ignore_max_via_stack_rule</a></p>                     |
| ignore_pin_error_in_test_cell_function |                                                                                                                                                                                                                                                    |
|                                        | <p>Ignore pin error in the test cell function.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ignore_pin_error_in_test_cell_function</a></p>                   |
| ignore_scan_combinational_arcs         |                                                                                                                                                                                                                                                    |
|                                        | <p>Should we ignore combinational arcs involving scan pins.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ignore_scan_combinational_arcs</a></p>               |
| ignore_sigma_arc_mismatch              |                                                                                                                                                                                                                                                    |
|                                        | <p>Ignores sigma arc inconsistency across views during mmmc consistency check.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ignore_sigma_arc_mismatch</a></p> |
| ilm_auto_use_sdc_files                 |                                                                                                                                                                                                                                                    |
|                                        | <p>autoUseSdcFiles</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                               |
| ilm_filter_internal_path               |                                                                                                                                                                                                                                                    |

|                            |                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>filterInternalPath</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ilm_filter_internal_path</a></p>                      |
| ilm_keep_async             |                                                                                                                                                                                                                |
|                            | <p>Keep paths from asynchronous input ports</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ilm_keep_async</a></p>          |
| ilm_keep_flatten           |                                                                                                                                                                                                                |
|                            | <p>Instruct Innovus to keep in flattened state</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ilm_keep_flatten</a></p>      |
| ilm_keep_high_fanout_ports |                                                                                                                                                                                                                |
|                            | <p>Keep high fanout input port paths</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ilm_keep_high_fanout_ports</a></p>      |
| ilm_keep_inst_in_sdc       |                                                                                                                                                                                                                |
|                            | <p>Keep instances which have constraints on it</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ilm_keep_inst_in_sdc</a></p> |
| ilm_keep_loopback          |                                                                                                                                                                                                                |
|                            | <p>Keep loop back paths</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ilm_keep_loopback</a></p>                           |
| ilm_max_num_insts          |                                                                                                                                                                                                                |

|                                                          |                                                                                                                                                                                                                              |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | Specify the maximum number of instances per port (used to define high_fanout_ports)<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 1000<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">ilm_max_num_insts</a>   |
| ilm_max_num_registers                                    |                                                                                                                                                                                                                              |
|                                                          | Specify the maximum number of registers per port (used to define high_fanout_ports)<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 40<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">ilm_max_num_registers</a> |
| ilm_slack_driven                                         |                                                                                                                                                                                                                              |
|                                                          | slackDriven<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">ilm_slack_driven</a>                                                                          |
| incremental_joules_instance_threshold_count_multi_thread |                                                                                                                                                                                                                              |
|                                                          | Incremental joules instance threshold count in MT.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 1000<br><b>Edit:</b> Yes                                                                                           |
| incrsyn_ispatial_support                                 |                                                                                                                                                                                                                              |
|                                                          | Enable incremental synthesis support in ispatial.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                          |
| incrsyn_pod_effort                                       |                                                                                                                                                                                                                              |
|                                                          | Effort level incremental POD. Possible values are - low, medium and high.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> low<br><b>Edit:</b> Yes                                                                  |
| incrsyn_read_cache                                       |                                                                                                                                                                                                                              |

|                                     |                                                                                                                                                                                                                         |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | Read cache directory for incremental synthesis.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                        |
| incrsyn_read_cache_dir_after_fv_map |                                                                                                                                                                                                                         |
|                                     | Read cache directory after fv map for incremental synthesis when two pass lec is enabled.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                              |
| incrsyn_use_cache_fv_map            |                                                                                                                                                                                                                         |
|                                     | Use the cache produced at the same time formal verification collateral is produced in base<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                            |
| incrsyn_write_cache                 |                                                                                                                                                                                                                         |
|                                     | Write cache directory for incremental synthesis.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                       |
| incrsyn_write_cache_fv_map          |                                                                                                                                                                                                                         |
|                                     | Write incremental synthesis cache at the same time formal verification collateral is produce<br>two-pass LEC flow.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                    |
| init_blackbox_for_undefined         |                                                                                                                                                                                                                         |
|                                     | When set to true, empty modules are treated as unresolved references.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">init_blackbox_for_undefined</a> |
| init_check_netlist                  |                                                                                                                                                                                                                         |



|                                |                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>After reading in the netlist, will do the same checks as check_design -netlist</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">init_check_netlist</a></p>                                                                                                                                                          |
| init_check_output_pin_constant |                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                | <p>Stops the Verilog reader if any output (or inout) signal pins have any 1'b0 or 1'b1 constants would cause shorts to power/ground.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">init_check_output_pin_constant</a></p>                                                                                            |
| init_delete_floating_hnets     |                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                | <p>If true, an hnet (Verilog wire statement) that has nothing connected to it, is deleted from the The default value is false in Innovus to allow for netlist repair, but true in Tempus to delete r timed nets and reduce memory.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">init_delete_floating_hnets</a></p> |
| init_design_done               |                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                | <p>dummy attribute, always return true in case of innovus</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                |
| init_design_mmmc_skip_inactive |                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                | <p>Enables runtime improvement during loading libraries.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">init_design_mmmc_skip_inactive</a></p>                                                                                                                                                                       |
| init_design_netlist_type       |                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                | <p>Specifies the source of the design netlist. You can specify either "Verilog" or "OA".</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> Verilog</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">init_design_netlist_type</a></p>                                                                                                                                         |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| init_design_uniquify              |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                   | <p>Specifies whether the top is uniquified. When set to 1 (true) the design will be uniquified during read and flatten process. Optimization is not allowed on a non-unique design. For use in a master/clone partitioning flow, the value should be 0 (false). Default = 0 (false).</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_design_uniquify</a></p> |
| init_enable_drc_region_layer      |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                   | <p>List of drc_region_layer to restore active DRC region layer.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_enable_drc_region_layer</a></p>                                                                                                                                                                                                               |
| init_ground_nets                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                   | <p>List of global Ground nets.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_ground_nets</a></p>                                                                                                                                                                                                                                                            |
| init_hdl_search_path              |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                   | <p>Specifies the search path for HDL files.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> .<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_hdl_search_path</a></p>                                                                                                                                                                                                                                            |
| init_ignore_pg_pin_polarity_check |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                   | <p>List of leaf cell pin names to ignore power/ground pin polarity check. By default connect_global_net and CPF check for matching polarity between the net and pin.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_ignore_pg_pin_polarity_check</a></p>                                                                                                     |
| init_keep_empty_modules           |                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>By default, usage of an empty Verilog module that has no matching library cell name becomes empty hinst. If this value is false, a dummy library cell is created, and usage of it becomes a You should then use other commands to 'fix' the dummy library cell before proceeding in the</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_keep_empty_modules</a></p>                                                                                                                                            |
| init_lef_files          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | <p>List of LEF files to be read. Mutually exclusive with init_oa_ref_lib.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_lef_files</a></p>                                                                                                                                                                                                                                                                                                                                                                      |
| init_min_dbu_per_micron |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | <p>Minimum DBU per micron</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_min_dbu_per_micron</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| init_mmmc_files         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | <p>Path to MMMC View Definition file.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_mmmc_files</a></p>                                                                                                                                                                                                                                                                                                                                                                                                         |
| init_no_new_assigns     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | <p>Turns on an assign-free flow. By default, applications like optimization can add Verilog assigns remove unnecessary buffers and reduce power. If you have external tools that do not support Verilog assigns, you can set this value to true to prevent applications from adding any new statements to the Verilog netlist. If there are existing assigns in the Verilog netlist you can remove them with the delete_assigns command.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_no_new_assigns</a></p> |
| init_oa_abstract_views  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>List of OpenAccess view names to be processed as abstracts (LEF MACRO equivalent). The read_netlist command uses init_oa_abstract_views list for on-demand loading of missing c</p> <p>Used by the read_physical and read_netlist commands.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_oa_abstract_views</a></p>      |
| init_oa_default_rule |                                                                                                                                                                                                                                                                                                                                                                                                                |
|                      | <p>The default rule to be used in OpenAccess. Typically used to specify a metal stack LEFDefaultRouteSpec type constraint group to read from the technology library if the library contains multiple metal stacks. Used by the read_physical command.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_oa_default_rule</a></p> |
| init_oa_design_cell  |                                                                                                                                                                                                                                                                                                                                                                                                                |
|                      | <p>OpenAccess design cell name that was read. Set by the read_netlist -oa_cell_view comma</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_oa_design_cell</a></p>                                                                                                                                                              |
| init_oa_design_lib   |                                                                                                                                                                                                                                                                                                                                                                                                                |
|                      | <p>OpenAccess design lib name that was read. Set by the read_netlist -oa_cell_view comman</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_oa_design_lib</a></p>                                                                                                                                                               |
| init_oa_design_view  |                                                                                                                                                                                                                                                                                                                                                                                                                |
|                      | <p>OpenAccess design view name that was read. Set by the read_netlist -oa_cell_view comm</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_oa_design_view</a></p>                                                                                                                                                               |
| init_oa_foundry_rule |                                                                                                                                                                                                                                                                                                                                                                                                                |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>The foundry constraint group to be used in OpenAccess. Typically used to specify a foundry_innovus_XXXX constraint group to read from the technology library if the library contains multiple foundry_innovus_XXXX constraint groups. Used by the read_physical command.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_oa_foundry_rule</a></p>                                                                                                                                                                                                                                                                                                                                        |
| init_oa_layout_views |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>List of OpenAccess view names to be processed for cell layout viewing (GDSII equivalent). used in the case where a Verilog netlist is read and views from the init_oa_layout_views list are used in the Quick Abstract Inference flow to derive abstract information from layout views for that were not processed from the libraries in the read_physical -oa_ref_libs list. The read_physical command uses init_oa_layout_view list for on-demand loading of missing cells. Used by the read_physical and read_netlist commands.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_oa_layout_views</a></p>                                                                              |
| init_oa_ref_libs     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>List of OpenAccess reference libraries. The first library in the list indicates the library that was used for technology information unless read_physical -oa_tech_lib is specified. All cells from the specified libraries are read. It is important to include the standard cell library in the list to make sure that cells that are not in the current netlist are read and available for optimization. The init_oa_abstract_views and init_oa_layout_views lists specify which views of the cells are read. Set by the read_physical -oa_ref_libs command.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_oa_ref_libs</a></p>                                                     |
| init_oa_search_libs  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>List of OpenAccess reference libraries to be searched when cells are used in the Verilog netlist that are not found in the list of cells read from the libraries specified by read_physical -oa_ref_libs command, by default all libraries in the cds.lib are searched. The init_oa_abstract_views and init_oa_layout_views lists are used to indicate which view names are looked for during the search. Applies only when read_netlist command is used to process a Verilog netlist specify the connectivity to be read. Set by the read_physical -oa_search_libs command and used by the read_netlist command.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> *<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_oa_search_libs</a></p> |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| init_oa_special_rule                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                       | <p>The special rule to be used in OpenAccess to define the via search order for power routing. Typically used to specify a metal stack LEFSpecialRouteRouteSpec type constraint group from the technology library if the library contains multiple metal stacks. Used by the read_pl command.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_oa_special_rule</a></p>                                                                                                         |
| init_oa_tech_lib                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                       | <p>OpenAccess library to use as source of technology information. If not specified, the first library in the read_physical -oa_ref_libs list will be used. Applies only when read_netlist is used to process a Verilog netlist. Set by the read_physical -oa_tech_lib command.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_oa_tech_lib</a></p>                                                                                                                            |
| init_power_intent_files               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                       | <p>Power intent files(s) read by last 'read_power_intent' command.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No<br/> <b>Reference:</b> <a href="#">init_power_intent_files</a></p>                                                                                                                                                                                                                                                                                                                                  |
| init_power_nets                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                       | <p>List of global Power nets.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_power_nets</a></p>                                                                                                                                                                                                                                                                                                                                                                              |
| init_read_netlist_allow_port_mismatch |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                       | <p>If false, a Verilog instance with ports that do not match the Verilog module or library cell definition will cause an error and exit. If true, an error message occurs, but the extra ports are added to the module or library cell definition to allow the inconsistent netlist to be read in and debugged. The default value is false i.e. 'dirty netlists' are not tolerated.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_read_netlist_allow_port_mismatch</a></p> |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| init_read_netlist_allow_undefined_cells | <p>If false, a Verilog reference to an undefined cell (e.g. a Verilog module name that has no m library cell and no Verilog module definition) causes an error and an exit after parsing the r true, an empty Verilog module is created with a warning. The default value is true in Innovu allow repair of 'dirty netlists' while it is false in Tempus to stop on 'dirty netlists'. An empty m will become an hinst or inst depending on the value of init_keep_empty_modules.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_read_netlist_allow_undefined_cells</a></p> |
| init_read_netlist_files                 | <p>A list of Verilog files read in during initialization. read_netlist will set this root attribute to the input files given to it. The original input file names are retained thru the flow for reference, a used by write_do_lec for the default golden netlist comparison file names.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_read_netlist_files</a></p>                                                                                                                                                                                                         |
| init_sync_relative_path                 | <p>Synchronize Relative paths</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">init_sync_relative_path</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| init_tiers                              | <p>Specifies the orientation of each tier, either 'up' or 'down', starting with the bottommost, for a tier design.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| init_timing_enabled                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>If read_physical is issued before read_mmmc or read_libs in RC and Tempus, an error will issued. In EDI, doing this will result in init_timing_enabled being set to false by the tool. ED in physical only mode, and any attempt to set timing data will result in an error. By default, init_timing_enabled is set to true.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> No<br/> <b>Reference:</b> <a href="#">init_timing_enabled</a></p> |
| input_assert_one_cold_pragma    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>Specifies the list of alias names of the pragma 'one_cold'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> assert_one_cold one_cold<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">input_assert_one_cold_pragma</a></p>                                                                                                                                                                                                                                  |
| input_assert_one_hot_pragma     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>Specifies the list of alias names of the pragma 'one_hot'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> assert_one_hot one_hot<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">input_assert_one_hot_pragma</a></p>                                                                                                                                                                                                                                      |
| input_asynchro_reset_blk_pragma |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>Specifies the list of alias names of the pragma 'asynchro_reset_blk'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> async_set_reset_local asynchro_reset_blk<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">input_asynchro_reset_blk_pragma</a></p>                                                                                                                                                                                                     |
| input_asynchro_reset_pragma     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>Specifies the list of alias names of the pragma 'asynchro_reset'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> async_set_reset asynchro_reset<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">input_asynchro_reset_pragma</a></p>                                                                                                                                                                                                                       |
| input_case_cover_pragma         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p>Specifies the list of alias names of the pragma 'full_case'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> full_case<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">input_case_cover_pragma</a></p>                                                                                                                                                                                                                                                     |



|                                 |                                                                                                                                                                                                                                                                                      |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input_case_decode_pragma        |                                                                                                                                                                                                                                                                                      |
|                                 | <p>Specifies the list of alias names of the pragma 'parallel_case'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> parallel_case</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">input_case_decode_pragma</a></p>                                    |
| input_map_to_mux_pragma         |                                                                                                                                                                                                                                                                                      |
|                                 | <p>Specifies the list of alias names of the pragma 'map_to_mux'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> map_to_mux infer_mux_override</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">input_map_to_mux_pragma</a></p>                        |
| input_pragma_keyword            |                                                                                                                                                                                                                                                                                      |
|                                 | <p>Specifies pragma keywords.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> cadence synopsys ambit pragma synthesis</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">input_pragma_keyword</a></p>                                                    |
| input_synchro_enable_blk_pragma |                                                                                                                                                                                                                                                                                      |
|                                 | <p>Specifies the list of alias names of the pragma 'synchro_enable_blk'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> synchro_enable_blk</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">input_synchro_enable_blk_pragma</a></p>                   |
| input_synchro_enable_pragma     |                                                                                                                                                                                                                                                                                      |
|                                 | <p>Specifies the list of alias names of the pragma 'synchro_enable'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> synchro_enable</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">input_synchro_enable_pragma</a></p>                               |
| input_synchro_reset_blk_pragma  |                                                                                                                                                                                                                                                                                      |
|                                 | <p>Specifies the list of alias names of the pragma 'synchro_reset_blk'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> sync_set_reset_local synchro_reset_blk</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">input_synchro_reset_blk_pragma</a></p> |

|                            |                                                                                                                                                                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input_synchro_reset_pragma |                                                                                                                                                                                                                                                                    |
|                            | <p>Specifies the list of alias names of the pragma 'synchro_reset'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> sync_set_reset synchro_reset</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">input_synchro_reset_pragma</a></p> |
| insert_clock_gating        |                                                                                                                                                                                                                                                                    |
|                            | <p>Enable/disable RTL clock gating.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 0 1 auto false true</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">insert_clock_gating</a></p>                  |
| insts                      |                                                                                                                                                                                                                                                                    |
|                            | <p>insts of current design</p> <p><b>Type:</b> <a href="#">obj(inst)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                     |
| internal_scale_factor      |                                                                                                                                                                                                                                                                    |
|                            | <p>The scaling factor for internal power of a design.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p>                                                                                                            |
| io_constraints             |                                                                                                                                                                                                                                                                    |
|                            | <p>Short-cut for [get_db current_design .io_constraints]</p> <p><b>Type:</b> <a href="#">obj(io_constraint)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                              |
| is_ilm_flattened           |                                                                                                                                                                                                                                                                    |
|                            | <p>Is true if you have ILMs, and the flatten_ilm command has flattened them so the internal ILM are visible. unflatten_ilm will set it back to false.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>             |
| is_ilm_read                |                                                                                                                                                                                                                                                                    |

|                            |                                                                                                                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | Is true if you have read in any ILMs with read_ilm. It is not affected by flatten_ilm/unflatten_i<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> No         |
| jedai_catalog_token        |                                                                                                                                                                                            |
|                            | <b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">jedai_catalog_token</a>                                                     |
| jedai_catalog_url          |                                                                                                                                                                                            |
|                            | <b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">jedai_catalog_url</a>                                                       |
| joules_incremental_quiet   |                                                                                                                                                                                            |
|                            | To silence the incremental joules info/warning messages.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes                                                  |
| joules_quiet               |                                                                                                                                                                                            |
|                            | To silence the joules info/warning messages.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                             |
| large_cell_arc_threshold   |                                                                                                                                                                                            |
|                            | Arc threshold count to create separate bdd's.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 200<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">large_cell_arc_threshold</a> |
| layer_name_no_abbreviation |                                                                                                                                                                                            |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>By default this value is true, and most command -layer &lt;layer&gt; options require a LEF/OA layer name, or a routing-layer index value (e.g. 1 for the first routing layer, 2 for the second, etc.) as input. If this global is false, then an older style abbreviation is also allowed (e.g. M1 for the first routing layer, M2 for the second routing layer, etc.). This layer abbreviation style is not recommended, and is only allowed for backward compatibility to migrate older scripts. Eventually this global will be removed, so you should modify your scripts to use LEF/OA layer names, or routing-layer index values for -layer options to avoid problems in the future.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">layer_name_no_abbreviation</a></p> |
| layers               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>All the layers defined in the LEF or OA techfile.</p> <p><b>Type:</b> <a href="#">obj(layer)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| leakage_scale_factor |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>The scaling factor for leakage power of a design.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 1.0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| lib_arcs             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>lib_arcs</p> <p><b>Type:</b> <a href="#">obj(lib_arc)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| lib_cells            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>lib_cells</p> <p><b>Type:</b> <a href="#">obj(lib_cell)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| lib_pins             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>lib_pins</p> <p><b>Type:</b> <a href="#">obj(lib_pin)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| libraries            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>Returns the information about the libraries in the design.</p> <p><b>Type:</b> <a href="#">obj(library)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| library_sets      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                   | <p>Returns the information about the library sets in the design.</p> <p><b>Type:</b> <a href="#">obj(library_set)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| link_library      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                   | <p>Link library for instantiated cells in the design.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">link_library</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| log_file          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                   | <p>The log file name of the program. It can be changed in the session to direct log file output to different file. Type 'man log_file' for more details.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">log_file</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| log_verbose_style |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                   | <p>When verbose logging is enabled (see <a href="#">source_verbose</a>, <a href="#">flow_verbose</a>, or <a href="#">set_proc_verbose</a>), logging style for each command is controllable. It is sometimes helpful to include line number comments for script debug and other times this is too much information. You can switch between settings in the middle of a script. 'debug' will output the file name, line number, commands, comments, and empty lines. 'echo' will output only the commands and line number. 'echo_comments' will output commands, comments, empty lines, and line number.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> debug echo echo_comments</p> <p><b>Default:</b> echo</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">log_verbose_style</a></p> |
| log_verbose_type  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>When verbose logging is enabled (see <code>source_verbose</code>, <code>flow_verbose</code>, or <code>set_proc_verbose</code>) default logging of each commands is pre-Tcl expansion. In some cases it is helpful to see pre-Tcl expansion, or even both pre-Tcl and post-Tcl expansion when doing detailed debugging. You can switch between settings in the middle of a script. Type 'man log_file' for more details.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> pre post both<br/> <b>Default:</b> pre<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">log_verbose_type</a></p> |
| logv_file                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                    | <p>The verbose log file name of the program. It can be changed in the session to direct logv file to a different file. Type 'man log_file' for more details.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">logv_file</a></p>                                                                                                                                                                                                                                                                                                            |
| lp_display_negative_internal_power |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                    | <p>For displaying negative Internal power results.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| map_clock_tree                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                    | <p>Enables mapping of all clock logic to a specified set of library cells.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">map_clock_tree</a></p>                                                                                                                                                                                                                                                                                                                                                                                        |
| map_drc_first                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                    | <p>Give preference to 'drc' cost during mapping.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">map_drc_first</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                   |
| map_latch_allow_async_decomp       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Allow implementing the asynchronous control logic externally to the latch.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">map_latch_allow_async_decomp</a></p>                                                                                                                                                                                      |
| map_prefer_non_inverted_clock_line |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                    | <p>The tool will try to avoid inserting inverters in the clock line, but if that is not possible still can them. If set to false, then the tool can chose the solution which is best from QoR point of view has no preference for solutions without inverter.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">map_prefer_non_inverted_clock_line</a></p> |
| map_respect_rtl_clk_phase          |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                    | <p>Respect the clock phase specification of registers from the RTL.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">map_respect_rtl_clk_phase</a></p>                                                                                                                                                                                                   |
| mark_async_pin_using_timing_arcs   |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                    | <p>Mark library pin having recovery or removal timing arcs as asynchronous.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">mark_async_pin_using_timing_arcs</a></p>                                                                                                                                                                                     |
| mark_macro_as_power_switch_cell    |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                    | <p>Power switch cell defined as macro is also marked as power switch cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">mark_macro_as_power_switch_cell</a></p>                                                                                                                                                                                      |
| mark_valid_lp_cell_as_usable       |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                    | <p>Mark valid low power cells as usable.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">mark_valid_lp_cell_as_usable</a></p>                                                                                                                                                                                                                           |

|                                    |                                                                                                                                                                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| markers                            |                                                                                                                                                                                                                                                             |
|                                    | <p>Short-cut to all the marker objects in the design.</p> <p><b>Type:</b> <a href="#">obj(marker)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                  |
| max_cpus_per_server                |                                                                                                                                                                                                                                                             |
|                                    | <p>Maximum number of active CPUs allowed to be used on one server. Both super-threading multi-threading respect this limit.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 16</p> <p><b>Edit:</b> Yes</p>                                   |
| merge_combinational_hier_instances |                                                                                                                                                                                                                                                             |
|                                    | <p>Allow merging of combinational hierarchical instances.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">merge_combinational_hier_instances</a></p>                      |
| merge_non_scan_to_scan_flops       |                                                                                                                                                                                                                                                             |
|                                    | <p>Controls whether non scan single bit flops can be merged to multibit scan flops.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">merge_non_scan_to_scan_flops</a></p> |
| mesg_severity_downgrade            |                                                                                                                                                                                                                                                             |
|                                    | <p>Allow message severity to be downgraded.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">mesg_severity_downgrade</a></p>                                              |
| messages                           |                                                                                                                                                                                                                                                             |
|                                    | <p>A list of all the message objects.</p> <p><b>Type:</b> <a href="#">obj(message)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                |
| metric_advanced_url_endpoint       |                                                                                                                                                                                                                                                             |



|                                                      |                                                                                                                                                                                                                                                               |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | <p>Base URL for the advanced metric server.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_advanced_url_endpoint</a></p>                                            |
| metric_capture_3d_hotspots                           |                                                                                                                                                                                                                                                               |
|                                                      | <p>Capture 3d hotspot metrics.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_3d_hotspots</a></p>                                                           |
| metric_capture_depth                                 |                                                                                                                                                                                                                                                               |
|                                                      | <p>Depth for capturing hinst design and power metrics.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_depth</a></p>                                             |
| metric_capture_design_image                          |                                                                                                                                                                                                                                                               |
|                                                      | <p>Capture design image.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_design_image</a></p>                                                                |
| metric_capture_design_image_blockages                |                                                                                                                                                                                                                                                               |
|                                                      | <p>Capture blockages in design image.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_design_image_blockages</a></p>                                         |
| metric_capture_design_image_blockages_exclude_layers |                                                                                                                                                                                                                                                               |
|                                                      | <p>Layers to exclude when capturing blockages in metric image.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_design_image_blockages_exclude_layers</a></p> |
| metric_capture_design_image_blockages_threshold      |                                                                                                                                                                                                                                                               |

|                                          |                                                                                                                                                                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Threshold for capturing blockages in metric image.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_design_image_blockages_threshold</a></p> |
| metric_capture_design_image_power_intent |                                                                                                                                                                                                                                               |
|                                          | <p>Capture power intent objects in design image.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_design_image_power_intent</a></p>           |
| metric_capture_design_image_route_drc    |                                                                                                                                                                                                                                               |
|                                          | <p>Capture route drc markers in design image.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_design_image_route_drc</a></p>                 |
| metric_capture_max_drc_markers           |                                                                                                                                                                                                                                               |
|                                          | <p>Maximum number of DRC markers to include in metric image.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_max_drc_markers</a></p>          |
| metric_capture_min_count                 |                                                                                                                                                                                                                                               |
|                                          | <p>Minimum instance count for capturing hinst design and power metrics.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_min_count</a></p>     |
| metric_capture_overwrite                 |                                                                                                                                                                                                                                               |
|                                          | <p>Overwrite pending metrics during create_snapshot category capture.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_overwrite</a></p>     |
| metric_capture_pba_tns_histogram         |                                                                                                                                                                                                                                               |

|                                     |                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Capture path based analysis histogram data for TNS.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_pba_tns_histogram</a></p>                                                                                                     |
| metric_capture_per_view             |                                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>Capture timing metrics per analysis_view.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_per_view</a></p>                                                                                                                         |
| metric_capture_reg2reg_metrics      |                                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>Force the capture of reg2reg metrics regardless of path groups.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_reg2reg_metrics</a></p>                                                                                           |
| metric_capture_timing_analysis_mode |                                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>Timing analysis mode for metric capture during create_snapshot categories setup and hold pba, epba, ipba.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> gba pba epba ipba</p> <p><b>Default:</b> gba</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_timing_analysis_mode</a></p> |
| metric_capture_timing_path_groups   |                                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>Defines user path groups to capture for detailed display.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_timing_path_groups</a></p>                                                                                               |
| metric_capture_timing_paths         |                                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>Number of paths to capture for detailed display.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 10</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_timing_paths</a></p>                                                                                                                 |

|                                        |                                                                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metric_capture_tns_histogram           |                                                                                                                                                                                                                                   |
|                                        | <p>Capture histogram data for TNS.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_tns_histogram</a></p>                         |
| metric_capture_tns_histogram_buckets   |                                                                                                                                                                                                                                   |
|                                        | <p>Number of buckets for the TNS histogram.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 50</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_tns_histogram_buckets</a></p>           |
| metric_capture_tns_histogram_max_slack |                                                                                                                                                                                                                                   |
|                                        | <p>Maximum slack for the TNS histogram.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_tns_histogram_max_slack</a></p>              |
| metric_capture_tns_histogram_paths     |                                                                                                                                                                                                                                   |
|                                        | <p>Number of paths to capture for the TNS histogram.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 10000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_tns_histogram_paths</a></p> |
| metric_capture_vth_metrics             |                                                                                                                                                                                                                                   |
|                                        | <p>Capture vth metrics.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_vth_metrics</a></p>                                      |
| metric_capture_vth_per_power_domain    |                                                                                                                                                                                                                                   |
|                                        | <p>Capture vth metrics per power domain.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_vth_per_power_domain</a></p>           |

|                                  |                                                                                                                                                                                                                                                                                |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metric_capture_wns_average_paths |                                                                                                                                                                                                                                                                                |
|                                  | <p>Defines average top-N paths for WNS.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_capture_wns_average_paths</a></p>                                                             |
| metric_category_default          |                                                                                                                                                                                                                                                                                |
|                                  | <p>Default metric categories to capture if none are provided.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> design</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_category_default</a></p>                                            |
| metric_current_run_id            |                                                                                                                                                                                                                                                                                |
|                                  | <p>Current unique run ID returned by the advanced metric server.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_current_run_id</a></p>                                               |
| metric_enable                    |                                                                                                                                                                                                                                                                                |
|                                  | <p>enable metric snapshot capture with create_snapshot</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_enable</a></p>                                                                |
| metric_summary_metrics           |                                                                                                                                                                                                                                                                                |
|                                  | <p>Summary metrics to show when creating snapshots.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> flow.cputime flow.realtime timing.setup.tns timing.setup.wns</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">metric_summary_metrics</a></p> |
| minimize_uniquify                |                                                                                                                                                                                                                                                                                |
|                                  | <p>Try to avoid uniquification of subdesigns.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">minimize_uniquify</a></p>                                                                     |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| modules                                         |                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                 | <p>Short-cut for [get_db current_design .modules]</p> <p><b>Type:</b> <a href="#">obj(module)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                              |
| multibit_allow_async_phase_map                  |                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                 | <p>Allow interchanging async pins for multibit mapping.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">multibit_allow_async_phase_map</a></p>                                                                                                                                                                     |
| multibit_allow_unused_bits                      |                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                 | <p>Allows unused bits in multibit instance.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">multibit_allow_unused_bits</a></p>                                                                                                                                                                                     |
| multibit_area_power_scoring                     |                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                 | <p>Controls whether area and power are to be scored in merging.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none area_only area_power</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">multibit_area_power_scoring</a></p>                                                                                                           |
| multibit_auto_exclude_registers_with_exceptions |                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                 | <p>Enable this to automatically apply dont_infer_multibit on all sequential instances with exceptions. By default exceptional instances are merged only when all exceptions are trans to multibit formed.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">multibit_auto_exclude_registers_with_exceptions</a></p> |
| multibit_aware_seq_mapping                      |                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Mapper prefers flops which have corresponding multibit variants.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto true false</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">multibit_aware_seq_mapping</a></p>                                                                                                                                                                                                                    |
| multibit_aware_seq_mapping_higher_priority     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                | <p>Multibit-aware mapping has higher priority than other constraints like avoid_tied_inputs, map_prefer_non_inverted_clock_line, exact_match_seq_sync_ctrls, exact_match_seq_async_ctrls, etc. Nonetheless, it remains lower priority than map_to_reg map_to_ser_register, scan and state retention constraints.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">multibit_aware_seq_mapping_higher_priority</a></p> |
| multibit_cells_from_different_busses           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                | <p>Allows to merge 1-bit sequential, tristate and combinational instances from different busses single multibit cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">multibit_cells_from_different_busses</a></p>                                                                                                                                                                                                   |
| multibit_combinational_name_concatenate_string |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                | <p>Separator string to be used for concatenation when naming multibit combinational and spe power cells. Recommended values for Genus-LEC flow are '_MB_' and '_mb_'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> _MB_</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                               |
| multibit_debug                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                | <p>Logs Multibit Inferencing debug messages.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">multibit_debug</a></p>                                                                                                                                                                                                                                                                                                 |
| multibit_invert_clock_phase                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                       |                                                                                                                                                                                                                                              |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | Used for enabling merging of cells with inverted clock phase.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">multibit_invert_clock_phase</a>                             |
| multibit_mapping_effort_level         |                                                                                                                                                                                                                                              |
|                                       | Effort level(high, auto, low) for multibit mapping.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> high auto low<br><b>Default:</b> high<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">multibit_mapping_effort_level</a> |
| multibit_predefined_allow_unused_bits |                                                                                                                                                                                                                                              |
|                                       | Allows unused bits in predefined multibit.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">multibit_predefined_allow_unused_bits</a>                                      |
| multibit_preserve_inferred_instances  |                                                                                                                                                                                                                                              |
|                                       | Preserve size_delete_ok inferred multibit instances.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">multibit_preserve_inferred_instances</a>                             |
| multibit_preserved_net_check          |                                                                                                                                                                                                                                              |
|                                       | Merge instances into multibit only if no preserve on connected nets.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">multibit_preserved_net_check</a>                     |
| multibit_seqs_instance_naming_style   |                                                                                                                                                                                                                                              |
|                                       | Naming style to be used for newly created sequential multibit instance.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> concat<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">multibit_seqs_instance_naming_style</a>        |
| multibit_seqs_members_naming_style    |                                                                                                                                                                                                                                              |



|                             |                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>Naming style for sequential instances to be merged into multibit cell.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> %s%d %s%d_reg %s_%d_ %s_%d_reg {%s[%d]} {%s[%d]_reg}</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">multibit_seqs_members_naming_style</a></p>           |
| multibit_split_string       |                                                                                                                                                                                                                                                                                                                    |
|                             | <p>Naming style string to be used for naming single bit instances when source netlist already has multibit cells. Default value is '_split_'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> _split_</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">multibit_split_string</a></p> |
| multibit_unused_input_value |                                                                                                                                                                                                                                                                                                                    |
|                             | <p>Connects each unconnected input pin in Multibit cell to specified value unless 'none' is supplied.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">multibit_unused_input_value</a></p>                                         |
| net_groups                  |                                                                                                                                                                                                                                                                                                                    |
|                             | <p>Short-cut for [get_db current_design .net_groups]</p> <p><b>Type:</b> <a href="#">obj(net_group)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                      |
| nets                        |                                                                                                                                                                                                                                                                                                                    |
|                             | <p>Short-cut for [get_db current_design .nets]. All the nets inside this design, including logical nets, physical-only nets, Verilog supply0/supply1, and Verilog 1'b0/1'b1 nets.</p> <p><b>Type:</b> <a href="#">obj(net)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                               |
| non_dft_timing_mode_name    |                                                                                                                                                                                                                                                                                                                    |
|                             | <p>Non-DFT timing mode name (default: NON_DFT_MODE).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> NON_DFT_MODE</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">non_dft_timing_mode_name</a></p>                                                                                  |
| num_eeq_variants            |                                                                                                                                                                                                                                                                                                                    |

|                             |                                                                                                                                                                                                                                                    |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | Maximum number of EEQ variants of any cell in library, from LIBRARY LEF58_CELLVARIA property 'CELLVARIANTS variantTotalNum YFLIPMAP {flippedVariantNum siteVariantNum<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| oa_allow_analysis_only      |                                                                                                                                                                                                                                                    |
|                             | Allow analysis only flow<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">oa_allow_analysis_only</a>                                                                             |
| oa_allow_bit_connection     |                                                                                                                                                                                                                                                    |
|                             | Whether to create instance terminal creations bitwise, if bus not found in OA.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">oa_allow_bit_connection</a>                      |
| oa_allow_tech_update        |                                                                                                                                                                                                                                                    |
|                             | Whether to allow the tech.db to be updated or not.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">oa_allow_tech_update</a>                                                      |
| oa_bindkey_file             |                                                                                                                                                                                                                                                    |
|                             | Virtuoso bind key file<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">oa_bindkey_file</a>                                                                                       |
| oa_cell_view_dir            |                                                                                                                                                                                                                                                    |
|                             | System defined transient global referring to origin of restored design.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">oa_cell_view_dir</a>                                     |
| oa_convert_diagonal_path_to |                                                                                                                                                                                                                                                    |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Whether to convert non-conforming pathsegs/paths to polygon or convert the endStyle to tr on orthogonal or read pathsegs styleand to custom on diagonal pathsegs</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> polygon default_style style</p> <p><b>Default:</b> polygon</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_convert_diagonal_path_to</a></p>                                                                                                                                                                                   |
| oa_cut_rows                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                     | <p>Whether to use oa_cut_rows or not</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_cut_rows</a></p>                                                                                                                                                                                                                                                                                                                                                                                           |
| oa_display_resource_file            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                     | <p>Display resource file</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_display_resource_file</a></p>                                                                                                                                                                                                                                                                                                                                                                                           |
| oa_display_resource_file_in_library |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                     | <p>Whether to use display resource file in the tech library</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_display_resource_file_in_library</a></p>                                                                                                                                                                                                                                                                                                                                            |
| oa_drc_fill_purpose                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                     | <p>Whether to save drcFill shapes to gap_fill or drawing purpose. By default DEF routes with + SHAPE DRCFILL are mapped to the OA 'gapFill' purpose. Some OA techfiles do not have t purpose defined, and require these shapes be mapped to 'drawing' purpose. In that case, t an additional internal attribute so they can round-trip back to DRCFILL shapes properly.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> gap_fill drawing</p> <p><b>Default:</b> gap_fill</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_drc_fill_purpose</a></p> |
| oa_enclose_quick_abstract_pins      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                        |                                                                                                                                                                                                                                                                                                |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Include shapes that enclose layout pin shapes</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_enclose_quick_abstract_pins</a></p>                                                                      |
| oa_full_layer_list     |                                                                                                                                                                                                                                                                                                |
|                        | <p>Whether to support full list of layer import</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_full_layer_list</a></p>                                                                                   |
| oa_full_path           |                                                                                                                                                                                                                                                                                                |
|                        | <p>Whether to use full path or not</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_full_path</a></p>                                                                                                     |
| oa_hybrid_rows         |                                                                                                                                                                                                                                                                                                |
|                        | <p>Read hybrid rows from PDK</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_hybrid_rows</a></p>                                                                                                         |
| oa_inst_placed_if_none |                                                                                                                                                                                                                                                                                                |
|                        | <p>Whether to map OA Placement status oacNonePlacementStatus of an instance to Placed(true) or Unplaced(false).</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_inst_placed_if_none</a></p>               |
| oa_lib_create_mode     |                                                                                                                                                                                                                                                                                                |
|                        | <p>Library creation method during write_db for newly created libraries</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none attach copy reference</p> <p><b>Default:</b> reference</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_lib_create_mode</a></p> |

|                           |                                                                                                                                                                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oa_lock                   |                                                                                                                                                                                                                                                                             |
|                           | <p>Whether to lock the OA database during read_db</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true error_if_locked</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_lock</a></p>                  |
| oa_logic_only_import      |                                                                                                                                                                                                                                                                             |
|                           | <p>Whether to run read_oa after init_design or not</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_logic_only_import</a></p>                                                          |
| oa_merge_trim             |                                                                                                                                                                                                                                                                             |
|                           | <p>Merge the trim metal shapes during save when the trim metal gap less than or equal to the max_gap in micron.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_merge_trim</a></p>     |
| oa_new_lib_compress_level |                                                                                                                                                                                                                                                                             |
|                           | <p>OA Database compression level</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_new_lib_compress_level</a></p>                                                                            |
| oa_non_rectilinear_shapes |                                                                                                                                                                                                                                                                             |
|                           | <p>Translates non-rectilinear shapes to rectilinear shapes.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> bbox approximate</p> <p><b>Default:</b> bbox</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_non_rectilinear_shapes</a></p> |
| oa_pin_purpose            |                                                                                                                                                                                                                                                                             |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Whether to set Pin Purpose as Pin or not</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_pin_purpose</a></p>                                                                                                                                                                                    |
| oa_push_pin_constraint              |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>whether write_db to push interoperable pin constraint into its corresponding blackbox abstract cellview.</p> <p>Note: Interoperable pin constraint means pin constraints that are understood by Innovus, \ and PVS.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_push_pin_constraint</a></p> |
| oa_quick_abstract_for_custom_cells  |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>Trim design shapes of Custom Cells</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> pcells stdcells all off</p> <p><b>Default:</b> all</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_quick_abstract_for_custom_cells</a></p>                                                                                                                     |
| oa_read_system_reserved_layers      |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>Read system reserved layers from PDK</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> all text off</p> <p><b>Default:</b> all</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_read_system_reserved_layers</a></p>                                                                                                                                  |
| oa_silently_ignore_unsupported_vias |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                     | <p>Whether read_db should issue messages about vias that are not supported in Innovus.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_silently_ignore_unsupported_vias</a></p>                                                                                                                    |
| oa_text_purpose                     |                                                                                                                                                                                                                                                                                                                                                                                          |

|                         |                                                                                                                                                                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Specify the purpose name that will be used when text labels (add_text) are written.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> drawing</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_text_purpose</a></p>               |
| oa_tie_net              |                                                                                                                                                                                                                                                                     |
|                         | <p>tieHigh tieLow net names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_tie_net</a></p>                                                                                    |
| oa_update_mode          |                                                                                                                                                                                                                                                                     |
|                         | <p>Whether to use (save/restore)OaDesign in update mode or not</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true auto</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_update_mode</a></p> |
| oa_use_virtuoso_bindkey |                                                                                                                                                                                                                                                                     |
|                         | <p>Whether to use Virtuoso bind keys</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_use_virtuoso_bindkey</a></p>                                                             |
| oa_use_virtuoso_color   |                                                                                                                                                                                                                                                                     |
|                         | <p>Whether to use Virtuoso color and stipple</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_use_virtuoso_color</a></p>                                                       |
| oa_view_sub_type        |                                                                                                                                                                                                                                                                     |
|                         | <p>Subtype of view of the OA design</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> vce vxl none</p> <p><b>Default:</b> vxl</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_view_sub_type</a></p>                               |

|                           |                                                                                                                                                                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oa_write_cds_fixed_vias   |                                                                                                                                                                                                                                                               |
|                           | <p>Save fixed via</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> new_vias_only off</p> <p><b>Default:</b> new_vias_only</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_write_cds_fixed_vias</a></p>                     |
| oa_write_mask_data_locked |                                                                                                                                                                                                                                                               |
|                           | <p>Whether to save colored shapes with locked attribute</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_write_mask_data_locked</a></p>                                  |
| oa_write_net_voltage      |                                                                                                                                                                                                                                                               |
|                           | <p>Whether to allow write_db to write voltage information to the oaBitNet's voltage attribute</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_write_net_voltage</a></p> |
| oa_write_relative_path    |                                                                                                                                                                                                                                                               |
|                           | <p>Whether to save relative path in config file.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">oa_write_relative_path</a></p>                                             |
| obj_type                  |                                                                                                                                                                                                                                                               |
|                           | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (root)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                          |
| obj_types                 |                                                                                                                                                                                                                                                               |
|                           | <p>object definitions</p> <p><b>Type:</b> <a href="#">obj(obj_type)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                 |
| opconds                   |                                                                                                                                                                                                                                                               |



|                                        |                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Returns the information about the opconds in the design.</p> <p><b>Type:</b> <a href="#">obj(opcond)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                 |
| operating_conditions                   |                                                                                                                                                                                                                                                                                                                   |
|                                        | <p>Operating condition for timing (from technology library).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">operating_conditions</a></p>                                                                                       |
| opt_activity_refresh_args              |                                                                                                                                                                                                                                                                                                                   |
|                                        | <p>Specify the list of setStim options to be passed for Joules Xreplay. Call "setOptMode - activity_refresh_args help" to get detailed usage.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_activity_refresh_args</a></p> |
| opt_activity_refresh_flow              |                                                                                                                                                                                                                                                                                                                   |
|                                        | <p>Switch between xreplay_stimulus and flash_replay_stimulus</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> xreplay flash_replay</p> <p><b>Default:</b> xreplay</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_activity_refresh_flow</a></p>                               |
| opt_activity_refresh_stages            |                                                                                                                                                                                                                                                                                                                   |
|                                        | <p>Specify the list of stages for refreshing activity. Flow stages : elab pre_map post_map</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_activity_refresh_stages</a></p>                                                  |
| opt_adaptive_subd1_flow                |                                                                                                                                                                                                                                                                                                                   |
|                                        | <p>Adaptively use weak cells with driving strength smaller or equal to D1 in optimization flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                  |
| opt_add_always_on_feed_through_buffers |                                                                                                                                                                                                                                                                                                                   |

|                                      |                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Activates adding Always On Feed Thru buffers</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_add_always_on_feed_through_buffers</a></p>                                                                |
| opt_add_insts                        |                                                                                                                                                                                                                                                                                                  |
|                                      | <p>Enables adding of new instances</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_add_insts</a></p>                                                                                                       |
| opt_add_ports                        |                                                                                                                                                                                                                                                                                                  |
|                                      | <p>Enables adding ports for optimal buffering</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_add_ports</a></p>                                                                                            |
| opt_all_end_points                   |                                                                                                                                                                                                                                                                                                  |
|                                      | <p>Enables the all end points optimization mode</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_all_end_points</a></p>                                                                                    |
| opt_allow_floating_outputs           |                                                                                                                                                                                                                                                                                                  |
|                                      | <p>Allow floating outputs for non-sequential cells during optimization.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_allow_floating_outputs</a></p>                                                    |
| opt_allow_multi_bit_on_flop_with_sdc |                                                                                                                                                                                                                                                                                                  |
|                                      | <p>allow merge or split works on flops which have SDC</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false merge_only split_only</p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_allow_multi_bit_on_flop_with_sdc</a></p> |
| opt_allow_only_cell_swapping         |                                                                                                                                                                                                                                                                                                  |

|                                                     |                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | <p>Forces the post_route optimization to perform only same-size Vth swapping for timing improvements</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_allow_only_cell_swapping</a></p>               |
| opt_area_recovery                                   |                                                                                                                                                                                                                                                                                            |
|                                                     | <p>Enables area reclamation step</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false default</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_area_recovery</a></p>                                              |
| opt_area_recovery_setup_target_slack                |                                                                                                                                                                                                                                                                                            |
|                                                     | <p>Specifies target slack value (in nano secs) for area and power reclaim during setup timing optimization</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_area_recovery_setup_target_slack</a></p> |
| opt_boundary_optimize_invert_hpins_rename_extension |                                                                                                                                                                                                                                                                                            |
|                                                     | <p>Extension to be appended on pin or net name, when boundary pin is inverted.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> _BAR</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_boundary_optimize_invert_hpins_rename_extension</a></p>             |
| opt_clock_traverse_by_level                         |                                                                                                                                                                                                                                                                                            |
|                                                     | <p>Traverse Instances in clock path 1 level at a time.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_clock_traverse_by_level</a></p>                                                              |
| opt_clone_insts_list                                |                                                                                                                                                                                                                                                                                            |
|                                                     | <p>list of instance list</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_clone_insts_list</a></p>                                                                                                    |

|                                           |                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_concatenate_default_and_user_prefixes |                                                                                                                                                                                                                                                                                                                                                                    |
|                                           | <p>Enables combining default prefix for instances(nets) with the prefix specified by opt_new_inst_prefix(opt_new_net_prefix)</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_concatenate_default_and_user_prefixes</a></p>                                                   |
| opt_consider_routing_congestion           |                                                                                                                                                                                                                                                                                                                                                                    |
|                                           | <p>Enables routing congestion check at the post_route stage during optimization of setup, hold, glitch. It also impacts post_cts hold fixing.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto false true</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_consider_routing_congestion</a></p> |
| opt_constant_inputs                       |                                                                                                                                                                                                                                                                                                                                                                    |
|                                           | <p>Force optimization of instances with constant inputs, even if it worsens objective gain</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_constant_inputs</a></p>                                                                                                          |
| opt_constant_nets                         |                                                                                                                                                                                                                                                                                                                                                                    |
|                                           | <p>Enables optimization of constant nets</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_constant_nets</a></p>                                                                                                                                                               |
| opt_delete_insts                          |                                                                                                                                                                                                                                                                                                                                                                    |
|                                           | <p>Enables instance deletion on all paths</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_delete_insts</a></p>                                                                                                                                                               |
| opt_detail_drv_failure_reason             |                                                                                                                                                                                                                                                                                                                                                                    |

|                                            |                                                                                                                                                                                                                                        |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>To enable printing detailed drv failure reason report file</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_detail_drv_failure_reason</a></p> |
| opt_detail_drv_failure_reason_max_num_nets |                                                                                                                                                                                                                                        |
|                                            | <p>Specify how many violating nets printed</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 50</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_detail_drv_failure_reason_max_num_nets</a></p>           |
| opt_dont_use_qn_sequential_pins            |                                                                                                                                                                                                                                        |
|                                            | <p>Do not use inverted outputs of sequential cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_dont_use_qn_sequential_pins</a></p>         |
| opt_down_size_insts                        |                                                                                                                                                                                                                                        |
|                                            | <p>Enables instance downsizing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_down_size_insts</a></p>                                           |
| opt_drv                                    |                                                                                                                                                                                                                                        |
|                                            | <p>Enables fixing max cap/max tran violations on constrained nets</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_drv</a></p>                    |
| opt_drv_margin                             |                                                                                                                                                                                                                                        |
|                                            | <p>Used for scaling max cap/max tran constraints</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_drv_margin</a></p>                             |
| opt_drv_with_miller_cap                    |                                                                                                                                                                                                                                        |

|                                            |                                                                                                                                                                                                                                                           |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_drv_with_miller_cap</a>                                                                                                               |
| opt_duplicate_cte_constrained_hport        |                                                                                                                                                                                                                                                           |
|                                            | allow buffering to duplicate hterms/hports with CTE constraints.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_duplicate_cte_constrained_hport</a>                                |
| opt_early_hold_fixing                      |                                                                                                                                                                                                                                                           |
|                                            | set the ideal or propagaged clock mode for hold fixing in preCTS<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> none ideal propagated<br><b>Default:</b> none<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_early_hold_fixing</a> |
| opt_enable_always_on_type_in_remove_assign |                                                                                                                                                                                                                                                           |
|                                            | Enable always on type in remove assign.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_enable_always_on_type_in_remove_assign</a>                                                 |
| opt_enable_clock_pulse_width_checks        |                                                                                                                                                                                                                                                           |
|                                            | Fix and honor pulse width check violations<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_enable_clock_pulse_width_checks</a>                                                     |
| opt_enable_data_to_data_checks             |                                                                                                                                                                                                                                                           |
|                                            | <b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> true false<br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_enable_data_to_data_checks</a>                                                                      |

|                                    |                                                                                                                                                                                                                                                                                       |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_enable_drv_failure_reason_warn |                                                                                                                                                                                                                                                                                       |
|                                    | <p>To enable printing warning message of drv failure reason in log</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                  |
| opt_enable_podv2_clock_opt_flow    |                                                                                                                                                                                                                                                                                       |
|                                    | <p>enable PodV2 and clock_opt_design flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_enable_podv2_clock_opt_flow</a></p>                                                                   |
| opt_enable_restructure             |                                                                                                                                                                                                                                                                                       |
|                                    | <p>Enables netlist restructuring</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_enable_restructure</a></p>                                                                                     |
| opt_enable_targeted_synthesis      |                                                                                                                                                                                                                                                                                       |
|                                    | <p>Enable targeted synthesis mode</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_enable_targeted_synthesis</a></p>                                                                            |
| opt_fix_fanout_load                |                                                                                                                                                                                                                                                                                       |
|                                    | <p>Enables fixing fanOut load violations</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_fix_fanout_load</a></p>                                                                               |
| opt_flop_pins_report               |                                                                                                                                                                                                                                                                                       |
|                                    | <p>print full hierarchical flop pins names for flops deleted in simplifyNetlist</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_flop_pins_report</a></p> |

|                           |                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_flops_report          |                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>print full hierarchical flop names for flops deleted in simplifyNetlist</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_flops_report</a></p>                                                                                        |
| opt_hier_add_antenna_cell |                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>add antenna diode to interface load</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_hier_add_antenna_cell</a></p>                                                                                                                                                         |
| opt_hier_opt_stage        |                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>Turn on default settings for hierarchical flows at different optimization stage.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> pre_cts post_cts unset</p> <p><b>Default:</b> unset</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_hier_opt_stage</a></p>                                                                 |
| opt_high_effort_cells     |                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>Provide a list of cells to be used for high effort optimization. Dont-use and Dont-touch const for these cells will be ignored for high effort optimization.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_high_effort_cells</a></p>                                     |
| opt_hold_allow_overlap    |                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>Enables hold optimization to insert cells with overlaps when inadequate free space is avail The placement will then be legalized before ECO routing.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto false true</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_hold_allow_overlap</a></p> |
| opt_hold_allow_resize     |                                                                                                                                                                                                                                                                                                                                                                     |



|                                      |                                                                                                                                                                                                                                                                                             |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Enables the resizing capability inside hold optimization.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true</p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_hold_allow_resize</a></p>                          |
| opt_hold_allow_setup_tns_degradation |                                                                                                                                                                                                                                                                                             |
|                                      | <p>Allows setup total negative slack to degrade during hold optimization.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_hold_allow_setup_tns_degradation</a></p>                                    |
| opt_hold_cells                       |                                                                                                                                                                                                                                                                                             |
|                                      | <p>Provide a list of buffer &amp; delay cells to be used for hold buffering. The dont_use constraints these cells will be ignored.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_hold_cells</a></p> |
| opt_hold_ignore_path_groups          |                                                                                                                                                                                                                                                                                             |
|                                      | <p>Specifies path groups to be excluded in hold timing optimization</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_hold_ignore_path_groups</a></p>                                                   |
| opt_hold_on_excluded_clock_nets      |                                                                                                                                                                                                                                                                                             |
|                                      | <p>Do hold fixing on the excluded clock nets</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_hold_on_excluded_clock_nets</a></p>                                                                     |
| opt_hold_slack_threshold             |                                                                                                                                                                                                                                                                                             |
|                                      | <p>Specifies slack threshold value beyond which the hold timing violations will not be fixed</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1000.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_hold_slack_threshold</a></p>                        |

|                                |                                                                                                                                                                                                                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_hold_target_slack          |                                                                                                                                                                                                                                                                                                      |
|                                | <p>Specifies target slack value (in nano secs) for hold timing optimization</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_hold_target_slack</a></p>                                                         |
| opt_honor_density_screen       |                                                                                                                                                                                                                                                                                                      |
|                                | <p>Control density screen</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_honor_density_screen</a></p>                                                                                                        |
| opt_honor_fences               |                                                                                                                                                                                                                                                                                                      |
|                                | <p>Specifies that the timing optimization takes fences constraints into account. Placement of cells must be legalized within each fence.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_honor_fences</a></p> |
| opt_icg_enable_pin_rebuffering |                                                                                                                                                                                                                                                                                                      |
|                                | <p>Rebuffer nets to enable pin of ICG instances after clock tree insertion</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true auto false</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_icg_enable_pin_rebuffering</a></p>      |
| opt_leakage_to_dynamic_ratio   |                                                                                                                                                                                                                                                                                                      |
|                                | <p>relative effort for leakage and dynamic power optimization in the range [0.0 - 1.0]</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_leakage_to_dynamic_ratio</a></p>                                       |
| opt_max_density                |                                                                                                                                                                                                                                                                                                      |

|                                                 |                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>Specifies maximum area utilization</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.95</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_max_density</a></p>                                                                                                                   |
| opt_max_length                                  |                                                                                                                                                                                                                                                                                                                     |
|                                                 | <p>Enables max length constraints in opt_design</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_max_length</a></p>                                                                                                          |
| opt_move_insts                                  |                                                                                                                                                                                                                                                                                                                     |
|                                                 | <p>Enables moving instances while fixing setup violations</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_move_insts</a></p>                                                                                                  |
| opt_multi_bit_combinational_merge_timing_effort |                                                                                                                                                                                                                                                                                                                     |
|                                                 | <p>Specifies the timing effort level for multibit combinational merging</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> medium</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_combinational_merge_timing_effort</a></p>      |
| opt_multi_bit_combinational_mode                |                                                                                                                                                                                                                                                                                                                     |
|                                                 | <p>Enables the user specific multi bit combination cell merge and split based on area and pow</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto power area</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_combinational_mode</a></p> |
| opt_multi_bit_combinational_opt                 |                                                                                                                                                                                                                                                                                                                     |
|                                                 | <p>enables or disables the multi-bit combinational optimization flow</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true mergeOnly splitOnly</p> <p><b>Default:</b> splitOnly</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_combinational_opt</a></p>       |

|                                                 |                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_multi_bit_combinational_split_timing_effort |                                                                                                                                                                                                                                                                                                               |
|                                                 | <p>Specifies the timing effort level for multibit combinational splitting</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> low</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_combinational_split_timing_effort</a></p> |
| opt_multi_bit_flop_merge_bank_label_inference   |                                                                                                                                                                                                                                                                                                               |
|                                                 | <p>Enables soft merging of flops with the same bank label</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_flop_merge_bank_label_inference</a></p>                                                            |
| opt_multi_bit_flop_merge_timing_effort          |                                                                                                                                                                                                                                                                                                               |
|                                                 | <p>Timing effort level for MBFF merging</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> medium</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_flop_merge_timing_effort</a></p>                                         |
| opt_multi_bit_flop_name_prefix                  |                                                                                                                                                                                                                                                                                                               |
|                                                 | <p>User to specify multi bit flop name prefix</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> CDN_MBIT_</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_flop_name_prefix</a></p>                                                                                 |
| opt_multi_bit_flop_name_separator               |                                                                                                                                                                                                                                                                                                               |
|                                                 | <p>User to specify flop merge and split name separator</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> _MB_</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_flop_name_separator</a></p>                                                                          |
| opt_multi_bit_flop_name_suffix                  |                                                                                                                                                                                                                                                                                                               |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>User to specify multi bit flop name suffix, For example: _FF</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_flop_name_suffix</a></p>                                                                                                                                                                                                                |
| opt_multi_bit_flop_opt                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                | <p>specifies multi bit flop merge/split opt</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true mergeOnly splitOnly</p> <p><b>Default:</b> splitOnly</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_flop_opt</a></p>                                                                                                                                                                             |
| opt_multi_bit_flop_reorder_bits                |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                | <p>Enable Swap MBFF bits to take advantage of mixed-driving strength cells. false: Disable; true: enable for timing and power optimization; timing: enable for timing optimization; power: enable for power optimization.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true timing power</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_flop_reorder_bits</a></p> |
| opt_multi_bit_flop_split_post_cts              |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                | <p>Enable MBFF split in post-CTS stage. false: Disable; wns: Enable in WNS optimization; tns: Enable in TNS optimization; true: Enable in both WNS and TNS optimization</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true wns tns</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                 |
| opt_multi_bit_flop_split_post_route            |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                | <p>Enable MBFF split in post-route stage. false: Disable; wns: Enable in WNS optimization, tns: Enable in tTNS optimization; true: Enable in both WNS and TNS optimization.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true wns tns</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                             |
| opt_multi_bit_flop_split_report_failure_reason |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                        |                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>To enable printing split failure reasons</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_flop_split_report_failure_reason</a></p>                                                                              |
| opt_multi_bit_flop_split_timing_effort |                                                                                                                                                                                                                                                                                                                   |
|                                        | <p>Timing effort level for MBFF splitting</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> low</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_flop_split_timing_effort</a></p>                                              |
| opt_multi_bit_merge_scan_head          |                                                                                                                                                                                                                                                                                                                   |
|                                        | <p>merge of the scan chain head/tail</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_merge_scan_head</a></p>                                                                                                     |
| opt_multi_bit_split_scan_head          |                                                                                                                                                                                                                                                                                                                   |
|                                        | <p>split of the scan chain head/tail</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_split_scan_head</a></p>                                                                                                     |
| opt_multi_bit_unused_bit_count         |                                                                                                                                                                                                                                                                                                                   |
|                                        | <p>specifies how many maximum unused bits to be allowed in each multibit cell</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_unused_bit_count</a></p>                                                                |
| opt_multi_bit_unused_bits              |                                                                                                                                                                                                                                                                                                                   |
|                                        | <p>specifies whether a mapped multibit instance can have one or more unused bits</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true pre_place pre_cts</p> <p><b>Default:</b> pre_place</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multi_bit_unused_bits</a></p> |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_multibit_allow_shift_register_head_flop_merge  |                                                                                                                                                                                                                                                                                                                                                                      |
|                                                    | <p>Controls whether head flops of shift register can be merged to multibit.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multibit_allow_shift_register_head_flop_merge</a></p>                                                                                             |
| opt_multibit_combinational_name_concatenate_string |                                                                                                                                                                                                                                                                                                                                                                      |
|                                                    | <p>Separator string to be used for concatenation when naming multibit combinational and spe power cells. Recommended values for Genus-LEC flow are '_MB_' and '_mb_'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> _MB_</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multibit_combinational_name_concatenate_string</a></p> |
| opt_multibit_seqs_name_concatenate_string          |                                                                                                                                                                                                                                                                                                                                                                      |
|                                                    | <p>Separator string to be used for concatenation when naming multibit sequential instances. Recommended values for Genus-LEC flow are '_MB_' and '_mb_'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> _MB_</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_multibit_seqs_name_concatenate_string</a></p>                       |
| opt_new_inst_prefix                                |                                                                                                                                                                                                                                                                                                                                                                      |
|                                                    | <p>Prefix string for name of new instances</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_new_inst_prefix</a></p>                                                                                                                                                             |
| opt_new_net_prefix                                 |                                                                                                                                                                                                                                                                                                                                                                      |
|                                                    | <p>Prefix string for name of new nets</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_new_net_prefix</a></p>                                                                                                                                                                   |
| opt_one_pass_lec                                   |                                                                                                                                                                                                                                                                                                                                                                      |

|                              |                                                                                                                                                                                                                                                                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Set options one pass lec opt support</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_one_pass_lec</a></p>                                                                                                                                       |
| opt_pin_swapping             |                                                                                                                                                                                                                                                                                                                                           |
|                              | <p>Enables pin swapping</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_pin_swapping</a></p>                                                                                                                                                        |
| opt_podv2_flow_effort        |                                                                                                                                                                                                                                                                                                                                           |
|                              | <p>Effort level for PodV2 flow</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto standard extreme</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_podv2_flow_effort</a></p>                                                                                           |
| opt_post_route_allow_overlap |                                                                                                                                                                                                                                                                                                                                           |
|                              | <p>Enables post_route optimization to insert cells with overlaps when inadequate free space is available. The placement will then be legalized before ECO routing.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_post_route_allow_overlap</a></p> |
| opt_post_route_area_reclaim  |                                                                                                                                                                                                                                                                                                                                           |
|                              | <p>To enable area reclaim in post_route optimization</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none setup_aware hold_and_setup_aware</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_post_route_area_reclaim</a></p>                                               |
| opt_post_route_art_flow      |                                                                                                                                                                                                                                                                                                                                           |
|                              | <p>Enables optimizing the top-level interface logic for a hierarchical floorplan</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_post_route_art_flow</a></p>                                                                                       |



|                                    |                                                                                                                                                                                                                                      |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_post_route_check_antenna_rules |                                                                                                                                                                                                                                      |
|                                    | Allows optimization to check/ignore antenna rules in lefsafe swaps<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_post_route_check_antenna_rules</a>          |
| opt_post_route_drv_recovery        |                                                                                                                                                                                                                                      |
|                                    | Enables post_route drv recovery in gigaopt.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> auto true false<br><b>Default:</b> auto<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_post_route_drv_recovery</a> |
| opt_post_route_fix_clock_drv       |                                                                                                                                                                                                                                      |
|                                    | enables DRV fixing on clock net<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_post_route_fix_clock_drv</a>                                                   |
| opt_post_route_fix_glitch          |                                                                                                                                                                                                                                      |
|                                    | Enables glitch fixing in AAE flow<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_post_route_fix_glitch</a>                                                    |
| opt_post_route_fix_si_transitions  |                                                                                                                                                                                                                                      |
|                                    | Enables sislew fixing in AAE flow<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_post_route_fix_si_transitions</a>                                           |
| opt_post_route_hold_recovery       |                                                                                                                                                                                                                                      |

|                                      |                                                                                                                                                                                                                                                    |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> auto true false<br><b>Default:</b> auto<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_post_route_hold_recovery</a>                                                             |
| opt_post_route_report_si_transitions |                                                                                                                                                                                                                                                    |
|                                      | report sislew violations<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_post_route_report_si_transitions</a>                                                               |
| opt_post_route_setup_recovery        |                                                                                                                                                                                                                                                    |
|                                      | Enables post_route SI setup timing recovery in gigaopt.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> auto true false<br><b>Default:</b> auto<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_post_route_setup_recovery</a> |
| opt_power_effort                     |                                                                                                                                                                                                                                                    |
|                                      | Effort level for power aware mode in optimization<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> none low high<br><b>Default:</b> none<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_power_effort</a>                      |
| opt_pre_route_ndr_aware              |                                                                                                                                                                                                                                                    |
|                                      | Provide a list of NDR names that will be used NDR-based optimization<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_pre_route_ndr_aware</a>                                 |
| opt_preserve_all_sequential          |                                                                                                                                                                                                                                                    |
|                                      | Preserve sequential element during the simplify netlist optimization step<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_preserve_all_sequential</a>                       |

|                                             |                                                                                                                                                                                                                                                                                       |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_preserve_hpin_function                  |                                                                                                                                                                                                                                                                                       |
|                                             | <p>Enables preserving logical functions at hierarchical ports</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_preserve_hpin_function</a></p>                                                   |
| opt_print_ports_nets_preserved_for_callback |                                                                                                                                                                                                                                                                                       |
|                                             | <p>Print nets and pins preserved for verification.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_print_ports_nets_preserved_for_callback</a></p>                                             |
| opt_remove_redundant_insts                  |                                                                                                                                                                                                                                                                                       |
|                                             | <p>Enables simplifying the netlist during optimization</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_remove_redundant_insts</a></p>                                                           |
| opt_report_multi_bit_unmerged_reasons       |                                                                                                                                                                                                                                                                                       |
|                                             | <p>When enabled, the reasons for unmerged multibit flops/latches will be printed in the MBFF summary</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_report_multi_bit_unmerged_reasons</a></p> |
| opt_resize_flip_flops                       |                                                                                                                                                                                                                                                                                       |
|                                             | <p>Enables FF resizing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_resize_flip_flops</a></p>                                                                                                |
| opt_resize_level_shifter_and_iso_insts      |                                                                                                                                                                                                                                                                                       |
|                                             | <p>Enables resizing of shifters and isolation instances</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_resize_level_shifter_and_iso_insts</a></p>                                             |

|                                                        |                                                                                                                                                                                                                                  |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_resize_power_switch_insts                          |                                                                                                                                                                                                                                  |
|                                                        | Enables resizing of power switch instances during optimization<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_resize_power_switch_insts</a>              |
| opt_retime_move_mux_loop_with_register                 |                                                                                                                                                                                                                                  |
|                                                        | Move the feedback mux loop along with the register during retiming.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_retime_move_mux_loop_with_register</a> |
| opt_retime_register_naming_suffix                      |                                                                                                                                                                                                                                  |
|                                                        | Suffix that gets appended to the name of the retimed flops.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> _reg<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_retime_register_naming_suffix</a>            |
| opt_route_opt_recovery                                 |                                                                                                                                                                                                                                  |
|                                                        | <b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> auto true false<br><b>Default:</b> auto<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_route_opt_recovery</a>                                                 |
| opt_sequential_genus_restructure_report_failure_reason |                                                                                                                                                                                                                                  |
|                                                        | To enable printing SeqCRR failure reasons<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_sequential_genus_restructure_report_failure_reason</a>          |
| opt_setup_target_slack                                 |                                                                                                                                                                                                                                  |
|                                                        | Specifies target slack value (in nano secs) for setup timing optimization<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_setup_target_slack</a>          |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_signoff_add_inst              |                                                                                                                                                                                                                                                                                                                                                                                              |
|                                   | <p>Specifies whether timing optimizer is allowed to add buffer/delay/inverter cell instances.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_add_inst</a></p>                                                                                                                                 |
| opt_signoff_add_load              |                                                                                                                                                                                                                                                                                                                                                                                              |
|                                   | <p>Specifies whether Hold optimizer is allowed to add dummy cell instances</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_add_load</a></p>                                                                                                                                                   |
| opt_signoff_allow_skewing         |                                                                                                                                                                                                                                                                                                                                                                                              |
|                                   | <p>useful skew based setup opt</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_allow_skewing</a></p>                                                                                                                                                                                          |
| opt_signoff_along_route_buffering |                                                                                                                                                                                                                                                                                                                                                                                              |
|                                   | <p>Allow buffer/inverter to be inserted along the route and not only at driver or sink terms.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_along_route_buffering</a></p>                                                                                                                    |
| opt_signoff_buffer_cell_list      |                                                                                                                                                                                                                                                                                                                                                                                              |
|                                   | <p>Specifies a list of buffer/delay and inverter cells that can be used for DRV opt and list of buffer/delays cells that can be used for Hold optimizer, regardless of whether those cells have "dont_use" attribute.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_buffer_cell_list</a></p> |
| opt_signoff_check_type            |                                                                                                                                                                                                                                                                                                                                                                                              |

|                                     |                                                                                                                                                                                                                                                                            |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Check type for path based analysis.<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> early late both<br/> <b>Default:</b> both<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">opt_signoff_check_type</a></p>                                   |
| opt_signoff_clock_cell_list         |                                                                                                                                                                                                                                                                            |
|                                     | <p>Restricts the list of buffers/inverter/gating cells that are allowed to be used by ECO on the clock tree.<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">opt_signoff_clock_cell_list</a></p> |
| opt_signoff_clock_max_level         |                                                                                                                                                                                                                                                                            |
|                                     | <p>max level allowed for clock tree<br/> <b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">opt_signoff_clock_max_level</a></p>                                                                              |
| opt_signoff_delete_inst             |                                                                                                                                                                                                                                                                            |
|                                     | <p>delete buffer for optimization<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">opt_signoff_delete_inst</a></p>                                                                                |
| opt_signoff_disable_geometry_checks |                                                                                                                                                                                                                                                                            |
|                                     | <p>Performs Vth swapping without checking pins geometry.<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">opt_signoff_disable_geometry_checks</a></p>                                            |
| opt_signoff_drv_margin              |                                                                                                                                                                                                                                                                            |
|                                     | <p>Specifies a margin value by which ratio the drv check would be relaxed or tightened<br/> <b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">opt_signoff_drv_margin</a></p>                           |

|                             |                                                                                                                                                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_signoff_eco_file_prefix |                                                                                                                                                                                                                                                                                     |
|                             | <p>Specifies the prefix name for the ECO files generated.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_eco_file_prefix</a></p>                                                     |
| opt_signoff_em_type         |                                                                                                                                                                                                                                                                                     |
|                             | <p>Specifies the current waveform calculation method when doing EM fixing.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> all avg rms peak</p> <p><b>Default:</b> all</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_em_type</a></p> |
| opt_signoff_fix_cell_em     |                                                                                                                                                                                                                                                                                     |
|                             | <p>Enables Cell EM violations fixing during DRV optimization.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_cell_em</a></p>                                                    |
| opt_signoff_fix_clock_drv   |                                                                                                                                                                                                                                                                                     |
|                             | <p>fix max_cap/max_tran violations on clock nets</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_clock_drv</a></p>                                                               |
| opt_signoff_fix_data_drv    |                                                                                                                                                                                                                                                                                     |
|                             | <p>fix max_cap/max_tran violations on data nets</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_data_drv</a></p>                                                                  |
| opt_signoff_fix_glitch      |                                                                                                                                                                                                                                                                                     |
|                             | <p>fix nets with glitch</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_glitch</a></p>                                                                                           |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_signoff_fix_hold_allow_setup_degrade      |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                               | <p>Allow setup slack fixing in hold fixing.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none low medium high</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_hold_allow_setup_degrade</a></p>                                                                                                                                         |
| opt_signoff_fix_hold_allow_setup_optimization |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                               | <p>simultaneous hold and setup fixing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_hold_allow_setup_optimization</a></p>                                                                                                                                                                                         |
| opt_signoff_fix_hold_allow_setup_tns_degrade  |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                               | <p>When active, the Setup Total Negative Slack can be degraded during Hold time violations while still maintaining the Setup Worst Negative Slack in every Setup active views.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_hold_allow_setup_tns_degrade</a></p>                                                 |
| opt_signoff_fix_hold_variation_aware          |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                               | <p>Variation aware hold fixing.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_hold_variation_aware</a></p>                                                                                                                                                                                                        |
| opt_signoff_fix_hold_with_margin              |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                               | <p>Specifies a slack margin value in nanoseconds for Hold time violations fixing. The purpose overfix any Hold violations by the amount of the margin value. This margin is not applied or that are already meeting the Hold target slack.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_hold_with_margin</a></p> |
| opt_signoff_fix_ir_drop                       |                                                                                                                                                                                                                                                                                                                                                                                                                        |



|                                            |                                                                                                                                                                                                                                                             |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | fix ir-drop hotspots<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_fix_ir_drop</a>                                                                                         |
| opt_signoff_fix_max_cap                    |                                                                                                                                                                                                                                                             |
|                                            | Enables max_cap violations fixing during DRV optimization.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_fix_max_cap</a>                                                    |
| opt_signoff_fix_max_transition             |                                                                                                                                                                                                                                                             |
|                                            | Enables max_tran violations fixing during DRV optimization.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_fix_max_transition</a>                                            |
| opt_signoff_fix_osus_glitch                |                                                                                                                                                                                                                                                             |
|                                            | fix nets with osus glitch<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_fix_osus_glitch</a>                                                                                |
| opt_signoff_fix_si_slew                    |                                                                                                                                                                                                                                                             |
|                                            | Performs SI Slew fixing during DRV fixing using resizing and buffering techniques.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_fix_si_slew</a>                           |
| opt_signoff_fix_slack_in_range_lower_limit |                                                                                                                                                                                                                                                             |
|                                            | Sets the lower slack limit for a violated path to be considered during timing closure.<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> -1e+30<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_fix_slack_in_range_lower_limit</a> |
| opt_signoff_fix_slack_in_range_upper_limit |                                                                                                                                                                                                                                                             |

|                                        |                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Sets the upper slack limit for a violated path to be considered during timing closure.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1e+30</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_slack_in_range_upper_limit</a></p> |
| opt_signoff_fix_xtalk                  |                                                                                                                                                                                                                                                                                   |
|                                        | <p>Reduces the crosstalk on the net that violates the thresholds set by the user.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_xtalk</a></p>                                |
| opt_signoff_fix_xtalk_preserve_hold    |                                                                                                                                                                                                                                                                                   |
|                                        | <p>preserve hold slack in xtalk fixing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_xtalk_preserve_hold</a></p>                                                             |
| opt_signoff_fix_xtalk_preserve_setup   |                                                                                                                                                                                                                                                                                   |
|                                        | <p>preserve setup slack in xtalk fixing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_fix_xtalk_preserve_setup</a></p>                                                           |
| opt_signoff_high_effort_cell_list      |                                                                                                                                                                                                                                                                                   |
|                                        | <p>High effort cells.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                            |
| opt_signoff_hold_target_slack          |                                                                                                                                                                                                                                                                                   |
|                                        | <p>Specifies a target slack value in nanoseconds for hold timing optimize</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_hold_target_slack</a></p>                                |
| opt_signoff_hold_target_slack_per_view |                                                                                                                                                                                                                                                                                   |

|                                        |                                                                                                                                                                                                                                                                                    |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Specifies view-wise target slack values in nanoseconds for hold timing optimization</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_hold_target_slack_per_view</a></p>            |
| opt_signoff_hold_xtalk_delta_threshold |                                                                                                                                                                                                                                                                                    |
|                                        | <p>Crosstalk delta delay threshold for selecting nets in hold timing views when running Xtalk fi</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.3</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_hold_xtalk_delta_threshold</a></p> |
| opt_signoff_hold_xtalk_slack_threshold |                                                                                                                                                                                                                                                                                    |
|                                        | <p>Slack threshold for selecting nets in hold timing views when running Xtalk fixing.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1000.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_hold_xtalk_slack_threshold</a></p>         |
| opt_signoff_ignore_drv_checks          |                                                                                                                                                                                                                                                                                    |
|                                        | <p>Ignores DRV checks during timing/area/power optimization.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_ignore_drv_checks</a></p>                                              |
| opt_signoff_keep_tmp_files             |                                                                                                                                                                                                                                                                                    |
|                                        | <p>To keep all the temporary files generated by D-MMMC and Tempus clients for debugging purposes.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_keep_tmp_files</a></p>            |
| opt_signoff_legal_only                 |                                                                                                                                                                                                                                                                                    |
|                                        | <p>Enables/disables legal location checks during resizing and buffering</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_legal_only</a></p>                                           |
| opt_signoff_load_cell_list             |                                                                                                                                                                                                                                                                                    |

|                               |                                                                                                                                                                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Specifies the list of load cells</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_load_cell_list</a></p>                                                            |
| opt_signoff_max_cap_margin    |                                                                                                                                                                                                                                                                     |
|                               | <p>Specifies the ratio by which the max_cap check would be relaxed or tightened</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_max_cap_margin</a></p>               |
| opt_signoff_max_local_density |                                                                                                                                                                                                                                                                     |
|                               | <p>threshold value for local density check. To enable give a value more than 0</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_max_local_density</a></p>            |
| opt_signoff_max_opt_runtime   |                                                                                                                                                                                                                                                                     |
|                               | <p>If set to a positive value, would restrict the optimization runtime by that value, in minutes.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_max_opt_runtime</a></p> |
| opt_signoff_max_paths         |                                                                                                                                                                                                                                                                     |
|                               | <p>Specifies how many paths are getting retimed for ECO DB generation in PBA mode.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 5000000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_max_paths</a></p>                |
| opt_signoff_max_runtime       |                                                                                                                                                                                                                                                                     |
|                               | <p>If set to a positive value, would restrict the command runtime by that value, in minutes.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_max_runtime</a></p>          |
| opt_signoff_max_slack         |                                                                                                                                                                                                                                                                     |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Specifies the maximum GBA slack value to be used when selecting paths to be retimed for DB generation in PBA mode.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_max_slack</a></p>                                                                                                                                                                |
| opt_signoff_max_transition_margin       |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <p>Specifies the ratio by which the max_tran check would be relaxed or tightened</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_max_transition_margin</a></p>                                                                                                                                                                                         |
| opt_signoff_num_report_paths            |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <p>Number of paths printed after signoff timing analysis per view</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 50</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_num_report_paths</a></p>                                                                                                                                                                                                                 |
| opt_signoff_nworst                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <p>Specifies the number of paths to be retimed per endpoints for ECO DB generation in PBA n</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 50</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_nworst</a></p>                                                                                                                                                                                                 |
| opt_signoff_optimize_core_only          |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <p>Enables the tool to optimize timing or recover area/power only on register to register paths.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_optimize_core_only</a></p>                                                                                                                                                                            |
| opt_signoff_optimize_replicated_modules |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <p>Allows the tool to perform timing optimization inside replicated hierarchies while generating one ECO file per sets of clones. The cell name of the replicated hierarchies have to be listed in the file pointed by the opt_signoff_partition_list_file option.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_optimize_replicated_modules</a></p> |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_signoff_optimize_sequential_cells |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                       | <p>Specifies whether the timing/area/power optimizer is allowed to optimize sequential cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_optimize_sequential_cells</a></p>                                                                                                                                                    |
| opt_signoff_partition_list_file       |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                       | <p>File containing list of partitions in the design</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_partition_list_file</a></p>                                                                                                                                                                                                    |
| opt_signoff_pba_eco_route             |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                       | <p>Enables PBA timing to be passed to ecoRoute for higher timing driven routing accuracy</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_pba_eco_route</a></p>                                                                                                                                                                    |
| opt_signoff_pba_effort                |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                       | <p>When enabling "high" effort mode, all the optimization engines will perform more aggressive timing closure or area/power recovery with respect to PBA timing and the expectation is to get better quality of results.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> medium high</p> <p><b>Default:</b> high</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_pba_effort</a></p> |
| opt_signoff_post_mask                 |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                       | <p>enable post mask ECO mode</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_post_mask</a></p>                                                                                                                                                                                                                                    |
| opt_signoff_post_mask_decap_cells     |                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                               |                                                                                                                                                                                                                                                               |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | Specify list of PF decap cells that are allowed to be removed in post mask mode<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_post_mask_decap_cells</a>                       |
| opt_signoff_post_mask_filler_cells            |                                                                                                                                                                                                                                                               |
|                                               | Specify list of PF filler cells that are allowed to be removed/reinserted in post mask mode<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_post_mask_filler_cells</a>          |
| opt_signoff_post_mask_gate_array_filler_cells |                                                                                                                                                                                                                                                               |
|                                               | original of enable post mask mode to use cells from the specified list at GA filler sites<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_post_mask_gate_array_filler_cells</a> |
| opt_signoff_post_mask_logic_cells             |                                                                                                                                                                                                                                                               |
|                                               | Specify list of PF filler logic/ECO cells that are allowed to be used during optimization in po: mode<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_post_mask_logic_cells</a> |
| opt_signoff_post_sta_tcl                      |                                                                                                                                                                                                                                                               |
|                                               | File containing the TCL code that will be executed immediately after signoff timing analysis view<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_post_sta_tcl</a>              |
| opt_signoff_power_aware                       |                                                                                                                                                                                                                                                               |
|                                               | power awareness in all optimizers and transforms<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_signoff_power_aware</a>                                                               |

|                             |                                                                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opt_signoff_power_opt_focus |                                                                                                                                                                                                                                                                                 |
|                             | <p>focus for power optimization</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> total leakage dynamic leakageOnly dynamicOnly</p> <p><b>Default:</b> total</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_power_opt_focus</a></p> |
| opt_signoff_pre_sta_tcl     |                                                                                                                                                                                                                                                                                 |
|                             | <p>File containing the TCL code that will be executed immediately prior to signoff timing analysis</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_pre_sta_tcl</a></p>            |
| opt_signoff_prefix          |                                                                                                                                                                                                                                                                                 |
|                             | <p>Specifies the prefix to be assigned to the new instances inserted.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ESO</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_prefix</a></p>                                             |
| opt_signoff_preserve_filler |                                                                                                                                                                                                                                                                                 |
|                             | <p>To enable preserve and restore of fillers</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_preserve_filler</a></p>                                                              |
| opt_signoff_read_eco_opt_db |                                                                                                                                                                                                                                                                                 |
|                             | <p>Path of directory where ECO Timing DB files must be loaded from.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_read_eco_opt_db</a></p>                                       |
| opt_signoff_read_irdrop_db  |                                                                                                                                                                                                                                                                                 |



|                                |                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Path of directory where command debug_irdrop generated IR drop DB files are kept</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_read_irdrop_db</a></p>                                                                                         |
| opt_signoff_resize_inst        |                                                                                                                                                                                                                                                                                                                                                  |
|                                | <p>enable/disable resizing transform</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_resize_inst</a></p>                                                                                                                                           |
| opt_signoff_retime             |                                                                                                                                                                                                                                                                                                                                                  |
|                                | <p>To enable PBA-based timing report and optimization for of all optimizers.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none aocv path_slew_propagation aocv_path_slew_propagation</p> <p><b>Default:</b> path_slew_propagation</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_retime</a></p> |
| opt_signoff_retime_depth       |                                                                                                                                                                                                                                                                                                                                                  |
|                                | <p>Retime depth for exhaustive path based analysis.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> default infinite</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_retime_depth</a></p>                                                                            |
| opt_signoff_retime_depth_early |                                                                                                                                                                                                                                                                                                                                                  |
|                                | <p>Retime depth for exhaustive path based early analysis.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> default infinite</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_retime_depth_early</a></p>                                                                |
| opt_signoff_retime_mode        |                                                                                                                                                                                                                                                                                                                                                  |

|                                        |                                                                                                                                                                                                                                                            |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Retime mode for path based analysis.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none path exhaustive</p> <p><b>Default:</b> path</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_retime_mode</a></p>  |
| opt_signoff_route_congestion_threshold |                                                                                                                                                                                                                                                            |
|                                        | <p>threshold value for route congestion buffering, value between 0 and 1</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.7</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_route_congestion_threshold</a></p> |
| opt_signoff_routing_congestion_aware   |                                                                                                                                                                                                                                                            |
|                                        | <p>Check routing congestion before adding buffers</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_routing_congestion_aware</a></p>                          |
| opt_signoff_select_drv_net_file        |                                                                                                                                                                                                                                                            |
|                                        | <p>list of nets to be optimized during DRV fixing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_select_drv_net_file</a></p>                                |
| opt_signoff_select_drv_pin_file        |                                                                                                                                                                                                                                                            |
|                                        | <p>list of pins to be optimized during DRV fixing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_select_drv_pin_file</a></p>                                |
| opt_signoff_select_hold_endpoints      |                                                                                                                                                                                                                                                            |
|                                        | <p>select end points for hold fixing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_select_hold_endpoints</a></p>                                           |
| opt_signoff_select_setup_endpoints     |                                                                                                                                                                                                                                                            |

|                                         |                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>select end points for setup fixing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_select_setup_endpoints</a></p>                                                                                                                                |
| opt_signoff_set_hold_endpoints_margin   |                                                                                                                                                                                                                                                                                                                                                   |
|                                         | <p>specify end point specific margin for hold fixing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_set_hold_endpoints_margin</a></p>                                                                                                              |
| opt_signoff_set_setup_endpoints_margin  |                                                                                                                                                                                                                                                                                                                                                   |
|                                         | <p>specify end point specific margin for setup fixing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_set_setup_endpoints_margin</a></p>                                                                                                            |
| opt_signoff_setup_recovery              |                                                                                                                                                                                                                                                                                                                                                   |
|                                         | <p>Forces Setup fixing to perform Setup timing recovery by using Vth swapping only. Useful for recover timing degradation after large ECO change during Power optimization.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_setup_recovery</a></p> |
| opt_signoff_setup_target_slack          |                                                                                                                                                                                                                                                                                                                                                   |
|                                         | <p>Specifies a target slack value in nanoseconds for setup timing optimization</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_setup_target_slack</a></p>                                                                                          |
| opt_signoff_setup_target_slack_per_view |                                                                                                                                                                                                                                                                                                                                                   |
|                                         | <p>Specifies view-wise target slack values in nanoseconds for setup timing optimization</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_setup_target_slack_per_view</a></p>                                                                         |
| opt_signoff_setup_xtalk_delta_threshold |                                                                                                                                                                                                                                                                                                                                                   |

|                                         |                                                                                                                                                                                                                                                                                   |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Crosstalk delta delay threshold for selecting nets in setup timing views when running Xtalk</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.3</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_setup_xtalk_delta_threshold</a></p> |
| opt_signoff_setup_xtalk_slack_threshold |                                                                                                                                                                                                                                                                                   |
|                                         | <p>Slack threshold for selecting nets in setup timing views when running Xtalk fixing.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1000.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_setup_xtalk_slack_threshold</a></p>      |
| opt_signoff_skew_core_only              |                                                                                                                                                                                                                                                                                   |
|                                         | <p>dont skew flops which eventually launches/captures any IO path</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_skew_core_only</a></p>                                           |
| opt_signoff_skip_drv_net_file           |                                                                                                                                                                                                                                                                                   |
|                                         | <p>list of nets to be excluded during DRV fixing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_skip_drv_net_file</a></p>                                                          |
| opt_signoff_swap_inst                   |                                                                                                                                                                                                                                                                                   |
|                                         | <p>Enables/disables VT swap resizing. Setup/leakage optimizers do not honor this option and perform VT-swapping.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_swap_inst</a></p>  |
| opt_signoff_upsize_in_power_opt         |                                                                                                                                                                                                                                                                                   |
|                                         | <p>Allow upsizing of cells for total power</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_upsize_in_power_opt</a></p>                                                             |
| opt_signoff_verbose                     |                                                                                                                                                                                                                                                                                   |

|                                          |                                                                                                                                                                                                                                                                          |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Specifies whether the command should output log file reporting with verbosity or not.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_verbose</a></p>                  |
| opt_signoff_write_eco_opt_db             |                                                                                                                                                                                                                                                                          |
|                                          | <p>Path of directory where ECO Timing DB files are saved.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ecoTimingDB</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_signoff_write_eco_opt_db</a></p>                                |
| opt_skew                                 |                                                                                                                                                                                                                                                                          |
|                                          | <p>Enables triggering the skew_clock command from within opt_design</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_skew</a></p>                                                   |
| opt_skew_apply_delay_limits_to_full_flow |                                                                                                                                                                                                                                                                          |
|                                          | <p>delay limits apply to whole flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_skew_apply_delay_limits_to_full_flow</a></p>                                                  |
| opt_skew_ccopt                           |                                                                                                                                                                                                                                                                          |
|                                          | <p>Effort for useful skew optimization within ccopt_design</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none standard extreme</p> <p><b>Default:</b> standard</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_skew_ccopt</a></p> |
| opt_skew_delay_pre_cts                   |                                                                                                                                                                                                                                                                          |
|                                          | <p>delaying mode in pre_cts</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_skew_delay_pre_cts</a></p>                                                                             |
| opt_skew_macro_only                      |                                                                                                                                                                                                                                                                          |

|                            |                                                                                                                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>skew hard blocks</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_skew_macro_only</a></p>                                                       |
| opt_skew_max_allowed_delay |                                                                                                                                                                                                                                          |
|                            | <p>max allowed skewing delay, unit is 'ns'</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_skew_max_allowed_delay</a></p>                         |
| opt_skew_min_allowed_delay |                                                                                                                                                                                                                                          |
|                            | <p>min allowed skewing delay, unit is 'ns'</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_skew_min_allowed_delay</a></p>                         |
| opt_skew_no_boundary       |                                                                                                                                                                                                                                          |
|                            | <p>consider boundary sequential elements</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_skew_no_boundary</a></p>                                 |
| opt_skew_post_route        |                                                                                                                                                                                                                                          |
|                            | <p>Enables useful skew optimization during post_route timing optimization.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_skew_post_route</a></p> |
| opt_skew_pre_cts           |                                                                                                                                                                                                                                          |
|                            | <p>Enables useful skew optimization during pre_cts timing optimization.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_skew_pre_cts</a></p>       |
| opt_target_based_opt_file  |                                                                                                                                                                                                                                          |

|                                  |                                                                                                                                                                                                                                                                        |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Target based opt input file</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_target_based_opt_file</a></p>                                                                     |
| opt_target_based_opt_file_only   |                                                                                                                                                                                                                                                                        |
|                                  | <p>Use data from target based opt file for optimization</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true</p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_target_based_opt_file_only</a></p> |
| opt_target_based_opt_hold_file   |                                                                                                                                                                                                                                                                        |
|                                  | <p>Target based opt input file for hold</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_target_based_opt_hold_file</a></p>                                                       |
| opt_tied_inputs                  |                                                                                                                                                                                                                                                                        |
|                                  | <p>Force optimization of instances with inputs tied together, even if it worsens objective gain</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_tied_inputs</a></p>             |
| opt_time_design_compress_reports |                                                                                                                                                                                                                                                                        |
|                                  | <p>save reports in compressed format</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_time_design_compress_reports</a></p>                                                        |
| opt_time_design_expanded_view    |                                                                                                                                                                                                                                                                        |
|                                  | <p>To report expanded views in time_design</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_time_design_expanded_view</a></p>                                                    |
| opt_time_design_num_paths        |                                                                                                                                                                                                                                                                        |

|                                              |                                                                                                                                                                                                                                                |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | Report specified number of paths in time_design<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 50<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_time_design_num_paths</a>                                                   |
| opt_time_design_report_net                   |                                                                                                                                                                                                                                                |
|                                              | To report net delays in time_design<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_time_design_report_net</a>                                                           |
| opt_time_design_report_path_type             |                                                                                                                                                                                                                                                |
|                                              | <b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> end summary full full_clock end_slack_only summary_slack_only<br><b>Default:</b> full_clock<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_time_design_report_path_type</a> |
| opt_time_design_vertical_timing_summary      |                                                                                                                                                                                                                                                |
|                                              | Prints vertical timing summary after time_design and final summary report<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_time_design_vertical_timing_summary</a>       |
| opt_tns                                      |                                                                                                                                                                                                                                                |
|                                              | Optimize all negative slack endpoints.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_tns</a>                                                                           |
| opt_unfix_clock_insts                        |                                                                                                                                                                                                                                                |
|                                              | Enables unfixing clock instances flow<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">opt_unfix_clock_insts</a>                                                              |
| opt_use_default_related_pg_pin_for_always_on |                                                                                                                                                                                                                                                |



|                                                     |                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | <p>Use default power or ground pin for the input or output pin's related_power_pin or related_ground_pin of always on buffer or inverter cell respectively.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_use_default_related_pg_pin_for_always_on</a></p> |
| opt_use_multi_output_gates                          |                                                                                                                                                                                                                                                                                                                                                    |
|                                                     | <p>Controls use of multiple output gates.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1 0 true false auto</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                          |
| opt_use_next_state_type_only_to_assign_sync_control |                                                                                                                                                                                                                                                                                                                                                    |
|                                                     | <p>Forces to use only nextstate_type for sync control assignment.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_use_next_state_type_only_to_assign_sync_control</a></p>                                                                                   |
| opt_verbose                                         |                                                                                                                                                                                                                                                                                                                                                    |
|                                                     | <p>Enables verbose information logging</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">opt_verbose</a></p>                                                                                                                                                      |
| opt_xor_gating_custom_banks                         |                                                                                                                                                                                                                                                                                                                                                    |
|                                                     | <p>The list of flops for XorGating (user specified banks)</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                         |
| opt_xor_gating_custom_banks_only                    |                                                                                                                                                                                                                                                                                                                                                    |
|                                                     | <p>The list of flops for XorGating (only use custom banks)</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                       |
| optimize_constant_0_flops                           |                                                                                                                                                                                                                                                                                                                                                    |

|                                    |                                                                                                                                                                                                                                          |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Allow constant 0 propagation through flip-flops.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">optimize_constant_0_flops</a></p>                  |
| optimize_constant_1_flops          |                                                                                                                                                                                                                                          |
|                                    | <p>Allow constant 1 propagation through flip-flops.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">optimize_constant_1_flops</a></p>                  |
| optimize_constant_across_preserved |                                                                                                                                                                                                                                          |
|                                    | <p>Enable constant optimization across preserved instances.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">optimize_constant_across_preserved</a></p> |
| optimize_constant_feedback_seqs    |                                                                                                                                                                                                                                          |
|                                    | <p>Allow optimization of uninitialized feedback flip-flops.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">optimize_constant_feedback_seqs</a></p>    |
| optimize_constant_latches          |                                                                                                                                                                                                                                          |
|                                    | <p>Allow constant propagation through latches.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">optimize_constant_latches</a></p>                       |
| optimize_merge_flops               |                                                                                                                                                                                                                                          |
|                                    | <p>Allow merging of equivalent flip-flops.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">optimize_merge_flops</a></p>                                |
| optimize_merge_latches             |                                                                                                                                                                                                                                          |

|                            |                                                                                                                                                                                                                                                                     |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>Allow merging of equivalent latches.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">optimize_merge_latches</a></p>                                                            |
| optimize_seq_x_to          |                                                                                                                                                                                                                                                                     |
|                            | <p>Value that should be used to optimize constant X sequential instances.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 0 1 x</p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">optimize_seq_x_to</a></p> |
| optimize_yield             |                                                                                                                                                                                                                                                                     |
|                            | <p>Optimize design for yield.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">optimize_yield</a></p>                                                                             |
| override_library_max_drc   |                                                                                                                                                                                                                                                                     |
|                            | <p>Allow user to override design rule constraints set on library.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">override_library_max_drc</a></p>                               |
| package_objects            |                                                                                                                                                                                                                                                                     |
|                            | <p>Short-cut for [get_db current_design .package_objects]</p> <p><b>Type:</b> <a href="#">obj(package_object)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                             |
| parse_lib_moments_table    |                                                                                                                                                                                                                                                                     |
|                            | <p>Parse library moments table while loading library file.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">parse_lib_moments_table</a></p>                                       |
| partition_floorplan_export |                                                                                                                                                                                                                                                                     |

|                                   |                                                                                                                                                                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Export setting</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> macro special_route pin_constraint</p> <p><b>Default:</b> special_route</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">partition_floorplan_export</a></p>        |
| partition_floorplan_import        |                                                                                                                                                                                                                                                                      |
|                                   | <p>Import setting</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> macro special_route pin_constraint</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">partition_floorplan_import</a></p>                   |
| partitions                        |                                                                                                                                                                                                                                                                      |
|                                   | <p>Short-cut for [get_db current_design .partitions]</p> <p><b>Type:</b> <a href="#">obj(partition)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                        |
| pbs_db_directory                  |                                                                                                                                                                                                                                                                      |
|                                   | <p>Specifies the directory where partition based synthesis can write temporary files.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> .</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">pbs_db_directory</a></p>                      |
| pbs_mim_flow                      |                                                                                                                                                                                                                                                                      |
|                                   | <p>Use folded hierarchies for partition based synthesis flow.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">pbs_mim_flow</a></p>                                                |
| percent_control_tolerance_for_map |                                                                                                                                                                                                                                                                      |
|                                   | <p>Tolerance to add for tall and short limits for mapping (does not apply to doubles).</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 10.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">percent_control_tolerance_for_map</a></p> |
| pg_base_pins                      |                                                                                                                                                                                                                                                                      |

|                      |                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>All the base_pins in all the base_cells.</p> <p><b>Type:</b> <a href="#">obj(pg_base_pin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                      |
| pg_nets              |                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>Short-cut for [get_db current_design .pg_nets]. List of power/ground nets in the design. This include physical-only PG nets, and Verilog supply0/supply1 nets. Note that Verilog nets as: to 1'b0/1'b1 are returned by the nets attribute, not the pg_nets attribute.</p> <p><b>Type:</b> <a href="#">obj(net)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| pg_pins              |                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>Short-cut for [get_db current_design .pg_pins].</p> <p><b>Type:</b> <a href="#">obj(pg_pin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                    |
| pg_ports             |                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>Short-cut for [get_db current_design .ports]</p> <p><b>Type:</b> <a href="#">obj(port)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                         |
| phys_early_skew_flow |                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>Enable early clock flow based early skewing.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">phys_early_skew_flow</a></p>                                                                                                                                                             |
| phys_insts           |                                                                                                                                                                                                                                                                                                                                                                             |
|                      | <p>Short-cut for [get_db current_design .phys_insts]</p> <p><b>Type:</b> <a href="#">obj(inst)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                    |
| pin_blockages        |                                                                                                                                                                                                                                                                                                                                                                             |

|                                |                                                                                                                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | Short-cut for [get_db current_design .pin_blockages]<br><b>Type:</b> <a href="#">obj(pin_blockage)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                     |
| pin_groups                     |                                                                                                                                                                                      |
|                                | Short-cut for [get_db current_design .pin_groups]<br><b>Type:</b> <a href="#">obj(pin_group)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                           |
| pin_guides                     |                                                                                                                                                                                      |
|                                | Short-cut for [get_db current_design .pin_guides]<br><b>Type:</b> <a href="#">obj(pin_guide)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                           |
| pins                           |                                                                                                                                                                                      |
|                                | Short-cut for [get_db current_design .pins]<br><b>Type:</b> <a href="#">obj(pin)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                       |
| place_3d_use_existing_bump_tsv |                                                                                                                                                                                      |
|                                | reuse pre-placed bumps or TSVs<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_3d_use_existing_bump_tsv</a> |
| place_blockages                |                                                                                                                                                                                      |
|                                | Short-cut for [get_db current_design .place_blockages]<br><b>Type:</b> <a href="#">obj(place_blockage)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                 |
| place_cell_edge_spacing        |                                                                                                                                                                                      |

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>List of cell edge-type spacing values from LEF CELLEDGESPACINGTABLE or OA techfile that affects the placer. It specifies the min-spacing between different edge-types of std-cells, in a format: {cell_edge_type1 cell_edge_type2 spacing}. The edge-type names match the base right_edge_type and left_edge_type attribute values.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No<br/> <b>Reference:</b> <a href="#">place_cell_edge_spacing</a></p> |
| place_check_report_fixed_inst_on_blockage |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                           | <p>check_place report fixed insts on placement blockage as violations</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                               |
| place_design_enable_3d                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                           | <p>enable 3d placement</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">place_design_enable_3d</a></p>                                                                                                                                                                                                                                                                                                                |
| place_design_floorplan_mode               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                           | <p>run placement in floorplan mode</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">place_design_floorplan_mode</a></p>                                                                                                                                                                                                                                                                                               |
| place_design_integrity_ir_fix_effort      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                           | <p>set power integrity placement IR fixing effort</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> low high<br/> <b>Default:</b> low<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">place_design_integrity_ir_fix_effort</a></p>                                                                                                                                                                                                                                       |
| place_design_refine_macro                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                           | <p>specify whether to run refine_macro_place after place_design -concurrent_macro</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">place_design_refine_macro</a></p>                                                                                                                                                                                                                                                  |

|                                                 |                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| place_design_refine_place                       |                                                                                                                                                                                                                                                                                        |
|                                                 | <p>if set to false, calls to refinePlace from other apps (such as place_opt_design, opt_design) to run refinePlace</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_design_refine_place</a></p> |
| place_detail_activity_power_driven              |                                                                                                                                                                                                                                                                                        |
|                                                 | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_activity_power_driven</a></p>                                                                                                              |
| place_detail_allow_border_pin_abut              |                                                                                                                                                                                                                                                                                        |
|                                                 | <p>allow instance abut with its neighbors when it has pins near by the cell boundary</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_allow_border_pin_abut</a></p>                     |
| place_detail_allow_single_height_row_symmetry_x |                                                                                                                                                                                                                                                                                        |
|                                                 | <p>allow single height row to be symmetryX</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_allow_single_height_row_symmetry_x</a></p>                                                   |
| place_detail_check_cut_spacing                  |                                                                                                                                                                                                                                                                                        |
|                                                 | <p>Check via cut spacing during DRC checking</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_check_cut_spacing</a></p>                                                                 |
| place_detail_check_inst_space_group             |                                                                                                                                                                                                                                                                                        |
|                                                 | <p>check inst space group</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_check_inst_space_group</a></p>                                                                               |



|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| place_detail_check_route         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                  | <p>do DRV checks against FIXED wires during legalization, along with preroutes</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_check_route</a></p>                                                                                                                                                                                                               |
| place_detail_color_aware_legal   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                  | <p>enable adjacent cell color conflict check in legalization</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_color_aware_legal</a></p>                                                                                                                                                                                                                           |
| place_detail_context_aware_legal |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                  | <p>specify the types of cell context rules the placer needs to honor. The argument values [optic required, user ignore_soft] can be used separately or combined, but they are exclusive to a none (default=all)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none all optional required user ignore_soft</p> <p><b>Default:</b> all</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_context_aware_legal</a></p> |
| place_detail_eco_max_distance    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                  | <p>specify max distance (in micron) for refinePlace ECO mode</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_eco_max_distance</a></p>                                                                                                                                                                                                                            |
| place_detail_eco_priority_insts  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                  | <p>select instance priority for refinePlace ECO mode</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> placed fixed eco adaptive</p> <p><b>Default:</b> placed</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_eco_priority_insts</a></p>                                                                                                                                                                            |
| place_detail_fixed_shifter       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                          |                                                                                                                                                                                                                                                                                                |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>mark shifters FIXED once placed</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_fixed_shifter</a></p>                                                                                       |
| place_detail_honor_inst_pad              |                                                                                                                                                                                                                                                                                                |
|                                          | <p>honor padding from set_inst_padding in detail placement</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_honor_inst_pad</a></p>                                                              |
| place_detail_io_pin_blockage             |                                                                                                                                                                                                                                                                                                |
|                                          | <p>ioPins from top-level are treated as pre-routes during DRV checks in legalization</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_io_pin_blockage</a></p>                                   |
| place_detail_ir_aware_max_drive_strength |                                                                                                                                                                                                                                                                                                |
|                                          | <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_ir_aware_max_drive_strength</a></p>                                                                                                                |
| place_detail_irdrop_aware_timing_effort  |                                                                                                                                                                                                                                                                                                |
|                                          | <p>selects which timing effort level to use during refinePlace</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none standard high</p> <p><b>Default:</b> high</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_irdrop_aware_timing_effort</a></p> |
| place_detail_irdrop_region_number        |                                                                                                                                                                                                                                                                                                |
|                                          | <p>value for debug_irdrop -nregion in native API</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_irdrop_region_number</a></p>                                                                     |
| place_detail_legalization_inst_gap       |                                                                                                                                                                                                                                                                                                |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>minimum gap between instances (unit sites)</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_legalization_inst_gap</a></p>                                                                                                                                                                                                      |
| place_detail_m3_stripe_push_down      |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                       | <p>Width threshold in site-units for 'virtual' push-down of M3 stripe to M2 for M1 pin-access. Value means all M3 stripes will be pushed-down.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> -1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_m3_stripe_push_down</a></p>                                                                                                      |
| place_detail_m3_stripe_shrink         |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                       | <p>Shrink the virtual M2 stripe (see spgM3StripePushDown) by given total site units, splitting v each side, starting on left. e.g. value of 1 shrinks by 1 site on left, 2 shrinks by 1 site on each value of 3 shrinks 2 sites on left and 1 on right.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_m3_stripe_shrink</a></p> |
| place_detail_max_shifter_column_depth |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                       | <p>maximum distance from vertical power domain boundary that a shifter can be placed</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 9999.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_max_shifter_column_depth</a></p>                                                                                                                                                    |
| place_detail_max_shifter_depth        |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                       | <p>maximum distance from a horizontal or vertical power domain boundary that a shifter can be placed</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 9999.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_max_shifter_depth</a></p>                                                                                                                                           |
| place_detail_max_shifter_row_depth    |                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                        |                                                                                                                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>maximum distance from a horizontal power domain bounary that a shifter can be placed</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 9999.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_max_shifter_row_depth</a></p> |
| place_detail_no_filler_without_implant |                                                                                                                                                                                                                                                                          |
|                                        | <p>there is no cell which has no implant obs</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_no_filler_without_implant</a></p>                                           |
| place_detail_pad_fixed_insts           |                                                                                                                                                                                                                                                                          |
|                                        | <p>honor cell-padding for FIXED instances in refinePlace</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_pad_fixed_insts</a></p>                                         |
| place_detail_pad_physical_cells        |                                                                                                                                                                                                                                                                          |
|                                        | <p>padding overlap with physcial cells would be honored</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_pad_physical_cells</a></p>                                       |
| place_detail_pgftv_insertion_cell_list |                                                                                                                                                                                                                                                                          |
|                                        | <p>Specify list of cells as PG FTV cells</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_pgftv_insertion_cell_list</a></p>                                                |
| place_detail_preroute_as_obs           |                                                                                                                                                                                                                                                                          |
|                                        | <p>specify layers on which preroute are taken as OBS</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_preroute_as_obs</a></p>                                              |
| place_detail_preserve_routing          |                                                                                                                                                                                                                                                                          |

|                                               |                                                                                                                                                                                                                                                        |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | do not delete routed wires during refinePlace<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_detail_preserve_routing</a>                                                      |
| place_detail_remove_affected_routing          |                                                                                                                                                                                                                                                        |
|                                               | delete only wires whose nets were touched due to moved cells<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_detail_remove_affected_routing</a>                               |
| place_detail_swap_eeq_cells                   |                                                                                                                                                                                                                                                        |
|                                               | replace master cells by its EEQ cell during legalization, to keep max-dist move low<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_detail_swap_eeq_cells</a>                 |
| place_detail_use_check_drc                    |                                                                                                                                                                                                                                                        |
|                                               | Use FGC based DRC engine in legalization<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_detail_use_check_drc</a>                                                             |
| place_detail_use_diffusion_transition_fill    |                                                                                                                                                                                                                                                        |
|                                               | insert OD transition filler in 1X gap<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_detail_use_diffusion_transition_fill</a>                                                |
| place_detail_use_gate_array_filler_groups     |                                                                                                                                                                                                                                                        |
|                                               | filler cell insts will only be replaced by logic insts that belong to same GA groups.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_detail_use_gate_array_filler_groups</a> |
| place_detail_use_no_diffusion_one_site_filler |                                                                                                                                                                                                                                                        |

|                                           |                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>override -fillerGapMinGap to 0 and all source-drain spacing rule to 1 site during legalizing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_use_no_diffusion_one_site_filler</a></p>                     |
| place_detail_wire_length_opt_effort       |                                                                                                                                                                                                                                                                                                              |
|                                           | <p>selects which wire-length optimization effort level to use during refinePlace</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none medium high</p> <p><b>Default:</b> medium</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_detail_wire_length_opt_effort</a></p> |
| place_global_activity_power_driven        |                                                                                                                                                                                                                                                                                                              |
|                                           | <p>identifies and constrains power-critical nets to reduce switching power</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_activity_power_driven</a></p>               |
| place_global_activity_power_driven_effort |                                                                                                                                                                                                                                                                                                              |
|                                           | <p>-place_global_activity_power_driven_effort mode</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> standard high</p> <p><b>Default:</b> standard</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_activity_power_driven_effort</a></p>                          |
| place_global_adaptive_subd1               |                                                                                                                                                                                                                                                                                                              |
|                                           | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                |
| place_global_align_macro                  |                                                                                                                                                                                                                                                                                                              |
|                                           | <p>enable mixed-placer to align nearby same-size macros during global placement</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_align_macro</a></p>                                                           |

|                                        |                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| place_global_allow_3d_stack            |                                                                                                                                                                                                                                                                                                 |
|                                        | <b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_global_allow_3d_stack</a>                                                                                                                                                 |
| place_global_auto_blockage_in_channel  |                                                                                                                                                                                                                                                                                                 |
|                                        | Placement will (temporarily) block channels between areas with limited routing capacity<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> none soft partial<br><b>Default:</b> partial<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_global_auto_blockage_in_channel</a> |
| place_global_clock_gate_aware          |                                                                                                                                                                                                                                                                                                 |
|                                        | find better placement for clock gating elements towards the center of gravity for fanout<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_global_clock_gate_aware</a>                                                    |
| place_global_clock_power_driven        |                                                                                                                                                                                                                                                                                                 |
|                                        | clock power driven<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_global_clock_power_driven</a>                                                                                                                        |
| place_global_clock_power_driven_effort |                                                                                                                                                                                                                                                                                                 |
|                                        | -place_global_clock_power_driven_effort mode<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> low standard high<br><b>Default:</b> low<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">place_global_clock_power_driven_effort</a>                                               |
| place_global_cong_effort               |                                                                                                                                                                                                                                                                                                 |

|                                            |                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>level of effort for congestion driven global placer</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high extreme auto</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_cong_effort</a></p> |
| place_global_cpg_effort                    |                                                                                                                                                                                                                                                                                   |
|                                            | <p>level of effort for CPG</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high extreme</p> <p><b>Default:</b> low</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_cpg_effort</a></p>                                    |
| place_global_cpg_file                      |                                                                                                                                                                                                                                                                                   |
|                                            | <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_cpg_file</a></p>                                                                                                                       |
| place_global_enable_advanced_pipeline      |                                                                                                                                                                                                                                                                                   |
|                                            | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_enable_advanced_pipeline</a></p>                                                                                                      |
| place_global_enable_distributed_place      |                                                                                                                                                                                                                                                                                   |
|                                            | <p>enable distributed placement platform</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_enable_distributed_place</a></p>                                                         |
| place_global_ignore_low_effort_path_groups |                                                                                                                                                                                                                                                                                   |
|                                            | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_ignore_low_effort_path_groups</a></p>                                                                                                 |



|                                 |                                                                                                                                                                                                                                                                    |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| place_global_ignore_scan        |                                                                                                                                                                                                                                                                    |
|                                 | <p>ignore scan net during placement</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true 1 false 0 auto</p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_ignore_scan</a></p>              |
| place_global_ignore_spare       |                                                                                                                                                                                                                                                                    |
|                                 | <p>discard spare cell connections during global placement</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_ignore_spare</a></p>                                     |
| place_global_max_density        |                                                                                                                                                                                                                                                                    |
|                                 | <p>placement strives to not let density exceed given value, in any part of design</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_max_density</a></p>             |
| place_global_module_aware_spare |                                                                                                                                                                                                                                                                    |
|                                 | <p>Spare insts are placed randomly along with the containing module's insts</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_module_aware_spare</a></p>             |
| place_global_module_padding     |                                                                                                                                                                                                                                                                    |
|                                 | <p>define the padding factor for the given module to reduce the local density and congestion</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_module_padding</a></p> |
| place_global_place_io_pins      |                                                                                                                                                                                                                                                                    |
|                                 | <p>place IO Pins concurrently with std.cell placement, and do layer assignment</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_place_io_pins</a></p>               |

|                                  |                                                                                                                                                                                                                                                                                           |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| place_global_reorder_scan        |                                                                                                                                                                                                                                                                                           |
|                                  | <p>turn on reorder scan during placement</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_reorder_scan</a></p>                                                                              |
| place_global_soft_guide_strength |                                                                                                                                                                                                                                                                                           |
|                                  | <p>level of effort for user defined softGuide for global placer</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> low</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_soft_guide_strength</a></p>      |
| place_global_timing_effort       |                                                                                                                                                                                                                                                                                           |
|                                  | <p>level of effort for timing driven global placer</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> medium high</p> <p><b>Default:</b> medium</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_timing_effort</a></p>                          |
| place_global_uniform_density     |                                                                                                                                                                                                                                                                                           |
|                                  | <p>enable even cell distribution for designs with less than 70% utilization</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_global_uniform_density</a></p> |
| place_hard_fence                 |                                                                                                                                                                                                                                                                                           |
|                                  | <p>honor fence and region constraints in refinePlace</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_hard_fence</a></p>                                                                           |
| place_hierarchical_flow          |                                                                                                                                                                                                                                                                                           |

|                               |                                                                                                                                                                                                                                                                             |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>enable hierarchical pin assignment flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_hierarchical_flow</a></p>                                                               |
| place_opt_post_place_tcl      |                                                                                                                                                                                                                                                                             |
|                               | <p>a tcl script to be sourced after initial placement and before preCTS optimization in place_opt_design</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_opt_post_place_tcl</a></p> |
| place_opt_run_global_place    |                                                                                                                                                                                                                                                                             |
|                               | <p>place opt run global place</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none seed full</p> <p><b>Default:</b> full</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_opt_run_global_place</a></p>                                |
| place_sdp_clock_location      |                                                                                                                                                                                                                                                                             |
|                               | <p>ICG cell location</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> center</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_sdp_clock_location</a></p>                                                                                 |
| place_sdp_disable_extend_core |                                                                                                                                                                                                                                                                             |
|                               | <p>disable the extended core flag</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_sdp_disable_extend_core</a></p>                                                                   |
| place_sdp_honor_dont_use      |                                                                                                                                                                                                                                                                             |
|                               | <p>honor dont_use on base_cells during EEQ swapping in SDP legalization</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_sdp_honor_dont_use</a></p>                                 |

|                               |                                                                                                                                                                                                                                                                                     |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| place_sdp_honor_orient        |                                                                                                                                                                                                                                                                                     |
|                               | <p>honor user specified orient during legalization</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_sdp_honor_orient</a></p>                                                                |
| place_sdp_legalization        |                                                                                                                                                                                                                                                                                     |
|                               | <p>Specify fixed corner of sdp box during legalizing sdp instance</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> NONE SW SE NW NE AUTO</p> <p><b>Default:</b> NONE</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_sdp_legalization</a></p> |
| place_sdp_legalization_effort |                                                                                                                                                                                                                                                                                     |
|                               | <p>Specifies the legalization effort in placing SDP groups</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> low</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_sdp_legalization_effort</a></p>        |
| place_sdp_max_move_distance   |                                                                                                                                                                                                                                                                                     |
|                               | <p>max sdp group move distance, default is half perimeter of the core box</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_sdp_max_move_distance</a></p>                                    |
| place_sdp_num_column          |                                                                                                                                                                                                                                                                                     |
|                               | <p>number of column for sdp group</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 2</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_sdp_num_column</a></p>                                                                                        |
| place_sdp_place_report        |                                                                                                                                                                                                                                                                                     |

|                                              |                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | <p>Output a detailed sdp placement data. Use empty string to reset to no output</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_sdp_place_report</a></p>                                                                |
| place_sdp_pre_fixed_cells_blockage_direction |                                                                                                                                                                                                                                                                                                                 |
|                                              | <p>Specify the direction when covering preplaced std cells using placement blockages</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> NONE X Y</p> <p><b>Default:</b> NONE</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_sdp_pre_fixed_cells_blockage_direction</a></p> |
| place_spare_update_timing_graph              |                                                                                                                                                                                                                                                                                                                 |
|                                              | <p>update timing graph during place_spare_modules</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">place_spare_update_timing_graph</a></p>                                                                                     |
| plan_3dic_power_map_unit                     |                                                                                                                                                                                                                                                                                                                 |
|                                              | <p>power unit for power map file</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> mw</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                               |
| plan_design_domain                           |                                                                                                                                                                                                                                                                                                                 |
|                                              | <p>Format: -domain {{PD1 &lt;value1&gt; net1} {PD2 &lt;value2&gt; net2}..}</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">plan_design_domain</a></p>                                                                         |
| plan_design_rail_model                       |                                                                                                                                                                                                                                                                                                                 |
|                                              | <p>Select rail model: virtual/existing.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> virtual existing</p> <p><b>Default:</b> existing</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">plan_design_rail_model</a></p>                                                        |
| plan_design_total_power                      |                                                                                                                                                                                                                                                                                                                 |

|                                          |                                                                                                                                                                                      |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | total power defined by user.<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">plan_design_total_power</a>          |
| planner_exclusive_one_level_logics_cells |                                                                                                                                                                                      |
|                                          | Specifies cells to keep full netlist instead of one-level logics when -keep_one_level_logics is on.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| planner_keep_one_level_logics            |                                                                                                                                                                                      |
|                                          | Specifies the option to keep one level logics<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                      |
| planner_keep_one_level_logics_cells      |                                                                                                                                                                                      |
|                                          | Specifies cells to keep one-level logics when -keep_one_level_logics is turned on.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> *<br><b>Edit:</b> Yes                   |
| pmbist_enable_multiple_views             |                                                                                                                                                                                      |
|                                          | Check whether support for multiple views of the same memory is requested or not.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                   |
| pmbist_full_async_reset                  |                                                                                                                                                                                      |
|                                          | Allow all PMBIST logic to have full asynchronous reset.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes                                             |
| port_busses                              |                                                                                                                                                                                      |
|                                          | Short-cut for [get_db current_design .port_busses]<br><b>Type:</b> <a href="#">obj(port_bus)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                           |

|                                           |                                                                                                                                                                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| port_shapes                               |                                                                                                                                                                                                                                                                   |
|                                           | Short-cut for [get_db current_design .port_shapes]<br><b>Type:</b> <a href="#">obj(port_shape)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                      |
| ports                                     |                                                                                                                                                                                                                                                                   |
|                                           | Short-cut for [get_db current_design .ports]<br><b>Type:</b> <a href="#">obj(port)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                  |
| power_add_simulation                      |                                                                                                                                                                                                                                                                   |
|                                           | Add simulation for design coverage<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_add_simulation</a>                                                                                    |
| power_adjust_input_activity_in_iterations |                                                                                                                                                                                                                                                                   |
|                                           | Adjust input activity in state propagation.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_adjust_input_activity_in_iterations</a>                                                       |
| power_adjust_macro_activity_in_iterations |                                                                                                                                                                                                                                                                   |
|                                           | Adjust macro activity in state propagation.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_adjust_macro_activity_in_iterations</a>                                                       |
| power_analysis_temperature                |                                                                                                                                                                                                                                                                   |
|                                           | Sets the temperature for static power calculation. If not set, then uses the current opcond temperature<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> -274.000000<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_analysis_temperature</a> |
| power_annotation_detail_report            |                                                                                                                                                                                                                                                                   |

|                                   |                                                                                                                                                                                                                                                                         |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Generate detailed annotation report showing objects in their specific type blocks.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_annotation_detail_report</a></p>         |
| power_auto_twf_delay_annotation   |                                                                                                                                                                                                                                                                         |
|                                   | <p>Auto detect zero delay vectors</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_auto_twf_delay_annotation</a></p>                                                            |
| power_average_rise_fall_cap       |                                                                                                                                                                                                                                                                         |
|                                   | <p>When set to true, the software uses the average of rise and fall capacitance from the liberty</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_average_rise_fall_cap</a></p> |
| power_block_independent_peakpower |                                                                                                                                                                                                                                                                         |
|                                   | <p>Power profiling done independently for each vector, design block</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_block_independent_peakpower</a></p>                        |
| power_bulk_pins                   |                                                                                                                                                                                                                                                                         |
|                                   | <p>Defines power and ground bulk LEF pins for the design.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_bulk_pins</a></p>                                                     |
| power_capacity                    |                                                                                                                                                                                                                                                                         |
|                                   | <p>Specifies the segment count for profiler</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> low</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_capacity</a></p>                          |
| power_case_insensitive_mapping    |                                                                                                                                                                                                                                                                         |



|                                 |                                                                                                                                                                                                                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>Case insensitive auto-mapping, ignoring the case(upper or lower) of a character while matching with design names.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_case_insensitive_mapping</a></p> |
| power_clock_source_as_clock     |                                                                                                                                                                                                                                                                                                |
|                                 | <p>Specifies to use the correct clock frequency for activity calculation.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_clock_source_as_clock</a></p>                                               |
| power_comprehensive_automapping |                                                                                                                                                                                                                                                                                                |
|                                 | <p>Comprehensively perform instance name mapping between the RTL netlist and GATE level.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_comprehensive_automapping</a></p>                            |
| power_compress_reports          |                                                                                                                                                                                                                                                                                                |
|                                 | <p>choose whether to compress power reports and set compression ratio</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_compress_reports</a></p>                                                      |
| power_constant_override         |                                                                                                                                                                                                                                                                                                |
|                                 | <p>Specifies to give higher precedence to propagated constants defined using the set_case_expression command.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_constant_override</a></p>               |
| power_corner                    |                                                                                                                                                                                                                                                                                                |
|                                 | <p>Defines the library corner (for non-MMMC setup).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_corner</a></p>                                                                                     |

|                                 |                                                                                                                                                                                                                                                             |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_create_driver_db          |                                                                                                                                                                                                                                                             |
|                                 | <p>Specifies to determine all the drivers/receivers on the ground net.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_create_driver_db</a></p>                    |
| power_current_generation_method |                                                                                                                                                                                                                                                             |
|                                 | <p>choose avg or peak current generation method</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> avg</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_current_generation_method</a></p>                                  |
| power_db_name                   |                                                                                                                                                                                                                                                             |
|                                 | <p>Specifies the name of the binary power database file ( filename.db ).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_db_name</a></p>                            |
| power_debug_instance            |                                                                                                                                                                                                                                                             |
|                                 | <p>print debug messages for instances</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                      |
| power_decap_cell_list           |                                                                                                                                                                                                                                                             |
|                                 | <p>Specifies the physical only cells, such as decap cells, to include in the power report.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_decap_cell_list</a></p>  |
| power_default_frequency         |                                                                                                                                                                                                                                                             |
|                                 | <p>Specifies to set the default frequency in the MHz unit for net not annotated by TWF.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_default_frequency</a></p> |
| power_default_slew              |                                                                                                                                                                                                                                                             |

|                                   |                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Specifies to set the default slew in the ns unit for net not annotated by TWF.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_default_slew</a></p>                                                                                        |
| power_default_supply_voltage      |                                                                                                                                                                                                                                                                                                                                       |
|                                   | <p>Specifies to set a default voltage for power nets in a scenario where the power engine can't determine the voltage of these nets.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_default_supply_voltage</a></p>                           |
| power_detailed_view_current_only  |                                                                                                                                                                                                                                                                                                                                       |
|                                   | <p>Ignore switching current for instances with detailed view</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_detailed_view_current_only</a></p>                                                                                               |
| power_disable_clock_gate_clipping |                                                                                                                                                                                                                                                                                                                                       |
|                                   | <p>Specifies to enable clock gate output clipping so that the output transition density of an ICG does not exceed the input clock pin transition density.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_disable_clock_gate_clipping</a></p> |
| power_disable_leakage_scaling     |                                                                                                                                                                                                                                                                                                                                       |
|                                   | <p>Specifies to exclude leakage power during scaling factor computation when target power is specified using the set_power command.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_disable_leakage_scaling</a></p>                          |
| power_disable_static              |                                                                                                                                                                                                                                                                                                                                       |

|                                         |                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>When set to true in the dynamic vector-based or dynamic vectorless flows, the command pe only dynamic analysis and turns-off static power calculation.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">power_disable_static</a></p> |
| power_display_negative_internal_power   |                                                                                                                                                                                                                                                                                                                     |
|                                         | <p>For displaying negative Internal power results.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">power_display_negative_internal_power</a></p>                                                                                        |
| power_distribute_switching_power        |                                                                                                                                                                                                                                                                                                                     |
|                                         | <p>distribute switching power to all connected I/O instances if the net does not have a driver in</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">power_distribute_switching_power</a></p>                                             |
| power_distributed_combine_report_format |                                                                                                                                                                                                                                                                                                                     |
|                                         | <p>control which power report files need to be combined</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> reduced<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">power_distributed_combine_report_format</a></p>                                                                            |
| power_distributed_setup                 |                                                                                                                                                                                                                                                                                                                     |
|                                         | <p>Specifies a file containing the customized setup details for running power analysis in the distributed mode.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">power_distributed_setup</a></p>                                         |
| power_domain_based_clipping             |                                                                                                                                                                                                                                                                                                                     |
|                                         | <p>Determines the frequency to be used to clip propagated activity that is too high.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">power_domain_based_clipping</a></p>                                                               |

|                                          |                                                                                                                                                                                                                                                                                |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_domains                            |                                                                                                                                                                                                                                                                                |
|                                          | <p>Short-cut for [get_db current_design .power_domains]</p> <p><b>Type:</b> <a href="#">obj(power_domain)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                            |
| power_dynamic_glitch_filter              |                                                                                                                                                                                                                                                                                |
|                                          | <p>set absolute time threshold value (in ns) for dynamic glitch filter</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_dynamic_glitch_filter</a></p>                                 |
| power_dynamic_scale_clock_by_frequency   |                                                                                                                                                                                                                                                                                |
|                                          | <p>set clock frequency scaling ratio by frequency value</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_dynamic_scale_clock_by_frequency</a></p>                                       |
| power_dynamic_scale_clock_by_name        |                                                                                                                                                                                                                                                                                |
|                                          | <p>set clock frequency scaling ratio by clock name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_dynamic_scale_clock_by_name</a></p>                                                 |
| power_dynamic_vectorless_ranking_methods |                                                                                                                                                                                                                                                                                |
|                                          | <p>choose ranking options from: load, slack, clock and vector_activity</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_dynamic_vectorless_ranking_methods</a></p>                      |
| power_enable_auto_mapping                |                                                                                                                                                                                                                                                                                |
|                                          | <p>Specifies to automatically perform instance name mapping between the RTL netlist and GA level netlist.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_auto_mapping</a></p> |

|                                                      |                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_enable_auto_queue                              |                                                                                                                                                                                                                                                                                                                                      |
|                                                      | <p>Automatically assign power job to proper queue</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_auto_queue</a></p>                                                                                                                 |
| power_enable_disk_mapping                            |                                                                                                                                                                                                                                                                                                                                      |
|                                                      | <p>Enable disk caching of toggles.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_disk_mapping</a></p>                                                                                                                              |
| power_enable_duty_propagation_with_global_activity   |                                                                                                                                                                                                                                                                                                                                      |
|                                                      | <p>Specifies to propagate duty cycle with the global switching activity setting for all data nets in dynamic vectorless analysis flow.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_duty_propagation_with_global_activity</a></p> |
| power_enable_dynamic_current_slew_load_interpolation |                                                                                                                                                                                                                                                                                                                                      |
|                                                      | <p>enable dynamic current slew load interpolation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_dynamic_current_slew_load_interpolation</a></p>                                                                                    |
| power_enable_dynamic_scaling                         |                                                                                                                                                                                                                                                                                                                                      |
|                                                      | <p>Enables set_power scaling for dynamic power analysis.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_dynamic_scaling</a></p>                                                                                                     |
| power_enable_flop_state_propagation                  |                                                                                                                                                                                                                                                                                                                                      |
|                                                      | <p>Enable flop state propagation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_flop_state_propagation</a></p>                                                                                                                      |

|                                   |                                                                                                                                                                                                                                                                        |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_enable_force_propagation    |                                                                                                                                                                                                                                                                        |
|                                   | <p>Enable force propagation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_force_propagation</a></p>                                                                  |
| power_enable_generated_clock      |                                                                                                                                                                                                                                                                        |
|                                   | <p>Specifies to get generated clock frequency during activity propagation for static power anal</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_generated_clock</a></p> |
| power_enable_input_net_power      |                                                                                                                                                                                                                                                                        |
|                                   | <p>Specifies to calculate the switching power of input nets.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_input_net_power</a></p>                                   |
| power_enable_interactive_reports  |                                                                                                                                                                                                                                                                        |
|                                   | <p>Enable voltus interactive reports</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_interactive_reports</a></p>                                                        |
| power_enable_mt_state_propagation |                                                                                                                                                                                                                                                                        |
|                                   | <p>Enable Multithreading in state propagation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_mt_state_propagation</a></p>                                              |
| power_enable_multithread_reports  |                                                                                                                                                                                                                                                                        |
|                                   | <p>to enable multithreaded power report generation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_multithread_reports</a></p>                                         |

|                                       |                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_enable_pba_for_tempus_pi        |                                                                                                                                                                                                                                                                                                    |
|                                       | <p>ensure path based timing analysis in tempus-pi flow.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_pba_for_tempus_pi</a></p>                                                                   |
| power_enable_power_target_flow        |                                                                                                                                                                                                                                                                                                    |
|                                       | <p>enable vectorless state propagation with power target</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_power_target_flow</a></p>                                                                 |
| power_enable_rtl_dynamic_vector_based |                                                                                                                                                                                                                                                                                                    |
|                                       | <p>Specifies to take an RTL or partial VCD/FSDB file as input and use that for dynamic vector-l flow.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_rtl_dynamic_vector_based</a></p>             |
| power_enable_scan_report              |                                                                                                                                                                                                                                                                                                    |
|                                       | <p>enable scan report automatical generation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_scan_report</a></p>                                                                                   |
| power_enable_single_cycle_scheduling  |                                                                                                                                                                                                                                                                                                    |
|                                       | <p>Stage-propagation based vectorless flow schedules new flops/macros every cycle to achie faster toggle coverage.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enable_single_cycle_scheduling</a></p> |
| power_enable_slew_based_ccs_pin_cap   |                                                                                                                                                                                                                                                                                                    |



|                                     |                                                                                                                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | honor ccs pin cap<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_enable_slew_based_ccs_pin_cap</a>                                                   |
| power_enable_stable_clock_gating    |                                                                                                                                                                                                                                |
|                                     | Use stable clock gate in vless flow<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_enable_stable_clock_gating</a>                                    |
| power_enable_stable_flop_scheduling |                                                                                                                                                                                                                                |
|                                     | Enable stable flop scheduling<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_enable_stable_flop_scheduling</a>                                       |
| power_enable_state_propagation      |                                                                                                                                                                                                                                |
|                                     | enable vectorless state propagation dynamic analysis<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_enable_state_propagation</a>                     |
| power_enable_tempus_pi              |                                                                                                                                                                                                                                |
|                                     | enable timing critical path analysis in state propagation based vectorless flow.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_enable_tempus_pi</a> |
| power_enable_xp                     |                                                                                                                                                                                                                                |
|                                     | standalone power analysis in voltus-XP<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_enable_xp</a>                                                  |
| power_engine                        |                                                                                                                                                                                                                                |

|                                 |                                                                                                                                                                                                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>For selecting legacy, joules power engine.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> legacy joules</p> <p><b>Default:</b> joules</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_engine</a></p>                         |
| power_enhanced_blackbox_avg     |                                                                                                                                                                                                                                                                         |
|                                 | <p>Specifies to use the average toggle rate of related inputs for enhanced blackbox propagation.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enhanced_blackbox_avg</a></p> |
| power_enhanced_blackbox_max     |                                                                                                                                                                                                                                                                         |
|                                 | <p>Specifies to use the maximum toggle rate of related inputs for enhanced blackbox propagation.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_enhanced_blackbox_max</a></p> |
| power_equivalent_annotation     |                                                                                                                                                                                                                                                                         |
|                                 | <p>Enable Qbar annotation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_equivalent_annotation</a></p>                                                                        |
| power_event_based_leakage_power |                                                                                                                                                                                                                                                                         |
|                                 | <p>Event based leakage power</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_event_based_leakage_power</a></p>                                                                 |
| power_external_load_config_file |                                                                                                                                                                                                                                                                         |
|                                 | <p>Allow user to provide external load file to set external load manually.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_external_load_config_file</a></p>                    |
| power_extraction_tech_file      |                                                                                                                                                                                                                                                                         |

|                                        |                                                                                                                                                                                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>technology file for exaction</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_extraction_tech_file</a></p>                                       |
| power_extractor_include                |                                                                                                                                                                                                                                            |
|                                        | <p>extraction include file</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_extractor_include</a></p>                                               |
| power_fanout_limit                     |                                                                                                                                                                                                                                            |
|                                        | <p>Specifies the fanout limit for nets.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> -1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_fanout_limit</a></p>                                          |
| power_force_library_merging            |                                                                                                                                                                                                                                            |
|                                        | <p>force library merging even if there is some error during merging</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_force_library_merging</a></p> |
| power_from_x_transition_factor         |                                                                                                                                                                                                                                            |
|                                        | <p>activity count for x to 0/1 transition</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.5</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_from_x_transition_factor</a></p>                        |
| power_from_z_transition_factor         |                                                                                                                                                                                                                                            |
|                                        | <p>activity count for z to 0/1 transition</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.25</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_from_z_transition_factor</a></p>                       |
| power_generate_activity_mapping_report |                                                                                                                                                                                                                                            |

|                                                              |                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                              | <p>turn on/off generation of activity_mapping.rpt</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_generate_activity_mapping_report</a></p>                                     |
| power_generate_current_for_rail                              |                                                                                                                                                                                                                                                                         |
|                                                              | <p>Generates current files for the specified rail.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_generate_current_for_rail</a></p>                                            |
| power_generate_flop_ranking_data                             |                                                                                                                                                                                                                                                                         |
|                                                              | <p>Specifies the name of the dir containing the ranking data.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_generate_flop_ranking_data</a></p>                                |
| power_generate_leakage_power_map_based_on_calculated_leakage |                                                                                                                                                                                                                                                                         |
|                                                              | <p>Print leakage power table based on calculated leakage power.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_generate_leakage_power_map_based_on_calculated_leakage</a></p> |
| power_generate_static_report_from_state_propagation          |                                                                                                                                                                                                                                                                         |
|                                                              | <p>generate static report from state propagation flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_generate_static_report_from_state_propagation</a></p>                    |
| power_get_state_dependent_leakage_power                      |                                                                                                                                                                                                                                                                         |
|                                                              | <p>To get state independent cell leakage power.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                                                                        |
| power_grid_libraries                                         |                                                                                                                                                                                                                                                                         |

|                                              |                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | <p>Specifies the name of the Cadence power cell libraries ( .cl ).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_grid_libraries</a></p>                                                                                                                                |
| power_handle_glitch                          |                                                                                                                                                                                                                                                                                                                                                                  |
|                                              | <p>A transition is defined as a glitch when the time difference between two sequential toggles than half of the rise transition time + fall transition time in the vector-based static power calc</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_handle_glitch</a></p> |
| power_handle_transport_glitch                |                                                                                                                                                                                                                                                                                                                                                                  |
|                                              | <p>Enable transport glitch support</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_handle_transport_glitch</a></p>                                                                                                                                                      |
| power_handle_tristate                        |                                                                                                                                                                                                                                                                                                                                                                  |
|                                              | <p>When set to true , all tristate device enable pin values will be taken into account when determining the propagation of the activity through tristate gates.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_handle_tristate</a></p>                                 |
| power_hier_delimiter                         |                                                                                                                                                                                                                                                                                                                                                                  |
|                                              | <p>Specifies the hierarchical delimiter in the DEF file. The default hierarchical delimiter is a forward slash ( / ) but can be changed by setting the -hier_delimiter parameter.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_hier_delimiter</a></p>                 |
| power_honor_combinational_logic_on_clock_net |                                                                                                                                                                                                                                                                                                                                                                  |
|                                              | <p>Specifies the behavior of combination logic on clock net.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_honor_combinational_logic_on_clock_net</a></p>                                                                                                              |

|                                                      |                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_honor_default_switching_activity_in_scheduling |                                                                                                                                                                                                                                                                                               |
|                                                      | <p>Enable default switching activity hier/domain baisted setup</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                              |
| power_honor_negative_energy                          |                                                                                                                                                                                                                                                                                               |
|                                                      | <p>A value of true specifies that Voltus will keep negative internal energy numbers from the .lit internal power table.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_honor_negative_energy</a></p> |
| power_honor_net_activity                             |                                                                                                                                                                                                                                                                                               |
|                                                      | <p>honor net activity from activity file</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_honor_net_activity</a></p>                                                                                   |
| power_honor_non_mission_leakage                      |                                                                                                                                                                                                                                                                                               |
|                                                      | <p>Option to activate Non-Mission leakage flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_honor_non_mission_leakage</a></p>                                                                     |
| power_hybrid_analysis                                |                                                                                                                                                                                                                                                                                               |
|                                                      | <p>Vbased + Vless runs done separately and results combined</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_hybrid_analysis</a></p>                                                                  |
| power_ignore_control_signals                         |                                                                                                                                                                                                                                                                                               |
|                                                      | <p>A value of true specifies that control signals will be ignored when propagating activity.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_ignore_control_signals</a></p>                           |

|                                                  |                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_ignore_data_phase_for_clock                |                                                                                                                                                                                                                                                                                                                            |
|                                                  | <b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_ignore_data_phase_for_clock</a>                                                                                                                                                                      |
| power_ignore_end_toggles_in_profile              |                                                                                                                                                                                                                                                                                                                            |
|                                                  | ignore toggles on end boundary in profiler analysis<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_ignore_end_toggles_in_profile</a>                                                                                                             |
| power_ignore_glitches_at_same_time_stamp         |                                                                                                                                                                                                                                                                                                                            |
|                                                  | Specifies to ignore glitches at the same transient time in the vector-based static and dynam power analysis flow.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_ignore_glitches_at_same_time_stamp</a>                                           |
| power_ignore_inout_pin_cap                       |                                                                                                                                                                                                                                                                                                                            |
|                                                  | When set to true , ignores the bidirectional pin capacitances ( direction : inout ) defined for I in the .lib file, when calculating switching power and internal power.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_ignore_inout_pin_cap</a> |
| power_ignore_macro_leakage_scale_for_temperature |                                                                                                                                                                                                                                                                                                                            |
|                                                  | Option for ignoring leakage power scaling on macro cell<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_ignore_macro_leakage_scale_for_temperature</a>                                                                                            |
| power_ignore_unconstraint_net_with_timing_data   |                                                                                                                                                                                                                                                                                                                            |

|                                             |                                                                                                                                                                                                                                                                                     |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>Enable ignore net/pin without twf time constraint when vless staeprop</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_ignore_unconstraint_net_with_timing_data</a></p>                  |
| power_include_file                          |                                                                                                                                                                                                                                                                                     |
|                                             | <p>Used for adding additional setting for power analysis</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_include_file</a></p>                                                               |
| power_include_initial_x_transitions         |                                                                                                                                                                                                                                                                                     |
|                                             | <p>Controls how the initial X state toggle should be counted.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_include_initial_x_transitions</a></p>                                         |
| power_include_sequential_clock_pin_power    |                                                                                                                                                                                                                                                                                     |
|                                             | <p>Reports the clock pin power of flip-flops as part of the clock network power.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_include_sequential_clock_pin_power</a></p>                |
| power_include_timing_in_current_file        |                                                                                                                                                                                                                                                                                     |
|                                             | <p>enable timing bit writing in current file</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_include_timing_in_current_file</a></p>                                                        |
| power_intent_add_pin_name_to_power_instance |                                                                                                                                                                                                                                                                                     |
|                                             | <p>Add interface pin name to the low power instance that commit_power_intent inserts for 180</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_add_pin_name_to_power_instance</a></p> |
| power_intent_allow_back_to_back_isolation   |                                                                                                                                                                                                                                                                                     |



|                                              |                                                                                                                                                                                                                                                                                   |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | <p>Supports back-to-back isolation insertion along cross-domain net based on power intent IS</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_allow_back_to_back_isolation</a></p> |
| power_intent_allow_dont_touch_cells          |                                                                                                                                                                                                                                                                                   |
|                                              | <p>Allow use of dont touch cells for commit_power_intent.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_allow_dont_touch_cells</a></p>                                          |
| power_intent_allow_dont_use_cells            |                                                                                                                                                                                                                                                                                   |
|                                              | <p>Allow use of avoided cells for commit_power_intent.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_allow_dont_use_cells</a></p>                                               |
| power_intent_allow_inverted_level_shifter    |                                                                                                                                                                                                                                                                                   |
|                                              | <p>Allow use of level shifters with inverter functionality.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_allow_inverted_level_shifter</a></p>                                   |
| power_intent_allow_nested_default_domain     |                                                                                                                                                                                                                                                                                   |
|                                              | <p>Allows a non-default power domain member to be logically nested in a default power domain</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_allow_nested_default_domain</a></p>   |
| power_intent_allow_power_domain_min_gap_zero |                                                                                                                                                                                                                                                                                   |
|                                              | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_allow_power_domain_min_gap_zero</a></p>                                                                                                |
| power_intent_allow_shifter_voltage_mismatch  |                                                                                                                                                                                                                                                                                   |

|                                                           |                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | <b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_intent_allow_shifter_voltage_mismatch</a>                                                                                                 |
| power_intent_assume_iso_enable_pin_is_always_on           |                                                                                                                                                                                                                                                                |
|                                                           | To mark all Isolation cell's enable pin as always on<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_intent_assume_iso_enable_pin_is_always_on</a>                                     |
| power_intent_check_all_nets_for_domain_crossing           |                                                                                                                                                                                                                                                                |
|                                                           | Checks all the nets for domain crossings<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_intent_check_all_nets_for_domain_crossing</a>                                                 |
| power_intent_disable_always_on_buffering                  |                                                                                                                                                                                                                                                                |
|                                                           | Disable always on buffering for some hierarchies where always on cell may not be available<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_intent_disable_always_on_buffering</a>     |
| power_intent_do_not_use_top_domain_for_port_voltage       |                                                                                                                                                                                                                                                                |
|                                                           | Controls voltage using top fterm domain for IO pin voltage<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_intent_do_not_use_top_domain_for_port_voltage</a>                          |
| power_intent_enable_third_domain_buffering                |                                                                                                                                                                                                                                                                |
|                                                           | Enable buffering in third domain on data to output path of isolation or level shifter cells.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_intent_enable_third_domain_buffering</a> |
| power_intent_honor_power_domain_for_domain_crossing_route |                                                                                                                                                                                                                                                                |

|                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                        | <p>to handle nets belong to multiple power domains</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_honor_power_domain_for_domain_crossing_route</a></p>                                                                                                                                                                                                                                                                                     |
| power_intent_honor_power_domain_for_intra_domain_route |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                        | <p>To route intra-power domain nets only within their power domains</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_honor_power_domain_for_intra_domain_route</a></p>                                                                                                                                                                                                                                                                       |
| power_intent_include_dot_lib_related_pg_pin            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                        | <p>Honors timing library related PG pins</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_include_dot_lib_related_pg_pin</a></p>                                                                                                                                                                                                                                                                                                             |
| power_intent_insert_auto_ls                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                        | <p>Enable capability to infer LS and to merge ISO into ELS depending on shifting requirement</p> <p>Possible values are :</p> <ul style="list-style-type: none"> <li>combo_ls : insert LS and merge existing ISOs to ELS if required</li> <li>ls_only : insert only LS</li> <li>none : do not insert auto-ls</li> </ul> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none combo_ls ls_only</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_insert_auto_ls</a></p> |
| power_intent_insert_on_switch_network                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                        | <p>Enable or disable insertion of isolation and level shifter cells on switch network.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_insert_on_switch_network</a></p>                                                                                                                                                                                                                                                                    |
| power_intent_invert_preserved_net                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                                |                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                | <p>Enable inversion on preserved nets for isolation insertion.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_invert_preserved_net</a></p>                                              |
| power_intent_iso_level_shifter_skip_constant_propagation_loads |                                                                                                                                                                                                                                                                                          |
|                                                                | <p>Type of loads of isolation or level shifters to skip constant propagation.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_iso_level_shifter_skip_constant_propagation_loads</a></p> |
| power_intent_iso_nor                                           |                                                                                                                                                                                                                                                                                          |
|                                                                | <p>Prefer isonor for cell selection.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_iso_nor</a></p>                                                                                     |
| power_intent_parse_error_on_missing_objects                    |                                                                                                                                                                                                                                                                                          |
|                                                                | <p>Treat missing objects as error.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_parse_error_on_missing_objects</a></p>                                                                 |
| power_intent_parse_honor_dont_use_cells                        |                                                                                                                                                                                                                                                                                          |
|                                                                | <p>Power intent parser to honor avoid attribute on library cell(s).</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_parse_honor_dont_use_cells</a></p>                                   |
| power_intent_read_enable_exhaustive_search                     |                                                                                                                                                                                                                                                                                          |
|                                                                | <p>Set this attribute to control the searches done in power intent parser.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_intent_read_enable_exhaustive_search</a></p>                        |
| power_intent_share_well_always_on_buffering_support            |                                                                                                                                                                                                                                                                                          |

|                                                             |                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | <b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_intent_share_well_always_on_buffering_support</a>                                                                                |
| power_intent_upf_insert_on_floating_pins                    |                                                                                                                                                                                                                                                       |
|                                                             | <b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_intent_upf_insert_on_floating_pins</a>                                                                                           |
| power_intent_use_cpf_global_connect_for_always_on_buffer    |                                                                                                                                                                                                                                                       |
|                                                             | Honors connection specs for AO buffer based on the power intent rules<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_intent_use_cpf_global_connect_for_always_on_buffer</a> |
| power_intent_use_cpf_global_connect_for_shifter             |                                                                                                                                                                                                                                                       |
|                                                             | Honors connection specs for shifters based on the CPF rules<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_intent_use_cpf_global_connect_for_shifter</a>                    |
| power_intent_use_effective_domain_for_iso_shifter_insertion |                                                                                                                                                                                                                                                       |
|                                                             | Honors effective domains for ISO/LS insertions<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_intent_use_effective_domain_for_iso_shifter_insertion</a>                      |
| power_ir_derated_timing_view                                |                                                                                                                                                                                                                                                       |
|                                                             | set timing view in tempus-pi flow.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">power_ir_derated_timing_view</a>                                                                 |
| power_keep_clock_gate_ratio_in_iterations                   |                                                                                                                                                                                                                                                       |

|                                            |                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Keep clock gate ratio in state propagation.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_keep_clock_gate_ratio_in_iterations</a></p>                                                |
| power_leakage_scale_factor_for_temperature |                                                                                                                                                                                                                                                                                    |
|                                            | <p>Performs a linear scaling of the leakage power for temperature in all libraries.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_leakage_scale_factor_for_temperature</a></p>          |
| power_lib_files                            |                                                                                                                                                                                                                                                                                    |
|                                            | <p>Specify the lib files to be used for power Analysis. Libs specified through this option are not during design loading</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_lib_files</a></p> |
| power_library_preference                   |                                                                                                                                                                                                                                                                                    |
|                                            | <p>option to choose instance binding to closest voltage or ecdm/ccsp library.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> voltage</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_library_preference</a></p>                              |
| power_macro_pgv_leakage                    |                                                                                                                                                                                                                                                                                    |
|                                            | <p>Use PG pin based leakage current from PGV instead of from liberty.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_macro_pgv_leakage</a></p>                                            |
| power_macro_toggle_pin_percentage          |                                                                                                                                                                                                                                                                                    |
|                                            | <p>Set percentage of the number of Macro bits to schedule</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_macro_toggle_pin_percentage</a></p>                                             |
| power_match_state_for_logic_x              |                                                                                                                                                                                                                                                                                    |

|                                     |                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Controls how the logic X will be evaluated in the boolean function of the 'when' state of a power table.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> x<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">power_match_state_for_logic_x</a></p>                                                  |
| power_memory_current_scaling_method |                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>NLPM Current Scaling Method for Memories [1/2/3]</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">power_memory_current_scaling_method</a></p>                                                                                                       |
| power_merge_switched_net_currents   |                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Specifies to merge switched net currents into always-on currents.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">power_merge_switched_net_currents</a></p>                                                                                   |
| power_method                        |                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Specifies the type of analysis to be performed.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> static dynamic dynamic_vectorless dynamic_vectorbased dynamic_mixed_vector_profile event_based<br/> <b>Default:</b> static<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">power_method</a></p> |
| power_min_leaf_count                |                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>min number of leaf instances count</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">power_min_leaf_count</a></p>                                                                                                                                    |
| power_modes                         |                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>List of the power mode names in the design.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                    |

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_multi_scenario_simulation           |                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                           | <p>Enable multi-scenario simulation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_multi_scenario_simulation</a></p>                                                                                                                                                                                  |
| power_multibit_flop_toggle_behavior       |                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                           | <p>select mbff toggle: simultaneous   independent(default)   sbff   pin_percentage during vecto state propagation dynamic analysis</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> simultaneous independent sbff pin_percentage</p> <p><b>Default:</b> independent</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_multibit_flop_toggle_behavior</a></p> |
| power_multibit_flop_toggle_pin_percentage |                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                           | <p>The percentage of MBFF bits to schedule</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_multibit_flop_toggle_pin_percentage</a></p>                                                                                                                                                                 |
| power_off_pg_bulk_leakage                 |                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                           | <p>Contribute the bulk pin leakage to the connected always on rail and zero leakage to the off</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_off_pg_bulk_leakage</a></p>                                                                                                                             |
| power_off_pg_nets                         |                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                           | <p>The name of the power rails that will be turned off.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_off_pg_nets</a></p>                                                                                                                                                                             |
| power_output_current_data_prefix          |                                                                                                                                                                                                                                                                                                                                                                                                 |



|                                   |                                                                                                                                                                                                                                                                              |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Specifies a prefix to the static or dynamic current files generated by the software.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_output_current_data_prefix</a></p>           |
| power_output_dir                  |                                                                                                                                                                                                                                                                              |
|                                   | <p>Used to save power engine output to the defined directory. If not set, will use the current working directory.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_output_dir</a></p> |
| power_partition_count             |                                                                                                                                                                                                                                                                              |
|                                   | <p>Number of physical partitions for distributed run.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_partition_count</a></p>                                                            |
| power_partition_spef              |                                                                                                                                                                                                                                                                              |
|                                   | <p>turn on/off spef partitioning in distributed flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                       |
| power_partition_twf               |                                                                                                                                                                                                                                                                              |
|                                   | <p>turn on/off twf partitioning in distributed flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_partition_twf</a></p>                                                           |
| power_pin_based_twf               |                                                                                                                                                                                                                                                                              |
|                                   | <p>turn on/off generation of pin based TWF in dynamic method</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_pin_based_twf</a></p>                                                  |
| power_pre_simulation_empty_period |                                                                                                                                                                                                                                                                              |

|                                           |                                                                                                                                                                                                                                                                               |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>Set pre-simulation empty period for power analysis</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_pre_simulation_empty_period</a></p>                                             |
| power_pre_simulation_period               |                                                                                                                                                                                                                                                                               |
|                                           | <p>Set pre-simulation period for power analysis</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_pre_simulation_period</a></p>                                                         |
| power_pre_simulation_power_exclude_period |                                                                                                                                                                                                                                                                               |
|                                           | <p>To specify pre simulation power exclude duration</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_pre_simulation_power_exclude_period</a></p>                                       |
| power_precision                           |                                                                                                                                                                                                                                                                               |
|                                           | <p>Specifies precision of decimal range [1-8] to ensure consistent decimal places are displayed each power component.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 8</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_precision</a></p>   |
| power_quit_on_activity_coverage_threshold |                                                                                                                                                                                                                                                                               |
|                                           | <p>Specifies the activity (VCD/FSDB/TCF) coverage threshold value.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_quit_on_activity_coverage_threshold</a></p>                       |
| power_read_rcdb                           |                                                                                                                                                                                                                                                                               |
|                                           | <p>Specifies to write a SPEF file containing only the total C for signal nets and pass it to the Dye Power engine.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_read_rcdb</a></p> |

|                                            |                                                                                                                                                                                                                                                                  |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_relax_arc_match                      |                                                                                                                                                                                                                                                                  |
|                                            | <p>Fallback arc lookup when no exact match</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_relax_arc_match</a></p>                                                      |
| power_report_black_boxes                   |                                                                                                                                                                                                                                                                  |
|                                            | <p>Specify this parameter to report cells that are used as black boxes.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_black_boxes</a></p>                      |
| power_report_clock_instance_switching_info |                                                                                                                                                                                                                                                                  |
|                                            | <p>Generate a report for clock instance switched during state propagation</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> None</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_clock_instance_switching_info</a></p> |
| power_report_idle_instances                |                                                                                                                                                                                                                                                                  |
|                                            | <p>report non-switching instances in vectorless state propagation dynamic analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_idle_instances</a></p>        |
| power_report_instance_switching_info       |                                                                                                                                                                                                                                                                  |
|                                            | <p>all or output_logic or none</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> all output_logic none</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_instance_switching_info</a></p>   |
| power_report_instance_switching_list       |                                                                                                                                                                                                                                                                  |
|                                            | <p>specify file which is having list of instances</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_instance_switching_list</a></p>                                 |

|                                        |                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_report_library_usage             |                                                                                                                                                                                                                                                                                                                                          |
|                                        | <p>generate report for lib usage</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_library_usage</a></p>                                                                                                                                   |
| power_report_missing_bulk_connectivity |                                                                                                                                                                                                                                                                                                                                          |
|                                        | <p>Specifies to report missing bulk pins in the missing input report ( -report_missing_input ).</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_missing_bulk_connectivity</a></p>                                                        |
| power_report_missing_input             |                                                                                                                                                                                                                                                                                                                                          |
|                                        | <p>Specifies to report missing netlist information in input files, such as library, LEF/DEF, PGV, TWF, and missing logical connectivity for cell instance (PGNET).</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_missing_input</a></p> |
| power_report_missing_nets              |                                                                                                                                                                                                                                                                                                                                          |
|                                        | <p>Controls reporting of missing nets in both static and dynamic power analysis.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_missing_nets</a></p>                                                                                    |
| power_report_scan_chain_statistics     |                                                                                                                                                                                                                                                                                                                                          |
|                                        | <p>report scan chain analysis statistics</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_scan_chain_statistics</a></p>                                                                                                                   |
| power_report_statistics                |                                                                                                                                                                                                                                                                                                                                          |

|                                           |                                                                                                                                                                                                                                                                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>Reports a set of statistics on the instance power, instance power density, clock power, or tra density.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_statistics</a></p>                 |
| power_report_time_display_fraction_digits |                                                                                                                                                                                                                                                                                               |
|                                           | <p>specify display fraction digits of reporting time</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> -1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_time_display_fraction_digits</a></p>                                                         |
| power_report_twf_attributes               |                                                                                                                                                                                                                                                                                               |
|                                           | <p>TWF attributes report format</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_report_twf_attributes</a></p>                                                                                         |
| power_reuse_flop_ranking_data             |                                                                                                                                                                                                                                                                                               |
|                                           | <p>Specifies the name of the dir containing the ranking data.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_reuse_flop_ranking_data</a></p>                                                         |
| power_reuse_flop_ranking_data_hier        |                                                                                                                                                                                                                                                                                               |
|                                           | <p>A hierarchy name from the user that will be used to search the ranking data for the block lev being analyzed.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_reuse_flop_ranking_data_hier</a></p> |
| power_scale_to_sdc_clock_frequency        |                                                                                                                                                                                                                                                                                               |
|                                           | <p>Specifies to enable the VCD clock frequency to be scaled up to the SDC clock frequency.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_scale_to_sdc_clock_frequency</a></p>                      |

|                                      |                                                                                                                                                                                                                                                            |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_scan_chain_activity            |                                                                                                                                                                                                                                                            |
|                                      | <p>define scan chain name and activity</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_scan_chain_activity</a></p>                                                 |
| power_scan_chain_name_pattern        |                                                                                                                                                                                                                                                            |
|                                      | <p>define scan chain name and pattern</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_scan_chain_name_pattern</a></p>                                              |
| power_scan_control_file              |                                                                                                                                                                                                                                                            |
|                                      | <p>Specifies the scan control file name required to run the Scan Mode Analysis flow.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_scan_control_file</a></p>     |
| power_scan_multi_bit_flop_chain_type |                                                                                                                                                                                                                                                            |
|                                      | <p>Specifies the type of multi-bit scan flip-flop.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> liberty</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_scan_multi_bit_flop_chain_type</a></p>                     |
| power_set_power_event_based          |                                                                                                                                                                                                                                                            |
|                                      | <p>Constrained power analysis in event based power</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_set_power_event_based</a></p>                                  |
| power_settling_buffer                |                                                                                                                                                                                                                                                            |
|                                      | <p>Specifies the settling buffer time (in ps unit) between multiple windows of a VCD file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_settling_buffer</a></p> |

|                               |                                                                                                                                                                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_show_cell_usage_stats   |                                                                                                                                                                                                                                                                     |
|                               | <p>show cell usage stats in power report</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_show_cell_usage_stats</a></p>                                                     |
| power_shutoff_aware_estimate  |                                                                                                                                                                                                                                                                     |
|                               | <p>Enable power calculation considering power domains which are shutoff.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_shutoff_aware_estimate</a></p>                     |
| power_smart_window            |                                                                                                                                                                                                                                                                     |
|                               | <p>Max power metric window extraction</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_smart_window</a></p>                                                                 |
| power_split_bus_power         |                                                                                                                                                                                                                                                                     |
|                               | <p>A value of false applies the internal power number to each individual bit.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_split_bus_power</a></p>                      |
| power_start_time_alignment    |                                                                                                                                                                                                                                                                     |
|                               | <p>Specifies to align all the signals of the activity file at time zero.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_start_time_alignment</a></p>                       |
| power_state_dependent_leakage |                                                                                                                                                                                                                                                                     |
|                               | <p>When set to false , power analysis performs state independent leakage power calculation.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_state_dependent_leakage</a></p> |

|                                                    |                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_stateprop_ignore_unannot_pins                |                                                                                                                                                                                                                                                                                           |
|                                                    | <p>Enable ignore unannotated pins in state propagation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_stateprop_ignore_unannot_pins</a></p>                                                     |
| power_static_multi_mode_scenario_file              |                                                                                                                                                                                                                                                                                           |
|                                                    | <p>set static multi-mode scenario file name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_static_multi_mode_scenario_file</a></p>                                                               |
| power_static_netlist                               |                                                                                                                                                                                                                                                                                           |
|                                                    | <p>Specifies to perform static power analysis using a Verilog or DEF only netlist.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> verilog</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_static_netlist</a></p>                                    |
| power_switching_power_on_rise_only                 |                                                                                                                                                                                                                                                                                           |
|                                                    | <p>Switching power analysis done only for rise toggles</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_switching_power_on_rise_only</a></p>                                                      |
| power_thermal_input_file                           |                                                                                                                                                                                                                                                                                           |
|                                                    | <p>Specifies to read a power map file to perform thermal analysis.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_thermal_input_file</a></p>                                                     |
| power_thermal_leakage_temperature_scale_table_file |                                                                                                                                                                                                                                                                                           |
|                                                    | <p>Defines library/cell/instance based leakage power scaling factors on each temperature poi</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_thermal_leakage_temperature_scale_table_file</a></p> |



|                                    |                                                                                                                                                                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_to_x_transition_factor       |                                                                                                                                                                                                                                                                                                           |
|                                    | <p>activity count for 0/1 to x transition</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.5</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_to_x_transition_factor</a></p>                                                                                         |
| power_to_z_transition_factor       |                                                                                                                                                                                                                                                                                                           |
|                                    | <p>activity count for 0/1 to z transition</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.25</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_to_z_transition_factor</a></p>                                                                                        |
| power_toggle_rate_unit             |                                                                                                                                                                                                                                                                                                           |
|                                    | <p>Toggle rate time unit.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> /ns</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_toggle_rate_unit</a></p>                                                                                                               |
| power_transition_factor_based_duty |                                                                                                                                                                                                                                                                                                           |
|                                    | <p>Duty calculation using transition factor of toggle</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_transition_factor_based_duty</a></p>                                                                       |
| power_transition_time_method       |                                                                                                                                                                                                                                                                                                           |
|                                    | <p>Specifies the minimum, maximum, or average transition time method that will be used with integrated timer or the external TWF.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> max</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_transition_time_method</a></p> |
| power_twf_delay_annotation         |                                                                                                                                                                                                                                                                                                           |
|                                    | <p>Allows you to choose between min, max, or avg arrival times from the timing window file (TW)</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> avg</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_twf_delay_annotation</a></p>                                     |

|                                                |                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_twf_load_cap                             |                                                                                                                                                                                                                                                                                                                                   |
|                                                | <p>Allows you to ignore or to select minimum, maximum, or average value of TWF external load capacitance for power calculation.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> max</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_twf_load_cap</a></p>                                     |
| power_unified_power_switch_flow                |                                                                                                                                                                                                                                                                                                                                   |
|                                                | <p>if enable ups flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_unified_power_switch_flow</a></p>                                                                                                                                   |
| power_use_cell_leakage_power_density           |                                                                                                                                                                                                                                                                                                                                   |
|                                                | <p>Specifies to use library leakage density times area (default_leakage_power_density).</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_use_cell_leakage_power_density</a></p>                                                            |
| power_use_fastest_clock_for_dynamic_scheduling |                                                                                                                                                                                                                                                                                                                                   |
|                                                | <p>Specifies to use only the respective fastest clock associated with each net or pin to schedule events for a given simulation period.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_use_fastest_clock_for_dynamic_scheduling</a></p> |
| power_use_lef_for_missing_cells                |                                                                                                                                                                                                                                                                                                                                   |
|                                                | <p>Allows a combination of LEF and PGV in the dynamic analysis flow.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_use_lef_for_missing_cells</a></p>                                                                                   |
| power_use_only_ddv_current                     |                                                                                                                                                                                                                                                                                                                                   |

|                                               |                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | <p>use only DDV current from PGV</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_use_only_ddv_current</a></p>                                                                                                       |
| power_use_physical_partition                  |                                                                                                                                                                                                                                                                                                             |
|                                               | <p>use geometric partition in DP</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_use_physical_partition</a></p>                                                                                                    |
| power_use_twf_stable_randomized_arrival_delay |                                                                                                                                                                                                                                                                                                             |
|                                               | <p>Allows tool to choose among all, data_only and none for stable randomized delay annotation timing window file.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_use_twf_stable_randomized_arrival_delay</a></p> |
| power_use_zero_delay_vector_file              |                                                                                                                                                                                                                                                                                                             |
|                                               | <p>Enables zero delay mode in vector-based dynamic analysis to avoid pessimism in current estimation.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_use_zero_delay_vector_file</a></p>                           |
| power_vector_based_multithread                |                                                                                                                                                                                                                                                                                                             |
|                                               | <p>Specifies to enable multi-threading for the VCD/FSDB based dynamic vector-based flows.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_vector_based_multithread</a></p>                                          |
| power_vector_profile_mode                     |                                                                                                                                                                                                                                                                                                             |
|                                               | <p>activity or density</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> activity event_based power_density transient</p> <p><b>Default:</b> event_based</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_vector_profile_mode</a></p>                                   |

|                                  |                                                                                                                                                                                                                                                                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_worst_case_vector_activity |                                                                                                                                                                                                                                                                                                    |
|                                  | <p>Specifies to use the worst activity value when multiple vectors are specified in the static power calculation flow.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_worst_case_vector_activity</a></p> |
| power_worst_step_size            |                                                                                                                                                                                                                                                                                                    |
|                                  | <p>Set worst step size for power analysis</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_worst_step_size</a></p>                                                                                          |
| power_worst_window_count         |                                                                                                                                                                                                                                                                                                    |
|                                  | <p>Set worst window count for power analysis</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_worst_window_count</a></p>                                                                                        |
| power_worst_window_coverage      |                                                                                                                                                                                                                                                                                                    |
|                                  | <p>Set worst window coverage type</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_worst_window_coverage</a></p>                                                                                            |
| power_worst_window_reports       |                                                                                                                                                                                                                                                                                                    |
|                                  | <p>generate Power reports for full simulation and worst power window</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> full</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_worst_window_reports</a></p>                                                        |
| power_worst_window_size          |                                                                                                                                                                                                                                                                                                    |
|                                  | <p>Set worst window size for power analysis</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_worst_window_size</a></p>                                                                                      |

|                               |                                                                                                                                                                                                                                                                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_worst_window_type       |                                                                                                                                                                                                                                                                                                                          |
|                               | <p>Set worst window type for power analysis</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_worst_window_type</a></p>                                                                                                            |
| power_write_bbox              |                                                                                                                                                                                                                                                                                                                          |
|                               | <p>save the instance bbox to a text file when output power db</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_write_bbox</a></p>                                                                                                |
| power_write_db                |                                                                                                                                                                                                                                                                                                                          |
|                               | <p>A value of true creates a binary database of power results.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_write_db</a></p>                                                                                                 |
| power_write_default_pti_files |                                                                                                                                                                                                                                                                                                                          |
|                               | <p>For instances that are not hooked to any power/ground rail, the current for these instances is reflected in the default static current files.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_write_default_pti_files</a></p> |
| power_write_dynamic_currents  |                                                                                                                                                                                                                                                                                                                          |
|                               | <p>Current write with method event_based</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_write_dynamic_currents</a></p>                                                                                                         |
| power_write_gui_db            |                                                                                                                                                                                                                                                                                                                          |
|                               | <p>A value of true creates a binary database of power results for gui.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_write_gui_db</a></p>                                                                                      |

|                                         |                                                                                                                                                                                                                                                        |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| power_write_profiling_db                |                                                                                                                                                                                                                                                        |
|                                         | <p>turn on/off profiling database writing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_write_profiling_db</a></p>                                          |
| power_write_query_db                    |                                                                                                                                                                                                                                                        |
|                                         | <p>A value of true creates query and netlist databases for UI to query power results.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                |
| power_write_simulation_db               |                                                                                                                                                                                                                                                        |
|                                         | <p>Save functional waveform foreach instance when parsing FSDB/VCD</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_write_simulation_db</a></p>                |
| power_write_static_currents             |                                                                                                                                                                                                                                                        |
|                                         | <p>A value of true tells Voltus to generate the current data files per net.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_write_static_currents</a></p>     |
| power_x_count_transition_using_3_states |                                                                                                                                                                                                                                                        |
|                                         | <p>Toggle count using 3 states in VCD</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_x_count_transition_using_3_states</a></p>                               |
| power_x_transition_factor               |                                                                                                                                                                                                                                                        |
|                                         | <p>When using VCD/FSDB one can specify how transitions to and from X are counted.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.5</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_x_transition_factor</a></p> |
| power_z_transition_factor               |                                                                                                                                                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>When using VCD/FSDB one can specify how transitions to and from Z are counted.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.25</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_z_transition_factor</a></p>                                  |
| power_zero_delay_vector_toggle_shift   |                                                                                                                                                                                                                                                                                          |
|                                        | <p>Specifies to shift the current waveform in the zero-delay VCD flow.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">power_zero_delay_vector_toggle_shift</a></p>                                    |
| preserve_combinational_loop_ports_nets |                                                                                                                                                                                                                                                                                          |
|                                        | <p>Preserve nets and pins involved in combinational loops as a reference for verification.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">preserve_combinational_loop_ports_nets</a></p>              |
| print_full_message_summary             |                                                                                                                                                                                                                                                                                          |
|                                        | <p>Used to control the format of message summary. If true, print full message, otherwith, print truncated message.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">print_full_message_summary</a></p> |
| program_major_version                  |                                                                                                                                                                                                                                                                                          |
|                                        | <p>The major version plus one digit. So if program_version = 19.11-e062_1 it would return '19</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">program_major_version</a></p>                             |
| program_name                           |                                                                                                                                                                                                                                                                                          |
|                                        | <p>name of the program</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">program_name</a></p>                                                                                                             |
| program_short_name                     |                                                                                                                                                                                                                                                                                          |

|                                        |                                                                                                                                                                                                                                           |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>short name of the program</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">program_short_name</a></p>                                                  |
| program_version                        |                                                                                                                                                                                                                                           |
|                                        | <p>version of the program</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">program_version</a></p>                                                        |
| propagate_constant_from_timing_model   |                                                                                                                                                                                                                                           |
|                                        | <p>Allow constant propagation from timing model instances.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">propagate_constant_from_timing_model</a></p> |
| proto_allow_model_with_io              |                                                                                                                                                                                                                                           |
|                                        | <p>Supports model with I/Os.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_allow_model_with_io</a></p>                                         |
| proto_allow_power_domain_in_flexmodel  |                                                                                                                                                                                                                                           |
|                                        | <p>Allow powerdomain in flexmodel</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_allow_power_domain_in_flexmodel</a></p>                        |
| proto_create_dir                       |                                                                                                                                                                                                                                           |
|                                        | <p>Specifies the directory to save created models.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> proto_model</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_dir</a></p>                    |
| proto_create_high_fanout_ps_per_micron |                                                                                                                                                                                                                                           |



|                                         |                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Create high fanout psPM models.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_high_fanout_ps_per_micron</a></p>                                                                                            |
| proto_create_lib                        |                                                                                                                                                                                                                                                                                                                |
|                                         | <p>Specifies the OA library to save created models.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> proto_model_lib</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_lib</a></p>                                                                                    |
| proto_create_metal_fill_ndr             |                                                                                                                                                                                                                                                                                                                |
|                                         | <p>Will be used by create_proto_net_delay_model for creating psPM model for nets with non c rules.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_metal_fill_ndr</a></p>                                      |
| proto_create_metal_fill_nominal         |                                                                                                                                                                                                                                                                                                                |
|                                         | <p>Will be used by create_proto_net_delay_model for creating psPM model for normal nets.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.5</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_metal_fill_nominal</a></p>                                            |
| proto_create_multi_corner_ps_per_micron |                                                                                                                                                                                                                                                                                                                |
|                                         | <p>Create multi corner psPM models.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_multi_corner_ps_per_micron</a></p>                                                                                         |
| proto_create_ndr_ps_per_micron          |                                                                                                                                                                                                                                                                                                                |
|                                         | <p>Control whether create_proto_net_delay_model should generate psPM model for non defa or not</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto on off</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_ndr_ps_per_micron</a></p> |

|                                         |                                                                                                                                                                                                                                                          |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proto_create_no_flex_filler             |                                                                                                                                                                                                                                                          |
|                                         | <p>Create proto model without adding flex filler.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_no_flex_filler</a></p>                                 |
| proto_create_partition_as_flexmodel     |                                                                                                                                                                                                                                                          |
|                                         | <p>Create partition specified as flexmodel automatically.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_partition_as_flexmodel</a></p>                 |
| proto_create_pipeline_flop              |                                                                                                                                                                                                                                                          |
|                                         | <p>Specifies a list of pipeline flop instance patterns that are used in the design.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_pipeline_flop</a></p> |
| proto_create_power_domain_ps_per_micron |                                                                                                                                                                                                                                                          |
|                                         | <p>Create power domain psPM models.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_power_domain_ps_per_micron</a></p>                                   |
| proto_create_timing_budget              |                                                                                                                                                                                                                                                          |
|                                         | <p>Keep consistent setting of create_timing_budget</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_create_timing_budget</a></p>                                 |
| proto_design_cover_fixed_macros         |                                                                                                                                                                                                                                                          |
|                                         | <p>force a proto seed cover fixed macros belong to it in proto_design</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_design_cover_fixed_macros</a></p>         |

|                                        |                                                                                                                                                                                                                                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proto_design_effort                    |                                                                                                                                                                                                                                                                                            |
|                                        | <p>Specifies proto_design effort</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> low</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_design_effort</a></p>                                                   |
| proto_design_keep_guide                |                                                                                                                                                                                                                                                                                            |
|                                        | <p>-keep_guide {true false} #default=true</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_design_keep_guide</a></p>                                                                                |
| proto_design_place_macro               |                                                                                                                                                                                                                                                                                            |
|                                        | <p>control if do macro placement in proto_design</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_design_place_macro</a></p>                                                                        |
| proto_design_remove_overlap            |                                                                                                                                                                                                                                                                                            |
|                                        | <p>control if do remove overlap in proto_design</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_design_remove_overlap</a></p>                                                                      |
| proto_feasible_target                  |                                                                                                                                                                                                                                                                                            |
|                                        | <p>Controls the feasible target feature to ignore huge negative slack endpoints from target computation and optimization.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_feasible_target</a></p> |
| proto_feasible_target_adjust_slack_pct |                                                                                                                                                                                                                                                                                            |

|                                           |                                                                                                                                                                                                                                                                                        |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | Controls percentage of slack value to be used as positive adjust value in path_adjust except<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 100<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">proto_feasible_target_adjust_slack_pct</a>                                |
| proto_feasible_target_threshold           |                                                                                                                                                                                                                                                                                        |
|                                           | Controls minimum threshold delay for which path_adjust should be applied.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> no_value<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">proto_feasible_target_threshold</a>                                                  |
| proto_feasible_target_threshold_clock_pct |                                                                                                                                                                                                                                                                                        |
|                                           | Controls slack to clock period percentage for which path_adjust should be applied.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 75<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">proto_feasible_target_threshold_clock_pct</a>                                        |
| proto_hdl                                 |                                                                                                                                                                                                                                                                                        |
|                                           | Continue elaboration, in presence of certain types of inconsistencies(e.g. module name case mismatch between module instance and module definition).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">proto_hdl</a>   |
| proto_identify_engine                     |                                                                                                                                                                                                                                                                                        |
|                                           | identify_proto algorithm <default=auto><br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> auto module_based instgroup_based<br><b>Default:</b> auto<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">proto_identify_engine</a>                                           |
| proto_identify_estimated_flexmodel_number |                                                                                                                                                                                                                                                                                        |
|                                           | Specified the estimated flexModel number to identify for the design when using instance group based identify_proto_model.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 20<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">proto_identify_estimated_flexmodel_number</a> |

|                                          |                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proto_identify_exclude_module            |                                                                                                                                                                                                                                                                                                                       |
|                                          | <p>Excludes specified module(s) but not its sub-modules for being marked as models.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_identify_exclude_module</a></p>                                                           |
| proto_identify_exclude_module_and_parent |                                                                                                                                                                                                                                                                                                                       |
|                                          | <p>Excludes specified module(s) and their parents for being marked as models.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_identify_exclude_module_and_parent</a></p>                                                      |
| proto_identify_exclude_module_tree       |                                                                                                                                                                                                                                                                                                                       |
|                                          | <p>Excludes specified module(s) and its sub-modules for being marked as models.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_identify_exclude_module_tree</a></p>                                                          |
| proto_identify_honor_objects_hierarchy   |                                                                                                                                                                                                                                                                                                                       |
|                                          | <p>Specified the list of objects which instance group based identify_proto_model will honor its hierarchy.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_identify_honor_objects_hierarchy</a></p>                           |
| proto_identify_max_inst                  |                                                                                                                                                                                                                                                                                                                       |
|                                          | <p>Specifies the maximum instance count threshold of a module to be marked as model.(default of total instance count and no lower than 300K instances)</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> -1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_identify_max_inst</a></p> |
| proto_identify_min_inst                  |                                                                                                                                                                                                                                                                                                                       |

|                                    |                                                                                                                                                                                                                                                                         |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Specifies the minimum instance count threshold of a module to be marked as model.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_identify_min_inst</a></p>                   |
| proto_keep_inst_file_only          |                                                                                                                                                                                                                                                                         |
|                                    | <p>FlexModel only contains inst in keep_inst_file.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_keep_inst_file_only</a></p>                                                 |
| proto_keep_instance_defined_in_sdc |                                                                                                                                                                                                                                                                         |
|                                    | <p>FlexModel will keep inst defined in sdc file</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_keep_instance_defined_in_sdc</a></p>                                           |
| proto_keep_slack_improve_ndr       |                                                                                                                                                                                                                                                                         |
|                                    | <p>Will be used by report_NDR_WNS_gain to keep top critical nets as non default rule nets .</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_keep_slack_improve_ndr</a></p>     |
| proto_max_report_ndr_net           |                                                                                                                                                                                                                                                                         |
|                                    | <p>Will be used by create_proto_net_delay_model for creating psPM model for nets with non c rules.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.05</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_max_report_ndr_net</a></p> |
| proto_route_net_ndr                |                                                                                                                                                                                                                                                                         |
|                                    | <p>Will be used by create_proto_net_delay_model for creating psPM model for nets with non c rules.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_route_net_ndr</a></p>        |

|                                  |                                                                                                                                                                                                                                                                                                                            |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proto_timing_net_delay           |                                                                                                                                                                                                                                                                                                                            |
|                                  | <p>Specifies the estimated delay per micron for each routing layer. Layer range is from 1 to max routing layers.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_timing_net_delay</a></p>                                          |
| proto_use_timing_net_delay_model |                                                                                                                                                                                                                                                                                                                            |
|                                  | <p>set mode of calculating wire delay</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> best_layer_no_detour best_layer_blockage_aware use_actual_wire</p> <p><b>Default:</b> best_layer_no_detour</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_use_timing_net_delay_model</a></p> |
| proto_verbose                    |                                                                                                                                                                                                                                                                                                                            |
|                                  | <p>Output detailed information into .log file.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">proto_verbose</a></p>                                                                                                                    |
| rc_corners                       |                                                                                                                                                                                                                                                                                                                            |
|                                  | <p>Returns the information about the rc corners in the design.</p> <p><b>Type:</b> <a href="#">obj(rc_corner)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                    |
| read_db_directory                |                                                                                                                                                                                                                                                                                                                            |
|                                  | <p>Specifies the directory of current loaded DB.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">read_db_directory</a></p>                                                                                                               |
| read_db_file_check               |                                                                                                                                                                                                                                                                                                                            |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>By default, read_db will check all files inside the saved DB directory are unchanged. If any have been edited, renamed or deleted, then an error will occur unless this value is set to false.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_db_file_check</a></p>                                                                                                                                                             |
| read_db_keep_updated_macro_lefs  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                  | <p>If set true, the lef files of updated macros that has been saved in DB would be honored, even if there are new lef files for these macros are specified in read_db -lef_files {xxx}. If set false, the updated macros' lef files in db would be ignored. Only the lef files that are specified in read_db -lef_files {xxx} would be updated.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_db_keep_updated_macro_lefs</a></p> |
| read_db_stop_at_design_in_memory |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                  | <p>By default, tool will error and stop if call read_db more than once in the same Innovus session unless this value is set to false.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_db_stop_at_design_in_memory</a></p>                                                                                                                                                                                                           |
| read_db_tool_name                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                  | <p>Indicates the tool name of the tool that created the restored db on the disk database</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_db_tool_name</a></p>                                                                                                                                                                                                                                                                       |
| read_db_version                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                  | <p>Indicates the version of the tool that generates an on disk database</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_db_version</a></p>                                                                                                                                                                                                                                                                                          |
| read_def_check_mask_shifts       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>If on/bypass, will check for conflicts between DEF COMPONENT MASKSHIFT and MACRO MASKSHIFT values with the LEF FIXEDMASK and LAYERMASKSHIFT values in the techn and MACRO sections, error out if on and error without out if bypass</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> off on bypass<br/> <b>Default:</b> off<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_def_check_mask_shifts</a></p>                                                     |
| read_physical_check_antenna                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                 | <p>If true, LEF (or OA cells) will have process antenna consistency checks done for LEF MACF ANTENNA* data (or equivalent OA data) attached to their pins. Input pins with 0 gate-area, pins with 0 diff-area, and inout pins with 0 gate-area or 0 diff-area values will generate warn false, no consistency checking is done.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_physical_check_antenna</a></p> |
| read_physical_check_colored_shape               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                 | <p>Specify how to check and ignore colored shapes on colorless layers during reading lef or d If true, error out if a colored shape is on a colorless layer; if false, give error message and ig the color.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_physical_check_colored_shape</a></p>                                                                                                              |
| read_physical_check_mask_shifts                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                 | <p>If on/bypass, will check for conflicts between LEF FIXEDMASK and LAYERMASKSHIFT val the technology section and MACRO section when reading in any LEF files, error out if on a without out if bypass</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> off on bypass<br/> <b>Default:</b> off<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_physical_check_mask_shifts</a></p>                                                                                |
| read_physical_extend_cell_obs_shapes_under_trim |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | <p>A list of routing layer. LEF OBS shapes on each &lt;layer&gt; that are overlapped partially by a s on TRIMMETAL layer that trims the routing layer, will be extended in the preferred routing d to be fully covered by the trim shape. This is a workaround for some advanced node libraries avoid trim/OBS short violations.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_physical_extend_cell_obs_shapes_under_trim</a></p> |
| read_physical_extend_polygon_cell_shapes      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                               | <p>A list of {{&lt;layer&gt; &lt;value&gt;} ...} pairs. LEF PIN/OBS shapes on each &lt;layer&gt; that are polygon simple rects), will be extended with OBS SPACING 0 rects by &lt;value&gt; (in microns) in the pr routing direction. This is a workaround for some advanced node libraries to avoid spacing violations during routing.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_physical_extend_polygon_cell_shapes</a></p> |
| read_physical_ignore_layer_names              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                               | <p>Specify the layer list that needs to be ignored.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_physical_ignore_layer_names</a></p>                                                                                                                                                                                                                                                                                             |
| read_physical_merge_multi_port_to_single_port |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                               | <p>If true, multiple ports with the same class in a LEF PIN will be merged into a single port.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_physical_merge_multi_port_to_single_port</a></p>                                                                                                                                                                                                                                    |
| read_physical_must_join_all_ports             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                               | <p>If false, the LEF_MUSTJOINALLPORTS property on a LEF PIN auto-creates MUSTJOIN pins for each PORT. If true, the MUSTJOIN pins are not created, but the router must connect to every with a signal wire or signal via that physically overlaps all the ports.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_physical_must_join_all_ports</a></p>                                                                                |
| read_write_lef_check_uncolored_shapes         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Specifies whether to check for uncolored LEF PIN and OBS shapes on multi-mask routing I when reading or writing LEF files. If the check is on, LEF read or write commands will error there are any uncolored routing layer shapes (cut layers are not checked). If set to pin_only check LEF PIN shapes and ignore LEF OBS shapes.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> on off pin_only<br/> <b>Default:</b> off<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">read_write_lef_check_uncolored_shapes</a></p> |
| redn_mopt_compute_redundant_toggles |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                     | <p>Runs redundant toggle analysis for memory.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                             |
| redn_mopt_generate_detailed_circuit |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                     | <p>Generates detailed stability detection logic for memory.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                               |
| redn_scrub_strict_clock_check       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                     | <p>Implement strict clock check for ODC/STB.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                              |
| redn_stb_trace_rtl_limit            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                     | <p>Pre-compute limit for stability optimization expression trace rtl.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 10<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                         |
| reload_when_for_macro_cell          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                     | <p>Reload when condition processing for macro library cell.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">reload_when_for_macro_cell</a></p>                                                                                                                                                                                                                                                                                                                              |
| reorder_scan_add_scan_port_prefix   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                   |                                                                                                                                                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>prefix of scan hport</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">reorder_scan_add_scan_port_prefix</a></p>                                                      |
| reorder_scan_clock_aware          |                                                                                                                                                                                                                                                          |
|                                   | <p>Clock tree aware reorder</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">reorder_scan_clock_aware</a></p>                                                          |
| reorder_scan_comp_logic           |                                                                                                                                                                                                                                                          |
|                                   | <p>trace considering multiple input logic</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">reorder_scan_comp_logic</a></p>                                             |
| reorder_scan_effort               |                                                                                                                                                                                                                                                          |
|                                   | <p>Specifies effort level for reorder_scan</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high auto</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">reorder_scan_effort</a></p> |
| reorder_scan_enable_for_partition |                                                                                                                                                                                                                                                          |
|                                   | <p>enable scanReorder for design with partition</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">reorder_scan_enable_for_partition</a></p>                             |
| reorder_scan_keep_hinst_port_name |                                                                                                                                                                                                                                                          |
|                                   | <p>Specifies filename which contains HInsts</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">reorder_scan_keep_hinst_port_name</a></p>                                  |
| reorder_scan_keep_hport           |                                                                                                                                                                                                                                                          |

|                                |                                                                                                                                                                                                                                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Maintains hierarchical ports without adding or deleting shifters</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">reorder_scan_keep_hport</a></p>                                                              |
| reorder_scan_prefer_horizontal |                                                                                                                                                                                                                                                                                                    |
|                                | <p>Makes horizontal scan chain connections if possible</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">reorder_scan_prefer_horizontal</a></p>                                                                   |
| reorder_scan_prefer_vertical   |                                                                                                                                                                                                                                                                                                    |
|                                | <p>Makes vertical scan chain connections if possible</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">reorder_scan_prefer_vertical</a></p>                                                                       |
| reorder_scan_skip_mode         |                                                                                                                                                                                                                                                                                                    |
|                                | <p>Specifies method for handling buffers and inverters</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> skip_none skip_buffer skip_floating_buffer</p> <p><b>Default:</b> skip_none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">reorder_scan_skip_mode</a></p> |
| reorder_scan_swap_effort       |                                                                                                                                                                                                                                                                                                    |
|                                | <p>Specifies effort level for reorder_scan different chains swapping</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> high</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">reorder_scan_swap_effort</a></p>                 |
| report_library_message_summary |                                                                                                                                                                                                                                                                                                    |
|                                | <p>Enables library message summary report.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">report_library_message_summary</a></p>                                                                                |

|                                     |                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| report_obj_display_limit            |                                                                                                                                                                                                                                                                                                                                                         |
|                                     | <p>Controls the number of objects that will be displayed when using report_obj. If the value is 'report_obj insts' will only display 10 inst objects with all their non-default attributes.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 1000<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">report_obj_display_limit</a></p> |
| report_pin_density_map_display_step |                                                                                                                                                                                                                                                                                                                                                         |
|                                     | <p>range step for displaying the pin density map</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.05<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">report_pin_density_map_display_step</a></p>                                                                                                                              |
| report_pin_density_map_grid_size    |                                                                                                                                                                                                                                                                                                                                                         |
|                                     | <p>specify the value (in row-height) of side of square grid for pin density calculations.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 10<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">report_pin_density_map_grid_size</a></p>                                                                                             |
| report_pin_density_map_grid_unit    |                                                                                                                                                                                                                                                                                                                                                         |
|                                     | <p>specify the value (in micron) of side of square grid for pin density calculations.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 50.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">report_pin_density_map_grid_unit</a></p>                                                                                            |
| report_pin_density_map_threshold    |                                                                                                                                                                                                                                                                                                                                                         |
|                                     | <p>threshold value over which the densities must be reported.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.5<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">report_pin_density_map_threshold</a></p>                                                                                                                     |
| report_place_density_map_grid_size  |                                                                                                                                                                                                                                                                                                                                                         |
|                                     | <p>specify the value (in row-height) of side of square grid for density calculations.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 10<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">report_place_density_map_grid_size</a></p>                                                                                               |

|                                        |                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| report_place_density_map_grid_unit     |                                                                                                                                                                                                                                                                                                                                                       |
|                                        | <p>specify the value (in micron) of side of square grid for density calculations.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 50.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">report_place_density_map_grid_unit</a></p>                                                                                      |
| report_place_density_map_ignore_block  |                                                                                                                                                                                                                                                                                                                                                       |
|                                        | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">report_place_density_map_ignore_block</a></p>                                                                                                                                                                          |
| report_place_density_map_ignore_filler |                                                                                                                                                                                                                                                                                                                                                       |
|                                        | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">report_place_density_map_ignore_filler</a></p>                                                                                                                                                                         |
| report_place_density_map_threshold     |                                                                                                                                                                                                                                                                                                                                                       |
|                                        | <p>threshold value over which the densities must be reported.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.75</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">report_place_density_map_threshold</a></p>                                                                                                          |
| resize_blockages                       |                                                                                                                                                                                                                                                                                                                                                       |
|                                        | <p>Short-cut for [get_db current_design .resize_blockages]</p> <p><b>Type:</b> <a href="#">obj(resize_blockage)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                             |
| resize_floorplan_cong_aware            |                                                                                                                                                                                                                                                                                                                                                       |
|                                        | <p>resizes and shifts the floorplan objects by estimating the congestion for the floorplan and automatically deciding where to draw a resize line to avoid the congested area.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">resize_floorplan_cong_aware</a></p> |

|                                          |                                                                                                                                                                                                                                                                          |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| resize_floorplan_honor_halo              |                                                                                                                                                                                                                                                                          |
|                                          | <p>honors placement halo and preserves the space between macros held by halo after the floorplan resize.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> true<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">resize_floorplan_honor_halo</a></p>    |
| resize_floorplan_io_fix                  |                                                                                                                                                                                                                                                                          |
|                                          | <p>fixes I/Os at current location.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">resize_floorplan_io_fix</a></p>                                                                             |
| resize_floorplan_io_move_with_edge       |                                                                                                                                                                                                                                                                          |
|                                          | <p>moves pins on the moved edge orthogonally to edge direction so pins stay on edge after movement.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">resize_floorplan_io_move_with_edge</a></p> |
| resize_floorplan_io_relative             |                                                                                                                                                                                                                                                                          |
|                                          | <p>resize the space among I/Os proportionally.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> true<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">resize_floorplan_io_relative</a></p>                                                             |
| resize_floorplan_maintain_resource_ratio |                                                                                                                                                                                                                                                                          |
|                                          | <p>Enable to maintain the resource ratio after resize_floorplan in shift-based mode.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">resize_floorplan_maintain_resource_ratio</a></p>          |
| resize_floorplan_relative                |                                                                                                                                                                                                                                                                          |



|                                     |                                                                                                                                                                                                                                                    |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | resizes the space among floorplan objects proportionally.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">resize_floorplan_relative</a>                                         |
| resize_floorplan_shift_based        |                                                                                                                                                                                                                                                    |
|                                     | shifts floorplan objects at appropriate location(s) without changing the proportional spacing<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">resize_floorplan_shift_based</a>   |
| resize_floorplan_shrink_fence       |                                                                                                                                                                                                                                                    |
|                                     | enables resized lines to go through fences and regions in floorplan.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">resize_floorplan_shrink_fence</a>                          |
| resize_floorplan_snap_to_track      |                                                                                                                                                                                                                                                    |
|                                     | snaps resize values (shrink/expand) of the floorplan to a multiple integer of the metal layer<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">resize_floorplan_snap_to_track</a> |
| resolve_show_tcl_command_with_value |                                                                                                                                                                                                                                                    |
|                                     | If user set the value to true, print the Tcl variable value in the resolve log.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                  |
| retime_async_reset                  |                                                                                                                                                                                                                                                    |
|                                     | Should flops with asynchronous reset be retimed.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">retime_async_reset</a>                                                          |
| retime_effort_level                 |                                                                                                                                                                                                                                                    |

|                               |                                                                                                                                                                                                                                                                                 |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Set effort level during retiming (high, medium(default), or low).</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> medium</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">retime_effort_level</a></p> |
| retime_optimize_reset         |                                                                                                                                                                                                                                                                                 |
|                               | <p>Determines whether the reset should be dropped if the computation results in a dont_care</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">retime_optimize_reset</a></p>                    |
| retime_verification_flow      |                                                                                                                                                                                                                                                                                 |
|                               | <p>Control the annotation in the netlist for retiming verification.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">retime_verification_flow</a></p>                                          |
| route_add_antenna_inst_prefix |                                                                                                                                                                                                                                                                                 |
|                               | <p>prefix for the diode insts added to fix antenna</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_add_antenna_inst_prefix</a></p>                                                      |
| route_adjust_auto_via_weight  |                                                                                                                                                                                                                                                                                 |
|                               | <p>adjust auto via weight for via swapping</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_adjust_auto_via_weight</a></p>                                                               |
| route_allow_inst_overlaps     |                                                                                                                                                                                                                                                                                 |
|                               | <p>don't do instance overlap check</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_allow_inst_overlaps</a></p>                                                                          |
| route_allow_pin_as_feedthru   |                                                                                                                                                                                                                                                                                 |

|                                            |                                                                                                                                                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>allow pin as feedthrough</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_allow_pin_as_feedthru</a></p>                                           |
| route_antenna_cell_name                    |                                                                                                                                                                                                                                               |
|                                            | <p>specify the antenna diode cell names for antenna fixing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_antenna_cell_name</a></p>                  |
| route_antenna_diode_insertion              |                                                                                                                                                                                                                                               |
|                                            | <p>insert antenna diode to fix antenna violation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_antenna_diode_insertion</a></p>                     |
| route_blockages                            |                                                                                                                                                                                                                                               |
|                                            | <p>Short-cut for [get_db current_design .route_blockages]</p> <p><b>Type:</b> <a href="#">obj(route_blockage)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                       |
| route_concurrent_minimize_via_count_effort |                                                                                                                                                                                                                                               |
|                                            | <p>specify the effort of concurrent via minimization</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> medium</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_concurrent_minimize_via_count_effort</a></p> |
| route_connect_to_bumps                     |                                                                                                                                                                                                                                               |
|                                            | <p>connect to bumps in native NanoRoute in Innovus</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_connect_to_bumps</a></p>                          |
| route_design_high_freq_bus_mixed_layer     |                                                                                                                                                                                                                                               |

|                                                |                                                                                                                                                                                                                         |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | bus mixed layer {false true auto only}<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_design_high_freq_bus_mixed_layer</a>                  |
| route_design_top_level_design                  |                                                                                                                                                                                                                         |
|                                                | specify this is a top level design<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                                    |
| route_detail_add_passive_fill_only_on_layers   |                                                                                                                                                                                                                         |
|                                                | specify layers where passive fill will be added<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_detail_add_passive_fill_only_on_layers</a>      |
| route_detail_allow_passive_fill_only_in_layers |                                                                                                                                                                                                                         |
|                                                | specify layers where do passive fill only<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_detail_allow_passive_fill_only_in_layers</a>       |
| route_detail_antenna_eco_list_file             |                                                                                                                                                                                                                         |
|                                                | set file name for antenna eco list<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_detail_antenna_eco_list_file</a>                             |
| route_detail_auto_stop                         |                                                                                                                                                                                                                         |
|                                                | control whether Nanoroute continues routing if there are many violations<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_detail_auto_stop</a> |
| route_detail_check_mar_on_cell_pin             |                                                                                                                                                                                                                         |

|                                                      |                                                                                                                                                                                                                                                                |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | <p>Enable mar checking on used cell pins</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_check_mar_on_cell_pin</a></p>                                         |
| route_detail_end_iteration                           |                                                                                                                                                                                                                                                                |
|                                                      | <p>specify the last iteration pass in a detailed routing step</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_end_iteration</a></p>                           |
| route_detail_fix_antenna                             |                                                                                                                                                                                                                                                                |
|                                                      | <p>fix antenna violation by jumping metal layers</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> auto true false</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_fix_antenna</a></p> |
| route_detail_fix_antenna_on_secondary_pg_nets        |                                                                                                                                                                                                                                                                |
|                                                      | <p>fix antenna violation on sencondary pg nets</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_fix_antenna_on_secondary_pg_nets</a></p>                        |
| route_detail_fix_antenna_with_gate_array_filler_mode |                                                                                                                                                                                                                                                                |
|                                                      | <p>detect GA filler flow while doing antenna violation fixing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_fix_antenna_with_gate_array_filler_mode</a></p>  |
| route_detail_merge_abutting_cut                      |                                                                                                                                                                                                                                                                |
|                                                      | <p>merge two abutting cell cut shapes into a new one</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_merge_abutting_cut</a></p>                                 |
| route_detail_min_length_for_spread_wire              |                                                                                                                                                                                                                                                                |

|                                        |                                                                                                                                                                                                                                                                 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>specify the min. length of a spreaded wire</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_min_length_for_spread_wire</a></p>                                  |
| route_detail_min_length_for_widen_wire |                                                                                                                                                                                                                                                                 |
|                                        | <p>specify the min. length of a widened wire</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_min_length_for_widen_wire</a></p>                                  |
| route_detail_min_slack_for_opt_wire    |                                                                                                                                                                                                                                                                 |
|                                        | <p>specify the min slack for a net wire optimization</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_min_slack_for_opt_wire</a></p>                             |
| route_detail_no_taper_in_layers        |                                                                                                                                                                                                                                                                 |
|                                        | <p>Specifies the range of the layer where taper is not allowed</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_no_taper_in_layers</a></p>                        |
| route_detail_no_taper_on_output_pin    |                                                                                                                                                                                                                                                                 |
|                                        | <p>forbid wire tapering at output pin</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true auto</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_no_taper_on_output_pin</a></p> |
| route_detail_on_grid_only              |                                                                                                                                                                                                                                                                 |
|                                        | <p>route on grid only</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_on_grid_only</a></p>                                                                     |
| route_detail_post_route_litho_repair   |                                                                                                                                                                                                                                                                 |

|                                           |                                                                                                                                                                                                                            |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | litho hotspot repair in post route stage<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_detail_post_route_litho_repair</a>                       |
| route_detail_post_route_spread_wire       |                                                                                                                                                                                                                            |
|                                           | spread wire in post route stage<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> auto<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_detail_post_route_spread_wire</a>                                |
| route_detail_post_route_swap_via          |                                                                                                                                                                                                                            |
|                                           | to enforce post route via swapping mode in detail route command.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_detail_post_route_swap_via</a> |
| route_detail_post_route_via_pillar_effort |                                                                                                                                                                                                                            |
|                                           | Set opt effort/range for opportunistic via pillar<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> high<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_detail_post_route_via_pillar_effort</a>        |
| route_detail_post_route_wire_widen        |                                                                                                                                                                                                                            |
|                                           | widen wire in post route stage<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> none<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_detail_post_route_wire_widen</a>                                  |
| route_detail_post_route_wire_widen_rule   |                                                                                                                                                                                                                            |
|                                           | specify the rule used to widen wire<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_detail_post_route_wire_widen_rule</a>                          |
| route_detail_postroute_via_priority       |                                                                                                                                                                                                                            |

|                                                            |                                                                                                                                                                                                                                                               |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            | <p>Set net priority for via swapping based on timing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_postroute_via_priority</a></p>                          |
| route_detail_search_and_repair                             |                                                                                                                                                                                                                                                               |
|                                                            | <p>run search-and-repair step after the initial detailed routing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_search_and_repair</a></p>                     |
| route_detail_signoff_effort                                |                                                                                                                                                                                                                                                               |
|                                                            | <p>specify when to give up on irresolvable violations</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> high</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_signoff_effort</a></p>                                 |
| route_detail_stub_routing_in_first_layer                   |                                                                                                                                                                                                                                                               |
|                                                            | <p>to use Minimum routing in the first routing layer defined in the LEF</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_stub_routing_in_first_layer</a></p> |
| route_detail_use_multi_cut_via_effort                      |                                                                                                                                                                                                                                                               |
|                                                            | <p>specify the effort to use multi-cut via during routing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> low</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_detail_use_multi_cut_via_effort</a></p>                    |
| route_diode_insertion_for_clock_nets                       |                                                                                                                                                                                                                                                               |
|                                                            | <p>allow diode insertion on clock nets</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_diode_insertion_for_clock_nets</a></p>                                        |
| route_disable_route_rule_on_via_pillar_to_special_net_wire |                                                                                                                                                                                                                                                               |



|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | <p>Disable non-default rules on via pillar to special net wire</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_disable_route_rule_on_via_pillar_to_special_net_wire</a></p>                                                                                                                                                                                                                                                     |
| route_early_global_effort_level                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                   | <p>Specifies the congestion effort level. Default is "standard", where congestion is more accurate; runtime is larger. If set to "medium", runtime will be much smaller and congestion report will be less accurate. Setting to "low", it will be faster than "medium" however less accurate.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium standard</p> <p><b>Default:</b> standard</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_early_global_effort_level</a></p> |
| route_early_global_honor_partition_allow_feedthru |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                   | <p>Honor partition fences with feedthrough for &lt;list_of_ptn_cell_names&gt;.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_early_global_honor_partition_allow_feedthru</a></p>                                                                                                                                                                                                                                                 |
| route_early_global_honor_partition_fence          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                   | <p>Honor partition fences for &lt;list_of_ptn_cell_names&gt;.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_early_global_honor_partition_fence</a></p>                                                                                                                                                                                                                                                                           |
| route_early_global_honor_partition_pin            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                   | <p>Honor partition fences and single-entry constraint for &lt;list_of_ptn_cell_names&gt;.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_early_global_honor_partition_pin</a></p>                                                                                                                                                                                                                                                 |
| route_early_global_honor_partition_pin_guide      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                   | <p>Honor partition pin guides. Default is true</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_early_global_honor_partition_pin_guide</a></p>                                                                                                                                                                                                                                                                                      |

|                                              |                                                                                                                                                                                                                                                                                                |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_early_global_honor_power_domain        |                                                                                                                                                                                                                                                                                                |
|                                              | consider MSV constraints while routing<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_early_global_honor_power_domain</a>                                                                                            |
| route_early_global_num_tracks_per_clock_wire |                                                                                                                                                                                                                                                                                                |
|                                              | Specify special number of tracks per clock wires.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_early_global_num_tracks_per_clock_wire</a>                                                                               |
| route_early_global_reverse_direction_regions |                                                                                                                                                                                                                                                                                                |
|                                              | Reverse routing direction in the given region on the specified layer-range. Example: "(x1 y1 M1:M2 (x3 y3 x4 y4) M3:M4 ..."<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_early_global_reverse_direction_regions</a> |
| route_early_global_route_bump_nets           |                                                                                                                                                                                                                                                                                                |
|                                              | Do not ignore special nets if they are connecting bumps<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes                                                                                                                                                       |
| route_early_global_route_selected_net_only   |                                                                                                                                                                                                                                                                                                |
|                                              | Only route the nets which are selected in the DB.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_early_global_route_selected_net_only</a>                                                                            |
| route_early_global_secondary_pg              |                                                                                                                                                                                                                                                                                                |
|                                              | Specifies whether secondary PG pins needs to be routed or not<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_early_global_secondary_pg</a>                                                                           |

|                                              |                                                                                                                                                                                                                                                                  |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_early_global_secondary_pg_max_fanout   |                                                                                                                                                                                                                                                                  |
|                                              | <p>Specifies the max fanout limit for each secondary PG subnet</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_early_global_secondary_pg_max_fanout</a></p>           |
| route_early_global_stripe_layer_range        |                                                                                                                                                                                                                                                                  |
|                                              | <p>Route secondary PG pins to access to the PG stripes in the given layer range.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_early_global_stripe_layer_range</a></p> |
| route_eco_ignore_existing_route              |                                                                                                                                                                                                                                                                  |
|                                              | <p>Ignore existing route in eco, mode: {all   signal}</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_eco_ignore_existing_route</a></p>                                  |
| route_enable_mixed_size_tsv_via              |                                                                                                                                                                                                                                                                  |
|                                              | <p>Enable mixed size tsv vias in routing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_enable_mixed_size_tsv_via</a></p>                                              |
| route_enable_route_rule_si_limit_length      |                                                                                                                                                                                                                                                                  |
|                                              | <p>use MAR wire length as NDR PRL requirement for metal spacing check</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_enable_route_rule_si_limit_length</a></p>       |
| route_enforce_route_rule_on_special_net_wire |                                                                                                                                                                                                                                                                  |
|                                              | <p>enforce ndr rule on all special wire segments of specified nets</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_enforce_route_rule_on_special_net_wire</a></p>     |

|                                        |                                                                                                                                                                                                                                                                                                            |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_extra_via_enclosure              |                                                                                                                                                                                                                                                                                                            |
|                                        | <p>specify an extra via enclosure to use when connecting to block pins and special net wires</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_extra_via_enclosure</a></p>                                          |
| route_fix_clock_nets                   |                                                                                                                                                                                                                                                                                                            |
|                                        | <p>set clock nets routing status to fixed or routed</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_fix_clock_nets</a></p>                                                                                        |
| route_high_freq_constraint_groups      |                                                                                                                                                                                                                                                                                                            |
|                                        | <p>only route specified constraint groups, legal values are {net match bus pair shield}, add "or" front to control route order</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_high_freq_constraint_groups</a></p> |
| route_high_freq_match_report_file      |                                                                                                                                                                                                                                                                                                            |
|                                        | <p>specify match report file name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_high_freq_match_report_file</a></p>                                                                                              |
| route_high_freq_num_reserved_layers    |                                                                                                                                                                                                                                                                                                            |
|                                        | <p>number of layers reserved for standard cell pin access</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_high_freq_num_reserved_layers</a></p>                                                                        |
| route_high_freq_remove_floating_shield |                                                                                                                                                                                                                                                                                                            |
|                                        | <p>remove floating shield segments for high frequency nets</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_high_freq_remove_floating_shield</a></p>                                                               |

|                                    |                                                                                                                                                                                                                                                |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_high_freq_search_repair      |                                                                                                                                                                                                                                                |
|                                    | run search and repair to remove violations, legal value is one of {auto false true only}<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> auto<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_high_freq_search_repair</a> |
| route_high_freq_shield_trim_length |                                                                                                                                                                                                                                                |
|                                    | specify minimum length of shielding wire to be kept<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_high_freq_shield_trim_length</a>                                  |
| route_honor_exclusive_region       |                                                                                                                                                                                                                                                |
|                                    | honor exclusive region routing<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_honor_exclusive_region</a>                                                              |
| route_honor_power_domain           |                                                                                                                                                                                                                                                |
|                                    | honor power domain routing<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_honor_power_domain</a>                                                                     |
| route_ignore_antenna_top_cell_pin  |                                                                                                                                                                                                                                                |
|                                    | ignore antenna check on block I/O pins<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_ignore_antenna_top_cell_pin</a>                                                 |
| route_ignore_follow_pin_shapes     |                                                                                                                                                                                                                                                |
|                                    | Ignore follow pin via shapes<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_ignore_follow_pin_shapes</a>                                                             |

|                                               |                                                                                                                                                                                                  |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_interposer_allow_diagonal_trunk         |                                                                                                                                                                                                  |
|                                               | allow_diagonal_trunk<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_interposer_allow_diagonal_trunk</a>                 |
| route_interposer_control_options              |                                                                                                                                                                                                  |
|                                               | control options<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_interposer_control_options</a>                           |
| route_interposer_interlayer_shielding_layers  |                                                                                                                                                                                                  |
|                                               | interlayer shielding layers<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_interposer_interlayer_shielding_layers</a>   |
| route_interposer_interlayer_shielding_nets    |                                                                                                                                                                                                  |
|                                               | interlayer shielding nets<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_interposer_interlayer_shielding_nets</a>       |
| route_interposer_interlayer_shielding_offsets |                                                                                                                                                                                                  |
|                                               | interlayer shielding offsets<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_interposer_interlayer_shielding_offsets</a> |
| route_interposer_interlayer_shielding_widths  |                                                                                                                                                                                                  |
|                                               | interlayer shielding widths<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_interposer_interlayer_shielding_widths</a>   |

|                                                     |                                                                                                                                                                                                              |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_interposer_same_layer_shielding_net           |                                                                                                                                                                                                              |
|                                                     | same_layer_shielding_net<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_interposer_same_layer_shielding_net</a>                     |
| route_interposer_same_layer_shielding_width_spacing |                                                                                                                                                                                                              |
|                                                     | same_layer_shielding_width_spacing<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_interposer_same_layer_shielding_width_spacing</a> |
| route_interposer_trunk_routing_layers               |                                                                                                                                                                                                              |
|                                                     | trunk_routing_layers<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_interposer_trunk_routing_layers</a>                             |
| route_interposer_trunk_routing_width_spacing        |                                                                                                                                                                                                              |
|                                                     | trunk_routing_width_spacing<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_interposer_trunk_routing_width_spacing</a>               |
| route_number_fail_limit                             |                                                                                                                                                                                                              |
|                                                     | set limit for number of fails<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_number_fail_limit</a>                                      |
| route_number_info_limit                             |                                                                                                                                                                                                              |
|                                                     | set limit for number of info messages<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes                                                                                           |
| route_number_thread                                 |                                                                                                                                                                                                              |

|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | <p>set the number of processors to be used in one workstation for multi-threading</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_number_thread</a></p>                                                                                                                                                                                                                 |
| route_number_warning_limit                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                       | <p>set limit for number of warnings</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_number_warning_limit</a></p>                                                                                                                                                                                                                                                        |
| route_process_node                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                       | <p>Specify the process node</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_process_node</a></p>                                                                                                                                                                                                                                                                    |
| route_rc_extraction_corner                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                       | <p>specific which RC extraction corner to use in routing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_rc_extraction_corner</a></p>                                                                                                                                                                                                                               |
| route_relaxed_route_rule_spacing_to_power_ground_nets |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                       | <p>relax the spacing requirement from NDR spacing for the layers</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_relaxed_route_rule_spacing_to_power_ground_nets</a></p>                                                                                                                                                                                          |
| route_reserve_space_for_multi_cut                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                       | <p>Reserves space to insert multicut vias in postroute stage. This option has to be set before routing. After routing with this parameter specified, you can add double-cut vias or larger overhang using the "route_design -via_opt" command. For examples, see route_design.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_reserve_space_for_multi_cut</a></p> |



|                               |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_reverse_direction       |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>reverse routing direction in area: (lx ly ux uy [bot_lyr : top_lyr]) ... , top_lyr and bot_lyr could either layer name or layer routing id</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_reverse_direction</a></p>                                                                                                         |
| route_route_clock_nets_first  |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>route clock nets first</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_route_clock_nets_first</a></p>                                                                                                                                                                                                                        |
| route_rules                   |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>All the route_rules. It is initialized from LEF NONDEFAULTRULES section or from OA equivalent. It includes the reserved name "default" for the implicit LEF default rule or the OA LEFDefaultRouteSpec. It also includes design specific route_rules from create_route_rule in from DEF or an OA cellview.</p> <p><b>Type:</b> <a href="#">obj(route_rule)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| route_selected_net_only       |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>route selected net only</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_selected_net_only</a></p>                                                                                                                                                                                                                           |
| route_shield_crosstie_offset  |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                               | <p>Specifies the offset in terms of number of tracks for adding crossties. The default is 0 for all layers. The syntax is 'layer_name:numTrack1 layer_name2:numTrack2... '.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_shield_crosstie_offset</a></p>                                                                      |
| route_shield_length_threshold |                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                                   |                                                                                                                                                                                                                         |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | min route segment length below which there is no shielding needed<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> -1.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_shield_length_threshold</a> |
| route_shield_report_skip_status                   |                                                                                                                                                                                                                         |
|                                                   | generate report file on if a net is skipped for shielding<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_shield_report_skip_status</a>        |
| route_shield_stripe_layer_range                   |                                                                                                                                                                                                                         |
|                                                   | specify the target layer range of stripes for tie net connection<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_shield_stripe_layer_range</a>  |
| route_shield_tap_cell_insertion                   |                                                                                                                                                                                                                         |
|                                                   | insert tap cell for shielding<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_shield_tap_cell_insertion</a>                                    |
| route_shield_tap_cell_name                        |                                                                                                                                                                                                                         |
|                                                   | specify the tap cell names for shielding<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_shield_tap_cell_name</a>                               |
| route_skip_analog                                 |                                                                                                                                                                                                                         |
|                                                   | skip routing nets or pins marked + USE ANALOG in the DEF file<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_skip_analog</a>                  |
| route_special_allow_non_preferred_direction_route |                                                                                                                                                                                                                         |

|                                                  |                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | <p>allow wrong way route</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_allow_non_preferred_direction_route</a></p>                                                                                         |
| route_special_avoid_over_core_row_layer          |                                                                                                                                                                                                                                                                                                               |
|                                                  | <p>specify number of layers to avoid routing over core row area</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> max layer of standard cell pin</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_avoid_over_core_row_layer</a></p>                                 |
| route_special_block_pin_connect_ring_pin_corners |                                                                                                                                                                                                                                                                                                               |
|                                                  | <p>Extend the ring pins at the corners</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_block_pin_connect_ring_pin_corners</a></p>                                                                            |
| route_special_block_pin_route_with_pin_width     |                                                                                                                                                                                                                                                                                                               |
|                                                  | <p>Connect using pin width only</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_block_pin_route_with_pin_width</a></p>                                                                                       |
| route_special_connect_broken_core_pin            |                                                                                                                                                                                                                                                                                                               |
|                                                  | <p>Break followpin for standard cell pin or obs</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_connect_broken_core_pin</a></p>                                                                              |
| route_special_core_pin_ignore_obs                |                                                                                                                                                                                                                                                                                                               |
|                                                  | <p>ignore specified blockage when connecting follow pin</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none placement_blockage overlap_obs block_halo</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_core_pin_ignore_obs</a></p> |
| route_special_core_pin_length                    |                                                                                                                                                                                                                                                                                                               |

|                                            |                                                                                                                                                                                                                                    |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | connect core pins of at least specified length<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_core_pin_length</a>                                |
| route_special_core_pin_length_as_inst      |                                                                                                                                                                                                                                    |
|                                            | connect core pins if their lengths are the same as the instances<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_core_pin_length_as_inst</a>      |
| route_special_core_pin_max_via_scale       |                                                                                                                                                                                                                                    |
|                                            | Maximum via width and height at stripe crossover for standcell pins (%)<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_core_pin_max_via_scale</a> |
| route_special_core_pin_merge_limit         |                                                                                                                                                                                                                                    |
|                                            | gaps smaller than this variable cause followpin rails to be joined<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_core_pin_merge_limit</a>       |
| route_special_core_pin_refer_to_follow_pin |                                                                                                                                                                                                                                    |
|                                            | M2 followpin generation uses follow pin wire<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_core_pin_refer_to_follow_pin</a>                     |
| route_special_core_pin_reference_macro     |                                                                                                                                                                                                                                    |
|                                            | use specified macro as reference for followpin creation<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_core_pin_reference_macro</a>               |
| route_special_core_pin_site_rail_width     |                                                                                                                                                                                                                                    |

|                                     |                                                                                                                                                                                                                                                                                                   |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>follow pin width at layer for specified site</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_core_pin_site_rail_width</a></p>                                                                  |
| route_special_core_pin_snap_to      |                                                                                                                                                                                                                                                                                                   |
|                                     | <p>snap M2 followpin referred to M1 pin to position</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> m1_pin opt_routing_track grid half_grid</p> <p><b>Default:</b> m1_pin</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_core_pin_snap_to</a></p> |
| route_special_core_pin_stop_route   |                                                                                                                                                                                                                                                                                                   |
|                                     | <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> RowEnd CellPinEnd</p> <p><b>Default:</b> RowEnd</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_core_pin_stop_route</a></p>                                                                            |
| route_special_endcap_as_core        |                                                                                                                                                                                                                                                                                                   |
|                                     | <p>route_special treat Endcap macro as Core macro</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_endcap_as_core</a></p>                                                                         |
| route_special_extend_nearest_target |                                                                                                                                                                                                                                                                                                   |
|                                     | <p>extend nearest target so that it can be connected</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_extend_nearest_target</a></p>                                                               |
| route_special_jog_threshold_ratio   |                                                                                                                                                                                                                                                                                                   |
|                                     | <p>distance ratio between a jogging target and a straight target</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 10.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_jog_threshold_ratio</a></p>                                                    |

|                                                  |                                                                                                                                                                                                              |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_special_layer_non_preferred_direction_cost |                                                                                                                                                                                                              |
|                                                  | layer cost of non-preferred direction<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_layer_non_preferred_direction_cost</a> |
| route_special_layer_preferred_direction_cost     |                                                                                                                                                                                                              |
|                                                  | layer cost of preferred direction<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_layer_preferred_direction_cost</a>         |
| route_special_pad_pin_min_via_size               |                                                                                                                                                                                                              |
|                                                  | Minimum via width percent between pad and ring.<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 20.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_pad_pin_min_via_size</a>   |
| route_special_pad_pin_split                      |                                                                                                                                                                                                              |
|                                                  | Split width and spacing for wide pad pins<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_pad_pin_split</a>                  |
| route_special_pad_pin_split_from_center          |                                                                                                                                                                                                              |
|                                                  | split pad pin from pin center<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_pad_pin_split_from_center</a>                 |
| route_special_pad_ring_use_lef                   |                                                                                                                                                                                                              |
|                                                  | Keep user LEF input for pad rings<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_special_pad_ring_use_lef</a>                       |

|                                        |                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_special_pg_pin_as_signal         |                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>List of cell/pin information NOT extract from CPF</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_pg_pin_as_signal</a></p>                                                                                                                    |
| route_special_secondary_pin_max_gap    |                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>gaps bigger than this variable cause level shifter rail to break for them</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_secondary_pin_max_gap</a></p>                                                                                      |
| route_special_secondary_pin_rail_width |                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>width of followpin rail</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_secondary_pin_rail_width</a></p>                                                                                                                                     |
| route_special_signal_pin_as_pg         |                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>signal pins as PG pins</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_signal_pin_as_pg</a></p>                                                                                                                                              |
| route_special_split_long_via           |                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>Split vias longer than &lt;threshold&gt; into smaller vias with specified &lt;step&gt; and bottom/left en &lt;offset&gt; and vertical/horizontal &lt;height&gt;</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 0 0 -1 -1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_split_long_via</a></p> |
| route_special_target_number            |                                                                                                                                                                                                                                                                                                                                                  |
|                                        | <p>number of target to connect</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_target_number</a></p>                                                                                                                                                 |

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_special_target_search_distance      |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                           | <p>target search distance</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_target_search_distance</a></p>                                                                                                                                                                                                                             |
| route_special_time_limit                  |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                           | <p>time limit (in seconds) that the router is allowed in routing each port</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_time_limit</a></p>                                                                                                                                                                                        |
| route_special_via_connect_to_shape        |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                           | <p>Make via connection to specified shapes</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> padring ring stripe blockring blockpin coverpin noshape blockwire corewire followpin iowire</p> <p><b>Default:</b> padring ring stripe blockring blockpin coverpin noshape blockwire corewire followpin iowire</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_via_connect_to_shape</a></p> |
| route_special_via_through_to_closest_ring |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                           | <p>A via should connect only to closest ring layer when multiple rings overlap</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_via_through_to_closest_ring</a></p>                                                                                                                                                                   |
| route_special_welltap_as_endcap           |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                           | <p>route_special treat Welltap macro as Endcap macro</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_special_welltap_as_endcap</a></p>                                                                                                                                                                                                        |
| route_strict_honor_route_rule             |                                                                                                                                                                                                                                                                                                                                                                                                                                       |



|                                        |                                                                                                                                                                                                                                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>Strictly enforce non-default rules</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_strict_honor_route_rule</a></p>                                                                           |
| route_strictly_honor_1d_routing        |                                                                                                                                                                                                                                                                                            |
|                                        | <p>Strictly honor 1d routing</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_strictly_honor_1d_routing</a></p>                                                                                  |
| route_stripe_layer_range               |                                                                                                                                                                                                                                                                                            |
|                                        | <p>specify the target layer range of stripes for tie net connection</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_stripe_layer_range</a></p>                                                     |
| route_third_party_data                 |                                                                                                                                                                                                                                                                                            |
|                                        | <p>Allow third party data as input</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_third_party_data</a></p>                                                                                       |
| route_tieoff_to_shapes                 |                                                                                                                                                                                                                                                                                            |
|                                        | <p>specify the target special wire shapes or target instance pin for tie net connection: [auto stri<br/>powergroundpin]</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_tieoff_to_shapes</a></p> |
| route_top_level_design                 |                                                                                                                                                                                                                                                                                            |
|                                        | <p>specify this is a top level design</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                    |
| route_trial_max_print_ignored_pad_nets |                                                                                                                                                                                                                                                                                            |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>Number of nets to be printed connecting a pad term to a fterm without geometry</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_trial_max_print_ignored_pad_nets</a></p>                                                                                                                                                                                                                                                                                                                                                                          |
| route_trim_pull_back_distance_from_boundary |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                             | <p><a href="#">route_trim_pull_back_distance_from_boundary</a> {&lt;layer&gt;:&lt;value&gt; ...}</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_trim_pull_back_distance_from_boundary</a></p>                                                                                                                                                                                                                                                                                                                                                  |
| route_trunk_with_cluster_target_size        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                             | <p>global control of trunk routing pattern for nets with TRUNK pattern attribute. 0: nets with TRI pattern will be routed as Steiner tree; 1: default, nets will be routed as traditional trunk pattern each pin directly connects to SNET trunk separately; &gt;1: define a proximate max cluster size (number of pins), and nets will be routed as fishbone style, i.e., several pins in a column are clustered together, and then connects to SNET trunk.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_trunk_with_cluster_target_size</a></p> |
| route_types                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                             | <p>All the route_types from create_route_type command.</p> <p><b>Type:</b> <a href="#">obj(route_type)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| route_use_auto_via                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                             | <p>allow to use internal generated vias</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_use_auto_via</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| route_via_weight                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                             | <p>set weight for vias</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_via_weight</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                          |                                                                                                                                                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_with_eco           |                                                                                                                                                                                                   |
|                          | enable eco routing<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_with_eco</a>                                          |
| route_with_litho_driven  |                                                                                                                                                                                                   |
|                          | enable the litho-driven routing<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_with_litho_driven</a>                    |
| route_with_si_driven     |                                                                                                                                                                                                   |
|                          | enable si driven routing<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_with_si_driven</a>                              |
| route_with_timing_driven |                                                                                                                                                                                                   |
|                          | enable timing driven routing<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_with_timing_driven</a>                      |
| route_with_trim_metal    |                                                                                                                                                                                                   |
|                          | set cut metal dgrid for short pin extension generated<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_with_trim_metal</a> |
| route_with_via_in_pin    |                                                                                                                                                                                                   |
|                          | enclose via in pin shape<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">route_with_via_in_pin</a>                           |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_with_via_only_for_block_cell_pin        |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                               | <p>enclose via in pin shape for macro cell pins</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_with_via_only_for_block_cell_pin</a></p>                                                                                                                                                                                                               |
| route_with_via_only_for_stdcell_pin           |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                               | <p>disable planar access to standard cell pins</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">route_with_via_only_for_stdcell_pin</a></p>                                                                                                                                                                                                                   |
| rows                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                               | <p>Short-cut for [get_db current_design .rows]</p> <p><b>Type:</b> <a href="#">obj(row)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                 |
| run_abstract_abstract_blockage_cut_around_pin |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                               | <p>For std cell, we do not want cover obs on pin. So:</p> <p>If -input_cell_type {std}, setting the value to all metal and via layer name list.</p> <p>If -input_cell_type {block} or -input_cell_type {io}, setting the value to all via layer name list</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">run_abstract_abstract_blockage_cut_around_pin</a></p> |
| run_abstract_antenna_connectivity             |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                               | <p>Specifies the layers for which connectivity is to be extracted. The syntax is {&lt;layer1&gt; &lt;connect_layer&gt; &lt;layer2&gt;}. </p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">run_abstract_antenna_connectivity</a></p>                                                                                                                              |
| run_abstract_antenna_diffusion_geometry       |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>Specifies diffusion layer, followed by expression to remove gate channel area from diffusion region.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">run_abstract_antenna_diffusion_geometry</a></p>                                                                                                                                                                                                                                                                                           |
| run_abstract_antenna_gate_geometry    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                       | <p>Specifies poly layer that forms a gate, and then the layer expression to derive the gate area option also triggers the creation of antenna models.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">run_abstract_antenna_gate_geometry</a></p>                                                                                                                                                                                                                                                  |
| run_abstract_blockage_detailed_layers |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                       | <p>Specifies the layers for which a detailed blockage model is to be created. The detailed blockage model generates blockages only where there are real shapes in the cell on the layer. This option applies to IO cells and block cells, it does not apply to standard cells.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">run_abstract_blockage_detailed_layers</a></p>                                                                                                                      |
| run_abstract_boundary_layers          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                       | <p>By default this list is empty, and the prBoundary is used. The prBoundary comes from an existing GDS layer that is used to compute the LEF SIZE (and OVERLAP OBS shapes for rectilinear boundaries). If there is no explicit GDS prBoundary layer, the boundary can be computed from the bounding box of geometries on this list of layers. Note, this option is almost never correct for standard cells.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">run_abstract_boundary_layers</a></p> |
| run_abstract_cell_symmetry            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                       | <p>Specifies a property value for cell symmetry. The valid values for the property symmetry include {R0   X   Y   R90   X Y   X R90   Y R90   X Y R90}. Default {X Y}</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">run_abstract_cell_symmetry</a></p>                                                                                                                                                                                                                                          |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| run_abstract_export_lef_version        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Version of LEF to export. The default version is the same as the version of technology LEF being imported.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 5.3 5.4 5.5 5.6 5.7 5.8</p> <p><b>Default:</b> 5.7</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">run_abstract_export_lef_version</a></p>                                                                                                                                              |
| run_abstract_extract_layers_power      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Specifies the layers through which the extractor extracts each power net. This is used to reduce the search space if abstract is taking too much run time for each cell. Usually not applicable to standard cells. For blocks it is common to only extract the power nets on the top one or two layers.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">run_abstract_extract_layers_power</a></p> |
| run_abstract_extract_layers_signal     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Specifies the layers through which the extractor extracts each net. This is used to reduce the search space if abstract is taking too much run time for each cell. Usually not applicable to standard cells.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">run_abstract_extract_layers_signal</a></p>                                                                                           |
| run_abstract_extract_pin_layers_power  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Determines which of the shapes found by the extractor should be turned into pins. This argument is valid only if the associated layer, geometry specification and connectivity have been provided in the technology LEF. Usually not applicable to standard cells.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">run_abstract_extract_pin_layers_power</a></p>                                  |
| run_abstract_extract_pin_layers_signal |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Determines which of the shapes found by the extractor should be turned into pins. This argument is valid only if the associated layer, geometry specification and connectivity have been provided in the technology LEF. Usually not applicable to standard cells.</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">run_abstract_extract_pin_layers_signal</a></p> |
| run_abstract_input_cell_type             |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                          | <p>Choose the cell type being translated with abstract. The tool adjusts its settings based on the selection. Optional.</p> <p><b>Type:</b> <a href="#">enum</a><br/><b>Enum Values:</b> std io block<br/><b>Default:</b> std<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">run_abstract_input_cell_type</a></p>                                                                                                                     |
| run_abstract_input_lef_tech_file         |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                          | <p>Input technology LEF file. This is used to get all the layer names, their types (routing, cut, etc.) and the layer order for extracting connectivity. Required.</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">run_abstract_input_lef_tech_file</a></p>                                                                                                          |
| run_abstract_input_stream_layer_map_file |                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Specify the file that maps GDS layer number / data type to LEF layer names in the Virtuoso Required.</p> <p>For example, the gds file is not colorized, you could use the common virtuoso format gds layer map to stream gds file.</p> <pre>#layer name #purpose name #stream layer #stream data type V0 drawing 35 0 V0 drawing 35 235 V0 drawing 35 236 M1 drawing 15 0 M1 drawing 15 235 M1 drawing 15 236 M1 drawing 15 239</pre> <p>For example, the gds file is colorized, you could specify the mask color in gds layer map to stream gds file.</p> <pre>#layer name #purpose name #stream layer #stream data type #photo mask color #color stream V0 drawing 35 0 V0 drawing 35 235 mask1Color unlocked V0 drawing 35 236 mask2Color unlocked M1 drawing 15 0 M1 drawing 15 235 mask1Color unlocked M1 drawing 15 236 mask2Color unlocked M1 drawing 15 239 mask3Color unlocked</pre> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">run_abstract_input_stream_layer_map_file</a></p> |
| run_abstract_keep_output_files |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                | <p>Do not clean up temporary files/libraries that are created in the process after command finishes</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">run_abstract_keep_output_files</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| run_abstract_pins_analog_names |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                | <p>Identifies analog pin names based on label strings in the Abstract to set the LEF "USE ANALOG" property. Enter a list of possible names for the analog pins. This can be a list of regular expressions, each separated by a space.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">run_abstract_pins_analog_names</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



|                                |                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| run_abstract_pins_clock_names  |                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                | <p>Identifies clock pin names based on label strings in the Abstract to set the LEF "USE CLOC" property. Enter a list of possible names for the clock pins. This can be a list of regular expressions, each separated by a space.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">run_abstract_pins_clock_names</a></p> |
| run_abstract_pins_ground_names |                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                | <p>Identifies ground pins based on label strings in the Abstract to set the LEF "USE GROUND" property. Enter a list of possible names for the ground pins. This can be a list of regular expressions, each separated by a space.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">run_abstract_pins_ground_names</a></p> |
| run_abstract_pins_output_names |                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                | <p>Identifies output pins based on label strings in the GDS to set the LEF direction to OUTPUT a list of regular expressions separated by spaces.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">run_abstract_pins_output_names</a></p>                                                                                |
| run_abstract_pins_power_names  |                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                | <p>Identifies power pin names based on label strings in the Abstract to set the LEF "USE POW" property. Enter a list of possible names for the power pins. This can be a list of regular expressions, each separated by a space.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">run_abstract_pins_power_names</a></p>  |
| run_abstract_pins_text_pin_map |                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                | <p>Maps text labels to pins on nets. Specify the layer purpose pairs based on which you want the abstract generator to search for geometry of any given layer-purpose pair with a text label.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">run_abstract_pins_text_pin_map</a></p>                                    |

|                                  |                                                                                                                                                                                                                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| run_abstract_pre_abstract_script |                                                                                                                                                                                                                                                                                                            |
|                                  | <p>a skill file that will be executed in abstract before the auto-generated script is executed. Opt</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">run_abstract_pre_abstract_script</a></p>                                      |
| run_abstract_selected_cells      |                                                                                                                                                                                                                                                                                                            |
|                                  | <p>Select cells to create abstract for using regex style expressions. Optional.</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">run_abstract_selected_cells</a></p>                                                               |
| run_abstract_site_name           |                                                                                                                                                                                                                                                                                                            |
|                                  | <p>Site to be used for macros. This site will be added to all the cells.</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">run_abstract_site_name</a></p>                                                                           |
| run_abstract_verbose             |                                                                                                                                                                                                                                                                                                            |
|                                  | <p>Verbose mode with extra print.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">run_abstract_verbose</a></p>                                                                                                                   |
| safety_dcls_isolate_halo_type    |                                                                                                                                                                                                                                                                                                            |
|                                  | <p>The type of isolation halo required for dcls (dual core lock step) groups.</p> <p><b>Supported values:</b><br/><b>Type:</b> <a href="#">enum</a><br/><b>Enum Values:</b> soft hard<br/><b>Default:</b> soft<br/><b>Edit:</b> No<br/><b>Reference:</b> <a href="#">safety_dcls_isolate_halo_type</a></p> |
| safety_dcls_isolate_type         |                                                                                                                                                                                                                                                                                                            |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>The type of isolation required for dcls (dual core lock step) groups.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> soft hard</p> <p><b>Default:</b> soft</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">safety_dcls_isolate_type</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| safety_dcls_route_types        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                | <p>The type of nets on which to enforce routing rules for dcls (dual core lock step) groups. Supported values are:</p> <p>internal: this net's pins are all logically inside a single dcls group. This net cannot route outside the group boundary.</p> <p>interface: this net has pins logically inside a dcls group, and pins logically outside of that group AND logically outside of its exclusive_groups. This net cannot route over the boundary of its exclusive_groups.</p> <p>common: this net has pins logically inside two or more dcls exclusive_groups. This net cannot route inside the boundary of those groups to complete the connections, but a violation should be reported.</p> <p>top: this net's pins are all logically outside of dcls groups. This net can route over the boundary of any dcls group, but cannot route over the boundary of two or more exclusive_groups.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> internal interface common top</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">safety_dcls_route_types</a></p> |
| safety_dmr_isolate_clock       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                | <p>The type of isolation required for the clock signals driving dmr (dual modular redundancy) flip-flops.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none split_network split_net</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| safety_dmr_isolate_reset       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                | <p>The type of isolation required for the reset signals driving dmr (dual modular redundancy) flip-flops.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none split_network split_net</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| safety_dmr_isolate_scan_enable |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>The type of isolation required for the scan_enable signals driving dmr (dual modular redun flops.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none split_network split_net</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p>                                                                                                                                                               |
| safety_dmr_spacing   |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                      | <p>This is the radial spacing in any direction that should be maintained between the 2 flops in (dual modular redundancy) group. The value is measured in microns. Attributes "safety_dmr_spacing_x   safety_dmr_spacing_y" and "safety_dmr_spacing" are mutually exclusive.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> No</p>                                                                               |
| safety_dmr_spacing_x |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                      | <p>This is the horizontal spacing that should be maintained between the 2 flops in a dmr (dual modular redundancy) group. The value is measured in microns. Only one of either "safety_dmr_spacing_x" or "safety_dmr_spacing_y" needs to be met. Attributes "safety_dmr_spacing_x   safety_dmr_spacing_y" and "safety_dmr_spacing" are mutually exclusive.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> No</p> |
| safety_dmr_spacing_y |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                      | <p>This is the vertical spacing that should be maintained between the 2 flops in a dmr (dual modular redundancy) group. The value is measured in microns. Only one of either "safety_dmr_spacing_x" or "safety_dmr_spacing_y" needs to be met. Attributes "safety_dmr_spacing_x   safety_dmr_spacing_y" and "safety_dmr_spacing" are mutually exclusive.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> No</p>   |
| safety_flow_enable   |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                      | <p>Master switch to enable the implementation and analysis of the safety-mechanisms defined in the USF file..</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">safety_flow_enable</a></p>                                                                                                                                                                                 |

|                          |                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| safety_midas_enable      |                                                                                                                                                                                                                                                                                                                                                             |
|                          | <p>Enables USF (Unified Safety Format) technology from Midas.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none fmeda physical</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">safety_midas_enable</a></p>                                                  |
| safety_parity_bit_cells  |                                                                                                                                                                                                                                                                                                                                                             |
|                          | <p>The list of base_cells allowed for those flops storing the parity-bits of the parity-groups.</p> <p><b>Type:</b> <a href="#">obj(base_cell)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                    |
| safety_ser_cells         |                                                                                                                                                                                                                                                                                                                                                             |
|                          | <p>The list of base_cells allowed for those insts with an ser (soft error resilience) safety mecha<br/>This root attribute is for designs without power_domains.</p> <p><b>Type:</b> <a href="#">obj(base_cell)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">safety_ser_cells</a></p>                         |
| safety_ser_type          |                                                                                                                                                                                                                                                                                                                                                             |
|                          | <p>The type of ser (soft error resilience) safety mechanism.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> hard soft</p> <p><b>Default:</b> hard</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">safety_ser_type</a></p>                                                                  |
| safety_tmr_isolate_clock |                                                                                                                                                                                                                                                                                                                                                             |
|                          | <p>The type of isolation required for the clock signals driving tmr (triple modular redundancy) f</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none split_net split_network</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">safety_tmr_isolate_clock</a></p> |
| safety_tmr_isolate_reset |                                                                                                                                                                                                                                                                                                                                                             |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>The type of isolation required for the reset signals driving tmr (triple modular redundancy) flops.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none split_network split_net</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">safety_tmr_isolate_reset</a></p>                                                                                                                                                           |
| safety_tmr_isolate_scan_enable |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                | <p>The type of isolation required for the scan_enable signals driving tmr (triple modular redundancy) flops.</p> <p><b>Supported values:</b></p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none split_network split_net</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">safety_tmr_isolate_scan_enable</a></p>                                                                                                                                               |
| safety_tmr_spacing             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                | <p>This is the radial spacing in any direction that should be maintained between the 3 flops in (triple modular redundancy) group. The value is measured in microns. Attributes "safety_tmr_spacing_x   safety_tmr_spacing_y" and "safety_tmr_spacing" are mutually exclusive.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">safety_tmr_spacing</a></p>                                                                                 |
| safety_tmr_spacing_x           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                | <p>This is the horizontal spacing that should be maintained between the 3 flops in a tmr (triple modular redundancy) group. The value is measured in microns. Only one of either "safety_tmr_spacing_x" or "safety_tmr_spacing_y" needs to be met. Attributes "safety_tmr_spacing_x   safety_tmr_spacing_y" and "safety_tmr_spacing" are mutually exclusive.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">safety_tmr_spacing_x</a></p> |
| safety_tmr_spacing_y           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>This is the vertical spacing that should be maintained between the 3 flops in a tmr (triple modular redundancy) group. The value is measured in microns. Only one of either "safety_tmr_spacing_x" or "safety_tmr_spacing_y" needs to be met. Attributes "safety_tmr_spacing_x"   "safety_tmr_spacing_y" and "safety_tmr_spacing" are mutually exclusive.</p> <p><b>Type:</b> <a href="#">coord</a><br/> <b>Default:</b> 0.0<br/> <b>Edit:</b> No<br/> <b>Reference:</b> <a href="#">safety_tmr_spacing_y</a></p> |
| safety_tmr_well_tap_cells |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                           | <p>The well-tap cells to be placed next to the left and right edges of tmr (triple modular redundancy) flops. This root attribute is for designs without power_domains.</p> <p><b>Type:</b> <a href="#">obj(base_cell)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No<br/> <b>Reference:</b> <a href="#">safety_tmr_well_tap_cells</a></p>                                                                                                                                                                        |
| scan_chains               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                           | <p>List of all scan_chains</p> <p><b>Type:</b> <a href="#">obj(scan_chain)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                     |
| scan_segments             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                           | <p>List of all scan_segments</p> <p><b>Type:</b> <a href="#">obj(scan_segment)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                 |
| script_begin              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                           | <p>Specifies the list of alias names of the pragma 'script_script'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> dc_script_begin dc_tcl_script_begin script_begin<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">script_begin</a></p>                                                                                                                                                                                                                                                   |
| script_end                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                           | <p>Specifies the list of alias names of the pragma 'script_end'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> dc_script_end dc_tcl_script_end script_end<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">script_end</a></p>                                                                                                                                                                                                                                                              |

|                                       |                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| script_search_path                    |                                                                                                                                                                                                                                                                                                                                                                   |
|                                       | <p>The variable provides a set of directories for the software to search for files that sourced using the Tcl 'source' command. The software will search in each search path directory for the specified file.</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> .<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">script_search_path</a></p> |
| selected                              |                                                                                                                                                                                                                                                                                                                                                                   |
|                                       | <p>The list of currently selected objects. It can be set to a list of objects as long as they are selected in the design.</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">selected</a></p>                                                                                               |
| set_db_verbose                        |                                                                                                                                                                                                                                                                                                                                                                   |
|                                       | <p>Specifies if additional messages are issued describing what attribute is being set on what object.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">set_db_verbose</a></p>                                                                                                            |
| setup_views                           |                                                                                                                                                                                                                                                                                                                                                                   |
|                                       | <p>Returns the information about the setup analysis views in the design.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">analysis_view</a>)*<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                   |
| si_accumulated_small_aggressor_factor |                                                                                                                                                                                                                                                                                                                                                                   |
|                                       | <p>Specifies the multiplication factor that controls the effect of the cap mode accumulated small attacker.</p> <p><b>Type:</b> <a href="#">double</a><br/><b>Default:</b> 1.0<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">si_accumulated_small_aggressor_factor</a></p>                                                                               |
| si_accumulated_small_aggressor_mode   |                                                                                                                                                                                                                                                                                                                                                                   |



|                                          |                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Specifies the multiplication factor that controls the effect of the cap mode accumulated small attacker.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> cap current zero_mean numerical</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_accumulated_small_aggressor_mode</a></p> |
| si_accumulated_small_aggressor_threshold |                                                                                                                                                                                                                                                                                                                                                    |
|                                          | <p>Specifies whether to use the current-matching based method for accumulated small attacker.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 10.01</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_accumulated_small_aggressor_threshold</a></p>                                                                |
| si_aggressor_alignment                   |                                                                                                                                                                                                                                                                                                                                                    |
|                                          | <p>Specifies the attacker alignment.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> path path_overlap timing_aware_edge input_arrival</p> <p><b>Default:</b> path</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_aggressor_alignment</a></p>                                                                 |
| si_clock_synchronicity                   |                                                                                                                                                                                                                                                                                                                                                    |
|                                          | <p>Specifies the value above which the clock delta delay should be calculated.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> asynchronous synchronous</p> <p><b>Default:</b> synchronous</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_clock_synchronicity</a></p>                                         |
| si_delay_clock_delta_threshold           |                                                                                                                                                                                                                                                                                                                                                    |
|                                          | <p>Specifies the value above which the clock delta delay should be calculated.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_delay_clock_delta_threshold</a></p>                                                                                          |
| si_delay_delta_annotation_mode           |                                                                                                                                                                                                                                                                                                                                                    |

|                                       |                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>Calculates the delay for each arc (cell arc and net arc) in a path.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> lumpedOnNet arc</p> <p><b>Default:</b> arc</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_delay_delta_annotation_mode</a></p>       |
| si_delay_delta_threshold              |                                                                                                                                                                                                                                                                                                 |
|                                       | <p>delta delay analysis threshold</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_delay_delta_threshold</a></p>                                                                                          |
| si_delay_double_switching             |                                                                                                                                                                                                                                                                                                 |
|                                       | <p>Specifies type for which double switching analysis will happen</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> none clock data_sequential all</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_delay_double_switching</a></p> |
| si_delay_enable_double_clocking_check |                                                                                                                                                                                                                                                                                                 |
|                                       | <p>Enables double clocking check in AAE.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_delay_enable_double_clocking_check</a></p>                                 |
| si_delay_enable_logical_correlation   |                                                                                                                                                                                                                                                                                                 |
|                                       | <p>Enables the AAE-SI logical correlation.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_delay_enable_logical_correlation</a></p>                                 |
| si_delay_enable_report                |                                                                                                                                                                                                                                                                                                 |

|                                         |                                                                                                                                                                                                                                                        |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Enables SI delay reports.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_delay_enable_report</a></p>                                                         |
| si_delay_separate_on_data               |                                                                                                                                                                                                                                                        |
|                                         | <p>Separates the delta delay on data (delta delay is always separated for clock).</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_delay_separate_on_data</a></p> |
| si_enable_bus_attacker_correlation      |                                                                                                                                                                                                                                                        |
|                                         | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_enable_bus_attacker_correlation</a></p>                                                                              |
| si_enable_drv_with_delta_slew           |                                                                                                                                                                                                                                                        |
|                                         | <p>Enables SI slews reporting.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_enable_drv_with_delta_slew</a></p>                                                |
| si_enable_glitch_overshoot_undershoot   |                                                                                                                                                                                                                                                        |
|                                         | <p>Enables overshoot/undershoot analysis and reporting.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_enable_glitch_overshoot_undershoot</a></p>               |
| si_enable_glitch_propagation            |                                                                                                                                                                                                                                                        |
|                                         | <p>Enables glitch propagation and driver weakening through single stage cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_enable_glitch_propagation</a></p> |
| si_enable_glitch_propagation_spice_deck |                                                                                                                                                                                                                                                        |

|                                               |                                                                                                                                                                                                                                                                      |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | <p>Enables glitch propagation in spice deck.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_enable_glitch_propagation_spice_deck</a></p>                                      |
| si_enable_logical_correlation_for_bus         |                                                                                                                                                                                                                                                                      |
|                                               | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                        |
| si_enable_two_stage_driver_weakening          |                                                                                                                                                                                                                                                                      |
|                                               | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_enable_two_stage_driver_weakening</a></p>                                                                                           |
| si_enable_user_defined_bus_merging            |                                                                                                                                                                                                                                                                      |
|                                               | <p>Enables bus merging feature - if a net is in two groups, then the groups are merged</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_enable_user_defined_bus_merging</a></p> |
| si_enable_virtual_attacker_constituent_report |                                                                                                                                                                                                                                                                      |
|                                               | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_enable_virtual_attacker_constituent_report</a></p>                                                                                 |
| si_glitch_accuracy_level                      |                                                                                                                                                                                                                                                                      |
|                                               | <p>Specifies the accuracy levels for glitch analysis</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_glitch_accuracy_level</a></p>                                                  |
| si_glitch_analysis_type                       |                                                                                                                                                                                                                                                                      |

|                                                         |                                                                                                                                                                                                           |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | <b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> source full_propagation<br><b>Default:</b> source<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_glitch_analysis_type</a>               |
| si_glitch_constrained_input_threshold_failure_point     |                                                                                                                                                                                                           |
|                                                         | <b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> input both<br><b>Default:</b> input<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_glitch_constrained_input_threshold_failure_point</a> |
| si_glitch_enable_dynamic_receiver_peak_limits           |                                                                                                                                                                                                           |
|                                                         | <b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_glitch_enable_dynamic_receiver_peak_limits</a>                                         |
| si_glitch_enable_report                                 |                                                                                                                                                                                                           |
|                                                         | Enables AAE SI glitch analysis.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_glitch_enable_report</a>                            |
| si_glitch_input_full_propagation_threshold              |                                                                                                                                                                                                           |
|                                                         | <b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.75<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_glitch_input_full_propagation_threshold</a>                                           |
| si_glitch_input_full_propagation_voltage_high_threshold |                                                                                                                                                                                                           |
|                                                         | <b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.75<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_glitch_input_full_propagation_voltage_high_threshold</a>                              |

|                                                        |                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| si_glitch_input_full_propagation_voltage_low_threshold |                                                                                                                                                                                                                                                                               |
|                                                        | <b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.75<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_glitch_input_full_propagation_voltage_low_threshold</a>                                                                                                   |
| si_glitch_input_threshold                              |                                                                                                                                                                                                                                                                               |
|                                                        | <p>Specifies the input glitch failure threshold value during AAE analysis.</p> <b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.4<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_glitch_input_threshold</a>                                                  |
| si_glitch_input_voltage_high_threshold                 |                                                                                                                                                                                                                                                                               |
|                                                        | <p>Specifies the input glitch vh failure threshold value during AAE analysis.</p> <b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.4<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_glitch_input_voltage_high_threshold</a>                                  |
| si_glitch_input_voltage_low_threshold                  |                                                                                                                                                                                                                                                                               |
|                                                        | <p>Specifies the input glitch vl failure threshold value during AAE analysis.</p> <b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.4<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_glitch_input_voltage_low_threshold</a>                                   |
| si_glitch_receiver_clock_peak_limit                    |                                                                                                                                                                                                                                                                               |
|                                                        | <p>Defines the glitch check limit for clock inputs in receiver peak mode. This parameter is spec a ratio of vdd.</p> <b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.05<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_glitch_receiver_clock_peak_limit</a> |
| si_glitch_receiver_latch_peak_limit                    |                                                                                                                                                                                                                                                                               |

|                                           |                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p>Defines the glitch check limit for latch inputs in receiver peak mode. This parameter is specified as a ratio of vdd.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_glitch_receiver_latch_peak_limit</a></p> |
| si_glitch_receiver_peak_limit             |                                                                                                                                                                                                                                                                                                         |
|                                           | <p>Defines the receiver peak/glitch peak limit at all nodes. The value specified for this parameter is a ratio of vdd.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.15</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_glitch_receiver_peak_limit</a></p>        |
| si_individual_aggressor_clock_threshold   |                                                                                                                                                                                                                                                                                                         |
|                                           | <p>Sets the glitch tolerance for individual attacker nets on victim clock nets.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.015</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_individual_aggressor_clock_threshold</a></p>                                    |
| si_individual_aggressor_simulation_filter |                                                                                                                                                                                                                                                                                                         |
|                                           | <p>Enables electrical filtering based on attackers' simulated glitch peaks.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_individual_aggressor_simulation_filter</a></p>                                         |
| si_individual_aggressor_threshold         |                                                                                                                                                                                                                                                                                                         |
|                                           | <p>AAE virtual attacker threshold</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.015</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_individual_aggressor_threshold</a></p>                                                                                        |
| si_max_bus_attacker_count                 |                                                                                                                                                                                                                                                                                                         |
|                                           | <p>Max number of bus attacker per victim net</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 999</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                             |
| si_max_virtual_attacker_constituents      |                                                                                                                                                                                                                                                                                                         |

|                                |                                                                                                                                                                                                                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <b>Type:</b> <a href="#">int</a><br><b>Default:</b> 5<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_max_virtual_attacker_constituents</a>                                                                                                                            |
| si_nonlinear_aggressor_slew    |                                                                                                                                                                                                                                                                                |
|                                | <p>Uses a non-linear attacker. If set to false , it uses a linear attacker.</p> <b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> true false<br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_nonlinear_aggressor_slew</a>               |
| si_num_iteration               |                                                                                                                                                                                                                                                                                |
|                                | <p>Specifies the maximum number of timing window iterations that should be performed during delay analysis.</p> <b>Type:</b> <a href="#">int</a><br><b>Default:</b> 2<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_num_iteration</a>                                |
| si_pessimistic_mode            |                                                                                                                                                                                                                                                                                |
|                                | <p>Pessimistic Mode Settings</p> <b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> advanced advanced_node reduced_pessimism increased_pessimism high_pessimism<br><b>Default:</b> advanced<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_pessimistic_mode</a> |
| si_reselection                 |                                                                                                                                                                                                                                                                                |
|                                | <p>Specifies the SI reselection criteria during timing window iterations.</p> <b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> delta_delay slack xcap_ratio<br><b>Default:</b> slack<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">si_reselection</a>           |
| si_reselection_delay_threshold |                                                                                                                                                                                                                                                                                |



|                                         |                                                                                                                                                                                                                                                  |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Allows to change the delay reselection criteria.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1e-11</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_reselection_delay_threshold</a></p>                  |
| si_reselection_hold_slack               |                                                                                                                                                                                                                                                  |
|                                         | <p>Specifies the slack reselection threshold for hold.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1e-12</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_reselection_hold_slack</a></p>                   |
| si_reselection_setup_slack              |                                                                                                                                                                                                                                                  |
|                                         | <p>Specifies the slack reselection threshold for setup.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> -1e-12</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_reselection_setup_slack</a></p>                 |
| si_secondary_attacker_decoupling_factor |                                                                                                                                                                                                                                                  |
|                                         | <p>Specifies decoupling factor for secondary aggressors.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_secondary_attacker_decoupling_factor</a></p>      |
| si_skip_noise_model_check               |                                                                                                                                                                                                                                                  |
|                                         | <p>Specifies a list of pins for which noise model check will not take place.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_skip_noise_model_check</a></p> |
| si_skip_timing_window                   |                                                                                                                                                                                                                                                  |
|                                         | <p>Specifies the list of nets for which timing window is to be skipped.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_skip_timing_window</a></p>          |
| si_switch_probability                   |                                                                                                                                                                                                                                                  |

|                                                 |                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>Sets the attacker switching probability.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.3</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_switch_probability</a></p>                                                                                              |
| si_unconstrained_net_use_infinite_timing_window |                                                                                                                                                                                                                                                                                                           |
|                                                 | <p>Specifies that unconstrained nets will have an infinite timing window.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false</p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_unconstrained_net_use_infinite_timing_window</a></p> |
| si_use_infinite_timing_window                   |                                                                                                                                                                                                                                                                                                           |
|                                                 | <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_use_infinite_timing_window</a></p>                                                                                                |
| si_view_specific_settings                       |                                                                                                                                                                                                                                                                                                           |
|                                                 | <p>Spicifies a global variable value or macro settings for particular view using {viewName:eng_global macro:value}</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                       |
| si_virtual_inter_die_attacker_slew              |                                                                                                                                                                                                                                                                                                           |
|                                                 | <p>This setting specifies the virtual inter die attacker slew value, default is 25ps</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 2.5e-11</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">si_virtual_inter_die_attacker_slew</a></p>                                    |
| sites                                           |                                                                                                                                                                                                                                                                                                           |
|                                                 | <p>All the sites defined in LEF or OA.</p> <p><b>Type:</b> <a href="#">obj(site)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                |
| skew_groups                                     |                                                                                                                                                                                                                                                                                                           |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>list of skew_group</p> <p><b>Type:</b> <a href="#">obj(skew_group)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">skew_groups</a></p>                                                                                                                                                                                                                                                                                                                                                              |
| socv_library             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                          | <p>To specify SOCV library.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">socv_library</a></p>                                                                                                                                                                                                                                                                                                                                                                |
| soft_stack_size_limit    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                          | <p>Soft stacksize limit in mbytes. When the tool is launched, auto-enlarge the soft stacksize limit to 0.2%RAM. If 0.2%RAM is larger than hard limit, set to the hard limit.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> 8</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">soft_stack_size_limit</a></p>                                                                                                                                                                                                       |
| source_continue_on_error |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                          | <p>Enable/Disable 'source' to continue the script on TCL_ERROR. If true, the source command ignores Tcl errors, and continues processing the script files. If false, a Tcl error will halt the script.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">source_continue_on_error</a></p>                                                                                                                                                                        |
| source_echo_filename     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                          | <p>If this is true, and verbose 'source' command logging is off (e.g. source_verbose is false), the 'source' command will be logged with 'Sourcing &lt;filename&gt;'. This can be useful to find which source files are calling other source files, or to log all the startup Tcl files that are loaded even though verbose logging is off. Type 'man log_file' for more details.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">source_echo_filename</a></p> |
| source_verbose           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Writes out each command inside the source script to the log and cmd files. Type 'man log_f more details.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">source_verbose</a></p>                                                                                                                                                                                       |
| source_verbose_line_length_limit |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                  | <p>Limits the command printed to stdout and the .log file to the specified number of characters verbose script source. This is normally only needed if you log commands "post Tcl expansi when long Tcl lists may occur in the command args. Type 'man log_file' for more details.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> -1<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">source_verbose_line_length_limit</a></p> |
| st_launch_wait_time              |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                  | <p>Maximum waiting time in minute for super thread launching.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 10<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">st_launch_wait_time</a></p>                                                                                                                                                                                                                                   |
| stack_via_rules                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                  | <p>The stack_via_rules available in this design from the LEF or create_stack_via_rules.</p> <p><b>Type:</b> <a href="#">obj(stack_via_rule)*</a><br/> <b>Default:</b> {}<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                    |
| startup_done                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                  | <p>dummy attribute, always return true in case of innovus</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                |
| stim_auto_mapping                |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                  | <p>Enable the auto mapping feature. Default its disabled (0), verilog(1), vhdl(2).</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                          |
| stim_leaf_modules                |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                             |                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>List of leaf level modules in simulation which have hierarchies in synthesis.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                |
| stim_min_mda_annotation     |                                                                                                                                                                                                                                  |
|                             | <p>Sets minimum annotation percentage for MDA nets.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 50.0</p> <p><b>Edit:</b> Yes</p>                                                                           |
| stim_name_map_file          |                                                                                                                                                                                                                                  |
|                             | <p>File name for dumping mapping between stim and netlist names using 'read_stimulus - export_map_file_data'.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> %w/stim_name_map.rpt</p> <p><b>Edit:</b> Yes</p> |
| stim_phy_child_host_setup   |                                                                                                                                                                                                                                  |
|                             | <p>Set multi host environment for PHY in Joules.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                |
| stim_phy_connection_timeout |                                                                                                                                                                                                                                  |
|                             | <p>Set connection timeout environment for PHY in Joules in seconds.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p>                                                               |
| stim_phy_host_setup         |                                                                                                                                                                                                                                  |
|                             | <p>Set multi host environment for PHY in Joules.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                |
| stim_phy_thread_count       |                                                                                                                                                                                                                                  |
|                             | <p>Set multi thread environment for PHY in Joules.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 8</p> <p><b>Edit:</b> Yes</p>                                                                                  |

|                              |                                                                                                                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| stim_track_dump_off_regions  |                                                                                                                                                                                                                               |
|                              | <p>Track and account for the dump off regions in Simulation. Currently this flow is supported on FSDB and VCD.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                  |
| support_internal_pg_pins     |                                                                                                                                                                                                                               |
|                              | <p>Support internal power and ground pins.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">support_internal_pg_pins</a></p>                       |
| support_master_slave_flop    |                                                                                                                                                                                                                               |
|                              | <p>Support master slave flop library cell.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">support_master_slave_flop</a></p>                     |
| support_multi_seq_elements   |                                                                                                                                                                                                                               |
|                              | <p>Support library cell having multiple sequential elements.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">support_multi_seq_elements</a></p>   |
| support_multi_seq_scan_latch |                                                                                                                                                                                                                               |
|                              | <p>Support scan latch library cell defined using 2 latches.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">support_multi_seq_scan_latch</a></p> |
| support_ui_units             |                                                                                                                                                                                                                               |
|                              | <p>Support for ui_units_timing and ui_units_capacitance attributes.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">support_ui_units</a></p>     |

|                                          |                                                                                                                                                                                                                                                                                              |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| syn_exact_match_sequential_async_control |                                                                                                                                                                                                                                                                                              |
|                                          | <p>Do not tie off (degenerate) asynchronous controls of sequential cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">syn_exact_match_sequential_async_control</a></p>                                |
| syn_exact_match_sequential_sync_control  |                                                                                                                                                                                                                                                                                              |
|                                          | <p>Forces the synchronous inputs of flop-flops to be mapped to corresponding library pins where possible.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">syn_exact_match_sequential_sync_control</a></p> |
| synthesis_off_command                    |                                                                                                                                                                                                                                                                                              |
|                                          | <p>Specifies the list of alias names of the 'synthesis_off' pragma.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> translate_off synthesis_off</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">synthesis_off_command</a></p>                                 |
| synthesis_on_command                     |                                                                                                                                                                                                                                                                                              |
|                                          | <p>Specifies the list of alias names of the 'synthesis_on' pragma.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> translate_on synthesis_on</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">synthesis_on_command</a></p>                                     |
| target_library                           |                                                                                                                                                                                                                                                                                              |
|                                          | <p>Target library for technology mapping.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">target_library</a></p>                                                                                           |
| tcl_error_info                           |                                                                                                                                                                                                                                                                                              |
|                                          | <p>If user set the value to true, application can print following error info if meet a tcl error.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">tcl_error_info</a></p>                                  |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tcl_partial_cmd_argument_matching | <p>Use this switch to control command argument partial matching as well as enum value partial matching. When it's turned on, argument '-opt' will match '-option' if no other argument name begins with -opt. Default is 'quiet', which means to silently allow partial-matching when there is ambiguity. 'warn' means allow partial-matching with a warning message. 'error' means to disallow partial-matching, and error out.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> quiet warn error<br/> <b>Default:</b> quiet<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">tcl_partial_cmd_argument_matching</a></p> |
| tcl_report_header_with_value      | <p>If user set the value to true, print the Tcl variable value in the report header.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">tcl_report_header_with_value</a></p>                                                                                                                                                                                                                                                                                                                                                                                            |
| tcl_return_display_length_limit   | <p>Limits the number of characters displayed to stdout and the .log file. The Tcl return value is not affected--only the display is truncated. A value of -1 indicates that there is no display limit. It is useful to limit the output to the display of any command that can return long lists of object values.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 10000<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">tcl_return_display_length_limit</a></p>                                                                                                                                                            |
| tech_db_units                     | <p>Database units per micron from LEF or OA techfile, unless init_min_dbu_per_micron is large.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No<br/> <b>Reference:</b> <a href="#">tech_db_units</a></p>                                                                                                                                                                                                                                                                                                                                                                                                      |
| tech_finfet_grid_direction        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



|                             |                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>An enum value indicating the direction of the finfet grid from the LEF FINFET statement or C</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> vertical horizontal</p> <p><b>Default:</b> vertical</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">tech_finfet_grid_direction</a></p>                                                                         |
| tech_finfet_grid_offset     |                                                                                                                                                                                                                                                                                                                                                                                                 |
|                             | <p>The finfet grid offset from 0 in microns from the LEF FINFET statement or OA.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">tech_finfet_grid_offset</a></p>                                                                                                                                         |
| tech_finfet_grid_pitch      |                                                                                                                                                                                                                                                                                                                                                                                                 |
|                             | <p>Returns a single floating point number indicating the pitch of the FINFET grid as described</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">tech_finfet_grid_pitch</a></p>                                                                                                                            |
| tech_inst_mask_shift_layers |                                                                                                                                                                                                                                                                                                                                                                                                 |
|                             | <p>Ordered list of layer pointers to layers that allow instance mask shifting. This is the list of lay will show up in the DEF COMPONENTMASK statement for layers that can have mask value shifted in standard cells.</p> <p><b>Type:</b> <a href="#">obj(layer)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">tech_inst_mask_shift_layers</a></p> |
| tech_mfg_grid               |                                                                                                                                                                                                                                                                                                                                                                                                 |
|                             | <p>Manufacturing grid from LEF or OA techfile. Note, this is the default value for each layer, bu be overridden for specific layers. In some technologies the mfg_grid for higher metal layers coarser than for the lower layers.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> <p><b>Reference:</b> <a href="#">tech_mfg_grid</a></p>         |
| techs                       |                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            | <p>All of the tech objects in the DB..</p> <p><b>Type:</b> <a href="#">obj(tech)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                     |
| test_clock_domains                                         |                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                            | <p>List of all test_clock_domains</p> <p><b>Type:</b> <a href="#">obj(test_clock_domain)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                             |
| test_clocks                                                |                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                            | <p>List of all test_clocks</p> <p><b>Type:</b> <a href="#">obj(test_clock)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                           |
| test_signals                                               |                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                            | <p>List of all test_signals</p> <p><b>Type:</b> <a href="#">obj(test_signal)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                         |
| texts                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                            | <p>Short-cut to all the text objects in the current_design.</p> <p><b>Type:</b> <a href="#">obj(text)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                 |
| timing_all_registers_filter_clock_pins_by_clock            |                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                            | <p>When enabled, and -clocks/-clocks_rise/-clocks_fall is specified along with -clock_pins to all_registers, command will filter and return only those clock_pins where atleast 1 clock from specified list arrives</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_all_registers_filter_clock_pins_by_clock</a></p> |
| timing_all_registers_identify_macros_include_is_macro_cell |                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                          |                                                                                                                                                                                                                                                                                             |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Enable to include cells as macro cells if is_macro_cell is true in library</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_all_registers_identify_macros_include_is_macro_cell</a></p>          |
| timing_all_registers_include_icg_cells   |                                                                                                                                                                                                                                                                                             |
|                                          | <p>Controls whether to include ICG cells as register in all_registers -edge_triggered</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_all_registers_include_icg_cells</a></p>                      |
| timing_allow_input_delay_on_clock_source |                                                                                                                                                                                                                                                                                             |
|                                          | <p>When set to true allows you to apply input delay constraints on a pin where a clock was pre asserted</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_allow_input_delay_on_clock_source</a></p> |
| timing_analysis_aging                    |                                                                                                                                                                                                                                                                                             |
|                                          | <p>Aging Analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                         |
| timing_analysis_aocv                     |                                                                                                                                                                                                                                                                                             |
|                                          | <p>AOCV Analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_analysis_aocv</a></p>                                                                                                            |
| timing_analysis_async_checks             |                                                                                                                                                                                                                                                                                             |
|                                          | <p>Async checks</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> async no_async async_only auto</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_analysis_async_checks</a></p>                                            |
| timing_analysis_case_analysis            |                                                                                                                                                                                                                                                                                             |

|                                        |                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>caseAnalysis<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_analysis_case_analysis</a></p>                                                                                                                   |
| timing_analysis_check_type             |                                                                                                                                                                                                                                                                                                   |
|                                        | <p>Report Violations for hold, setup<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> setup hold<br/> <b>Default:</b> setup<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_analysis_check_type</a></p>                                                            |
| timing_analysis_clock_gating           |                                                                                                                                                                                                                                                                                                   |
|                                        | <p>clockGatingCheck<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_analysis_clock_gating</a></p>                                                                                                                |
| timing_analysis_clock_net_marking_mode |                                                                                                                                                                                                                                                                                                   |
|                                        | <p>clkNetsMarking<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> before_constant_propagation after_constant_propagation<br/> <b>Default:</b> before_constant_propagation<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_analysis_clock_net_marking_mode</a></p> |
| timing_analysis_clock_propagation_mode |                                                                                                                                                                                                                                                                                                   |
|                                        | <p>Timing arc attributes effect on timing analysis<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> sdc_control forced_ideal<br/> <b>Default:</b> sdc_control<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_analysis_clock_propagation_mode</a></p>              |
| timing_analysis_clock_source_paths     |                                                                                                                                                                                                                                                                                                   |
|                                        | <p>clkSrcPath<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_analysis_clock_source_paths</a></p>                                                                                                                |

|                                            |                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_analysis_cppr                       |                                                                                                                                                                                                                                                                                                                                                              |
|                                            | <p>Removes pessimism from clock paths that have a portion of the clock network in common between the clock source and clock destination paths</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> both none setup hold</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_analysis_cppr</a></p> |
| timing_analysis_enable_inter_power_domain  |                                                                                                                                                                                                                                                                                                                                                              |
|                                            | <p>Inter Power Domain Analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                             |
| timing_analysis_enable_robustness_mode     |                                                                                                                                                                                                                                                                                                                                                              |
|                                            | <p>Timing Robustness Analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                              |
| timing_analysis_engine                     |                                                                                                                                                                                                                                                                                                                                                              |
|                                            | <p>Statistical or Static or Rql</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> statistical static rql</p> <p><b>Default:</b> static</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_analysis_engine</a></p>                                                                                                         |
| timing_analysis_honor_active_logic_view    |                                                                                                                                                                                                                                                                                                                                                              |
|                                            | <p>Honor Active Logic View</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_analysis_honor_active_logic_view</a></p>                                                                                                                                                |
| timing_analysis_multi_input_switching_mode |                                                                                                                                                                                                                                                                                                                                                              |

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Specify MIS mode: 0,Simultaneous input switching is not considered 1,MIS analysis consid<br/>for which multiple inputs of a cell tied together (tied-inputs) 2,MIS analysis consider both tie<br/>inputs and inputs of a cells switching independent to each other 3,derate based MIS analys</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_analysis_multi_input_switching_mode</a></p> |
| timing_analysis_precision_ps             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                          | <p>Control delay precision 0.1ps or 0.01ps</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.01<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_analysis_precision_ps</a></p>                                                                                                                                                                                                                                                              |
| timing_analysis_self_loops_paths_no_skew |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                          | <p>Eliminates clock skew due to clock uncertainty for a path starting and ending at the same re<br/>If the clock skew is not eliminated, the timing for such paths is pessimistic</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_analysis_self_loops_paths_no_skew</a></p>                                                                                                            |
| timing_analysis_socv                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                          | <p>SOCV Analysis</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_analysis_socv</a></p>                                                                                                                                                                                                                                                                                                 |
| timing_analysis_type                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                          | <p>Single or BCWC or OCV TimingAnalysis type</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> single best_case_worst_case ocv<br/> <b>Default:</b> best_case_worst_case<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_analysis_type</a></p>                                                                                                                                                                                             |
| timing_aocv_analysis_mode                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                    |                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>Controls handling of AOCV analysis modes</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> launch_capture clock_only separate_data_clock combine_launch_capture</p> <p><b>Default:</b> launch_capture</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_aocv_analysis_mode</a></p> |
| timing_aocv_chip_size                              |                                                                                                                                                                                                                                                                                                                           |
|                                                    | <p>Specifies the diagonal length of the chip, in microns</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1e+30</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_aocv_chip_size</a></p>                                                                                               |
| timing_aocv_core_size                              |                                                                                                                                                                                                                                                                                                                           |
|                                                    | <p>Specifies the diagonal length of the core area, in microns</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1e+30</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_aocv_core_size</a></p>                                                                                          |
| timing_aocv_derate_mode                            |                                                                                                                                                                                                                                                                                                                           |
|                                                    | <p>Controls the AOCV derating mode</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> aocv_multiplicative aocv_additive</p> <p><b>Default:</b> aocv_multiplicative</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_aocv_derate_mode</a></p>                                          |
| timing_aocv_slack_threshold                        |                                                                                                                                                                                                                                                                                                                           |
|                                                    | <p>Analyze aocv slack based at specified threshold</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                      |
| timing_aocv_stage_count_update_on_timing_reset     |                                                                                                                                                                                                                                                                                                                           |
|                                                    | <p>When set to true, the aocv stage counts are recalculated when timing is reset.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_aocv_stage_count_update_on_timing_reset</a></p>                                               |
| timing_apply_check_derate_to_external_output_delay |                                                                                                                                                                                                                                                                                                                           |

|                                                              |                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                              | <p>Applies check derate to output external delay</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_apply_check_derate_to_external_output_delay</a></p>                                                                                                              |
| timing_apply_default_primary_input_assertion                 |                                                                                                                                                                                                                                                                                                                                                             |
|                                                              | <p>When set to true, primary input and bidirectional ports that do not have an explicit arrival time specified are provided a default arrival time</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_apply_default_primary_input_assertion</a></p>                   |
| timing_apply_exceptions_to_data_check_related_pin            |                                                                                                                                                                                                                                                                                                                                                             |
|                                                              | <p>When set to true, any false path assertion which blocks the data path to the related (-from) pin the data-to-data check also causes the check to be disabled</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_apply_exceptions_to_data_check_related_pin</a></p> |
| timing_apply_setup_hold_exceptions_to_data_check_related_pin |                                                                                                                                                                                                                                                                                                                                                             |
|                                                              | <p>When set to false, application of path exception with only -setup/hold on reference path does not affect signal path</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_apply_setup_hold_exceptions_to_data_check_related_pin</a></p>                              |
| timing_cap_unit                                              |                                                                                                                                                                                                                                                                                                                                                             |
|                                                              | <p>If set, this value is passed to set_library_unit -cap to set the capacitance units used in timing and .sdc files and timing reports. Legal values are 1pf and 1ff.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_cap_unit</a></p>                             |
| timing_case_analysis_for_icg_propagation                     |                                                                                                                                                                                                                                                                                                                                                             |



|                                                        |                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                        | <p>Determines whether constant propagation continues through integrated clock gating (ICG)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false require_seq_prop always case_only constant_only</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_case_analysis_for_icg_propagation</a></p> |
| timing_case_analysis_for_sequential_propagation        |                                                                                                                                                                                                                                                                                                                                                                 |
|                                                        | <p>When set to true, calculates constants on the outputs of sequential elements</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true 0 1 case_only constant_only</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_case_analysis_for_sequential_propagation</a></p>                    |
| timing_case_analysis_propagation                       |                                                                                                                                                                                                                                                                                                                                                                 |
|                                                        | <p>Overrides sequential and ICG propagation behaviors.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_case_analysis_propagation</a></p>                                                                                                                               |
| timing_check_timing_report_all_checks                  |                                                                                                                                                                                                                                                                                                                                                                 |
|                                                        | <p>Enables reporting of all checks performed by check_timing</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_check_timing_report_all_checks</a></p>                                                                                                                   |
| timing_check_timing_signal_level_high_to_low_threshold |                                                                                                                                                                                                                                                                                                                                                                 |
|                                                        | <p>Threshold for warnig about difference in signal level between hight_drv and low_rcv</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_check_timing_signal_level_high_to_low_threshold</a></p>                                                                        |
| timing_check_timing_signal_level_low_to_high_threshold |                                                                                                                                                                                                                                                                                                                                                                 |
|                                                        | <p>Threshold for warnig about difference in signal level between low_drv and high_rcv</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_check_timing_signal_level_low_to_high_threshold</a></p>                                                                         |

|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_clock_phase_propagation                                  |                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                 | <p>Lets you select appropriate clock phases at register clock pins when both positive and negative phases of the same clock signal are seen in the clock network</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> positive negative both</p> <p><b>Default:</b> both</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_clock_phase_propagation</a></p> |
| timing_clock_pulse_width_sensitivity_checks_threshold           |                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                 | <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                               |
| timing_clock_source_paths_unconstrained_mark_clock_used_as_data |                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                 | <p>global is set to true, is_clock_used_as_data returns true for unconstrained clock source net</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_clock_source_paths_unconstrained_mark_clock_used_as_data</a></p>                                                                                   |
| timing_clock_source_use_driving_cell                            |                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                 | <p>When set to false, the create_clock command for output pins of cells uses default slew instead of the propagated slew from the primary input ports</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_clock_source_use_driving_cell</a></p>                                                        |
| timing_clock_uncertainty_from_to_precedence                     |                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                 | <p>Gives higher precedence to clock-to-clock uncertainty when set to true</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_clock_uncertainty_from_to_precedence</a></p>                                                                                                                            |
| timing_collection_all_fanin_fanout_traversal_mode               |                                                                                                                                                                                                                                                                                                                                                                                             |

|                                                                   |                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                   | <p>Controls the way software processes netlist for all_fanin/all_fanout command processing</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> hierarchical flat</p> <p><b>Default:</b> flat</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_collection_all_fanin_fanout_traversal_mode</a></p> |
| timing_collection_result_display_limit                            |                                                                                                                                                                                                                                                                                                                                     |
|                                                                   | <p>Limits the number of objects of a collection to be displayed in the output report</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_collection_result_display_limit</a></p>                                                                 |
| timing_collection_variable_assignment_compatibility               |                                                                                                                                                                                                                                                                                                                                     |
|                                                                   | <p>Enables printing names of output of get objects commands when set to false</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_collection_variable_assignment_compatibility</a></p>                                                         |
| timing_conditions                                                 |                                                                                                                                                                                                                                                                                                                                     |
|                                                                   | <p>Returns the information about the timing conditions in the design.</p> <p><b>Type:</b> <a href="#">obj(timing_condition)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                               |
| timing_consider_late_arrival_for_ssi_derate                       |                                                                                                                                                                                                                                                                                                                                     |
|                                                                   | <p>Using late arrival time for window overlapping evaluation during SSI analysis.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                 |
| timing_constraint_disable_min_max_input_delay_worst_casing        |                                                                                                                                                                                                                                                                                                                                     |
|                                                                   | <p>Used for disabling the override of -max value from -min value in set_input_delay.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_constraint_disable_min_max_input_delay_worst_casing</a></p>                                          |
| timing_constraint_enable_detailed_report_invalid_begin_end_points |                                                                                                                                                                                                                                                                                                                                     |

|                                                          |                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | <p>Enables detailed reporting of invalid begin points/end points specified with set_false_path/set_multicycle_path constraints else only checks structural end points</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_constraint_enable_detailed_report_invalid_begin_end_points</a></p> |
| timing_constraint_enable_logging                         |                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                          | <p>Enables dumping of constraint command in log file</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_constraint_enable_logging</a></p>                                                                                                                                                   |
| timing_constraint_enable_report_invalid_begin_end_points |                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                          | <p>Enables reporting of invalid begin points/end points specified with set_false_path/set_multicycle_path constraints</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_constraint_enable_report_invalid_begin_end_points</a></p>                                                          |
| timing_constraint_enable_search_path                     |                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                          | <p>Enable searching of constraint files in search_path directory variable</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_constraint_enable_search_path</a></p>                                                                                                                          |
| timing_constraint_enable_separate_multicycle_data_checks |                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                          | <p>When set to true, disables multicycle path constraint application on data checks</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_constraint_enable_separate_multicycle_data_checks</a></p>                                                                                            |
| timing_constraint_load_minimal_set_for_automated_eco     |                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                          | <p>Enable/disable constraints while loading for opt signoff</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_constraint_load_minimal_set_for_automated_eco</a></p>                                                                                                                        |

|                                                                            |                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_latency |                                                                                                                                                                                                                                                                                                             |
|                                                                            | Excludes check delays from set_max/min_delay -ignore_clock_latency slack computation<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_late</a>                         |
| timing_constraint_path_delay_exclude_io_delay_from_ignore_clock_latency    |                                                                                                                                                                                                                                                                                                             |
|                                                                            | Excludes input/output delays from set_max/min_delay -ignore_clock_latency slack comput<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_constraint_path_delay_exclude_io_delay_from_ignore_clock_latency</a>                       |
| timing_constraint_path_delay_exclude_unconstrained_endpoints               |                                                                                                                                                                                                                                                                                                             |
|                                                                            | When set to true, does not constrain unconstrained path with path delay.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_constraint_path_delay_exclude_unconstrained_endpoints</a>                                                |
| timing_constraint_path_delay_include_clock_pin_endpoints                   |                                                                                                                                                                                                                                                                                                             |
|                                                                            | When set to true, considers register clock pins as endpoint's for path delay constraints appl<br>upstream<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_constraint_path_delay_include_clock_pin_endpoints</a>                   |
| timing_constraint_update_io_latency_averaging_mode                         |                                                                                                                                                                                                                                                                                                             |
|                                                                            | Controls the analysis to update io latency for rise and fall both or for rise only<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> rise_fall rise_only<br><b>Default:</b> rise_fall<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_constraint_update_io_latency_averaging_mode</a> |
| timing_constraint_warn_for_timing_derate_exceeding_max_limit               |                                                                                                                                                                                                                                                                                                             |

|                                                                         |                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                         | <p>Sets the limit on derate value for giving the warning in case the larger derate is applied</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 100.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_constraint_warn_for_timing_derate_exceeding_max_limit</a></p> |
| timing_constraints_enable_path_exception_efficient_file_line_debug_mode |                                                                                                                                                                                                                                                                                                         |
|                                                                         | <p>Reports the honored and ignored exceptions that were applicable on the path with the file i and line number with compression On</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                    |
| timing_context_apply_default_primary_input_assertion                    |                                                                                                                                                                                                                                                                                                         |
|                                                                         | <p>propogates @(D) phase at block level for unmapped clocks</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_context_apply_default_primary_input_assertion</a></p>                                             |
| timing_context_apply_port_sdc_exceptions                                |                                                                                                                                                                                                                                                                                                         |
|                                                                         | <p>honor path exceptions on block constraints in HC flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_context_apply_port_sdc_exceptions</a></p>                                                            |
| timing_context_clock_mapping_check_waveform                             |                                                                                                                                                                                                                                                                                                         |
|                                                                         | <p>Controls the waveform checking</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_context_clock_mapping_check_waveform</a></p>                                                                                 |
| timing_context_clock_mapping_percentage_tolerance                       |                                                                                                                                                                                                                                                                                                         |
|                                                                         | <p>Controls the tolerance value in clock mapping</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_context_clock_mapping_percentage_tolerance</a></p>                                                           |
| timing_context_clock_phase_based_clock_mapping                          |                                                                                                                                                                                                                                                                                                         |

|                                                |                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Enable clock mapping on block data ports based on clock mapped on block clock ports</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_context_clock_phase_based_clock_mapping</a></p>                               |
| timing_context_continue_on_invalid_module_list |                                                                                                                                                                                                                                                                                                                |
|                                                | <p>When all specified modules/instances are invalid, continue with timing analysis without generating timing context</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_context_continue_on_invalid_module_list</a></p> |
| timing_context_data_phase_based_clock_mapping  |                                                                                                                                                                                                                                                                                                                |
|                                                | <p>Enable clock mapping on block data ports based on clock mapped on other block data ports</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_context_data_phase_based_clock_mapping</a></p>                           |
| timing_context_port_based_clock_mapping        |                                                                                                                                                                                                                                                                                                                |
|                                                | <p>Enable clock mapping based on block ports.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_context_port_based_clock_mapping</a></p>                                                                               |
| timing_continue_on_error                       |                                                                                                                                                                                                                                                                                                                |
|                                                | <p>When set to true, directs software to skip the error and continue processing when an error occurs during timing analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_continue_on_error</a></p>                |
| timing_cppr_opposite_edge_mean_scale_factor    |                                                                                                                                                                                                                                                                                                                |
|                                                | <p>Controls mean delay component for CPPR credit for opposite transition edges at common port</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_cppr_opposite_edge_mean_scale_factor</a></p>                           |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_cppr_opposite_edge_sigma_scale_factor      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                   | <p>Controls sigma delay component for CPPR credit for opposite transition edges at common</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 1.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_cppr_opposite_edge_sigma_scale_factor</a></p>                                                                                                                                                                                                                             |
| timing_cppr_opposite_edge_sigma_scale_factor_cell |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                   | <p>Controls cell sigma delay component for CPPR credit for opposite transition edges at common</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 1.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_cppr_opposite_edge_sigma_scale_factor_cell</a></p>                                                                                                                                                                                                                   |
| timing_cppr_opposite_edge_sigma_scale_factor_net  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                   | <p>Controls net sigma delay component for CPPR credit for opposite transition edges at common</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 1.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_cppr_opposite_edge_sigma_scale_factor_net</a></p>                                                                                                                                                                                                                     |
| timing_cppr_propagate_thru_latches                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                   | <p>When set to true, the cppr of a timing path crossing a latch (or latches) is calculated using the common pin of its origin flop and the capture device at the end of the path. Otherwise the usual segment-based cppr calculation is performed. This cppr setting is considered only when the thru analysis mode is enabled.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_cppr_propagate_thru_latches</a></p> |
| timing_cppr_remove_clock_to_data_pessimism        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                   | <p>When set to true, removes clock reconvergence pessimism (CRP) for clock source paths</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_cppr_remove_clock_to_data_pessimism</a></p>                                                                                                                                                                                                                                 |
| timing_cppr_self_loop_mode                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



|                                                           |                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | <p>When set to true in case of self-loop paths, computes CPPR adjustment by taking the difference between early and late clock arrival time of the common point</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_cppr_self_loop_mode</a></p>                                                           |
| timing_cppr_skip_clock_reconvergence                      |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                           | <p>Specifies the branch point to use for computing clock path pessimism removal (CPPR) adjustment when there is reconvergence in the clock tree</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_cppr_skip_clock_reconvergence</a></p>                                                                |
| timing_cppr_skip_clock_reconvergence_for_unmatched_clocks |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                           | <p>When set to true, this enables CPPR branch point search in reconverging clock tree only when launching and capturing clocks are different.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_cppr_skip_clock_reconvergence_for_unmatched_clocks</a></p>                                             |
| timing_cppr_threshold_ps                                  |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                           | <p>Specifies the maximum amount of pessimism that clock path pessimism removal (CPPR) adjustment is allowed to leave in the path</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 20.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_cppr_threshold_ps</a></p>                                                                                          |
| timing_cppr_transition_sense                              |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                           | <p>Specifies the transition sense of the launching and capturing clocks at the common node, to calculate clock path pessimism removal (CPPR)</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> normal same_transition same_transition_expanded<br/> <b>Default:</b> normal<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_cppr_transition_sense</a></p> |
| timing_create_clock_default_propagated                    |                                                                                                                                                                                                                                                                                                                                                                                          |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Enables clocks to be created in propagated mode</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_create_clock_default_propagated</a></p>                                                                                                                                                                                                                                                                   |
| timing_default_opcond_per_lib       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>When set to true, use default operating conditions for each lib</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_default_opcond_per_lib</a></p>                                                                                                                                                                                                                                                             |
| timing_defer_mmmc_obj_updates       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>You can set the timing_defer_mmmc_object_updates global to true, so that the software all sequence of MMMC updates to be accumulated before new data is loaded and analyzed</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_defer_mmmc_obj_updates</a></p>                                                                                                                                                |
| timing_derate_aocv_dynamic_delays   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>This global controls whether or not AOCV derating factors are applied to the dynamic, SI-in delay component or not. With a value of '1', the AOCV derating factor will be applied to both static and dynamic components of the delay arc. When set to '0', the AOCV derate will only the static component of the delay.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_derate_aocv_dynamic_delays</a></p> |
| timing_derate_aocv_reference_point  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                     | <p>Allows specification of reference point for AOCV factors</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_derate_aocv_reference_point</a></p>                                                                                                                                                                                                                                                                   |
| timing_derate_dynamic_compatibility |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | <p>This global controls how the set_timing_derate factors that are not specified using either -s' dynamic options are applied to the delay. When this global is set to '1' the behavior will be compatible with previous releases. In this mode, the static and dynamic components will be summed before applying derating. With a setting of '0', the static and dynamic components derated separately and then combined. A setting of '0' is equivalent to using separate derat assertions with the -dynamic and -static options explicitly specified.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_derate_dynamic_compatibility</a></p> |
| timing_derate_enable_incremental_multiplication_mode |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                      | <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| timing_derate_incremental_adjust_additive_mode       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                      | <p>By default or with a setting of 'false', successive application of incremental derating using th increment option (or deprecated -incremental_adjust option) will overwrite any existing der value with the new value. When this global is set to 'true', successive application of the der be accumulated into a final derate value.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_derate_incremental_adjust_additive_mode</a></p>                                                                                                                                                                                                     |
| timing_derate_incremental_multiply_accumulative_mode |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                      | <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_derate_incremental_multiply_accumulative_mode</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| timing_derate_negative_delay_backward_compatibility  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                      | <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_derate_negative_delay_backward_compatibility</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| timing_derate_ocv_reference_point                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                                          |                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | <p>Allows specification of reference point for OCV factors</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_derate_ocv_reference_point</a></p>                                  |
| timing_derate_spatial_distance_unit                      |                                                                                                                                                                                                                                                                     |
|                                                          | <p>Set the distance unit.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> default 1um 1nm</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_derate_spatial_distance_unit</a></p>               |
| timing_derate_voltage_scaling_mode                       |                                                                                                                                                                                                                                                                     |
|                                                          | <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> first_library snap_to_nearest interpolate</p> <p><b>Default:</b> first_library</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_derate_voltage_scaling_mode</a></p>              |
| timing_disable_black_box_endpoints                       |                                                                                                                                                                                                                                                                     |
|                                                          | <p>Ignore any boundary model endpoints on black box instances</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                     |
| timing_disable_bus_contention_check                      |                                                                                                                                                                                                                                                                     |
|                                                          | <p>Checks for setup and hold violations in three-state bus designs</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_bus_contention_check</a></p>                   |
| timing_disable_clock_period_checks                       |                                                                                                                                                                                                                                                                     |
|                                                          | <p>When set to true, disables timing model clock period checks during timing analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_clock_period_checks</a></p> |
| timing_disable_constant_propagation_for_sequential_cells |                                                                                                                                                                                                                                                                     |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>When set to true, disables constant prop. across sequential cells if sequential_prop global i</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_disable_constant_propagation_for_sequential_cells</a></p>                                                                                     |
| timing_disable_drv_report_on_constant_nets     |                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                | <p>Disables DRV checks for nets having disabled constant propagation:</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_disable_drv_report_on_constant_nets</a></p>                                                                                                                              |
| timing_disable_floating_bus_check              |                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                | <p>When set to true, disables propagation of minimum delay through three state disable timing and maximum delay through three state enable arcs. These checks are only valid during floating bus conditions</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_disable_floating_bus_check</a></p> |
| timing_disable_genclk_combinational_blocking   |                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                | <p>Disables blocking of generated clocks with '-combinational' option which are downstream of another generated clock</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_disable_genclk_combinational_blocking</a></p>                                                                            |
| timing_disable_get_clocks_for_current_instance |                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                | <p>disable current_instance support on get_clocks</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                    |
| timing_disable_inferred_clock_gating_checks    |                                                                                                                                                                                                                                                                                                                                                                                    |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>When set to true, disables clock gating checks that are inferred on combinational elements clock path. Explicit clock gating checks that are described in the timing library are not affect this global variable</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_inferred_clock_gating_checks</a></p> |
| timing_disable_inout_output_side_timing_checks |                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                | <p>When set to false, timing checks on both the input and output sides of the bidirectional pins are analyzed</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_inout_output_side_timing_checks</a></p>                                                                                                    |
| timing_disable_internal_inout_cell_paths       |                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                | <p>When set to false, enables internal bidirectional feedback paths that are completely contained within one instance</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_internal_inout_cell_paths</a></p>                                                                                                   |
| timing_disable_internal_inout_net_arcs         |                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                | <p>When set to true, this global disables internal bidirectional feedback paths that span multiple instances</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_internal_inout_net_arcs</a></p>                                                                                                              |
| timing_disable_lib_pulse_width_checks          |                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                | <p>When set to true, disables timing model pulse width checks during timing analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_lib_pulse_width_checks</a></p>                                                                                                                                      |
| timing_disable_library_data_to_data_checks     |                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                      |                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>When set to true, this global disables data-to-data checks that are coded in the library as no sequential timing checks</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_library_data_to_data_checks</a></p> |
| timing_disable_library_tieoffs       |                                                                                                                                                                                                                                                                                                                  |
|                                      | <p>Disables constant functions in a library</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_library_tieoffs</a></p>                                                                                            |
| timing_disable_netlist_constants     |                                                                                                                                                                                                                                                                                                                  |
|                                      | <p>When set to true, ignores constants defined in the Verilog netlist</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_netlist_constants</a></p>                                                                |
| timing_disable_nochange_checks       |                                                                                                                                                                                                                                                                                                                  |
|                                      | <p>When set to true, disables no change timing model checks during timing analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_nochange_checks</a></p>                                                     |
| timing_disable_non_sequential_checks |                                                                                                                                                                                                                                                                                                                  |
|                                      | <p>When set to true, disables the timing arcs between any data-to-clock or clock-to-clock check</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_non_sequential_checks</a></p>                                  |
| timing_disable_output_as_clock_port  |                                                                                                                                                                                                                                                                                                                  |
|                                      | <p>Controls clock to output port to be treated as data or clock irrespective of constraints set on</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_output_as_clock_port</a></p>                                |
| timing_disable_parallel_arcs         |                                                                                                                                                                                                                                                                                                                  |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | <p>to control enabling/disabling both CTE's parallel arc reduction, and the related task compression on the AAE side. Setting this variable to 'true' should yield AAE task compression mode '2'. it to 'false' will result in task compression mode '0'</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_disable_parallel_arcs</a></p> |
| timing_disable_pulse_width_same_edge_si_cpvr_mode |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                   | <p>Enables SI CPVR credit up to the last divergent pin for pulse width checks for zero width pulse clocks</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_disable_pulse_width_same_edge_si_cpvr_mode</a></p>                                                                                                                           |
| timing_disable_report_header_info                 |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                   | <p>Controls whether timing reports are generated using a common report header.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_disable_report_header_info</a></p>                                                                                                                                                                     |
| timing_disable_retime_clock_path_slew_propagation |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                   | <p>Controls whether analysis is performed on data or clock paths</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_disable_retime_clock_path_slew_propagation</a></p>                                                                                                                                                                    |
| timing_disable_sdf_retain_arc_merging             |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                   | <p>Controls handling of retain arcs in SDF2.1</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_disable_sdf_retain_arc_merging</a></p>                                                                                                                                                                                                  |
| timing_disable_skew_checks                        |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                   | <p>When set to true, disables library skew checks for timing analysis</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_disable_skew_checks</a></p>                                                                                                                                                                                     |



|                                               |                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_disable_test_signal_arc                |                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>When set to true, timing analysis will not analyze the signal arcs coming from or going to the pin</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_test_signal_arc</a></p>                                                                                       |
| timing_disable_timing_model_latch_inferencing |                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>Controls whether latch behavior is inferred for cell descriptions tagged with the Liberty timing_model_type attribute, including all values: abstracted, extracted, or qtm</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_timing_model_latch_inferencing</a></p> |
| timing_disable_tristate_disable_arcs          |                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>When set to true, disables all 0/1-&gt;Z transitions of tristate arcs during timing analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_tristate_disable_arcs</a></p>                                                                                       |
| timing_disable_user_data_to_data_checks       |                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>When set to true, disables data-to-data checks that are created by the set_data_check command</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_disable_user_data_to_data_checks</a></p>                                                                                   |
| timing_driving_cell_override_library          |                                                                                                                                                                                                                                                                                                                                                                       |
|                                               | <p>Controls the selection of library cells specified using the set_driving_cell command</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_driving_cell_override_library</a></p>                                                                                               |
| timing_drv_analysis_ignore_latency_phases     |                                                                                                                                                                                                                                                                                                                                                                       |

|                                                    |                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>Enables ignoring latency phases in drv analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_drv_analysis_ignore_latency_phases</a></p>                                                                                                    |
| timing_enable_aocv_slack_based                     |                                                                                                                                                                                                                                                                                                                                            |
|                                                    | <p>Enable aocv analysis using slack based method</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_aocv_slack_based</a></p>                                                                                                                 |
| timing_enable_cap_fanout_in_copy_drv_assertions    |                                                                                                                                                                                                                                                                                                                                            |
|                                                    | <p>Enables copying of max/min capacitance and max/min fanout assertions</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                   |
| timing_enable_case_analysis_conflict_warning       |                                                                                                                                                                                                                                                                                                                                            |
|                                                    | <p>Controls to report case analysis conflict warnings to CTE_constant_mismatch.rpt file.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_case_analysis_conflict_warning</a></p>                                                            |
| timing_enable_clock_phase_based_rise_fall_derating |                                                                                                                                                                                                                                                                                                                                            |
|                                                    | <p>Determines the interpretation of edge-specific derating factors for clock paths, specified using set_timing_derate -rise/-fall parameters</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_clock_phase_based_rise_fall_derating</a></p> |
| timing_enable_data_through_clock_gating            |                                                                                                                                                                                                                                                                                                                                            |
|                                                    | <p>When set to false, blocks signals arriving on the enable of the clock-gating check</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_data_through_clock_gating</a></p>                                                                    |
| timing_enable_db_attribute_query                   |                                                                                                                                                                                                                                                                                                                                            |

|                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                | <p>Enables querying DB attribute through CTE commands</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_db_attribute_query</a></p>                                                                                                                                                                                                                                    |
| timing_enable_derating_for_pulse_width_checks                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                | <p>Applies check derating for pulse width checks</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_derating_for_pulse_width_checks</a></p>                                                                                                                                                                                                                           |
| timing_enable_early_late_data_slews_for_setuptools_mode_checks |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                | <p>When set to true, the global variable enables propagation of early and late slews on data path delay calculation then uses these early and late slews to calculate timing checks. You can enable this global in simultaneous setup and hold mode only.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_early_late_data_slews_for_setuptools_mode_checks</a></p> |
| timing_enable_edge_based_path_adjust_group_mode                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                | <p>Enables transition-based set_path_adjust_groups.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                |
| timing_enable_genclk_divide_by_inherit_parent_duty_cycle       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                | <p>Controls inheritance of duty cycle from master clock for generated clocks with divide by option.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_genclk_divide_by_inherit_parent_duty_cycle</a></p>                                                                                                                                                             |
| timing_enable_genclk_source_path_register_limit                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                | <p>Limits generated clock source latency path to traverse across one register only</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_genclk_source_path_register_limit</a></p>                                                                                                                                                                                       |
| timing_enable_generated_clock_edge_based_source_latency        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                                        |                                                                                                                                                                                                                                                            |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                        | Controls how the software chooses generated clock source latency paths<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_generated_clock_edge_based_source_latency</a>       |
| timing_enable_get_obj_escaped_name_backward_compatible |                                                                                                                                                                                                                                                            |
|                                                        | Enables printing of escape characters in the get_object_name command output.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_get_obj_escaped_name_backward_compatible</a> |
| timing_enable_get_objects_regexp_compatibility         |                                                                                                                                                                                                                                                            |
|                                                        | Enables the -regexp option compatibility for get_* commands<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_get_objects_regexp_compatibility</a>                          |
| timing_enable_get_pins_of_lib_cell_pins                |                                                                                                                                                                                                                                                            |
|                                                        | Enable get_pins -of lib cell pins<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes                                                                                                                                         |
| timing_enable_hier_context_unmapped_clock_analysis     |                                                                                                                                                                                                                                                            |
|                                                        | For unmapped block clocks, enable timing analysis based on blok SDC constraints set on I/O ports.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                        |
| timing_enable_hierarchical_get_nets_support            |                                                                                                                                                                                                                                                            |
|                                                        | Enable the hierarchical segment support for get_nets.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_hierarchical_get_nets_support</a>                                   |
| timing_enable_individual_distance_inter_die_analysis   |                                                                                                                                                                                                                                                            |

|                                                           |                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | <p>Enables individual inter die calculation in 3DIC analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_individual_distance_inter_die_analysis</a></p>                                                                                       |
| timing_enable_latch_thru_mode                             |                                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>Enable the latch thru analysis mode in which timing paths can propagate across latches, depending on their arrival time with respect to the latch transparency window.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_latch_thru_mode</a></p> |
| timing_enable_latency_through_clock_gating                |                                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>Controls propagation of latency phases across a clock gating element.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_latency_through_clock_gating</a></p>                                                                                      |
| timing_enable_minimal_constraints_loading_for_opt_signoff |                                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>Enable/disable constraints while loading for opt signoff</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                      |
| timing_enable_mmmc_loop_breaking                          |                                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>When set to true, loop breaking is handled independently per analysis view</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_mmmc_loop_breaking</a></p>                                                                                           |
| timing_enable_multi_drive_net_reduction_with_assertions   |                                                                                                                                                                                                                                                                                                                                                    |

|                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                   | <p>Controls whether multi-drive net reduction (if enabled) will attempt to reduce multi-drive nets also have only either/both set_annotated_delay or set_annotated_transitions assertions present. By default, any assertions present on different drivers of multi-drive nets will prevent reduction of the multi-drive net.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> none all delay<br/> <b>Default:</b> none<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_enable_multi_drive_net_reduction_with_assertions</a></p> |
| timing_enable_multi_frequency_latch_analysis                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                   | <p>Enables multi-frequency latch timing analysis of the latch time borrowing for when a data signal coming to a latch is controlled by a clock with a frequency different to a clock of the latch enable signal</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_enable_multi_frequency_latch_analysis</a></p>                                                                                                                                                                 |
| timing_enable_multicycle_data_check_compatibility                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                   | <p>Enables checking for SDC compatible data checks when set_multicycle_path -start parameter is specified</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_enable_multicycle_data_check_compatibility</a></p>                                                                                                                                                                                                                                                                  |
| timing_enable_path_delay_to_unconstrained_endpoints_compatibility |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                   | <p>When set to true, does not constrain path with path delay if path ends at unconnected input or output of a combo cell.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_enable_path_delay_to_unconstrained_endpoints_compatibility</a></p>                                                                                                                                                                                                                                  |
| timing_enable_pessimistic_cprr_for_reconvergent_clock_paths       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                   | <p>When set to true, enables pessimistic cprr adjustment for re-convergent clock paths</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_enable_pessimistic_cprr_for_reconvergent_clock_paths</a></p>                                                                                                                                                                                                                                                                           |
| timing_enable_power_ground_constants                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                                 |                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | Controls whether case analysis is inferred from power and ground rail connections.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_power_ground_constants</a>                                                                                            |
| timing_enable_preset_clear_arcs                 |                                                                                                                                                                                                                                                                                                                                           |
|                                                 | Determines whether timing arcs are created to model the transition to active state (assertion preset or clear pin, and the subsequent transition of the output to controlled state<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_preset_clear_arcs</a> |
| timing_enable_pulse_latch                       |                                                                                                                                                                                                                                                                                                                                           |
|                                                 | Enables pulse-latch analysis<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_pulse_latch</a>                                                                                                                                                              |
| timing_enable_set_reset_latch_preset_clear_arcs |                                                                                                                                                                                                                                                                                                                                           |
|                                                 | Enable timing through clear/preset arcs of SR latches.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_set_reset_latch_preset_clear_arcs</a>                                                                                                             |
| timing_enable_si_cpvr                           |                                                                                                                                                                                                                                                                                                                                           |
|                                                 | Enables more accurate CPVR analysis when incremental delays are present and the timing_remove_clock_reconvergence_pessimism global variable is set to true<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_si_cpvr</a>                                    |
| timing_enable_simultaneous_setup_hold_mode      |                                                                                                                                                                                                                                                                                                                                           |
|                                                 | Controls whether setup and hold checks are analyzed separately, or together on the same graph<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_simultaneous_setup_hold_mode</a>                                                                           |

|                                                  |                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_enable_spatial_derate_mode                |                                                                                                                                                                                                                                                                                                                                                      |
|                                                  | <p>When set to true, enables spatial derate feature.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_spatial_derate_mode</a></p>                                                                                                                    |
| timing_enable_timing_window_pessimism_removal    |                                                                                                                                                                                                                                                                                                                                                      |
|                                                  | <p>When set to true, removes common clock path pessimism between aggressor and victim pi the design</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_timing_window_pessimism_removal</a></p>                                                         |
| timing_enable_tristate_clock_gating              |                                                                                                                                                                                                                                                                                                                                                      |
|                                                  | <p>When this global is set to true, inferred gated-clock checks are added when clock and data converge through the tristate enable and data input of tristate buffers</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_tristate_clock_gating</a></p> |
| timing_enable_uncertainty_for_clock_checks       |                                                                                                                                                                                                                                                                                                                                                      |
|                                                  | <p>Specifies clock uncertainty check type true   all   false   none   pulse_width period skew</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_uncertainty_for_clock_checks</a></p>                                                                 |
| timing_enable_uncertainty_for_pulse_width_checks |                                                                                                                                                                                                                                                                                                                                                      |
|                                                  | <p>When set to true, considers clock uncertainty when performing minimum pulse width check</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_enable_uncertainty_for_pulse_width_checks</a></p>                                                               |
| timing_enable_unique_vt_mode_for_gba             |                                                                                                                                                                                                                                                                                                                                                      |
|                                                  | <p>when true Enables Unique VtMode for GBA analysis</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                |



|                                                |                                                                                                                                                                                                                                                                           |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_enable_vtskew_derate_mode               |                                                                                                                                                                                                                                                                           |
|                                                | <b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_vtskew_derate_mode</a>                                                                                                                      |
| timing_enable_zero_delay_analysis_mode         |                                                                                                                                                                                                                                                                           |
|                                                | When set to true, uses zero delays in timing analysis.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_enable_zero_delay_analysis_mode</a>                                                      |
| timing_extract_model_aocv_mode                 |                                                                                                                                                                                                                                                                           |
|                                                | This global sets the AOCV mode to be used during model extraction<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> none graph_based path_based<br><b>Default:</b> none<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_extract_model_aocv_mode</a> |
| timing_extract_model_case_analysis_in_library  |                                                                                                                                                                                                                                                                           |
|                                                | When set to false, specifies that port propagated constants are written to the generated con file<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_extract_model_case_analysis_in_library</a>     |
| timing_extract_model_check_arcs_as_lvf         |                                                                                                                                                                                                                                                                           |
|                                                | With this global turned on, check arcs will be modeled as LVF arcs<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_extract_model_check_arcs_as_lvf</a>                                           |
| timing_extract_model_consider_design_level_drv |                                                                                                                                                                                                                                                                           |

|                                                                               |                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                               | <p>When set to false, specifies that user asserted design level DRVs should not be considered writing to the extracted timing model</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_consider_design_level_drv</a></p> |
| timing_extract_model_disable_cycle_adjustment                                 |                                                                                                                                                                                                                                                                                                                              |
|                                                                               | <p>When set to true, the cycle adjustment for the multicycle-paths that could not be pushed out, disabled</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_disable_cycle_adjustment</a></p>                           |
| timing_extract_model_enable_combinational_arc_to_clock_source_on_output_ports |                                                                                                                                                                                                                                                                                                                              |
|                                                                               | <p>When ON, models the combinational arc ending on an end point with clock asserted on it.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_enable_combinational_arc_to_clock_source_on_output_ports</a></p>          |
| timing_extract_model_enable_unconstrained_port_modelling                      |                                                                                                                                                                                                                                                                                                                              |
|                                                                               | <p>When ON, extracts unconstrained path to the output ports.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_enable_unconstrained_port_modelling</a></p>                                                             |
| timing_extract_model_exhaustive_validation_dir                                |                                                                                                                                                                                                                                                                                                                              |
|                                                                               | <p>Specifies the output directory where the validation reports will be written</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_exhaustive_validation_dir</a></p>                                                      |
| timing_extract_model_exhaustive_validation_mode                               |                                                                                                                                                                                                                                                                                                                              |
|                                                                               | <p>Enables ETM validation at the minimum and maximum indices of the slew / load indices</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_exhaustive_validation_mode</a></p>                                           |

|                                                                     |                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_extract_model_gating_as_nochange_arc                         |                                                                                                                                                                                                                                                                                                                                                        |
|                                                                     | <p>When set to false, disables conversion of clock gating checks to nochange arcs in the extra timing model</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_gating_as_nochange_arc</a></p>                                                      |
| timing_extract_model_half_cycle_non_interface_paths                 |                                                                                                                                                                                                                                                                                                                                                        |
|                                                                     | <p>When ON, models half cycle non interface paths as MPW checks in ETM</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_half_cycle_non_interface_paths</a></p>                                                                                  |
| timing_extract_model_ideal_clock_latency_arc                        |                                                                                                                                                                                                                                                                                                                                                        |
|                                                                     | <p>Specifies if latency arc from ideal master clock need to be extracted in timing model</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_ideal_clock_latency_arc</a></p>                                                                        |
| timing_extract_model_include_applied_load_in_characterization_range |                                                                                                                                                                                                                                                                                                                                                        |
|                                                                     | <p>When set to true, specifies that actual load visible to timer, will be included in the load characterization range for path endpoints</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_include_applied_load_in_characterization_range</a></p> |
| timing_extract_model_include_applied_slew_in_characterization_range |                                                                                                                                                                                                                                                                                                                                                        |
|                                                                     | <p>When set to true, specifies that actual timer slew must be included in the characterization range timing model.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_include_applied_slew_in_characterization_range</a></p>                       |
| timing_extract_model_max_feedthru_characterization_load             |                                                                                                                                                                                                                                                                                                                                                        |

|                                                          |                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | <p>Load specified will be used by model extractor for feed through path characterization.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                             |
| timing_extract_model_non_borrowing_latch_path_as_setup   |                                                                                                                                                                                                                                                                                                                                         |
|                                                          | <p>If turned ON, models the non borrowing interface latch paths as setup check.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_non_borrowing_latch_path_as_setup</a></p>                                                       |
| timing_extract_model_slew_propagation_mode               |                                                                                                                                                                                                                                                                                                                                         |
|                                                          | <p>Specifies the type of slew propagation to use for generating extracted timing model</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> worst_slew path_based_slew</p> <p><b>Default:</b> worst_slew</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_slew_propagation_mode</a></p> |
| timing_extract_model_write_clock_checks_as_arc           |                                                                                                                                                                                                                                                                                                                                         |
|                                                          | <p>Allows timing model to write min pulse width or min period checks as arcs.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_write_clock_checks_as_arc</a></p>                                                                  |
| timing_extract_model_write_clock_checks_as_scalar_tables |                                                                                                                                                                                                                                                                                                                                         |
|                                                          | <p>Allows timing model to write min pulse width or min period checks as scalar arcs.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_write_clock_checks_as_scalar_tables</a></p>                                                |
| timing_extract_model_write_lvf                           |                                                                                                                                                                                                                                                                                                                                         |
|                                                          | <p>When set to true, generated ETM contains LVF model.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_write_lvf</a></p>                                                                                                        |
| timing_extract_model_write_min_max_clock_tree_path       |                                                                                                                                                                                                                                                                                                                                         |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>When set to true, writes the min max worst latency for each clock source.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_extract_model_write_min_max_clock_tree_path</a></p>                                                                                          |
| timing_generate_normalized_driver_waveform      |                                                                                                                                                                                                                                                                                                                                                                    |
|                                                 | <p>To generate ndw if it is not defined in the library.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_generate_normalized_driver_waveform</a></p>                                                                                                                       |
| timing_generated_clocks_allow_nested_assertions |                                                                                                                                                                                                                                                                                                                                                                    |
|                                                 | <p>Enables clock latency handling for nested generated clocks created on the same timing pin as their master clocks</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_generated_clocks_allow_nested_assertions</a></p>                                                     |
| timing_generated_clocks_inherit_ideal_latency   |                                                                                                                                                                                                                                                                                                                                                                    |
|                                                 | <p>Controls generated clock's inheritance of parent's ideal network latency.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true actual_partial_latency_path ideal_partial_latency_path</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_generated_clocks_inherit_ideal_latency</a></p> |
| timing_get_of_objects_hier_compatibility        |                                                                                                                                                                                                                                                                                                                                                                    |
|                                                 | <p>When set to true, the get_pins -of_objects parameter will return hierarchical leaf pins connected to the specified objects</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_get_of_objects_hier_compatibility</a></p>                                                  |
| timing_hier_obj_name_compatibility              |                                                                                                                                                                                                                                                                                                                                                                    |

|                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                  | <p>Controls how hierarchical delimiters are interpreted in the search pattern, when the -hier parameter is used with the get_* collection command</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_hier_obj_name_compatibility</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| timing_hierarchical_context_continue_on_top_block_clock_mismatch |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                  | <p>By default or when timing_hierarchical_context_continue_on_top_block_clock_mismatch is 'false' the read_timing_context command will assert an error condition if any of the top-level could not be matched with a block-level clock. Processing of the current script will stop. If timing_hierarchical_context_continue_on_top_block_clock_mismatch is set to 'true', script processing will be allowed to continue despite any mismatch between top and block-level clock. Mismatches will be reported in the file: timing_context_clock_mapping_data.rpt. You should examine this file to determine if corrections or additions are required to the clock mapping file.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> |
| timing_ignore_lumped_rc_assertions                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                  | <p>When set to false, the set_load and set_resistance values override the actual extracted representation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_ignore_lumped_rc_assertions</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| timing_io_use_clock_network_latency                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                  | <p>Controls whether network latency of a reference clock is added or not to the data arrival time at the port</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> always ideal</p> <p><b>Default:</b> ideal</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_io_use_clock_network_latency</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| timing_ipd_allow_paths_for_asynchronous_clocks                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                  | <p>When enabled, ensure the paths between the asynchronous clocks are allowed to be treated as IPD paths.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                                          |                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_ipd_xdomain_path_filtering_threshold              | <p>The specified threshold in ns will be used for path filtering criterion during native ipd xdomain solution.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.003</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                              |
| timing_library_allow_invalid_primary_power_pins_lib_cell | <p>Library cell having invalid primary_power pins is treated as unusable.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_allow_invalid_primary_power_pins_lib_cell</a></p>                                                                                                                   |
| timing_library_allow_multiple_sync_control               | <p>Allow multiple synchronous clear and preset pins.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_allow_multiple_sync_control</a></p>                                                                                                                                                      |
| timing_library_build_async_deassert_arc                  | <p>Controls whether input to output arcs from the preset or clear pins transitioning to inactives included when the timing system is initialized</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_build_async_deassert_arc</a></p>                                                              |
| timing_library_convert_async_setuphold_to_recrem         | <p>Controls whether single-edged setup and hold checks on asynchronous pins in the Liberty are inferred as recovery and removal checks. This is mainly to support older modeling style legacy libraries.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_convert_async_setuphold_to_recrem</a></p> |
| timing_library_generated_clock_use_group_name            |                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                                                |                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                | <p>When set to true, the software uses the generated_clock group name when creating a generated clock from a library-generated clock group</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_library_generated_clock_use_group_name</a></p> |
| timing_library_hold_constraint_corner_sigma_multiplier         |                                                                                                                                                                                                                                                                                                                               |
|                                                                | <p>User control multiplier to generate constraint table applied for hold arc.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_library_hold_constraint_corner_sigma_multiplier</a></p>                                                     |
| timing_library_hold_sigma_multiplier                           |                                                                                                                                                                                                                                                                                                                               |
|                                                                | <p>User control multiplier to generate constraint table applied for hold arc.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 0.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_library_hold_sigma_multiplier</a></p>                                                                       |
| timing_library_ignore_attribute_check_during_pin_convert       |                                                                                                                                                                                                                                                                                                                               |
|                                                                | <p>Ignore liberty attributes check while converting library's logical pin defined as power or ground to LEF.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_library_ignore_attribute_check_during_pin_convert</a></p>                     |
| timing_library_ignore_combinational_arc_mismatch_tables        |                                                                                                                                                                                                                                                                                                                               |
|                                                                | <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                     |
| timing_library_infer_async_pins_from_timing_arcs               |                                                                                                                                                                                                                                                                                                                               |
|                                                                | <p>This global marks the pins as async pins based on the timing arcs associated with the pin.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_library_infer_async_pins_from_timing_arcs</a></p>                                           |
| timing_library_infer_cap_range_from_c1cn_dynamic_pin_cap_model |                                                                                                                                                                                                                                                                                                                               |



|                                                           |                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | <p>Enable Syntech support to infer cap range from n-piece dynamic pincap.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_infer_cap_range_from_c1cn_dynamic_pin_cap_model</a></p>             |
| timing_library_infer_cap_range_from_ccs_receiver_model    |                                                                                                                                                                                                                                                                                                 |
|                                                           | <p>When true infer the rise/fall capacitance range values from ccs receiver capacitance model</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_infer_cap_range_from_ccs_receiver_model</a></p> |
| timing_library_infer_cap_range_from_dynamic_pin_cap_model |                                                                                                                                                                                                                                                                                                 |
|                                                           | <p>Enable Syntech support to infer cap range from dynamic pincap.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_infer_cap_range_from_dynamic_pin_cap_model</a></p>                          |
| timing_library_infer_cap_range_from_ecsm_receiver_model   |                                                                                                                                                                                                                                                                                                 |
|                                                           | <p>When this global is enabled would infer cap range from ecsm capacitance group of library.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_infer_cap_range_from_ecsm_receiver_model</a></p> |
| timing_library_infer_socv_from_aocv                       |                                                                                                                                                                                                                                                                                                 |
|                                                           | <p>Infer sensitivity data from AOCV libraries</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_infer_socv_from_aocv</a></p>                                                                    |
| timing_library_interpolate_drv_values                     |                                                                                                                                                                                                                                                                                                 |
|                                                           | <p>Allows the software to use a range of trilib DRV values for performing delay calculations</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_interpolate_drv_values</a></p>                   |
| timing_library_limit_messages                             |                                                                                                                                                                                                                                                                                                 |

|                                                         |                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | Controls the printing of LBR messages.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_library_limit_messages</a>                                                                      |
| timing_library_look_up_drv_per_frequency                |                                                                                                                                                                                                                                                                 |
|                                                         | Interpolation method for 'max_cap' 1D table (linear worst).<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> worst linear<br><b>Default:</b> worst<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_library_look_up_drv_per_frequency</a> |
| timing_library_multibit_force_merge_combinational_cells |                                                                                                                                                                                                                                                                 |
|                                                         | Merge combinational instances into multibit ignoring qor change.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_library_multibit_force_merge_combinational_cells</a>                 |
| timing_library_multibit_force_merge_iso_cells           |                                                                                                                                                                                                                                                                 |
|                                                         | Merge combinational instances into multibit ignoring qor change.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_library_multibit_force_merge_iso_cells</a>                           |
| timing_library_multibit_force_merge_sequential_cells    |                                                                                                                                                                                                                                                                 |
|                                                         | Merge sequential instances into multibit ignoring qor change.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_library_multibit_force_merge_sequential_cells</a>                       |
| timing_library_multibit_use_combinational_cells         |                                                                                                                                                                                                                                                                 |
|                                                         | Allows to merge 1-bit combinational cells into multibit cells.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_library_multibit_use_combinational_cells</a>                           |
| timing_library_read_ccs_noise_data                      |                                                                                                                                                                                                                                                                 |

|                                               |                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | <p>can read ccs noise construct from library when enabled.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_read_ccs_noise_data</a></p>                                               |
| timing_library_read_without_ecsm              |                                                                                                                                                                                                                                                                                       |
|                                               | <p>Switch off loading of ECSM data</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_read_without_ecsm</a></p>                                                                        |
| timing_library_read_without_power             |                                                                                                                                                                                                                                                                                       |
|                                               | <p>can read library without power by setting this global to true</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_read_without_power</a></p>                                         |
| timing_library_read_without_sensitivity       |                                                                                                                                                                                                                                                                                       |
|                                               | <p>Switch off loading of ECSM timing sensitivity data</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_read_without_sensitivity</a></p>                                              |
| timing_library_respect_async_control_priority |                                                                                                                                                                                                                                                                                       |
|                                               | <p>Consider the liberty description of asynchronous inputs priority for sequential lib_cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_respect_async_control_priority</a></p> |
| timing_library_scale_aocv_to_socv_to_n_sigma  |                                                                                                                                                                                                                                                                                       |
|                                               | <p>AOCV derates are expected to be derived based on 3-sigma variation</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 3.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_scale_aocv_to_socv_to_n_sigma</a></p>                         |
| timing_library_sequential_in_out_phase_opt    |                                                                                                                                                                                                                                                                                       |

|                                                         |                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | <p>Allows to propagate inverters through sequential instances.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_sequential_in_out_phase_opt</a></p>                                                |
| timing_library_setup_constraint_corner_sigma_multiplier |                                                                                                                                                                                                                                                                                                     |
|                                                         | <p>User control multiplier to generate constraint table using sigma values applied for setup arc</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_setup_constraint_corner_sigma_multiplier</a></p> |
| timing_library_setup_sigma_multiplier                   |                                                                                                                                                                                                                                                                                                     |
|                                                         | <p>User control multiplier to generate constraint table using sigma values applied for setup arc</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_setup_sigma_multiplier</a></p>                   |
| timing_library_support_3d_table_power_arc               |                                                                                                                                                                                                                                                                                                     |
|                                                         | <p>Support three dimensional power arc.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_support_3d_table_power_arc</a></p>                                                                        |
| timing_library_support_combinational_clock              |                                                                                                                                                                                                                                                                                                     |
|                                                         | <p>Support data-pins marked as clock on combinational cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_support_combinational_clock</a></p>                                                  |
| timing_library_support_merge_libs                       |                                                                                                                                                                                                                                                                                                     |
|                                                         | <p>Enables appending libraries with same names &amp; PVT conditions in different .lib files.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_support_merge_libs</a></p>                           |
| timing_library_support_serial_scan_in_multibit_cell     |                                                                                                                                                                                                                                                                                                     |

|                                                |                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Support the serial scan input multibit cell in which serial scan input is specified using intern instead of scan input pin for all the outputs expect the first.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_library_support_serial_scan_in_multibit_cell</a></p> |
| timing_library_support_transparent_latch_group |                                                                                                                                                                                                                                                                                                                                                             |
|                                                | <p>Support library cell having tlatch group.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_library_support_transparent_latch_group</a></p>                                                                                                                             |
| timing_library_suppress_library_messages       |                                                                                                                                                                                                                                                                                                                                                             |
|                                                | <p>Controls the printing of library parsing messages.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_library_suppress_library_messages</a></p>                                                                                                                         |
| timing_library_term_voltage_from_lib_pin       |                                                                                                                                                                                                                                                                                                                                                             |
|                                                | <p>Use Library pin voltage for reporting the term voltage.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_library_term_voltage_from_lib_pin</a></p>                                                                                                                         |
| timing_library_ui_respect_preserve             |                                                                                                                                                                                                                                                                                                                                                             |
|                                                | <p>Cause netlist editing commands to fail if they would modify preserved logic.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_library_ui_respect_preserve</a></p>                                                                                                      |
| timing_library_use_max_cap_look_up_table       |                                                                                                                                                                                                                                                                                                                                                             |
|                                                | <p>Honor frequency based max_cap lut values from library for design rule violations.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_library_use_max_cap_look_up_table</a></p>                                                                                           |
| timing_library_use_sequential_test_cell        |                                                                                                                                                                                                                                                                                                                                                             |

|                                                    |                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>Use test cell function if main cell function is unusable.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_use_sequential_test_cell</a></p>                                                                            |
| timing_library_use_two_piece_receiver_cap          |                                                                                                                                                                                                                                                                                                                            |
|                                                    | <p>When libraries have both 2 piece and multi piece pin caps, give priority to two piece model:</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_use_two_piece_receiver_cap</a></p>                                       |
| timing_library_zero_negative_timing_check_arcs     |                                                                                                                                                                                                                                                                                                                            |
|                                                    | <p>When true check for negative values in timing arcs</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_library_zero_negative_timing_check_arcs</a></p>                                                                            |
| timing_max_transition_use_si_transition            |                                                                                                                                                                                                                                                                                                                            |
|                                                    | <p>When set to true, enables noisy slew reporting in DRV flows</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_max_transition_use_si_transition</a></p>                                                                          |
| timing_multi_frequency_clock_rounding_factor       |                                                                                                                                                                                                                                                                                                                            |
|                                                    | <p>Rounds off irrational numbers so that they can be read appropriately</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1e-05</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_multi_frequency_clock_rounding_factor</a></p>                                                          |
| timing_normalized_driver_waveform_clip_linear_part |                                                                                                                                                                                                                                                                                                                            |
|                                                    | <p>To control the clipping of the generated ndw which is generated by the global timing_generate_normalized_driver_waveform.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_normalized_driver_waveform_clip_linear_part</a></p> |
| timing_normalized_driver_waveform_weight_factor    |                                                                                                                                                                                                                                                                                                                            |

|                                                                     |                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     | Decides the weight factor for the exponential and the linear part in the waveform generated global timing_generate_normalized_driver_waveform.<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.5<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_normalized_driver_waveform_weight_factor</a> |
| timing_nsigma_multiplier                                            |                                                                                                                                                                                                                                                                                                                       |
|                                                                     | Controls sigma multiplier to be used in SOCV mode<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 3.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_nsigma_multiplier</a>                                                                                                                     |
| timing_null_collection_return_compatibility                         |                                                                                                                                                                                                                                                                                                                       |
|                                                                     | Allows you to migrate previous release scripts to the new use model<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_null_collection_return_compatibility</a>                                                                                |
| timing_path_based_enable_bounding_box_for_io_paths                  |                                                                                                                                                                                                                                                                                                                       |
|                                                                     | when true, enables bounding box mode for i/o paths<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_path_based_enable_bounding_box_for_io_paths</a>                                                                                          |
| timing_path_based_enable_closest_common_pin_driver_for_bounding_box |                                                                                                                                                                                                                                                                                                                       |
|                                                                     | when TRUE, enables bounding box calculation using closest common pin driver from clock<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_path_based_enable_closest_common_pin_driver_for_bounding_box</a>                                     |
| timing_path_based_enable_exhaustive_depth_bounded_by_gba            |                                                                                                                                                                                                                                                                                                                       |
|                                                                     | Enable the tool to bound the PBA exhaustive analysis by next worst GBA slack incase the c exhausted for any given endpoint<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> true<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_path_based_enable_exhaustive_depth_bounded_by_gba</a>             |

|                                                     |                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_path_based_enable_report_launch_clock_path   |                                                                                                                                                                                                                                                                                                                                  |
|                                                     | <p>set it to false to skip reporting launch clock path in pba path_type full report</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_path_based_enable_report_launch_clock_path</a></p>                                                  |
| timing_path_based_enable_verbose_mode               |                                                                                                                                                                                                                                                                                                                                  |
|                                                     | <p>controls the verbosity of path based analysis messages printed during and at the end of the analysis</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 0 1 2 true false</p> <p><b>Default:</b> 1</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_path_based_enable_verbose_mode</a></p> |
| timing_path_based_epba_bounded_mode                 |                                                                                                                                                                                                                                                                                                                                  |
|                                                     | <p>Turn on EPBA Bounded mode</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_path_based_epba_bounded_mode</a></p>                                                                                                                      |
| timing_path_based_exhaustive_enable_design_coverage |                                                                                                                                                                                                                                                                                                                                  |
|                                                     | <p>Enables the exhaustive path based analysis to evaluate all violating endpoints</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_path_based_exhaustive_enable_design_coverage</a></p>                                                 |
| timing_path_based_exhaustive_max_paths_limit        |                                                                                                                                                                                                                                                                                                                                  |
|                                                     | <p>Set the maximum number of paths which can be retimed during exhaustive path based analysis</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 2000000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_path_based_exhaustive_max_paths_limit</a></p>                                           |
| timing_path_based_ipba_endpoints_max_limit          |                                                                                                                                                                                                                                                                                                                                  |



|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | <p>Set endpoints max limit, for infinite PBA analysis of top max GBA endpoints</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 100000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_path_based_ipba_endpoints_max_limit</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                           |
| timing_path_based_low_memory_mode                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                           | <p>A lower value for this variable will reduce the peak memory footprint of path based analysis at the cost of some addition runtime.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 10.0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| timing_path_based_report_analysis_summary_max_paths_limit |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                           | <p>specifies the limit of maximum PBA FEPs reported</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 2000000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_path_based_report_analysis_summary_max_paths_limit</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| timing_pba_exhaustive_path_nworst_limit                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                           | <p>Specify the nworst exhaustive limit to be honored by tool during PBA exhaustive limit</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 10000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_pba_exhaustive_path_nworst_limit</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| timing_pll_clock_use_driving_cell                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                           | <p>When set to false, the software considers the PLL clock output to be electrically isolated from input-side reference clock, and creates the output clock with an idealized transition. You can set <code>set_annotated_transition</code> to customize the output clock transition. When set to true, delay calculation of all enabled arcs to the PLL clock output will be used to compute the transition driving arcs are available in the model or if they are disabled, the same idealized transition used.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_pll_clock_use_driving_cell</a></p> |
| timing_prefix_module_name_with_library_generated_clock    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                                 |                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                 | <p>When set to true, the software appends the instance name to the clock pin name when creating a generated clock</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_prefix_module_name_with_library_generated_clock</a></p>                           |
| timing_propagate_latch_data_uncertainty                         |                                                                                                                                                                                                                                                                                                                                              |
|                                                                 | <p>When set to true, uses the clock phase associated with a flush latch's data pin as the from clock phase for downstream uncertainty timing calculations</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_propagate_latch_data_uncertainty</a></p> |
| timing_property_arrival_clocks_consider_clock_source_data_phase |                                                                                                                                                                                                                                                                                                                                              |
|                                                                 | <p>When set to true, the arrival_clocks property will consider clock source data phases</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_property_arrival_clocks_consider_clock_source_data_phase</a></p>                                            |
| timing_property_arrival_clocks_consider_data_phase              |                                                                                                                                                                                                                                                                                                                                              |
|                                                                 | <p>When set to true, the arrival_clocks property will return pure data phase</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_property_arrival_clocks_consider_data_phase</a></p>                                                                    |
| timing_property_arrival_clocks_consider_latency_phase           |                                                                                                                                                                                                                                                                                                                                              |
|                                                                 | <p>When set to true, the arrival_clocks property will return latency phase</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_property_arrival_clocks_consider_latency_phase</a></p>                                                                   |
| timing_property_arrival_window_enable_tcl_dict_format           |                                                                                                                                                                                                                                                                                                                                              |
|                                                                 | <p>When set to TRUE, new format for the arrival_windows property will be used to print.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_property_arrival_window_enable_tcl_dict_format</a></p>                                                     |

|                                                   |                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_property_return_null_collection_with_quiet |                                                                                                                                                                                                                                                                                                |
|                                                   | <p>Enables returning properties values for all objects when -quiet option is used in get_property</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_property_return_null_collection_with_quiet</a></p> |
| timing_rail_swing_checks_high_voltage_threshold   |                                                                                                                                                                                                                                                                                                |
|                                                   | <p>user to specify voltage thresholds for high rail swing checks</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.95</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_rail_swing_checks_high_voltage_threshold</a></p>                                   |
| timing_rail_swing_checks_low_voltage_threshold    |                                                                                                                                                                                                                                                                                                |
|                                                   | <p>user to specify voltage thresholds for low rail swing checks</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.05</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_rail_swing_checks_low_voltage_threshold</a></p>                                     |
| timing_rcdb_allow_mismatch_option                 |                                                                                                                                                                                                                                                                                                |
|                                                   | <p>Add the -allow_mismatch option to read_parasitics for BM rcdb loading.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_rcdb_allow_mismatch_option</a></p>                                          |
| timing_recompute_sdf_in_setuphold_mode            |                                                                                                                                                                                                                                                                                                |
|                                                   | <p>Controls the recomputing of SDF delays when the software is in simultaneous setup and hold analysis mode</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_recompute_sdf_in_setuphold_mode</a></p>  |
| timing_reduce_multi_drive_net_arcs                |                                                                                                                                                                                                                                                                                                |

|                                                                     |                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     | <p>Controls the reduction of the number of net arcs created for timing analysis for nets driven by parallel buffers</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_reduce_multi_drive_net_arcs</a></p>                                                   |
| timing_reduce_multi_drive_net_arcs_threshold                        |                                                                                                                                                                                                                                                                                                                                                    |
|                                                                     | <p>Sets a threshold number used by the tool to trigger the reduction of timing arcs of nets driven by parallel buffers</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 10000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_reduce_multi_drive_net_arcs_threshold</a></p>                                      |
| timing_report_analysis_summary_csv_extended_new_format              |                                                                                                                                                                                                                                                                                                                                                    |
|                                                                     | <p>This global keeps the report_analysis_summary_csv_extended report in sync with report_analysis_summary report.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_analysis_summary_csv_extended_new_format</a></p>                                |
| timing_report_arrival_property_worstcase_mode                       |                                                                                                                                                                                                                                                                                                                                                    |
|                                                                     | <p>To be set to 0, 1, 2 or 3 to differentiate clock and data phase on properties arrival_*. </p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_arrival_property_worstcase_mode</a></p>                                                                   |
| timing_report_begin_end_pair_max_path_limit                         |                                                                                                                                                                                                                                                                                                                                                    |
|                                                                     | <p>Controls the number of max_paths to be reported by report_timing in begin_end_pair options. Higher max_paths limit may lead to high memory and runtime.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 2000000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_begin_end_pair_max_path_limit</a></p> |
| timing_report_check_timing_unconstrained_endpoints_due_to_constants |                                                                                                                                                                                                                                                                                                                                                    |

|                                                       |                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | <p>Will split uncons endpoint warning into another warning uncons endpoint due to case anal</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_check_timing_unconstrained_endpoints_due_to_constants</a></p> |
| timing_report_clock_pin_as_begin_point                |                                                                                                                                                                                                                                                                                                            |
|                                                       | <p>Enable whether to show CLK as start point</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_clock_pin_as_begin_point</a></p>                                                                             |
| timing_report_constraint_enable_extended_drv_format   |                                                                                                                                                                                                                                                                                                            |
|                                                       | <p>Enables reporting of report_constraint as per clock/data phases and rise/fall</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_constraint_enable_extended_drv_format</a></p>                            |
| timing_report_constraint_extended_cell_em_flow        |                                                                                                                                                                                                                                                                                                            |
|                                                       | <p>Enables extended Cell EM flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_constraint_extended_cell_em_flow</a></p>                                                                                 |
| timing_report_constraint_format                       |                                                                                                                                                                                                                                                                                                            |
|                                                       | <p>Enables custom drv reporting in specified format</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_constraint_format</a></p>                                                                              |
| timing_report_constraint_max_paths_limit              |                                                                                                                                                                                                                                                                                                            |
|                                                       | <p>Set the maximum number of violations (Data/Clock checks) to be reported.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 2000000</p> <p><b>Edit:</b> Yes</p>                                                                                                                             |
| timing_report_constraint_rise_fall_clock_period_check |                                                                                                                                                                                                                                                                                                            |

|                                                               |                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                               | <p>Enables rise-fall check for clock_period check type in report_constraint</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_constraint_rise_fall_clock_period_check</a></p>                                                                                                           |
| timing_report_constraint_use_infinity_slack_for_unconstrained |                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                               | <p>Enables reporting of timing slack as INFINITY for unconstrained paths for configurable DR\</p> <p>The default behavior is to report timing slacks using star "*" character</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_constraint_use_infinity_slack_for_unconstrained</a></p> |
| timing_report_default_formatting                              |                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                               | <p>Setting to control the formatting output of report_timing</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> stage_delay split_delay</p> <p><b>Default:</b> stage_delay</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                |
| timing_report_default_frequency_for_unconstrained_nets        |                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                               | <p>Defines frequency for unconstrained nets to be used in max_cap drv reporting</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1e+09</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_default_frequency_for_unconstrained_nets</a></p>                                                                                                    |
| timing_report_disable_max_paths_per_group                     |                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                               | <p>This global controls whether or not group-based mode semantics are used when -group ar</p> <p>max_paths are used with report_timing.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_disable_max_paths_per_group</a></p>                                                           |
| timing_report_drv_enable_clock_source_as_clock                |                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                               | <p>Enables clock drv for pure clock source path in place of data drv</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_drv_enable_clock_source_as_clock</a></p>                                                                                                                         |

|                                                      |                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_report_drv_enable_frequency_per_view          |                                                                                                                                                                                                                                                                                                                                                                |
|                                                      | <p>Enables view based frequency drv flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_drv_enable_frequency_per_view</a></p>                                                                                                                                 |
| timing_report_drv_enable_ghz_notation_for_drv_fields |                                                                                                                                                                                                                                                                                                                                                                |
|                                                      | <p>Enables reporting of clock frequency in Ghz notation for custom drv reporting</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_drv_enable_ghz_notation_for_drv_fields</a></p>                                                                               |
| timing_report_drv_enable_slew_threshold_scaling      |                                                                                                                                                                                                                                                                                                                                                                |
|                                                      | <p>Enables transition constraints scaling as per slew threshold</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_drv_enable_slew_threshold_scaling</a></p>                                                                                                     |
| timing_report_drv_per_frequency                      |                                                                                                                                                                                                                                                                                                                                                                |
|                                                      | <p>Enables worst case reporting of max_transition/max_capacitance and considers set_max_cap(tran)_per_freq, set_max_transition(capacitance) and library defined values in comparison</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_drv_per_frequency</a></p> |
| timing_report_drv_per_frequency_per_input_slew       |                                                                                                                                                                                                                                                                                                                                                                |
|                                                      | <p>Enables frequency dependent drv lookup from library</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_drv_per_frequency_per_input_slew</a></p>                                                                                                               |
| timing_report_drv_use_worst_timing_slack             |                                                                                                                                                                                                                                                                                                                                                                |

|                                                       |                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | <p>Enables reporting of the worst DRV slack across all views for configurable DRV report</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_drv_use_worst_timing_slack</a></p>                        |
| timing_report_enable_clock_to_for_unconstrained_paths |                                                                                                                                                                                                                                                                                                     |
|                                                       | <p>Enables Unconstrained Path reporting for -to_clock option</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_clock_to_for_unconstrained_paths</a></p>                                       |
| timing_report_enable_cpvr_point                       |                                                                                                                                                                                                                                                                                                     |
|                                                       | <p>Enable Cpvr Point reporting in report timing command</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_cpvr_point</a></p>                                                                  |
| timing_report_enable_em_index_clipping_report         |                                                                                                                                                                                                                                                                                                     |
|                                                       | <p>Enables index clipping report in em drv reports</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_em_index_clipping_report</a></p>                                                         |
| timing_report_enable_flag_field_symbols               |                                                                                                                                                                                                                                                                                                     |
|                                                       | <p>Enables reporting of Instance/Net related bits in report_timing Flags column</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_flag_field_symbols</a></p>                                  |
| timing_report_enable_lead_trail_to_rise_fall_map      |                                                                                                                                                                                                                                                                                                     |
|                                                       | <p>Enable reporting of Rise/Fall edge times instead of Lead/Trail edge times of clock in report command</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_lead_trail_to_rise_fall_map</a></p> |
| timing_report_enable_markers                          |                                                                                                                                                                                                                                                                                                     |



|                                                               |                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                               | <p>Enables marker for report_timing for Pin/timing points columns</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_markers</a></p>                                                  |
| timing_report_enable_max_capactiance_drv_for_constant_nets    |                                                                                                                                                                                                                                                                                           |
|                                                               | <p>Enables max capactiance DRV checks for nets having disabled constant propagation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_max_capactiance_drv_for_constant_nets</a></p> |
| timing_report_enable_max_path_limit_warning                   |                                                                                                                                                                                                                                                                                           |
|                                                               | <p>warning message if max_paths option specified is not enough for coverage</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_max_path_limit_warning</a></p>                        |
| timing_report_enable_qpin_as_start_for_begin_end_pair         |                                                                                                                                                                                                                                                                                           |
|                                                               | <p>Enables reporting of multiple paths with different q-pins but same clock pin</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_qpin_as_start_for_begin_end_pair</a></p>          |
| timing_report_enable_report_clock_timing_across_clock_pin     |                                                                                                                                                                                                                                                                                           |
|                                                               | <p>Enables fanout tracing across clock pins if any combinational arc is present</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_report_clock_timing_across_clock_pin</a></p>      |
| timing_report_enable_si_debug                                 |                                                                                                                                                                                                                                                                                           |
|                                                               | <p>For decision regarding SI attribute debugging through path collection objects</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_si_debug</a></p>                                 |
| timing_report_enable_unique_pins_multiple_capture_clock_paths |                                                                                                                                                                                                                                                                                           |

|                                                           |                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | <p>enables reporting different paths w.r.t ref clocks in unique_pins</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_unique_pins_multiple_capture_clock_paths</a></p>                                                                                      |
| timing_report_enable_verbose_ssta_mode                    |                                                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>Enables new format for header in report_timing SOCV mode</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_verbose_ssta_mode</a></p>                                                                                                                      |
| timing_report_enable_write_twf_third_party_for_input_pins |                                                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>Enable writing of timing information in third party twf for input pins</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_enable_write_twf_third_party_for_input_pins</a></p>                                                                                     |
| timing_report_fields                                      |                                                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>Specifies a report_timing format</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> instance arc cell delay arrival required</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_fields</a></p>                                                                                                                           |
| timing_report_generated_clock_info                        |                                                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>When set to true (the default), generated clock information is automatically added to the report. Generated clocks are encountered in either the launching or latching clock paths</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_generated_clock_info</a></p> |
| timing_report_group_based_mode                            |                                                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>Groups paths by clock domain</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_group_based_mode</a></p>                                                                                                                                                          |
| timing_report_limit_max_path_count                        |                                                                                                                                                                                                                                                                                                                                                                    |

|                                                          |                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | <p>Set the maximum number of violations (Data/Clock checks) to be reported.<br/> <b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 2000000<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_report_limit_max_path_count</a></p>                                |
| timing_report_max_transition_check_using_nsigma_slew     |                                                                                                                                                                                                                                                                                   |
|                                                          | <p>Enables report_constraint drv checking to use mean+nsigma slew values in socv flow<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_report_max_transition_check_using_nsigma_slew</a></p>     |
| timing_report_property_fastest_clock_consider_data_phase |                                                                                                                                                                                                                                                                                   |
|                                                          | <p>When on, it will consider data phase also while finding the fastest clock.<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_report_property_fastest_clock_consider_data_phase</a></p>         |
| timing_report_pulse_width_matching_launch_capture_paths  |                                                                                                                                                                                                                                                                                   |
|                                                          | <p>Reports only those pulse_width paths that have same pins on launch and capture path<br/> <b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_report_pulse_width_matching_launch_capture_paths</a></p> |
| timing_report_redirect_message_types                     |                                                                                                                                                                                                                                                                                   |
|                                                          | <p>Redirect reporting messages to report file<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> info warning error all none<br/> <b>Default:</b> none<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_report_redirect_message_types</a></p>         |
| timing_report_retime_formatting_mode                     |                                                                                                                                                                                                                                                                                   |

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | <p>This variable can be used to manage the retiming fields in reporting format. Based upon giving setting it would automatically replaces (or add) the default columns with respective retiming columns. For example in case of delay column it will add(or replace) the 'Retime Delay' automatically.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> manual retime_compare retime_replace<br/> <b>Default:</b> manual<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_report_retime_formatting_mode</a></p> |
| timing_report_skip_constraint_loop_check                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                         | <p>skip reporting of loop in case timing graph inside loop is broken due to create_clock, set_input_delay, set_output_delay defined inside the loop</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_report_skip_constraint_loop_check</a></p>                                                                                                                                                                                                               |
| timing_report_skip_icg_from_all_fanin_fanout_collection |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                         | <p>Skip ICG clock pins from all_fanin/fanout collection when queried with -startpoint/endpoint</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_report_skip_icg_from_all_fanin_fanout_collection</a></p>                                                                                                                                                                                                                                                     |
| timing_report_socv_summary_mean_sigma                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                         | <p>Enables new format for header in report_timing SOCV mode</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_report_socv_summary_mean_sigma</a></p>                                                                                                                                                                                                                                                                                                          |
| timing_report_split_other_end_arrival                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                         | <p>This global splits the printing of other end arrival time in timing reports into other end path delay and clock edge time.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                     |
| timing_report_timing_header_detail_info                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                                    |                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>Controls whether the report_timing reports are generated using the default or extended report header</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> default extended</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_timing_header_detail_info</a></p> |
| timing_report_unconstrained_path_early_late_header |                                                                                                                                                                                                                                                                                                                                          |
|                                                    | <p>When set to true, prints late and early type for unconstrained path</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_unconstrained_path_early_late_header</a></p>                                                                     |
| timing_report_unconstrained_paths                  |                                                                                                                                                                                                                                                                                                                                          |
|                                                    | <p>When set to true, the report_timing command reports unconstrained paths if it cannot find a constrained path to report</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_unconstrained_paths</a></p>                                   |
| timing_report_use_receiver_model_capacitance       |                                                                                                                                                                                                                                                                                                                                          |
|                                                    | <p>Uses ECSM pin capacitance values for load or DRV calculation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_use_receiver_model_capacitance</a></p>                                                                                  |
| timing_report_use_worst_parallel_cell_arc          |                                                                                                                                                                                                                                                                                                                                          |
|                                                    | <p>When set to false, setting the -nworst parameter reports several paths by selecting parallel the library cell between two pins</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_report_use_worst_parallel_cell_arc</a></p>                   |
| timing_resolve_driver_conflicts                    |                                                                                                                                                                                                                                                                                                                                          |

|                                         |                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Resolves values of nets having multiple drivers</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> conservative aggressive</p> <p><b>Default:</b> aggressive</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_resolve_driver_conflicts</a></p>                                       |
| timing_scaling_for_negative_checks      |                                                                                                                                                                                                                                                                                                                             |
|                                         | <p>Modifies the scaling of the negative timing analysis check value</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> default divider multiplier</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_scaling_for_negative_checks</a></p>                   |
| timing_scaling_for_negative_delays      |                                                                                                                                                                                                                                                                                                                             |
|                                         | <p>Modifies the scaling of the negative delay value to be used during timing analysis</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> default divider multiplier</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_scaling_for_negative_delays</a></p> |
| timing_sdf_adjust_negative_setuphold    |                                                                                                                                                                                                                                                                                                                             |
|                                         | <p>Controls how Setup and Hold check values are adjusted when both are initially negative.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_sdf_adjust_negative_setuphold</a></p>                                                  |
| timing_sdf_enable_setuphold_scond_ccond |                                                                                                                                                                                                                                                                                                                             |
|                                         | <p>To enable proper generation of SDF (Standard Delay Format) with scond and ccond qualifi SETUPHOLD and RECREM timing checks</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_sdf_enable_setuphold_scond_ccond</a></p>            |
| timing_sdf_nochange_verilog_semantics   |                                                                                                                                                                                                                                                                                                                             |
|                                         | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_self_loop_paths_no_skew_max_depth        |                                                                                                                                                                                                                                                                                                                                                                            |
|                                                 | <p>Identifies self-loop paths at the specified depth.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 10</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_self_loop_paths_no_skew_max_depth</a></p>                                                                                                                                      |
| timing_self_loop_paths_no_skew_max_slack        |                                                                                                                                                                                                                                                                                                                                                                            |
|                                                 | <p>Identifies self-loop paths at the specified threshold</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_self_loop_paths_no_skew_max_slack</a></p>                                                                                                                               |
| timing_set_clock_source_to_output_as_data       |                                                                                                                                                                                                                                                                                                                                                                            |
|                                                 | <p>When set to true, causes a clock source path leading to an output or bidi port to be treated as a data path if there is a set_output_delay or set_data_check assertion on the port</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_set_clock_source_to_output_as_data</a></p> |
| timing_socv_preserve_variation_with_annotations |                                                                                                                                                                                                                                                                                                                                                                            |
|                                                 | <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_socv_preserve_variation_with_annotations</a></p>                                                                                                                                                                                     |
| timing_socv_rc_variation_mode                   |                                                                                                                                                                                                                                                                                                                                                                            |
|                                                 | <p>Enable interconnect variation mode in SOCV mode</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_socv_rc_variation_mode</a></p>                                                                                                                                                 |
| timing_socv_sigma_spatial_multiplier            |                                                                                                                                                                                                                                                                                                                                                                            |

|                                               |                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | <p>The value specified for this global provides a fractional percentage multiplier which allows user to select that all, none, or some of the spatial derate be applied</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1.0</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_socv_sigma_spatial_multiplier</a></p> |
| timing_socv_view_based_nsigma_multiplier_mode |                                                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>Enables setting of view-based separate setup and hold sigma multipliers</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_socv_view_based_nsigma_multiplier_mode</a></p>                                                                                      |
| timing_spatial_derate_chip_size               |                                                                                                                                                                                                                                                                                                                                                         |
|                                               | <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 1e+30</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_spatial_derate_chip_size</a></p>                                                                                                                                                                                |
| timing_spatial_derate_distance_mode           |                                                                                                                                                                                                                                                                                                                                                         |
|                                               | <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> bounding_box chip_size arc_bounding_box</p> <p><b>Default:</b> bounding_box</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_spatial_derate_distance_mode</a></p>                                                                                                    |
| timing_suppress_escape_characters             |                                                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>Suppresses the reporting of escape characters in timing object names</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_suppress_escape_characters</a></p>                                                                                                     |
| timing_suppress_ilm_constraint_mismatches     |                                                                                                                                                                                                                                                                                                                                                         |
|                                               | <p>When set to true, the software suppresses all error and warning messages related to objects found when loading SDC constraint files for the ILM flow</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">timing_suppress_ilm_constraint_mismatches</a></p>            |



|                                                      |                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timing_time_unit                                     | <p>This value is passed to set_library_unit-time to set the time units used in timing library and files and timing reports. Legal values are none, 1ns, 1ps, 10ps, 100ps.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> none<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_time_unit</a></p> |
| timing_use_clock_pin_attribute_for_clock_net_marking | <p>To allow the propagation of clocks to pins with clock attributes regardless of the presence of arcs or trigger arcs at the pins</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_use_clock_pin_attribute_for_clock_net_marking</a></p>     |
| timing_use_incremental_si_transition                 | <p>When set to true, enables usage of incremental slew during DRV violation reporting</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_use_incremental_si_transition</a></p>                                                                  |
| timing_use_latch_early_launch_edge                   | <p>Enables the latch use early launch edge feature.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_use_latch_early_launch_edge</a></p>                                                                                                       |
| timing_use_latch_time_borrow                         | <p>When set to false, does not consider time borrowing during timing analysis. Time borrowing amount of time borrowed by a previous logic</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">timing_use_latch_time_borrow</a></p>                       |
| timing_use_verilog_for_model_netlist                 |                                                                                                                                                                                                                                                                                                                                  |

|                                                             |                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | Whether to write scope and boundary model netlists in verilog format rather than binary<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_use_verilog_for_model_netlist</a>                          |
| timing_waveform_aware_pulse_width_checks_high_voltage_level |                                                                                                                                                                                                                                                                         |
|                                                             | User to specify high voltage level to be used for waveform aware pulse-width checks<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.95<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_waveform_aware_pulse_width_checks_high_voltage_level</a> |
| timing_waveform_aware_pulse_width_checks_low_voltage_level  |                                                                                                                                                                                                                                                                         |
|                                                             | User to specify low voltage level to be used for waveform aware pulse-width checks<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> 0.05<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_waveform_aware_pulse_width_checks_low_voltage_level</a>   |
| timing_write_sdf_no_escape_backslash                        |                                                                                                                                                                                                                                                                         |
|                                                             | Supports various patterns of the escape character.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">timing_write_sdf_no_escape_backslash</a>                                                          |
| tns_critical_range                                          |                                                                                                                                                                                                                                                                         |
|                                                             | Slack-delta to consider critical (relative to the worst endpoint slack).<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> 0.0<br><b>Edit:</b> Yes<br><b>Reference:</b> <a href="#">tns_critical_range</a>                                                      |
| top_sdps                                                    |                                                                                                                                                                                                                                                                         |
|                                                             | Short-cut to all top level structured data path (sdp) objects.<br><b>Type:</b> <a href="#">obj(sdp)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                       |
| trace_obj_macro_pins                                        |                                                                                                                                                                                                                                                                         |

|                                    |                                                                                                                                                                                                                                                                                                                         |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>specify macro pin name patterns in regular expression</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">trace_obj_macro_pins</a></p>                                                                                                 |
| trace_obj_max_fanin_fanout         |                                                                                                                                                                                                                                                                                                                         |
|                                    | <p>Specify the maximum number of a pin's fanin/fanout,a pin will be ignored in tracing connect if its fanin/fanout larger than the specified number</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 1000</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">trace_obj_max_fanin_fanout</a></p> |
| trace_obj_register_inputs          |                                                                                                                                                                                                                                                                                                                         |
|                                    | <p>Specify register input pin names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">trace_obj_register_inputs</a></p>                                                                                                                 |
| trace_obj_register_outputs         |                                                                                                                                                                                                                                                                                                                         |
|                                    | <p>Specify register output pin names</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">trace_obj_register_outputs</a></p>                                                                                                               |
| track_patterns                     |                                                                                                                                                                                                                                                                                                                         |
|                                    | <p>Short-cut to all the track_pattern objects in the design.</p> <p><b>Type:</b> <a href="#">obj(track_pattern)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                               |
| treat_non_seq_arc_cell_as_unusable |                                                                                                                                                                                                                                                                                                                         |
|                                    | <p>Libcell having non sequential setup arc is treated as unusable if 'timing_disable_non_sequential_checks' is set to false.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">treat_non_seq_arc_cell_as_unusable</a></p>               |
| trim_grids                         |                                                                                                                                                                                                                                                                                                                         |

|                          |                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Short-cut to all the trim_grid objects in the design from the LEF TRIMMETALTRACK statement.</p> <p><b>Type:</b> <a href="#">obj(trim_grid)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                          |
| turbo_lib_loading        |                                                                                                                                                                                                                                                                                                                                                  |
|                          | <p>Populate timing arcs, power arcs and library establishment data only on need basis for run speedup.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 2</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">turbo_lib_loading</a></p>                                                                                   |
| ui_precision             |                                                                                                                                                                                                                                                                                                                                                  |
|                          | <p>Specifies the number of significant digits to be displayed in timing reports for data with no s type.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 3</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ui_precision</a></p>                                                                                      |
| ui_precision_capacitance |                                                                                                                                                                                                                                                                                                                                                  |
|                          | <p>Specifies the number of significant digits to be displayed in timing reports for data of type capacitance.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 3</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ui_precision_capacitance</a></p>                                                                     |
| ui_precision_derating    |                                                                                                                                                                                                                                                                                                                                                  |
|                          | <p>Specifies the number of significant digits to be displayed in timing reports for derating factor as those specified via set_timing_derate constraints or AOCV derating libraries.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 3</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ui_precision_derating</a></p> |
| ui_precision_power       |                                                                                                                                                                                                                                                                                                                                                  |
|                          | <p>Specifies the number of significant digits to be displayed in timing reports for data of type power.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 3</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">ui_precision_power</a></p>                                                                                 |

|                                |                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ui_precision_sensitivities     |                                                                                                                                                                                                                                                                                                                                                |
|                                | <p>Specifies the number of significant digits to be displayed in timing reports for statistical sensitivity values and SOCV sigma values.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 3<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">ui_precision_sensitivities</a></p>                                           |
| ui_precision_timing            |                                                                                                                                                                                                                                                                                                                                                |
|                                | <p>Specifies the number of significant digits to be displayed for timing reports for delay type values including cell and net delays, transitions, arrival and required times, and slacks.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 3<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">ui_precision_timing</a></p> |
| ui_units_capacitance           |                                                                                                                                                                                                                                                                                                                                                |
|                                | <p>capacitance unit for CUI</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> 1pf 1ff 10ff 100ff<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                             |
| ui_units_capacitance_reporting |                                                                                                                                                                                                                                                                                                                                                |
|                                | <p>Capacitance reporting unit for CUI</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> 1pf 1ff 10ff 100ff<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                    |
| ui_units_power_reporting       |                                                                                                                                                                                                                                                                                                                                                |
|                                | <p>Power reporting unit for CUI</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                    |
| ui_units_timing                |                                                                                                                                                                                                                                                                                                                                                |
|                                | <p>timing unit for CUI</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> 1ns 1ps 10ps 100ps<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                  |

|                                             |                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ui_units_timing_reporting                   |                                                                                                                                                                                                                                                                                                           |
|                                             | <p>Time reporting unit for CUI</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> 1ns 1ps 10ps 100ps</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                |
| uniquify_naming_style                       |                                                                                                                                                                                                                                                                                                           |
|                                             | <p>the default uniquify naming style as defined by this tcl global</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> %s_%d</p> <p><b>Edit:</b> Yes</p>                                                                                                                                    |
| unmap_scan_flops                            |                                                                                                                                                                                                                                                                                                           |
|                                             | <p>Whether it is OK to unmap scan flops {false   0   not_mapped_for_dft   true   1}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false not_mapped_for_dft true</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">unmap_scan_flops</a></p> |
| use_area_from_lef                           |                                                                                                                                                                                                                                                                                                           |
|                                             | <p>Controls whether the cell area from the LEF files should be used.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">use_area_from_lef</a></p>                                                                       |
| use_compatibility_based_grouping            |                                                                                                                                                                                                                                                                                                           |
|                                             | <p>Allows to merge 1-bit sequential, combinational and tristate cells into multibit cells based on compatible groups.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">use_compatibility_based_grouping</a></p>          |
| use_main_cell_output_function_for_test_cell |                                                                                                                                                                                                                                                                                                           |
|                                             | <p>Use main cell's output function for test cell.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">use_main_cell_output_function_for_test_cell</a></p>                                                                   |

|                                     |                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| use_multibit_cells                  |                                                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Allows to merge 1-bit sequential, combinational, tristate and iso/ls cells into multibit cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">use_multibit_cells</a></p>                                                                                              |
| use_multibit_iso_cells              |                                                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Allows to merge 1-bit iso cells into multibit cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">use_multibit_iso_cells</a></p>                                                                                                                                     |
| use_multibit_seq_and_tristate_cells |                                                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Allows to merge 1-bit sequential and tristate cells into multibit cells.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">use_multibit_seq_and_tristate_cells</a></p>                                                                                                    |
| use_scan_seqs_for_non_dft           |                                                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Allow the use of scan sequential cells for non-DFT purposes (true false degenerated_only)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> true false degenerated_only</p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">use_scan_seqs_for_non_dft</a></p>                                       |
| verification_directory_naming_style |                                                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>Format of directory in which verification files are written when the 'write_verification_files' at is 'true'. The directory will be created if it does not already exist.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> fv/%s</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">verification_directory_naming_style</a></p> |
| via_def_rules                       |                                                                                                                                                                                                                                                                                                                                                               |
|                                     | <p>All the via_def_rules defined in the LEF or OpenAccess techfile.</p> <p><b>Type:</b> <a href="#">obj(via_def_rule)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                               |

|                                  |                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| via_defs                         |                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>All the via_defs (via master definitions).</p> <p><b>Type:</b> <a href="#">obj(via_def)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                  |
| violations                       |                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>List of all violations</p> <p><b>Type:</b> <a href="#">obj(violation)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                    |
| vt_swap_out_dir                  |                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>directory of vt swap report file, default is current directroy ./</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                 |
| vt_swap_out_file                 |                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>file name of vt swap report</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                       |
| vt_swap_to_fix_filler_violations |                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>invoke swap cell to fix filler violation</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                         |
| wcdc_clock_dom_comb_propagation  |                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>To control clock domain propagation through combinational elements in CDC. Valid values: exact_phase or either_phase or wire or logic_phase or logic_nophase.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> logic_nophase</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">wcdc_clock_dom_comb_propagation</a></p> |
| write_clp_lib_statetable         |                                                                                                                                                                                                                                                                                                                                                       |



|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>Whether 'read library' has the '-statetable' switch.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_clp_lib_statetable</a></p>                                                                                                                                                                                                                                                                         |
| write_db_auto_save_user_globals         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                         | <p>If true, write_db automatically saves user created Tcl global variables. User globals are defined as any "info globals" names that do not exist at startup (e.g. created by user executed code). See the <code>define_variables</code> command to add or delete specific Tcl variables from the automatic save list.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_db_auto_save_user_globals</a></p> |
| write_db_binary_timing_constraints      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                         | <p>If true, write_db will save timing constraints in binary format. The value of the global should be persistent in subsequent write_db and read_db operations.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_db_binary_timing_constraints</a></p>                                                                                                                                                       |
| write_db_cmd_file_limit                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                         | <p>write_db saves the complete .cmd history across multiple sessions in the inn.cmd.gz file in the save directory up to this file length limit in units of Mb. If it is set to 0, no history is saved at all.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 10</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_db_cmd_file_limit</a></p>                                                                                                                       |
| write_db_copy_timing_constraints_always |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                         | <p>Specifies whether SDC files are always copied to the DB directory rather than links to external SDC files. By default they are only copied if the constraints were modified.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_db_copy_timing_constraints_always</a></p>                                                                                                                                 |
| write_db_create_read_file               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <p>If true, 'write_db my_db' will also create a 'my_db.read' file with a 'read_db my_db' command inside it. This is a Tcl file that can be passed to the Linux executable to make it easy to load at startup like this: 'innovus -files my_db.read'. For OA usage, it would have the appropriate 'read_db -oa_lib_cell_view {&lt;lib&gt; &lt;cell&gt; &lt;view&gt;}' command inside it.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_db_create_read_file</a></p> |
| write_db_include_metal_fill_rules      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>specify whether to save metalfill settings</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_db_include_metal_fill_rules</a></p>                                                                                                                                                                                                                                                                                                                                        |
| write_db_restrict_under_batch_mode     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>By default, a new db can be saved under batch mode, tool errors out when this attribute is set true.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_db_restrict_under_batch_mode</a></p>                                                                                                                                                                                                                                                                        |
| write_db_restrict_under_eco_batch_mode |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Controls whether to restrict db saving in eco batch mode.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_db_restrict_under_eco_batch_mode</a></p>                                                                                                                                                                                                                                                                                                               |
| write_db_save_unused_lef_block_names   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                        | <p>Specifies whether to remember a list of unused macros during saving</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_db_save_unused_lef_block_names</a></p>                                                                                                                                                                                                                                                                                                       |
| write_def_compress_vias                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>Compress the via statement in SPECIALNETS section. This variable controls write_def to s with two dimension array, so it can use 'DO numX BY numY STEP stepX stepY' statement.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_def_compress_vias</a></p>                                                                                        |
| write_def_hierarchy_delimiter |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                               | <p>Specify the write_def hierarchy delimiter.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_def_hierarchy_delimiter</a></p>                                                                                                                                                                                                                  |
| write_def_include_lef_ndr     |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                               | <p>Output LEF NONDEFAULTRULES in DEF. This variable controls write_def to output LEF nondefault rule information. By default, write_def does not write out nondefault rules defined in LEF. Set this variable to true to output nondefault rules defined in LEF .</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_def_include_lef_ndr</a></p> |
| write_def_include_lef_vias    |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                               | <p>Output LEF vias in DEF. This variable controls write_def to output LEF via information. By default, write_def does not write out via defined in LEF. Set this variable to true to output LEF via.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_def_include_lef_vias</a></p>                                                             |
| write_def_lef_out_version     |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                               | <p>Specify LEF/DEF output version. Possible string value are 5.5, 5.6, 5.7, 5.8 and 6.0. The default is 5.8.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> 5.5 5.6 5.7 5.8 6.0<br/> <b>Default:</b> 5.8<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_def_lef_out_version</a></p>                                                                                                           |
| write_def_polygon_die_area    |                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>By default write_def writes out polygon DIEAREA if the design is rectilinear. To write out rectangular DIEAERA along with blockages in cut-out area for a rectilinear design, set this to false.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_def_polygon_die_area</a></p>                                                                                                                                                                                                                     |
| write_def_stream_check_uncolored               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                | <p>Specify how to check and ignore uncolored shapes on DPT layers during DEF out and stream. If true or 1, checks uncolored shapes on all DPT layers; if false or 0, doesn't do check; if metal_only, only checks uncolored shapes on metal layers, uncolored shapes on cut layers ignored. It's automatically set to metal_only for TSMC N5 and below nodes.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> true 1 false 0 metal_only<br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_def_stream_check_uncolored</a></p> |
| write_do_lec_add_no_black_box_retime_subdesign |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                | <p>Whether to write 'add noblack box -submodules' for the retime modules in the dofile.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_add_no_black_box_retime_subdesign</a></p>                                                                                                                                                                                                                                                                                                             |
| write_do_lec_analyze_abort                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                | <p>Whether to enable 'analyze_abort' flow in LEC.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_analyze_abort</a></p>                                                                                                                                                                                                                                                                                                                                                                      |
| write_do_lec_analyze_setup                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                | <p>Whether to enable 'analyze setup' flow in LEC.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_analyze_setup</a></p>                                                                                                                                                                                                                                                                                                                                                                      |
| write_do_lec_auto_analyze                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                    |                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Whether to enable 'auto analyze' in LEC.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_do_lec_auto_analyze</a></p>                                                                                                                            |
| write_do_lec_black_box_ilm_modules |                                                                                                                                                                                                                                                                                                                                            |
|                                    | <p>Black box the ILM modules.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_do_lec_black_box_ilm_modules</a></p>                                                                                                                                |
| write_do_lec_compare_threads       |                                                                                                                                                                                                                                                                                                                                            |
|                                    | <p>Specifies number of threads for 'set compare option -threads'.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 4</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_do_lec_compare_threads</a></p>                                                                                                       |
| write_do_lec_composite_compare     |                                                                                                                                                                                                                                                                                                                                            |
|                                    | <p>Specifies two-step comparison: RTL-to-fv_map and fv_map-to-revised. This attribute has no effect when root attribute 'one_pass_formal_verification' is 'true'.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_do_lec_composite_compare</a></p> |
| write_do_lec_dft_constraint_file   |                                                                                                                                                                                                                                                                                                                                            |
|                                    | <p>DFT constraints file. This file must exist when verification is performed.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_do_lec_dft_constraint_file</a></p>                                                                                   |
| write_do_lec_gzip_lec_json         |                                                                                                                                                                                                                                                                                                                                            |
|                                    | <p>Compress the implementation information files. Conformal 19.10-s300 or newer is required to execute the dofile.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_do_lec_gzip_lec_json</a></p>                                                   |

|                                  |                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| write_do_lec_hier_append_string  |                                                                                                                                                                                                                                                                                                                                               |
|                                  | <p>When not empty, the value is used as the '-append_string' argument to the Conformal LEC 'write_hier_compare_dofile' command written to the dofile by 'write_do_lec'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_hier_append_string</a></p>   |
| write_do_lec_hier_comp_threshold |                                                                                                                                                                                                                                                                                                                                               |
|                                  | <p>The threshold limit for minimum number of instances in a module in order to perform hierarchical comparison.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 50<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_hier_comp_threshold</a></p>                                                             |
| write_do_lec_hier_compare_string |                                                                                                                                                                                                                                                                                                                                               |
|                                  | <p>When not empty, the value is used as the '-compare_string' argument to the Conformal LEC 'write_hier_compare_dofile' command written to the dofile by 'write_do_lec'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_hier_compare_string</a></p> |
| write_do_lec_hier_prepend_string |                                                                                                                                                                                                                                                                                                                                               |
|                                  | <p>When not empty, the value is used as the '-prepend_string' argument to the Conformal LEC 'write_hier_compare_dofile' command written to the dofile by 'write_do_lec'.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_hier_prepend_string</a></p> |
| write_do_lec_lib_statetable      |                                                                                                                                                                                                                                                                                                                                               |
|                                  | <p>Whether 'read library' has the '-statetable' switch.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_lib_statetable</a></p>                                                                                                                       |
| write_do_lec_low_power_analysis  |                                                                                                                                                                                                                                                                                                                                               |

|                                            |                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>Adds the Conformal Low Power commands to compare the power domains between the go and revised designs. Power intent must be present.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_low_power_analysis</a></p>                         |
| write_do_lec_multithread_license_list      |                                                                                                                                                                                                                                                                                                                                      |
|                                            | <p>Specifies the license, or a list of licenses, to use for multi-threaded processing. If an empty string is specified, the LEC default list will be used.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_multithread_license_list</a></p> |
| write_do_lec_no_dft_constraints            |                                                                                                                                                                                                                                                                                                                                      |
|                                            | <p>Do not include automatically generated DFT constraints in the dofile.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_no_dft_constraints</a></p>                                                                                        |
| write_do_lec_parallel_threads              |                                                                                                                                                                                                                                                                                                                                      |
|                                            | <p>Specifies number of threads for 'set parallel option -threads'.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 4<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_parallel_threads</a></p>                                                                                                     |
| write_do_lec_post_add_no_translate_modules |                                                                                                                                                                                                                                                                                                                                      |
|                                            | <p>An extra dofile to be source'd from dofile right after any 'add_notranslate_modules' command before reading the library.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_post_add_no_translate_modules</a></p>                           |
| write_do_lec_run_hier_check_non_equivalent |                                                                                                                                                                                                                                                                                                                                      |
|                                            | <p>Adds '-check_noneq' option to run_hier_compare / go_hier_compare.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_do_lec_run_hier_check_non_equivalent</a></p>                                                                                 |

|                               |                                                                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| write_do_lec_set_cdn_syn_root |                                                                                                                                                                                                                                                                                                                 |
|                               | <p>Whether to write out the setting for CDN_SYNTH_ROOT in the dofile.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">write_do_lec_set_cdn_syn_root</a></p>                                                                           |
| write_do_lec_uniquify         |                                                                                                                                                                                                                                                                                                                 |
|                               | <p>Whether to have a 'uniquify -all' command.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> true<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">write_do_lec_uniquify</a></p>                                                                                                            |
| write_do_lec_use_lec_model    |                                                                                                                                                                                                                                                                                                                 |
|                               | <p>Whether to use LEC built-in CW/DW models.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> true<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">write_do_lec_use_lec_model</a></p>                                                                                                        |
| write_do_lec_use_smart_lec    |                                                                                                                                                                                                                                                                                                                 |
|                               | <p>Adds adaptive-proof commands and information on massively parallel hierarchical comparison to the dofile. Conformal Smart LEC license required.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">write_do_lec_use_smart_lec</a></p> |
| write_do_lec_verbose          |                                                                                                                                                                                                                                                                                                                 |
|                               | <p>Add more verbose reports to the dofile.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> false<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">write_do_lec_verbose</a></p>                                                                                                               |
| write_lec_dft_constraints     |                                                                                                                                                                                                                                                                                                                 |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>This attribute is a TCL dict and holds the LEC dft pin constraints from the last LEC run by Genus. The constraints are passed forward through write_design. The write_do_lec command will automatically include the constraints in the do file.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_lec_dft_constraints</a></p>                                                                                                                                                                                           |
| write_lec_directory_naming_style     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      | <p>The directory name where 'write_do_lec' will write verification files when the 'write_lec_files' attribute is 'true'. The directory will be created if it does not already exist, and will overwrite any existing directory of the same name. A %s in the string is replaced with the design name (e.g., 'design_name'). A %d in the string is replaced with a unique integer to avoid overwriting any existing directory.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> fv/invs/%s<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_lec_directory_naming_style</a></p> |
| write_lec_files                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      | <p>Specifies whether 'write_do_lec' should write intermediate files to the verification directory specified by the 'write_lec_directory_naming_style' attribute. If false, no files are written.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> true<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_lec_files</a></p>                                                                                                                                                                                                                                                       |
| write_lef_abstract_customer_header   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      | <p>Write lef abstract files with customer specified header information.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_lef_abstract_customer_header</a></p>                                                                                                                                                                                                                                                                                                                                                             |
| write_netlist_full_pin_out           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      | <p>Output all pins.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_netlist_full_pin_out</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                        |
| write_netlist_port_association_style |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p>Do implicit port mapping.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_netlist_port_association_style</a></p>                                                                                                                                                                                          |
| write_stream_allow_path_type4    |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>If false, converts PATHTYPE4 to PATHTYPE0.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_allow_path_type4</a></p>                                                                                                                                                                                 |
| write_stream_cell_instance_color |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>stream out the instance based color. The instance mask shift is honored in write_stream. W option is on and -merge option is used, the merged gds files with mask shift suffix must be provided through option -merge.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_cell_instance_color</a></p> |
| write_stream_cell_master_color   |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>stream out the cell based color from innovus color engine.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_cell_master_color</a></p>                                                                                                                                                                |
| write_stream_cell_name_prefix    |                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                  | <p>Add a prefix to ALL cell names being written out, use user specified prefix can avoid potent conflict when integrating different macros, which potentially 2 different cells (in 2 different lil with the same name.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_cell_name_prefix</a></p>       |
| write_stream_cell_name_suffix    |                                                                                                                                                                                                                                                                                                                                                                                                       |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>Instead of appending color pattern to normal cell names, use user specified suffix. This app fixed color shift (flip) pattern only. i.e., only single shift (flip) pattern applied to the design.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_stream_cell_name_suffix</a></p>                                        |
| write_stream_check_map_file    |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Check the specified map file. If the objects existed in innovus, but the objects have no object mapping in map file, print the missed objects mapping and missed objects in log file, only the objects mapping are checked by the option.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_stream_check_map_file</a></p> |
| write_stream_compatible        |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Output instance only is FOREIGN is specified.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_stream_compatible</a></p>                                                                                                                                                                                                 |
| write_stream_define_via_name   |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Specifies the naming convention that the write_stream command uses for via cells declared in DEF file.</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> default<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_stream_define_via_name</a></p>                                                                                                                               |
| write_stream_flatten_via_cells |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Flatten via instance its via cell name matches list of via cell names</p> <p><b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                         |
| write_stream_ignore_fixed_mask |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                | <p>Ignore macro FIXEDMASK keywords, enable color flip function.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_stream_ignore_fixed_mask</a></p>                                                                                                                                                                           |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| write_stream_label_all_pin_shape  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                   | <p>If true, label pin name to its all shapes.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_label_all_pin_shape</a></p>                                                                                                                                                                                                                                                                                                                                                                       |
| write_stream_merge_append         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                   | <p>Append cell color information in current design to cell geometries that are from merging file option is for TSMC 10nm fixed-mask design. It is exclusive with write_stream_ignore_inst_color attribute and write_stream_ignore_inst_cce_color attribute, it will set those two attributes to automatically when it is set to true.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_merge_append</a></p>                                                                                      |
| write_stream_no_pin_label_on_nets |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                   | <p>Do not label pin text on specified nets.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_no_pin_label_on_nets</a></p>                                                                                                                                                                                                                                                                                                                                                                         |
| write_stream_oasis_cell_offset    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                   | <p>Include cell offset table to declare the byte offset from the beginning of the file (byte 0) to where corresponding CELL record appears in the file.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_oasis_cell_offset</a></p>                                                                                                                                                                                                                                                               |
| write_stream_oasis_compression    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                   | <p>CBLOCK compression is partial-selective encryption technology uses OASIS CBLOCK records to encrypt arbitrarily encapsulated sets of data records within an OASIS layout file. REPETITION compression supports an 'array' of cell placements, geometries, or text elements instead of write each the element separate only different in position, this can save duplicate information for an OASIS layout file size.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_oasis_compression</a></p> |

|                                                      |                                                                                                                                                                                                                                                                |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| write_stream_oasis_layer_name                        |                                                                                                                                                                                                                                                                |
|                                                      | <p>Support LAYERNAME in OASIS file.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_oasis_layer_name</a></p>                                                   |
| write_stream_pin_text_orientation                    |                                                                                                                                                                                                                                                                |
|                                                      | <p>Output pin name on the boundary of design to improve readability.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_pin_text_orientation</a></p>               |
| write_stream_remove_nets                             |                                                                                                                                                                                                                                                                |
|                                                      | <p>Don't output nets in the list.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_remove_nets</a></p>                                                           |
| write_stream_remove_short_metal_within_cell_boundary |                                                                                                                                                                                                                                                                |
|                                                      | <p>The trim metal short inside FTV_cell_list should be cut off.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_remove_short_metal_within_cell_boundary</a></p> |
| write_stream_snap_to_mfg                             |                                                                                                                                                                                                                                                                |
|                                                      | <p>Snap net's wire to manufacture grid if it is not.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> <p><b>Reference:</b> <a href="#">write_stream_snap_to_mfg</a></p>                                       |
| write_stream_stream_convert_rect_to_path             |                                                                                                                                                                                                                                                                |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>When enabled, DEF SPECIALNETS path/rect and FILLS shapes are converted to GDSII/str PATH data instead of BOUNDARY data. Conversion to path data can result in significantly output database when there are many rectangle shapes (example: after metal fill is include</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_stream_stream_convert_rect_to_path</a></p> |
| write_stream_stream_version             |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                         | <p>Specify GDSII version to output.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 3<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_stream_stream_version</a></p>                                                                                                                                                                                                                                                         |
| write_stream_text_size                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                         | <p>Changes the size of the text used in the text labels.</p> <p><b>Type:</b> <a href="#">double</a><br/> <b>Default:</b> 1.0<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_stream_text_size</a></p>                                                                                                                                                                                                                                    |
| write_stream_uniquify_cell_names_prefix |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                         | <p>Adds a prefix, instead of a suffix, to uniquified cell names.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_stream_uniquify_cell_names_prefix</a></p>                                                                                                                                                                                                           |
| write_stream_via_names                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                         | <p>Determines the naming convention used for vias declared in the LEF file.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> false<br/> <b>Edit:</b> Yes<br/> <b>Reference:</b> <a href="#">write_stream_via_names</a></p>                                                                                                                                                                                                                 |
| write_stream_virtual_connection         |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                          |                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Specifies whether a colon (:) label is appended to pin names for DEF pins with the .extraN : and to LEF pins with multiple ports (if -output_macros is specified for LEF pin ports that have disjoint shapes).</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> true<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">write_stream_virtual_connection</a></p> |
| write_verification_files |                                                                                                                                                                                                                                                                                                                                                                                    |
|                          | <p>Specifies whether to write intermediate files to the verification directory, which is specified by 'verification_directory_naming_style' attribute.</p> <p><b>Type:</b> <a href="#">bool</a><br/><b>Default:</b> true<br/><b>Edit:</b> Yes<br/><b>Reference:</b> <a href="#">write_verification_files</a></p>                                                                   |
| xedebug_executable       |                                                                                                                                                                                                                                                                                                                                                                                    |
|                          | <p>Set xedebug path for PHY in Joules.</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                  |

## route\_blockage

### Parent Objects

[design](#), [root](#)

### Definition

Routing blockage

| Attribute | Description                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------|
| base_name |                                                                                                                |
|           | <p>name of blockage</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p> |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| density           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                   | <p>The density percentage allowed for a partial routing blockage. It causes the global router to only use up to this percent of the routing resource on the layer in the blockage area, so the global router will see higher congestion and put fewer routes on that layer in the area.</p> <p><b>Type:</b> <code>int</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> Yes</p>                                                                                                                                                |
| design_rule_width |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                   | <p>Specifies that the blockage has an effective width for the purposes of spacing calculations to other shapes on the same layer. A value of 0 indicates that there is no specified design_rule_width value. The design_rule_width and spacing attributes are not allowed to have non-default values at the same time, so the design_rule_width value cannot be changed when the spacing attribute's value is not the default.</p> <p><b>Type:</b> <code>coord</code><br/> <b>Default:</b> <code>no_value</code><br/> <b>Edit:</b> Yes</p> |
| inst              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                   | <p>The instance that the routing blockage is associated with (equivalent to DEF BLOCKAGES + COMPONENT)</p> <p><b>Type:</b> <code>obj(inst)</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                            |
| is_except_pg_net  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                   | <p>Indicates that Power/Ground routing is ignored when checking for DRC violations (including shorts) involving the current shape (equivalent to DEF BLOCKAGES + EXCEPTPGNET)</p> <p><b>Type:</b> <code>bool</code><br/> <b>Default:</b> <code>false</code><br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                      |
| is_fixed          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                   | <p>Disable set_db to change any attribute for route blockages with .is_fixed=1.</p> <p><b>Type:</b> <code>bool</code><br/> <b>Default:</b> <code>false</code><br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                    |
| is_no_wrong_way   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



|                |                                                                                                                                                                                                                                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>Block wrong way routing. Only real routing wires are honored to not have jogs in this region.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                  |
| is_only_pg_net |                                                                                                                                                                                                                                                                                                                 |
|                | <p>Indicate that the routing blockage applies only to power or ground SPECIALNETS routing. (equivalent to DEF BLOCKAGES + ONLYPGNET)</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                          |
| is_pushdown    |                                                                                                                                                                                                                                                                                                                 |
|                | <p>Indicates that routing blockage has been pushed down from a higher level in the design hierarchy. The idea is that the routing blockage is owned by a higher level(equivalent to DEF BLOCKAGES + PUSHDOWN)</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p> |
| layer          |                                                                                                                                                                                                                                                                                                                 |
|                | <p>layer of blockage</p> <p><b>Type:</b> <a href="#">obj(layer)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                       |
| name           |                                                                                                                                                                                                                                                                                                                 |
|                | <p>name of blockage</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                            |
| obj_type       |                                                                                                                                                                                                                                                                                                                 |
|                | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum (route_blockage)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                  |
| rects          |                                                                                                                                                                                                                                                                                                                 |

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>List of non-overlapping rectangles that defines the shape of the route_blockage. If the blockage was defined with a polygon, it is broken up into non-overlapping rectangles</p> <p><b>Type:</b> <code>rect*</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| shapes  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|         | <p>List of shapes that define the blockage area</p> <p><b>Type:</b> <code>obj(shape)*</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| spacing |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|         | <p>Specifies the minimum spacing allowed between the blockage and any other shape on the same layer (e.g. DEF BLOCKAGES with +SPACING). If there is no spacing value defined, the default is -1 in DB units, which will appear like a small value &lt; 0 (e.g.-0.0005 if the get_db db_units is 2000). The design_rule_width and spacing attributes are not allowed to have non-default values at the same time, so the spacing value cannot be changed when the design_rule_width attribute's value is also set. To reset the spacing value use [set_db -dbu route_blockages .spacing -1].</p> <p><b>Type:</b> <code>coord</code><br/> <b>Default:</b> <code>no_value</code><br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| type    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|         | <p>The routing blockage type. default: means the blockage is completely blocked and check_drc will treat it like min-width routing for spacing checks. partial: means a percentage of the routing resource is available on the layer (see density attribute). fills: means do not add any metal fill in the area. except_fills: means do not add objects except metal fills in the area.slots: means don't add slots to the area. slots is not used in Innovus but is allowed to match DEF. The type corresponds to the DEF BLOCKAGES + FILLS or + SLOTS value. There is no DEF syntax for 'partial', so partial blockages are currently lost when written to DEF. core_eol is an end-of-line blockage added by finish_floorplan -add_core_end_of_line_blockage using LEF COREEOLBLOCKAGE statements. block_eol: An end-of-line blockage added by finish_floorplan - add_block_eol_blockage using LEF BLOCKEOLBLOCKAGE statements.</p> <p><b>Type:</b> <code>enum</code><br/> <b>Enum Values:</b> default slots fills except_fills partial core_eol block_eol boundary_eol<br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> Yes</p> |

# route\_rule

## Parent Objects

[patch\\_wire](#), [track\\_pattern](#), [wire](#), [pg\\_base\\_pin](#), [net](#), [tech](#), [via](#), [route\\_type](#), [root](#), [base\\_pin](#)

## Definition

Rule information

| Attribute       | Description                                                                                                                                                                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name       |                                                                                                                                                                                                                                                |
|                 | Name of non-default rule.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                      |
| is_from_lib     |                                                                                                                                                                                                                                                |
|                 | Indicates whether the rule came from the library technology (true: from LEF or OA tech) or from the design (false: from DEF, OA database, or create_route_rule).<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| is_hard_spacing |                                                                                                                                                                                                                                                |
|                 | Indicates that any spacing values that exceed the LEF LAYER spacing requirements are 'hard' rules instead of 'soft' rules.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                       |
| layer_rules     |                                                                                                                                                                                                                                                |
|                 | List of layers rules<br><b>Type:</b> <a href="#">obj(layer_rule)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                 |
| min_cuts        |                                                                                                                                                                                                                                                |

|          |                                                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | List of cut layer and minimum number of cuts allowed for any via using the specified cut layer<br><b>Type:</b> <a href="#">string</a> *<br><b>Default:</b> ""<br><b>Edit:</b> No                          |
| name     |                                                                                                                                                                                                           |
|          | Name of non-default rule.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                 |
| obj_type |                                                                                                                                                                                                           |
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">route_rule</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| tech     |                                                                                                                                                                                                           |
|          | The tech for this route_rule.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">tech</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                       |
| via_defs |                                                                                                                                                                                                           |
|          | List of via, default or USEVIA or derived from mincut.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">via_def</a> )*<br><b>Default:</b> ""<br><b>Edit:</b> No                                          |

## route\_type

### Parent Objects

[root](#)

### Definition

route type

| Attribute                 | Description                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name                 |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>Name of route_type object.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                       |
| bottom_mask_layer_num     |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>Specify the bottom layer number that the mask constraint should be applied.</p> <p><b>Type:</b> int</p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                         |
| bottom_one_side_layer_num |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>Specify the bottom layer number that one side spacing constrain should be applied on. By default, NDR spacing is applied to both sides of an NDR net or wire. Use this to specify the layer range for wires that have only one neighboring wire with minimum spacing, which means only one side needs to follow larger NDR spacing.</p> <p><b>Type:</b> int</p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> |
| bottom_preferred_layer    |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>The preferred lowest routing layer. This attribute is a soft limit; that is, NanoRoute might use a layer below the specified layer if necessary to complete routing.</p> <p><b>Type:</b> obj(layer)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                         |
| driver_use_multi_cut_via  |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                           | <p>If true, then driver pins will use multi-cut vias.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                             |
| em_route_rule             |                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Specifies the EM route_rule to associate with this route type. When routing within the distance specified in route_em_rule_distance from the output pin, the router will use this route_rule to route. By default no route_em_rule is used.</p> <p><b>Type:</b> string<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                       |
| em_route_rule_distance |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                        | <p>Specifies the distance from the output pin, when the route_em_rule is applied to the net. This allows a larger width route_rule for a short distance to avoid EM violations near the output pin. When routing outside this distance from the output pin, the router uses the normal net route_rule.</p> <p><b>Type:</b> double<br/><b>Default:</b> no_value<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                         |
| is_table               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                        | <p>Indicates that whether the route type is table based.</p> <p><b>Type:</b> bool<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| mask                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                        | <p>Indicates mask number for multiple mask layer usage. Refer to layers .numMask attribute for valid range, 0 indicates unconstrained. Layers that do not support the specified value will be treated as unconstrained. (Legal range: 0-3).</p> <p><b>Type:</b> int<br/><b>Default:</b> 0<br/><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                              |
| min_stack_layer        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                        | <p>The net should use a stacked via from output pins up to the given layer before starting normal routing. This is normally used to force the routing to higher layers with wider widths to reduce wiring resistance or avoid EM current limits for high-drive outputs. If output_stack_via_rule or input_stack_via_rule is also specified, the specified stack via rule is used for the input or output pins accordingly. If both output_stack_via_rule/input_stack_via_rule are not specified, a single-cut stacked-via will be used for the output pins only.</p> <p><b>Type:</b> obj(layer)<br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p> |
| name                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                      |                                                                                                                                                                                                                                             |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Name of route_type object.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                               |
| obj_type                             |                                                                                                                                                                                                                                             |
|                                      | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">route_type</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                  |
| route_effort                         |                                                                                                                                                                                                                                             |
|                                      | <p>route effort</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> low medium high</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                   |
| route_rule                           |                                                                                                                                                                                                                                             |
|                                      | <p>The non-default rule corresponding to the net, nets with the default routing rule will return NULL (0x0).</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">route_rule</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>       |
| shield_net                           |                                                                                                                                                                                                                                             |
|                                      | <p>shield net</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">net</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                             |
| shield_side                          |                                                                                                                                                                                                                                             |
|                                      | <p>Specifies whether to perform one sided or two sided shielding for the route type specified.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> one_side both_side</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| shield_tap_instance_insertion_effort |                                                                                                                                                                                                                                             |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>different route type for clock shielding, including high frequency(high) and low frequency(standard) that main difference is what ground tie cell instance/via used to achieve shield segments electrical connection.</p> <p>None: default</p> <p>Standard: use existing standard cell vss m0 pins as much as possible, create a new instance of a ground tie cell under the shield route if cannot find an existing vss m0 pin within a user controlled search distance of a required connection, the new cell vss pins can have a "max fanout" of 2 shield nets.</p> <p>High: must create ground tie cell instances, and user can specify the physical distance of these new ground tie cells along the clock segments. new instance m0 vss pin to the clock shield segments will be a via stack of single vias on ever layer to the shield nets. It is acceptable for the router to create a n-1 metal shape to connect adjacent vss shield segments together and then route this single vss shape down to m0 ground tie pin. Shield segments may be electrically connected between adjacent route layers through vias. If this is done, there shall be no Manhattan distance between ground tie cells that is larger than the user provided value. Newly inserted ground tie cells cannot be shared between clocks.</p> <p><b>Type:</b> enum</p> <p><b>Enum Values:</b> none standard high</p> <p><b>Default:</b> none</p> <p><b>Edit:</b> Yes</p> |
| stack_distance         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                        | <p>Specifies that the cut distance of cuts on adjacent layers in the stacked vias are defined in min_stack_layer.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| top_mask_layer_num     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                        | <p>Specify the top layer number that the mask constraint should be applied.</p> <p><b>Type:</b> int</p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| top_one_side_layer_num |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                        | <p>Specify the top layer number that one side spacing constrain should be applied on. By default, NDR spacing is applied to both sides of an NDR net or wire. Use this to specify the layer range for wires that have only one neighboring wire with minimum spacing, which means only one side needs to follow larger NDR spacing.</p> <p><b>Type:</b> int</p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



|                     |                                                                                                                                                                                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| top_preferred_layer |                                                                                                                                                                                                                                                                        |
|                     | <p>The preferred highest routing layer. This attribute is a soft limit; that is, NanoRoute might use a layer above the specified layer if necessary to complete routing.</p> <p><b>Type:</b> <a href="#">obj(layer)</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |

## row

### Parent Objects

[design](#), [root](#)

### Definition

Row (core), constructed from sites (equivalent to DEF ROWS)

| Attribute | Description                                                                                                                                         |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name |                                                                                                                                                     |
|           | <p>Name of row (generated name)</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                           |
| name      |                                                                                                                                                     |
|           | <p>Name of row (generated name)</p> <p><b>Type:</b> <a href="#">string</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                           |
| num_x     |                                                                                                                                                     |
|           | <p>Number of sites in X direction (refer to DEF ROW syntax)</p> <p><b>Type:</b> <a href="#">int</a><br/><b>Default:</b> ""<br/><b>Edit:</b> Yes</p> |
| num_y     |                                                                                                                                                     |

|          |                                                                                                                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <p>Number of sites in Y direction (refer to DEF ROW syntax)</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                           |
| obj_type |                                                                                                                                                                                                                     |
|          | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">row</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| orient   |                                                                                                                                                                                                                     |
|          | <p>Orientation of the sites in the row</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> r0 r90 r180 r270 mx mx90 my my90</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                   |
| rect     |                                                                                                                                                                                                                     |
|          | <p>Rectangle that defines the row shape</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                               |
| site     |                                                                                                                                                                                                                     |
|          | <p>The site used in the row</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">site</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                     |
| step_x   |                                                                                                                                                                                                                     |
|          | <p>Step in the X direction (refer to DEF ROW syntax)</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                          |
| step_y   |                                                                                                                                                                                                                     |
|          | <p>Step in the Y direction (refer to DEF ROW syntax)</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                          |

# sdp

## Parent Objects

inst, design, root

## Definition

A structured datapath object. Each sdp is formed hierarchically from a list of sdps below it. Each sdp can be a row, column, space or inst (see .type). At the leaf-level, an sdp can only be a space or inst. See the create\_sdp\_group command for more help.

| Attribute | Description                                                                                                                                                                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bbox      |                                                                                                                                                                                                                                                                                    |
|           | Bounding box of the data path.<br><b>Type:</b> rect<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                       |
| flip      |                                                                                                                                                                                                                                                                                    |
|           | Specifies if an sdp group or an sdp element is flipped in vertical, horizontal, or both directions. If an sdp group is flipped, its members will also be flipped.<br><b>Type:</b> enum<br><b>Enum Values:</b> none flip_x flip_y flip_xy<br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| hier_name |                                                                                                                                                                                                                                                                                    |
|           | Specifies the hierarchical path name prepended to inst names inside the sdp file. This attribute is only available when .is_top = 1.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                              |
| insts     |                                                                                                                                                                                                                                                                                    |
|           | All the insts from a hierarchical descent below this sdp to the bottom of the tree.<br><b>Type:</b> obj(inst)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                            |

|            |                                                                                                                                                                                                                                                                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| is_placed  |                                                                                                                                                                                                                                                                                                                                                        |
|            | <p>Specifies if the data path is placed. This attribute is only available when .is_top = 1.</p> <p><b>Type:</b> <code>bool</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                        |
| is_top     |                                                                                                                                                                                                                                                                                                                                                        |
|            | <p>Indicates that the data path is a top data path group.</p> <p><b>Type:</b> <code>bool</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                                                          |
| justify_by |                                                                                                                                                                                                                                                                                                                                                        |
|            | <p>Specifies the anchor point that will be used for aligning an sdp group or an sdp element. If the justify_by constraint is not specified at current level, it will be inherited from its parent level.</p> <p><b>Type:</b> <code>enum</code><br/> <b>Enum Values:</b> sw se nw ne mid<br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> Yes</p> |
| local_list |                                                                                                                                                                                                                                                                                                                                                        |
|            | <p>An ordered list of sdp objects and inst objects contained by this sdp (e.g. just one-level of the sdp hierarchy).</p> <p><b>Type:</b> <code>obj(sdp)* obj(inst)*</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                               |
| location   |                                                                                                                                                                                                                                                                                                                                                        |
|            | <p>Lower left location of the data path.</p> <p><b>Type:</b> <code>point</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                         |
| name       |                                                                                                                                                                                                                                                                                                                                                        |
|            | <p>Name of Data Path.</p> <p><b>Type:</b> <code>string</code><br/> <b>Default:</b> <code>""</code><br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                            |
| obj_type   |                                                                                                                                                                                                                                                                                                                                                        |

|               |                                                                                                                                                                                                                                                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (sdp)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                  |
| orient        |                                                                                                                                                                                                                                                                                                                                                  |
|               | <p>The orientation of an sdp group or an sdp element. If the orientation is specified at sdp group level, it will be applied to instances that belong to this sdp group.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> r0 r180 mx my none<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                  |
| padding       |                                                                                                                                                                                                                                                                                                                                                  |
|               | <p>Edge extend value of top SDP bounding box.</p> <p><b>Type:</b> <a href="#">point</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                        |
| parent        |                                                                                                                                                                                                                                                                                                                                                  |
|               | <p>Return parent sdp group that the sdp belongs to.</p> <p><b>Type:</b> <a href="#">obj</a>(sdp)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                |
| pin_alignment |                                                                                                                                                                                                                                                                                                                                                  |
|               | <p>The sdp edge to use for pin alignment. The w, e, mid values are only available for sdp objects in a column sdp group (.type = column), that have .pin_names set. All others will have a value of unknown.</p> <p><b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> unknown w e mid<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| pin_names     |                                                                                                                                                                                                                                                                                                                                                  |
|               | <p>The pin names by which sdps are aligned using the .pin_alignment setting.</p> <p><b>Type:</b> <a href="#">string</a>*<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                       |
| sdps          |                                                                                                                                                                                                                                                                                                                                                  |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>List of all the sdps from a hierarchical descent inside and below this sdp.<br/> <b>Type:</b> <a href="#">obj(sdp)*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                    |
| snap_row_site_index |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     | <p>Specifies the single row site that SDP bbox should snap to.<br/> <b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> -1<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                         |
| space               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     | <p>Specifies a space value to be skipped. If the space value is defined in a column, then this value is for row skipping and represents the number of skipped rows. If the space value is defined in a row, then this value is for column skipping and represents the number of M2 tracks (pitch of first vertical routing layer). This attribute is only valid when .type = space.<br/> <b>Type:</b> <a href="#">string</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p> |
| type                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     | <p>Type of data path object. A row or column sdp has a .local_list with an ordered list of sdp and inst objects, while a space type only has a .space value.<br/> <b>Type:</b> <a href="#">enum</a><br/> <b>Enum Values:</b> row column space<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                 |

## shape

### Parent Objects

[route\\_blockage](#), [place\\_blockage](#), [layer\\_shape](#)

### Definition

A LEF PIN or OBS single-layer shape (RECT, POLYGON or PATH statement). Note that PATH statements are converted to one or more equivalent rect shapes. See shape\_vias for a LEF VIA used in a PIN or OBS statement.

| Attribute | Description                                                                                                                                                                                                       |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mask      |                                                                                                                                                                                                                   |
|           | Indicates mask number for multiple mask layer usage. Refer to layer's .num_masks attribute for legal range, 0 indicates uncolored.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 0<br><b>Edit:</b> Yes   |
| obj_type  |                                                                                                                                                                                                                   |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">shape</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No              |
| polygon   |                                                                                                                                                                                                                   |
|           | Points of the polygon for either rect or polygon shapes.<br><b>Type:</b> <a href="#">polygon</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                         |
| rect      |                                                                                                                                                                                                                   |
|           | Bounding box for either rect or polygon shapes.<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                     |
| type      |                                                                                                                                                                                                                   |
|           | The shape type (rect or polygon). A LEF PATH statement is converted to one or more rect shapes.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> rect polygon<br><b>Default:</b> ""<br><b>Edit:</b> No |

## shape\_via

### Parent Objects

[base\\_cell](#), [physical\\_pin](#)

## Definition

layer shape via

| Attribute    | Description                                                                                                                                                                                                                                                                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bottom_mask  |                                                                                                                                                                                                                                                                                                                                                                    |
|              | <p>Indicates mask number for bottom layer for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p>                                                                                                                       |
| bottom_rects |                                                                                                                                                                                                                                                                                                                                                                    |
|              | <p>List of rectangles (typically only one) on bottom routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master)</p> <p><b>Type:</b> <a href="#">rect</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                     |
| cut_mask     |                                                                                                                                                                                                                                                                                                                                                                    |
|              | <p>Indicates mask number for cut layer for multiple mask layer usage. Applies to lower left cut of the via, other cuts are rotated from the reference cut in the lower left corner. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored},cutMask,,</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p> |
| cut_rects    |                                                                                                                                                                                                                                                                                                                                                                    |
|              | <p>List of rectangles on cut layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master).</p> <p><b>Type:</b> <a href="#">rect*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                   |
| location     |                                                                                                                                                                                                                                                                                                                                                                    |
|              | <p>Via location</p> <p><b>Type:</b> <a href="#">point</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                         |
| obj_type     |                                                                                                                                                                                                                                                                                                                                                                    |



|           |                                                                                                                                                                                                                                                                                 |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">shape_via</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                       |
| top_mask  |                                                                                                                                                                                                                                                                                 |
|           | <p>Indicates mask number for top layer for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p>                                   |
| top_rects |                                                                                                                                                                                                                                                                                 |
|           | <p>List of rectangles (typically only one) on top routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master)</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| via_def   |                                                                                                                                                                                                                                                                                 |
|           | <p>The via master of this via</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">via_def</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                             |

## site

### Parent Objects

[base\\_cell](#), [design](#), [tech](#), [row](#), [power\\_domain](#), [root](#)

### Definition

LEF SITE

| Attribute | Description |
|-----------|-------------|
| base_name |             |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>Name of site (equivalent to LEF SITE NAME)</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                  |
| class             |                                                                                                                                                                                                                                                                                                                                                                                                |
|                   | <p>Site class (equivalent to LEF SITE CLASS)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> pad core</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                 |
| density_in_design |                                                                                                                                                                                                                                                                                                                                                                                                |
|                   | <p>Returns the density of the specified tech site in the design. User should query density info through get_db current_design .core_site.density_in_design . If user queries this attribute through get_db sites .density_in_design when no design is loaded, the result would be meaningless.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| name              |                                                                                                                                                                                                                                                                                                                                                                                                |
|                   | <p>Name of site (equivalent to LEF SITE NAME)</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                  |
| obj_type          |                                                                                                                                                                                                                                                                                                                                                                                                |
|                   | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">site</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                           |
| row_pattern       |                                                                                                                                                                                                                                                                                                                                                                                                |
|                   | <p>The row pattern for the site. (equivalent to LEF SITE ROWPATTERN)</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                           |
| size              |                                                                                                                                                                                                                                                                                                                                                                                                |

|          |                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------|
|          | Size of the site (equivalent to LEF SITE SIZE)<br><b>Type:</b> <a href="#">point</a><br><b>Default:</b> ""<br><b>Edit:</b> No        |
| symmetry |                                                                                                                                      |
|          | site symmetry.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> none x y xy any<br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| tech     |                                                                                                                                      |
|          | The tech for this site.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">tech</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No        |

## skew\_group

### Parent Objects

[pin](#), [root](#), [port](#)

### Definition

skew group

| Attribute                 | Description |
|---------------------------|-------------|
| cts_skew_group_constrains |             |

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | <p>Specifies how this skew group constrains the balancing of sinks during CCOpt.<br/>         If set to "default", this skew group will constrain sinks during "clock_design" and during the initial global balancing step of a regular "ccopt_design" run.<br/>         If set to "all", this skew group will constrain both "clock_design" and the whole of "ccopt_design" - not just the initial solution.<br/>         If set to "none" this specifies that the skew group will only be used for reporting purposes.<br/>         Valid values: none default all<br/> <b>Type:</b> string<br/> <b>Default:</b> default<br/> <b>Edit:</b> Yes</p> |
| cts_skew_group_created_from_clock            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                              | <p>This contains the name of the SDC clock that this skew group has been created to represent the balancing constraints in CCOpt.<br/>         Valid values: string<br/> <b>Type:</b> string<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                       |
| cts_skew_group_created_from_constraint_modes |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                              | <p>This contains the names of the constraint mode that this skew_group has been created to represent the balancing constraints in CCOpt.<br/>         Valid values: string<br/> <b>Type:</b> string<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                |
| cts_skew_group_created_from_delay_corners    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                              | <p>This contains the delay corners associated with this skew_group that has been created to represent the balancing constraints in CTS.<br/>         Valid values: string<br/> <b>Type:</b> string<br/> <b>Default:</b> ""<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                 |
| cts_skew_group_include_source_latency        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <p>Specifies whether clock tree source latency should be included when timing the skew group.</p> <p>Valid values: true false</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| cts_skew_group_target_insertion_delay |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                       | <p>The target insertion delay used for clock tree synthesis. This may be set to the following values:</p> <p>auto - Allow the minimum clustered insertion delay to be pushed up a little (around 5%) to facilitate clock tree power reduction.</p> <p>A numeric value - Attempt to balance the clock tree to the specified insertion delay (specified in library units). CTS will attempt to have a longest clock path delay of no more than this value plus half of the skew target, and a shortest path delay of no less than this value minus half the skew target.</p> <p>The value must be positive and real valued.</p> <p>Valid values: auto   double</p> <p><b>Type:</b> string</p> <p><b>Default:</b> auto</p> <p><b>Edit:</b> Yes</p>                                      |
| cts_target_skew                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                       | <p>This specifies the target skew for clock tree balancing. This may be set to a numeric value, or one of 'auto', 'ignore' or 'default'.</p> <p>If set to 'auto' this indicates that an appropriate skew target should be computed.</p> <p>If set to 'ignore' this indicates that skew should not be balanced for this corner/path combination.</p> <p>If unspecified then the value of this attribute is 'default'.</p> <p>If the value of the attribute is 'default' the target skew for late delays in the primary delay corner is interpreted as 'auto' and as 'ignore' otherwise.</p> <p>Valid values: default   auto   ignore   double</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> delay_corner</p> <p><b>Default:</b> default</p> <p><b>Edit:</b> Yes</p> |
| ignore_pins                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|              |                                                                                                                                                                                                           |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | ignored pins within the skew group<br><b>Type:</b> <a href="#">obj(pin)*</a> <a href="#">obj(port)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                          |
| name         |                                                                                                                                                                                                           |
|              | name of skew_group<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                        |
| obj_type     |                                                                                                                                                                                                           |
|              | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">skew_group</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| sinks        |                                                                                                                                                                                                           |
|              | sinks (pins or ports) of the skew group<br><b>Type:</b> <a href="#">obj(pin)*</a> <a href="#">obj(port)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                     |
| sinks_active |                                                                                                                                                                                                           |
|              | active sinks (pins or ports) of the skew group<br><b>Type:</b> <a href="#">obj(pin)*</a> <a href="#">obj(port)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                              |
| sources      |                                                                                                                                                                                                           |
|              | source(s) (pins or ports) of the skew group<br><b>Type:</b> <a href="#">obj(pin)*</a> <a href="#">obj(port)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                 |

## special\_via

### Parent Objects

[net](#)

## Definition

A special\_via is normally used for power-vias, flip-chip routing, and sometimes pre-routed nets. special\_vias (along with special\_wires) are not changed during signal routing. They appear in the DEF SPECIALNETS section.

| Attribute         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bottom_mask       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                   | <p>Is the mask number for the lower, left shape on the bottom layer of the via. Normally there is only one shape on the bottom layer of a via, but if there are two or more bottom layer shapes, then the mask for the other shapes on the bottom layer are derived from the corresponding via_def mask values by "shifting" the via_def's mask values to match. See the DEF manual section on 'Multi-Mask Layers with Special Wiring' for figures and examples. A value of 0 indicates the bottom layer is uncolored, or the layer is not a multi-mask layer.</p> <p><b>Type:</b> int<br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p> |
| bottom_rects      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                   | <p>List of rectangles (typically only one) on bottom routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master)</p> <p><b>Type:</b> rect*<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                         |
| bottom_rects_mask |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                   | <p>List of mask values for each rect in bottom_rects in the same order as bottom_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer.</p> <p><b>Type:</b> int*<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                          |
| cut_mask          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>Is the mask number for the lower, left cut of the via. The mask for the other cuts of the special_via are derived from the via_def by "shifting" the via_def's cut masks to match. So, if the via_def lower, left cut is mask 1, and the special_via cut_mask is set to 3, then all the via_def cuts on mask 1 become mask 3 for this special_via, and similarly cuts on 2 shift to 1, and cuts on 3 shift to 2. See the layer .num_masks attribute for the max mask value allowed. See the DEF manual section on 'Multi-Mask Layers with Special Wiring' for figures and examples. A value of 0 indicates the cut layer is uncolored, or the layer is not a multi-mask layer.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p> |
| cut_rects      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                | <p>List of rectangles on cut layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master)</p> <p><b>Type:</b> <a href="#">rect*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| cut_rects_mask |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                | <p>List of mask values for each rect in cut_rects in the same order as cut_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer.</p> <p><b>Type:</b> <a href="#">int*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| net            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                | <p>The net that the special_via belongs to</p> <p><b>Type:</b> <a href="#">obj(net)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| obj_type       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum (special_via)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| point          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                | <p>Location of Via</p> <p><b>Type:</b> <a href="#">point</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |



|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| shape          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                | <p>DEF SPECIALNETS + SHAPE equivalents (ring, stripe, etc.)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> notype ring stripe followpin iowire corewire blockwire fillwire blockagewire padring blocking drcfill fillwireopc</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                      |
| shield_net     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                | <p>The net that is shielded if status is shield</p> <p><b>Type:</b> <a href="#">obj(net)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| status         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                | <p>Wiring status (equivalent to DEF SPECIALNETS special wiring status)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> routed fixed cover shield unknown</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                           |
| top_mask       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                | <p>Is the mask number for the lower, left shape on the top layer of the via. Normally there is only one shape on the top layer of a via, but if there are two or more top layer shapes, then the mask for the other shapes on the top layer are derived from the corresponding via_def mask values by "shifting" the via_def's mask values to match. See the DEF manual section on 'Multi-Mask Layers with Special Wiring' for figures and examples. A value of 0 indicates the top layer is uncolored, or the layer is not a multi-mask layer.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> |
| top_rects      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                | <p>List of rectangles (typically only one) on top routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master)</p> <p><b>Type:</b> <a href="#">rect*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                             |
| top_rects_mask |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|            |                                                                                                                                                                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | List of mask values for each rect in top_rects in the same order as top_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer.<br><b>Type:</b> <code>int*</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No |
| user_class |                                                                                                                                                                                                                                                     |
|            | User_class value (empty string is returned if no user_class is specified)<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> Yes                                                                                |
| via_def    |                                                                                                                                                                                                                                                     |
|            | The via master<br><b>Type:</b> <code>obj(via_def)</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                                                                                                                                      |

## special\_wire

### Parent Objects

`net`

### Definition

Special wire (equivalent to DEF SPECIALNETS wiring)

| Attribute       | Description                                                                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| area            |                                                                                                                                                                          |
|                 | Area of the special wire as defined by the LEF MACRO SIZE or OVERLAP information<br><b>Type:</b> <code>area</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No |
| begin_extension |                                                                                                                                                                          |

|                |                                                                                                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>Extension of path at the first point (only on path type)</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                       |
| end_extension  |                                                                                                                                                                                                                                  |
|                | <p>Extension of path at the last point (only on path type)</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                        |
| has_trim_metal |                                                                                                                                                                                                                                  |
|                | <p>Indicate the special wire has trimm etal or not</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                 |
| is_patch       |                                                                                                                                                                                                                                  |
|                | <p>Indicate this secial wire is a patch wire</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                       |
| layer          |                                                                                                                                                                                                                                  |
|                | <p>The layer of special_wire</p> <p><b>Type:</b> <a href="#">obj(layer)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                |
| mask           |                                                                                                                                                                                                                                  |
|                | <p>Indicates mask number for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> |
| net            |                                                                                                                                                                                                                                  |
|                | <p>The net that the special_wire belongs to</p> <p><b>Type:</b> <a href="#">obj(net)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                    |

|            |                                                                                                                                                                                                                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| obj_type   |                                                                                                                                                                                                                                                                                                         |
|            | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">special_wire</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                            |
| path       |                                                                                                                                                                                                                                                                                                         |
|            | <p>2 points for pathSeg center-line, n points for path center-line, n points for polygon, null for rect</p> <p><b>Type:</b> <a href="#">point*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                 |
| polygon    |                                                                                                                                                                                                                                                                                                         |
|            | <p>Polygon boundary for the object, the first point is not repeated as the last point in the list</p> <p><b>Type:</b> <a href="#">polygon</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                      |
| rect       |                                                                                                                                                                                                                                                                                                         |
|            | <p>Rectangle that defines the special_wire shape if type = rect, or if type = path_seg and it is orthogonal. If type = polygon, or 45-degree path_seg, this is the bounding box of the shape</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>             |
| shape      |                                                                                                                                                                                                                                                                                                         |
|            | <p>DEF SPECIALNETS + SHAPE equivalents (ring, stripe, etc.)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> notype ring stripe followpin iowire corewire blockwire fillwire blockagewire padding blocking drofill fillwireopc</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| shield_net |                                                                                                                                                                                                                                                                                                         |
|            | <p>The net that is shielded if status is shieldNet</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">net</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                    |
| status     |                                                                                                                                                                                                                                                                                                         |

|                  |                                                                                                                                                                                                                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <p>Wiring status (equivalent to DEF SPECIALNETS special wiring status)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> routed fixed cover shield unknown</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                   |
| trim_metal_color |                                                                                                                                                                                                                                                                                                                      |
|                  | <p>Color of this special wire trim metal</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                |
| trim_metal_rect  |                                                                                                                                                                                                                                                                                                                      |
|                  | <p>The trim_metal rect if the special_wire has a trim_metal shape attached. This only occurs for some advanced node layers that use self-aligned patterning. {0 0 0 0} is returned if there is no trim_metal attached.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| type             |                                                                                                                                                                                                                                                                                                                      |
|                  | <p>Type of shape (rect, polygon, etc.)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> rect polygon path_seg path</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                           |
| user_class       |                                                                                                                                                                                                                                                                                                                      |
|                  | <p>User_class value (empty string is returned if no user_class is specified)</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                        |
| width            |                                                                                                                                                                                                                                                                                                                      |
|                  | <p>Width of path/pathSeg type</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                  |

# stack\_via\_rule

## Parent Objects

[pin](#), [pg\\_base\\_pin](#), [tech](#), [root](#), [pg\\_pin](#), [base\\_pin](#)

## Definition

Stack via rule information

| Attribute | Description                                                                                                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name      |                                                                                                                                                                                                               |
|           | Stack Via Rule name.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                          |
| obj_type  |                                                                                                                                                                                                               |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">stack_via_rule</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

# tech

## Parent Objects

[via\\_def](#), [route\\_rule](#), [base\\_cell](#), [design](#), [pg\\_base\\_pin](#), [root](#), [site](#), [via\\_def\\_rule](#), [layer](#), [base\\_pin](#)

## Definition

The LEF or OA technology and library data used by a given design. It has all the layers, base\_cells, route\_rules, via\_defs, etc. for a given process technology node. It may be used by one or more design objects.

| Attribute | Description |
|-----------|-------------|
|-----------|-------------|

|              |                                                                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_cells   |                                                                                                                                                                                   |
|              | All of the base_cells for this tech.<br><b>Type:</b> <a href="#">obj(base_cell)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                     |
| base_pins    |                                                                                                                                                                                   |
|              | All of the base_pins for this tech.<br><b>Type:</b> <a href="#">obj(base_pin)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                       |
| layers       |                                                                                                                                                                                   |
|              | All of the layers for this tech.<br><b>Type:</b> <a href="#">obj(layer)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                             |
| lef_files    |                                                                                                                                                                                   |
|              | All of the LEF files loaded to create this tech object.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                           |
| name         |                                                                                                                                                                                   |
|              | The name of the tech.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                             |
| obj_type     |                                                                                                                                                                                   |
|              | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (tech)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| pg_base_pins |                                                                                                                                                                                   |
|              | All of the pg_base_pins for this tech.<br><b>Type:</b> <a href="#">obj(pg_base_pin)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                 |

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_rule         |                                                                                                                                                            |
|                    | <p>All of the route_rules for this tech.</p> <p><b>Type:</b> <a href="#">obj(route_rule)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>          |
| sites              |                                                                                                                                                            |
|                    | <p>All of the sites for this tech.</p> <p><b>Type:</b> <a href="#">obj(site)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                     |
| stack_via_rules    |                                                                                                                                                            |
|                    | <p>All of the stack_via_rules for this tech.</p> <p><b>Type:</b> <a href="#">obj(stack_via_rule)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| tap_distance_rules |                                                                                                                                                            |
|                    | <p>Returns the LEF TAPDISTANCERULE info.</p> <p><b>Type:</b> <a href="#">obj(tap_distance_rule)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| via_def_rules      |                                                                                                                                                            |
|                    | <p>All of the via_def_rules for this tech.</p> <p><b>Type:</b> <a href="#">obj(via_def)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>          |
| via_defs           |                                                                                                                                                            |
|                    | <p>All of the via_defs for this tech.</p> <p><b>Type:</b> <a href="#">obj(via_def)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>               |



# text

## Parent Objects

design, root

## Definition

Interoperable text labels

| Attribute   | Description                                                                                                                                                                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alignment   |                                                                                                                                                                                                                                                                          |
|             | Horizontal and vertical alignment of the text string<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> upper_left center_left lower_left upper_center center_center lower_center upper_right center_right lower_right<br><b>Default:</b> ""<br><b>Edit:</b> No |
| drafting    |                                                                                                                                                                                                                                                                          |
|             | Indicates if the text is always displayed left-to-right or top-to-bottom. Text will remain readable even if rotated and mirrored if this value is true.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                    |
| font_name   |                                                                                                                                                                                                                                                                          |
|             | Font name<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> euro_style gothic math roman script stick fixed swedish mil_spec<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                          |
| font_number |                                                                                                                                                                                                                                                                          |
|             | Font Number<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> 0 1 2 3 4 5 6 7 8<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                       |

|            |                                                                                                                                                                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| height     |                                                                                                                                                                                                                                                                |
|            | Text height<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> no_value<br><b>Edit:</b> Yes                                                                                                                                                              |
| label      |                                                                                                                                                                                                                                                                |
|            | Text string value<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                                             |
| layer      |                                                                                                                                                                                                                                                                |
|            | The layer of text<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                                                                                                         |
| oa_purpose |                                                                                                                                                                                                                                                                |
|            | User specified purpose name for OA text layer purpose pair support. Only values available that exist in the library's tech graph are allowed. The default value is 'drawing'.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| obj_type   |                                                                                                                                                                                                                                                                |
|            | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">text</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                            |
| orient     |                                                                                                                                                                                                                                                                |
|            | Text Orientation<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> r0 r90 r180 r270 mx mx90 my my90<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                         |
| point      |                                                                                                                                                                                                                                                                |

|  |                                                                                              |
|--|----------------------------------------------------------------------------------------------|
|  | Text location<br><b>Type:</b> <a href="#">point</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
|--|----------------------------------------------------------------------------------------------|

## timing\_condition

### Parent Objects

[delay\\_corner](#), [root](#)

### Definition

A timing\_condition represents a set of libraries at a specific operating condition - effectively defining a device corner. When assigned to a power domain - the domain uses the library and PVT bindings defined by the timing\_condition. Timing conditions may be assigned to one or several power domains in the design via the delay\_corner object's timing condition attributes. Use the create\_timing\_condition and update\_timing\_condition commands to create and modify timing\_conditions.

| Attribute    | Description                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| library_sets |                                                                                                                                                                                                                     |
|              | Specifies the associated library_set object(s). A list of library_sets (a bundle) can be used for voltage interpolation.<br><b>Type:</b> <a href="#">obj(library_set)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| name         |                                                                                                                                                                                                                     |
|              | Provides the name of this timing_condition object as specified by create_timing_condition.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                          |
| obj_type     |                                                                                                                                                                                                                     |
|              | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (timing_condition)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                       |

|                |                                                                                                                                                                                                                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| opcond         |                                                                                                                                                                                                                                                                                                                              |
|                | <p>Specifies the associated opcond object. If this is not set, the opcond is derived from timing library itself. The root attribute timing_default_opcond_per_lib can also affect which opcond is used if it is not set here.</p> <p><b>Type:</b> <a href="#">obj(opcond)</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| opcond_library |                                                                                                                                                                                                                                                                                                                              |
|                | <p>The library of the opcond for this timing condition. Can be empty if this is a virtual opcond.</p> <p><b>Type:</b> <a href="#">obj(library)</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>                                                                                                                            |

## timing\_path

### Parent Objects

### Definition

cte timing path

| Attribute           | Description                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| actual_clock_period |                                                                                                                                                         |
|                     | <p>Returns the actual_clock_period property for a timing_path</p> <p><b>Type:</b> <a href="#">double</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| actual_pulse_width  |                                                                                                                                                         |
|                     | <p>Returns the actual_pulse_width property for a timing_path</p> <p><b>Type:</b> <a href="#">double</a><br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>  |
| applied_exception   |                                                                                                                                                         |

|                                     |                                                                                                                                                                                                               |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Returns the applied exception on timing path.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                     |
| arrival                             |                                                                                                                                                                                                               |
|                                     | <p>arrival</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                    |
| arrival_mean                        |                                                                                                                                                                                                               |
|                                     | <p>Returns the arrival_mean property for a timing_path</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                        |
| arrival_sigma                       |                                                                                                                                                                                                               |
|                                     | <p>Returns the arrival_sigma property for a timing_path</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                       |
| asserted_network_insertion_delay    |                                                                                                                                                                                                               |
|                                     | <p>Returns the network insertion delay asserted on this timing path.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| asserted_source_insertion_delay     |                                                                                                                                                                                                               |
|                                     | <p>Returns the source insertion delay asserted on this timing path.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>  |
| borrowing_path_q_pin_and_transition |                                                                                                                                                                                                               |

|                                 |                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | Returns the borrowing_path_q_pin_and_transition property for a timing_path<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                        |
| borrowing_path_q_pin_transition |                                                                                                                                                                                                   |
|                                 | Returns the borrowing_path_q_pin_transition property for a timing_path<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                            |
| capture_clock_path              |                                                                                                                                                                                                   |
|                                 | Returns the capture_clock_path property for a timing_path<br><b>Type:</b> obj(timing_path)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                                              |
| capture_parasitics_capacitance  |                                                                                                                                                                                                   |
|                                 | Returns the parasitics info of launch and data nets on timing path post cpr.<br><b>Type:</b> string<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| capture_parasitics_resistance   |                                                                                                                                                                                                   |
|                                 | Returns the parasitics info of launch and data nets on timing path post cpr.<br><b>Type:</b> string<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| capturing_clock                 |                                                                                                                                                                                                   |
|                                 | Returns the capturing_clock property for a timing_path<br><b>Type:</b> obj(clock)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                       |
| capturing_clock_close_edge_time |                                                                                                                                                                                                   |
|                                 | Returns the capturing_clock_close_edge_time property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                            |

|                                  |                                                                                                                                         |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| capturing_clock_close_edge_type  |                                                                                                                                         |
|                                  | Returns the capturing_clock_close_edge_type property for a timing_path<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| capturing_clock_is_inverted      |                                                                                                                                         |
|                                  | Returns the capturing_clock_is_inverted property for a timing_path<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No        |
| capturing_clock_is_propagated    |                                                                                                                                         |
|                                  | Returns the capturing_clock_is_propagated property for a timing_path<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No      |
| capturing_clock_latency          |                                                                                                                                         |
|                                  | Returns the capturing_clock_latency property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| capturing_clock_latency_mean     |                                                                                                                                         |
|                                  | Returns the capturing_clock_latency_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No     |
| capturing_clock_latency_sigma    |                                                                                                                                         |
|                                  | Returns the capturing_clock_latency_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No    |
| capturing_clock_latency_skewness |                                                                                                                                         |
|                                  | Returns the capturing_clock_latency_skewness property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |

|                                         |                                                                                                                                                                                                                                  |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| capturing_clock_latency_stddev          |                                                                                                                                                                                                                                  |
|                                         | <p>Returns the capturing_clock_latency_stddev property for a timing_path</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                         |
| capturing_clock_network_insertion_delay |                                                                                                                                                                                                                                  |
|                                         | <p>Returns the network insertion delay of the capturing clock path of this timing path.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| capturing_clock_open_edge_time          |                                                                                                                                                                                                                                  |
|                                         | <p>capturing_clock_open_edge_time</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                |
| capturing_clock_open_edge_type          |                                                                                                                                                                                                                                  |
|                                         | <p>Returns the capturing_clock_open_edge_type property for a timing_path</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                         |
| capturing_clock_pin                     |                                                                                                                                                                                                                                  |
|                                         | <p>Returns the capturing_clock_pin property for a timing_path</p> <p><b>Type:</b> obj(pin)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                 |
| capturing_clock_reference_edge_type     |                                                                                                                                                                                                                                  |
|                                         | <p>capturing_clock_reference_edge_type</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                           |
| capturing_clock_root_open_edge_time     |                                                                                                                                                                                                                                  |



|                                           |                                                                                                                                                                                                          |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | capturing_clock_root_open_edge_time<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                      |
| capturing_clock_root_open_edge_type       |                                                                                                                                                                                                          |
|                                           | capturing_clock_root_open_edge_type<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                      |
| capturing_clock_source_arrival_time       |                                                                                                                                                                                                          |
|                                           | Returns the capturing_clock_source_arrival_time property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                               |
| capturing_clock_source_arrival_time_mean  |                                                                                                                                                                                                          |
|                                           | Returns the capturing_clock_source_arrival_time_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                          |
| capturing_clock_source_arrival_time_sigma |                                                                                                                                                                                                          |
|                                           | Returns the capturing_clock_source_arrival_time_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                         |
| capturing_clock_source_insertion_delay    |                                                                                                                                                                                                          |
|                                           | Returns the source insertion delay of the capturing clock path of this timing path.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| capturing_point                           |                                                                                                                                                                                                          |
|                                           | Returns a pointer to the end point of the timing path.<br><b>Type:</b> obj(port)* obj(pin)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                     |

|                                    |                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| capturing_point_is_level_sensitive |                                                                                                                                                                                |
|                                    | <p>Returns a value of true if the end point of the data path is level sensitive.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| check_delay                        |                                                                                                                                                                                |
|                                    | <p>Returns the check_delay property for a timing_path</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                          |
| check_delay_mean                   |                                                                                                                                                                                |
|                                    | <p>Returns the check_delay_mean property for a timing_path</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                     |
| check_delay_sigma                  |                                                                                                                                                                                |
|                                    | <p>Returns the check_delay_sigma property for a timing_path</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                    |
| check_type                         |                                                                                                                                                                                |
|                                    | <p>Returns the check_type property for a timing_path</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                           |
| clock_path_end_point               |                                                                                                                                                                                |
|                                    | <p>clock_path_end_point</p> <p><b>Type:</b> <a href="#">obj(timing_point)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                            |
| clock_source_jitter                |                                                                                                                                                                                |
|                                    | <p>clock_source_jitter</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                         |

|                              |                                                                                                                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| clock_uncertainty            |                                                                                                                                                                                                                      |
|                              | <p>clock_uncertainty</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                 |
| clock_uncertainty_offset     |                                                                                                                                                                                                                      |
|                              | <p>Returns the clock uncertainty offset of timing path.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                     |
| clock_uncertainty_offset_pin |                                                                                                                                                                                                                      |
|                              | <p>Returns the pin name of the clock uncertainty offset of the timing path.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| cpr_adjustment               |                                                                                                                                                                                                                      |
|                              | <p>Returns the CPR adjustment of the timing path.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                    |
| cpr_adjustment_mean          |                                                                                                                                                                                                                      |
|                              | <p>Returns the cpr_adjustment_mean property for a timing_path</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                        |
| cpr_adjustment_sigma         |                                                                                                                                                                                                                      |
|                              | <p>Returns the cpr_adjustment_sigma property for a timing_path</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                       |
| cpr_branch_point             |                                                                                                                                                                                                                      |

|                               |                                                                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | Returns the CPPR branch point of the given timing path.<br><b>Type:</b> obj(port)* obj(pin)*<br><b>Default:</b> ""<br><b>Edit:</b> No                    |
| cumulative_manhattan_length   |                                                                                                                                                          |
|                               | Returns the cumulative_manhattan_length property for a timing_path.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                      |
| cumulative_manhattan_length_x |                                                                                                                                                          |
|                               | Returns the cumulative_manhattan_length_x property for a timing_path.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                    |
| cumulative_manhattan_length_y |                                                                                                                                                          |
|                               | Returns the cumulative_manhattan_length_y property for a timing_path.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                    |
| cycle_adjustment              |                                                                                                                                                          |
|                               | Returns the cycle_adjustment property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                  |
| data_path                     |                                                                                                                                                          |
|                               | Returns the data path arrival.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No      |
| data_path_mean                |                                                                                                                                                          |
|                               | Returns the data path arrival mean.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |

|                        |                                                                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data_path_sigma        |                                                                                                                                                                                                           |
|                        | <p>Returns the data path arrival sigma.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                          |
| dominant_vt_group      |                                                                                                                                                                                                           |
|                        | <p>Returns the name of the dominant vt group of the timing path.</p> <p><b>Type:</b> string</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| drive_adjustment       |                                                                                                                                                                                                           |
|                        | <p>Returns the drive_adjustment property for a timing_path</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                |
| drive_adjustment_mean  |                                                                                                                                                                                                           |
|                        | <p>Returns the drive_adjustment_mean property for a timing_path</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                           |
| drive_adjustment_sigma |                                                                                                                                                                                                           |
|                        | <p>Returns the drive_adjustment_sigma property for a timing_path</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                          |
| external_delay         |                                                                                                                                                                                                           |
|                        | <p>external_delay</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                         |
| gba_slack              |                                                                                                                                                                                                           |

|                       |                                                                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Returns the gba_slack property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                     |
| gba_slack_mean        |                                                                                                                                                                                      |
|                       | Returns the gba_slack_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                |
| gba_slack_sigma       |                                                                                                                                                                                      |
|                       | Returns the gba_slack_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                               |
| hold                  |                                                                                                                                                                                      |
|                       | hold<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                 |
| inter_die_skew_adjust |                                                                                                                                                                                      |
|                       | Returns the inter die skew adjustment value of the timing path.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| ipba_slack_adj        |                                                                                                                                                                                      |
|                       | Returns the IPBA slack adjustment value.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No                        |
| is_borrow_edge_adjust |                                                                                                                                                                                      |

|                         |                                                                                                                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Returns true if time borrow adjustment is done for the timing path.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| is_clock_gating_hold    |                                                                                                                                                                                                                               |
|                         | <p>Returns a value of true if the path is a clock gating hold check.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                            |
| is_clock_gating_setup   |                                                                                                                                                                                                                               |
|                         | <p>Returns a value of true if the path is a clock gating setup check.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                           |
| is_epba_depth_exhausted |                                                                                                                                                                                                                               |
|                         | <p>Returns true if epba depth is exhausted for the timing path.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>        |
| is_invalid              |                                                                                                                                                                                                                               |
|                         | <p>Returns the is_invalid property for a timing_path</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                            |
| is_ipba_bounded_path    |                                                                                                                                                                                                                               |
|                         | <p>Returns true if the path is IPBA bounded path</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                       |
| is_path_borrowing       |                                                                                                                                                                                                                               |

|                               |                                                                                                                                                                                                    |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | Returns the is_path_borrowing property for a timing_path<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                             |
| is_retimed                    |                                                                                                                                                                                                    |
|                               | Returns the is_retimed property for a timing_path<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                    |
| is_time_given                 |                                                                                                                                                                                                    |
|                               | Returns the is_time_given property for a timing_path<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                 |
| is_transparent_latch          |                                                                                                                                                                                                    |
|                               | Returns a value of true if the path is a transparent latch path.<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                     |
| launch_clock_path             |                                                                                                                                                                                                    |
|                               | launch_clock_path<br><b>Type:</b> obj(timing_path)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                       |
| launch_parasitics_capacitance |                                                                                                                                                                                                    |
|                               | Returns the parasitics info of launch and data nets on timing path post cppr.<br><b>Type:</b> string<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| launch_parasitics_resistance  |                                                                                                                                                                                                    |
|                               | Returns the parasitics info of launch and data nets on timing path post cppr.<br><b>Type:</b> string<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |



|                                         |                                                                                                                                                                                                                                  |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| launching_clock                         |                                                                                                                                                                                                                                  |
|                                         | <p>launching_clock</p> <p><b>Type:</b> obj(clock)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                          |
| launching_clock_is_inverted             |                                                                                                                                                                                                                                  |
|                                         | <p>launching_clock_is_inverted</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                     |
| launching_clock_is_propagated           |                                                                                                                                                                                                                                  |
|                                         | <p>launching_clock_is_propagated</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                   |
| launching_clock_latency                 |                                                                                                                                                                                                                                  |
|                                         | <p>launching_clock_latency</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                       |
| launching_clock_network_insertion_delay |                                                                                                                                                                                                                                  |
|                                         | <p>Returns the network insertion delay of the launching clock path of this timing path.</p> <p><b>Type:</b> double</p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| launching_clock_open_edge_time          |                                                                                                                                                                                                                                  |
|                                         | <p>launching_clock_open_edge_time</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                |
| launching_clock_open_edge_type          |                                                                                                                                                                                                                                  |

|                                           |                                                                                                                                                                                                          |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | launching_clock_open_edge_type<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                           |
| launching_clock_source_arrival_time       |                                                                                                                                                                                                          |
|                                           | launching_clock_source_arrival_time<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                      |
| launching_clock_source_arrival_time_mean  |                                                                                                                                                                                                          |
|                                           | Returns the launching_clock_source_arrival_time_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                          |
| launching_clock_source_arrival_time_sigma |                                                                                                                                                                                                          |
|                                           | Returns the launching_clock_source_arrival_time_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                         |
| launching_clock_source_insertion_delay    |                                                                                                                                                                                                          |
|                                           | Returns the source insertion delay of the capturing clock path of this timing path.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| launching_input_delay                     |                                                                                                                                                                                                          |
|                                           | launching_input_delay<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                    |
| launching_point                           |                                                                                                                                                                                                          |
|                                           | Returns a pointer to the start point of the timing path.<br><b>Type:</b> obj(port)* obj(pin)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                   |

|                                    |                                                                                                                                                                                                                             |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| launching_point_is_level_sensitive |                                                                                                                                                                                                                             |
|                                    | <p>Returns a value of true if the begin point of the data path is level sensitive.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                            |
| lending_path_d_pin                 |                                                                                                                                                                                                                             |
|                                    | <p>Returns the lending_path_d_pin property for a timing_path</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                |
| lending_path_d_pin_transition      |                                                                                                                                                                                                                             |
|                                    | <p>Returns the lending_path_d_pin_transition property for a timing_path</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                     |
| nets                               |                                                                                                                                                                                                                             |
|                                    | <p>Returns a collection of nets of the timing path.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">net</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |
| num_cell_arcs                      |                                                                                                                                                                                                                             |
|                                    | <p>Returns the num_cell_arcs property for a timing_path</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                     |
| num_net_arcs                       |                                                                                                                                                                                                                             |
|                                    | <p>Returns the num_net_arcs property for a timing_path</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                      |
| obj_type                           |                                                                                                                                                                                                                             |
|                                    | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">timing_path</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

|                            |                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| other_end_arrival          |                                                                                                                                   |
|                            | Returns the other_end_arrival property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| other_end_arrival_mean     |                                                                                                                                   |
|                            | Returns the other_end_arrival_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No     |
| other_end_arrival_sigma    |                                                                                                                                   |
|                            | Returns the other_end_arrival_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No    |
| other_end_arrival_skewness |                                                                                                                                   |
|                            | Returns the other_end_arrival_skewness property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |
| other_end_arrival_stddev   |                                                                                                                                   |
|                            | Returns the other_end_arrival_stddev property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No   |
| path_adjust_value          |                                                                                                                                   |
|                            | Returns the path_adjust_value property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| path_cell_delay            |                                                                                                                                   |
|                            | Returns the path_cell_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No            |

|                       |                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------|
| path_cell_delay_mean  |                                                                                                                              |
|                       | Returns the path_cell_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| path_cell_delay_sigma |                                                                                                                              |
|                       | Returns the path_cell_delay_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |
| path_delay            |                                                                                                                              |
|                       | Returns the path_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No            |
| path_delay_adjustment |                                                                                                                              |
|                       | Returns the path_delay_adjustment property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |
| path_delay_mean       |                                                                                                                              |
|                       | Returns the path_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No       |
| path_delay_sigma      |                                                                                                                              |
|                       | Returns the path_delay_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No      |
| path_group            |                                                                                                                              |
|                       | path_group<br><b>Type:</b> obj(path_group)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                         |

|                      |                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| path_group_name      |                                                                                                                             |
|                      | Returns the name of the path group for the timing path.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No     |
| path_net_delay       |                                                                                                                             |
|                      | Returns the path_net_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No       |
| path_net_delay_mean  |                                                                                                                             |
|                      | Returns the path_net_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| path_net_delay_sigma |                                                                                                                             |
|                      | Returns the path_net_delay_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |
| path_ptr_loss        |                                                                                                                             |
|                      | Returns the path_ptr_loss property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No        |
| path_type            |                                                                                                                             |
|                      | Reports max type for late path and min type for early path<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| period               |                                                                                                                             |
|                      | Returns the period property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No               |

|                       |                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------|
| phase_shift           |                                                                                                                              |
|                       | Returns the phase_shift property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No           |
| ptr_mean              |                                                                                                                              |
|                       | Returns the ptr_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No              |
| ptr_sigma             |                                                                                                                              |
|                       | Returns the ptr_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No             |
| ptr_skewness          |                                                                                                                              |
|                       | Returns the ptr_skewness property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| ptr_slack             |                                                                                                                              |
|                       | Returns the PTR Slack Time property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No        |
| real_gba_arrival      |                                                                                                                              |
|                       | Returns the real_gba_arrival property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No      |
| real_gba_arrival_mean |                                                                                                                              |
|                       | Returns the real_gba_arrival_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |

|                           |                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| real_gba_arrival_sigma    |                                                                                                                                  |
|                           | Returns the real_gba_arrival_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No    |
| real_gba_arrival_skewness |                                                                                                                                  |
|                           | Returns the real_gba_arrival_skewness property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |
| real_gba_arrival_stddev   |                                                                                                                                  |
|                           | Returns the real_gba_arrival_stddev property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No   |
| real_gba_slack            |                                                                                                                                  |
|                           | Returns the real_gba_slack property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No            |
| real_gba_slack_mean       |                                                                                                                                  |
|                           | Returns the real_gba_slack_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No       |
| real_gba_slack_sigma      |                                                                                                                                  |
|                           | Returns the real_gba_slack_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No      |
| real_gba_slack_skewness   |                                                                                                                                  |
|                           | Returns the real_gba_slack_skewness property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No   |



|                       |                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| real_gba_slack_stddev |                                                                                                                                                 |
|                       | <p>Returns the real_gba_slack_stddev property for a timing_path</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| recovery              |                                                                                                                                                 |
|                       | <p>Returns the recovery time at the end point of the path.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| removal               |                                                                                                                                                 |
|                       | <p>removal</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |
| required_time         |                                                                                                                                                 |
|                       | <p>Returns the required time for the timing path.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>               |
| required_time_mean    |                                                                                                                                                 |
|                       | <p>Returns the required_time_mean property for a timing_path</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>    |
| required_time_sigma   |                                                                                                                                                 |
|                       | <p>Returns the required_time_sigma property for a timing_path</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>   |
| setup                 |                                                                                                                                                 |
|                       | <p>The setup time at the end point of the path.</p> <p><b>Type:</b> double</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                 |

|                  |                                                                                                                                                        |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| skew             |                                                                                                                                                        |
|                  | Returns the skew property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                            |
| slack            |                                                                                                                                                        |
|                  | Returns the slack of the timing path.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                  |
| slack_adjusted   |                                                                                                                                                        |
|                  | Returns the adjusted slack value.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| slack_mean       |                                                                                                                                                        |
|                  | Returns the slack_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                      |
| slack_sigma      |                                                                                                                                                        |
|                  | Returns the slack_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                     |
| spatial_distance |                                                                                                                                                        |
|                  | Returns the spatial_distance property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                |
| time_borrowed    |                                                                                                                                                        |

|                              |                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|                              | Returns the amount of time borrowed from the timing end point.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No      |
| time_lent                    |                                                                                                                                     |
|                              | Returns the amount of time lent to the timing start point.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| timing_points                |                                                                                                                                     |
|                              | timing_points<br><b>Type:</b> obj(timing_point)*<br><b>Default:</b> ""<br><b>Edit:</b> No                                           |
| total_cell_delta_delay       |                                                                                                                                     |
|                              | Returns the total_cell_delta_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No       |
| total_cell_delta_delay_mean  |                                                                                                                                     |
|                              | Returns the total_cell_delta_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| total_cell_delta_delay_sigma |                                                                                                                                     |
|                              | Returns the total_cell_delta_delay_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |
| total_delta_delay            |                                                                                                                                     |
|                              | Returns the total_delta_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No            |
| total_delta_delay_mean       |                                                                                                                                     |

|                             |                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | Returns the total_delta_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                             |
| total_delta_delay_sigma     |                                                                                                                                                                                                                                           |
|                             | Returns the total_delta_delay_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                            |
| total_net_delta_delay       |                                                                                                                                                                                                                                           |
|                             | Returns the total_net_delta_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                              |
| total_net_delta_delay_mean  |                                                                                                                                                                                                                                           |
|                             | Returns the total_net_delta_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                         |
| total_net_delta_delay_sigma |                                                                                                                                                                                                                                           |
|                             | Returns the total_net_delta_delay_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                        |
| underated_slack             |                                                                                                                                                                                                                                           |
|                             | Returns the underated slack time for the timing path. The underated slack value is available with timing path object only if it is generated using the -derate parameter.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |
| view_name                   |                                                                                                                                                                                                                                           |
|                             | The name of the analysis view for the timing_path<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                         |

|                              |                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| worst_cell_delay             |                                                                                                                                     |
|                              | Returns the worst_cell_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No             |
| worst_cell_delay_mean        |                                                                                                                                     |
|                              | Returns the worst_cell_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No        |
| worst_cell_delay_sigma       |                                                                                                                                     |
|                              | Returns the worst_cell_delay_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No       |
| worst_cell_delta_delay       |                                                                                                                                     |
|                              | Returns the worst_cell_delta_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No       |
| worst_cell_delta_delay_mean  |                                                                                                                                     |
|                              | Returns the worst_cell_delta_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| worst_cell_delta_delay_sigma |                                                                                                                                     |
|                              | Returns the worst_cell_delta_delay_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |
| worst_delay                  |                                                                                                                                     |
|                              | Returns the worst_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                  |

|                                 |                                                                                                                                         |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| worst_delay_mean                |                                                                                                                                         |
|                                 | Returns the worst_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                 |
| worst_delay_sigma               |                                                                                                                                         |
|                                 | Returns the worst_delay_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                |
| worst_delta_delay               |                                                                                                                                         |
|                                 | Returns the worst_delta_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                |
| worst_delta_delay_mean          |                                                                                                                                         |
|                                 | Returns the worst_delta_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No           |
| worst_delta_delay_sigma         |                                                                                                                                         |
|                                 | Returns the worst_delta_delay_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| worst_manhattan_length          |                                                                                                                                         |
|                                 | Returns the worst_manhattan_length property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No           |
| worst_manhattan_length_net_name |                                                                                                                                         |
|                                 | Returns the worst_manhattan_length_net_name property for a timing_path.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No |

|                             |                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| worst_net_delay             |                                                                                                                                   |
|                             | Returns the worst_net_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No            |
| worst_net_delay_mean        |                                                                                                                                   |
|                             | Returns the worst_net_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No       |
| worst_net_delay_sigma       |                                                                                                                                   |
|                             | Returns the worst_net_delay_sigma property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No      |
| worst_net_delta_delay       |                                                                                                                                   |
|                             | Returns the worst_net_delta_delay property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No      |
| worst_net_delta_delay_mean  |                                                                                                                                   |
|                             | Returns the worst_net_delta_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |
| worst_net_delta_delay_sigma |                                                                                                                                   |
|                             | Returns the worst_net_delta_delay_mean property for a timing_path<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No |

# timing\_point

## Parent Objects

[timing\\_path](#)

## Definition

cte timing point

| Attribute               | Description                                                                                                                                                                           |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| arrival                 |                                                                                                                                                                                       |
|                         | Returns the arrival time at the timing point.<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                         |
| arrival_mean            |                                                                                                                                                                                       |
|                         | Returns the arrival_mean property for a timing_point<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                  |
| arrival_sigma           |                                                                                                                                                                                       |
|                         | Returns the arrival_sigma property for a timing_point<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                 |
| arrival_skewness        |                                                                                                                                                                                       |
|                         | Returns the skewness in arrival at timing point.<br><b>Type:</b> <a href="#">double</a><br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| clock_unroll_adjustment |                                                                                                                                                                                       |



|                 |                                                                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <p>Returns the clock unrolling adjustment at the timing point.<br/> <b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| delay           |                                                                                                                                                                                                 |
|                 | <p>Returns the delay property for a timing_point.<br/> <b>Type:</b> double<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                     |
| delay_effective |                                                                                                                                                                                                 |
|                 | <p>Returns the effective delay at the timing point.<br/> <b>Type:</b> double<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>            |
| delay_mean      |                                                                                                                                                                                                 |
|                 | <p>Returns the delay_mean property of a timing_point<br/> <b>Type:</b> double<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                  |
| delay_sigma     |                                                                                                                                                                                                 |
|                 | <p>Returns the delay_sigma property for a timing_point.<br/> <b>Type:</b> double<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                               |
| delay_skewness  |                                                                                                                                                                                                 |
|                 | <p>Returns the skewness in delay at the timing point.<br/> <b>Type:</b> string<br/> <b>Allowed -index values:</b> analysis_view clock<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>          |
| delta_delay     |                                                                                                                                                                                                 |

|                       |                                                                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Returns the delta_delay property for a timing_point<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                             |
| direction             |                                                                                                                                                                                 |
|                       | Returns the direction property for a timing_point<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                               |
| hierarchical_name     |                                                                                                                                                                                 |
|                       | Returns the hierarchical_name property for a timing_point<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                       |
| is_fanin_arc_cell_arc |                                                                                                                                                                                 |
|                       | Returns cell arc in the fanin of the timing point<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                 |
| is_hierarchical       |                                                                                                                                                                                 |
|                       | Returns the is_hierarchical property for a timing_point<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No                                                           |
| load                  |                                                                                                                                                                                 |
|                       | Returns the total load seen at the timing point<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No            |
| local_ref_clock       |                                                                                                                                                                                 |
|                       | Returns the local reference clock name of the timing point<br><b>Type:</b> string<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |

|                      |                                                                                                                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| local_slack          |                                                                                                                                                                                                                   |
|                      | <p>Returns the local slack at the timing point.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>          |
| local_slack_adjusted |                                                                                                                                                                                                                   |
|                      | <p>Returns the local adjusted slack at the timing point.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Allowed -index values:</b> analysis_view clock</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| net                  |                                                                                                                                                                                                                   |
|                      | <p>Reports the nets connected to the timing point</p> <p><b>Type:</b> <a href="#">obj(net)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                              |
| obj_type             |                                                                                                                                                                                                                   |
|                      | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum (timing_point)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>      |
| pin                  |                                                                                                                                                                                                                   |
|                      | <p>Returns the pin property for a timing_point</p> <p><b>Type:</b> <a href="#">obj(port)* obj(pin)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |
| si_object            |                                                                                                                                                                                                                   |
|                      | <p>Returns the si_object property for a timing_point</p> <p><b>Type:</b> <a href="#">obj(delay_noise_obj)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                |
| slack                |                                                                                                                                                                                                                   |

|                |                                                                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | Returns the slack property for a timing_point<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                           |
| slack_mean     |                                                                                                                                                                         |
|                | Returns the slack_mean property for a timing_point<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                      |
| slack_sigma    |                                                                                                                                                                         |
|                | Returns the slack_sigma property for a timing_point<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                     |
| slack_skewness |                                                                                                                                                                         |
|                | Returns the skewness in slack at the timing point.<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| slew           |                                                                                                                                                                         |
|                | Returns the path-specific slew of the timing point.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                     |
| slew_mean      |                                                                                                                                                                         |
|                | Returns the slew_mean property for a timing_point.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                      |
| slew_sigma     |                                                                                                                                                                         |
|                | Returns the slew_sigma property for a timing_point<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                      |

|                      |                                                                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| slew_skewness        |                                                                                                                                                                       |
|                      | Returns the skewness in slew at the timing point<br><b>Type:</b> double<br><b>Allowed -index values:</b> analysis_view clock<br><b>Default:</b> ""<br><b>Edit:</b> No |
| spatial_derate       |                                                                                                                                                                       |
|                      | Returns the spatial_derate property for a timing_point<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                |
| spatial_derate_sigma |                                                                                                                                                                       |
|                      | Returns the spatial_derate_sigma property for a timing_point<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                          |
| ssi_derate           |                                                                                                                                                                       |
|                      | Returns the path-specific ssi_derate of the timing point.<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                             |
| total_derate         |                                                                                                                                                                       |
|                      | Returns the total_derate property for a timing_point<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                                  |
| total_derate_sigma   |                                                                                                                                                                       |
|                      | Returns the total_derate_sigma property for a timing_point<br><b>Type:</b> double<br><b>Default:</b> ""<br><b>Edit:</b> No                                            |
| transition_type      |                                                                                                                                                                       |

|                   |                                                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | Returns the transition type of the timing point: rise or fall .<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| user_derate       |                                                                                                                                                 |
|                   | Returns the user_derate property for a timing_point<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> ""<br><b>Edit:</b> No             |
| user_derate_sigma |                                                                                                                                                 |
|                   | Returns the user_derate_sigma property for a timing_point<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> ""<br><b>Edit:</b> No       |
| voltage           |                                                                                                                                                 |
|                   | Returns the voltage property for a timing_point<br><b>Type:</b> <a href="#">double</a><br><b>Default:</b> ""<br><b>Edit:</b> No                 |

## track\_pattern

### Parent Objects

[design](#), [root](#)

### Definition

Floorplan track information (DEF TRACKS equivalent)

| Attribute | Description |
|-----------|-------------|
| direction |             |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <p>Specifies the location and direction of the first track defined. x indicates vertical lines; y indicates horizontal lines.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> y x</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                      |
| layers     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|            | <p>List of layers</p> <p><b>Type:</b> <a href="#">obj(layer)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                          |
| mask       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|            | <p>Specifies the mask number for the first track.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                  |
| num_tracks |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|            | <p>Specifies the number of tracks to create for the grid</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                          |
| obj_type   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|            | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum (track_pattern)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                   |
| route_rule |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|            | <p>Pointer to route_rule object (DEF NONDEFAULTRULE) to be used as a constraint for wiring that can be created on the track. An empty value indicates that there is no constraint on the rules of the wires on the track. This value can be set with add_tracks and is intended for advanced nodes that do not allow different route_rules on the same track for lower routing layers.</p> <p><b>Type:</b> <a href="#">obj(route_rule)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| start      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|       |                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | Specifies the coordinate of the first line<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> no_value<br><b>Edit:</b> Yes                                                                                                                                                                                                                                                            |
| step  |                                                                                                                                                                                                                                                                                                                                                                                             |
|       | Specifies the spacing between the tracks<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> no_value<br><b>Edit:</b> Yes                                                                                                                                                                                                                                                              |
| width |                                                                                                                                                                                                                                                                                                                                                                                             |
|       | Width constraint for wiring that can be created on the track. A value of 0 indicates that there is no constraint on the width of wires on the track. This value can be set with add_tracks and is intended for advanced nodes that do not allow different widths on the same track for lower routing layers.<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No |

## trim\_grid

### Parent Objects

[design](#), [root](#)

### Definition

Trim grid pattern. Equivalent to LEF TRIMMETALTRACK statement that defines the grids that trim shapes must align to. See the LEF manual for more descriptions, figures and examples of how these attributes are defined.

| Attribute | Description                                                                                                                                                         |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| group     |                                                                                                                                                                     |
|           | Specifies the group names of the trim_grid from the LEF TRIMMETALSTACK GROUP value.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> {}<br><b>Edit:</b> No |



|             |                                                                                                                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| layer       |                                                                                                                                                                                                                                       |
|             | <p>The trim layer of this trim_grid from the LEF TRIMMETALSTACK trimLayer value</p> <p><b>Type:</b> <a href="#">obj(layer)</a></p> <p><b>Default:</b> {}</p> <p><b>Edit:</b> No</p>                                                   |
| mask        |                                                                                                                                                                                                                                       |
|             | <p>Specifies the mask number for the trim_grid, from the LEF TRIMMETALSTACK MASK value.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> {}</p> <p><b>Edit:</b> No</p>                                                  |
| obj_type    |                                                                                                                                                                                                                                       |
|             | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum(trim_grid)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                              |
| offset      |                                                                                                                                                                                                                                       |
|             | <p>Specifies the offset from the lower left point of the core box to the first trim_grid from the LEF TRIMMETALSTACK COREOFFSET value.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> {}</p> <p><b>Edit:</b> No</p> |
| on_track    |                                                                                                                                                                                                                                       |
|             | <p>True if the LEF TRIMMETALSTACK ONTRACK keyword is given for this trim_grid.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                       |
| pitch       |                                                                                                                                                                                                                                       |
|             | <p>Specifies the spacing between each trim_grid from the LEF TRIMMETALSTACK PITCH value.</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> {}</p> <p><b>Edit:</b> No</p>                                               |
| track_count |                                                                                                                                                                                                                                       |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>The number of times to repeat the pitch for this track_index from the LEF TRIMMETALSTACK METALTRACKPITCH value. See the LEF manual for figures and examples of how this is used.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> No</p>                                                                                                                                                                             |
| track_index |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|             | <p>Specifies the index value of the trim_grid from the LEF TRIMMETALSTACK METALTRACKOFFSET value. The offset and pitch values apply to this track_index for track_count times. See the LEF manual for figures and examples of how this is used to overlay different track_index settings on one layer on top of each other to create non-uniform grid patterns.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> No</p> |

## via

### Parent Objects

[net](#)

### Definition

DEF NETS via instance

| Attribute   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bottom_mask |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|             | <p>Is the mask number for the lower, left shape on the bottom layer of the via. Normally there is only one shape on the bottom layer of a via, but if there are two or more bottom layer shapes, then the mask for the other shapes on the bottom layer are derived from the corresponding via_def mask values by "shifting" the via_def's mask values to match. See the DEF manual section on 'Multi-Mask Patterns for Routing Points' for figures and examples. A value of 0 indicates the bottom layer is uncolored, or the layer is not a multi-mask layer.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> Yes</p> |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bottom_rects      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                   | <p>List of rectangles (typically only one) on bottom routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master)</p> <p><b>Type:</b> <a href="#">rect*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| bottom_rects_mask |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                   | <p>List of mask values for each rect in bottom_rects in the same order as bottom_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer.</p> <p><b>Type:</b> <a href="#">int*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| cut_mask          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                   | <p>Is the mask number for the lower, left cut of the via. The mask for the other cuts of the via are derived from the via_def by "shifting" the via_def's cut masks to match. So, if the via_def lower, left cut is mask 1, and the via cut_mask is set to 3, then all the via_def cuts on mask 1 become mask 3 for this via, and similarly cuts on 2 shift to 1, and cuts on 3 shift to 2. See the layer .num_masks attribute for the max mask value allowed. See the DEF manual section on 'Multi-Mask Patterns for Routing Points' for figures and examples. A value of 0 indicates the cut layer is uncolored, or the layer is not a multi-mask layer.</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> no_value<br/> <b>Edit:</b> Yes</p> |
| cut_rects         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                   | <p>List of rectangles on cut layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master)</p> <p><b>Type:</b> <a href="#">rect</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| cut_rects_mask    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                   | <p>List of mask values for each rect in cut_rects in the same order as cut_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer.</p> <p><b>Type:</b> <a href="#">int*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| location          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|            |                                                                                                                                                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <p>Location of Via</p> <p><b>Type:</b> <a href="#">point</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                             |
| net        |                                                                                                                                                                                                                                               |
|            | <p>The net that the via belongs to</p> <p><b>Type:</b> <a href="#">obj(net)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                          |
| obj_type   |                                                                                                                                                                                                                                               |
|            | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum (via)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                           |
| route_rule |                                                                                                                                                                                                                                               |
|            | <p>The non-default rule corresponding to the via, vias with the default routing rule will return an empty string.</p> <p><b>Type:</b> <a href="#">obj(route_rule)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                    |
| status     |                                                                                                                                                                                                                                               |
|            | <p>Wiring status (equivalent to DEF NETS regular wiring status)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> unknown routed fixed cover noshield</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                 |
| top_mask   |                                                                                                                                                                                                                                               |
|            | <p>Indicates mask number for top layer for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> |
| top_rects  |                                                                                                                                                                                                                                               |

|                |                                                                                                                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>List of rectangles (typically only one) on top routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master)</p> <p><b>Type:</b> <a href="#">rect</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| top_rects_mask |                                                                                                                                                                                                                                                                             |
|                | <p>List of mask values for each rect in top_rects in the same order as top_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer.</p> <p><b>Type:</b> <a href="#">int*</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                    |
| via_def        |                                                                                                                                                                                                                                                                             |
|                | <p>The via cell</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">via_def</a>)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                           |

## via\_def

### Parent Objects

[route\\_rule](#), [tech](#), [shape\\_via](#), [via](#), [special\\_via](#), [root](#)

### Definition

A via definition. This is equivalent to the LEF/DEF VIA section statements. A via\_def can be either a parameterized via (VIA with VIARULE parameters), or a fixed via (VIA with just RECT statements). See the LEF/DEF manual VIA section for more details.

| Attribute    | Description                                                                                                                                 |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| bottom_layer |                                                                                                                                             |
|              | <p>The bottom routing layer</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">layer</a>)<br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| bottom_rects |                                                                                                                                             |

|                   |                                                                                                                                                                                                                                                                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>List of rectangles (typically only one) on bottom routing layer.</p> <p><b>Type:</b> <a href="#">rect*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                            |
| bottom_rects_mask |                                                                                                                                                                                                                                                                                                                                                               |
|                   | <p>List of mask values for each rect in bottom_rects in the same order as bottom_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer.</p> <p><b>Type:</b> <a href="#">int*</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> Yes</p>                                                                                     |
| cut_class         |                                                                                                                                                                                                                                                                                                                                                               |
|                   | <p>Returns the name of the CUTCLASS definition (from LEF or OA tech) for the cut_rects in this via_def. It returns an empty string if no CUTCLASS exists for the cut layer, or no CUTCLASS matches the size of the cut_rects.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                 |
| cut_columns       |                                                                                                                                                                                                                                                                                                                                                               |
|                   | <p>The number of cut columns. It is only set for generated vias created from a via_def_rule. See the LEF/DEF manual VIA definition with VIARULE and ROWCOL values for more details. It is 0 for fixed vias (e.g. a LEF/DEF VIA definition with only RECT values).</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> No</p> |
| cut_layer         |                                                                                                                                                                                                                                                                                                                                                               |
|                   | <p>The cut layer of via</p> <p><b>Type:</b> <a href="#">obj(layer)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                   |
| cut_rects         |                                                                                                                                                                                                                                                                                                                                                               |
|                   | <p>List of rectangles on cut layer</p> <p><b>Type:</b> <a href="#">rect*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                             |
| cut_rects_mask    |                                                                                                                                                                                                                                                                                                                                                               |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>List of mask values for each rect in cut_rects in the same order as cut_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer.</p> <p><b>Type:</b> <a href="#">int*</a><br/> <b>Default:</b> no_value<br/> <b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                          |
| cut_rows    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|             | <p>The number of cut rows. It is only set for generated vias created from a via_def_rule. See the LEF/DEF manual VIA definition with VIARULE and ROWCOL values for more details. It is 0 for fixed vias (e.g. a LEF/DEF VIA definition with only RECT values).</p> <p><b>Type:</b> <a href="#">int</a><br/> <b>Default:</b> 0<br/> <b>Edit:</b> No</p>                                                                                                                                                                   |
| cut_size    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|             | <p>The {width height} of the first rect in cut_rects in microns. It is only set for generated vias created from a via_def_rule. See the LEF/DEF manual VIA definition with VIARULE and CUTSIZE values for more details. It is {0 0} for fixed vias (e.g. a LEF/DEF VIA definition with only RECT values).</p> <p><b>Type:</b> <a href="#">point</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                     |
| from_design |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|             | <p>Indicates that the via is from design</p> <p>from_design = true &amp;&amp; from_lib = false: The via is generated from DEF or create_via_definition.</p> <p>from_design = false &amp;&amp; from_lib = true: The via is generated from LEF.</p> <p>from_design = true &amp;&amp; from_lib = true: The via is generated by NanoRoute.</p> <p>from_design = false &amp;&amp; from_lib = false: The via is generated by VIAGEN.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| from_lib    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|             | <p>Indicates that the via is from library. See the from_design attribute for more details.</p> <p><b>Type:</b> <a href="#">bool</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                     |
| is_default  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>Indicates that the via is a default via (LEF VIA DEFAULT)</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| is_non_default |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                | <p>Indicates that the via is declared in a LEF NONDEFAULTRULE</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| name           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                | <p>Via name</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| obj_type       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">via_def</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| resistance     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                | <p>Via resistance in ohms that is derived from LEF or OA data. This may not match the resistance of RC extraction results derived from extraction coefficient data. If the via is a fixed via with a resistance value defined in the LEF VIA definition statement or OA via_def, that value is returned. For vias without a resistance value, the resistance is computed from the cut-layer resistance_per_cut value and the number of cuts in the via (or equivalent cuts for a LEF CUTCLASS or OA cut_class via with different cut sizes). If both the via definition, and the cut-layer has no resistance value, then 0.0 is returned</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| tech           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                | <p>The tech for this via_def.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">tech</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| top_layer      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



|                |                                                                                                                                                                                                                                                                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | The top routing layer of via<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                                                   |
| top_rects      |                                                                                                                                                                                                                                                                                                                                                    |
|                | List of rectangles (typically only one) on top routing layer.<br><b>Type:</b> <a href="#">rect*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                                                                                       |
| top_rects_mask |                                                                                                                                                                                                                                                                                                                                                    |
|                | List of mask values for each rect in top_rects in the same order as top_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer.<br><b>Type:</b> <a href="#">int*</a><br><b>Default:</b> no_value<br><b>Edit:</b> Yes                                                                                                   |
| via_def_rule   |                                                                                                                                                                                                                                                                                                                                                    |
|                | The via_def_rule for this via_def. It is only set for generated vias created from via_def_rule parameters. See the LEF/DEF manual VIA statement with VIARULE for more details. It returns {} for fixed vias (e.g. a LEF/DEF VIA with only RECT values).<br><b>Type:</b> <a href="#">obj(via_def_rule)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |

## via\_def\_rule

### Parent Objects

[via\\_def](#), [root](#)

### Definition

Equivalent of one LEF VIARULE GENERATE statement that has the rules to create new generated vias (e.g. LEF/DEF VIA statements with VIARULE parameters). See the LEF manual for more details.

| Attribute        | Description |
|------------------|-------------|
| bottom_enclosure |             |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <p>Two minimum enclosure values for the cuts in the via. The order of the two values does not matter. The bottom layer shape must enclose all the cuts by one of the enclosure values in one direction (e.g. either X or Y), and by the other enclosure value in the other direction. If it is {0 0}, which is recommended for newer technologies, then only the DRC rules are used to compute the minimum enclosure..</p> <p><b>Type:</b> <a href="#">point</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p> |
| bottom_layer |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|              | <p>The bottom routing layer for the via.</p> <p><b>Type:</b> <a href="#">obj(layer)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                          |
| bottom_width |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|              | <p>Optional min and max width. If given, this rule should only be used if the bottom wire width is greater than or equal to the first value (min-width), and less than or equal to the second value (max-width). For example, {1 2} means min-width is <math>\geq 1.0</math>, and max_width is <math>\leq 2.0</math>. If not given, the default is {0 0}, which means this rule can be used for any width.</p> <p><b>Type:</b> <a href="#">point</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>             |
| cut_layer    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|              | <p>The cut layer for the via.</p> <p><b>Type:</b> <a href="#">obj(layer)</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                     |
| cut_rect     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|              | <p>The size of one cut rectangle.</p> <p><b>Type:</b> <a href="#">rect</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                       |
| cut_spacing  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|              | <p>Minimum center-to-center spacing in the X and Y directions..</p> <p><b>Type:</b> <a href="#">point</a><br/> <b>Default:</b> ""<br/> <b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                        |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                    | <p>Via rule name.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                         |
| obj_type           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                    | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">via_def_rule</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                              |
| resistance_per_cut |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                    | <p>Optional via resistance per cut in ohms that is defined in LEF or OA data. This value is useful for estimation, but will not match the resistance extracted by RC extraction commands that use more accurate coefficient files. It is 0.0 if not given in the library data.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                            |
| tech               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                    | <p>The tech for this via_def_rule.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">tech</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                     |
| top_enclosure      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                    | <p>Two minimum enclosure values for the cuts in the via. The order of the two values does not matter. The top layer shape must enclose all the cuts by one of the enclosure value in one direction (e.g. either X or Y), and by the other enclosure value in the other direction. If it is {0 0}, which is recommended, then only the DRC rules are used to compute the minimum enclosure.</p> <p><b>Type:</b> <a href="#">point</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| top_layer          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                    | <p>The top routing layer for the via.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">layer</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                 |

| top_width |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | <p>Optional min and max width. If given, this rule should only be used if the top wire width is greater than or equal to the first value (min-width), and less than or equal to the second value (max-width). For example,. {1 2} means min-width is &gt;= 1.0, and max_width is &lt;= 2.0). If not given, the default is {0 0}, which means this rule can be used for any width.</p> <p><b>Type:</b> point<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |

## virtual\_wire

### Parent Objects

net

### Definition

DEF NETS VIRTUAL wire

| Attribute   | Description                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| begin_layer |                                                                                                                             |
|             | <p>The begin layer for the virtual connection</p> <p><b>Type:</b> obj(layer)<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p> |
| begin_point |                                                                                                                             |
|             | <p>Reference point to the begin symbolic location</p> <p><b>Type:</b> point<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>  |
| end_layer   |                                                                                                                             |
|             | <p>The end layer for the virtual connection</p> <p><b>Type:</b> obj(layer)<br/><b>Default:</b> ""<br/><b>Edit:</b> No</p>   |

|           |                                                                                                                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| end_point |                                                                                                                                                                                                             |
|           | Reference point to the end symbolic location<br><b>Type:</b> <a href="#">point</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                 |
| net       |                                                                                                                                                                                                             |
|           | The net that the virtual connection belongs to<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">net</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                         |
| obj_type  |                                                                                                                                                                                                             |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">virtual_wire</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| status    |                                                                                                                                                                                                             |
|           | Wiring status ( equivalent to DEF NET regular wiring status)<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> unknown routed fixed cover noshield<br><b>Default:</b> ""<br><b>Edit:</b> No       |

## what\_if\_via

### Parent Objects

[net](#)

### Definition

After loading a Voltus IR-drop analysis result, you can use `add_what_if_shapes` to add 'what if' power vias to see how they would improve the IR-drop results without modifying the real power-mesh.

| Attribute    | Description |
|--------------|-------------|
| bottom_layer |             |

|           |                                                                                                                                                                                                            |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | The bottom routing layer for this what_if_via.<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                         |
| cut_layer |                                                                                                                                                                                                            |
|           | The cut_layer for this what_if_via.<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                    |
| cut_rect  |                                                                                                                                                                                                            |
|           | The rect on the cut_layer for this what_if_via.<br><b>Type:</b> <a href="#">rect</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                              |
| net       |                                                                                                                                                                                                            |
|           | The net for this what_if_via.<br><b>Type:</b> <a href="#">obj(net)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                            |
| obj_type  |                                                                                                                                                                                                            |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">what_if_via</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| top_layer |                                                                                                                                                                                                            |
|           | The top routing layer for this what_if_via.<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                            |

## what\_if\_wire

### Parent Objects

[net](#)

## Definition

After loading a Voltus IR-drop analysis result, you can use `add_what_if_shapes` to add 'what if' power wires to see how they would improve the IR-drop results without modifying the real power-mesh.

| Attribute | Description                                                                                                                                                                           |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| layer     |                                                                                                                                                                                       |
|           | The layer for the wire.<br><b>Type:</b> <code>obj(layer)</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                              |
| net       |                                                                                                                                                                                       |
|           | The net of the wire.<br><b>Type:</b> <code>obj(net)</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                   |
| obj_type  |                                                                                                                                                                                       |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <code>enum(what_if_wire)</code><br><b>Default:</b> ""<br><b>Edit:</b> No |
| rect      |                                                                                                                                                                                       |
|           | The rect for the wire.<br><b>Type:</b> <code>rect</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                     |
| width     |                                                                                                                                                                                       |
|           | The width of the wire.<br><b>Type:</b> <code>coord</code><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                    |

# wire

## Parent Objects

net

## Definition

A 2-point wire segment intended for symbolic routing where the end-points normally align with other wire or via end-points, or end on a pin shape or special\_wire. The wire, via, and patch\_wire objects correspond to the DEF NETS symbolic routing data.

| Attribute       | Description                                                                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| begin_extension |                                                                                                                                                                                       |
|                 | Extension of wire at the first point<br><b>Type:</b> coord<br><b>Default:</b> no_value<br><b>Edit:</b> Yes                                                                            |
| direction       |                                                                                                                                                                                       |
|                 | Direction of wire, consistent with layer direction from LEF/OpenAccess<br><b>Type:</b> enum<br><b>Enum Values:</b> vertical other horizontal<br><b>Default:</b> ""<br><b>Edit:</b> No |
| end_extension   |                                                                                                                                                                                       |
|                 | Extension of wire at the second point<br><b>Type:</b> coord<br><b>Default:</b> no_value<br><b>Edit:</b> Yes                                                                           |
| layer           |                                                                                                                                                                                       |
|                 | The layer of wire<br><b>Type:</b> obj(layer)<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                 |
| length          |                                                                                                                                                                                       |



|            |                                                                                                                                                                                                                                  |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <p>The center-line length of the wire between the two end points. It does not include the extension values</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                        |
| mask       |                                                                                                                                                                                                                                  |
|            | <p>Indicates mask number for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> Yes</p> |
| net        |                                                                                                                                                                                                                                  |
|            | <p>The net that the wire belongs to</p> <p><b>Type:</b> <a href="#">obj(net)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                            |
| obj_type   |                                                                                                                                                                                                                                  |
|            | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum(wire)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                              |
| points     |                                                                                                                                                                                                                                  |
|            | <p>2 points (center-line) for the wire</p> <p><b>Type:</b> <a href="#">point*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                           |
| rect       |                                                                                                                                                                                                                                  |
|            | <p>Rectangle that defines the wire shape, including any begin extension or end extension.</p> <p><b>Type:</b> <a href="#">rect</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                          |
| route_rule |                                                                                                                                                                                                                                  |

|        |                                                                                                                                                                                                                               |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | <p>The non-default rule corresponding to the wire, wires with the default routing rule will return empty string.</p> <p><b>Type:</b> <a href="#">obj(route_rule)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>    |
| status |                                                                                                                                                                                                                               |
|        | <p>Wiring status (equivalent to DEF NETS regular wiring status)</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> unknown routed fixed cover noshield</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| width  |                                                                                                                                                                                                                               |
|        | <p>Width of wire</p> <p><b>Type:</b> <a href="#">coord</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                               |

## port\_bus

### Parent Objects

[design](#), [root](#), [port](#)

### Definition

The bus on port

| Attribute | Description                                                                                                                                                      |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name |                                                                                                                                                                  |
|           | <p>Non-vector portion of the bus name. (e.g. A for bus A[7:0] ).</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| bits      |                                                                                                                                                                  |

|              |                                                                                                                                                                                          |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <p>A list of individual ports that constitute the bus.</p> <p><b>Type:</b> <a href="#">obj(port)*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                               |
| design       |                                                                                                                                                                                          |
|              | <p>The design of the port bus.</p> <p><b>Type:</b> <a href="#">obj(design)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |
| direction    |                                                                                                                                                                                          |
|              | <p>Returns the direction of the port bus.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> internal in out inout</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| escaped_name |                                                                                                                                                                                          |
|              | <p>The escaped name of the port bus.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                     |
| lsb          |                                                                                                                                                                                          |
|              | <p>Least significant bit. Returns the bottom bit index of the bus.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                          |
| msb          |                                                                                                                                                                                          |
|              | <p>Most significant bit. Returns the top bit index of the bus.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                              |
| name         |                                                                                                                                                                                          |
|              | <p>The name of the port bus</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                              |

|          |                                                                                                                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| obj_type |                                                                                                                                                                                                                          |
|          | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">port_bus</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |

## hport\_bus

### Parent Objects

[hinst](#), [design](#), [root](#), [hport](#)

### Definition

The bus on hport

| Attribute    | Description                                                                                                                                                                                        |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name    |                                                                                                                                                                                                    |
|              | <p>Non-vector portion of the bus name (eg. A for bus A[7:0])</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                       |
| bits         |                                                                                                                                                                                                    |
|              | <p>The ports of the bus. All the bits of one bus will have the same obj_type.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">hport</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| direction    |                                                                                                                                                                                                    |
|              | <p>Returns the direction of the hport_bus.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> internal in out inout</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>          |
| escaped_name |                                                                                                                                                                                                    |

|          |                                                                                                                                                                                        |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | The escaped name of the hport_bus<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                      |
| hinst    |                                                                                                                                                                                        |
|          | Returns the hierarchical instance to which the hport_bus belongs.<br><b>Type:</b> <a href="#">obj(hinst)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                  |
| lsb      |                                                                                                                                                                                        |
|          | Bottom bit index of hport_bus.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                            |
| msb      |                                                                                                                                                                                        |
|          | Top bit index of hport_bus.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                               |
| name     |                                                                                                                                                                                        |
|          | The name of hport_bus.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                 |
| obj_type |                                                                                                                                                                                        |
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (hport_bus)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |

## hpin\_bus

### Parent Objects

[hinst](#), [design](#), [hpin](#), [root](#)

## Definition

The bus on hpin

| Attribute    | Description                                                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name    |                                                                                                                                                                   |
|              | Non-vector portion of the bus name (eg. A for bus A[7:0])<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                         |
| bits         |                                                                                                                                                                   |
|              | The ports of the bus. All the bits of one bus will have the same obj_type.<br><b>Type:</b> <a href="#">obj(hpin)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No    |
| direction    |                                                                                                                                                                   |
|              | Returns the direction of the hpin_bus.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> internal in out inout<br><b>Default:</b> ""<br><b>Edit:</b> No |
| escaped_name |                                                                                                                                                                   |
|              | The escaped name of the hpin_bus<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                  |
| hinst        |                                                                                                                                                                   |
|              | Returns the hierarchical instance to which the hpin_bus belongs.<br><b>Type:</b> <a href="#">obj(hinst)</a><br><b>Default:</b> ""<br><b>Edit:</b> No              |
| lsb          |                                                                                                                                                                   |
|              | Bottom bit index of bus.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                             |

|          |                                                                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| msb      |                                                                                                                                                                                                         |
|          | Top bit index of bus.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                      |
| name     |                                                                                                                                                                                                         |
|          | The name of hpin_bus.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                   |
| obj_type |                                                                                                                                                                                                         |
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">hpin_bus</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

## test\_signal

### Parent Objects

[inst](#), [design](#), [test\\_clock](#), [root](#), [dft\\_configuration\\_mode](#)

### Definition

test signal

| Attribute | Description                                                                                                                                           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| active    |                                                                                                                                                       |
|           | Value of test signal during test mode.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> high low<br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| atpg_use  |                                                                                                                                                       |

|                        |                                                                                                                                                                                                                                    |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | ATPG usage of test signal.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> none opcg_enable opcg_trigger opcg_edge_mode opcg_custom_se opcg_misr_reset_clock<br><b>Default:</b> none<br><b>Edit:</b> Yes               |
| base_name              |                                                                                                                                                                                                                                    |
|                        | Name of test_signal<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                                |
| dedicated_pin          |                                                                                                                                                                                                                                    |
|                        | Whether a dedicated test pin is used for the test signal.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                        |
| default_shift_enable   |                                                                                                                                                                                                                                    |
|                        | When 'true', indicates this test signal is the default shift enable.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                             |
| dft_compression_signal |                                                                                                                                                                                                                                    |
|                        | Type of Compression test signal.<br><b>Type:</b> <a href="#">enum</a><br><b>Enum Values:</b> none compression_enable spreader_mask_enable mask_load<br><b>Default:</b> none<br><b>Edit:</b> Yes                                    |
| dft_hookup_pin         |                                                                                                                                                                                                                                    |
|                        | Port/pin where the test signal actually hooks up inside the core.<br><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| dft_hookup_polarity    |                                                                                                                                                                                                                                    |



|               |                                                                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------------------|
|               | Value of test signal during test mode at the hookup pin.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No |
| divide_fall   |                                                                                                                          |
|               | Falling edge denominator.<br><b>Type:</b> int<br><b>Default:</b> no_value<br><b>Edit:</b> No                             |
| divide_period |                                                                                                                          |
|               | Clock period denominator.<br><b>Type:</b> int<br><b>Default:</b> no_value<br><b>Edit:</b> No                             |
| divide_rise   |                                                                                                                          |
|               | Rising edge denominator.<br><b>Type:</b> int<br><b>Default:</b> no_value<br><b>Edit:</b> No                              |
| escaped_name  |                                                                                                                          |
|               | Name of test_signal including escape characters<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| fall          |                                                                                                                          |
|               | Falling edge numerator (fraction of period).<br><b>Type:</b> int<br><b>Default:</b> no_value<br><b>Edit:</b> No          |
| function      |                                                                                                                          |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <p>Function of the test signal object.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> parallel_sdi parallel_sdo compress_sdi compress_sdo serial_sdi serial_sdo shift_enable test_mode scan_clock lssd_scan_clock_a lssd_scan_clock_b aux_scan_clock compression_clock compression_enable spread_enable pipe_sen mask_load mask_enable out_pipe_bypass decomp_pipe_bypass comp_pipe_bypass codec_pipe_bypass cpi_overwrapper_clock wrapper_and_compression_clock shift_wr capture_wr update_wr wir_test_c elastic_reset smartscan_enable smartscan_broadcast opcg_enable opcg_trigger opcg_load opcg_load_enable lbist_test_enable lbist_direct_reset lbist_direct_done lbist_direct_fail lbist_runbist lbist_run misr_bypass mbist_retention_test_out lp_staggered_shift_mode misr_lp_gating_load swizzle_enable async_hold segment_bypass</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| has_fanout    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|               | <p>Whether a dedicated test pin is used for the test signal by fanout analysis.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> false true test_only</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ideal         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|               | <p>Whether the test signal network should be treated as ideal.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| index         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|               | <p>Index of the test signal object.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> 0</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| lec_value     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|               | <p>Whether a lec value is attached to the test signal.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| master_signal |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|            |                                                                                                                                                                                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <p>Master test signal.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> true</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                                  |
| name       |                                                                                                                                                                                                                                                                                          |
|            | <p>Name of test_signal</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                   |
| obj_type   |                                                                                                                                                                                                                                                                                          |
|            | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">test_signal</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                              |
| period     |                                                                                                                                                                                                                                                                                          |
|            | <p>Clock period numerator.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> 0.0</p> <p><b>Edit:</b> No</p>                                                                                                                                                              |
| pin        |                                                                                                                                                                                                                                                                                          |
|            | <p>Port/pin associated to test signal.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">pin</a>) <a href="#">obj</a>(<a href="#">hpin</a>) <a href="#">obj</a>(<a href="#">port</a>) <a href="#">obj</a>(<a href="#">hport</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p> |
| pmbist_use |                                                                                                                                                                                                                                                                                          |
|            | <p>Test signal used during ATPG to control programmable MBIST logic (possible values are: {none test_block_async_reset test_clock_select test_rambypass test_ramsequential retent</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>       |
| rise       |                                                                                                                                                                                                                                                                                          |
|            | <p>Rising edge numerator (fraction of period).</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> No</p>                                                                                                                                        |

|                     |                                                                                                                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| scan_shift          |                                                                                                                                                                                                                   |
|                     | Whether the test signal will be captured as a clock in the ATPG file.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                              |
| type                |                                                                                                                                                                                                                   |
|                     | Type of test signal.<br><b>Type:</b> <code>enum</code><br><b>Enum Values:</b> test_mode shift_enable scan_clock scan_clock_a scan_clock_b aux_scan.<br><b>Default:</b> ""<br><b>Edit:</b> Yes                     |
| user_defined_signal |                                                                                                                                                                                                                   |
|                     | Whether the test signal was defined by the user or automatically generated by the tool.<br><b>Type:</b> <code>enum</code><br><b>Enum Values:</b> true false simulated<br><b>Default:</b> true<br><b>Edit:</b> Yes |
| wir_reset_value     |                                                                                                                                                                                                                   |
|                     | Value of the WIR output pin when the wir reset signal is asserted.<br><b>Type:</b> <code>enum</code><br><b>Enum Values:</b> high low<br><b>Default:</b> ""<br><b>Edit:</b> Yes                                    |
| wir_signal          |                                                                                                                                                                                                                   |
|                     | Whether test signal is a WIR signal.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                                                               |
| wir_tm_value        |                                                                                                                                                                                                                   |
|                     | Value of the WIR output pin when the wir is being scan tested.<br><b>Type:</b> <code>enum</code><br><b>Enum Values:</b> high low<br><b>Default:</b> ""<br><b>Edit:</b> Yes                                        |

# formal\_verification\_constraint

## Parent Objects

[design](#), [root](#)

## Definition

formal verification constraint

| Attribute    | Description                                                                                                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name    |                                                                                                                                                                                                                               |
|              | Name of formal_verification_constraint<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                        |
| escaped_name |                                                                                                                                                                                                                               |
|              | Name of formal_verification_constraint including escape characters<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                            |
| name         |                                                                                                                                                                                                                               |
|              | Name of formal_verification_constraint<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                        |
| obj_type     |                                                                                                                                                                                                                               |
|              | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">formal_verification_constraint</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

# test\_clock

## Parent Objects

[inst](#), [design](#), [root](#)

## Definition

test clock

| Attribute         | Description                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| at_speed          |                                                                                                                                                                          |
|                   | Whether the test clock is an at-speed test clock.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                      |
| base_name         |                                                                                                                                                                          |
|                   | Name of test_clock<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                       |
| blocking_se       |                                                                                                                                                                          |
|                   | The blocking shift enable corresponding to the test clock.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">test_signal</a> )<br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| controllable      |                                                                                                                                                                          |
|                   | Whether the test clock is controllable in test mode.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                   |
| controllable_from |                                                                                                                                                                          |

|                     |                                                                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | Source the test clock is controllable from.<br><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                      |
| dft_hookup_pin      |                                                                                                                                                                                                                                   |
|                     | Port/pin where the test clock actually hooks up inside the core.<br><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| dft_hookup_polarity |                                                                                                                                                                                                                                   |
|                     | Value of test clock during test mode at the hookup pin.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                           |
| divide_fall         |                                                                                                                                                                                                                                   |
|                     | Falling edge denominator.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                            |
| divide_period       |                                                                                                                                                                                                                                   |
|                     | Clock period denominator.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> 1<br><b>Edit:</b> No                                                                                                                             |
| divide_rise         |                                                                                                                                                                                                                                   |
|                     | Rising edge denominator.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                                             |
| domain_se           |                                                                                                                                                                                                                                   |
|                     | The domain shift enable corresponding to the test clock.<br><b>Type:</b> <a href="#">obj(test_signal)</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes                                                                               |
| escaped_name        |                                                                                                                                                                                                                                   |

|           |                                                                                                                                                                                                                            |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | <p>Name of test_clock including escape characters</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                          |
| fall      |                                                                                                                                                                                                                            |
|           | <p>Falling edge numerator (fraction of period).</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                               |
| function  |                                                                                                                                                                                                                            |
|           | <p>Function of the test clock object.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> test_clock dft_clock compressor_clock wrck</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                  |
| name      |                                                                                                                                                                                                                            |
|           | <p>Name of test_clock</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                      |
| obj_type  |                                                                                                                                                                                                                            |
|           | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">test_clock</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| off_state |                                                                                                                                                                                                                            |
|           | <p>Off state during shift mode.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                               |
| period    |                                                                                                                                                                                                                            |
|           | <p>Clock period numerator.</p> <p><b>Type:</b> <a href="#">double</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                 |



|                      |                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rise                 |                                                                                                                                                                                   |
|                      | Rising edge numerator (fraction of period).<br><b>Type:</b> <code>int</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                                                |
| root_source_pins     |                                                                                                                                                                                   |
|                      | Root source pin(s) of the test clock.<br><b>Type:</b> <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No              |
| root_source_polarity |                                                                                                                                                                                   |
|                      | Polarity between root pin and clock pin of the test clock.<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                              |
| sources              |                                                                                                                                                                                   |
|                      | Source pin(s) of the test clock.<br><b>Type:</b> <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                   |
| user_defined_signal  |                                                                                                                                                                                   |
|                      | Whether the test clock was defined by the user or automatically generated by the tool.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> <code>true</code><br><b>Edit:</b> Yes |

## scan\_chain

### Parent Objects

[hinst](#), [inst](#), [design](#), [hpin](#), [pin](#), [actual\\_scan\\_segment](#), [root](#), [violation](#), [hport](#), [port](#)

### Definition

scan chain

| Attribute          | Description                                                                                                                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name          |                                                                                                                                                                                                |
|                    | Name of scan_chain<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                             |
| body               |                                                                                                                                                                                                |
|                    | Body (segment) of scan chain.<br><b>Type:</b> <a href="#">object*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                 |
| complete           |                                                                                                                                                                                                |
|                    | Scan chain was completely specified by user or analysis.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> No                                                      |
| compressed         |                                                                                                                                                                                                |
|                    | Identifies if a chain has undergone compression.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes                                                             |
| dft_hookup_pin_sdi |                                                                                                                                                                                                |
|                    | Core side hookup pin for SDI.<br><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| dft_hookup_pin_sdo |                                                                                                                                                                                                |
|                    | Core side hookup pin for SDO.<br><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| dft_partition      |                                                                                                                                                                                                |

|              |                                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <p>Determine the DFT partition of the specified actual_scan_chain/actual_scan_segment/scan_chain/scan_segment.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| domain       |                                                                                                                                                                                                |
|              | <p>Clock domain of scan chain.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                 |
| edge         |                                                                                                                                                                                                |
|              | <p>Edge of clock domain in scan chain.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> any</p> <p><b>Edit:</b> No</p>                                                                        |
| escaped_name |                                                                                                                                                                                                |
|              | <p>Name of scan_chain including escape characters</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                              |
| head         |                                                                                                                                                                                                |
|              | <p>Head (segment) of scan chain.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                               |
| max_length   |                                                                                                                                                                                                |
|              | <p>User specified length for this scan chain (might be slightly different than the actual length).</p> <p><b>Type:</b> int</p> <p><b>Default:</b> no_value</p> <p><b>Edit:</b> No</p>          |
| name         |                                                                                                                                                                                                |
|              | <p>Name of scan_chain</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                          |

|                        |                                                                                                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| non_shared_scan_out    |                                                                                                                                                                                                                            |
|                        | <p>Non shared scan out of scan chain is on functional data path.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                         |
| obj_type               |                                                                                                                                                                                                                            |
|                        | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">scan_chain</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| scan_clock_a           |                                                                                                                                                                                                                            |
|                        | <p>Scan clock a (signal) of scan chain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                    |
| scan_clock_b           |                                                                                                                                                                                                                            |
|                        | <p>Scan clock b (signal) of scan chain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                    |
| scan_in                |                                                                                                                                                                                                                            |
|                        | <p>Scan in pin of scan chain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                              |
| scan_out               |                                                                                                                                                                                                                            |
|                        | <p>Scan out pin of scan chain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                             |
| sdi_compression_signal |                                                                                                                                                                                                                            |
|                        | <p>Whether the scan in of the chain is shared with mask enable.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                           |

|                 |                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| shared_input    |                                                                                                                                                                                |
|                 | <p>Scan chain shares scan input with another functional pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                  |
| shared_output   |                                                                                                                                                                                |
|                 | <p>Scan chain shares scan output with another functional pin.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                 |
| shared_select   |                                                                                                                                                                                |
|                 | <p>Control signal to the multiplexer added because the scan out pin is shared.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| shift_enable    |                                                                                                                                                                                |
|                 | <p>Scan shift enable (signal) of scan chain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                   |
| tail            |                                                                                                                                                                                |
|                 | <p>Tail (segment) of scan chain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                               |
| terminal_lockup |                                                                                                                                                                                |
|                 | <p>Type of the terminal lockup element in the scan chain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p>                    |

# actual\_scan\_chain

## Parent Objects

[hinst](#), [inst](#), [design](#), [hpin](#), [pin](#), [actual\\_scan\\_segment](#), [root](#), [violation](#), [hport](#), [port](#)

## Definition

actual scan chain

| Attribute              | Description                                                                                                                                |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| analyzed               |                                                                                                                                            |
|                        | Whether this is an analyzed chain.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> No                        |
| base_name              |                                                                                                                                            |
|                        | Name of actual_scan_chain<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                  |
| compressed             |                                                                                                                                            |
|                        | Identifies if a chain has undergone compression.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> Yes         |
| connected_shift_enable |                                                                                                                                            |
|                        | Connected shift enable attribute of an actual scan chain.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> No |
| dft_hookup_pin_sdi     |                                                                                                                                            |

|                    |                                                                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | Core side hookup pin for SDI.<br><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| dft_hookup_pin_sdo |                                                                                                                                                                                                |
|                    | Core side hookup pin for SDO.<br><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| dft_partition      |                                                                                                                                                                                                |
|                    | Determine the DFT partition of the specified actual_scan_chain/actual_scan_segment/scan_chain/scan_segment.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No    |
| domain             |                                                                                                                                                                                                |
|                    | Clock domain of scan chain.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                    |
| edge               |                                                                                                                                                                                                |
|                    | Edge of clock domain in scan chain.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> any<br><b>Edit:</b> No                                                                           |
| elements           |                                                                                                                                                                                                |
|                    | Elements of scan chain (a Tcl list).<br><b>Type:</b> <a href="#">object*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                          |
| escaped_name       |                                                                                                                                                                                                |
|                    | Name of actual_scan_chain including escape characters<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                          |

|                     |                                                                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| head_lockup         |                                                                                                                                                                                                                                   |
|                     | <p>Instance of the head lockup element in the scan chain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> none</p> <p><b>Edit:</b> No</p>                                                                       |
| mode_name           |                                                                                                                                                                                                                                   |
|                     | <p>Mode to which the scan chain belongs to.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                       |
| name                |                                                                                                                                                                                                                                   |
|                     | <p>Name of actual_scan_chain</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                      |
| non_shared_scan_out |                                                                                                                                                                                                                                   |
|                     | <p>Non shared scan out of scan chain is on functional data path.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                |
| obj_type            |                                                                                                                                                                                                                                   |
|                     | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">actual_scan_chain</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| other_clocks        |                                                                                                                                                                                                                                   |
|                     | <p>Other clock pins of the segment and their active values as a Tcl list.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                         |
| power_domain        |                                                                                                                                                                                                                                   |
|                     | <p>All the power domains of this scan chain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                      |



|                        |                                                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reg_count              |                                                                                                                                                                            |
|                        | <p>Number of flip-flops in scan chain.</p> <p><b>Type:</b> <code>int</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                              |
| scan_clock_a           |                                                                                                                                                                            |
|                        | <p>Scan clock a (signal) of scan chain.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                          |
| scan_clock_b           |                                                                                                                                                                            |
|                        | <p>Scan clock b (signal) of scan chain.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                          |
| scan_in                |                                                                                                                                                                            |
|                        | <p>Scan in pin of scan chain.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                    |
| scan_out               |                                                                                                                                                                            |
|                        | <p>Scan out pin of scan chain.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> No</p>                                   |
| sdi_compression_signal |                                                                                                                                                                            |
|                        | <p>Whether the scan in of the chain is shared with mask enable.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> <code>""</code></p> <p><b>Edit:</b> Yes</p> |
| shared_input           |                                                                                                                                                                            |
|                        | <p>Scan chain shares scan input with another functional pin.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> <code>false</code></p> <p><b>Edit:</b> No</p>    |

|                 |                                                                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| shared_output   |                                                                                                                                                             |
|                 | Scan chain shares scan output with another functional pin.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> No                 |
| shared_select   |                                                                                                                                                             |
|                 | Control signal to the multiplexer added because the scan out pin is shared.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| shift_enable    |                                                                                                                                                             |
|                 | Scan shift enable (signal) of scan chain.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                   |
| terminal_lockup |                                                                                                                                                             |
|                 | Instance of the terminal lockup element in the scan chain.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> none<br><b>Edit:</b> No                |

# violation

## Parent Objects

[design](#), [root](#)

## Definition

violation

| Attribute | Description |
|-----------|-------------|
| base_name |             |

|              |                                                                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
|              | Name of violation<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                                |
| description  |                                                                                                                                            |
|              | Violation description.<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                           |
| endpoints    |                                                                                                                                            |
|              | End points.<br><b>Type:</b> <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No |
| escaped_name |                                                                                                                                            |
|              | Name of violation including escape characters<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No    |
| fixed        |                                                                                                                                            |
|              | Whether fixed by fix_dft_violations.<br><b>Type:</b> <code>bool</code><br><b>Default:</b> <code>false</code><br><b>Edit:</b> No            |
| id           |                                                                                                                                            |
|              | Violation ID.<br><b>Type:</b> <code>int</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                                       |
| name         |                                                                                                                                            |
|              | Name of violation<br><b>Type:</b> <code>string</code><br><b>Default:</b> <code>""</code><br><b>Edit:</b> No                                |
| obj_type     |                                                                                                                                            |

|                      |                                                                                                                                                                                                                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">violation</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                             |
| reg_count            |                                                                                                                                                                                                                                                                                                                       |
|                      | <p>Register count.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                                                                                       |
| registers            |                                                                                                                                                                                                                                                                                                                       |
|                      | <p>Affected registers.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">inst</a>)* <a href="#">obj</a>(<a href="#">hinst</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                                                                |
| root_node            |                                                                                                                                                                                                                                                                                                                       |
|                      | <p>Root port/pin.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">pin</a>) <a href="#">obj</a>(<a href="#">hpin</a>) <a href="#">obj</a>(<a href="#">port</a>) <a href="#">obj</a>(<a href="#">hport</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                    |
| segments             |                                                                                                                                                                                                                                                                                                                       |
|                      | <p>Affected abstract segments.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">scan_chain</a>)* <a href="#">obj</a>(<a href="#">scan_segment</a>)* <a href="#">obj</a>(<a href="#">actual_scan_segment</a>)* <a href="#">obj</a>(<a href="#">actual_scan</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| tristate_net_drivers |                                                                                                                                                                                                                                                                                                                       |
|                      | <p>Driver pins for tristate net.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">pin</a>)* <a href="#">obj</a>(<a href="#">hpin</a>)* <a href="#">obj</a>(<a href="#">port</a>)* <a href="#">obj</a>(<a href="#">hport</a>)*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                 |
| tristate_net_load    |                                                                                                                                                                                                                                                                                                                       |
|                      | <p>Load pin for tristate net.</p> <p><b>Type:</b> <a href="#">obj</a>(<a href="#">pin</a>) <a href="#">obj</a>(<a href="#">hpin</a>) <a href="#">obj</a>(<a href="#">port</a>) <a href="#">obj</a>(<a href="#">hport</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                        |
| type                 |                                                                                                                                                                                                                                                                                                                       |

|  |                                                                                                 |
|--|-------------------------------------------------------------------------------------------------|
|  | Violation type.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
|--|-------------------------------------------------------------------------------------------------|

## dft\_configuration\_mode

### Parent Objects

[design](#), [root](#)

### Definition

dft configuration mode

| Attribute    | Description                                                                                                                                                                                       |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name    |                                                                                                                                                                                                   |
|              | Name of dft_configuration_mode<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                    |
| current_mode |                                                                                                                                                                                                   |
|              | Current scan mode.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> No                                                                                               |
| decoded_pin  |                                                                                                                                                                                                   |
|              | Configuration mode decoded pin.<br><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| escaped_name |                                                                                                                                                                                                   |

|                  |                                                                                                                                                                                                     |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | Name of dft_configuration_mode including escape characters<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                          |
| mode_enable_high |                                                                                                                                                                                                     |
|                  | Test Signals that are held to active high value for the mode.<br><b>Type:</b> <a href="#">obj(test_signal)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                            |
| mode_enable_low  |                                                                                                                                                                                                     |
|                  | Test Signals that are held to active low value for the mode.<br><b>Type:</b> <a href="#">obj(test_signal)*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                             |
| name             |                                                                                                                                                                                                     |
|                  | Name of dft_configuration_mode<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                      |
| obj_type         |                                                                                                                                                                                                     |
|                  | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (dft_configuration_mode)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| type             |                                                                                                                                                                                                     |
|                  | Configuration mode type.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                            |
| usage            |                                                                                                                                                                                                     |
|                  | Configuration mode usage.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                           |
| user_defined     |                                                                                                                                                                                                     |

|  |                                                                                                                      |
|--|----------------------------------------------------------------------------------------------------------------------|
|  | Is user defined configuration mode.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> No |
|--|----------------------------------------------------------------------------------------------------------------------|

## tap\_distance\_rule

### Parent Objects

[tech](#)

### Definition

Returns the LEF TAPDISTANCERULE info.

| Attribute           | Description                                                                                                                                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| default             |                                                                                                                                                                                                                  |
|                     | The effective default tap distance of the rule.<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                   |
| name                |                                                                                                                                                                                                                  |
|                     | The name of the tap_distance_rule defined in LEF.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                |
| obj_type            |                                                                                                                                                                                                                  |
|                     | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">tap_distance_rule</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| tap_layer_distances |                                                                                                                                                                                                                  |

|  |                                                                                                                                                              |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Returns the layer and its tap distance.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">tap_layer_distance</a> )*<br><b>Default:</b> ""<br><b>Edit:</b> No |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

## tap\_layer\_distance

### Parent Objects

[tap\\_distance\\_rule](#)

### Definition

Returns layer's distance info in LEF TAPDISTANCERULE.

| Attribute | Description                                                                                                                                                                                                       |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| distance  |                                                                                                                                                                                                                   |
|           | Returns the tap distance of the specified layer.<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                   |
| layer     |                                                                                                                                                                                                                   |
|           | Returns the layer object.<br><b>Type:</b> <a href="#">obj</a> ( <a href="#">layer</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                  |
| obj_type  |                                                                                                                                                                                                                   |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">tap_layer_distance</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |



# interposer\_route\_rule

## Parent Objects

[net](#)

## Definition

Return net's interposer route rule

| Attribute | Description                                                                                                                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| layer     |                                                                                                                                                                                                    |
|           | The layer rule.<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                |
| net       |                                                                                                                                                                                                    |
|           | The net rule.<br><b>Type:</b> <a href="#">obj(net)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                    |
| obj_type  |                                                                                                                                                                                                    |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (interposer_route_rule)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| offset    |                                                                                                                                                                                                    |
|           | The offset rule.<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                    |
| spacing   |                                                                                                                                                                                                    |

|       |                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------|
|       | The offset rule.<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| width |                                                                                                 |
|       | The width rule.<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No  |

## seq\_function

### Parent Objects

[lib\\_cell](#)

### Definition

cte seq\_function

| Attribute    | Description                                                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_name    |                                                                                                                                                                               |
|              | Returns base name of the seq function object.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No  |
| d_function   |                                                                                                                                                                               |
|              | Returns d function of the seq function object.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No |
| escaped_name |                                                                                                                                                                               |

|          |                                                                                                                                                                                                             |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Returns escaped name of the seq function object.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                             |
| lib_cell |                                                                                                                                                                                                             |
|          | Returns lib cell associated with that seq function object.<br><b>Type:</b> <a href="#">obj(lib_cell)</a> *<br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| name     |                                                                                                                                                                                                             |
|          | Returns name of the seq function object.<br><b>Type:</b> <a href="#">string</a><br><b>Allowed -index values:</b> analysis_view<br><b>Default:</b> ""<br><b>Edit:</b> No                                     |
| obj_type |                                                                                                                                                                                                             |
|          | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">seq_function</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

## test\_clock\_domain

### Parent Objects

[design](#), [root](#)

### Definition

test clock domain

| Attribute | Description |
|-----------|-------------|
| base_name |             |

|              |                                                                                                                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | Name of test_clock_domain<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                        |
| escaped_name |                                                                                                                                                                                                                  |
|              | Name of test_clock_domain including escape characters<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                            |
| name         |                                                                                                                                                                                                                  |
|              | Name of test_clock_domain<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                        |
| obj_type     |                                                                                                                                                                                                                  |
|              | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">test_clock_domain</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |

## scan\_segment

### Parent Objects

[hinst](#), [inst](#), [design](#), [hpin](#), [pin](#), [actual\\_scan\\_segment](#), [root](#), [violation](#), [hport](#), [port](#)

### Definition

scan segment

| Attribute | Description |
|-----------|-------------|
| active    |             |

|                           |                                                                                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | Shift enable active value for an abstract segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                    |
| base_name                 |                                                                                                                                                                       |
|                           | Name of scan_segment<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                  |
| clock                     |                                                                                                                                                                       |
|                           | Head clock pin for an abstract segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                               |
| clock_edge                |                                                                                                                                                                       |
|                           | Active (head) clock edge at the local clock pin for an abstract segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                              |
| clock_gating_shift_enable |                                                                                                                                                                       |
|                           | Clock gating shift enable pin for an abstract, preserved or shift_reg segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                        |
| connected_scan_clock_a    |                                                                                                                                                                       |
|                           | Whether an abstract or preserved segment has its common scan clock a pin already globally connected.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> No |
| connected_scan_clock_b    |                                                                                                                                                                       |
|                           | Whether an abstract or preserved segment has its common scan clock b pin already globally connected.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> No |

|                        |                                                                                                                                                                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| connected_shift_enable |                                                                                                                                                                                                              |
|                        | <p>Whether an abstract or preserved segment has its common shift enable pin already globally connected.</p> <p><b>Type:</b> <code>bool</code><br/><b>Default:</b> <code>false</code><br/><b>Edit:</b> No</p> |
| core_wrapper           |                                                                                                                                                                                                              |
|                        | <p>Whether the segment is for a core wrapper cell.</p> <p><b>Type:</b> <code>bool</code><br/><b>Default:</b> <code>false</code><br/><b>Edit:</b> Yes</p>                                                     |
| core_wrapper_ports     |                                                                                                                                                                                                              |
|                        | <p>Specifies the ports associated with the wrapper.</p> <p><b>Type:</b> <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code><br/><b>Default:</b> <code>""</code><br/><b>Edit:</b> Yes</p>                |
| core_wrapper_type      |                                                                                                                                                                                                              |
|                        | <p>Specifies the wrapper type.</p> <p><b>Type:</b> <code>string</code><br/><b>Default:</b> <code>""</code><br/><b>Edit:</b> Yes</p>                                                                          |
| core_wrapper_usage     |                                                                                                                                                                                                              |
|                        | <p>Specifies the wrapper usage.</p> <p><b>Type:</b> <code>string</code><br/><b>Default:</b> <code>""</code><br/><b>Edit:</b> Yes</p>                                                                         |
| ctl_defined            |                                                                                                                                                                                                              |
|                        | <p>Scan segment defined in ctl abstract model.</p> <p><b>Type:</b> <code>bool</code><br/><b>Default:</b> <code>false</code><br/><b>Edit:</b> Yes</p>                                                         |
| dft_dont_scan          |                                                                                                                                                                                                              |

|                                                |                                                                                                                                                                                                                                                                                  |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Exclude this abstract segment from scan-synthesis.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                             |
| dft_exclude_abstract_segment_from_los_pipeline |                                                                                                                                                                                                                                                                                  |
|                                                | <p>Avoid insertion of launch-off-shift pipelined logic to shift enable of abstract segment.</p> <p><b>Type:</b> <code>bool</code></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                       |
| dft_hookup_pin_sdi                             |                                                                                                                                                                                                                                                                                  |
|                                                | <p>Core side hookup pin for SDI.</p> <p><b>Type:</b> <code>obj(pin)</code> <code>obj(hpin)</code> <code>obj(port)</code> <code>obj(hport)</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                            |
| dft_hookup_pin_sdo                             |                                                                                                                                                                                                                                                                                  |
|                                                | <p>Core side hookup pin for SDO.</p> <p><b>Type:</b> <code>obj(pin)</code> <code>obj(hpin)</code> <code>obj(port)</code> <code>obj(hport)</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                            |
| dft_part_of_segment                            |                                                                                                                                                                                                                                                                                  |
|                                                | <p>The type of segment this segment belongs to {abstract   fixed   floating   preserved   shift_register}.</p> <p><b>Type:</b> <code>enum</code></p> <p><b>Enum Values:</b> fixed floating preserve abstract shift_register</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| dft_partition                                  |                                                                                                                                                                                                                                                                                  |
|                                                | <p>Determine the DFT partition of the specified actual_scan_chain/actual_scan_segment/scan_chain/scan_segment.</p> <p><b>Type:</b> <code>string</code></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                      |
| dft_status                                     |                                                                                                                                                                                                                                                                                  |

|                                   |                                                                                                                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | DFT rule check status as analyzed by check_dft_rules {Passes DFT rules   Fails DFT rules   Abstract Segment Dont scan   Dont scan   Misc. non scan}.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No |
| dft_tail_test_clock               |                                                                                                                                                                                                                      |
|                                   | Top level tail test_clock for an abstract segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                   |
| dft_tail_test_clock_edge          |                                                                                                                                                                                                                      |
|                                   | Active top level tail test_clock edge for an abstract segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                       |
| dft_tail_test_clock_waveform_edge |                                                                                                                                                                                                                      |
|                                   | Active top level tail test_clock waveform leading or trailing edge for an abstract segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                          |
| dft_test_clock                    |                                                                                                                                                                                                                      |
|                                   | Top level (head) test_clock for an abstract segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                 |
| dft_test_clock_edge               |                                                                                                                                                                                                                      |
|                                   | Active top level (head) test_clock edge for an abstract segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                     |
| dft_test_clock_waveform_edge      |                                                                                                                                                                                                                      |
|                                   | Active top level (head) test_clock waveform leading or trailing edge for an abstract segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                        |



|                |                                                                                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dft_violation  |                                                                                                                                                                    |
|                | <p>DFT rule violation as analyzed by check_dft_rules.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>              |
| elements       |                                                                                                                                                                    |
|                | <p>Elements of scan segment (a Tcl list).</p> <p><b>Type:</b> <a href="#">object*</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                         |
| escaped_name   |                                                                                                                                                                    |
|                | <p>Name of scan_segment including escape characters</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                |
| head_skew_safe |                                                                                                                                                                    |
|                | <p>Whether an abstract segment has a lockup element at the head.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>  |
| instance       |                                                                                                                                                                    |
|                | <p>Instance for an abstract segment.</p> <p><b>Type:</b> <a href="#">obj(inst)</a> <a href="#">obj(hinst)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| model_defined  |                                                                                                                                                                    |
|                | <p>Scan segment defined in abstract model.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                       |
| name           |                                                                                                                                                                    |
|                | <p>Name of scan_segment</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                            |

|              |                                                                                                                                                                                                                              |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| obj_type     |                                                                                                                                                                                                                              |
|              | <p>The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a></p> <p><b>Type:</b> <a href="#">enum</a> (<a href="#">scan_segment</a>)</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| other_clocks |                                                                                                                                                                                                                              |
|              | <p>Other clock pins of the segment and their active values as a Tcl list.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                    |
| power_domain |                                                                                                                                                                                                                              |
|              | <p>All the power domains of this scan chain.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                 |
| reg_count    |                                                                                                                                                                                                                              |
|              | <p>Number of flip-flops in scan segment.</p> <p><b>Type:</b> <a href="#">int</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                                        |
| reorderable  |                                                                                                                                                                                                                              |
|              | <p>Whether the preserved segment is reorderable for scanDEF purposes.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p>                                                       |
| scan_in      |                                                                                                                                                                                                                              |
|              | <p>SDI pin for an abstract, preserved or shift_reg segment.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                  |
| scan_out     |                                                                                                                                                                                                                              |
|              | <p>SDO pin for an abstract, preserved or shift_reg segment.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                  |

|                      |                                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| shift_enable         |                                                                                                                                                      |
|                      | Shift enable pin for an abstract, preserved or shift_reg segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                    |
| skew_safe            |                                                                                                                                                      |
|                      | Whether an abstract segment has a lockup element at the end.<br><b>Type:</b> bool<br><b>Default:</b> false<br><b>Edit:</b> No                        |
| tail_clock           |                                                                                                                                                      |
|                      | Tail clock pin for an abstract segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                              |
| tail_clock_edge      |                                                                                                                                                      |
|                      | Active tail clock edge at the local clock pin for an abstract segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No               |
| test_modes           |                                                                                                                                                      |
|                      | Test mode pins of the segment and their active values as a Tcl list.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                 |
| type                 |                                                                                                                                                      |
|                      | Type of scan segment.<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                |
| user_defined_segment |                                                                                                                                                      |
|                      | Whether the segment was defined by the user or automatically identified by the tool.<br><b>Type:</b> bool<br><b>Default:</b> true<br><b>Edit:</b> No |

# actual\_scan\_segment

## Parent Objects

[hinst](#), [inst](#), [design](#), [hpin](#), [pin](#), [root](#), [violation](#), [hport](#), [port](#)

## Definition

scan segment

| Attribute                 | Description                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| active                    |                                                                                                                                                          |
|                           | Shift enable active value for an abstract segment.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                       |
| base_name                 |                                                                                                                                                          |
|                           | Name of actual_scan_segment<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                              |
| clock                     |                                                                                                                                                          |
|                           | Head clock pin for an abstract segment.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                  |
| clock_edge                |                                                                                                                                                          |
|                           | Active (head) clock edge at the local clock pin for an abstract segment.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| clock_gating_shift_enable |                                                                                                                                                          |

|                        |                                                                                                                                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>Clock gating shift enable pin for an abstract, preserved or shift_reg segment.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                      |
| connected_scan_clock_a |                                                                                                                                                                                        |
|                        | <p>Whether an abstract or preserved segment has its common scan clock a pin already global connected.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p> |
| connected_scan_clock_b |                                                                                                                                                                                        |
|                        | <p>Whether an abstract or preserved segment has its common scan clock b pin already global connected.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p> |
| connected_shift_enable |                                                                                                                                                                                        |
|                        | <p>Whether an abstract or preserved segment has its common shift enable pin already global connected.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> No</p> |
| core_wrapper           |                                                                                                                                                                                        |
|                        | <p>Whether the segment is for a core wrapper cell.</p> <p><b>Type:</b> bool</p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                   |
| core_wrapper_type      |                                                                                                                                                                                        |
|                        | <p>Specifies the wrapper type.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                        |
| core_wrapper_usage     |                                                                                                                                                                                        |

|                                                |                                                                                                                                                                                                                                                                                     |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Specifies the wrapper usage.</p> <p><b>Type:</b> <a href="#">string</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> Yes</p>                                                                                                                                                    |
| ctl_defined                                    |                                                                                                                                                                                                                                                                                     |
|                                                | <p>Scan segment defined in ctl abstract model.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                                                                    |
| dft_exclude_abstract_segment_from_los_pipeline |                                                                                                                                                                                                                                                                                     |
|                                                | <p>Avoid insertion of launch-off-shift pipelined logic to shift enable of abstract segment.</p> <p><b>Type:</b> <a href="#">bool</a></p> <p><b>Default:</b> false</p> <p><b>Edit:</b> Yes</p>                                                                                       |
| dft_hookup_pin_sdi                             |                                                                                                                                                                                                                                                                                     |
|                                                | <p>Core side hookup pin for SDI.</p> <p><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                   |
| dft_hookup_pin_sdo                             |                                                                                                                                                                                                                                                                                     |
|                                                | <p>Core side hookup pin for SDO.</p> <p><b>Type:</b> <a href="#">obj(pin)</a> <a href="#">obj(hpin)</a> <a href="#">obj(port)</a> <a href="#">obj(hport)</a></p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                   |
| dft_part_of_segment                            |                                                                                                                                                                                                                                                                                     |
|                                                | <p>The type of segment this segment belongs to {abstract   fixed   floating   preserved   shift_register}.</p> <p><b>Type:</b> <a href="#">enum</a></p> <p><b>Enum Values:</b> fixed floating preserve abstract shift_register</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| dft_partition                                  |                                                                                                                                                                                                                                                                                     |

|                                   |                                                                                                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p>Determine the DFT partition of the specified actual_scan_chain/actual_scan_segment/scan_chain/scan_segment.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>           |
| dft_scan_chain                    |                                                                                                                                                                                                          |
|                                   | <p>The actual scan chain this segment belongs to.</p> <p><b>Type:</b> obj(scan_chain) obj(scan_segment) obj(actual_scan_segment) obj(actual_scan_ch</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |
| dft_tail_test_clock               |                                                                                                                                                                                                          |
|                                   | <p>Top level tail test_clock for an abstract segment.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                    |
| dft_tail_test_clock_edge          |                                                                                                                                                                                                          |
|                                   | <p>Active top level tail test_clock edge for an abstract segment.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                        |
| dft_tail_test_clock_waveform_edge |                                                                                                                                                                                                          |
|                                   | <p>Active top level tail test_clock waveform leading or trailing edge for an abstract segment.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                           |
| dft_test_clock                    |                                                                                                                                                                                                          |
|                                   | <p>Top level (head) test_clock for an abstract segment.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                                  |
| dft_test_clock_edge               |                                                                                                                                                                                                          |
|                                   | <p>Active top level (head) test_clock edge for an abstract segment.</p> <p><b>Type:</b> string</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p>                                                      |

|                              |                                                                                                                                                                                                                    |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dft_test_clock_waveform_edge |                                                                                                                                                                                                                    |
|                              | Active top level (head) test_clock waveform leading or trailing edge for an abstract segmen<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                        |
| elements                     |                                                                                                                                                                                                                    |
|                              | Elements of scan segment (a Tcl list).<br><b>Type:</b> <a href="#">object*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                            |
| escaped_name                 |                                                                                                                                                                                                                    |
|                              | Name of actual_scan_segment including escape characters<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                            |
| head_skew_safe               |                                                                                                                                                                                                                    |
|                              | Whether an abstract segment has a lockup element at the head.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> No                                                                     |
| instance                     |                                                                                                                                                                                                                    |
|                              | Instance for an abstract segment.<br><b>Type:</b> <a href="#">obj(inst)</a> <a href="#">obj(hinst)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                    |
| name                         |                                                                                                                                                                                                                    |
|                              | Name of actual_scan_segment<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                        |
| obj_type                     |                                                                                                                                                                                                                    |
|                              | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">actual_scan_segment</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |



|              |                                                                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| other_clocks |                                                                                                                                                        |
|              | Other clock pins of the segment and their active values as a Tcl list.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| power_domain |                                                                                                                                                        |
|              | All the power domains of this scan chain.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                              |
| reg_count    |                                                                                                                                                        |
|              | Number of flip-flops in scan segment.<br><b>Type:</b> <a href="#">int</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                     |
| reorderable  |                                                                                                                                                        |
|              | Whether the preserved segment is reorderable for scanDEF purposes.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> No    |
| scan_in      |                                                                                                                                                        |
|              | SDI pin for an abstract, preserved or shift_reg segment.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No               |
| scan_out     |                                                                                                                                                        |
|              | SDO pin for an abstract, preserved or shift_reg segment.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No               |
| shift_enable |                                                                                                                                                        |
|              | Shift enable pin for an abstract, preserved or shift_reg segment.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No      |

|                 |                                                                                                                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| skew_safe       |                                                                                                                                                        |
|                 | Whether an abstract segment has a lockup element at the end.<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> false<br><b>Edit:</b> No          |
| tail_clock      |                                                                                                                                                        |
|                 | Tail clock pin for an abstract segment.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                |
| tail_clock_edge |                                                                                                                                                        |
|                 | Active tail clock edge at the local clock pin for an abstract segment.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| type            |                                                                                                                                                        |
|                 | Type of scan segment.<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                  |

## isolation\_rule

### Parent Objects

[power\\_domain](#)

### Definition

power\_intent isolation rule

| Attribute  | Description |
|------------|-------------|
| applies_to |             |

|                     |                                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------|
|                     | isolation_rule's applies_to<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No          |
| applies_to_boundary |                                                                                                      |
|                     | isolation_rule's applies_to_boundary<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No |
| clamp_value         |                                                                                                      |
|                     | isolation_rule's clamp_value<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No         |
| diff_supply_only    |                                                                                                      |
|                     | isolation_rule's diff_supply_only<br><b>Type:</b> bool<br><b>Default:</b> ""<br><b>Edit:</b> No      |
| domain              |                                                                                                      |
|                     | isolation_rule's domain<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No              |
| elements            |                                                                                                      |
|                     | isolation_rule's elements<br><b>Type:</b> string*<br><b>Default:</b> ""<br><b>Edit:</b> No           |
| exclude_elements    |                                                                                                      |
|                     | isolation_rule's exclude_elements<br><b>Type:</b> string*<br><b>Default:</b> ""<br><b>Edit:</b> No   |
| isolation_sense     |                                                                                                      |

|                  |                                                                                                                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | isolation_rule's isolation_sense<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                              |
| isolation_signal |                                                                                                                                                                                                               |
|                  | isolation_rule's isolation_signal<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                             |
| isolation_supply |                                                                                                                                                                                                               |
|                  | isolation_rule's isolation_supply<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                             |
| location         |                                                                                                                                                                                                               |
|                  | isolation_rule's location<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                     |
| name             |                                                                                                                                                                                                               |
|                  | isolation_rule's name<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                         |
| no_isolation     |                                                                                                                                                                                                               |
|                  | isolation_rule's no_isolation<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                   |
| obj_type         |                                                                                                                                                                                                               |
|                  | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">isolation_rule</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| sink             |                                                                                                                                                                                                               |

|        |                                                                                                         |
|--------|---------------------------------------------------------------------------------------------------------|
|        | isolation_rule's sink<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No   |
| source |                                                                                                         |
|        | isolation_rule's source<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |

## level\_shifter\_rule

### Parent Objects

[power\\_domain](#)

### Definition

power\_intent level\_shifter rule

| Attribute           | Description                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------|
| applies_to          |                                                                                                                          |
|                     | level_shifter_rule's applies_to<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No          |
| applies_to_boundary |                                                                                                                          |
|                     | level_shifter_rule's applies_to_boundary<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| domain              |                                                                                                                          |

|                  |                                                                                                                         |
|------------------|-------------------------------------------------------------------------------------------------------------------------|
|                  | level_shifter_rule's domain<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No             |
| elements         |                                                                                                                         |
|                  | level_shifter_rule's elements<br><b>Type:</b> <a href="#">string</a> *<br><b>Default:</b> ""<br><b>Edit:</b> No         |
| exclude_elements |                                                                                                                         |
|                  | level_shifter_rule's exclude_elements<br><b>Type:</b> <a href="#">string</a> *<br><b>Default:</b> ""<br><b>Edit:</b> No |
| input_supply     |                                                                                                                         |
|                  | level_shifter_rule's input_supply<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No       |
| location         |                                                                                                                         |
|                  | level_shifter_rule's location<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No           |
| name             |                                                                                                                         |
|                  | level_shifter_rule's name<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No               |
| no_shift         |                                                                                                                         |
|                  | level_shifter_rule's no_shift<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No             |
| obj_type         |                                                                                                                         |

|               |                                                                                                                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">level_shifter_rule</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| output_supply |                                                                                                                                                                                                                   |
|               | level_shifter_rule's output_supply<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                |
| rule          |                                                                                                                                                                                                                   |
|               | level_shifter_rule's rule<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                         |
| sink          |                                                                                                                                                                                                                   |
|               | level_shifter_rule's sink<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                         |
| source        |                                                                                                                                                                                                                   |
|               | level_shifter_rule's source<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                       |

## retention\_rule

### Parent Objects

[power\\_domain](#)

### Definition

power\_intent retention rule

| Attribute         | Description                                                                                                                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| domain            |                                                                                                                                                                                                               |
|                   | retention_rule's domain<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                       |
| elements          |                                                                                                                                                                                                               |
|                   | retention's elements<br><b>Type:</b> <a href="#">string*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                         |
| exclude_elements  |                                                                                                                                                                                                               |
|                   | retention's exclude_elements<br><b>Type:</b> <a href="#">string*</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                 |
| name              |                                                                                                                                                                                                               |
|                   | retention_rule's name<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                         |
| no_retention      |                                                                                                                                                                                                               |
|                   | retention_rule's no_retention<br><b>Type:</b> <a href="#">bool</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                   |
| obj_type          |                                                                                                                                                                                                               |
|                   | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">retention_rule</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| restore_condition |                                                                                                                                                                                                               |



|                                          |                                                                                                                            |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|                                          | retention_rule's restore_condition<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                         |
| restore_signal                           |                                                                                                                            |
|                                          | retention_rule's restore_signal<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                            |
| retention_supply                         |                                                                                                                            |
|                                          | retention_rule's retention_supply<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                          |
| save_condition                           |                                                                                                                            |
|                                          | retention_rule's save_condition<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                            |
| save_signal                              |                                                                                                                            |
|                                          | retention_rule's save_signal<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> No                               |
| zero_pin_live_slave_clock_isolation_cell |                                                                                                                            |
|                                          | retention_rule's zero_pin_live_slave_clock_isolation_cell<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| zero_pin_live_slave_reset_isolation_cell |                                                                                                                            |
|                                          | retention_rule's zero_pin_live_slave_reset_isolation_cell<br><b>Type:</b> string<br><b>Default:</b> ""<br><b>Edit:</b> Yes |
| zero_pin_live_slave_set_isolation_cell   |                                                                                                                            |

|  |                                                                                                                                          |
|--|------------------------------------------------------------------------------------------------------------------------------------------|
|  | retention_rule's zero_pin_live_slave_set_isolation_cell<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> Yes |
|--|------------------------------------------------------------------------------------------------------------------------------------------|

## power\_switch\_rule

### Parent Objects

[power\\_domain](#)

### Definition

power\_intent power\_switch rule

| Attribute         | Description                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------------|
| ack_port          |                                                                                                                   |
|                   | power_switch_rule's ack_port<br><b>Type:</b> <a href="#">string*</a><br><b>Default:</b> ""<br><b>Edit:</b> No     |
| control_port      |                                                                                                                   |
|                   | power_switch_rule's control_port<br><b>Type:</b> <a href="#">string*</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| domain            |                                                                                                                   |
|                   | power_switch_rule's domain<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No        |
| input_supply_port |                                                                                                                   |

|                    |                                                                                                                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | power_switch_rule's input_supply_port<br><b>Type:</b> <a href="#">string</a> *<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                          |
| name               |                                                                                                                                                                                                                  |
|                    | power_switch_rule's name<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                         |
| obj_type           |                                                                                                                                                                                                                  |
|                    | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum</a> ( <a href="#">power_switch_rule</a> )<br><b>Default:</b> ""<br><b>Edit:</b> No |
| off_state          |                                                                                                                                                                                                                  |
|                    | power_switch_rule's off_state<br><b>Type:</b> <a href="#">string</a> *<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                  |
| on_state           |                                                                                                                                                                                                                  |
|                    | power_switch_rule's on_state<br><b>Type:</b> <a href="#">string</a> *<br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                   |
| output_supply_port |                                                                                                                                                                                                                  |
|                    | power_switch_rule's output_supply_port<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                           |
| supply_set         |                                                                                                                                                                                                                  |
|                    | power_switch_rule's supply_set<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                                   |

# ndr\_eol\_spacing

## Parent Objects

## Definition

Return NDR eol spacing

| Attribute | Description                                                                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| layer     |                                                                                                                                                                                              |
|           | The value of LAYER in EOLSPACING property.<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                               |
| obj_type  |                                                                                                                                                                                              |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (ndr_eol_spacing)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |
| spacing   |                                                                                                                                                                                              |
|           | The value of SPACING in EOLSPACING property.<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                  |
| within    |                                                                                                                                                                                              |
|           | The value of WITHIN in EOLSPACING property.<br><b>Type:</b> <a href="#">coord</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                   |

# inst\_physical\_pin

## Parent Objects

## Definition

This presents inst physical pin selected from GUI, It can only be accessed by selecting the object with the GUI.

| Attribute | Description                                                                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base_pin  |                                                                                                                                                                                                |
|           | The pointer to equivalent base pin<br><b>Type:</b> <a href="#">obj(base_pin)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                      |
| inst      |                                                                                                                                                                                                |
|           | The inst of this physical pin<br><b>Type:</b> <a href="#">obj(inst)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                               |
| layer     |                                                                                                                                                                                                |
|           | The layer of the physical pin<br><b>Type:</b> <a href="#">obj(layer)</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                              |
| location  |                                                                                                                                                                                                |
|           | Location of the physical pin.<br><b>Type:</b> <a href="#">point</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                   |
| name      |                                                                                                                                                                                                |
|           | The name of this physical pin<br><b>Type:</b> <a href="#">string</a><br><b>Default:</b> ""<br><b>Edit:</b> No                                                                                  |
| obj_type  |                                                                                                                                                                                                |
|           | The obj_type name for this object. To see all obj_type names, go to <a href="#">Contents</a><br><b>Type:</b> <a href="#">enum (inst_physical_pin)</a><br><b>Default:</b> ""<br><b>Edit:</b> No |

|      |                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------|
| rect |                                                                                                                                       |
|      | <p>The bounding box of all shapes of the physical pin.</p> <p><b>Type:</b> rect*</p> <p><b>Default:</b> ""</p> <p><b>Edit:</b> No</p> |